Alexandrov geometry

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Introduction

In this book, the term Alexandrov space will refer to one of the following types of metric spaces:

- ◇ spaces with curvature bounded below;
- ◇ spaces with curvature bounded above;
- ◇ spaces with two-sided bounded curvature.

All these classes of spaces are defined using metric comparison to a model space, without using any analytical tools.

The goal of this book is to give a comprehensive description of theory of Alexandrov spaces with curvature bounded below (CBB); and in the theory of Alexandrov spaces with curvature bounded above (CBA), about which more is currently available in book form, to cover all the basic material as well as selected topics inspired by considering the two contexts simultaneously. Although the two fields developed quite independently, they have many similar guiding intuitions and technical tools. Our approach is novel in its attention to the interrelatedness of the two fields, and its emphasis on the way each illuminates the other. We only consider inner theory, leaving most applications aside.

History

The first synthetic description of curvature is due to Abraham Wald; it was given in a lonely publication on a "coordinateless description of Gauss surfaces" published in 1936, see [164]. In 1941, similar definitions were rediscovered independently by Aleksandr Danilovich Aleksandrov, see [12]. In Alexandrov's work the first fruitful applications of this approach were given. Mainly:

- ◇ Alexandrov's embedding theorem — *metrics of non-negative curvature on the sphere, and only they, are isometric to closed convex surfaces in Euclidean 3-space.* This theorem appears first in [12] and later in the book [18].

- ◇ Gluing theorem, which gave a flexible tool to modify non-negativly curved metrics on a sphere, see [19].

These two results together gave a very intuitive geometric tool to study embeddings and bending of surfaces in Euclidean space, and changed this subject dramatically. They formed the foundation of branch of geometry now called *Alexandrov geometry.*

Alexandrov geometry can use "back to Euclid" as a slogan. Alexandrov spaces are defined via axioms similar to the one given by Euclid, but certain equalities exchanged to inequalities. Depending on the sign of inequalities we get Alexandrov spaces with *curvature bounded above* and *curvature bounded below.* Althogh the definitions of two classes of spaces are similar, their properties and known applications are qute different.

Curvature bounded below. The theory was developed mostly in the two-dimensional case by Alexandrov and his students Yurii Fedorovich Borisov, Yurii Dmitrievich Burago, Iosif Meerovich Liberman, Sergey Pantelemonovich Olovyanishnikov, Aleksey Vasilyevich Pogorelov, Yurii Grigorievich Reshetnyak, Yurii Alexandrovich Volkov, Viktor Abramovich Zalgaller and others.

Before the 80's there were only few papers on the higher dimensional analog.

- ◇ The work of Anatoliy Dmitrievich Milka, who started to consider multidimensional Alexandrov's spaces with lower curvature bound in 60's; Milka used the global definition similarly to what we do in this book. In particular he proved the splitting theorem for such spaces. Milka was developing the theory of polyhedral spaces with curvature bounded below.
- ◇ The work of Bruce Kleiner on non-negatively curved manifolds with infinite group of symmetries. This work can be considered as the first application of Alexandrov geometry in dimension grater than 2, but Kleiner used Alexandrov geometry only implicitly,

In the 80's the interest in convergence of Riemannian manifolds created by Gromov's compactness theorem ([60]) turned attention toward the singular spaces that can occur as limits of Riemannian manifolds. Immediately it was recognized that if the manifolds have a uniform lower sectional curvature bound, then the limit spaces have a lower curvature bound in the sense of Alexandrov. There followed throughout the 90's an explosion of work starting with papers of Yurij Dmitrievich Burago, Grigori Yakovlevich Perelman and Mikhail Leonidovich Gromov [34] and [124].

The above activity was very much related to comparison geometry. The latter approach is not that maximalistic — it is a branch of differential geometry which compares Riemannian manifolds to spaces of constant curvature. In addition to the already-mentioned Gromov's compactness theorem, the following results had a big influence on the development of Alexandrov's geometry:

- ⋄ Toponogov comparison and splitting theorems; these are generalizations of Cohn-Vossen's theorem,
- ⋄ Finiteness theorems of Cheeger and Grove–Petersen–Wu,
- ⋄ Yamaguchi fibration theorem.

Here is a list of available introductions to Alexandrov spaces with curvature bounded below:

- ⋄ The original paper of Burago, Gromov and Perelman [34] and its extension [124] written by Perelman. It is the first introduction to Alexandrov's geometry.
- ⋄ Shiohame's introduction to Alexandrov's geometry [154]. It is designed to be reader-friendly.
- ⋄ Plaut's survey in Alexandrov's geometry [140] written mostly for topologists. The first 8 sections can be used as an introduction, but it is a bit brief.
- ⋄ [32, Chapter 10] is yet another reader-friendly introduction.

Curvature bounded above. The study of the spaces with curvature bounded above started later. The first paper on the subject was written by Alexandrov, appearing in 1951, see [15]. An analogous weaker definition was considered earlier by Busemann in [35].

The fundamental results in this direction were obtained by Yurii Grigorievich Reshetnyak. It includes his majorization theorem and gluing theorem. The gluing theorem states that if two non-positively curved spaces have isometric convex sets, then the space obtained by gluing these sets along the isometry is also non-positively curved.

One of the most beautiful applications of the gluing theorem was given recently by Dmitry Burago, Sergei Ferleger and Alexey Kanonenko in [33]; in particular it gives an estimate for the number of collisions of a billiard without walls.

Here is a list of available introductions to the Alexandrov spaces with curvature bounded above:

- ⋄ [30]
- ⋄ [23] Lecture notes of Werner Ballmann
- ⋄ [32, Chapter 9] a reader-friendly introduction.

Further development. The theory of spaces with two sided curvature bound was developed by V. N. Berestovskii and I. G. Nikolaev. It turns out that these are in fact non-smooth Riemannian manifolds.

A spin-off of the idea of synthetically defining upper curvature bounds was given by Gromov. He gave an analogous definition for discrete metric spaces, so called δ-hyperbolic spaces. It provided an intuitive geometric tool for studying small cancellation groups.

The curvature dimension condition by Lott, Villani and Sturm is a further generalization of Alexandrov spaces with curvature bounded below; see the book of Villani [162] and references there in.

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Part I

Preliminaries

Chapter 1

Model plane

1.1 Trigonometry

Given a real number κ, the *model* κ*-plane* will be a complete simply connected 2-dimensional Riemannian manifold of constant curvature κ.

The model κ-plane will be denoted by $\mathbb{M}^2[\kappa]$. If $\kappa > 0$, $\mathbb{M}^2[\kappa]$ is isometric to a sphere of radius $\frac{1}{\sqrt{\kappa}}$; if $\kappa = 0$, $\mathbb{M}^2[\kappa]$ is the Euclidean plane; and if $\kappa < 0$, $\mathbb{M}^2[\kappa]$ is the Lobachevsky plane.

Set $\varpi^\kappa = \operatorname{diam} \mathbb{M}^2[\kappa]$, so $\varpi^\kappa = \infty$ if $\kappa \leqslant 0$ and $\varpi^\kappa = \pi/\sqrt{\kappa}$ if $\kappa > 0$.

The distance between points $x, y \in \mathbb{M}^2[\kappa]$ will be denoted by $|x\,y|$, and $[xy]$ will denote the segment connecting x and y. The segment $[xy]$ is uniquely defined if $\kappa \leqslant 0$ and if $\kappa > 0$ we require that $|x\,y| < \varpi^\kappa = \pi/\sqrt{\kappa}$.

A triangle in $\mathbb{M}^2[\kappa]$, with vertices x, y, z will be denoted by $[xyz]$[1]. The angle of $[xyz]$ at x will be denoted by $\measuredangle[x\,{}^{y}_{z}]$.

By $\tilde{\triangle}^\kappa\{a, b, c\}$ we denote a triangle in $\mathbb{M}^2[\kappa]$ with sidelengths a, b, c, so $[xyz] = \tilde{\triangle}^\kappa\{a, b, c\}$ means that $x, y, z \in \mathbb{M}^2[\kappa]$ are such that $|x\,y| = c$, $|y\,z| = a$, $|z\,x| = b$. For $\tilde{\triangle}^\kappa\{a, b, c\}$ to be defined, the sides a, b, c must satisfy the triangle inequality. If $\kappa > 0$, we require in addition that $a + b + c < 2\cdot\varpi^\kappa$; otherwise $\tilde{\triangle}^\kappa\{a, b, c\}$ is considered to be undefined.

Trigonometric functions. We will need three "trigonometric functions" in $\mathbb{M}^2[\kappa]$: sn^κ, cs^κ and md^κ; sn stands for sine; cs, for cosine; and md, for modified distance. They are defined as follows: Let sn^κ,

[1] Formally, a triangle is an ordered set of its sides, so $[xyz]$ is just a short notation for a triple $([yz], [zx], [xy])$.

cs^κ, $\overline{\mathrm{md}^\kappa}\colon \mathbb{R}_{\geqslant} \to \mathbb{R}$ be the solutions of the following IVPs respectively

$$\begin{cases} x'' + kx = 0 \\ x(0) = 0 \\ x''(0) = 1 \end{cases} \qquad \begin{cases} y'' + ky = 0 \\ y(0) = 1 \\ y'(0) = 0 \end{cases} \qquad \begin{cases} z'' + kz = 1 \\ z(0) = 0 \\ z'(0) = 0 \end{cases}$$

Then set

$$\mathrm{md}^\kappa(t) = \begin{cases} \overline{\mathrm{md}^\kappa} \text{ if } 0 \leqslant t \leqslant \varpi^\kappa \\ \frac{2}{k} \text{ if } t > \varpi^\kappa \end{cases}$$

Here are the tables which relate our trigonometric functions to the standard ones, where we take $\kappa > 0$:

$$\begin{aligned} &\mathrm{sn}^{\pm\kappa} = \tfrac{1}{\sqrt{\kappa}}\cdot\mathrm{sn}^{\pm 1}(x\cdot\sqrt{\kappa}); && \mathrm{cs}^{\pm\kappa} = \mathrm{cs}^{\pm 1}(x\cdot\sqrt{\kappa}); \\ &\mathrm{sn}^{-1}x = \sinh x; && \mathrm{cs}^{-1}x = \cosh x; \\ &\mathrm{sn}^{0}x = x; && \mathrm{cs}^{0}x = 1; \\ &\mathrm{sn}^{1}x = \sin x; && \mathrm{cs}^{1}x = \cos x. \end{aligned}$$

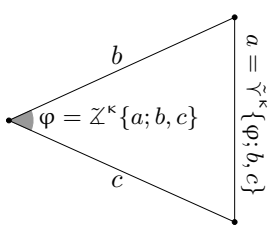
$$\begin{aligned} &\mathrm{md}^{\pm\kappa} = \tfrac{1}{\kappa}\cdot\mathrm{md}^{\pm 1}(x\cdot\sqrt{\kappa}); && \mathrm{md}^{-1}x = \cosh x - 1; \\ &\mathrm{md}^{0}x = \tfrac{1}{2}\cdot x^2; && \mathrm{md}^{1}x = \left[\begin{matrix} 1 - \cos x & \text{for} & x \leqslant \pi \\ 2 & \text{for} & x > \pi \end{matrix}\right.; \end{aligned}$$

Note that

$$\mathrm{md}^\kappa(x) = \int\limits_0^x \mathrm{sn}^\kappa(x)\cdot \boldsymbol{d}x \quad \text{for} \quad x \leqslant \varpi^\kappa$$

Let φ be the angle of $\tilde{\triangle}^\kappa\{a,b,c\}$ which is opposite to a. In this case we will write

$$a = \tilde{\gamma}^\kappa\{\varphi; b, c\} \quad \text{or} \quad \varphi = \measuredangle^\kappa\{a; b, c\}.$$



The functions $\tilde{\gamma}^\kappa$ and $\measuredangle^\kappa$ will be called correspondingly the *model side* and the *model angle*. Set

$$\tilde{\gamma}^\kappa\{\varphi; b, -c\} = \tilde{\gamma}^\kappa\{\varphi; -b, c\} \overset{def}{=\!=} \tilde{\gamma}^\kappa\{\pi - \varphi; b, c\},$$

this way we define $\tilde{\gamma}^\kappa\{\varphi; b, c\}$ for some negative values of b and c.

1.1.1. Properties of standard functions.

a) *For fixed a and φ, the function*

$$y(t) = \mathrm{md}^{\kappa}\left(\tilde{\Upsilon}^{\kappa}\{\varphi; a, t\}\right)$$

satisfies the following differential equation:

$$y'' + \kappa \cdot y = 1.$$

b) *Let $\alpha\colon [a,b] \to \mathbb{M}^2[\kappa]$ be a unit-speed geodesic, and A be the image of a complete geodesic. If $f(t)$ is the distance from $\alpha(t)$ to A, the function*

$$y(t) = \mathrm{sn}^{\kappa}(f(t))$$

satisfies the following differential equation:

$$y'' + \kappa \cdot y = 0.$$

c) *For fixed κ, b and c, the function*

$$a \mapsto \mathcal{Z}^{\kappa}\{a; b, c\}$$

is increasing and defined on a real interval. Equivalently, the function

$$\varphi \mapsto \tilde{\Upsilon}^{\kappa}\{\varphi; b, c\}$$

is increasing and defined if $b, c < \varpi^{\kappa}$ and $\varphi \in [0,\pi]$[2].

d) *For fixed φ, a, b, c, the functions*

$$\kappa \mapsto \mathcal{Z}^{\kappa}\{a; b, c\} \quad \text{and} \quad \kappa \mapsto \tilde{\Upsilon}^{\kappa}\{\varphi; b, c\}$$

are respectively non-decreasing (in fact, decreasing, if $a \notin \{|b-c|, b+c\}$) and non-increasing (in fact, increasing, if $\varphi \notin \{0, \pi\}$).

e) *(Alexandrov's lemma) Assume that for real numbers a, b, a', b', x and κ, the following two expressions are defined*

1. $\mathcal{Z}^{\kappa}\{a; b, x\} + \mathcal{Z}^{\kappa}\{a'; b', x\} - \pi$,
2. $\mathcal{Z}^{\kappa}\{a'; b + b', a\} - \mathcal{Z}^{\kappa}\{x; a, b\}$,

Then they have the same sign.

[2] Formally speaking, if $\kappa > 0$ and $b + c \geqslant \varpi^{\kappa}$, it is defined only for $\varphi \in [0, \pi)$, but $\tilde{\Upsilon}^{\kappa}\{\varphi; b, c\}$ can be extended to $[0, \pi]$ as a continuos function.

Proof. All the properties except Alexandrov's lemma (e) can be shown by direct calculation. Alexandrov's lemma is reformulated in 6.2.1 and is proved there. □

Cosine law.

The above formulas easily imply the cosine law in $\mathbb{M}^2[\kappa]$, which can be expressed as follows

$$\cos\varphi = \begin{cases} \dfrac{b^2+c^2-a^2}{2\cdot b\cdot c} & \text{if } \kappa = 0 \\ \dfrac{\operatorname{cs}^\kappa a - \operatorname{cs}^\kappa b\cdot\operatorname{cs}^\kappa c}{\kappa\cdot\operatorname{sn}^\kappa b\cdot\operatorname{sn}^\kappa c} & \text{if } \kappa \neq 0 \end{cases}$$

However, rather than using these explicit formulas, we mainly will use the properties of $\tilde{\measuredangle}^\kappa$ and $\tilde{\Upsilon}^\kappa$ listed in 1.1.1.

1.2 Hemisphere lemma

1.2.1. Hemisphere lemma. *For $\kappa > 0$, any closed path of length $< 2\cdot\varpi^\kappa$ (respectively, $\leqslant 2\cdot\varpi^\kappa$) in $\mathbb{M}^2[\kappa]$ lies in an open (respectively, closed) hemisphere.*

Proof. By rescaling, we may assume that $\kappa = 1$ and thus $\varpi^\kappa = \pi$ and $\mathbb{M}^2[\kappa] = \mathbb{S}^2$. Let α be a closed curve in $\mathbb{S}^2$ of length $2\cdot\ell$.

Assume $\ell < \pi$. Let $\check{\alpha}$ be a subarc of α of length ℓ, with endpoints p and q. Since $|pq| \leqslant \ell < \pi$, there is a unique geodesic $[pq]$ in $\mathbb{S}^2$. Let z be the midpoint of $[pq]$. We claim that α lies in the open hemisphere centered at z. If not, α intersects the boundary great circle in a point say r. Without loss of generality we may assume that $r \in \check{\alpha}$. The arc $\check{\alpha}$ together with its reflection in z form a closed curve of length $2\cdot\ell$ which passes through r and its antipodal point r'. Thus $\ell = \operatorname{length}\check{\alpha} \geqslant$ $\geqslant |r\,r'| = \pi$, a contradiction.

If $\ell = \pi$, then either α is a local geodesic, and hence a great circle, or α may be strictly shortened by substituting a geodesic arc for a subarc of α whose endpoints p^1, p^2 are arbitrarily close to some point p on α. In the latter case, α lies in a closed hemisphere obtained as a limit of closures of open hemispheres containing the shortened curves as p^1, p^2 approach p. □

1.2.2. Exercise. *Build a proof of Hemisphere lemma 1.2.1 based on Crofton's formula.*

Chapter 2

Metric spaces

In this chapter we fix some conventions assuming that the reader is familiar with basic notions in metric geometry.

2.1 Definitions

2.1.1. Definition. *Let $\underline{\mathcal{X}}$ be a set and*

$$|{*}{*}|\colon \underline{\mathcal{X}} \times \underline{\mathcal{X}} \to [0,\infty]$$

be a function, $(x,y) \mapsto |x\,y|$ which satisfies the following conditions:

a) $|x\,x| = 0$ for any $x \in \underline{\mathcal{X}}$;

b) (symmetry) $|x\,y| = |y\,x|$ for any $x, y \in \underline{\mathcal{X}}$;

c) (triangle inequality) $|x\,z| \leqslant |x\,y| + |y\,z|$ for any $x, y, z \in \underline{\mathcal{X}}$.

Then

- *The pair $\mathcal{X} = (\underline{\mathcal{X}}, |{*}{*}|)$ is called a* $[0,\infty]$-metric space.
- *The set $\underline{\mathcal{X}}$ is called* underlying set *of $\mathcal{X}$. The elements of $\underline{\mathcal{X}}$ are called* points *of the metric space $\mathcal{X}$.*
- *The function $|{*}{*}|$ is called* $[0,\infty]$-metric *of $\mathcal{X}$,*

With a slight abuse of notations we will write $x \in \mathcal{X}$ meaning that x is a point of $\mathcal{X}$.

The value $|x\,y|$ is called the *distance* from x to y. If we need to emphasize that the distance is taken in the space $\mathcal{X}$, we will write $|x\,y|_{\mathcal{X}}$;

The function $|x\,{*}|\colon \underline{\mathcal{X}} \to \mathbb{R}$ defined as $|x\,{*}|\colon y \mapsto |x\,y|$ will be called the *distance function* from x.

We will need three variations of the above definition, which we denote by $(0,\infty]$-metric, $[0,\infty)$-metric and $(0,\infty)$-metric, the intervals

$(0,\infty]$, $[0,\infty)$, $(0,\infty)$ describe possible values for the distance between distinct points in the space.

2.1.2. Definition. *A $[0,\infty]$-metric space is called a $(0,\infty]$-metric space if $|xy| = 0$ if and only if $x = y$. In this case the function $(x,y) \mapsto |xy|$ is called a $(0,\infty]$-metric.*

2.1.3. Definition. *A $[0,\infty]$-metric space is called a $[0,\infty)$-metric space if $|xy| < \infty$ for any two points x, y. In this case the function $(x,y) \mapsto |xy|$ is called a $[0,\infty)$-metric.*

2.1.4. Definition. *A $[0,\infty]$-metric space is called a $(0,\infty)$-metric space if $0 < |xy| < \infty$ for any two points $x \neq y$. In this case the function $(x,y) \mapsto |xy|$ is called a $(0,\infty)$-metric.*

The main object of study in this book are $(0,\infty)$-metric spaces which will be also called *metric spaces*. However, the notion of $[0,\infty]$-metric spaces which allow for infinite distances between points is useful in various definitions and constructions such as the following one

- Assume $\{\mathcal{X}_\alpha\}_{\alpha\in\mathcal{A}}$ be a collection of $[0,\infty]$-metric spaces. The disjoint union

$$\boldsymbol{X} = \bigsqcup_{\alpha\in\mathcal{A}} \mathcal{X}_\alpha$$

  has a natural $[0,\infty]$-metric on it defined as follows: given two points $x \in \mathcal{X}_\alpha$ and $y \in \mathcal{X}_\beta$ set

$$\begin{aligned} |xy|_{\boldsymbol{X}} &= \infty && \text{if } \alpha \neq \beta, \\ |xy|_{\boldsymbol{X}} &= |xy|_{\mathcal{X}_\alpha} && \text{if } \alpha = \beta. \end{aligned}$$

  The obtained metric space $\boldsymbol{X}$ will be called the *disjoint union* of $\{\mathcal{X}_\alpha\}_{\alpha\in\mathcal{A}}$ and denoted as $\bigsqcup_{\alpha\in\mathcal{A}} \mathcal{X}_\alpha$.
- Another example is the construction of length-metric given in Section 2.3.

Now let us give some constructions which show that allowing distances between distinct point might to vanish can also be useful.

- Suppose a set $\mathcal{X}$ comes with a set of $[0,\infty]$-metrics $|{*}{*}|_\alpha$ for $\alpha \in \mathcal{A}$. Then set

$$|xy| = \inf\{\, |xy|_\alpha \mid \alpha \in \mathcal{A} \,\}.$$

  Clearly, $|{*}{*}|$ is a $[0,\infty]$-metric. But it need not be a $(0,\infty]$-metric even if all $|{*}{*}|_\alpha$ are $(0,\infty)$-metrics.

- ◇ Let $\mathcal{X}$ be a set, $\mathcal{Y}$ be a metric space and let $\Phi\colon \mathcal{X} \to \mathcal{Y}$ be a map. Then one can consider the *pullback metric* on $\mathcal{X}$ defined as
$$|x\,y|_{\mathcal{X}} = |\Phi(x)\ \Phi(y)|_{\mathcal{Y}}$$
for any $x, y \in \mathcal{X}$.
Note that the pullback metric $|{*}{*}|_{\mathcal{X}}$ is only $[0,\infty)$ even if the metric on $\mathcal{Y}$ is $(0,\infty)$, since Φ need not be injective.

On the other hand, as we will see below, nearly any question about the other 3 types of metric spaces can be reduced to a question about $(0,\infty)$-metric spaces.

- ◇ If $\mathcal{X}$ is a $[0,\infty]$-metric space and $x, x' \in \mathcal{X}$ are such that $|x\,x'| = 0$ then $|y\,x| = |y\,x'|$ for any $y \in \mathcal{X}$. Thus, $|{*}{*}|$ defines a metric on the quotient set $\underline{\mathcal{X}}/{\sim}$ with the equivalence relation $x \sim y$ iff $|x\,y| = 0$. This way we obtain a $(0,\infty]$-metric space $\mathcal{X}'$. Often we will not distinguish between the $(0,\infty]$-metric space $\mathcal{X}'$ and the $[0,\infty]$-metric space $\mathcal{X}$.
- ◇ The relation $x \sim y$ iff $|x\,y| < \infty$ is an equivalence relation. The equivalence class of a point $x \in \mathcal{X}$ will be called the *metric component* of x; it will be denoted as $\mathcal{X}_x = \mathrm{B}(x,\infty)$. Thus, any metric space is a disjoint union of its metric components in the sense defined above.

Related definitions. For a metric space as above,

- ◇ Given $R \in [0,\infty]$ and $x \in \mathcal{X}$, the sets
$$\mathrm{B}(x,R) = \{y \in \mathcal{X} \mid |x\,y| < R\},$$
$$\overline{\mathrm{B}}[x,R] = \{y \in \mathcal{X} \mid |x\,y| \leqslant R\}$$
are called respectively the *open* and the *closed balls* of radius R with center x.
  - ◦ If we need to emphasize that these balls are taken in the metric space $\mathcal{X}$, we write $\mathrm{B}(x,R)_{\mathcal{X}}$ and $\overline{\mathrm{B}}[x,R]_{\mathcal{X}}$ correspondingly.
  - ◦ In case $\mathcal{X} = \mathbb{M}^m[\kappa]$, we often will not need to specify the center of the ball, so we may write $\mathrm{B}(R)_{\mathbb{M}^m[\kappa]}$ and $\overline{\mathrm{B}}[R]_{\mathbb{M}^m[\kappa]}$ correspondingly.
- ◇ A set $A \subset \mathcal{X}$ is called *bounded* if $A \subset \mathrm{B}(x,R)$ for some $x \in \mathcal{X}$ and $R < \infty$.
- ◇ For subsets $A, B \subset \mathcal{X}$, we define the distance between A and B as
$$|A\,B| \overset{def}{=\!=} \inf\{\,|a\,b| \mid a \in A,\ b \in B\,\}\,.$$

- For any subset $A \subset \mathcal{X}$, the sets

$$\mathrm{B}(A,R) = \{y \in \mathcal{X} \mid |Ay| < R\},$$
$$\overline{\mathrm{B}}[A,R] = \{y \in \mathcal{X} \mid |Ay| \leqslant R\}$$

  are called respectively the *open* and *closed R-neighborhoods* of A.
- Similarly, define

$$\mathrm{S}[A,R] = \{y \in \mathcal{X} \mid |Ay| = R\},$$

  In particular, if $A = \{x\}$ is a one-point set in $\mathcal{X}$ then $\mathrm{S}[x,R]$ is the metric sphere of radius R around x.
- For $0 < r < R$, let

$$\mathrm{A}[A,r,R] = \{x \in \mathcal{L} | \text{ such that } r \leqslant |xA| \leqslant R\}$$

  be the closed annulus around A.
- Any subset A in a metric space $\mathcal{X}$ will be also considered as a *subspace*; i.e. a metric space with the metric defined as restriction of the metric of $\mathcal{X}$ to $A \times A \subset \mathcal{X} \times \mathcal{X}$.

Types of metric spaces.

- A metric space $\mathcal{X}$ is called *complete* if any *Cauchy sequence* in $\mathcal{X}$ converges. Namely, a sequence of points $x_n \in \mathcal{X}$ is called a *Cauchy sequence* if for any $\varepsilon > 0$, we have $|x_n x_m| < \varepsilon$ for all large enough n and m. A sequence of points $x_n \in \mathcal{X}$ is called *convergent* if there is $x \in \mathcal{X}$ such that for any $\varepsilon > 0$ we have $|x_n x| < \varepsilon$ for all large enough n.
  Let $\mathcal{X}$ be arbitrary metric space. Given two Cauchy sequences (x_n) and (x'_n) in $\mathcal{X}$, the following limit is always defined:

$$\rho((x_n),(x'_n)) = \lim_{n\to\infty} |x_n x'_n|_{\mathcal{X}}$$

  and ρ gives a pseudometric on the set of all Cauchy sequences in $\mathcal{X}$. The corresponding metric space $\mathcal{Z}$ is called the *completion* of $\mathcal{X}$, briefly, $\mathcal{Z} = \overline{\mathcal{X}}$.
  Note that there is a natural embedding $\mathcal{X} \hookrightarrow \overline{\mathcal{X}}$; each point $x \in \mathcal{X}$ is mapped to a constant sequence $(x, x, \dots)$. Thus the space $\mathcal{X}$ can be (and will be) considered as a subset of $\overline{\mathcal{X}}$.
- A metric space $\mathcal{X}$ is called *locally complete* if for each point $p \in \mathcal{X}$ there is some $R > 0$ such that the closed ball $\overline{\mathrm{B}}[p,R] \subset \mathcal{X}$ is complete.

- ◇ A metric space $\mathcal{X}$ is called *proper* if all closed bounded sets in $\mathcal{X}$ are compact. This condition is equivalent to each of the following statements:
  1. For some (and therefore any) point $p \in \mathcal{X}$ and any $R < \infty$, the closed ball $\overline{\mathrm{B}}[p, R] \subset \mathcal{X}$ is compact.
  2. The function $|p_*| \colon \mathcal{X} \to \mathbb{R}$ is proper for some (and therefore any) point $p \in \mathcal{X}$.

2.1.5. Proposition. *Proper metric spaces are separable/second countable.*

Proof. Let $\mathcal{X}$ be proper metric space and let $p \in \mathcal{X}$. Since $\mathcal{X} = \cup \cup_{n=1}^{\infty} \overline{\mathrm{B}}[p, n]$, to see that X is separable it's clearly enough to construct a countable dense subset in each $\overline{\mathrm{B}}[p, n]$. For every $m \geqslant 1$ take an open cover of $\overline{\mathrm{B}}[p, n]$ by open $1/m$-balls. By compactness of $\overline{\mathrm{B}}[p, n]$ it admits a finite subcover $\mathrm{B}(q_1^m, 1/m), \dots, \mathrm{B}(q_{N_m}^m, 1/m)$. Then $\cup_{m,j}\{q_j^m\}$ is a countable dense subset of $\overline{\mathrm{B}}[p, n]$. Lastly, to see that $\mathcal{X}$ is second countable, let $\{p_i\}_{i=1}^{\infty}$ be a countable dense subset in $\mathcal{X}$.
Then $\cup_{i,j \geqslant 1} \mathrm{B}(p_i, \frac{1}{j})$ is a countable basis of topology of $\mathcal{X}$. □

2.1.6. Proposition. *Let $\mathcal{X}$ be a metric space. Then the following are equivalent*

i) *$\mathcal{X}$ is compact;*

ii) *$\mathcal{X}$ is* sequentially compact*, i.e. any sequence of points in $\mathcal{X}$ contains a convergent subsequence;*

iii) *$\mathcal{X}$ is complete and for any $\varepsilon > 0$ there is a finite ε-net in $\mathcal{X}$, i.e. there is a finite collection of points $p_1, \dots, p_N$ such that $\cup_i \mathrm{B}(p_i, \varepsilon) = \mathcal{X}$.*

Proof. i) $\Rightarrow$ ii) and ii) $\Rightarrow$ iii) are obvious. We will leave iii) $\Rightarrow$ ii) as an exercise.

Let us prove that ii) implies i). Let $\{U_\alpha\}_{\alpha \in \mathcal{A}}$ be an open cover of $\mathcal{X}$. We claim that there is an $\varepsilon > 0$ such that for any $p \in \mathcal{X}$ there is $\alpha \in \mathcal{A}$ such that $\mathrm{B}(p_i, \varepsilon) \subset U_\alpha$. Such ε is called the *Lebesgue number* of the cover $\{U_\alpha\}_{\alpha \in \mathcal{A}}$.

Suppose no such ε exists. Then there exists a sequence of points $p_i \in \mathcal{X}$ such that $\mathrm{B}(p_i, 1/i)$ is not contained in any U_α. By sequential compactness by passing to a subsequence we can assume that $p_i \to \to p$. Then $p \in U_{\alpha_0}$ for some α_0 and since U_{α_0} is open it follows that $\mathrm{B}(p_i, 1/i) \subset U_{\alpha_0}$ for all large i. This is a contradiction and hence the cover $\{U_\alpha\}_{\alpha \in \mathcal{A}}$ admits a positive Lebesgue number ε. Then by iii) there is a finite ε net $p_1, \dots, p_N$ in $\mathcal{X}$. By the choice of ε for each i there is α_i such that $\mathrm{B}(p_i, \varepsilon) \subset U_{\alpha_i}$. Then $\cup_{i=1}^{N} U_{\alpha_i} = \mathcal{X}$. □

A subset of a metric space is called G-delta set if it can be presented as an intersection of countable number of open subset.

2.1.7. Bair's theorem. *Let $\mathcal{X}$ be a complete metric space and $\{\Omega_n\}$, $n \in \mathbb{N}$ be a collection of open dense subsets of $\mathcal{X}$. Then $\bigcap_{n=1}^{\infty} \Omega_n$ is dense in $\mathcal{X}$.*

Proof. Let $U \subset \mathcal{X}$ be open. We need to show that $U \cap \left(\bigcap_{n=1}^{\infty} \Omega_n\right) \neq \varnothing$. Since Ω_1 is open and dense, there is $p_1 \in \mathcal{X}$ and $r_1 > 0$ such that $\overline{\mathrm{B}}[p_1, r_1] \subset U \cap \Omega_1$. Since Ω_2 is open and dense there is $p_2 \in \in \mathcal{X}, 0 < r_2 < r_1/2$ such that $\overline{\mathrm{B}}[p_2, r_2] \subset \mathrm{B}(p_1, r_1) \cap \Omega_2 \subset U \cap \Omega_1 \cap \Omega_2$. Proceeding by induction we can construct a sequence of balls $\overline{\mathrm{B}}(p_i, r_i)$ such that $0 < r_i < \frac{r_{i-1}}{2}$ satisfying $\overline{\mathrm{B}}[p_i, r_i] \subset \mathrm{B}(p_{i-1}, r_{i-1}) \cap \Omega_i \subset \subset U \cap \left(\bigcap_{k=1}^{i} \Omega_k\right)$. Then the sequence $\{p_i\}$ is Cauchy and hence, by completeness of $\mathcal{X}$ we have that $p_i \to p$. By construction, $p_i \in \in \overline{\mathrm{B}}[p_j, r_j]$ for all $i \geqslant j$ and hence the same is true for p. Thus, $p \in U \cap \left(\bigcap_{n=1}^{\infty} \Omega_n\right) \neq \varnothing$. □

Recall that a family of maps between metric spaces $\{f_\alpha \colon \mathcal{X} \to \to \mathcal{Y}\}_{\alpha \in \mathcal{A}}$ is called *equicontinuous* if for any $\varepsilon > 0$ there is $\delta > 0$ such that for any $p, q \in \mathcal{X}$ with $|pq| < \delta$ and any $\alpha \in \mathcal{A}$ it holds that $|f(p)\ f(q)| < \varepsilon$.

2.1.8. Arzelà-Ascoli theorem. *[30, Lemma 3.10]*

Let $\{f_\alpha \colon \mathcal{X} \to \mathcal{Y}\}_{\alpha \in \mathcal{A}}$ be an equicontinuous family of maps where $\mathcal{Y}$ is compact and $\mathcal{X}$ is separable.

Then $\{f_\alpha\}$ contains a uniformly convergent subsequence $f_{\alpha_i} \to \to f \colon \mathcal{X} \to \mathcal{Y}$.

2.2 Invariants of metric spaces

Let $\mathcal{X}$ be a metric space. Then

- ⋄ The *diameter* of $\mathcal{X}$ is defined as

$$\operatorname{diam} \mathcal{X} = \sup\{\, |xy| \mid x, y \in \mathcal{X} \,\}.$$

- ⋄ The *radius* of $\mathcal{X}$ is defined as

$$\operatorname{rad} \mathcal{X} = \inf\{\, R > 0 \mid \mathrm{B}(x, R) = \mathcal{X} \text{ for some } x \in \mathcal{X} \,\}.$$

- ⋄ The packing number *ε-pack* of $\mathcal{X}$ is the maximal number (possibly infinite) of points in $\mathcal{X}$ at distance $> \varepsilon$ from each other; it is denoted by $\operatorname{pack}_\varepsilon \mathcal{X}$. If $m = \operatorname{pack}_\varepsilon \mathcal{X} < \infty$, then a set $\{x^1, x^2, \dots, x^m\}$ in $\mathcal{X}$ such that $|x^i x^j| > \varepsilon$ is called a *maximal ε-packing* in $\mathcal{X}$.

2.3 Length spaces

2.3.1. Definition. *Let $\mathcal{X}$ be a metric space. Given $[a,b] \subset [-\infty,\infty]$ and a continuous curve $\alpha\colon [a,b] \to \mathcal{X}$ we define its* length *as*

$$\operatorname{length}\alpha \overset{def}{=\!=} \sup_{a\leqslant t_0\leqslant t_1\leqslant\ldots\leqslant t_n\leqslant b} \Sigma_i |\alpha(t_i)\ \alpha_{i+1}|$$

It is easy to see that if $\tau\colon [c,d] \to [a,b]$ is a homeomorphism then $\operatorname{length}\alpha = \operatorname{length}(\alpha\circ\tau)$.

Given two points x and y in a metric space $\mathcal{X}$ consider the value

$$\|x\,y\| = \inf_{\alpha}\{\operatorname{length}\alpha\},$$

where infimum is taken for all paths α from x to y.

It is easy to see that $\|{*}{*}\|$ defines a metric on $\mathcal{X}$; it will be called the *length-metric* on $\mathcal{X}$.

2.3.2. Definition. *If $\|x\,y\| = |x\,y|$ for any pair of points $x,y \in \mathcal{X}$ then $\mathcal{X}$ is called a* length space.

In other words, a metric space $\mathcal{X}$ is a *length space* if for any $\varepsilon > 0$ and any two points $x,y \in \mathcal{X}$ with $|x\,y| < \infty$ there is a path $\alpha\colon [0,1] \to \mathcal{X}$ connecting[1] x to y such that

$$\operatorname{length}\alpha < |x\,y| + \varepsilon.$$

Let us give a few more variations of the notion of length spaces.

- ◇ Assume $R > 0$. If $\|x\,y\| = |x\,y|$ for any pair $|x\,y| < R$ then $\mathcal{X}$ is called an *R-length space*.
- ◇ If any point in $\mathcal{X}$ admits a neighborhood Ω such that $\|x\,y\| = |x\,y|$ for any pair of points $x,y \in \Omega$ then $\mathcal{X}$ is called a *locally length space*.
- ◇ A metric space $\mathcal{X}$ is called *geodesic* if for any two points $x,y \in \mathcal{X}$ with $|x\,y| < \infty$ there is a geodesic $[xy]$ in $\mathcal{X}$.
- ◇ Assume $R > 0$. A metric space $\mathcal{X}$ is called *R-geodesic* if for any two points $x,y \in \mathcal{X}$ with $|x\,y| < R$ there is a geodesic $[xy]$ in $\mathcal{X}$.

Note that the notions of ∞-length spaces and length spaces are the same. Clearly, any geodesic space is a length space and any R-geodesic space is an R-length space.

Example. Let $\mathcal{X}$ be obtained by gluing a countable collection of disjoint intervals I_i of length $1 + 1/i$ where for each I_i one end is

[1] I.e., such that $\alpha(0) = x$ and $\alpha(1) = y$.

glued to $p = \{0\}$ and the other to $q = \{1\}$. Then $\mathcal{X}$ carries a natural complete length metric with respect to which $|pq| = 1$ but there is no geodesic connecting p to q.

2.3.3. Exercise. *Let $\mathcal{X}$ be a metric space and $\|{*}{*}\|$ is the length metric on it. Show the following*

a) If complete $(\mathcal{X}, \|{*}{*}\|)$ is complete.

b) If $\mathcal{X}$ is compact then $(\mathcal{X}, \|{*}{*}\|)$ is geodesic.

2.3.4. Exercise. *Give an example of a complete length space for which no pair of distinct points can be joined by a geodesic.*

In this book, most of the time we consider length spaces. If $\mathcal{X}$ is length space, and $A \subset \mathcal{X}$. The set A comes with the inherited metric from $\mathcal{X}$ which might be not a length-metric. The corresponding length-metric on A will be denoted as $|{*}{*}|_A$.

2.3.5. Definition. *Let $\mathcal{X}$ be a metric space and $x, y \in \mathcal{X}$.*

(i) A point $z \in \mathcal{X}$ is called a midpoint *of x and y if*

$$|xz| = |yz| = \tfrac{1}{2}\cdot|xy|.$$

(ii) Assume $\varepsilon \geqslant 0$. A point $z \in \mathcal{X}$ is called ε-midpoint *of x and y if*

$$|xz|, |yz| < \tfrac{1}{2}\cdot|xy| + \varepsilon.$$

Note that a 0-midpoint is the same as a midpoint.

2.3.6. Lemma. *Let $\mathcal{X}$ be a complete metric space and $R > 0$.*

a) Assume that for any pair of points $x, y \in \mathcal{X}$ such that $|xy| < R$ and any $\varepsilon > 0$ there is a ε-midpoint z. Then the space $\mathcal{X}$ is an R-length space.

b) Assume that for any pair of points $x, y \in \mathcal{X}$ such that $|xy| < R$ there is a midpoint z. Then the space $\mathcal{X}$ is an R-geodesic space.

Proof. Let $x, y \in \mathcal{X}$ be a pair of points such that $|xy| < R$.

Set $\varepsilon_n = \frac{\varepsilon}{2^{2\cdot n}}$.

Set $\alpha(0) = x$ and $\alpha(1) = y$.

Set $\alpha(\frac{1}{2})$ to be an ε_1-midpoint of $\alpha(0)$ and $\alpha(1)$. Further, set $\alpha(\frac{1}{4})$ and $\alpha(\frac{3}{4})$ to be ε_2-midpoints for the pairs $(\alpha(0), \alpha(\frac{1}{2})$ and $(\alpha(\frac{1}{2}), \alpha(1)$ respectively. Applying the above procedure recursively, on the n-th step we define $\alpha(\frac{k}{2^n})$ for every odd integer k such that $0 < \frac{k}{2^n} < 1$, as an ε_n-midpoint of the already defined $\alpha(\frac{k-1}{2^n})$ and $\alpha(\frac{k+1}{2^n})$.

In this way we define $\alpha(t)$ for $t \in W$, where W denotes the set of dyadic rationals in $[0, 1]$. For any $t \in [0, 1]$ consider a sequence of

$t_n \in W$ such that $t_n \to t$ as $n \to \infty$. Note that the sequence $\alpha(t_n)$ converges; define $\alpha(t)$ as its limit. It is easy to see that $\alpha(t)$ does not depend on the choice of the sequence t_n and $\alpha\colon [0,1] \to \mathcal{X}$ is a path from x to y. Moreover,

❶
$$\begin{aligned}\operatorname{length}\alpha &\leqslant |xy| + \sum_{n=1}^{\infty} 2^{n-1}\cdot\varepsilon_n \leqslant \\ &\leqslant |xy| + \tfrac{\varepsilon}{2}.\end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, we get (a).

To prove (b), one should repeat the same argument taking midpoints instead of ε_n-midpoints. In this case ❶ holds for $\varepsilon_n = \varepsilon = 0$. □

Since in a compact space a sequence of $1/n$-midpoints z_n contains a convergent subsequence, Lemma 2.3.6 immediately implies

2.3.7. Proposition. *A proper length space is geodesic.*

2.3.8. Hopf–Rinow theorem. *Any complete, locally compact length space is proper.*

Proof. Let $\mathcal{X}$ be a locally compact length space. Given $x \in \mathcal{X}$, denote by $\rho(x)$ the supremum of all $R > 0$ such that the closed ball $\overline{\mathrm{B}}[x,R]$ is compact. Since $\mathcal{X}$ is locally compact

❷
$$\rho(x) > 0 \quad \text{for any} \quad x \in \mathcal{X}.$$

It is sufficient to show that $\rho(x) = \infty$ for some (and therefore any) point $x \in \mathcal{X}$.

Assume the contrary; i.e. $\rho(x) < \infty$.

❸ $B = \overline{\mathrm{B}}[x, \rho(x)]$ *is compact for any* x.

Indeed, $\mathcal{X}$ is a length space; therefore for any $\varepsilon > 0$, the set $\overline{\mathrm{B}}[x, \rho(x) - \varepsilon]$ is a compact ε-net in B. Since B is closed and hence complete, it has to be compact by Proposition 2.1.6. △

❹ $|\rho(x) - \rho(y)| \leqslant |xy|_{\mathcal{X}}$, *in particular* $\rho\colon \mathcal{X} \to \mathbb{R}$ *is a continuous function.*

Indeed, assume the contrary; i.e., $\rho(x) + |x-y| < \rho(y)$ for some $x, y \in \mathcal{X}$. Then $\overline{\mathrm{B}}[x, \rho(x) + \varepsilon]$ is a closed subset of $\overline{\mathrm{B}}[y, \rho(y)]$ for some $\varepsilon > 0$. Then compactness of $\overline{\mathrm{B}}[y, \rho(y)]$ implies compactness of $\overline{\mathrm{B}}[x, \rho(x) + \varepsilon]$, a contradiction. △

Set $\varepsilon = \min_{y \in B}\{\rho(y)\}$; the minimum is defined since B is compact. From ❷, we have $\varepsilon > 0$.

Choose a finite $\frac{\varepsilon}{10}$-net $\{a_1, a_2, \dots, a_n\}$ in B. The union W of the closed balls $\overline{\mathrm{B}}[a_i, \varepsilon]$ is compact. Clearly $\overline{\mathrm{B}}[x, \rho(x)+\frac{\varepsilon}{10}] \subset W$. Therefore $\overline{\mathrm{B}}[x, \rho(x) + \frac{\varepsilon}{10}]$ is compact; a contradiction. $\square$

2.3.9. Exercise. *(from [30]) Construct a geodesic space which is locally compact, but whose completion is neither geodesic nor locally compact.*

2.4 Convex sets

2.4.1. Definition. *Let $\mathcal{X}$ be a metric space. A set $A \subset \mathcal{X}$ is called* convex *if for every two points $p, q \in A$ there every geodesic $[pq]$ of $\mathcal{X}$ lies in A. It is called* weakly convex *if for every two points $p, q \in A$ there is a geodesic $[pq]$ of $\mathcal{X}$ which lies in A. If in addition every local geodesic from p to q lies in A, then A is called* totally convex

If these definition is applied only when $|pq| < R$, $R \in (0, \infty]$ and only for the geodesics of length $< R$ then A is called correspondingly R-convex, weakly R-convex *and* totally R-convex *.*

A set $A \subset \mathcal{X}$ is called locally convex *if every point $a \in A$ admits an open neighborhood $\Omega \ni a$ suc that for every two points $p, q \in A \cap \Omega$ every geodesic $[pq] \subset \Omega$ lies in A. Similarly one defines locally weakly convex and locally totally convex sets.*

Remarks.

- ⋄ Note that *(weakly) convex sets* can be also regarded as *(weakly) ∞-convex sets.*
- ⋄ Note that inherited metric on a weakly convex set coinsides with its length-metric.
- ⋄ Note that any open set is locally convex by definition.

The following proposition states that weak convexity survive under ulralimit. An analogus statement about convexity does not hold; for example there is a sequence of convex discs in $\mathbb{S}^2$ which converges to the half sphere $\mathbb{S}^2_+$, which is not convex.

2.4.2. Proposition. *Let $(\mathcal{X}_n, \star_n)$ be a sequence of metric spaces with marked points. Assume $A_n \subset \mathcal{X}_n$ be a sequence of weakly convex sets and $\mathcal{X}_n \to \mathcal{X}_\omega$ and $A_n \to A_\omega \subset \mathcal{X}_\omega$ as $n \to \omega$. Then A_ω is a weakly convex set of $\mathcal{X}_\omega$.*

Proof. Fix $x_\omega, y_\omega \in A_\omega$. Consider the sequences $x_n, y_n \in A_n$ such that $x_n \to x_\omega$ and $y_n \to y_\omega$ as $n \to \omega$.

Denote by α_n a geodesic path from x_n to y_n which lies in A_n. Set

$$\alpha_\omega(t) = \lim_{n \to \omega} \alpha_n(t).$$

Note that α_ω is a geodesic path which lies in A_ω. Hence the result follows. $\square$

2.5 Quotient spaces

Quotient spaces. Assume $\mathcal{X}$ is a metric space with an equivalence relation $\sim$. Note that given two pseudometrics ρ_1 and ρ_2 on $\mathcal{X}/\!\sim$ their maximum

$$\rho(x,y) = \max\{\rho_1(x,y), \rho_2(x,y)\}$$

is also a pseudometric; also if for these two pseudometric ρ_1 and ρ_2 the projection $\mathcal{X} \to \mathcal{X}/\!\sim$ is short then so it true for $\rho(x,y)$.

It follows that the quotient space $\mathcal{X}/\!\sim$ admits natural pseudometric; this is the maximal pseudometric on $\mathcal{X}/\!\sim$ which makes the quotient map $\mathcal{X} \to \mathcal{X}/\!\sim$ to be short. The corresponding metric space will be also denoted as $\mathcal{X}/\!\sim$ and will be called the *quotient space* of $\mathcal{X}$ by the equivalence relation $\sim$.

In general case the points of the metric space $\mathcal{X}/\!\sim$ are formed by equivalence classes in $\mathcal{X}$ for a wider equivalence relation, say $\approx$; i.e., such that $x \sim y$ implies $x \approx y$ for any $x, y \in \mathcal{X}$. But in the most of the cases we will consider, the set of equivalence classes will coincide with the set of points in the metric space $\mathcal{X}/\!\sim$.

2.5.1. Proposition. *Let $\mathcal{X}$ be a length space and $\sim$ be an equivalence relation on $\mathcal{X}$ then $\mathcal{X}/\!\sim$ is an length space.*

Proof. Let $\mathcal{Y}$ be arbitrary metric space. Since $\mathcal{X}$ is a length space, the map $f\colon \mathcal{X} \to \mathcal{Y}$ is short if and only if

$$\operatorname{length}(f \circ \alpha) \leqslant \operatorname{length} \alpha$$

for any curve $\alpha\colon \mathbb{I} \to \mathcal{X}$.

Denote by $\|{*}{*}\|$ the length-metric on $\mathcal{Y}$. It follows that if $f\colon \mathcal{X} \to \mathcal{Y}$ is short then so is $f\colon \mathcal{X} \to (\mathcal{Y}, \|{*}{*}\|)$.

Consider the quotient map $f\colon \mathcal{X} \to \mathcal{X}/\!\sim$. Recall that the space $\mathcal{X}/\!\sim$ is defined by the maximal pseudometric which makes f to be short.

Denote by $\|{*}{*}\|$ the length-metric on $\mathcal{X}/\!\sim$. It follows that

$$f\colon \mathcal{X} \to (\mathcal{X}/\!\sim, \|{*}{*}\|)$$

is also short.

Note that

$$\|xy\| \geqslant |xy|_{\mathcal{X}/\sim}$$

for any $x, y \in \mathcal{X}/{\sim}$. From maximality of $|{**}|_{\mathcal{X}/\sim}$, we get

$$\|x\,y\| = |x\,y|_{\mathcal{X}/\sim}$$

for any $x, y \in \mathcal{X}/{\sim}$; i.e., $\mathcal{X}/{\sim}$ is a length space. □

Gluing. Recal that disjoint union of metric spaces can be also considered as a metric. Therefore the above construction works as well for an equivalence relation on a disjoint union of metric spaces.

Consider two metric spaces $\mathcal{X}_1$ and $\mathcal{X}_2$ with subset $A_1 \subset \mathcal{X}_1$ and $A_2 \subset \mathcal{X}_2$ and a bijection $\varphi\colon A_1 \to A_2$. Consider the minimal equivalence relation on $\mathcal{X}_1 \sqcup \mathcal{X}_2$ such that $a \sim \varphi(a)$ for any $a \in A_1$. In this case, we say that the corresponding quotient space $(\mathcal{X}_1 \sqcup \mathcal{X}_2)/{\sim}$ will be called *gluing* of $\mathcal{X}$ and $\mathcal{Y}$ along φ and denoted as

$$\mathcal{X}_1 \sqcup_\varphi \mathcal{X}_2.$$

Assume that the map $\varphi\colon A_1 \to A_2$ is distance preserving for the inherited metrics. Then the projections $\iota_i\colon \mathcal{X}_i \to \mathcal{X}_1 \sqcup_\varphi \mathcal{X}_2$ are also distance preserving and

$$|\iota_1(x_1)\ \iota_2(x_2)|_{\mathcal{X}_1 \sqcup_\varphi \mathcal{X}_2} = \inf\left\{\, |x_1\,a_1|_{\mathcal{X}_1} + |x_2\,a_2|_{\mathcal{X}_2} \;\middle|\; a_2 = \varphi(a_1) \,\right\}$$

for any $x_1 \in \mathcal{X}_1$ and $x_2 \in \mathcal{X}_2$.

Group actions. Assume a group G acts on a metric space $\mathcal{X}$. Consider relation $\sim$ on $\mathcal{X}$ defined as $x \sim y$ if there is $g \in G$ such that $x = g{\cdot}y$. Note that $\sim$ is an equivalence relation.

In this case the quotiont space $\mathcal{X}/{\sim}$ will ba also denoted as $\mathcal{X}/G$.

Assume G acts on $\mathcal{X}$ by isometrics and it has closed orbits.

The case of isometric actions will be important for us.

Assume that the group G acts on $\mathcal{X}$ by isometries and it has closed orbits. Let us denote by $\mathcal{X}/G$ the space of orbits of G.

Let us equip $\mathcal{X}/G$ with a metric. Given two orbits $G{\cdot}x$ and $G{\cdot}y$ in $\mathcal{X}/G$ set

$$|G{\cdot}x\ G{\cdot}y|_{\mathcal{X}/G} = \inf\left\{\, |x\,g{\cdot}y|_{\mathcal{X}} = |g^{-1}{\cdot}x\ y|_{\mathcal{X}} \;\middle|\; g \in G \,\right\}.$$

Note that since G-orbits assumed to be closed $|G{\cdot}x\ G{\cdot}y|_{\mathcal{X}/G} = 0$ if and only if $G{\cdot}x = G{\cdot}y$. It straightforward to check the remaining axioms of metric.

2.5.2. Proposition. *Assume $\mathcal{X}$ is metric space and group G acts on $\mathcal{X}$ by isometries. Then the projection $\mathcal{X} \to \mathcal{X}/G$ is a submetry.*

Proof. We need to show that the quotient map is short ans 1-co-Lipschitz. Both statements follow from the definition of quotient space. □

2.6 Doubling

Let $\mathcal{V}$ be a metric space and $A \subset \mathcal{V}$ be a closed subset. A metric space $\mathcal{W}$ glued from two copies of $\mathcal{V}$ along A is called *doubling of $\mathcal{V}$ in A*.

More formally the space $\mathcal{W}$ can be described by the following properties.

- ⋄ The space $\mathcal{W}$ contains $\mathcal{V}$ as a subspace; in particular the set A can be treated as a subset of $\mathcal{W}$.
- ⋄ There is an isometric involution of $\mathcal{W}$ which is called *reflection in A*; further it will be denoted as $x \mapsto x'$.
- ⋄ For any $x \in \mathcal{W}$ we have $x \in \mathcal{V}$ or $x' \in \mathcal{V}$ and

$$|x'y|_{\mathcal{W}} = |xy'|_{\mathcal{W}} = \inf_{a \in A}\{|xa|_{\mathcal{V}} + |ay|_{\mathcal{V}}\}.$$

  for any $x, y \in \mathcal{V}$.

The image of $\mathcal{V}$ under the reflection in A will be denoted by $\mathcal{V}'$. The subspace $\mathcal{V}'$ is an isometric copy of $\mathcal{V}$. Clearly $\mathcal{V} \cup \mathcal{V}' = \mathcal{W}$ and $\mathcal{V} \cap \mathcal{V}' = A$. Moreover $a = a' \iff a \in A$.

The following proposition follows directly from the definitions.

2.6.1. Proposition. *Assume $\mathcal{W}$ is the doubling of metric space $\mathcal{V}$ in its closed subset A. Then*

a) If $\mathcal{V}$ is a complete length space then so is $\mathcal{W}$.

b) If $\mathcal{V}$ is proper then so is $\mathcal{W}$. In this case, for any $x, y \in \mathcal{V}$ there is $a \in A$ such that

$$|xa|_{\mathcal{V}} + |ay|_{\mathcal{V}} = |xy'|_{\mathcal{W}}.$$

c) Given $x \in \mathcal{W}$ set $\bar{x} = x$ if $x \in \mathcal{V}$ and $\bar{x} = x'$ otherwise. The map $\mathcal{W} \to \mathcal{V}$ defined by $x \mapsto \bar{x}$ is short and length-preserving. In particular if γ is a geodesic in $\mathcal{W}$ with ends in $\mathcal{V}$ then $\bar{\gamma}$ is a geodesic in $\mathcal{V}$ with the same ends.

2.7 Kuratowsky embedding

Given a metric space $\mathcal{X}$, let us denote by $\mathrm{Bnd}(\mathcal{X}, \mathbb{R})$ the space of all bounded functions on $\mathcal{X}$ equipped with the supremum norm,

$$\|f\| = \sup_{x \in \mathcal{X}}\{|f(x)|\}.$$

Kuratowski embedding. Given a point $p \in \mathcal{X}$, consider map $\mathrm{kur}_p \colon \mathcal{X} \to \mathrm{Bnd}(\mathcal{X}, \mathbb{R})$, defined as $\mathrm{kur}_p\, x = |x_*| - |p_*|$. The map kur_p will be called *Kuratowski map at p*.

From the triangle inequality, we have

$$\|\operatorname{kur}_p x - \operatorname{kur}_p y\| = \sup_{z\in\mathcal{X}}\{||x\,z| - |y\,z||\} = |x\,y|.$$

Therefore, for any $p \in \mathcal{X}$, the Kuratowski map gives a distance preserving map $\operatorname{kur}_p\colon \mathcal{X} \hookrightarrow \operatorname{Bnd}(\mathcal{X},\mathbb{R})$. Thus, we can (and often will) consider space $\mathcal{X}$ as a subset of $\operatorname{Bnd}(\mathcal{X},\mathbb{R})$.

2.7.1. Exercise. *Show that any compact space is isometric to a subspace in a compact length space.*

Chapter 3

Convergence of metric spaces

In this section we discuss the Gromov–Hausdorff convergence of metric spaces.

It seems that Hausdorff convergence was first introduced by Hausdorff in [71] and couple of years later an equivalent definition was given by Blaschke in [31]. Further a refinement of this definition was introduced by Frolík in [56] and then rediscovered by Wijsman in [165], however this refinement was a step back to so called *closed convergence* introduced by Hausdorff in the original book. By that reason we call it Hausdorff convergence instead of *Hausdorff–Blascke–Frolík–Wijsman convergence.*

The definition of Gromov–Hausdorff convergence or metric spaces use the Hausdorff convergence. It means that the sequence of metric spaces admit a sequence of distance preserving embedding into huge metric space so that the their images converge in Hausdorff sense. The Gromov–Hausdorff convergence is also defined in a nonstandard way.

3.1 Convergence of subsets

Let $\mathcal{X}$ be a metric space and $A \subset \mathcal{X}$. We will denote by $|Ax| = |Ax|_{\mathcal{X}}$ the distance from A to a point x in $\mathcal{X}$; i.e.,

$$|Ax| \overset{def}{=\!=} \inf\{\, |ax| \mid a \in A \,\}\,.$$

3.1.1. Definition of Hausdorff convergence. *Given a sequence of closed sets $(A_n)_{n=1}^{\infty}$ in a metric space $\mathcal{X}$ a closed set $A_\infty \subset \mathcal{X}$*

is called Hausdorff limit of $(A_n)_{n=1}^\infty$*, briefly* $A_n \xrightarrow{\mathsf{H}} A_\infty$ *if*

$$|A_n\,x| \to |A_\infty\,x| \quad \text{as} \quad n \to \infty$$

for any fixed $x \in \mathcal{X}$*.*

In this case the sequence of closed sets $(A_n)_{n=1}^\infty$ *is called* converging *or* converging in the sense of Hausdorff*.*

3.1.2. Selection theorem. *Let* $\mathcal{X}$ *be a proper space and* $(A_n)_{n=1}^\infty$ *be a sequence of closed sets in* $\mathcal{X}$*. Assume that for some (and therefore any) point* $x \in \mathcal{X}$ *the sequence* $|A_n\,x|$ *is bounded. Then* $(A_n)_{n=1}^\infty$ *has a converging subsequence in the sense of Hausdorff.*

Proof. Since X is proper, we can choose a countable dense set $\{x_1, x_2, \dots \dots\}$ in $\mathcal{X}$. Note that the sequence $a_n = |A_n\,x_k|$ is bounded for each k. Therefore, passing to a subsequence of $(A_n)_{n=1}^\infty$, we can assume that $|A_n\,x_k|$ is converging as $n \to \infty$ for any fixed k.

Note that for each n, the function $|A_n\,*|\colon \mathcal{X} \to \mathbb{R}$ is 1-Lipschitz and nonnegative. Therefore the sequence $|A_n\,*|$ converges pointwise to a 1-Lipschitz nonnegative function $f\colon \mathcal{X} \to \mathbb{R}$.

Set $A_\infty = f^{-1}(0)$. Since f is 1-Lipschitz, $|A_\infty\,y| \geqslant f(y)$ for any $y \in \mathcal{X}$. It remains to show that $|A_\infty\,y| \leqslant f(y)$ for any y.

Assume contrary; i.e., $f(z) < R < |A_\infty\,z|$ for some $z \in \mathcal{X}$ and $R > 0$. Then for any large enough n there is a point $z_n \in A_n$ such that $|x\,z_n| \leqslant R$. Since $\mathcal{X}$ is proper, we can pass to a partial limit z_∞ of z_n as $n \to \infty$.

Clearly that $f(z_\infty) = 0$, i.e., $z_\infty \in A_\infty$. On the other hand,

$$|A_\infty\,y| \leqslant |z_\infty\,y| \leqslant R < |A_\infty\,y|,$$

a contradiction. □

3.2 Gromov–Hausdorff convergence

3.2.1. Definition. *Let* $\{\,\mathcal{X}_\alpha \mid \alpha \in \mathcal{A}\,\}$ *be a set of metric spaces. A metric space* $\boldsymbol{X}$ *is called* common space *of* $\{\,\mathcal{X}_\alpha \mid \alpha \in \mathcal{A}\,\}$ *if its underlying set is formed by the disjoint union*

$$\bigsqcup_{\alpha \in \mathcal{A}} \mathcal{X}_\alpha$$

and each inclusion $\iota_\alpha\colon \mathcal{X}_\alpha \hookrightarrow \boldsymbol{X}$ *is distance preserving.*

3.2.2. Definition. *Let* $\boldsymbol{X}$ *be a common space for proper metric spaces* $\mathcal{X}_1, \mathcal{X}_2, \dots$ *and* $\mathcal{X}_\infty$*. Assume that* $\mathcal{X}_n$ *forms an open set in* $\boldsymbol{X}$ *for each* $n < \infty$ *and* $\mathcal{X}_n \xrightarrow{\mathsf{H}} \mathcal{X}_\infty$ *in* $\boldsymbol{X}$ *as* $n \to \infty$*.*

In this case the topology τ *of* $\boldsymbol{X}$ *is called* Gromov–Hausdorff convergence *and we write* $\mathcal{X}_n \xrightarrow[\text{GH}]{} \mathcal{X}_\infty$ *or* $\mathcal{X}_n \xrightarrow[\text{GH}]{\tau} \mathcal{X}_\infty$*; the later notation is used if we need to consider specific Gromov–Hausdorff convergene* τ*. The space* $\mathcal{X}_\infty$ *is called the limit space of the sequence* $(\mathcal{X}_n)$ *along* τ*.*

Once we write $\mathcal{X}_n \xrightarrow[\text{GH}]{} \mathcal{X}_\infty$ we mean that we made a choice of Gromov–Hausdorff convergence.

Note that for a fixed sequence of metric spaces $\mathcal{X}_1, \mathcal{X}_2, \dots$ one may construct different Gromov–Hausdorff convergences, say $\mathcal{X}_n \xrightarrow[\text{GH}]{\tau} \mathcal{X}_\infty$ and $\mathcal{X}_n \xrightarrow[\text{GH}]{\tau'} \mathcal{X}'_\infty$ and their limit spaces $\mathcal{X}_\infty$ and $\mathcal{X}'_\infty$ need not to be isometric to each other. For example, for the constant sequence $\mathcal{X}_n \overset{iso}{=\!=} \mathbb{R}_{\geqslant 0}$, one may take $\mathcal{X}_\infty \overset{iso}{=\!=} \mathbb{R}_{\geqslant 0}$. In this case a point in the disjont space $\boldsymbol{X}$ can be thought as a pair $(x, n) \in \mathbb{R}_{\geqslant} \times (\mathbb{Z}_{>} \cup \{\infty\})$ and the metric on $\boldsymbol{X}$ can be defined the following way

$$|(x,n)\ (y,m)|_{\boldsymbol{X}} \overset{def}{=\!=} |\tfrac{1}{n} - \tfrac{1}{m}| + |x - y|,$$

where we assume that $0 = \frac{1}{\infty}$. On the other hand, one can take $\mathcal{X}'_\infty \overset{iso}{=\!=} \mathbb{R}$, and consider the metric

$$\begin{aligned} |(x,n)\ (y,m)|_{\boldsymbol{X}'} &= |\tfrac{1}{n} - \tfrac{1}{m}| + |(x-n) - (y-m)|, \\ |(x,n)\ (y,\infty)|_{\boldsymbol{X}'} &= \tfrac{1}{n} + |(x-n) - y|, \\ |(x,\infty)\ (y,\infty)|_{\boldsymbol{X}'} &= |x - y|. \end{aligned}$$

where $n, m < \infty$.

3.2.3. Induced convergences. *Assume* $\mathcal{X}_n \xrightarrow[\text{GH}]{\tau} \mathcal{X}_\infty$*, and* $\boldsymbol{X}$ *as in the definition 3.2.2 and* $\iota_n \colon \mathcal{X}_n \hookrightarrow \boldsymbol{X}$*,* $\iota_\infty \colon \mathcal{X}_\infty \hookrightarrow \boldsymbol{X}$ *are corresponding inclusions.*

a) A sequence of points $x_n \in \mathcal{X}_n$ *converges to* $x_\infty \in \mathcal{X}_\infty$ *(briefly,* $x_n \to x_\infty$ *or* $x_n \xrightarrow{\tau} x_\infty$*) if* $|x_n x_\infty|_{\boldsymbol{X}} \to 0$*.*

b) A sequence of closed sets $\mathfrak{C}_n \subset \mathcal{X}_n$ *converges to a closed set* $\mathfrak{C}_\infty \subset \mathcal{X}_\infty$ *(briefly,* $\mathfrak{C}_n \to \mathfrak{C}_\infty$ *or* $\mathfrak{C}_n \xrightarrow{\tau} \mathfrak{C}_\infty$*) if* $\mathfrak{C}_n \xrightarrow[\text{H}]{} \mathfrak{C}_\infty$ *as subsets of* $\boldsymbol{X}$*.*

c) A sequence of open sets $\Omega_n \subset \mathcal{X}_n$ *converges to an open set* $\Omega_\infty \subset \mathcal{X}_\infty$ *(briefly,* $\Omega_n \to \Omega_\infty$ *or* $\Omega_n \xrightarrow{\tau} \Omega_\infty$*) if the complements* $\mathcal{X}_n \backslash \Omega_n$ *converge to the complement* $\mathcal{X}_\infty \backslash \Omega_\infty$ *as closed sets.*

d) Let $\mathcal{X}_n \xrightarrow[\text{GH}]{\tau} \mathcal{X}_\infty$ *and* $\mathcal{Y}_n \xrightarrow[\text{GH}]{\vartheta} \mathcal{Y}_\infty$*. A sequence of submaps,* $\Phi_n \colon \mathcal{X}_n \mapsto \mathcal{Y}_n$ *converges to a submap* $\Phi_\infty \colon \mathcal{X}_\infty \mapsto \mathcal{Y}_\infty$ *if the following conditions holds*

- $\operatorname{Dom} \Phi_n \to \operatorname{Dom} \Phi_\infty$ *as a sequence of open sets.*

- *For any* $x_\infty \in \operatorname{Dom}\Phi_\infty$ *and any sequence* $x_n \in \mathcal{X}_n$ *such that* $x_n \to x_\infty$

$$\mathcal{Y}_n \ni \Phi_n(x_n) \xrightarrow{\vartheta} \Phi_\infty(x_\infty) \in \mathcal{Y}_\infty$$

as $n \to \infty$.

e) *Given a sequence of measures* μ_n *on* $\mathcal{X}_n$ *denote by* $\iota_n\#\mu_n$ *the pushforward measures on* $\boldsymbol{X}$. *We say that* (μ_n) *weakly converges to a measure* μ_∞ *on* $\mathcal{X}_\infty$ *(brefly,* $\mu_n \xrightarrow{\mu}_\infty$ *or* $\mu_n \xrightarrow{\tau} \mu_\infty$*) if the pushforward measures* $\iota_n\#\mu_n$ *weakly converge to the* $\iota_\infty\#\mu_\infty$.
In other words, if for any continuous function $\varphi\colon \boldsymbol{X} \to \mathbb{R}$ *with a compact support, we have*

$$\int\limits_{\mathcal{X}_n} \varphi\circ\iota_n(x)\cdot\boldsymbol{d}_x\mu_n \to \int\limits_{\mathcal{X}_\infty} \varphi\circ\iota_\infty(x)\cdot\boldsymbol{d}_x\mu_\infty$$

as $n \to \infty$.

Liftings. Given a Gromov–Hausdorff convergence $\mathcal{X}_n \xrightarrow[\text{GH}]{} \mathcal{X}_\infty$ and a point $p_\infty \in \mathcal{X}_\infty$ any sequence of points $p_n \in \mathcal{X}_n$ such that $p_n \xrightarrow[\text{GH}]{} p$ will be called *lifting* of p_∞. In this case the point $p_n \in \mathcal{X}_n$ will be called a lifting of p_∞ in $\mathcal{X}_n$. In this case we say that $|p_n\,*|\colon \mathcal{X}_n \to \mathbb{R}$ is a *lifting* of the distance function $|p\,*|\colon \mathcal{X} \to \mathbb{R}$. Obviousely $|p_n\,*| \xrightarrow[\text{GH}]{} |p\,*|$.

Note that liftings are not uniquely defined. We will be interested in the properties of liftings for sufficiently large n.

The same way we can talk about liftings of a point array $\boldsymbol{p}_\infty = (p^1_\infty, p^2_\infty, \dots, p^{\bar{k}}_\infty)$ in $\mathcal{X}_\infty$ and about the corresponding distance map $|\boldsymbol{p}_\infty\,*|\colon \mathcal{X}_\infty \to \mathbb{R}^{\bar{k}}$

$$|\boldsymbol{p}_\infty\,*|\colon x \mapsto (|p^1_\infty\,x|, |p^2_\infty\,x|, \dots, |p^k_\infty\,x|)$$

3.3 Gromov's selection theorem

3.3.1. Gromov's selection theorem. *Let* $\mathcal{X}_n$ *be a sequence of proper metric spaces with marked points* $x_n \in \mathcal{X}_n$. *Assume that for any fixed* $R, \varepsilon > 0$ *there is* $N = N(R,\varepsilon) \in \mathbb{Z}_>$ *such that for each* n *the ball* $\overline{\mathrm{B}}[x_n, R] \subset \mathcal{X}_n$ *admits a finite* ε*-net with at most* N *points. Then there is a subsequence of* $\mathcal{X}_n$ *which admit a Gromov–Hausdorff convergence such that the sequence of marked points* $x_n \in \mathcal{X}_n$ *converges.*

Proof. Note that there is a sequence of integers $M_1 < M_2 < \dots$ such that in each space $\mathcal{X}_n$ there is a sequence of points $z_{i,n} \in \mathcal{X}_n$ such that

$$|z_{i,n}\,x_n|_{\mathcal{X}_n} \leqslant \bar{k}+1 \quad \text{if} \quad i \leqslant M_{\bar{k}}$$

and the points $z_{1,n},\dots,z_{M_{\bar k},n}$ form an $\frac{1}{\bar k}$-net in $\overline{\mathrm{B}}[x_n,\bar k]_{\mathcal{X}_n}$. The existence of the sequence $M_1,M_2,\dots$ follows from the main assumption in the theorem.

Passing to a subsequence, we can assume that the sequence

$$\ell_n=|z_{i,n}\,z_{j,n}|_{\mathcal{X}_n}$$

converges for any i and j.

Let us consider countable set of points $\mathcal{W}=\{w_1,w_2,\dots\}$ equipped with the pseudometric defined as

$$|w_i,w_j|_{\mathcal{W}}=\lim_{n\to\infty}|z_{i,n}\,z_{j,n}|_{\mathcal{X}_n}.$$

Let $\hat{\mathcal{W}}$ be the metric space corresponding to $\mathcal{W}$. Denote by $\mathcal{X}_\infty$ the completion of $\hat{\mathcal{W}}$.

It remains to show that there is a Gromov–Hausdorff convergence $\mathcal{X}_n\xrightarrow[\mathrm{GH}]{}\mathcal{X}_\infty$ such that the sequence $x_n\in\mathcal{X}_n$ converges. To prove it, we need to construct a metric on the disjoint union of

$$\boldsymbol{X}=\mathcal{X}_\infty\sqcup\mathcal{X}_1\sqcup\mathcal{X}_2\sqcup\dots$$

satisfies definitions 3.2.1 and 3.2.2.

The metric can be defined as follows. Fix a sequence $\varepsilon_{\bar k}\to 0+$ and let $N_{\bar k}$ be the minimal integer such that

$$\big||w_i\,w_j|_{\mathcal{W}}-|z_{i,n}\,z_{j,n}|_{\mathcal{X}_n}\big|<\varepsilon_{\bar k}$$

if $i,j\leqslant N_{\bar k}$ and $n\geqslant N_{\bar k}$.

$$|x\,y|_{\boldsymbol{X}}=\begin{cases}|x\,y|_{\mathcal{X}_n} & x,y\in\mathcal{X}_n\\ |x\,y|_{\mathcal{X}_\infty} & x,y\in\mathcal{X}_\infty\end{cases}$$

Set,

$$\delta(z_{i,n},z_{i,\infty})=\sup\big\{\,\big||z_{i,m}\;z_{j,m}|_{\mathcal{X}_m}-|z_{i,\infty}\;z_{j,\infty}|_{\mathcal{X}_\infty}\big|\;\big|\;j\leqslant i,\,m\geqslant n\,\big\}.$$

Let us equip $\boldsymbol{X}$ with the maximal metric such that each inclusion $\iota_n\colon\mathcal{X}_n$ is short and the $|z_{i,n}\,z_{i,\infty}|_{\boldsymbol{X}}\leqslant\delta(z_{i,n},z_{i,\infty})$ for any i and n. In other words, let us define $\square$

3.3.2. Lemma. *Let $\mathcal{X}$ and $\mathcal{Y}$ be two metric spaces with two sequences of points $x_1,x_2,\dots\in\mathcal{X}$ and $y_1,y_2,\dots\in\mathcal{Y}$. Set*

$$\varepsilon_n=\max\big\{\,\big||x_n\,x_i|_{\mathcal{X}}-|y_n\,y_i|_{\mathcal{Y}}\big|\;\big|\;i<n\,\big\}.$$

Then for any $\varepsilon > 0$, the function $\rho\colon ((\mathcal{X} \sqcup \mathcal{Y}) \times (\mathcal{X} \sqcup \mathcal{Y}))$ defined below is a metric on the disjoint union $\mathcal{X} \sqcup \mathcal{Y}$.

$$\begin{aligned}
\rho(x,x') &= |x\,x'|_{\mathcal{X}} && \text{for any } \ x,x' \in \mathcal{X},\\
\rho(y,y') &= |y\,y'|_{\mathcal{Y}} && \text{for any } \ y,y' \in \mathcal{Y},\\
\rho(x,y) = \rho(y,x) &= \inf_i\{|x\,x_n|_{\mathcal{X}} + \varepsilon + \varepsilon_n + |y_n\,y|_{\mathcal{Y}}\} && \text{for any } \ x \in \mathcal{X} \text{ and } y \in \mathcal{Y}.
\end{aligned}$$

In particular, both inculsions $\iota_{\mathcal{X}}\colon \mathcal{X} \hookrightarrow (\mathcal{X} \sqcup \mathcal{Y}, \rho)$ and $\iota_{\mathcal{Y}}\colon \mathcal{Y} \hookrightarrow$ $\hookrightarrow (\mathcal{X} \sqcup \mathcal{Y}, \rho)$ are distance preserving.

3.3.3. Theorem. *Let $\mathcal{X}_n$ be a sequence of metric spaces which admit two Gromov–Hausdorff convergences τ and τ'. Assume $\mathcal{X}_n \xrightarrow[\text{GH}]{\tau} \mathcal{X}_\infty$ and $\mathcal{X}_n \xrightarrow[\text{GH}]{\tau'} \mathcal{X}'_\infty$.*

a) If $\mathcal{X}_\infty$ is compact then $\mathcal{X}_\infty \overset{iso}{=\!=} \mathcal{X}'_\infty$.

b) If $\mathcal{X}_\infty$ is proper and there is a sequence of points $x_n \in \mathcal{X}_n$ which converges in both τ and τ' then $\mathcal{X}_\infty \overset{iso}{=\!=} \mathcal{X}'_\infty$.

3.4 Convergence of compact spaces.

3.4.1. Definition. *Let $\mathcal{X}$ and $\mathcal{Y}$ be metric space, a map $f\colon \mathcal{X} \to \mathcal{Y}$ is called* ε-isometry *if the following two condition hold:*

a) $\operatorname{Im} f$ *is an ε-net of $\mathcal{Y}$*

b) $|f(x)\ f(x')|_{\mathcal{Y}} \lesseqgtr |x\,x'|_{\mathcal{X}} \pm \varepsilon$ *for any $x, x' \in \mathcal{X}$*

The following theorem roughly states that isometry class of Gromov–Hausdorff limit is uniquely defined once it is compact.

3.4.2. Theorem. *Let $\mathcal{X}_1, \mathcal{X}_2, \dots$ and $\mathcal{X}_\infty$ and $\bar{\mathcal{X}}_\infty$ be metric spaces such that $\mathcal{X}_n \xrightarrow{\tau} \mathcal{X}_\infty$, $\mathcal{X}_n \xrightarrow{\bar\tau} \bar{\mathcal{X}}_\infty$.*

Assume that $\bar{\mathcal{X}}_\infty$ is compact. Then $\mathcal{X}_\infty \overset{iso}{=\!=} \bar{\mathcal{X}}_\infty$.

Proof. For each $x_\infty \in \mathcal{X}_\infty$, fix its liftings $x_n \in \mathcal{X}_n$.

Choose some ultrafilter ω. Define $\bar{x}_\infty \in \bar{\mathcal{X}}_\infty$ such that $\bar{x}_n \xrightarrow[\text{GH}]{\bar\tau} \bar{x}_\infty$; since $\bar{\mathcal{X}}_\infty$ is compact, the ω-limit is defined. We claim that the map $x_\infty \to \bar{x}_\infty$ is an isometry.

Indeed, according to the definition of Gromov–Hausdorff convergence,

$$|\bar{x}_\infty\,\bar{y}_\infty|_{\bar{\mathcal{X}}_\infty} = \lim_{n\to\omega} |x_n\,y_n|_{\mathcal{X}_n} = |x_\infty\,y_\infty|_{\mathcal{X}_\infty}.$$

Thus, the map $x_\infty \to \bar{x}_\infty$ gives a distance preserving map $\Phi\colon \mathcal{X}_\infty \hookrightarrow$ $\hookrightarrow \bar{\mathcal{X}}_\infty$. In particular, $\mathcal{X}_\infty$ is compact. Switching $\mathcal{X}_\infty$ and $\bar{\mathcal{X}}_\infty$ and applying the same argument, we get an isometiric embedding $\bar{\mathcal{X}}_\infty \hookrightarrow$ $\hookrightarrow \mathcal{X}_\infty$. Thus, the result follows from Lemma 3.4.5. $\square$

3.4.3. Definition. *Given two compact spaces $\mathcal{X}$ and $\mathcal{Y}$, we will write*

- $\mathcal{X} \leqslant \mathcal{Y}$ *if there is a non-contracting map* $\Phi\colon \mathcal{X} \to \mathcal{Y}$.
- $\mathcal{X} \leqslant \mathcal{Y} + \varepsilon$ *if there is a map* $\Phi\colon \mathcal{X} \to \mathcal{Y}$ *such that for any* $x, x' \in \mathcal{X}$ *we have*

$$|x\,x'| \leqslant |\Phi(x)\ \Phi(x')| + \varepsilon.$$

Further, we define Gromov–Hausdorff distance *between* $\mathcal{X}$ *and* $\mathcal{Y}$, $\mathrm{D}_{\mathsf{GH}}(\mathcal{X},\mathcal{Y})$ *as infimum of all* $\varepsilon > 0$ *such that* $\mathcal{X} \leqslant \mathcal{Y}+\varepsilon$ *and* $\mathcal{Y} \leqslant \mathcal{X}+\varepsilon$.

Remark for AKP. This notation is a bit slang-like, maybe better to use $\succcurlyeq$ instead of $\geqslant$. Yet $\mathcal{Y}+\varepsilon$ migh be needed to be changed; one might think that $\mathcal{Y}+\varepsilon$ is a metric space $(\underline{\mathcal{Y}}, |{*}{*}| + \varepsilon\rho)$, where ρ is a discreate metric on $\underline{\mathcal{Y}}$ (i.e. $\rho(x,y) = 1$ if $x \neq y$). It will be totally correct to introduce some operator say $\mathrm{Add}_\varepsilon\,\mathcal{Y}$, but it is also less intuitive... One may also write something like $\mathcal{X} \succcurlyeq \mathcal{Y}^{+\varepsilon}$ or $\mathcal{X} \succcurlyeq^{\varepsilon+} \mathcal{Y}$...

The Gromov–Hausdorff distance turns the set of all isometry classes of compact metric spaces into a metric space. The following theorem shows that convergence in this space coinsides with the Gromov–Hausdorff convergence defined above.

3.4.4. Theorem. *Let* $\mathcal{X}_1, \mathcal{X}_2, \dots$ *and* $\mathcal{X}_\infty$ *be compact metric spaces. Then*

$$\mathcal{X}_n \to \mathcal{X}_\infty \quad \Leftrightarrow \quad \mathrm{D}_{\mathsf{GH}}(\mathcal{X}_n, \mathcal{X}_\infty) \to 0.$$

Proof; ($\Leftarrow$). Let $\mathcal{X}_n \xrightarrow{\tau} \mathcal{X}_\infty$ and $b_n\colon \mathcal{X}_n \to \mathcal{X}_\infty$ be a sequence of maps such that

$$|a_n(x)\ a_n(y)|_{\mathcal{X}_\infty} \geqslant |x\,y|_{\mathcal{X}_n} - \varepsilon_n,$$
$$|b_n(v)\ b_n(w)|_{\mathcal{X}_n} \geqslant |v\,w|_{\mathcal{X}_\infty} - \varepsilon_n$$

for any $x, y \in \mathcal{X}_n$ and $v, w \in \mathcal{X}_\infty$ and some sequence $\varepsilon_n \to 0+$.

Let us show that $\mathcal{X}_n \xrightarrow{\tau} \mathcal{X}_\infty$.

Fix $\varepsilon > 0$ and choose a maximal ε-packing $\{x^1, x^2, \dots, x^k\}$ in $\mathcal{X}_\infty$ such that the value $\sum_{i<j} |x^i\,x^j|$ is maximal. Note that

$$|a_n \circ b_n(x^i)\ a_n \circ b_n(x^j)| > |x^i\,x^j| - 2\cdot\varepsilon_n.$$

Thus, $|a_n \circ b_n(x^i)\ a_n \circ b_n(x^j)| - |x^i\,x^j| \to_{n\to\infty} 0$ for all i and j. For all large n, we have $2\cdot\varepsilon_n < |x^i\,x^j| - \varepsilon$. Thus $|b_n(x^i)\ b_n(x^j)|_{\mathcal{X}_n} > \varepsilon$ and

$$|a_n \circ b_n(x^i)\ a_n \circ b_n(x^j)|_{\mathcal{X}_n} > \varepsilon$$

for all $i \neq j$. Thus, for each large n, the set $\{a_n \circ b_n(x^i)\}$ forms a maximal ε-packing and therefore an ε-net in $\mathcal{X}_\infty$. Since $\varepsilon > 0$ is arbitrary, we get **??**.

Since $\{a_n \circ b_n(x^i)\}$ is an ε-net in $\mathcal{X}_\infty$, for any $y_n \in \mathcal{X}_n$, there is x^i such that $|a_n \circ b_n(x^i)\ a_n(y_n)| < \varepsilon$. Thus, $|b_n(x^i)\ y_n| < \varepsilon + \varepsilon_n$.

Further, note that the set $\{b_n(x^i)\}$ forms an $(\varepsilon + \varepsilon_n)$-net in $\mathcal{X}_n$. Otherwise, if for some $y_n \in \mathcal{X}_n$, we have $|y_n\, b_n(x^i)|$ for all i, then $|a_n \circ b_n(x^i)\ a_n(y_n)| > \varepsilon$; i.e. $\bar{k} < \operatorname{pack}_\varepsilon X_\infty$, a contradiction.

($\Rightarrow$). Assume $\mathcal{X}_n \xrightarrow{\tau} \mathcal{X}_\infty$. Fix $\varepsilon > 0$ and choose a maximal ε-packing $\{x^1, x^2, \dots, x^{\bar{k}}\}$ in $\mathcal{X}_\infty$. For each x^i, choose a sequence $x^i_n \in \mathcal{X}_n$ such that $a_n(x^i_n) \to x^i$. Note that for all large n, we have $|x^i_n\, x^j_n| > \varepsilon$. For each point $z \in \mathcal{X}_\infty$, choose x^i so that $|z\, x^i| < \varepsilon$ and define map $b_n \colon \mathcal{X}_\infty \to \mathcal{X}_n$ such that $b_n(z) = x^i_n$. $\square$

3.4.5. Lemma. *Let $\mathcal{X}$ and $\mathcal{Y}$ be two metric spaces and $\mathcal{X}$ is compact then*

$$\mathcal{X} \geqslant \mathcal{Y} \geqslant \mathcal{X} \quad \Leftrightarrow \quad \mathcal{X} \overset{iso}{=\!=} \mathcal{Y}.$$

The following proof was suggested by Travis Morrison.

Proof. Let $f \colon \mathcal{X} \to \mathcal{Y}$ and $g \colon \mathcal{Y} \to \mathcal{X}$ be non contracting mappings. It is sufficient to prove that $h = g \circ f \colon \mathcal{X} \to \mathcal{X}$ is an isometry.

Given any pair of points $x, y \in \mathcal{X}$, set $x_n = h^{\circ n}(x)$ and $y_n = h^{\circ n}(y)$. Since $\mathcal{X}$ is compact, one can choose an incresing sequence of integers $n_{\bar{k}}$ such that both sequences $(x_{n_i})_{i=1}^\infty$ and $(y_{n_i})_{i=1}^\infty$ converge. In particular, both of these sequences are Cauchy; i.e.,

$$|x_{n_i}\, x_{n_j}|, |y_{n_i}\, y_{n_j}| \to 0$$

as $\min\{i, j\} \to \infty$. Sinse h is nonconotracting, we get

$$|x\, x_{|n_i - n_j|}| \leqslant |x_{n_i}\, x_{n_j}|.$$

It follows that there is a sequence $m_i \to \infty$ such that

❶ $$x_{m_i} \to x \quad \text{and} \quad y_{m_i} \to y \quad \text{as} \quad \bar{k} \to \infty.$$

Let $\ell_n = |x_n\, y_n|$. Since h is noncontracting, (ℓ_n) is a nondecreasing sequence. On the other hand, from ❶, it follows that $\ell_{m_i} \to |x\, y| = \ell_0$ as $m_i \to \infty$; i.e., (ℓ_n) is a constant sequece. In particular $\ell_0 = \ell_1$ for any x and $y \in \mathcal{X}$; i.e., h is distance preserving map.

Therefore $h(X)$ is isometric to $\mathcal{X}$. From ❶, we get that $h(\mathcal{X})$ is everywhere dense. Sinse X is compact, we get $h(\mathcal{X}) = \mathcal{X}$. $\square$

3.5 Comments

One of the most remarcble revolution in geometry was made by seemingly very simple observation made by Gromov. He gave a transparent

intuitive notion of convergence for metric spaces. Although some types of convergences of metric spaces were considered in geometry before Gromov, all the considered metric were designed to be used in very speacific problems. A very good account for different types of convergences used in metric geometry can be found in ???.

The Gromov–Hausdorff convergence (under the name Hausdorff convergence) was introduced by Gromov in [59]. It was essential step in the proof that any group of polynomial growth has is a nilpontant subgroup of finite index. It is remarkable that the ultra-limits, an other important type of convergence, also appear first in the context of groups of polynomial growth; they were introduced by ??? in ???, where generalization of Gromov's theorem Both concepts quikly sprad into all branches of geometry.

The definition 3.1.1 was given by Frolík in [56] and was rediscovered by Wijsman in [165].

This definition is a slight modification of the convergence for subsets of metric space, first introduced by Hausdorff in [71], two years later the same convergence was introduced by Baschke in [31]. Blaschke does not refer to Hausdorff; it might mean that his discovery was independent or yet more likely that this type of convergence was used in mathematical folklore before these publications.

If in this definition, you exachange the pointwise convergence to uniform convergence, then you get the usual Hausdorff convergence. These two convergences, coinside on the set of bounded subsets and Frolík–Wijsman modification make it more sutable for working with unbounded sets.

Originaly, the Gromov–Hausdorff convergence was defined differently, see ???. Instead one consider two cases:

1. If the limit is compact, then one require existance of one GH-convergence.
2. In the non-compact case one makes a choice of point $x_\infty \in \in \mathcal{X}_\infty$ and a sequence $x_n \in \mathcal{X}_n$ and then to define pointed Gromov–Hausdorff convergence $(\mathcal{X}_n, x_n) \to (\mathcal{X}_\infty, x_\infty)$ one requires existance of GH-convergence τ, such that $x_n \xrightarrow{\tau} x_\infty$.

The later is called *pointed Gromov–Hausdorff convergence.* It is easy to see that if $\mathcal{X}_n \xrightarrow{\tau} \mathcal{X}_\infty$ then for any sequence $x_n \in \mathcal{X}_n$ such that $x_n \xrightarrow{\tau} x_\infty \in \mathcal{X}_\infty$, the seqence $(\mathcal{X}_n, x_n)$ converges to $(\mathcal{X}_\infty, x_\infty)$ in the sense of pointed Gromov–Hausdorff convergence.

The Gromov–Hausdorff convergence is defined only for proper spaces; thus, once we write $\mathcal{X}_n \to \mathcal{X}_\infty$, we implicitly assume that all $\mathcal{X}_1, \mathcal{X}_2, \dots$ and $\mathcal{X}_\infty$ are proper spaces.

Chapter 4

Ultralimits

Here we introduce ultralimits of sequence of points, metric spaces and functions. The ultralimits of metric spaces can be considered as an extension of Gromov–Hausdorff convergence since the ultralimit is isometric to the Gromov–Hausdorff limit once the later is defined. Our presentation is based on [86].

In this book, the use of ultralimits is very limited; we use them only as a canonical way to pass to a convergent subsequence. (In principle, we could avoid to sell our souls to the set-theoretical devel but then we had to say "pass to convergents subsequence" too many times.)

4.1 Ultrafilters

We will need existence of one selective ultrafilter ω, you can think that it is fixed once for all. The existence follows from axiom of choice and continuum hypothesis.

Recall that $\mathbb{N}$ denotes the set of natural numbers, $\mathbb{N} = \{1, 2, \dots\}$

4.1.1. Definition. *A finitely additive measure ω on $\mathbb{N}$ is called* ultrafilter *if it satisfies the following conditions:*

a) $\omega(S) = 0$ *or* 1 *for any subset* $S \subset \mathbb{N}$*,*

An ultrafilter ω is called nonprinciple *if in addition*

b) $\omega(F) = 0$ *for any finite subset* $F \subset \mathbb{N}$*.*

A nonprinciple ultrafilter ω is called selective *if in addition*

c) For any partition of $\mathbb{N}$ into sets $\{C_\alpha\}_{\alpha\in\mathcal{A}}$ such that $\omega(C_\alpha) = 0$ for each α there is a set $S \subset \mathbb{N}$ such that $\omega(S) = 1$ and $S \cap C_\alpha$ is a one-point set for each $\alpha \in \mathcal{A}$.

More commonly nonprinciple ultrafilter is defined as a collection, say $\mathfrak{F}$ of sets in $\mathbb{N}$ such that

1. if $P \in \mathfrak{F}$ and $Q \supset P$ then $Q \in \mathfrak{F}$,
2. if $P, Q \in \mathfrak{F}$ then $P \cap Q \in \mathfrak{F}$,
3. for any subset $P \subset \mathbb{N}$, either P or its complement is an element of $\mathfrak{F}$.
4. if $F \subset \mathbb{N}$ is finite then $F \notin \mathfrak{F}$.

Setting $P \in \mathfrak{F} \Leftrightarrow \omega(P) = 1$, makes these two definitions equivalent.

A nonempty collection of sets $\mathfrak{F}$ which does not include the empty set and satisfy only conditions 1 and 2 is called *filter*; if in addition it satisfies Condition 3 it is called *ultrafilter*. From Zorn's lemma, it follows that every filter contains an ultrafilter. Thus there is an ultrafilter $\mathfrak{F}$ which contained in the filter of all complements of finite sets; clearly it is nonprinciple.

The existence of selective ultrafilter follows from continuum hypothesis; it was proved by Rudin in [148].

Given a set $S \subset \mathbb{N}$, consider subset Ω_S of all ultrafilters ω such that $\omega(S) = 1$. It is straightforward to check that the sets Ω_S for all $S \subset \mathbb{N}$ form a topology on the set of ultrafilters on $\mathbb{N}$. The obtained space is called *Stone–Čech compactification* of $\mathbb{N}$; it is usually denoted as $\beta\mathbb{N}$. There is a natural embedding $\mathbb{N} \hookrightarrow \beta\mathbb{N}$ defined as following $n \mapsto \omega_n$, $\omega_n(S) = 1$ if and only if $n \in S$. (The image of $\mathbb{N}$ is formed by the principle ultrafilters.) Thus we can (and will) consider $\mathbb{N}$ as a subset of $\beta\mathbb{N}$.

The space $\beta\mathbb{N}$ is the maximal compact Hausdorff space which contains $\mathbb{N}$ as an everywhere dense subset; more precisely, for any compact Hausdorff space $\mathcal{X}$ and a map $f\colon \mathbb{N} \to \mathcal{X}$ there is unique continuous map $\bar{f}\colon \beta\mathbb{N} \to X$ such that the restriction $\bar{f}|\mathbb{N}$ coincides with f.

4.2 Ultralimits of points

Fix an ultrafilter ω. Assume (x_n) is a sequence of points in a metric space $\mathcal{X}$. Let us define the ω-*limit* of (x_n) as the point x_ω such that for any $\varepsilon > 0$, ω-almost all elements of (x_n) lie in $\mathrm{B}(x_\omega, \varepsilon)$; i.e.,

$$\omega\{\, n \in \mathbb{N} \mid |x_\omega x_n| < \varepsilon \,\} = 1.$$

In this case, we will write

$$x_\omega = \lim_{n\to\omega} x_n \quad \text{or} \quad x_n \to x_\omega \text{ as } n \to \omega.$$

For example if ω is the principle ultrafilter such that $\omega(\{n\}) = 1$ for some $n \in \mathbb{N}$ then $x_\omega = x_n$.

Note that ω-limits of a sequence and its subsequence may differ, say in general

$$\lim_{n\to\omega} x_n \neq \lim_{n\to\omega} x_{2\cdot n}.$$

4.2.1. Proposition. *Let ω be a nonprinciple ultrafilter. Assume (x_n) is a sequence of points in a metric space $\mathcal{X}$ and $x_n \to x_\omega$ as $n \to \omega$. Then x_ω is a partial limit of the sequence (x_n); i.e., there is a subsequence $(x_n)_{n\in S}$ which converges to x_ω in the usual sense.*

Moreover, if ω is selective, then the subsequence $(x_n)_{n\in S}$ can be chosen so that $\omega(S) = 1$.

Proof. Given $\varepsilon > 0$, set $S_\varepsilon = \{\, n \in \mathbb{N} \mid |x_n x_\omega| < \varepsilon \,\}$.

Note that $\omega(S_\varepsilon) = 1$ for any $\varepsilon > 0$. Since ω is nonprinciple, the set S_ε is infinite. Therefore we can choose an increasing sequence $(n_{\bar k})$ such that $n_{\bar k} \in S_{\frac{1}{\bar k}}$ for each $\bar k \in \mathbb{N}$. Clearly $x_{n_{\bar k}} \to x_\omega$ as $\bar k \to \infty$.

Now assume that ω is selective. Consider the sets

$$C_{\bar k} = \left\{\, n \in \mathbb{N} \;\middle|\; \tfrac{1}{n} < |x_n x_\omega| \leqslant \tfrac{1}{n-1} \,\right\},$$

where we assume $\frac{1}{0} = \infty$ and the set

$$C_\infty = \{\, n \in \mathbb{N} \mid x_n = x_\omega \,\},$$

Note that $\omega(C_{\bar k}) = 0$ for any $\bar k \in \mathbb{N}$.

If $\omega(C_\infty) = 1$, we can take the sequence $(x_n)_{n\in C_\infty}$.

Otherwise discarding all empty sets among $C_{\bar k}$ and C_∞ gives a partition of $\mathbb{N}$ into countable collection of ω-small sets. Since ω is selective, we can choose a set $S \subset \mathbb{N}$ such that S meets each set of the partition at one point and $\omega(S) = 1$. Clearly the sequence $(x_n)_{n\in S}$ converges to x_o in the usual sense. $\square$

The following proposition is analogous to the statement that any sequence in a compact metric space has a convergent subsequence; it can be proved the same way.

4.2.2. Proposition. *Let $\mathcal{X}$ be a compact metric space. Then any sequence of points (x_n) in $\mathcal{X}$ has unique ω-limit x_ω.*

In particular, a bounded sequence of real numbers has a unique ω-limit.

The following lemma is an ultralimit analog of Cauchy convergence test.

4.2.3. Lemma. *Let (x_n) be a sequence of points in a complete space $\mathcal{X}$. Assume for each subsequence (y_n) of (x_n), the ω-limit*

$$y_\omega = \lim_{n\to\omega} y_n \in \mathcal{X}$$

is defined and does not depend on the choice of subsequence, then the sequence (x_n) converges in the usual sense.

Proof. Assume that (x_n) is not a Cauchy sequence. Then for some $\varepsilon > 0$, there is a subsequence (y_n) of (x_n) such that $|x_n\,y_n| \geqslant \varepsilon$ for all n.

It follows that $|x_\omega\,y_\omega| \geqslant \varepsilon$, a contradiction. □

4.2.4. Corollary. *Assume $\mathcal{X}$ is a complete length space and $p, q \in \in \mathcal{X}$ cannot be joined by a geodesic in $\mathcal{X}$. Then there are at least two distinct geodesics between p and q in the ultrapower $\mathcal{X}^\omega$.*

4.3 Ultralimits of spaces

Let $\mathcal{X}_n$ be a sequence of metric spaces $\mathcal{X}_n$. Consider all sequences $x_n \in \mathcal{X}_n$. On the set of all such sequences, define a pseudometric on these subsequences by

❶ $$|(x_n)\ (y_n)| = \lim_{n\to\omega} |x_n\,y_n|.$$

Note that the ω-limit on the right hand side is always defined and takes a value in $[0, \infty]$.

Set $\mathcal{X}_\omega$ to be the corresponding metric space; i.e., the underlying set of $\mathcal{X}_\omega$ is formed by classes of equivalence of sequences of points $x_n \in \mathcal{X}_n$ defined by

$$(x_n) \sim (y_n) \ \Leftrightarrow\ \lim_{n\to\omega} |x_n\,y_n| = 0$$

and the distance is defined as in ❶.

The space $\mathcal{X}_\omega$ is called ω-*limit* of $\mathcal{X}_n$. Typically $\mathcal{X}_\omega$ will denote the ω-limit of sequence $\mathcal{X}_n$; we may also write

$$\mathcal{X}_n \to \mathcal{X}_\omega \ \text{ as } \ n \to \omega \ \text{ or } \ \mathcal{X}_\omega = \lim_{n\to\omega} \mathcal{X}_n.$$

Given a sequence $x_n \in \mathcal{X}_n$, we will denote by x_ω its equivalence class which is a point in $\mathcal{X}_\omega$; equivalently we will write

$$x_n \to x_\omega \ \text{ as } \ n \to \omega \ \text{ or } \ x_\omega = \lim_{n\to\omega} x_n.$$

Gromov–Hausdorff convergence versus ultralimits.

4.3.1. Theorem. *Assume $\mathcal{X}_n$ is a sequence of complete spaces. Let $\mathcal{X}_n \to \mathcal{X}_\omega$ as $n \to \omega$ and $\mathcal{Y}_n \subset \mathcal{X}_n$ is a sequence of subsets such that $\mathcal{Y}_n \xrightarrow[\text{GH}]{} \mathcal{Y}_\infty$. Then there is an distance preserving map $\iota : \mathcal{Y}_\infty \to \mathcal{X}_\omega$.*

Moreover:

a) If $\mathcal{X}_n \xrightarrow[\text{GH}]{} \mathcal{X}_\infty$ and $\mathcal{X}_\infty$ is compact then $\mathcal{X}_\infty$ is isometric to $\mathcal{X}_\omega$.

b) If $\mathcal{X}_n \xrightarrow[\text{GH}]{} \mathcal{X}_\infty$ and $\mathcal{X}_\infty$ is proper then $\mathcal{X}_\infty$ is isometric to a metric component of $\mathcal{X}_\omega$.

Proof. For each point $y_\infty \in \mathcal{Y}_\infty$ choose a lifting $y_n \in \mathcal{Y}_n$. Pass to the ω-limit $y_\omega \in \mathcal{X}_\omega$ of (y_n). Clearly for any $y_\infty, z_\infty \in \mathcal{Y}_\infty$, we have

$$|y_\infty z_\infty|_{\mathcal{Y}_\infty} = |y_\omega z_\omega|_{\mathcal{X}_\omega};$$

i.e., the map $y_\infty \mapsto y_\omega$ gives a distance preserving map $\iota : \mathcal{Y}_\infty \to \mathcal{X}_\omega$.

(a)+(b). Fix $x_\omega \in \mathcal{X}_\omega$. Choose a sequence $x_n \in \mathcal{X}_n$ such that $x_n \to x_\omega$ as $n \to \omega$.

Denote by $\boldsymbol{X}$ the common space for the convergence $\mathcal{X}_n \xrightarrow[\text{GH}]{} \mathcal{X}_\infty$; see the definition of Gromov–Hausdorff convergence (3.2.2). Consider the sequence (x_n) as a sequence of points in $\boldsymbol{X}$.

If the ω-limit x_∞ of (x_n) exists, it has to lie in $\mathcal{X}_\infty$.

The point x_∞, if defined, does not depend on the choice of (x_n). Indeed, if $y_n \in \mathcal{X}_n$ be an other sequence such that $y_n \to x_\omega$ as $n \to \omega$ then

$$|y_\infty x_\infty| = \lim_{n\to\omega} |y_n x_n| = 0;$$

i.e., $x_\infty = y_\infty$.

This way we constructed a map $\nu\colon x_\omega \to x_\infty$; it is defined on a subset of $\operatorname{Dom}\nu \subset \mathcal{X}_\omega$. By construction of ι, we get $\iota \circ \nu(x_\omega) = x_\omega$ for any $x_\omega \in \operatorname{Dom}\nu$.

Finally note that if $\mathcal{X}_\infty$ is compact then ν is defined on whole $\mathcal{X}_\omega$; this proves (a).

If $\mathcal{X}_\infty$ is proper, choose any point $z_\infty \in \mathcal{X}_\infty$ and set $z_\omega = \iota(z_\infty)$. For any point $x_\omega \in \mathcal{X}_\omega$ on finite distance from z_ω, for the sequence x_n as above we have that $|z_n x_n|$ is bounded for ω-almost all n. Since $\mathcal{X}_\infty$ is proper $\nu(x_\omega)$ is defined; in other words ν is defined on the metric component of z_ω. Hence (b) follows. □

4.3.2. Corollary. *ω-limit of a sequence of complete length spaces is geodesic.*

Proof. Given two points $x_\omega, y_\omega \in \mathcal{X}_\omega$, find two bounded sequences of points $x_n, y_n \in \mathcal{X}_n$, $x_n \to x_\omega$, $y_n \to y_\omega$ as $n \to \omega$. Consider a sequence of paths $\gamma_n\colon [0,1] \to \mathcal{X}_n$ from $_n$ to y_n such that

$$\operatorname{length}\gamma_n \leqslant |x_n y_n| + \tfrac{1}{n}$$

Apply Theorem 4.3.1 for the images $\mathcal{Y}_n = \gamma_n([0,1]) \subset \mathcal{X}_n$. □

Ultrapower. If all the metric spaces in the sequence are identical $\mathcal{X}_n = \mathcal{X}$, its ω-limit $\lim_{n\to\omega} \mathcal{X}_n$ is denoted by $\mathcal{X}^\omega$ and called ω-power of $\mathcal{X}$.

According to Theorem 4.3.1, there is a distance preserving map $\iota : \mathcal{X} \hookrightarrow \mathcal{X}^\omega$, here $\iota(y)$ is the equivalence class of constant sequence $y_n = y$.

The image $\iota(\mathcal{X})$ might be proper subset of $\mathcal{X}^\omega$. For example $\mathbb{R}^\omega$ has pairs of points on distance ∞ from each other although each metric component of $\mathbb{R}^\omega$ is isometric to $\mathbb{R}$.

According to Theorem 4.3.1, if $\mathcal{X}$ is compact then $\iota(\mathcal{X}) = \mathcal{X}^\omega$; in particular, $\mathcal{X}^\omega$ is isometric to $\mathcal{X}$. If $\mathcal{X}$ is proper then $\iota(\mathcal{X})$ forms a metric component of $\mathcal{X}^\omega$.

The embedding ι gives us right to treat $\mathcal{X}$ as a subset of its ultrapower $\mathcal{X}^\omega$.

4.3.3. Lemma. *Let $\mathcal{X}$ be a complete metric space. Then $\mathcal{X}^\omega$ is geodesic space if and only if $\mathcal{X}$ is an length space.*

4.4 Ultralimits of functions

Let $\mathcal{X}_n$ be a sequence of metric spaces with marked points $\star_n \in \mathcal{X}_n$. Assume $(\mathcal{X}_n, \star_n) \to (\mathcal{X}_\omega, \star_\omega)$ as $n \to \omega$.

Let $f_n \colon \mathcal{X}_n \multimap \mathbb{R}$ be a sequence of subfunctions. Consider the set Ω of all points $x_\omega \in \mathcal{X}_\omega$ such that if $x_n \to x_\omega$ as $n \to \omega$ for a sequence $x_n \in \mathcal{X}_n$ then $x_n \in \operatorname{Dom} f_n$ for ω-almost all n. Note that Ω is open.

Assume there is a function $f_\omega \colon \Omega \to \mathbb{R}$ which satisfies the following condition. For any $x_\omega \in \Omega$ and any sequence of points $x_n \in \mathcal{X}_n$ if $x_n \to x_\omega$ as $n \to \omega$ then $f_n(x_n) \to f_\omega(x_o)$ as $n \to \omega$. In this case the subfunction $f_\omega \colon \mathcal{L}_\omega \to \mathbb{R}$ is said to be ω-limit of $f_n \colon \mathcal{L}_n \to \mathbb{R}$.

The following lemma gives a mild condition on a sequence of functions f_n which guarantee the existence of ω-limit.

4.4.1. Lemma. *Let $\mathcal{X}_n$ be a sequence of metric spaces with marked points $\star_n \in \mathcal{X}_n$. Assume $(\mathcal{X}_n, \star_n) \to (\mathcal{X}_\omega, \star_\omega)$ as $n \to \omega$ and $f_n \colon \mathcal{X}_n \multimap \multimap \mathbb{R}$ be a sequence of subfunctions.*

Given $\varepsilon > 0$, consider open set

$$\Omega_n^\varepsilon = \left\{ x_n \in \mathcal{X}_n \;\middle|\; |(\mathcal{L} \setminus \operatorname{Dom} f_n)\; x_n| > \varepsilon, |\star_n x_n| < \tfrac{1}{\varepsilon} \right\}.$$

Then if for any $\varepsilon > 0$ the restrictions $f_n|\Omega_n^\varepsilon$ are uniformly bounded and uniformly continuous then ω-limit $f_\omega \colon \mathcal{L}_\omega \to \mathbb{R}$ of f_n is defined.

In particular if f_n are uniformly bounded and continuous then the ω-limit is defined.

The proof is straightforward.

If $f \colon \mathcal{X} \multimap \mathbb{R}$ be a subfunction. The ω-limit of constant sequence $f_n = f$ is called ω-power of f and denoted by f^ω. So

$$f^\omega \colon \mathcal{X} \multimap \mathbb{R}, \quad f^\omega(x_\omega) = \lim_{n \to \omega} f(x_n).$$

Recall that we treat $\mathcal{X}$ as a subset of its ω-power $\mathcal{X}^\omega$. Note that $\operatorname{Dom} f = \mathcal{X} \cap \operatorname{Dom} f^\omega$. Moreover, if $\mathrm{B}(x,\varepsilon)_{\mathcal{X}} \subset \operatorname{Dom} f$ then $\mathrm{B}(x,\varepsilon)_{\mathcal{X}^\omega} \subset \operatorname{Dom} f^\omega$.

4.4.2. Exercise. *Construct a sequence of compact length spaces $\mathcal{X}_n$ with a converging sequence of $\pounds$-Lipschitz concave functions $f_n\colon \mathcal{X}_n \to \mathbb{R}$ such that the ω-limit $\mathcal{X}_\omega$ of $\mathcal{X}_n$ is compact and the ω-limit $f_\omega\colon \mathcal{X}_\omega \to \mathbb{R}$ of f_n is not concave.*

Chapter 5

Maps and functions

Here we introduce some classes of maps between metric spaces and develop a language to describe different levels of convexity/concavity of real-valued functions on general metric space.

Changing sign changes a concave function into a convex function and other way around. Concave functions are used mainly in the CBB-spaces, while convex ones are used mainly in CAT-spaces.

5.1 Maps, functions and relatives

We will often need maps and functions defined on subsets of a metric space. We call them *submaps* and *subfunction*. Thus given a metric space $\mathcal{X}$ and $\mathcal{Y}$, a submap $\Phi\colon \mathcal{X} \circ\!\!\to \mathcal{Y}$ is a map defined on a subset $\operatorname{Dom}\Phi \subset \mathcal{X}$.

Almost all subfunctions and sub maps we consider are continuous (or at least semicontinuous). In this case the domain $\operatorname{Dom}\Phi$ is automatically open. Indeed, the continuous maps are defined as those for which preimage of open set is open. Assume $\Phi\colon \mathcal{X} \circ\!\!\to \mathcal{Y}$ is a continuous submap. Since $\mathcal{Y}$ is open then so is $\Phi^{-1}(\mathcal{Y})$ which is the same as $\operatorname{Dom}\Phi$. The same is true for upper and lower semicontinuous functions $f\colon \mathcal{X} \circ\!\!\to \mathbb{R}$ since it is a continuous function for special topology on $\mathbb{R}$.

5.1.1. Isometry. *Let $\mathcal{X}$ and $\mathcal{Y}$ be metric spaces and $\Phi\colon \mathcal{X} \to \mathcal{Y}$ be a map*

a) The map Φ is distance preserving *if*

$$|\Phi(x) - \Phi(x')|_{\mathcal{Y}} = |x - x'|_{\mathcal{X}}$$

for any $x, x' \in X$.

b) A distance preserving bijection Φ is called an isometry*.*

c) The spaces X and Y are called isometric *(briefly $X \overset{iso}{=\!=} Y$) if there is an isometry $\Phi\colon X \to Y$.*

5.1.2. Lipschitz maps. *Let $\mathcal{X}$ and $\mathcal{Y}$ be metric spaces $\Phi\colon \mathcal{X} \looparrowright \mathcal{Y}$ be a submap and $\pounds \in \mathbb{R}$.*

*a) The submap Φ is called $\pounds$-*Lipschitz *if*

$$|\Phi(x)\ \Phi(y)|_{\mathcal{Y}} \leqslant \pounds\cdot|x\,y|_{\mathcal{X}}$$

for any two points $x, y \in \operatorname{Dom}\Phi$.

- *The 1-Lipschitz maps will be also called* short*.*

b) We say that Φ is Lipschitz *if it is $\pounds$-Lipschitz for some constant $\pounds$. The minimal such constant is denoted by* $\operatorname{lip}\Phi$*.*

c) We say that Φ is locally Lipschitz *if any point $x \in \operatorname{Dom}\Phi$ admits a neighborhood $\Omega \subset \operatorname{Dom}\Phi$ such that the restriction $\Phi|\Omega$ is Lipschitz.*

d) Given $p \in \operatorname{Dom}\Phi$, we denote by $\operatorname{lip}_p\Phi$ the infimum of the real values $\pounds$ such that p admits a neighborhood $\Omega \subset$ $\subset \operatorname{Dom}\Phi$ such that the restriction $\Phi|\Omega$ is $\pounds$-Lipschitz.

Note that $\Phi\colon \mathcal{X} \to \mathcal{Y}$ is $\pounds$-Lipschitz if

$$\Phi(\mathrm{B}(x, R)_{\mathcal{X}}) \subset \mathrm{B}(\Phi(x), \pounds\cdot R)_{\mathcal{Y}}$$

for any $R \geqslant 0$ and $x \in \mathcal{X}$. The following definition gives a dual version of it.

5.1.3. Definitions. *Let $\mathcal{X}$ and $\mathcal{Y}$ be metric spaces, $\Phi\colon \mathcal{X} \to \mathcal{Y}$ be a map and $\pounds \in \mathbb{R}$.*

*a) The map Φ is called $\pounds$-*co-Lipshitz *if*

$$\Phi(\mathrm{B}(x, \pounds\cdot R)_{\mathcal{X}}) \supset \mathrm{B}(\Phi(x), R)_{\mathcal{Y}}$$

for any $x \in \mathcal{X}$ and $R > 0$.

b) The map Φ is called co-Lipshitz *if it is $\pounds$-co-Lipshitz for some constant $\pounds$. The minimal such constant is denoted by* $\operatorname{colip}\Phi$*.*

- *The map Φ is called* submetry *if it is short and 1-co-Lipschitz.*

The $\pounds$-co-Lipschitz maps can be considered as a quantitative version of open maps. By that reason they also called $\pounds$-open; see [34]. Also, be aware that some authors, refer to our $\pounds$-co-Lipschitz maps as to $\frac{1}{\pounds}$-co-Lipschitz; see ???.

5.1.4. Proposition. *Any co-Lipschits map is open and surjective.*

Proof. Follows from the definition directly. □

5.1.5. Proposition. *Let $\mathcal{X}$ and $\mathcal{Y}$ be metric spaces and $\Phi\colon \mathcal{X} \to \mathcal{Y}$ is a continuous co-Lipschitz map Then $\mathcal{Y}$ is complete.*

Proof. Choose a Cauchy sequence $y_1, y_2, \dots$ in $\mathcal{Y}$. Passing to a subsequence if nececury, we may assume that $|y_n\, y_{n+1}|_{\mathcal{Y}} < \frac{1}{2^n}$ for each n.

Denote by $\pounds$ a co-Lipschitz constant of Φ. Note that there is a sequence $x_1, x_2, \dots$ in $\mathcal{X}$ such that

❶ $$\Phi(x_n) = y_n \quad \text{and} \quad |x_n\, x_{n+1}|_{\mathcal{X}} < \tfrac{\pounds}{2^n}.$$

for each n. Indeed, such a sequence (x_n) can be constructed recursively. Assuming that the points $x_1, \dots, x_{n-1}$ are already constructed, the existence of x_n satisfying ❶ follows since Φ is $\pounds$-co-Lipschitz.

Notice that (x_n) is a Cauchy sequence in $\mathcal{X}$. Denote by x_∞ its limit and set $y_\infty = \Phi(x_\infty)$. Since Φ is continuous, $y_n \to y_\infty$ as $n \to \infty$. Hence the result follows. □

5.2 Sumbetry

A map $\sigma\colon \mathcal{L} \to \mathcal{M}$ between the metric spaces $\mathcal{L}$ and $\mathcal{M}$ is called *submetry* if

$$\sigma(\mathrm{B}(p,r)_{\mathcal{L}}) = \mathrm{B}(\sigma(p), r)_{\mathcal{M}}$$

for any $p \in \mathcal{L}$ and $r \geqslant 0$.

Equivalently, a map $\sigma\colon \mathcal{L} \to \mathcal{M}$ is called sumbetry if it is 1-Lipshitz and 1-co-Lipschitz at the same time.

Note that any submetry is an onto map.

The main souse of examples of submetries comes from isometric group actions.

Nemely, assume $\mathcal{L}$ is a metric space and the group G acts on $\mathcal{L}$. Denote by $\mathcal{L}/G$ the set of G-orbits; let us equip it with the Hausdorff metric.

In general, $\mathcal{L}/G$ is a premetric space, but if all the G-orbits form closed sets in $\mathcal{L}$ then $\mathcal{L}/G$ is a metric space.

5.2.1. Proposition. *Let $\mathcal{L}$ be a metric space and group G acts on $\mathcal{L}$ by isometries and in such a way that every G-orbit is closed. Then the projection map $\mathcal{L} \to \mathcal{L}/G$ is a sumbetry.*

5.2.2. Proposition. *Let $\mathcal{L}$ be a length space and $\sigma\colon \mathcal{L} \to \mathcal{M}$ is a submetry. Then $\mathcal{M}$ is a length space.*

Proof. Fix $\varepsilon > 0$ and a pair of points $x, y \in \mathcal{M}$.

Since σ is 1-co-Lipschitz, there are points $\hat{x}, \hat{y} \in \mathcal{L}$ such that $\sigma(\hat{x}) = x$, $\sigma(\hat{y}) = y$ and $|\hat{x}\,\hat{y}|_{\mathcal{L}} < |x\,y|_{\mathcal{M}} + \varepsilon$.

Since $\mathcal{L}$ is a length space, there is a curve γ joining $\hat{x}$ to $\hat{y}$ in $\mathcal{L}$ such that

$$\operatorname{length}\gamma \leqslant |x\,y|_{\mathcal{M}} + \varepsilon.$$

Since σ is 1-Lipschitz, there

$$\operatorname{length}\sigma\circ\gamma \leqslant \operatorname{length}\gamma.$$

Note that the curve $\sigma\circ\gamma$ joins x to y and from above

$$\operatorname{length}\sigma\circ\gamma < |x\,y|_{\mathcal{M}} + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, we have that $\mathcal{M}$ is a length space. □

5.3 Speed of curves

Let $\mathcal{X}$ be a metric space. A *curve* in $\mathcal{X}$ is a continuous map $\alpha\colon \mathbb{I} \to \mathcal{X}$, where $\mathbb{I}$ is a *real interval* (i.e. an arbitrary convex subset of $\mathbb{R}$); the curve it called *Lipschitz* or *locally Lipschitz* if the map $\alpha\colon \mathbb{I} \to \mathcal{X}$ is.

The length of the curve $\alpha\colon \mathbb{I} \to \mathcal{X}$, is defined by

$$\operatorname{length}\alpha = \sup\left\{ \sum_n |\alpha(t_{n-1})\ \alpha(t_n)|_{\mathcal{X}} \;\middle|\; t_0 < t_1 < \ldots < t_k \right\}.$$

The following lemma is an easy exercise.

5.3.1. Lower semicontinuity of length. *Assume $\alpha_n\colon \mathbb{I} \to \mathcal{X}$ is a sequence of curves which converges pointwise to a curve $\alpha_\infty\colon \mathbb{I} \to \mathcal{X}$. Then*

$$\operatorname{length}\alpha_\infty \leqslant \varliminf_{n\to\infty} \operatorname{length}\alpha_n.$$

The following theorem follows from [32, 2.7.4].

5.3.2. Theorem. *Let $\mathcal{X}$ be a metric space and $\alpha\colon \mathbb{I} \to \mathcal{X}$ be a locally Lipschitz curve. Then the speed function*

$$\operatorname{speed}\alpha(t_0) = \lim_{t\to t_0} \frac{|\alpha(t)\ \alpha(t_0)|}{|t - t_0|}$$

is defined for almost all $t_0 \in \mathbb{I}$ and

$$\operatorname{length}\alpha = \int_{\mathbb{I}} \operatorname{speed}\alpha(t)\cdot \boldsymbol{d}t,$$

where $\int$ *denotes the Lebesgue integral.*

A curve $\alpha\colon \mathbb{I} \to \mathcal{X}$ is called *unit-speed curve*, if for any subinterval $[a,b] \subset \mathbb{I}$, we have

$$b - a = \operatorname{length}(\alpha|[a,b]).$$

According to the above theorem, this is equivalent to the condition that α is Lipschitz and speed $\alpha \overset{a.e.}{=\!=} 1$.

The following theorem from [84] generalize the one above.

5.3.3. Theorem. *Let* $\mathcal{X}$ *be a metric space and* $f\colon \mathbb{R}^n \multimap \mathcal{X}$ *be* 1-*Lipschitz. Then for almost all* $x \in \operatorname{Dom} f$ *there is a prenorm* $\|*\|_x$ *on* $\mathbb{R}^n$ *such that we have*

$$|f(y)\ f(z)|_{\mathcal{X}} = \|z-y\|_x + o(|y-x| + |z-x|).$$

Moreover if $s(x) = \sup\left\{ \frac{\|w\|_x}{|w|} \;\middle|\; w \neq 0 \right\}$ *then*

$$\operatorname{HausMes}_n[\operatorname{Im} f] \leqslant \int\limits_{\operatorname{Dom} f} s(x).$$

The norm $\|*\|_x$ in the above theorem will be called *differential of induced metric* at x.

5.4 Convexity of real-to-real functions

We will be interested in the solutions of the following differential inequalities

❶ $$y'' + \kappa\cdot y \geqslant \lambda \quad \text{(and respectively} \quad y'' + \kappa\cdot y \leqslant \lambda)$$

for fixed $\kappa, \lambda \in \mathbb{R}$. The solution $y\colon \mathbb{R} \multimap \mathbb{R}$ are only assumed to be lower(respectively upper) semicontinuous subfunctions.

The inequalities ❶ are understood in the sense of distributions. I.e., for any smooth function φ with compact support $\operatorname{Supp}\varphi \subset \operatorname{Dom} y$ the following inequality should be satisfied:

❷ $$\int\limits_{\operatorname{Dom} y} [y(t)\cdot\varphi''(t) + \kappa\cdot y(t)\cdot\varphi(t) - \lambda]\cdot \boldsymbol{d}t \geqslant 0 \quad \text{(and respectively} \quad \leqslant 0).$$

The integral is understood in the sense of Lebesgue, in particular the inequality ❷ makes sense for any Borel-measurable subfunction y.

5.4.1. Proposition. *Let* $\mathbb{I} \subset \mathbb{R}$ *is an open interval and* $y_n\colon \mathbb{I} \to \mathbb{R}$ *be a sequence of solutions of one of the inequality in* ❶. *Assume* $y_n(t) \to$ $\to y_\infty(t)$ *as* $n \to \infty$ *for any* $t \in \mathbb{I}$. *Then* y_∞ *is a solution of the same inequality in* ❶.

Assume y is a solution of one of the inequality in ❶. For $t_0 \in \operatorname{Dom} y$, let us define right (left) derivative $y^+(t_0)$ ($y^-(t_0)$) at t_0 by

$$y^{\pm}(t_0) = \lim_{t \to t_0 \pm} \frac{y(t) - y(t_0)}{|t - t_0|}.$$

The sign convention we use for y^- is not standard, thus for $y(t) = t$ we have $y^{\pm}(t) = \pm 1$.

5.4.2. Proposition. *Let $\mathbb{I} \subset \mathbb{R}$ is an open interval and $y_n \colon \mathbb{I} \to \mathbb{R}$ be a sequence of solutions of one of the inequality in ❶. Then right and left derivatives $y^+(t_0)$ and $y^-(t_0)$ are defined for any $t_0 \in \mathbb{I}$.*

The next theorem gives a number of equivalent ways to define such a generalized solution.

5.4.3. Theorem. *Let $\mathbb{I}$ be an open real interval and $y \colon \mathbb{I} \to \mathbb{R}$ be a locally Lipschitz function then the following conditions are equivalent:*

a) $y'' \geqslant \lambda - \kappa \cdot y$ *(respectively* $y'' \leqslant \lambda - \kappa \cdot y$*)*

b) (barrier inequality) For any $t_0 \in \mathbb{I}$, there is a solution $\bar{y}$ of ordinary differential equation $\bar{y}'' = \lambda - \kappa \cdot \bar{y}$ with $\bar{y}(t_0) = y(t_0)$ such that $\bar{y} \geqslant y$ (respectively $\bar{y} \leqslant y$) for all $t \in [t_0 - \varpi^\kappa, t_0 + \varpi^\kappa] \cap \mathbb{I}$.
The function $\bar{y}$ is called lower *(respectively* upper*)* barrier *of y at t_0.*

c) Suppose $y'' \leqslant \lambda - \kappa \cdot y$. Let $t_0 \in \mathbb{I}$ and let $\bar{y}$ be a solution of the ordinary differential equation $\bar{y}'' = \lambda - \kappa \cdot \bar{y}$ such that $\bar{y}(t_0) = y(t_0)$ and $y^+(t_0) \leqslant y(t_0) \leqslant -y^-(t_0)$. (note that such $\bar{y}$ is unique if y is differentiable at t_0).
Then $\bar{y} \geqslant y$ for all $t \in [t_0 - \varpi^\kappa, t_0 + \varpi^\kappa] \cap \mathbb{I}$, i.e. $\bar{y}$ is a barrier of y at t_0.

d) (Jensen's inequality) For any pair of values $t_1 < t_2$ in $\mathbb{I}$, such that $|t_2 - t_1| < \varpi^\kappa$ the unique solutions $z(t)$ of

$$z'' = \lambda - \kappa \cdot z$$

such that

$$z(t_1) = y(t_1), \quad z(t_2) = y(t_2)$$

satisfies $y(t) \leqslant z(t)$ (respectively $y(t) \geqslant z(t)$) for all $t \in$ $\in [t_1, t_2]$.

Proof. It is left to the reader. □

We will often need the following simple fact about convergence of derivative of convex functions:

5.4.4. Lemma on equilibrium. *Let $\mathbb{I}$ be an open interval and $f_n\colon \mathbb{I} \to \mathbb{R}$ be a sequence of concave functions. Assume the functions f_n pointwise converge to a function $f\colon \mathbb{I} \to \mathbb{R}$. Then for any $t_0 \in \mathbb{I}$,*

$$f^{\pm}(t_0) \leqslant \varliminf_{n\to\infty} f_n^{\pm}(t_0).$$

Proof. Since f_n are convex, we have $f_n^+(t_0) + f_n^-(t_0) \geqslant 0$, and for any t,

$$f_n(t) \geqslant f_n(t_0) \pm f^{\pm}(t_0)\cdot(t - t_0).$$

Passing to the limit, we get

$$f(t) \geqslant f(t_0) + \left[\varlimsup_{n\to\infty} f_n^+(t_0)\right]\cdot(t - t_0)$$

for $t \geqslant t_0$, and

$$f(t) \geqslant f(t_0) - \left[\varlimsup_{n\to\infty} f_n^-(t_0)\right]\cdot(t - t_0)$$

for $t \leqslant t_0$. Hence the result. □

5.4.5. Corollary. *Let $\mathbb{I}$ be an open interval and $f_n\colon \mathbb{I} \to \mathbb{R}$ be a sequence of functions which satisfy $f_n'' \leqslant \lambda$ and converge pointwise to a function $f\colon \mathbb{I} \to \mathbb{R}$. Then:*

a) If f is differentiable at $t_0 \in \mathbb{I}$, then

$$f'(t_0) = \pm \lim_{n\to\infty} f_n^{\pm}(t_0).$$

b) If all f_n and f are differentiable at $t_0 \in \mathbb{I}$, then

$$f'(t_0) = \lim_{n\to\infty} f_n'(t_0).$$

Proof. Set $\hat{f}_n(t) = f_n(t) - \frac{\lambda}{2}\cdot t^2$ and $\hat{f}(t) = f(t) - \frac{\lambda}{2}\cdot t^2$. Note that $\hat{f}_n$ are concave and $\hat{f}_n \to \hat{f}$ pointwise. Thus, theorem follows from the lemma on equilibrium (5.4.4). □

5.5 Convexity of functions on metric space.

Although the following definitions has sense for general metric space, it only defines reasonable classes of function for reasonable spaces — Alexandrov spaces are included.

5.5.1. Definition. *Let $\mathcal{X}$ be a metric space. A subfunction $f\colon \mathcal{X} \looparrowright (-\infty, \infty]$ is called* semiconvex *if for any point $p \in \operatorname{Dom} f$ there*

is $\lambda \in \mathbb{R}$ *and a neighborhood* $\Omega_p \subset \operatorname{Dom} f$ *such for any unit-speed geodesic* γ *in* Ω_p *the real-to-real function* $t \mapsto f \circ \gamma(t) - \frac{\lambda}{2} \cdot t^2$ *is convex.*

A subfunction $f\colon \mathcal{X} \looparrowright [-\infty, \infty)$ *is called* semiconcave *if the subfunction* $(-f)$ *is semiconvex.*

Further, we give a meaning of certain type of second order ordinary differential inequalities in context of metric spaces.

5.5.2. Definition. *Let* $\mathcal{X}$ *be a metric space and* $\varphi\colon \mathbb{R}^2 \to \mathbb{R}$ *be a Lipschitz function. Given a semiconvex (semiconcave) subfunction* $f\colon \mathcal{X} \looparrowright \bar{\mathbb{R}}$*, we will write*

$$f'' \geqslant \varphi(f, |f'|) \quad (\textit{correspondingly} \quad f'' \leqslant \varphi(f, |f'|)\ )$$

if for any geodesic γ *in* $\operatorname{Dom} f$ *the real-to-real function* $y = f \circ \gamma$ *satisfies the differential inequality*

$$y'' \geqslant \varphi(y, |y'|) \quad (\textit{correspondingly} \quad y'' \leqslant \varphi(y, |y'|))$$

in the sense of ???.

For $\lambda \in \mathbb{R}$, the solutions of differential inequalities

$$f'' \geqslant \lambda \quad \text{and} \quad f'' \leqslant \lambda$$

will be called correspondingly λ-*convex* and λ-*concave* subfunctions. If $\lambda = 0$, 0-convex and 0-concave subfunctions will be also called *convex* and *concave* correspondingly. If f is λ-convex for some $\lambda > 0$ it will be called *strongly convex*; correspondingly if f is λ-concave for some $\lambda < 0$ it will be called *strongly concave.*

5.5.3. Proposition. *Let* $\mathcal{X}$ *be a metric space. Assume that* $f\colon \mathcal{X} \looparrowright \mathbb{R}$ *is a semiconvex subfunction and* $\varphi\colon \mathbb{R} \to \mathbb{R}$ *is a nondecreasing seimconvex function. Then the composition* $\varphi \circ f$ *is a semiconvex subfunction.*

NB! Various authors define the class of λ-convex (λ-concave) function differently. It may correspond to $\pm\lambda$-convex ($\pm\lambda$-concave) or $\pm\frac{\lambda}{2}$-convex ($\pm\frac{\lambda}{2}$-concave) function in our definitions.

Further, two classes of functions, satisfying differential inequalities

$$f'' \geqslant \lambda - \kappa \cdot f \quad \text{and} \quad f'' \leqslant \lambda - \kappa \cdot f$$

will play a very important role. The solutions of these differential inequalities should be understood to possess a special form of convexity and concavity that is optimized to reflect properties of spaces with curvature $\leqslant \kappa$ and $\geqslant \kappa$. If $\kappa \neq 0$, by adding the constant $-\lambda/\kappa$ to these solutions, we obtain the solutions of the differential inequalities

$$f'' \geqslant -\kappa \cdot f \quad \text{and} \quad f'' \leqslant -\kappa \cdot f,$$

which also arise naturally.

5.6 Miscellaneous

5.6.1. Convexity of the limit.

Tietze extension theorem???

5.6.2. Lemma. *Let $\mathcal{X}$ be a metric space and $f\colon \mathcal{X} \to \mathbb{R}$ be a continuous function. Then for any $\varepsilon > 0$ there is a locally Lipschitz function $f_\varepsilon\colon \mathcal{X} \to \mathbb{R}$ such that $|f(x) - f_\varepsilon(x)| < \varepsilon$ for any $x \in \mathcal{X}$.*

Proof. Assume that $f \geqslant 1$. Construct a continous positive function $\rho\colon \mathcal{X} \to \mathbb{R}_{>0}$ such that

$$|x\,y| < \rho(x) \quad \Rightarrow \quad |f(x) - f(y)| < \varepsilon.$$

Consider function

$$f_\varepsilon(x) = \sup\left\{ f(x)\cdot\left(1 - \tfrac{|x\,y|}{\rho(x)}\right) \;\middle|\; x \in \mathcal{X} \right\}.$$

It is straightforward to check that each f_ε is locally Lipschtz and $0 \leqslant$ $\leqslant f_\varepsilon - f < \varepsilon$.

Note that any continuous function can be presented as difference of two continuous functions bounded below by 1. Hence the result. $\square$

Chapter 6

The ghost of Euclid

6.1 Geodesics, triangles and hinges

Geodesics and their relatives. Let $\mathcal{X}$ be a metric space and $\mathbb{I}$ be a real interval. A globally isometric map $\gamma\colon \mathbb{I} \to \mathcal{X}$ is called a *unit-speed geodesic*[1]; in other words, $\gamma\colon \mathbb{I} \to \mathcal{X}$ is a unit-speed geodesic if

$$|\gamma(s)\ \gamma(t)|_{\mathcal{X}} = |s - t|$$

for any pair $s, t \in \mathbb{I}$.

A unit-speed geodesic $\gamma\colon \mathbb{R}_{\geqslant 0} \to \mathcal{X}$ is called a *ray*.

A unit-speed geodesic $\gamma\colon \mathbb{R} \to \mathcal{X}$ is called a *line*.

A unit-speed geodesic between p and q in $\mathcal{X}$ will be denoted by $\mathrm{geod}_{[pq]}$. We assume $\mathrm{geod}_{[pq]}$ is parametrized starting at p; i.e. $\mathrm{geod}_{[pq]}(0) = p$ and $\mathrm{geod}_{[pq]}(|pq|) = q$. The image of $\mathrm{geod}_{[pq]}$ will be denoted by $[pq]$ and called a *geodesic*. The term *geodesic* will also be used for a linear reparametrization of a unit-speed geodesic; when a confusion is possible we call the latter a *constant-speed geodesic*. With slight abuse of notation, we will use $[pq]$ also for the class of all linear reparametrizations of $\mathrm{geod}_{[pq]}$.

We may write $[pq]_{\mathcal{X}}$ to emphasize that the geodesic $[pq]$ is in the space $\mathcal{X}$. Also we use the following short-cut notation:

$$]pq[= [pq]\backslash\{p, q\}, \qquad]pq] = [pq]\backslash\{p\}, \qquad [pq[= [pq]\backslash\{q\}.$$

A curve that is a reparametrization, not necessarily linear, of a geodesic will be called a *pregeodesic*.

[1] Various authors call it differently: *shortest path*, *minimizing geodesic*.

In general, a geodesic between p and q need not exist and if it exists, it need not to be unique. However, once we write $\operatorname{geod}_{[pq]}$ or $[pq]$ we mean that we made a choice of geodesic.

A constant-speed geodesic $\gamma\colon [0,1] \to \mathcal{X}$ is called a *geodesic path*. Given a geodesic $[pq]$, we denote by $\operatorname{path}_{[pq]}$ the corresponding geodesic path; i.e.,

$$\operatorname{path}_{[pq]}(t) \equiv \operatorname{geod}_{[pq]}(t\cdot|pq|).$$

A curve $\gamma\colon \mathbb{I} \to \mathcal{X}$ is called a *local geodesic*, if for any $t \in \mathbb{I}$ there is a neighborhood $U \ni t$ in $\mathbb{I}$ such that the restriction $\gamma|_U$ is a constant-speed geodesic. If $\mathbb{I} = [0,1]$, then γ is called a *local geodesic path*.

6.1.1. Proposition. *Suppose $\mathcal{X}$ is a metric space and $\gamma\colon [0,\infty) \to \mathcal{X}$ is a ray. Then the* Busemann function $\operatorname{bus}_\gamma\colon \mathcal{X} \to \mathbb{R}$

❶ $$\operatorname{bus}_\gamma(x) = \lim_{t\to\infty} |\gamma(t)\ x| - t$$

is defined and 1*-Lipschitz.*

Proof. As follows from the triangle inequality, the function

$$t \mapsto |\gamma(t)\ x| - t$$

is nonincreasing in t. Clearly $|\gamma(t)\ x| - t \geqslant -|\gamma(0)\ x|$. Thus the limit in ❶ is defined. □

Triangles. For a triple of points $p,q,r \in \mathcal{X}$, a choice of triple of geodesics $([qr],[rp],[pq])$ will be called a *triangle* and we will use short notation $[pqr] = ([qr],[rp],[pq])$. Again given a triple $p,q,r \in \mathcal{X}$ it may be no triangle $[pqr]$ simply because one of the pairs of these points can not be joined by a geodesic, and also it maybe many different triangles with these vertexes, any of which can be denoted by $[pqr]$. Once we write $[pqr]$, it means we made a choice of such a triangle, i.e. a choice of each $[qr],[rp]$ and $[pq]$. The value $|pq| + |qr| + |rp|$ will be called *perimeter of triangle* $[pqr]$; it obviousely coinsides with perimeter of triple p, q, r defined below.

Hinges. Let $p,x,y \in \mathcal{X}$ be a triple of points such that p is distinct from x and y. A pair geodesics $([px],[py])$ will be called *hinge* and briefly, it will be denoted by $[p\,{}^x_y] = ([px],[py])$.

6.2 Model angles and triangles.

Let $\mathcal{X}$ be a metric space, $p,q,r \in \mathcal{X}$ and $\kappa \in \mathbb{R}$. Let us define its *model triangle* $[\tilde p\tilde q\tilde r]$ (briefly, $[\tilde p\tilde q\tilde r] = \tilde\triangle^\kappa(pqr)$) to be a triangle in the model plane $\mathbb{M}^2[\kappa]$ such that

$$|\tilde p\tilde q| = |pq|, \quad |\tilde q\tilde r| = |qr|, \quad |\tilde r\tilde p| = |rp|.$$

In the notation of Section 1.1, $\tilde\triangle^\kappa(pqr) = \tilde\triangle^\kappa\{|qr|, |rp|, |pq|\}$.

If $\kappa \leqslant 0$ model triangle allways defined, it exists and unique up to isometry of $\mathbb{M}^2[\kappa]$. If $\kappa > 0$, the model triangle is said to be defined if in addition

$$|pq| + |qr| + |rp| < 2\cdot\varpi^\kappa.$$

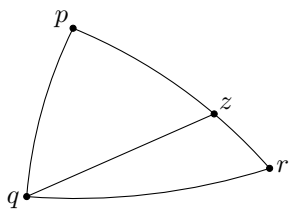
In this case it also exists and unique up to isometry of $\mathbb{M}^2[\kappa]$. The value $|pq| + |qr| + |rp|$ will be called *perimeter of triple* p, q, r.

If for $p, q, r \in \mathcal{X}$, $[\tilde p\tilde q\tilde r] = \tilde\triangle^\kappa(pqr)$ is defined and $|pq|, |pr| > 0$, the angle measure of $[\tilde p\tilde q\tilde r]$ at $\tilde p$ will be called *model angle* of triple p, q, r and it will be denoted by $\tilde\angle^\kappa(p\,{}^q_r)$.

In the notation of Section 1.1, $\tilde\angle^\kappa(p\,{}^q_r) = \tilde\angle^\kappa\{|qr|; |pq|, |pr|\}$.

6.2.1. Alexandrov's lemma. *Let p, q, r, z be distinct points in a metric space such that $z \in\,]pr[$ and*

$$|pq| + |qr| + |rp| < 2\cdot\varpi^\kappa.$$



Then the following expressions have the same sign:

a) $\tilde\angle^\kappa(p\,{}^q_z) - \tilde\angle^\kappa(p\,{}^q_r)$,

b) $\tilde\angle^\kappa(z\,{}^q_p) + \tilde\angle^\kappa(z\,{}^q_r) - \pi$.

Moreover,

$$\tilde\angle^\kappa(q\,{}^p_r) \geqslant \tilde\angle^\kappa(q\,{}^p_z) + \tilde\angle^\kappa(q\,{}^z_r),$$

with equality if and only if the expressions in (a) and (b) vanish.

Proof. By the triangle inequality,

$$|pq| + |qz| + |zp| \leqslant |pq| + |qr| + |rp| < 2\cdot\varpi^\kappa.$$

Therefore the model triangle $[\tilde p\tilde q\tilde z] = \tilde\triangle^\kappa pqz$ is defined. Take a point $\tilde r$ on the extension of $[\tilde p\tilde z]$ beyond $\tilde z$ so that $|\tilde p\tilde r| = |pr|$ (and therefore $|\tilde p\tilde z| = |pz|$).

From monotonicity of function $a \mapsto \tilde\angle^\kappa\{a; b, c\}$ (1.1.1c), the following expressions have the same sign:

(i) $\measuredangle[\tilde p\,{}^{\tilde q}_{\tilde r}] - \tilde\angle^\kappa(p\,{}^q_r)$;

(ii) $|\tilde p\tilde r| - |pr|$;

(iii) $\measuredangle[\tilde z\,{}^{\tilde q}_{\tilde r}] - \tilde\angle^\kappa(z\,{}^q_r)$.

Since

$$\measuredangle[\tilde p\,{}^{\tilde q}_{\tilde r}] = \measuredangle[\tilde p\,{}^{\tilde q}_{\tilde z}] = \tilde\angle^\kappa(p\,{}^q_z)$$

and

$$\measuredangle[\tilde z\,{}^{\tilde q}_{\tilde r}] = \pi - \measuredangle[\tilde z\,{}^{\tilde p}_{\tilde q}] = \pi - \tilde\angle^\kappa(z\,{}^p_q),$$

the first statement follows.

For the second statement, construct $[\tilde q\tilde z r'] = \tilde\triangle^{\kappa} qzr$ on the opposite side of $[\tilde q\tilde z]$ from $[\tilde p\tilde q\tilde z]$. Since

$$|\tilde p r'| \leqslant |\tilde p\,\tilde z| + |\tilde z\, r'| = |pz| + |zr| = |pr|,$$

then

$$\begin{aligned}\measuredangle^{\kappa}(q\,{}^{p}_{z}) + \measuredangle^{\kappa}(q\,{}^{z}_{r}) &= \measuredangle[\tilde q\,{}^{\tilde p}_{\tilde z}] + \measuredangle[\tilde q\,{}^{\tilde z}_{r'}] = \\ &= \measuredangle[\tilde q\,{}^{\tilde p}_{r'}] \leqslant \\ &\leqslant \measuredangle^{\kappa}(q\,{}^{p}_{r}).\end{aligned}$$

Equality holds if and only if $|\tilde p r'| = |pr|$, as required. □

6.3 Angles and the first variation.

Given a hinge $[p\,{}^{x}_{y}]$, we define its *angle* as follows:

$$\measuredangle[p\,{}^{x}_{y}] \overset{def}{=\!=} \lim_{\bar x,\bar y\to p} \measuredangle^{\kappa}(p\,{}^{\bar x}_{\bar y}),$$

where $\bar x \in\,]px]$ and $\bar y \in\,]py]$.

Similarly to $\measuredangle^{\kappa}(p\,{}^{x}_{y})$, we will use short notation

$$\tilde\Upsilon^{\kappa}[p\,{}^{x}_{y}] = \tilde\Upsilon^{\kappa}\left\{\measuredangle[p\,{}^{x}_{y}]; |px|, |py|\right\}.$$

The value $\tilde\Upsilon^{\kappa}[p\,{}^{x}_{y}]$ will be called the *model side* of hinge $[p\,{}^{x}_{y}]$.

Remark. We stick to this definition since in Alexandrov's geometry angles always defined (see Theorem 8.3.1*d* and Corollary 9.3.2*b*). For general metric spaces, the angle need not be defined and it is more natural to consider *upper angle* which is defined as

$$\measuredangle^{\mathrm{up}}[p\,{}^{x}_{y}] \overset{def}{=\!=} \varlimsup_{\bar x,\bar y\to p} \measuredangle^{\kappa}(p\,{}^{\bar x}_{\bar y}),$$

where $\bar x \in\,]px]$ and $\bar y \in\,]py]$. A good discussion of different definition of angles is given in ???.

6.3.1. Lemma. *For any* $\kappa, \mathrm{K} \in \mathbb{R}$*, there exists* $c \in \mathbb{R}$ *such that*

❶ $$|\measuredangle^{\mathrm{K}}(p\,{}^{x}_{y}) - \measuredangle^{\kappa}(p\,{}^{x}_{y})| \leqslant c\cdot|px|\cdot|py|,$$

whenever the lefthand side is defined.

Lemma 6.3.1 implies that the definition of angle is independent of κ. In particular, one can take $\kappa = 0$ in the definition, so that the angle can be calculated from the cosine law:

$$\cos\measuredangle^{0}(p\,{}^{x}_{y}) = \frac{|px|^2 + |py|^2 - |xy|^2}{2\cdot|px|\cdot|py|}.$$

Proof. The function $\kappa \mapsto \measuredangle^{\kappa}(p\,{}^{x}_{y})$ is nondecreasing (see 1.1.1d). Thus, for $\mathrm{K} > \kappa$, we have

$$\begin{aligned}0 \leqslant \measuredangle^{\mathrm{K}}(p\,{}^{x}_{y}) - \measuredangle^{\kappa}(p\,{}^{x}_{y}) \leqslant & \measuredangle^{\mathrm{K}}(p\,{}^{x}_{y}) + \measuredangle^{\mathrm{K}}(x\,{}^{p}_{y}) + \measuredangle^{\mathrm{K}}(y\,{}^{p}_{x}) - \\ & - \measuredangle^{\kappa}(p\,{}^{x}_{y}) - \measuredangle^{\kappa}(x\,{}^{p}_{y}) - \measuredangle^{\kappa}(y\,{}^{p}_{x}) = \\ = & \mathrm{K}\cdot \operatorname{area} \tilde{\triangle}^{\mathrm{K}}(pxy) - \kappa\cdot \operatorname{area} \tilde{\triangle}^{\kappa}(pxy).\end{aligned}$$

Thus, ❶ follows since

$$0 \leqslant \operatorname{area} \tilde{\triangle}^{\kappa}(pxy) \leqslant \operatorname{area} \tilde{\triangle}^{\mathrm{K}}(pxy),$$

where the latter area is at most $|px|\cdot|py|$.

□

6.3.2. Triangle inequality for angles. *Let $[px^1]$, $[px^2]$ and $[px^3]$ be three geodesics in a metric space. If all of the angles $\alpha^{ij} = \measuredangle[p\,{}^{x^i}_{x^j}]$ are defined then they satisfy the triangle inequality:*

$$\alpha^{13} \leqslant \alpha^{12} + \alpha^{23}.$$

Remark. The above theorem also holds for upper angles, see the remark on page 52.

Proof. Since $\alpha^{13} \leqslant \pi$, we can assume that $\alpha^{12} + \alpha^{23} < \pi$. Set $\gamma^i = \operatorname{geod}_{[px^i]}$. Given any $\varepsilon > 0$, for all sufficiently small $t, \tau, s \in \mathbb{R}_+$ we have

$$\begin{aligned}|\gamma^1(t)\ \gamma^3(\tau)| \leqslant & |\gamma^1(t)\ \gamma^2(s)| + |\gamma^2(s)\ \gamma^3(\tau)| < \\ < & \sqrt{t^2 + s^2 - 2\cdot t\cdot s\cdot \cos(\alpha^{12} + \varepsilon)} + \\ & + \sqrt{s^2 + \tau^2 - 2\cdot s\cdot \tau\cdot \cos(\alpha^{23} + \varepsilon)} \leqslant\end{aligned}$$

(Below we define $s(t,\tau)$ so that for $s = s(t,\tau)$, this chain of inequalities continues the following way.)

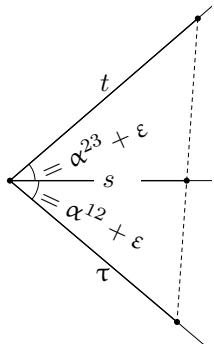
$$\leqslant \sqrt{t^2 + \tau^2 - 2\cdot t\cdot \tau\cdot \cos(\alpha^{12} + \alpha^{23} + 2\cdot\varepsilon)}.$$

Thus for any $\varepsilon > 0$,

$$\alpha^{13} \leqslant \alpha^{12} + \alpha^{23} + 2\cdot\varepsilon.$$

Hence the result.

To define $s(t,\tau)$, consider three rays $\tilde\gamma^1$, $\tilde\gamma^2$, $\tilde\gamma^3$ on a Euclidean plane starting at one point, such that $\measuredangle(\tilde\gamma^1,\tilde\gamma^2) = \alpha^{12}+\varepsilon$, $\measuredangle(\tilde\gamma^2,\tilde\gamma^3) = \alpha^{23}+\varepsilon$ and $\measuredangle(\tilde\gamma^1,\tilde\gamma^3) = \alpha^{12}+\alpha^{23}+2\cdot\varepsilon$. We parametrize each ray by length from the starting point. Given two positive numbers $t,\tau\in\mathbb{R}_+$, let $s = s(t,\tau)$ be the number such that $\tilde\gamma^2(s)\in$ $\in[\tilde\gamma^1(t)\ \tilde\gamma^3(\tau)]$. Clearly $s\leqslant\max\{t,\tau\}$, so t,τ,s may be taken sufficiently small. $\square$



6.3.3. Exercise. *Prove that the sum of adjacent angles is at least π.*

More precisely: let $\mathcal{X}$ be a complete length space and $p,x,y,z\in\mathcal{X}$. If $p\in\,]xy[$, then

$$\measuredangle[p\,{}^{x}_{z}] + \measuredangle[p\,{}^{y}_{z}] \geqslant \pi$$

whenever each angle on the left-hand side is defined.

6.3.4. First variation inequality. *Assume for hinge $[q\,{}^{p}_{x}]$ the angle $\alpha = \measuredangle[q\,{}^{p}_{x}]$ is defined then*

$$|p\,\mathrm{geod}_{[qx]}(t)| \leqslant |qp| - t\cdot\cos\alpha + o(t).$$

Proof. Take sufficiently small $\varepsilon>0$. For all sufficiently small $t>0$, we have that

$$\begin{aligned}|\mathrm{geod}_{[qp]}(t/\varepsilon)\ \mathrm{geod}_{[qx]}(t)| &\leqslant \tfrac{t}{\varepsilon}\cdot\sqrt{1+\varepsilon^2-2\cdot\varepsilon\cdot\cos\alpha} + o(t) \leqslant\\ &\leqslant \tfrac{t}{\varepsilon} - t\cdot\cos\alpha + t\cdot\varepsilon.\end{aligned}$$

Applying triangle inequality, we get

$$\begin{aligned}|p\,\mathrm{geod}_{[qx]}(t)| &\leqslant |p\,\mathrm{geod}_{[qp]}(t/\varepsilon)| + |\mathrm{geod}_{[qp]}(t/\varepsilon)\ \mathrm{geod}_{[qx]}(t)| \leqslant\\ &\leqslant |pq| - t\cdot\cos\alpha + t\cdot\varepsilon\end{aligned}$$

for any $\varepsilon>0$ and all sufficiently small t. Hence the result. $\square$

Chapter 7

Measure theory

7.1 Carathéodory's construction

Let $\mathcal{X}$ be a metric space. The function μ with values in $[0,\infty]$, defined on all subsets of $\mathcal{X}$ is called *outer measure* if

- $\mu\varnothing = 0$;
- If $A \subset B \subset \mathcal{X}$ then $\mu A \leqslant \mu B$;
- For any sequence $A_1, A_2, \dots$ of subsets of $\mathcal{X}$ we have

$$\mu\left(\bigcup_n A_n\right) \leqslant \sum_n \mu A_n.$$

A subset $E \subset \mathcal{X}$ is called μ-*measurable* if

$$\mu A = \mu(A \cap E) + \mu(A \backslash E)$$

for every subset $A \subset \mathcal{X}$.

Recall that given two subset A and B of a metric space $\mathcal{X}$, we define

$$|A\,B|_{\mathcal{X}} \overset{def}{=\!=} \inf\{\, |a\,b|_{\mathcal{X}} \mid a \in A \text{ and } b \in B \,\}.$$

The following is a classical lemma in measure theory, see [53, 2.1.3 and 2.3.2(9)].

7.1.1. Carathéodory's lemma. *Let μ be an outer measure on a metric space $\mathcal{X}$. Then the μ-measurable sets form a sigma-algebra.*

Moreover, if

❶ $$\mu(A \cup B) = \mu A + \mu B$$

for any two sets A and B such that $|A\,B| > 0$ then any Borel set in $\mathcal{X}$ is μ-measurable.

Carathéodory's construction. Fix a function ρ, which returns a value in $[0,\infty]$ for any closed subset of $\mathcal{X}$. Define outer measure Carath_ρ of set W in $\mathcal{X}$ in the following way

$$\mathrm{Carath}_\rho W \overset{def}{=\!=} \lim_{\varepsilon\to 0} \inf\left\{ \sum_{n\in\mathbb{N}} \rho A_n \;\middle|\; \begin{array}{l} \bigcup\limits_{n\in\mathbb{N}} A_n \supset W, \text{all } A_n \text{ are closed,} \\ \text{and } \operatorname{diam} A_n < \varepsilon \text{ for each } n. \end{array} \right\}.$$

Note that the value of the infimum above is non-decreasing in ε; in particular the limit is defined.

An outer measure μ on $\mathcal{X}$ is called *Borel regular* if any Borel set in $\mathcal{X}$ is μ-measurable and for any set $A\subset\mathcal{X}$ there is a Borel set B such that $A\subset B$ and $\mu A=\mu B$.

From Carathéodory's lemma (7.1.1), we get the following.

7.1.2. Corollary. *The Carathéodory's construction always produce a Borel regular outer measure.*

7.1.3. Examples. *Let us apply the Carathéodory's construction above for different choices of the function ρ. Assume that A is a closed subset of a metric space $\mathcal{X}$. Fix a real value $\alpha\geqslant 0$ and an integer value $\bar{k}\geqslant 0$.*

a) If we take $\rho A=(\operatorname{diam} A)^\alpha$, then the measure Carath_ρ *is called the* α-dimensional Hausdorff measure *and denoted as* $\mathrm{HausMes}_\alpha$.

b) If $\rho A=(\operatorname{rad} A)^\alpha$, the measure Carath_ρ *is called the* α-dimensional spherical measure *and denoted as* SphMes_α.

c) Assume ρA is the least lower bound for the Lebesgue measure of a compact set K in $\mathbb{E}^{\bar{k}}$ such that $f(K)\supset A$ for some short map $f\colon K\to\mathcal{X}$ (if no such K exists, we set $\rho A=\infty$). For this choice of ρ, the measure Carath_ρ *will be called $\bar{k}$-dimensional* long measure *and it will be denoted as* $\mathrm{LongMes}_{\bar{k}}$.

Everytime if we need to emphasize that this is a measure μ is defined on the space $\mathcal{X}$, we use $\mathcal{X}$ as the index. For example, we may write $(\mathrm{HausMes}_\alpha A)_{\mathcal{X}}$ *for the α-dimensional Hausdorff measure of set A in $\mathcal{X}$.*

The following proposition follows since

$$2\cdot\operatorname{rad} A\geqslant\operatorname{diam} A\geqslant\operatorname{rad} A$$

holds for any set A in a metric space.

7.1.4. Proposition. *Let $\mathcal{X}$ be a metric space and the subset $A\subset\mathcal{X}$ is arbitrary subset then*

a) For any $\alpha \geqslant 0$*, we have*

$$2^{\alpha}\cdot \operatorname{SphMes}_{\alpha} A \geqslant \operatorname{HausMes}_{\alpha} A \geqslant \operatorname{SphMes}_{\alpha} A.$$

b) For any integer $m \geqslant 0$ *we have*

$$\operatorname{LongMes}_{m} A \geqslant \tfrac{\alpha_m}{2^m}\cdot \operatorname{HausMes}_{m} A.$$

where α_m *is the Lebesgue measure of the unit ball in* $\mathbb{E}^m$*.*

The following proposition trivially follows from the definitions

7.1.5. Proposition. *Let* $\mathcal{X}$ *and* $\mathcal{Y}$ *be metric spaces,* $A \subset \mathcal{X}$ *and* $f\colon \mathcal{X} \to \mathcal{Y}$ *be a* $\pounds$*-Lipschitz map. Then*

$$\operatorname{HausMes}_{\alpha} f(A) \leqslant \pounds^{\alpha}\cdot \operatorname{HausMes}_{\alpha} A,$$

$$\operatorname{LongMes}_{\bar{k}} f(A) \leqslant \pounds^{\alpha}\cdot \operatorname{LongMes}_{\bar{k}} A$$

and

$$\operatorname{SphMes}_{\alpha} f(A) \leqslant \pounds^{\alpha}\cdot \operatorname{SphMes}_{\alpha} A.$$

If the map f *is* $\pounds^{\mp 1}$*-bi-Lipschitz then we have in addition*

$$\operatorname{HausMes}_{\alpha} f(A) \geqslant \pounds^{-\alpha}\cdot \operatorname{HausMes}_{\alpha} A,$$

$$\operatorname{LongMes}_{\bar{k}} f(A) \geqslant \pounds^{-\alpha}\cdot \operatorname{LongMes}_{\bar{k}} A$$

and

$$\operatorname{SphMes}_{\alpha} f(A) \leqslant (2\cdot\pounds)^{-\alpha}\cdot \operatorname{SphMes}_{\alpha} A.$$

7.2 Measurable functions

7.2.1. Definition. *Let* $\mathcal{X}$ *be a metric space equipped with Borel regular measure* μ*. A function* $f\colon \mathcal{X} \to \mathbb{R}$ *is called measurable if* $f^{-1}(C)$ *is* μ*-measurable for any Borel subset* $C \subset \mathbb{R}$*.*

7.2.2. Lusin's theorem. *Let* $\mathcal{X}$ *be a metric space equipped with Borel regular measure* μ*. Assume* $\mathcal{X}$ *can be covered by countable number of open sets with finite* μ*-measure. Then for any* μ*-measurable function* $f\colon \mathcal{X} \to \mathbb{R}$ *and* $\varepsilon > 0$ *there is a continuous function* $g\colon \mathcal{X} \to \mathbb{R}$ *such that*

$$\mu\{\, x \in \mathcal{X} \mid f(x) \neq g(x) \,\} < \varepsilon.$$

7.3 Volume

The following classical result follows directly from [53, 2.7.7].

7.3.1. Lebesgue's theorem. *There is a unique Borel regular measure on $\mathbb{E}^m$ which is invariant with respect to translations and such that the measure of unit cube $[0,1]^n$ is 1.*

The measure on $\mathbb{E}^m$ described in the Lebesgue's theorem is called *Euclidean volume*; it will be denoted by vol_m.

Let us denote by α_m the volume of unit ball in $\mathbb{E}^m$. Let us define volume in general metric space.

7.3.2. Definition. *Assume A is a Borel subset of metric space and m is a nonnegative integer. The m-dimesional voulme of A is defined as*

$$\mathrm{vol}_m A = \tfrac{\alpha_m}{2^m} \cdot \mathrm{HausMes}_m$$

The constant $\frac{\alpha_m}{2^m}$ is needed to make this definition agree with the the definition of volume on the Euclidean spaces.

7.4 Doubling spaces and measures

7.4.1. Definition. *A metric space $\mathcal{X}$ is called* doubling *if there is $r_0 > 0$ and a positive integer N such that*

$$\mathrm{pack}_{\frac{r}{2}} B < N$$

for any ball B of radius $r \leqslant r_0$ in $\mathcal{X}$.

7.4.2. Proposition. *Any complete doubling metric space is locally compact.*

Proof. Let $\mathcal{X}$ be a complete doubling metric space and the values r_0 and N are as in Definition 7.4.1.

Fix a ball B with radius $r \leqslant r_0$ in $\mathcal{X}$. Applying the definition of doubling space recursively, we get that for any positive integer n the ball B can be covered by N^n balls of radius $\frac{r}{2^n}$. In particular, B is uniformly bounded and therefore compact. □

Given a closed ball $B = \overline{\mathrm{B}}[x, R]$, and a positive real number $\mathfrak{z}$, let us denote by $\mathfrak{z}\cdot B$ the ball $\overline{\mathrm{B}}[x, \mathfrak{z}\cdot R]$. (When we speak of a ball B, it has to be understood that it comes with a fixed center and radius, although these in general are not uniquely determined by B as a set.)

7.4.3. Definition. *Let $\mathcal{X}$ be a metric space. A Borel regular measure μ on $\mathcal{X}$ is called* doubling *if (1) $\mu\Omega > 0$ for any open set Ω and (2) there are real numbers $r_0 > 0$ and c such that*

$$\mu(2{\cdot}B) < c{\cdot}\mu B$$

for any ball B in $\mathcal{X}$ with radius at most r_0.

Note that the definition implies that $2{\cdot}B$ has finite measure.

7.4.4. Proposition. *Let $\mathcal{X}$ be a metric space with a doubling measure μ. Then $\mathcal{X}$ is doubling.*

In particular, by Proposition 7.4.2, it is locally compact.

Proof. Assume that the values r_0 and c are as in the Definition 7.4.3.

Fix a ball $B = \overline{\mathrm{B}}[p, r]$ with the radius $r < \frac{r_0}{2}$. Assume the points $x_1, x_2, \dots, x_n \in B$ are such that $|x_i x_j| > \frac{r}{2}$ for any $i \neq j$. In other words the balls $B_i = \overline{\mathrm{B}}[x_i, \frac{r}{4}]$ are mutually disjoint.

Note that $8{\cdot}B_i \supset B$ for any i. Since μ is doubling, we get

❶ $$\mu B_i > \tfrac{1}{c^3}{\cdot}\mu B$$

On the other hand $B_i \subset 2{\cdot}B$ for each i. Therefore

$$\sum_{i=1}^{n} \mu B_i \leqslant \mu(2{\cdot}B) <$$
$$< c{\cdot}\mu B.$$

Together with ❶, the later implies

$$n < c^4.$$

□

7.4.5. Proposition. *Let $\mathcal{X}$ be a complete length space with doubling measure μ. Then any bounded set in $\mathcal{X}$ has finite μ-measure.*

Proof. Let A be a bounded set in $\mathcal{X}$; we need to show that $\mu A < \infty$. Without loss of generality we may assume that A is closed; otherwise pass to the closure of A.

By Proposition 7.4.2, $\mathcal{X}$ is locally compact. Since $\mathcal{X}$ is a complete length space, by Hopf–Rinow theorem (2.3.8), it has to be proper. In particular, A has to be compact.

From the definition of doubling measure, there is a positive real value r_0 such that any ball of radius r_0 in $\mathcal{X}$ has finite measure (see the remark after the Definition 7.4.3).

Since A is compact it can be covered by a finite number of such balls. Hence the result follows. $\square$

7.4.6. Proposition. *Let $\mathcal{X}$ be a complete length space with doubling measure μ then the ratio*

$$\frac{\mu(\overline{\mathrm{B}}[a,r_a])}{\mu(\overline{\mathrm{B}}[b,r_b])}$$

can be bounded in terms of $|a\,b|_{\mathcal{X}}$, r_a and r_b.

Proof. Let r_0 and c be as in the definition of doubling measure (7.4.3).

Note that if B is a ball of radius at most r then

$$1 \leqslant \frac{\mu B}{\mu\left(\frac{1}{2^{\bar{k}}}\cdot B\right)} \leqslant c^{\bar{k}}.$$

Therefore it is sufficient to prove the statement in the case $r_a = r_b = r_0$.

Set $A = \overline{\mathrm{B}}[a,r_0]$ and $B = \overline{\mathrm{B}}[b,r_0]$. Note that if $|a\,b| < r_0$ then $A \subset 2\cdot B$; in particular, $\mu A \leqslant \mu(2\cdot B)$. Since μ is doubling, we get

❷ $$\frac{\mu A}{\mu B} < c.$$

Since $\mathcal{X}$ is a length space, any two points x and y in $\mathcal{X}$ can be joint by a chain $x = x_0, x_1, \dots x_n = y$ such that $|x_{i-1}\,x_i| < r_0$ for all i. Applying the inequality ❷ n times, we get that if $r_a = r_b = r_0$ then

❸ $$\frac{\mu A}{\mu B} < c^N,$$

where $N = \lceil \frac{|x\,y|}{r_0} \rceil$. $\square$

7.5 Covering theorems

Let $A \subset \mathcal{X}$. A family of closed balls $\mathfrak{B}$ in $\mathcal{X}$ is called *Vitali covering*[1] of A if for every $a \in A$ and $\varepsilon > 0$, there is a ball from $\mathfrak{B}$ centered at a and radius $R < \varepsilon$.

In this section will discuss two closely related covering theorems.

7.5.1. Vitali covering theorem. *Let μ be a doubling measure on a metric space $\mathcal{X}$ and let $\mathfrak{B}$ be a Vitali covering of a set $A \subset \mathcal{X}$.*

[1] Our definition of Vitali covering is among the most restrictive ones. More general definitions are given in ???.

Assume $\mu A < \infty$*, then there is a countable collection of disjoint balls* $\{B_n\}_{n\in\mathcal{A}} \subset \mathfrak{B}$ *such that*

$$\mu\left(A \Big\backslash \bigcup_{n\in\mathcal{A}} B_n\right) = 0.$$

For the proof see for example [72, Theorem 1.6]; in a more general form it appears in [53, Section 2.8]

Next covering theorem makes a stronger condition on metric space but relax the condition on the measure. To formulate this theorem we need the notion of *directionally limited metric spaces.* This class of spaces was introduced in [53, 2.8.9]; essentially these are the spaces where proof of Besicovitch covering theorem works. We state the definition is a slightly less general form which will be sufficient for our needs; compare [72, page 7]).

7.5.2. Definition. *A metric space* $\mathcal{X}$ *is called* directionally limited *if there is* $\varepsilon > 0$ *and a positive integer* M *such that for any point* $p \in \mathcal{X}$ *there are at most* M *distinct points* $a_1, \dots a_n$ *such that if for some* $i \neq j$ *we have a point* x $x \in [pa_i]$ *and* $|p - x| = |p - a_j|$ *then* $|x - a_j| \geqslant \frac{1}{4}\cdot|p - a_j|$.

7.5.3. Besicovitch covering theorem. *Let* $\mathcal{X}$ *be a directionally limited metric space and* μ *be a Borel regular measure on* $\mathcal{X}$*. Assume* A *is a subset in* $\mathcal{X}$ *such that* $\mu A < \infty$ *and let* $\mathfrak{B}$ *be a Vitali covering of* A*. Then there is at most countable collection of disjointed balls* $\{B_n\}_{n\in\mathcal{A}} \subset \mathfrak{B}$ *such that*

$$\mu\left(A \Big\backslash \bigcup_{n\in\mathcal{A}} B_n\right) = 0.$$

7.6 Coarea formulas

In order to formulate the coarea formula, first we need to define Jacobians.

Linear algebra. Let V be a finite dimensional real vector space, equipped with inner product $\langle *, * \rangle$. Consider the norm on V is defined as $|v| \overset{def}{=\!=} \sqrt{\langle v, v\rangle}$.

We denote by $V^{\wedge n}$ the n-th *exterior power* of V; i.e.,

$$V^{\wedge n} = \underbrace{V \wedge V \wedge \dots \wedge V}_{n \text{ times}}.$$

Note that the inner product $\langle *,*\rangle$ on V leads to canonical inner product on $V^{\wedge n}$. This is the (necessary unique) inner product such that

$$\left\langle v^1\wedge\dots\wedge v^n, v^1\wedge\dots\wedge v^n\right\rangle = 1$$

for any orthonormal system of vectors $v^1,\cdots,v^n$ in V.

Assume W is an other finite dimensional real vector space, equipped with inner product. Let $L\colon V\to W$ be a linear operator. Then one can consider *operator norm* of L defined as

$$|L| \overset{def}{=\!=} \inf\{\,c\geqslant 0\mid |L(v)|\leqslant c\cdot|v| \text{ for any } v\in V\,\}\,.$$

In other words,

$$|L| = \operatorname{lip} L;$$

i.e., $|L|$ is the best Lipschitz constant of L.

The n-th exterior power of a linear map $L\colon V\to W$ is defined as the (necessary unique) linear map $L^{\wedge n}\colon V^{\wedge n}\to W^{\wedge n}$ such that the following identity

$$L^{\wedge n}(v^1\wedge v^2\wedge\ldots\wedge v^n) = L(v^1)\wedge L(v^2)\wedge\ldots\wedge L(v^n)$$

holds for any $v^1,v^2,\dots,v^n\in V$.

Note that

$$|L^{\wedge n}| = \sup_A\{\operatorname{vol}_n[L(A)]\}$$

where A runs along all n-dimesional parallelepipeds in V with unit n-volume.

Coarea formula. Let $\bar{k}\geqslant m$ and $f\colon\mathbb{E}^{\bar{k}}\looparrowright\mathbb{E}^m$ be smooth submap then

$$❶\qquad \int\limits_Q |(\boldsymbol{d}_x f)^{\wedge m}|\cdot\boldsymbol{d}_x\operatorname{vol}_{\bar{k}} = \int\limits_{f(Q)}\operatorname{vol}_{\bar{k}-m}(f^{-1}(y)\cap Q)\cdot\boldsymbol{d}_y\operatorname{vol}_m\,.$$

for any Borel set $Q\subset\operatorname{Dom} f$.

The identity ❶ is called *the coarea formula*; the right hand side in ❶ is called *the coarea of the restriction* $f|Q$.

The coarea formula is a standard result in multivariate calculus. It admits a number of generalizations, in particular Federer showed that this formula holds for Lipschitz maps. Namely he proved the following

7.6.1. Federer's coarea formula. *Let $f\colon\mathbb{E}^{\bar{k}}\looparrowright\mathbb{E}^m$ be a locally Lipschitz submap then coarea formula ❶ holds for any Borel set $Q\subset$ $\subset\operatorname{Dom} f$.*

The case $\bar{k}>m$ is proved in [53, 3.2.11] and for $\bar{k}=m$ it follows from [53, 3.2.3] since in this case $\operatorname{vol}_{\bar{k}-m}=\operatorname{vol}_0$ is the counting measure.

7.7 Singularities of convex functions on $\mathbb{R}^m$

7.7.1. Definition. *Let $\mathcal{X}$ be a metric space, $E \subset \mathcal{X}$ and n is an integer.*

a) The subset E is called n-rectifiable *if there exists a non-contacting map from E to a bounded subset in $\mathbb{E}^n$*

b) Let Ω be an open subset in $\mathcal{X}$. The subset E is called locally n-rectifiable *in Ω if for any point $x \in \Omega$ there is a neigborhood $N \ni x$ such that $E \cap N$ is rectifible.*

c) The subset E is called is called countably n-rectifiable *if it can be presented as a countable union of rectifible sets.*

The following proposition follows directly from definition above.

7.7.2. Proposition. *A finite uinon of n-rectifiable sets is n-rectifiable.*

Let $f\colon \mathbb{R}^m \multimap \mathbb{R}$ be a semiconvex subfunction. In this case the differential

$$\boldsymbol{d}_x f\colon \mathbb{R}^m \multimap \mathbb{R}$$

can be defined as the limit

$$\boldsymbol{d}_x f(v) = \lim_{\varepsilon\to 0} \frac{f(x+\varepsilon\cdot v) - f(x)}{\varepsilon}.$$

It is defined for any $x \in \operatorname{Dom} f$ and it is a convex homogineus function; i.e.,

$$\mathfrak{z}\cdot \boldsymbol{d}_x f(v) = \boldsymbol{d}_x f(\mathfrak{z}\cdot v)$$

for any $\mathfrak{z} \geqslant 0$.

A point $x \in \operatorname{Dom} f$ is called *regular point of f* if $\boldsymbol{d}_x f$ is linear. We say that x is *$\mathfrak{k}$-regular* if $\boldsymbol{d}_x f$ is linear on a $\mathfrak{k}$-dimensional subspace of $\mathbb{R}^m$. We say that x is *$(\mathfrak{k},\varepsilon)$-regular* if there is a $\mathfrak{k}$-dimensional subspace V of $\mathbb{R}^m$ such that the restriction $\boldsymbol{d}_x f|V$ is *ε-linear*; i.e., there is a linear function $\ell\colon V \to \mathbb{R}$ such that

$$\ell(v) \leqslant \boldsymbol{d}_x f(v) < \ell(v) + \varepsilon\cdot|v|$$

for any nonzero vector $v \in V$.

The subsets of regular, $\mathfrak{k}$-regular, $(\mathfrak{k},\varepsilon)$-regular, singular, $\mathfrak{k}$-singular, $(\mathfrak{k},\varepsilon)$-singular of $\operatorname{Dom} f$ will be denoted as Reg_f, $\operatorname{Reg}_f(\mathfrak{k})$, $\operatorname{Reg}_f(\mathfrak{k},\varepsilon)$, Sing_f, $\operatorname{Sing}_f(\mathfrak{k})$, $\operatorname{Sing}_f(\mathfrak{k},\varepsilon)$ correspondingly.

7.7.3. Zajíček's theorem. *Let $f\colon \mathbb{R}^m \multimap \mathbb{R}$ be a locally Lipschitz semiconvex subfunction. Then for any $\mathfrak{k} \leqslant m$ and $\varepsilon > 0$ the set $\operatorname{Sing}_f(\mathfrak{k},\varepsilon)$ is locally $(\mathfrak{k}-1)$-rectifiable in $\operatorname{Dom} f$.*

In particular, for any $\bar{k} \leqslant m$, the set $\operatorname{Sing}_f(\bar{k})$ *is countably* $(\bar{k}-1)$*-rectifiable and* Sing_f *is countably* $(m-1)$*-rectifiable.*

In the proof we will need the following definition.

7.7.4. Definition. *A function $f\colon \mathbb{R}^m \to \mathbb{R}$ is called DC-function if it can be presented as a* difference *$f = a - b$ of two* convex *functions $a, b\colon \mathbb{R}^m \to\!\!\!\!\circ\, \mathbb{R}$.*

A map $\boldsymbol{f} = (f^1, \dots, f^{\bar{k}})\colon \mathbb{R}^m \to \mathbb{R}^{\bar{k}}$ is called DC-map if each of its coordinates f^i is a DC-function.

Since convex function are locally Lipschiz, we get the following.

7.7.5. Proposition. *Any DC-function $f\colon \mathbb{R}^m \to \mathbb{R}$ is locally Lipschitz.*

We repeat Zajíček's proof from [167]. We show that $\operatorname{Sing}_f(\bar{k}, \varepsilon)$ locally lies in a finite collection of graphs of DC-functions $\mathbb{R}^{\bar{k}-1} \to \mathbb{R}^{m-k+1}$ and then apply Proposition 7.7.5.

Proof. Without loss of generality, we may assume that $\operatorname{Dom} f$ is convex bounded open set, f is λ-concave and $\pounds$-Lipschitz for some for some λ and $\pounds$. Further, passing to a new function

$$f_{\text{new}}(\boldsymbol{x}) \overset{def}{=\!=} f(\boldsymbol{x}) - \tfrac{\lambda}{2}\cdot\|\boldsymbol{x}\|_2^2,$$

if nesessury, we may assume that f is concave.

Given a linear subspace $V \subset \mathbb{R}^m$ and a linear function $\ell\colon V \to \mathbb{R}$ consider the envelop subfunction $h_{V,\ell}\colon \mathbb{R}^m \to\!\!\!\!\circ\, \mathbb{R}$ defined as

$$h_{V,\ell}(\boldsymbol{x}) = \sup\{\, f(\boldsymbol{x}+\boldsymbol{v}) + \ell(\boldsymbol{v}) \mid \boldsymbol{v} \in V \,\};$$

$h_{V,\ell}$ is set to be defined at $\boldsymbol{x}$ if $\boldsymbol{x}+\boldsymbol{v} \in \operatorname{Dom} f$ for some $\boldsymbol{v} \in V$. Let us prove couple of claims.

❶ *The subfunction $h_{V,\ell}(\boldsymbol{x})$ is concave.*

Indeed, let $\boldsymbol{x}_0, \boldsymbol{x}_1 \in \operatorname{Dom} h_{V,\ell}$. Set $\boldsymbol{x}_t = (1-t)\cdot\boldsymbol{x}_0 + t\cdot\boldsymbol{x}_1$. Clearly $x_t \in \operatorname{Dom} h_{V,\ell}$ for all $t \in [0,1]$.

We need to show that

❷ $$h_{V,\ell}(\boldsymbol{x}_t) \geqslant (1-t)\cdot h_{V,\ell}(\boldsymbol{x}_0) + t\cdot h_{V,\ell}(\boldsymbol{x}_1).$$

Fix $y_0, y_1 \in \mathbb{R}$ such that $y_i < h_{V,\ell}(\boldsymbol{x}_i)$ for $i \in \{1,2\}$. Then there are $\boldsymbol{z}_0, \boldsymbol{z}_1 \in \operatorname{Dom} f$ such that $\boldsymbol{z}_i - \boldsymbol{x}_i \in V$ and

$$f(\boldsymbol{z}_i) + \ell(\boldsymbol{z}_i - \boldsymbol{x}_i) > y_i.$$

Set $\boldsymbol{z}_t = (1-t)\cdot\boldsymbol{z}_0 + t\cdot\boldsymbol{z}_1$. Clearly $\boldsymbol{z}_t - \boldsymbol{x}_t \in V$. Since $\operatorname{Dom} f$ is convex, $\boldsymbol{z}_t \in \operatorname{Dom} f$ for any $t \in [0,1]$. Thus

$$\begin{aligned} h_{V,\ell}(\boldsymbol{x}_t) &\geqslant f(\boldsymbol{z}_t) + \ell(\boldsymbol{z}_t - \boldsymbol{x}_t) \geqslant \\ &\geqslant (1-t)\cdot[f(\boldsymbol{z}_0) + \ell(\boldsymbol{z}_0 - \boldsymbol{x}_0)] + t\cdot[f(\boldsymbol{z}_1) + \ell(\boldsymbol{z}_1 - \boldsymbol{x}_1)] > \\ &> (1-t)\cdot y_0 + t\cdot y_1. \end{aligned}$$

Recall that the last inequality holds for any y_0 and y_1 such that $y_i < h_{V,\ell}(\boldsymbol{x}_i)$. hence ❷ follows. △

❸ *Let V be a n-dimensional subspace of $\mathbb{E}^m$ and $\boldsymbol{\ell} = (\ell^0, \ell^1, \dots \dots \ell^n)\colon V \to \mathbb{R}^{n+1}$ be an array of linear functions such that $\ell^1, \dots, \ell^n$ are linearly independent and $\ell^0 + \ell^1 + \ldots + \ell^n = 0$. Then the set $Z[V, \boldsymbol{\ell}]$ of all solutions of*

$$h_{V,\ell^0}(\boldsymbol{x}) = h_{V,\ell^1}(\boldsymbol{x}) = \ldots = h_{V,\ell^n}(\boldsymbol{x})$$

forms a subset of a graph of a DC*-submap $H\colon V^\perp \looparrowright V$.*

Proof of ❸. Set $h^i = h_{V,\ell^i}$.

Note that for fixed $\boldsymbol{x} \in V^\perp$ and $\boldsymbol{v} \in V$,

$$h^i(\boldsymbol{x} + \boldsymbol{v}) = h^i(\boldsymbol{x}) - \ell^i(\boldsymbol{v}).$$

It follows that for each $\boldsymbol{x} \in V^\perp$ there is unique $\boldsymbol{v} \in V$ such that

❹ $$h^0(\boldsymbol{x} + \boldsymbol{v}) = h^1(\boldsymbol{x} + \boldsymbol{v}) = \ldots = h^n(\boldsymbol{x} + \boldsymbol{v})$$

More over there is a linear operator $L\colon \mathbb{R}^{n+1} \to V$ such that ❹ hols for

$$\boldsymbol{v} = L \circ \boldsymbol{h}(x).$$

Since the restrictions $h^i|V^\perp$ are concave, we get that $\boldsymbol{x} \mapsto \boldsymbol{v}(\boldsymbol{x})$ is a DC-map. It remains to apply Proposition 7.7.5. △

Note that for any point $z \in \operatorname{Sing}_f(\bar{k}, \varepsilon)$ there is a subspace V of dimension $n \overset{def}{=\!=} m - \bar{k} + 1$ and an array of linear functions $\ell^0, \ell^1, \dots \dots \ell^n$ as above such that $z \in Z(V, \boldsymbol{\ell})$. Moreover any point $x \in \operatorname{Dom} f$ has a neigborhood $\mathrm{B}(\varepsilon, x)$ such that all points in $\mathrm{B}(\varepsilon, x) \cap \operatorname{Sing}_f(\bar{k}, \varepsilon)$ are covered by finite number of sets $Z(V, \boldsymbol{\ell})$ as above. According to Claim ❸, sets $Z(V, \boldsymbol{\ell})$ is $(\bar{k}-1)$-rectifable. According to Proposition 7.7.2, $\operatorname{Sing}_f(\bar{k}, \varepsilon)$ is locally $(\bar{k}-1)$-rectifable in $\operatorname{Dom} f$.

Since

$$\operatorname{Sing}_f(\bar{k}) = \bigcup_{n \in \mathbb{Z}_>} \operatorname{Sing}_f(\bar{k}, \tfrac{1}{n}),$$

the set $\operatorname{Sing}_f(\bar{k})$ is countably $(\bar{k}-1)$-rectifable. Since

$$\operatorname{Sing}_f = \operatorname{Sing}_f(m),$$

the set Sing_f is countably $(m-1)$-rectifable. □

7.8 Exercises

7.8.1. Exercise. *Give an example of a set A in $\mathbb{R}^m$ space with finite such that* $\operatorname{LongMes}_{\bar{k}} A < \infty$*, but A is not $\bar{k}$-rectifiable for some $\bar{k} < m$.*

7.8.2. Besicovitch inequality. *Let $\mathcal{X}$ be a metric space and $\Phi\colon [0,1]^{\times m} \to \mathcal{X}$ be a continuous map. Set*

$$\begin{aligned} A^i &= \left\{ (x^1, \dots, x^m) \in [0,1]^{\times m} \;\middle|\; x^i = 0 \right\}, \\ B^i &= \left\{ (x^1, \dots, x^m) \in [0,1]^{\times m} \;\middle|\; x^i = 1 \right\}, \\ a^i &= |\Phi(A^i)\ \Phi(B^i)|_{\mathcal{X}} \end{aligned}$$

a) Show that

$$\operatorname{LongMes}_m(\operatorname{Im}\Phi) \geqslant a^1 \cdot a^2 \cdots a^m.$$

b) Show that in general, the inequality

$$\operatorname{vol}_m(\operatorname{Im}\Phi) \geqslant a^1 \cdot a^2 \cdots a^m$$

does not hold.

Part II

Definitions and constructions

Chapter 8

Definitions of curvature bounded below

In Section 8.1, we will start with a global definition of Alexandrov spaces via (1+3)-point comparison. In Section 8.3, we give a number of equivalent angle comparison definitions. These definitions maek it easy to adapt the Euclidean intuition to the Alexandrov's world. Later we give a few equivalent global definitions via concavity of distance functions and development (Section 8.4).

In Section 10.1 gives yet another definition via extension of short maps on 4-point subsets. This definition give a uniform approach to definitions of spaces with lower and upper curvature bound. It gives a conceptional point of view, but it almost not used in the proofs.

In Section 8.5, we give local versions of each definition and prove the globalization theorem.

Further, we discuss some basic corollaries of the definition: Section 8.6 contains first variation formula and non-splitting property of geodesics; Section 8.7 discusses specific properties of spaces with positive curvature bound are given.

8.1 (1+3)-point comparison

8.1.1. Definition. *A quadruple of points p, x^1, x^2, x^3 in a metric space satisfies* (1+3)-point κ-comparison *if*

❶ $$\measuredangle^\kappa(p\,{}^{x^1}_{x^2}) + \measuredangle^\kappa(p\,{}^{x^2}_{x^3}) + \measuredangle^\kappa(p\,{}^{x^3}_{x^1}) \leqslant 2\cdot\pi.$$

or at least one of the model angles $\measuredangle^\kappa(p\,{}^{x^i}_{x^j})$ is not defined.

8.1.2. Definition. *A complete length space $\mathcal{L}$ is called an Alexandrov space with curvature $\geqslant \kappa$ (briefly, $\mathcal{L} \in \mathrm{CBB}[\kappa]$) if any quadruple $p, x^1, x^2, x^3 \in \mathcal{L}$ satisfies (1+3)-point comparison.*

We say that $\mathcal{L}$ an Alexandrov space with curvature bounded below (briefly, $\mathcal{L}$ is a CBB *space or $\mathcal{L} \in \mathrm{CBB}$) if there is $\kappa \in \mathbb{R}$ such that $\mathcal{L} \in \mathrm{CBB}[\kappa]$.*

Remarks

- ◇ Alexandrov spaces with curvature bounded below usually will be denoted by $\mathcal{L}$, for $\mathcal{L}$ower curvature bound.
- ◇ In the definition of spaces with curvature $\geqslant \kappa$, when $\kappa > 0$ most authors assume in addition that $\operatorname{diam} \mathcal{L} \leqslant \varpi^\kappa$, or equivalently, that $\mathcal{L}$ is not isometric to one of the exceptional spaces, see 8.7.1. We do not make this assumption. In particular, we consider the real line to have curvature $\geqslant 1$.
- ◇ If $\kappa < \mathrm{K}$, the following inclusion holds:

$$\mathrm{CBB}[\kappa] \supset \mathrm{CBB}[\mathrm{K}].$$

However, in the case $\mathrm{K} > 0$ it is not at all trivial, and will be shown only in 8.5.5.

8.1.3. Exercise. *Let $\mathcal{L}$ be a complete length space. Show that $\mathcal{L} \in \mathrm{CBB}[\kappa]$ if for any quadrouple of points $p, x^1, x^2, x^3 \in \mathcal{L}$ such that*

$$|px^1|, |px^2|, |px^3| < \varpi^\kappa$$

there is a quadrouple of points $q, y^1, y^2, y^3 \in \mathbb{M}^3[\kappa]$ such that

$$|px^i| = |qy^i| \quad \text{and} \quad |x^i x^j| \leqslant |y^i y^j|$$

for all i and j.

The exercise above is a partial case of $(1{+}n)$-point comparison (10.2.1).

8.1.4. Exercise. *Let $\mathcal{L}$ be a length space. Show that $\mathcal{L} \in \mathrm{CBB}[0]$ if and only if*

$$\operatorname{area} \tilde{\triangle}^0(xyz) \leqslant \operatorname{area} \tilde{\triangle}^0(pxy) + \operatorname{area} \tilde{\triangle}^0(pyz) + \operatorname{area} \tilde{\triangle}^0(pzx)$$

for any 4 distinct points $p, x, y, z \in \mathcal{L}$.

The following exercise decribes the first known definition of spaces with curvature bounded below; it was given by Abraham Wald in [164].

8.1.5. Exercise. *Let $\mathcal{L}$ be a complete length space and $\kappa \leqslant 0$. Prove that $\mathcal{L} \in \mathrm{CBB}[\kappa]$ if and only if for any quadruple of points $p, q, r, s \in \mathcal{L}$ can be isometrically embedded into some $\mathbb{M}^2[\mathrm{K}]$ for some $\mathrm{K} \geqslant \kappa$.*

Is the same true for $\kappa > 0$, what is the difference?

8.1.6. Exercise. *Let $\mathcal{L}$ be $\mathbb{R}^m$ with the metric defined by a norm). Show that $\mathcal{L} \in \mathrm{CBB}$ if and only if $\mathcal{L} \overset{iso}{=\!=} \mathbb{E}^m$.*

It is straightforward to check the following claim (for definitions of ω-limit and ω-power, see Section 4.2).

8.1.7. Proposition. *Let ω be a selective ultrafilter on $\mathbb{N}$, $(\mathcal{L}_n, \star_n) \to$ $\to (\mathcal{L}_\omega, \star_\omega)$ and $\kappa_n \to \kappa_\omega$ as $n \to \omega$. Then*

$$\mathcal{L}_n \in \mathrm{CBB}[\kappa_n] \quad \Longrightarrow \quad \mathcal{L}_\omega \in \mathrm{CBB}[\kappa_\omega].$$

Moreover, for any complete metric space $\mathcal{L}$,

$$\mathcal{L} \in \mathrm{CBB}[\kappa] \quad \Longleftrightarrow \quad \mathcal{L}^\omega \in \mathrm{CBB}[\kappa],$$

where $\mathcal{L}^\omega$ denotes ultrapower of $\mathcal{L}$.

8.1.8. Theorem. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$ and $\mathcal{M}$ be a metric space. Assume there is a sumbetry $\sigma\colon \mathcal{L} \to \mathcal{M}$. Then $\mathcal{M} \in \mathrm{CBB}[\kappa]$.*

The theorem above together with Proposition 5.2.1 imply the following.

8.1.9. Corollary. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$ and the group G acts on $\mathcal{L}$ by isometries and it has closed orbits. Then $\mathcal{L}/G \in \mathrm{CBB}[\kappa]$.*

Proof of 8.1.8. Fix $\delta > 0$ and a quadruple of points $p, x^1, x^2, x^3 \in \mathcal{M}$. Choose arbitrary $\hat{p} \in \mathcal{L}$ such that $\sigma(\hat{p}) = p$. Next σ is submetry, we can choose the points $\hat{x}^1, \hat{x}^2, \hat{x}^3 \in \mathcal{M}$. such that $\sigma(\hat{x}_i) = x_i$ and

$$|\hat{p}\,\hat{x}^i|_{\mathcal{L}} \lesseqgtr |p\,x^i|_{\mathcal{M}} \pm \delta$$

for all i. Note that

$$|\hat{x}^i\,\hat{x}^j|_{\mathcal{L}} > |x^i\,x^j|_{\mathcal{M}} - 2{\cdot}\delta$$

for all i and j. Therefore given $\varepsilon > 0$, the value δ above can be chousen in such a way that the inequality

❷ $$\mathcal{Z}^\kappa(\hat{p}\,{}^{\hat{x}^1}_{\hat{x}^2}) < \mathcal{Z}^\kappa(p\,{}^{x^1}_{x^2}) + \varepsilon$$

holds for all i and j.

By (1+3)-point comparison in $\mathcal{L}$, we have

$$\measuredangle^{\kappa}(\hat{p}\,{}^{\hat{x}^1}_{\hat{x}^2}) + \measuredangle^{\kappa}(\hat{p}\,{}^{\hat{x}^2}_{\hat{x}^3}) + \measuredangle^{\kappa}(\hat{p}\,{}^{\hat{x}^3}_{\hat{x}^1}) \leqslant 2\cdot\pi$$

assuming all the angles on the left hand side are defined. Applying ❷, we get

$$\measuredangle^{\kappa}(p\,{}^{x^1}_{x^2}) + \measuredangle^{\kappa}(p\,{}^{x^2}_{x^3}) + \measuredangle^{\kappa}(p\,{}^{x^3}_{x^1}) < 2\cdot\pi + 3\cdot\varepsilon.$$

Since $\varepsilon > 0$ is arbitrary we have

$$\measuredangle^{\kappa}(p\,{}^{x^1}_{x^2}) + \measuredangle^{\kappa}(p\,{}^{x^2}_{x^3}) + \measuredangle^{\kappa}(p\,{}^{x^3}_{x^1}) \leqslant 2\cdot\pi;$$

that is, the (1+3)-point κ-comparison holds in $\mathcal{M}$. It remains to apply Proposition 5.2.2. □

8.2 Geodesics

Recall that general complete length spaces might have no geodesics; see Exercise 2.3.4.

8.2.1. Exercise. *Construct a space $\mathcal{L} \in \mathrm{CBB}[0]$ which is not geodesic.*

We are goinfg to show that all CBB spaces are G-delta geodesic in the following sense.

8.2.2. Definition. *A complete length space $\mathcal{X}$ is called* G-delta geodesic *if for any point $p \in \mathcal{X}$ there is a dense G-delta set $W_p \subset \mathcal{X}$ such that for any $q \in W_p$ there is a geodesic $[pq]$.*

8.2.3. Definition. *Let $\mathcal{X}$ be a metric space and $p \in \mathcal{X}$. A point $q \in \mathcal{X}$ is called p-*straight *(briefly, $q \in \mathrm{Str}(p)$) if*

$$\varlimsup_{r\to q} \frac{|pr| - |pq|}{|qr|} = 1.$$

For an array of points $x^1, x^2, \dots, x^{\text{k}}$, we use the notation

$$\mathrm{Str}(x^1, x^2, \dots, x^{\text{k}}) = \bigcap_{i=1}^{\text{k}} \mathrm{Str}(x^i).$$

8.2.4. Theorem. *Let $\mathcal{L} \in \mathrm{CBB}$ and $p \in \mathcal{L}$. Then the set $\mathrm{Str}(p)$ is a dense G-delta set, and for any $q \in \mathrm{Str}(p)$ there is a unique geodesic $[pq]$.*

In particular, $\mathcal{L}$ is G-delta geodesic.

The theorem above was proved by Plaut [140, Th. 27]. His proof requires the following lemma in metric geometry.

8.2.5. Lemma. *Let $\mathcal{X}$ be a complete length space. Then for any $p \in \mathcal{X}$ the set of* $\mathrm{Str}(p) \subset \mathcal{X}$ *is a dense G-delta set.*

Proof. Set

$$\Omega_n = \left\{ q \in \mathcal{X} \;\middle|\; (1-\tfrac{1}{n})\cdot|qr| < |pr| - |pq| < \tfrac{1}{n} \text{ for some } r \in \mathcal{X} \right\}.$$

Clearly Ω_n is open; let us show that it is dense in $\mathcal{X}$. Assuming the contrary, there is a point $x \in \mathcal{X}$ such that $\mathrm{B}(x,\varepsilon) \cap \Omega_n = \varnothing$ for some $\varepsilon > 0$. Since $\mathcal{L}$ is a length space, for any $\delta > 0$, there exists a point $y \in \mathcal{L}$ such that $|xy| < \frac{\varepsilon}{2} + \delta$ and $|py| < |px| - \frac{\varepsilon}{2} + \delta$. For small enough ε and δ, it follows that $(1-\frac{1}{n})\cdot|yx| < |px| - |py| < \frac{1}{n}$, i.e. $y \in \Omega_n$, a contradiction.

Since $\mathrm{Str}(p) = \bigcap_{n\in\mathbb{N}} \Omega_n$, the result follows. □

Proof of Theorem 8.2.4. Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$. We show that if $q \in \mathrm{Str}(p)$, then there is a unique geodesic connecting p and q. Thus the theorem will follow from Lemma 8.2.5.

Note that it is enough to show that for all sufficiently small $t > 0$ there is unique point z such that

❶ $$t = |qz| = |pq| - |pz|.$$

First let us show uniqueness. Assume z and z' both satisfy ❶. Take a sequence $r_n \to q$ so that $(|pr_n| - |pq|)/|qr_n| \to 1$. By the triangle inequality,

$$|zr| - |zq|, \quad |z'r| - |z'q| \geqslant |pr| - |pq|;$$

thus, as $n \to \infty$,

$$\frac{|zr_n| - |zq|}{|qr_n|}, \quad \frac{|z'r_n| - |z'q|}{|qr_n|} \to 1.$$

Therefore $\measuredangle^\kappa(q\,{}^{z}_{r_n}),\ \measuredangle^\kappa(q\,{}^{z'}_{r_n}) \to \pi$. (Here we use that t is small, otherwise if $\kappa > 0$ the angles might be undefined.) From (1+3)-point comparison (8.1.2), $\measuredangle^\kappa(q\,{}^{z}_{z'}) = 0$ and thus $z = z'$.

The proof of existence is similar. Choose a sequence r_n as above. Since $\mathcal{L}$ is a complete length space, there is a sequence $z_k \in \mathcal{L}$ such that $|qz_k|,\ |pq| - |pz_k| \to t$ as $k \to \infty$. Then

$$\lim_{n\to\infty} \lim_{k\to\infty} \measuredangle^\kappa(q\,{}^{z_k}_{r_n}) = \pi.$$

Thus, for any $\varepsilon > 0$ and large enough $n, \bar{k}$, we have $\measuredangle^{\kappa}(q\,{}^{z_{\bar{k}}}_{r_n}) > \pi - \varepsilon$. From (1+3)-point comparison (8.1.2), for all large $\bar{k}$ and j, we have $\measuredangle^{\kappa}(q\,{}^{z_{\bar{k}}}_{z_j}) < 2\cdot\varepsilon$ and thus

$$|z_{\bar{k}}\,z_j| < \varepsilon\cdot c(\kappa, t);$$

i.e. $\{z_n\}$ is a Cauchy sequence, and $z = \lim_n z_n$ satisfies ❶. □

8.2.6. Exercise. *Give an alternative proof of Theorem 8.2.4 built on the fact that geodesics do not split (8.6.1) and Corollary 4.2.4.*

8.2.7. Exercise. *Construct a space $\mathcal{L} \in \mathrm{CBB}$ which contains an everywhere dense G-delta set A such that $A \cap]xy[= \varnothing$ for any geodesic $[xy]$ in $\mathcal{L}$.*

8.2.8. Exercise. *Let $\mathcal{L} \in \mathrm{CBB}$ and $A \subset \mathcal{L}$ be a closed subset. Show that there is a dense G-delta set $W \subset \mathcal{L}$ such that for any $q \in W$ there is unique geodesic $[pq]$ with $p \in A$ and $|pq| = |Aq|$.*

8.3 More comparisons

Now we are ready to give a number of equivalent definitions of Alexandrov space:

8.3.1. Theorem. *Let $\mathcal{L}$ be a complete length space. Then*

a) $\mathcal{L} \in \mathrm{CBB}[\kappa]$

if and only if $\mathcal{L}$ is G-delta geodesic and one of the following conditions holds for all $p, x, y \in \mathcal{L}$, once $\tilde{\triangle}^{\kappa}(pxy)$ is defined:

b) (adjacent angle comparison) for any geodesic $[xy]$ and $z \in$ $\in]xy[$, $z \neq p$ we have

$$\measuredangle^{\kappa}(z\,{}^{p}_{x}) + \measuredangle^{\kappa}(z\,{}^{p}_{y}) \leqslant \pi.$$

c) (point-on-side comparison) for any geodesic $[xy]$ and $z \in$ $\in]xy[$, we have

$$\measuredangle^{\kappa}(x\,{}^{p}_{y}) \leqslant \measuredangle^{\kappa}(x\,{}^{p}_{z});$$

or, equivalently,

$$|\tilde{p}\,\tilde{z}| \leqslant |p\,z|,$$

where $[\tilde{p}\tilde{x}\tilde{y}] = \tilde{\triangle}^{\kappa}(pxy)$, $\tilde{z} \in]\tilde{x}\tilde{y}[$, $|\tilde{x}\,\tilde{z}| = |x\,z|$.

d) (hinge comparison) for any hinge $[x\,{}^{p}_{y}]$*, the angle* $\measuredangle[x\,{}^{p}_{y}]$ *is defined and*

$$\measuredangle[x\,{}^{p}_{y}] \geqslant \tilde{\measuredangle}^{\kappa}(x\,{}^{p}_{y}),$$

or equivalently

$$\tilde{\Upsilon}^{\kappa}[x\,{}^{p}_{y}] \geqslant |py|.$$

Moreover, if $z \in\,]xy[$*,* $z \neq p$ *then*

$$\measuredangle[z\,{}^{p}_{y}] + \measuredangle[z\,{}^{p}_{x}] \leqslant \pi$$

for any two hinges $[z\,{}^{p}_{y}]$ *and* $[z\,{}^{p}_{x}]$ *with common side* $[zp]$*.*

Remarks. The monotonicity of model angle with respect to adjacent sidelengths (a) was named the *convexity property* by Alexandrov.

A bit weaker form of (d) is given in 8.5.6. See also Problem 8.8.1.

Proof. By Theorem 8.2.4, we have that (a) imples that $\mathcal{L}$ is G-delta geodesic.

(a) $\Rightarrow$ *(b).* Since $z \in\,]xy[$, we have $\tilde{\measuredangle}^{\kappa}(z\,{}^{x}_{y}) = \pi$. Thus, (1+3)-point comparison

$$\tilde{\measuredangle}^{\kappa}(z\,{}^{x}_{y}) + \tilde{\measuredangle}^{\kappa}(z\,{}^{p}_{x}) + \tilde{\measuredangle}^{\kappa}(z\,{}^{p}_{y}) \leqslant 2\cdot\pi$$

implies

$$\tilde{\measuredangle}^{\kappa}(z\,{}^{p}_{x}) + \tilde{\measuredangle}^{\kappa}(z\,{}^{p}_{y}) \leqslant \pi.$$

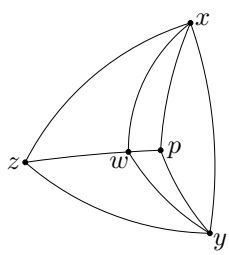
(b) $\Leftrightarrow$ *(c).* Follows directly from Alexandrov's lemma (6.2.1).

(b) + (c) $\Rightarrow$ *(d).* From (c) we get that for $\bar{p} \in\,]xp]$ and $\bar{y} \in\,]xy]$ the function $(|x\bar{p}|, |x\bar{y}|) \mapsto \tilde{\measuredangle}^{\kappa}(x\,{}^{\bar{p}}_{\bar{y}})$ is nonincreasing in each argument. In particular, $\measuredangle[x\,{}^{p}_{y}] = \sup\{\tilde{\measuredangle}^{\kappa}(x\,{}^{\bar{p}}_{\bar{y}})\}$. Thus, $\measuredangle[x\,{}^{p}_{y}]$ is defined and it is at least $\tilde{\measuredangle}^{\kappa}(x\,{}^{p}_{y})$.

From above and (b), it follows that $\measuredangle[z\,{}^{p}_{y}] + \measuredangle[z\,{}^{p}_{x}] \leqslant \pi$.

(d) $\Rightarrow$ *(a).* Assume first that $\mathcal{L}$ is geodesic. Consider point $w \in\,]pz[$ close to p. From (d), it follows that

$$\measuredangle[w\,{}^{x}_{z}] + \measuredangle[w\,{}^{x}_{p}] \leqslant \pi \quad\text{and}\quad \measuredangle[w\,{}^{y}_{z}] + \measuredangle[w\,{}^{y}_{p}] \leqslant \pi.$$



Since $\measuredangle[w\,{}^{x}_{y}] \leqslant \measuredangle[w\,{}^{x}_{p}] + \measuredangle[w\,{}^{y}_{p}]$ (see 6.3.2), we get

$$\measuredangle[w\,{}^{x}_{z}] + \measuredangle[w\,{}^{y}_{z}] + \measuredangle[w\,{}^{x}_{y}] \leqslant 2\cdot\pi.$$

Applying the first inequality in (d), we obtain

$$\tilde{\measuredangle}^{\kappa}(w\,{}^{x}_{z}) + \tilde{\measuredangle}^{\kappa}(w\,{}^{y}_{z}) + \tilde{\measuredangle}^{\kappa}(w\,{}^{x}_{y}) \leqslant 2\cdot\pi.$$

Passing to the limits $w \to p$, we obtain

$$\measuredangle^{\kappa}(p\,{}^{x}_{z}) + \measuredangle^{\kappa}(p\,{}^{y}_{z}) + \measuredangle^{\kappa}(p\,{}^{x}_{y}) \leqslant 2\cdot\pi.$$

Finally, if $\mathcal{L}$ is not geodesic, we can apply above arguments for sequences of points $p_n, w_n \to p$, $p_n \in \mathrm{Str}(z)$, $w_n \in\,]zp_n[$ and $x_n \to x$, $y_n \to y$, $x_n, y_n \in \mathrm{Str}(w_n)$. □

8.3.2. Angle-sidelength monotonicity. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$, $p, x, y \in$ $\in \mathcal{L}$, $\tilde{\triangle}^{\kappa}(pxy)$ is defined and there is a geodesic $[xy]$. Then for $\bar{y} \in\,]xy]$ the function*

$$|x\,\bar{y}| \mapsto \measuredangle^{\kappa}(x\,{}^{p}_{\bar{y}})$$

is nonincreasing.

In particular, if geodesic $[xp]$ exists and $\bar{p} \in\,]xp]$ then

a) the function

$$(|x\,\bar{y}|, |x\,\bar{p}|) \mapsto \measuredangle^{\kappa}(x\,{}^{\bar{p}}_{\bar{y}})$$

is nonincresing in each argument

b) The angle $\measuredangle[x\,{}^{p}_{y}]$ is defined and

$$\measuredangle[x\,{}^{p}_{y}] = \sup\left\{\, \measuredangle^{\kappa}(x\,{}^{\bar{p}}_{\bar{y}}) \;\middle|\; \bar{p} \in\,]xp]\,,\ \bar{y} \in\,]xy]\,\right\}.$$

Proof. See the first part of (b)+(c)⇒(d) in the proof of 8.3.1. □

8.3.3. Exercise. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$, $p, x, y \in \mathcal{L}$ and $v, w \in\,]xy[$. Prove that*

$$\measuredangle^{\kappa}(x\,{}^{y}_{p}) = \measuredangle^{\kappa}(x\,{}^{v}_{p}) \quad\Leftrightarrow\quad \measuredangle^{\kappa}(x\,{}^{y}_{p}) = \measuredangle^{\kappa}(x\,{}^{w}_{p}).$$

8.3.4. Advanced Exercise. *Construct a geodesic space $\mathcal{X} \notin \mathrm{CBB}[0]$, but which meets the following condition: for any 3 points $p, x, y \in \mathcal{X}$ there is a geodesic $[xy]$ such that for any $z \in\,]xy[$*

$$\measuredangle^{0}(z\,{}^{p}_{x}) + \measuredangle^{0}(z\,{}^{p}_{y}) \leqslant \pi.$$

8.3.5. Advanced Exercise. *Assume $\mathcal{L}$ be a complete length space such that for any quadrouple $p, x, y, z \in \mathcal{L}$ the following inequality holds*

❶ $$|p\,x|^2 + |p\,y|^2 + |p\,z|^2 \geqslant \tfrac{1}{3}\cdot\left[|x\,y|^2 + |y\,z|^2 + |z\,x|^2\right].$$

Prove that $\mathcal{L} \in \mathrm{CBB}[0]$.

Construct a metric on a 4-point set which satisfies inequality ❶ for any relabeling of its points by p, x, y, z, but which does not admit a distance preserving map into a $\mathrm{CBB}[0]$ space.

Assume that for a trianglle $[x^1x^2x^3]$ in a metric space its κ-model triangle $[\tilde{x}^1\tilde{x}^2\tilde{x}^3] = \tilde{\triangle}^\kappa(x^1x^2x^3)$ is defined. We say the triangle $[x^1x^2x^3]$ is κ-*fat* if the natural map (see definition 9.4.1) $[\tilde{x}^1\tilde{x}^2\tilde{x}^3] \to [x^1x^2x^3]$ is distance non contracting.

8.3.6. Exercise. *Prove that any triangle with perimeter $< \varpi^\kappa$ in a* CBB$[\kappa]$*-space is κ-fat.*

The following exercise is inspired by Busemann's definition given in [35].

8.3.7. Exercise.

a) Show that any space $\mathcal{L} \in$ CBB$[0]$ satisfies the following condition:

- *For any three points $p, q, r \in \mathcal{L}$, if $\bar{q}$ and $\bar{r}$ are midpoints of geodesics $[pq]$ and $[pr]$ correspondingly then $2\cdot|\bar{q}\bar{r}| \geqslant |qr|$.*

b) Show that $\mathbb{R}^2$ admits a metric defined by a norm such that it satisfies above condition, but which is not an Alexandrov space.

8.4 Function comparison and development

In this section we will translate the angle comparison definitions (Theorem 8.3.1) to a concavity-like property of the distance functions as defined in Section 5.5. Conceptually, this is quite important step; we reformulate a global geometric condition into an infinitesimal condition on distance functions. We will give two formulation of this comparison: analytical, through function concavity, and geometrical, using development.

8.4.1. Theorem. *Let $\mathcal{L}$ be a complete length space. Then the following statements are equivalent:*

a) $\mathcal{L} \in$ CBB$[\kappa]$.

b) (function comparison) $\mathcal{L}$ is G-delta geodesic and for any $p \in \mathcal{L}$, the function $f = \mathrm{md}^\kappa \circ |p_*|$ satisfies the differential inequality

$$f'' \leqslant 1 - \kappa\cdot f.$$

in $\mathrm{B}(p, \varpi^\kappa)$.

In particular, $\mathcal{L} \in$ CBB$[0]$ if and only if for any $p \in \mathcal{L}$, the function $|p_*|^2 \colon \mathcal{L} \to \mathbb{R}$ is 2-concave as defined in Section 5.5.

Proof. Given a geodesic $[xy]$ in $\mathrm{B}(p,\varpi^\kappa)$, let $\ell=|xy|$. Consider model triangle $[\tilde p\tilde x\tilde y]=\tilde\triangle^\kappa(pxy)$ and set

$$\tilde r(t)=|\tilde p\,\mathrm{geod}_{[\tilde x\tilde y]}(t)|,\qquad\qquad r(t)=|p\,\mathrm{geod}_{[xy]}(t)|.$$

Clearly $\tilde r(0)=r(0)$ and $\tilde r(\ell)=r(\ell)$. Set $\tilde f=\mathrm{md}^\kappa\circ\tilde r$ and $f=\mathrm{md}^\kappa\circ r$. From 1.1.1*a* we get $\tilde f''=1-\kappa\cdot\tilde f$.

Note that the point-on-side comparison (8.3.1*c*) for point p and geodesic $[xy]$ is equivalent to $\tilde r\leqslant r$. Since md^κ is increasing on $[0,\varpi^\kappa)$, $\tilde r\leqslant r$ is equivalent to $\tilde f\leqslant f$, which is Jensen's inequality (5.4.3*d*) for the function $t\mapsto\mathrm{md}^\kappa|p\,\mathrm{geod}_{[xy]}(t)|$ on interval $[0,\ell]$. Hence the result. □

Recall that Busemann functions defined in Proposition 6.1.1. The following exercise is analogous to Exercise 9.5.2.

8.4.2. Exercise. *Let* $\mathcal{L}\in\mathrm{CBB}[\kappa]$ *and* $\mathrm{bus}_\gamma\colon\mathcal{L}\to\mathbb{R}$ *is the Busemann function for a ray* $\gamma\colon[0,\infty)\to\mathcal{L}$.

a) If $\kappa=0$ *then the Busemann function* bus_γ *is concave.*

b) If $\kappa=-1$ *then the function*

$$f=\exp\circ\mathrm{bus}_\gamma$$

satisfies

$$f''-f\leqslant 0$$

We will translate the above comparison into more geometric language, using the following definition of κ-development. This definition appears in [16] and an earlier fom of it can be found in [100]. The definition is somewhat lengthy, but it defines a useful comparison object for a curve. Often it is easier to write proofs in terms of function comparison, but think in terms of κ-developments.

8.4.3. Lemma. *Let* $\kappa\in\mathbb{R}$, $\mathcal{X}$ *be a metric space,* $\gamma\colon\mathbb{I}\to\mathcal{X}$ *be a* 1*-Lipschitz curve,* $p\in\mathcal{X}$, $\tilde p\in\mathbb{M}^2[\kappa]$ *and* $0<|p\gamma(t)|<\varpi^\kappa$ *for all* $t\in\mathbb{I}$. *Then there exists a unique up to rotation curve* $\tilde\gamma\colon\mathbb{I}\to\mathbb{M}^2[\kappa]$, *parameterized by the arc-length such that* $|\tilde p\tilde\gamma(t)|=|p\gamma(t)|$ *for all* t *and the direction of* $[\tilde p\tilde\gamma(t)]$ *monotonically turns around* $\tilde p$ *conterclockwise, as* t *increases.*

8.4.4. Definition of development. *If* $p,\tilde p,\gamma,\tilde\gamma$ *be as in Lemma 8.4.3 then* $\tilde\gamma$ *is called the* κ-development of γ with respect to p*; the point* $\tilde p$ *is called* the basepoint of development. *When we say that the* κ*-development of* γ *with respect to* p *is* defined *we allways assume that* $0<|p\gamma(t)|<\varpi^\kappa$ *for all* $t\in\mathbb{I}$.

Proof of Lemma 8.4.3. Consider functions $\rho, \vartheta\colon \mathbb{I} \to \mathbb{R}$ defined as

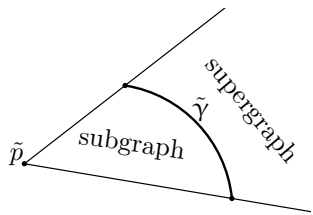
$$\rho(t) = |p\gamma(t)|, \qquad \vartheta(t) = \int\limits_{t_0}^{t} \frac{\sqrt{1-(\rho'(\mathfrak{t}))^2}}{\operatorname{sn}^\kappa \circ \rho(\mathfrak{t})} \cdot \boldsymbol{d}\mathfrak{t},$$

where $t_0 \in \mathbb{I}$ is a fixed number and $\int$ stays for Lebesgue integral. Since γ is 1-Lipshitz, so is $\rho(t)$ and thus, the function ϑ is defined and non decreasing.

It is straightforward to check that (ρ, ϑ) describe $\tilde\gamma$ in polar coordinates on $\mathbb{M}^2[\kappa]$ with center at $\tilde p$. $\square$

We need the following analogues of sub- and super-graphsand convex/concave functions, adapted to polar coordiantes in $\mathbb{M}^2[\kappa]$.

8.4.5. Definition. *Let $\tilde\gamma\colon \mathbb{I} \to \mathbb{M}^2[\kappa]$ be a curve and $\tilde p \in \mathbb{M}^2[\kappa]$ such that there is unique geodesic $[\tilde p\tilde\gamma(t)]$ for any $t \in \mathbb{I}$ and the direction of $[\tilde p\tilde\gamma(t)]$ turns monotonically as t grows.*



The set formed by all geodesics from $\tilde p$ to the points on $\tilde\gamma$ is called the subgraph *of $\tilde\gamma$ with respect to $\tilde p$.*

The set of all points $\tilde x \in \mathbb{M}^2[\kappa]$ such that a geodesic $[\tilde p\tilde x]$ intersect $\tilde\gamma$ is called the supergraph *of $\tilde\gamma$ with respect to $\tilde p$.*

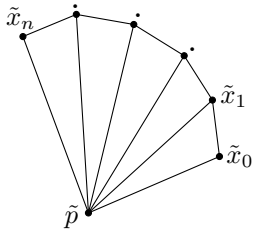
The curve $\tilde\gamma$ is called convex (concave) with respect to $\tilde p$ *if the subgraph (supergraph) of $\tilde\gamma$ with respect to $\tilde p$ is convex.*

The curve $\tilde\gamma$ is called locally convex (concave) with respect to $\tilde p$ *if for any interior value t_0 in $\mathbb{I}$ there is a subsegment $(a,b) \subset \mathbb{I}$, $(a,b) \ni t_0$, such that the restriction $\tilde\gamma|(a,b)$ is convex (concave) with respect to $\tilde p$.*

Note that if $\kappa > 0$ then the supergraph of a curve is a subgraph with respect to the opposite point.

For developments, all the notions above will be considered with respect to their basepoints. In particular, if $\tilde\gamma$ is a development, we will say it is *(locally) convex* if it is (locally) convex with respect to its basepoint.

8.4.6. Development comparison. *$\mathcal{L} \in \mathrm{CBB}[\kappa]$ if and only if $\mathcal{L}$ is G-delta geodesic and for any point $p \in \mathcal{L}$ and any geodesic γ in $\mathrm{B}(p, \varpi^\kappa)\backslash\{p\}$, its κ-development with respect to p is convex.*



Proof; (⇒). By Theorem 8.2.4, $\mathcal{L}$ is G-delta geodesic.

Let $\gamma\colon [0,T] \to \mathrm{B}(p,\varpi^\kappa)\backslash\{p\}$ be a unit-speed geodesic in $\mathcal{L}$. Consider partition $0 = t_0 < t_1 < \ldots < t_n = T$. Set $x_i = \gamma(t_i)$. Let us construct a chain of model triangles $[\tilde p\tilde x_{i-1}\tilde x_i] = \tilde\triangle^\kappa(px_{i-1}x_i)$ in such a way that direction $[\tilde p\tilde x_i]$ turns conterclockwise as i grows. Condition 8.3.1b implies that

❶ $$\measuredangle[\tilde x_i\,{}^{\tilde x_{i-1}}_{\tilde p}] + \measuredangle[\tilde x_i\,{}^{\tilde x_{i+1}}_{\tilde p}] \leqslant \pi.$$

Further, since γ is a unit-speed geodesic, we have that

❷ $$\sum_{i=1}^n |x_{i-1}\,x_i| \leqslant |px_0| + |px_n|.$$

If $\kappa \leqslant 0$ then ❷ immediately implies that

❸ $$\sum_{i=1}^n \measuredangle[\tilde p\,{}^{\tilde x_i}_{\tilde x_{i-1}}] \leqslant \pi.$$

If $\kappa > 0$ then ❸ require more work. Applying rescaling we can assume that $\kappa = 1$. Assume the contrary; i.e., $\sum_{i=1}^n \measuredangle[\tilde p\,{}^{\tilde x_i}_{\tilde x_{i-1}}] > \pi$, then for some $\bar k$ there is a point $\tilde y \in [\tilde x_{\bar k}\tilde x_{\bar k+1}]$ such that

$$\measuredangle[\tilde p\,{}^{\tilde x_0}_{\tilde x_1}] + \ldots + \measuredangle[\tilde p\,{}^{\tilde x_{\bar k-1}}_{\tilde x_{\bar k}}] + \measuredangle[\tilde p\,{}^{\tilde x_{\bar k}}_{\tilde y}] = \pi.$$

By ❷, the broken line $\tilde x_0\tilde x_1\ldots\tilde x_{\bar k}\tilde y$ is convex; in particular,

$$|\tilde p\tilde x_0|_{\mathbb{S}^2} + |\tilde p\tilde y|_{\mathbb{S}^2} \leqslant \pi$$

By triangle inequality

$$|\tilde x_0\,\tilde x_1|_{\mathbb{S}^2} + \ldots + |\tilde x_{\bar k-1}\,\tilde x_{\bar k}|_{\mathbb{S}^2} + |\tilde x_{\bar k}\,\tilde y|_{\mathbb{S}^2} \geqslant |\tilde p\tilde x_0|_{\mathbb{S}^2} + |\tilde p\tilde y|_{\mathbb{S}^2}$$

Applying the triangle inequality again, we get

$$|\tilde y\,\tilde x_{\bar k+1}|_{\mathbb{S}^2} + |\tilde x_{\bar k+1}\,\tilde x_{\bar k+2}|_{\mathbb{S}^2} + \ldots + |\tilde x_{n-1}\,\tilde x_n|_{\mathbb{S}^2} > |\tilde p\tilde x_n|_{\mathbb{S}^2} - |\tilde p\tilde y|_{\mathbb{S}^2}$$

Hence

$$T = \sum_{i=1}^n |\tilde x_i\,\tilde x_{i-1}|_{\mathbb{S}^2} > |\tilde p\tilde x_0|_{\mathbb{S}^2} + |\tilde p\tilde x_n|_{\mathbb{S}^2}.$$

The later contradicts that γ is length minimizing.

Inequalities ❶ and ❸ imply that the polygon $[\tilde{p}\tilde{x}_0\tilde{x}_1\dots\tilde{x}_n]$ is convex.

Let us take finer and finer partitions and pass to the limit of the polygon $\tilde{p}\tilde{x}_0\tilde{x}_1\dots\tilde{x}_n$. We obtain a convex curvelinear triangle formed by a curve $\tilde{\gamma}\colon[0,T]\to\mathbb{M}^2[\kappa]$ which is the limit of broken line $\tilde{x}_0\tilde{x}_1\dots$ $\dots\tilde{x}_n$ and two geodesics $[\tilde{p}\,\tilde{\gamma}(0)]$, $[\tilde{p}\,\tilde{\gamma}(T)]$. Since $[\tilde{p}\tilde{x}_0\tilde{x}_1\dots\tilde{x}_n]$ is convex, the natural parametrization of $\tilde{x}_0\tilde{x}_1\dots\tilde{x}_n$ converges to a natural parametrization of $\tilde{\gamma}$ (see **??**). Thus $\tilde{\gamma}$ is the κ-development of γ with respect to p.

($\Leftarrow$). Assuming convexity of the development, we will prove the point-on-side comparison (8.3.1c). We can assume that $p\notin[xy]$; otherwise the statement is trivial.

Set $T=|x\,y|$ and $\gamma(t)=\mathrm{geod}_{[xy]}(t)$ it is a geodesic in $\mathrm{B}(p,\varpi^\kappa)\backslash\{p\}$. Let $\tilde{\gamma}\colon[0,T]\to\mathbb{M}^2[\kappa]$ be κ-development with base $\tilde{p}$ of γ with respect to p. Take a partition $0=t_0<t_1<\dots<t_n=T$ and set

$$\tilde{y}_i=\tilde{\gamma}(t_i)\quad\text{and}\quad\tau_i=|\tilde{y}_0\,\tilde{y}_1|+|\tilde{y}_1\,\tilde{y}_2|+\dots+|\tilde{y}_{i-1}\;\tilde{y}_i|.$$

Since $\tilde{\gamma}$ is convex, for a fine partition we have that broken line $\tilde{y}_0\tilde{y}_1\dots$ $\dots\tilde{y}_n$ is also convex. Applying Alexandrov's lemma (6.2.1) inductively to pairs of triangles $\tilde{\triangle}^\kappa\{\tau_{i-1},|\tilde{p}\,\tilde{y}_0|,|\tilde{p}\,\tilde{y}_{i-1}|\}$ and $\tilde{\triangle}^\kappa\{|\tilde{y}_{i-1}\;\tilde{y}_i|,|\tilde{p}\,\tilde{y}_{i-1}|,|\tilde{p}\,\tilde{y}_i|\}$, we obtain that sequence $\tilde{\measuredangle}^\kappa\{|\tilde{p}\,\tilde{y}_i|;|\tilde{p}\,\tilde{y}_0|,\tau_i\}$ is non increasing.

For finer and finer partitions we have

$$\max_i\{|\tau_i-t_i|\}\to 0.$$

Thus, point-on-side comparison (8.3.1c) follows. □

8.5 Local definitions and globalization

The aim of this section is giving local analogs of CBB[κ] spaces and proving the equivalence to the global definitions — so called globalization theorem.

First we need a local analog of the notion of a G-delta geodesic space (definition 8.2.2):

8.5.1. Definition. *Let $\mathcal{X}$ be a complete length space. An open set $\Omega\subset\mathcal{X}$ is called G-delta geodesic if for every point $p\in\Omega$ there is a G-delta set W_p which is dense in Ω such that for any $q\in W_p$ there is a geodesic $[pq]$ in $\mathcal{X}$.*

8.5.2. Theorem. *Let $\mathcal{X}$ be a complete length space and $p\in\mathcal{X}$. Then the following conditions are equivalent:*

1) *(local (1+3)-point comparison) there is* $R_1 > 0$ *such that comparison*

$$\measuredangle^{\kappa}(q\,{}^{x^1}_{x^2}) + \measuredangle^{\kappa}(q\,{}^{x^2}_{x^3}) + \measuredangle^{\kappa}(q\,{}^{x^3}_{x^1}) \leqslant 2\cdot\pi$$

holds for any $q, x^1, x^2, x^3 \in \mathrm{B}(p, R_1)$.

2) *(local Kirszbraun property) there is* $R_2 > 0$*, such that for any 3-point subset* F_3 *and any 4-point subset* $F_4 \supset F_3$ *in* $\mathrm{B}(p, R_2)$*, any short map* $f\colon F_3 \to \mathbb{M}^2[\kappa]$ *can be extended as a short map* $\bar{f}\colon F_4 \to \mathbb{M}^2[\kappa]$ *(so* $f = \bar{f}|F_3$*).*

3) *(local function comparison) there is* $R_3 > 0$*, such that* $\mathrm{B}(p, R_3)$ *is G-delta geodesic and for any* $q \in \mathrm{B}(p, R_3)$*, the function* $f = \mathrm{md}^{\kappa} \circ |q_*|$ *satisfies* $f'' \leqslant 1 - \kappa\cdot f$ *in* $\mathrm{B}(p, R_3)$.

4) *(local adjacent angle comparison) there is* $R_4 > 0$ *such that* $\mathrm{B}(p, R_4)$ *is G-delta geodesic and if* q *and geodesic* $[xy]$ *lie in* $\mathrm{B}(p, R_4)$ *and* $z \in\,]xy[$ *then*

$$\measuredangle^{\kappa}(z\,{}^{q}_{x}) + \measuredangle^{\kappa}(z\,{}^{q}_{y}) \leqslant \pi.$$

5) *(local point-on-side comparison) there is* $R_5 > 0$*, such that* $\mathrm{B}(p, R_5)$ *is G-delta geodesic and if* q *and geodesic* $[xy]$ *lie in* $\mathrm{B}(p, R_5)$ *and* $z \in\,]xy[$*, we have*

$$\measuredangle^{\kappa}(x\,{}^{q}_{y}) \leqslant \measuredangle^{\kappa}(x\,{}^{q}_{z});$$

or, equivalently,

$$|\tilde{p}\,\tilde{z}| \leqslant |p\,z|,$$

where $[\tilde{p}\tilde{x}\tilde{y}] = \tilde{\triangle}^{\kappa}(pxy)$*,* $\tilde{z} \in\,]\tilde{x}\tilde{y}[$*,* $|\tilde{x}\,\tilde{z}| = |x\,z|$.

6) *(local hinge comparison) there is* $R_6 > 0$*, such that* $\mathrm{B}(p, R_6)$ *is G-delta geodesic and if* $x \in \mathrm{B}(p, R_6)$ *then (1) for any hinge* $[x\,{}^{q}_{y}]$*, the angle* $\measuredangle[x\,{}^{q}_{y}]$ *is defined and (2) if* $x \in\,]yz[$ *then*[1]

$$\measuredangle[x\,{}^{q}_{y}] + \measuredangle[x\,{}^{q}_{z}] \leqslant \pi.$$

Moreover, if hinge $[x\,{}^{q}_{y}]$ *lies in* $\mathrm{B}(p, R_6)$ *then*

$$\measuredangle[x\,{}^{q}_{y}] \geqslant \measuredangle^{\kappa}(x\,{}^{q}_{y}).$$

or equivalently

$$\tilde{\Upsilon}^{\kappa}[x\,{}^{q}_{y}] \geqslant |q\,y|.$$

[1] Let us remind that $[x\,{}^{q}_{y}]$ and $[x\,{}^{q}_{z}]$ are short notations for pairs $([xq], [xy])$ and $([xq], [xz])$, thus these two hinges automatically have common side $[xq]$.

7) *(local development comparison) there is* $R_7 > 0$*, such that* $\mathrm{B}(p, R_7)$ *is G-delta geodesic and if a geodesic* γ *lies in* $\mathrm{B}(p, R_7)$ *and* $q \in \mathrm{B}(p, R_7)\backslash\gamma$ *then the* κ*-development* $\tilde\gamma$ *with respect to* q *is convex.*

Moreover, for each pair $i, j \in \{1, 2, \dots, 7\}$ *we can assume that* $R_i > \frac{1}{9}\cdot R_j$.

Proof. The proof of each equivalence repeats the proof of corresponding global equivalence in localized form; see proofs of theorems 8.3.1, 8.4.1, 8.4.4, 10.1.1. □

Now, let us give a local definition of lower curvature bound (8.1.2).

8.5.3. Definition. *Let* $\mathcal{X}$ *be a metric space and* $p \in \mathcal{X}$*, we say that* $\mathcal{X}$ has curvature $\geqslant \mathrm{K}$ at p *(briefly,* $\operatorname{curv}_p \mathcal{X} \geqslant \mathrm{K}$*) if* p *satisfies any of the equivalent condition of Theorem 8.5.2 for any* $\kappa < \mathrm{K}$.

8.5.4. Globalization theorem. *Let* $\mathcal{L}$ *be a complete length metric space such that* $\operatorname{curv}_p \mathcal{L} \geqslant \mathrm{K}$ *for any point* $p \in \mathcal{L}$ *then* $\mathcal{L} \in \mathrm{CBB}[\mathrm{K}]$.

In the two-dimensional case this theorem was proved in [16], later, in [159] it was proved for Riemannian manifolds of all dimensions. In the above generality, the theorem first appears in [34]; simplifications and modifications were given in [139], [154], [32]. Our proof, is based on presentations in [139] and [32].

The following corollary says that the expression "space with curvature $\geqslant \kappa$" makes sense.

8.5.5. Corollary. $\mathcal{L} \in \mathrm{CBB}[\mathrm{K}]$ *if and only if* $\mathcal{L} \in \mathrm{CBB}[\kappa]$ *for any* $\kappa < \mathrm{K}$.

Proof. Note that if $\mathrm{K} \leqslant 0$, this statement follows directly from definition of Alexandrov space (8.1.2) and monotonicity of the function $\kappa \mapsto \measuredangle^\kappa(x\,{}^y_z)$ (1.1.1*d*).

The "if"-part also follows directly from definition.

For $\mathrm{K} > 0$, the angle $\measuredangle^{\mathrm{K}}(x\,{}^y_z)$ might be undefined, but, it is defined for sufficiently small triangles. Thus, if $\kappa < \mathrm{K}$ then

$$\mathcal{L} \in \mathrm{CBB}[\mathrm{K}] \quad\Longrightarrow\quad \operatorname{curv}_p \mathcal{L} \geqslant \kappa \ \text{ for any } \ p \in \mathcal{L}.$$

Applying globalization theorem (8.5.4), we get the "only if"-part. □

For the proof of globalization theorem, will need three lemmas. First we need the following characterization of Alexandrov space; it coinsides with 8.3.1*d* if $\kappa \leqslant 0$ and slightly weaker in case $\kappa > 0$.

8.5.6. Short hinge lemma. *Let $\mathcal{L}$ be a complete G-delta geodesic space such that for any hinge $[x\,{}^{p}_{y}]$ in $\mathcal{L}$ the angle $\measuredangle[x\,{}^{p}_{y}]$ is defined and moreover, if $x \in\,]yz[$ then*

$$\measuredangle[x\,{}^{p}_{y}] + \measuredangle[x\,{}^{p}_{z}] \leqslant \pi.$$

Assume that for any hinge $[x\,{}^{p}_{y}]$ in $\mathcal{L}$ we have

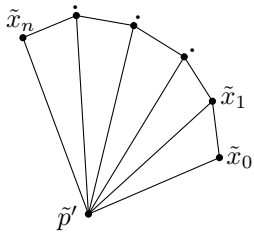
$$|px| + |xy| < \varpi^{\kappa} \quad \Rightarrow \quad \measuredangle[x\,{}^{p}_{y}] \geqslant \tilde{\measuredangle}^{\kappa}(x\,{}^{p}_{y}),$$

then $\mathcal{L} \in \mathrm{CBB}[\kappa]$.

Proof. If $\kappa \leqslant 0$, the lemma follows directly from the hinge comparison (8.3.1d). For $\kappa > 0$ the proof is very similar to the proof of equivalence of development comparison definition (8.4.6).

Applying rescaling, we can assume that $\kappa = 1$, therefore $\mathbb{M}^2[\kappa] = \mathbb{S}^2$ and $\varpi^{\kappa} = \pi$.

According to 8.4.6, it is enough to show that for any point $p \in \mathcal{L}$ and any geodesic $\gamma\colon [0,T] \to \mathrm{B}(p,\pi)\backslash\{p\}$, its κ-development with respect to p is convex.



Clearly $|p\gamma(t)| < \pi - \varepsilon$ for some fixed $\varepsilon > 0$. Consider partition $0 = t_0 < t_1 < \ldots < t_n = T$ such that $|t_i - t_{i-1}| < \varepsilon$. Set $x_i = \gamma(t_i)$ and choose a point $p' \in \mathrm{Str}(x_0, x_1, \ldots, x_n)$ sufficiently close to p. Construct a chain of model triangles $[\tilde{p}'\tilde{x}_{i-1}\tilde{x}_i] = \tilde{\triangle}^{\kappa}(p'x_{i-1}x_i)$ on such a way that direction $[\tilde{p}'\tilde{x}_i]$ turns conterclockwise as i grows. From the assumption of the lemma,

❶
$$\begin{aligned}\measuredangle[\tilde{x}_i\,{}^{\tilde{x}_{i-1}}_{\tilde{p}'}] + \measuredangle[\tilde{x}_i\,{}^{\tilde{x}_{i+1}}_{\tilde{p}'}] &= \tilde{\measuredangle}^{\kappa}(x_i\,{}^{x_{i-1}}_{p'}) + \tilde{\measuredangle}^{\kappa}(x_i\,{}^{x_{i+1}}_{p'}) \leqslant \\ &\leqslant \measuredangle[x_i\,{}^{x_{i-1}}_{p'}] + \measuredangle[x_i\,{}^{x_{i+1}}_{p'}] \leqslant \\ &\leqslant \pi.\end{aligned}$$

Note that

❷
$$\sum_{i=1}^{n} \measuredangle[\tilde{p}'\,{}^{\tilde{x}_i}_{\tilde{x}_{i-1}}] \leqslant \pi$$

Indeed, if $\sum_{i=1}^{n} \measuredangle[\tilde{p}'\,{}^{\tilde{x}_i}_{\tilde{x}_{i-1}}] > \pi$, then for some $\bar{k}$ there is a point $\tilde{y} \in [\tilde{x}_{\bar{k}}\tilde{x}_{\bar{k}+1}]$ such that

$$\measuredangle[\tilde{p}'\,{}^{\tilde{x}_0}_{\tilde{x}_1}] + \ldots + \measuredangle[\tilde{p}'\,{}^{\tilde{x}_{\bar{k}-1}}_{\tilde{x}_{\bar{k}}}] + \measuredangle[\tilde{p}'\,{}^{\tilde{x}_{\bar{k}}}_{\tilde{y}}] = \pi.$$

By ❶, the broken line $\tilde{x}_0\tilde{x}_1\ldots\tilde{x}_{\bar{k}}\tilde{y}$ is convex; in particular,

$$|\tilde{p}'\,\tilde{x}_0|_{\mathbb{S}^2} + |\tilde{p}'\,\tilde{y}|_{\mathbb{S}^2} \leqslant \pi$$

Therefore, by triangle inequality

$$|\tilde{x}_0\,\tilde{x}_1|_{\mathbb{S}^2} + \ldots + |\tilde{x}_{k-1}\;\tilde{x}_k|_{\mathbb{S}^2} + |\tilde{x}_k\,\tilde{y}|_{\mathbb{S}^2} \geqslant |\tilde{p}'\,\tilde{x}_0|_{\mathbb{S}^2} + |\tilde{p}'\,\tilde{y}|_{\mathbb{S}^2}$$

Applying the triangle inequality again, we get

$$|\tilde{y}\,\tilde{x}_{k+1}|_{\mathbb{S}^2} + |\tilde{x}_{k+1}\;\tilde{x}_{k+2}|_{\mathbb{S}^2} + \ldots + |\tilde{x}_{n-1}\;\tilde{x}_n|_{\mathbb{S}^2} > |\tilde{p}'\,\tilde{x}_n|_{\mathbb{S}^2} - |\tilde{p}'\,\tilde{y}|_{\mathbb{S}^2}$$

Hence

$$T = \sum_{i=1}^{n} |\tilde{x}_i\,\tilde{x}_{i-1}|_{\mathbb{S}^2} > |\tilde{p}'\,\tilde{x}_0|_{\mathbb{S}^2} + |\tilde{p}'\,\tilde{x}_n|_{\mathbb{S}^2}.$$

The later contradicts that γ is length minimizing.

From ❶ and ❷, it follows that the spherical polygon $[\tilde{p}'\tilde{x}_0\tilde{x}_1\ldots\tilde{x}_n]$ is convex.

Taking finer and finer partitions and passing to the limit $p' \to p$, the broken line $\tilde{x}_0\tilde{x}_1\ldots\tilde{x}_n$ approaches development of γ with respect to p and the statement follows. □

The following lemma says that if comparison holds for all small hinges then it holds for slightly bigger hinges near the given point.

8.5.7. Key lemma. *Let $\kappa \in \mathbb{R}$, $0 < \ell \leqslant \varpi^\kappa$, $\mathcal{X}$ be a complete geodesic space and $p \in \mathcal{X}$ be a point such that $\operatorname{curv}_x \mathcal{X} \geqslant \kappa$ for any $x \in \mathrm{B}(p, 2\cdot\ell)$.*

Assume that for any point $q \in \mathrm{B}(p,\ell)$, comparison

$$\measuredangle[x\,{}^{y}_{q}] \geqslant \measuredangle^\kappa(x\,{}^{y}_{q})$$

holds for any hinge $[x\,{}^{y}_{q}]$ with $|x\,y| + |x\,q| < \frac{2}{3}\cdot\ell$. Then comparison

$$\measuredangle[x\,{}^{p}_{q}] \geqslant \measuredangle^\kappa(x\,{}^{p}_{q})$$

holds for any hinge $[x\,{}^{p}_{q}]$ with $|x\,p| + |x\,q| < \ell$.

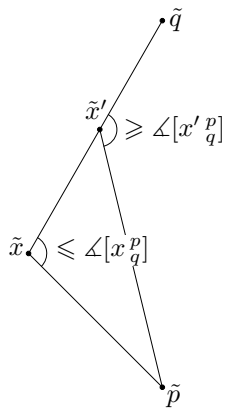
Proof of key lemma (8.5.7) . We will prove an equivalent inequality:

❸ $$\tilde{\Upsilon}^\kappa[x\,{}^{p}_{q}] \geqslant |p\,q|.$$

for any hinge $[x\,{}^{p}_{q}]$ with $|x\,p| + |x\,q| < \ell$.

Fix q. Given a hinge $[x\,{}^{p}_{q}]$ such that $\frac{2}{3}\cdot\ell \leqslant |p\,x| + |x\,q| < \ell$, let us construct a new hinge $[x'\,{}^{p}_{q}]$ which is smaller, i.e.

❹ $$|p\,x| + |x\,q| \geqslant |p\,x'| + |x'\,q|$$



and such that

❺ $$\tilde\Upsilon^\kappa[x\,{}^p_q] \geqslant \tilde\Upsilon^\kappa[x'\,{}^p_q].$$

Assume $|xq| \geqslant |xp|$, otherwise switch the roles of p and q in the following construction. Take $x' \in [xq]$ such that

❻ $$|px| + 3\cdot|xx'| = \tfrac{2}{3}\cdot\ell$$

Choose a geodesic $[x'p]$ and consider the hinge $[x'\,{}^p_q]$ fromed by $[x'p]$ and $[x'q] \subset [xq]$. (In fact the same argument as in 8.6.1 shows that condition $[x'q] \subset [xq]$ always holds.)

Then ❹ follows from the triangle inequality.

Further, note that we have $x, x' \in \mathrm{B}(p,\ell) \cap \mathrm{B}(q,\ell)$ and moreover $|px| + |xx'|, |px'| + |x'x| < \frac{2}{3}\cdot\ell$. In particular,

❼ $$\measuredangle[x\,{}^p_{x'}] \geqslant \tilde\measuredangle^\kappa(x\,{}^p_{x'}) \quad\text{and}\quad \measuredangle[x'\,{}^p_x] \geqslant \tilde\measuredangle^\kappa(x'\,{}^p_x).$$

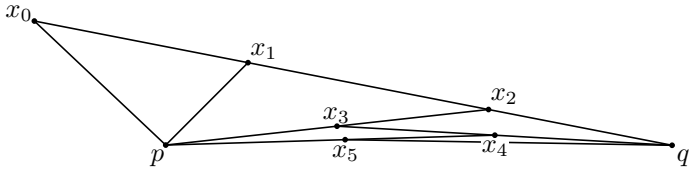
Now, let $[\tilde x\tilde x'\tilde p] = \tilde\triangle^\kappa(xx'p)$. Take $\tilde q$ on the extension of $[\tilde x\tilde x']$ beyond x' such that $|\tilde x\tilde q| = |xq|$ (and therefore $|\tilde x'\tilde q| = |x'q|$). From ❼,

$$\measuredangle[x\,{}^p_q] = \measuredangle[x\,{}^p_{x'}] \geqslant \tilde\measuredangle^\kappa(x\,{}^p_{x'}) \quad\Rightarrow\quad \tilde\Upsilon^\kappa[x\,{}^q_p] \geqslant |\tilde p\tilde q|.$$

From the assumptions of the lemma, we have $\measuredangle[x'\,{}^p_x] + \measuredangle[x'\,{}^p_q] \leqslant \pi$; thus ❼ implies

$$\pi - \tilde\measuredangle^\kappa(x'\,{}^p_x) \geqslant \pi - \measuredangle[x'\,{}^p_x] \geqslant \measuredangle[x'\,{}^p_q].$$

Therefore $|\tilde p\tilde q| \geqslant \tilde\Upsilon^\kappa[x'\,{}^q_p]$ and ❺ follows.



Let us continue the proof. Set $x_0 = x$. Let us apply inductively the above construction to get a sequence of hinges $[x_n\,{}^p_q]$ with $x_{n+1} = x'_n$.

The sequence might terminate at some n only if $|px_n| + |x_nq| < \frac{2}{3}\cdot\ell$. In this case, by the assumptions of the lemma, $\tilde\Upsilon^\kappa[x_n\,{}^p_q] \geqslant |pq|$. From ❺, we get that the sequence $s_n = \tilde\Upsilon^\kappa[x_n\,{}^p_q]$ is decreasing. Hence inequality ❸ follows.

Otherwise, the sequence $r_n = |px_n| + |x_nq|$ is non-increasing and $r_n \geqslant \frac{2}{3}\cdot\ell$ for all n. Hence $r_n \to r$ as $n \to \infty$. Pass to a subsequence

$y_k = x_{n_k}$ such that $y'_k = x_{n_k+1} \in [y_k q]$. (In particular, $|y_k\,q| \geqslant |y_k\,p|$.) Clearly,

❽ $$|p\,y_k| + |y_k\,y'_k| - |p\,y'_k| = r_k - r_{k+1} \underset{k\to\infty}{\to} 0.$$

From ❻, it follows that $|y_k\,y'_k| > \frac{\ell}{100}$. From ❻ and ❽, it follows that $|p\,y_k| > \frac{\ell}{100}$ for all large k. Therefore, $\angle^{\kappa}(y_k\,{}^{p}_{y'_k}) \to \pi$. Since $\measuredangle[y_k\,{}^{p}_{y'_k}] \geqslant$ $\geqslant \angle^{\kappa}(y_k\,{}^{p}_{y'_k})$, we have $\measuredangle[y_k\,{}^{p}_{q}] = \measuredangle[y_k\,{}^{p}_{y'_k}] \to \pi$. Therefore,

$$|p\,y_k| + |y_k\,q| - \tilde{\Upsilon}^{\kappa}[y_k\,{}^{p}_{q}] \to 0.$$

(Here we used that $\ell \leqslant \varpi^{\kappa}$.) Together with the triangle inequality

$$|p\,y_k| + |y_k\,q| \geqslant |p\,q|$$

this yields

$$\lim_{n\to\infty} \tilde{\Upsilon}^{\kappa}[y_k\,{}^{p}_{q}] \geqslant |p\,q|.$$

Applying monotonicity of sequence $s_n = \tilde{\Upsilon}^{\kappa}[y_k\,{}^{p}_{q}]$ we obtain ❸. □

The following lemma is analogous to the existence of minimum point on a compact.

8.5.8. Lemma on almost minimum. *Let $\mathcal{X}$ be a complete metric space, $p \in \mathcal{X}$ and $r\colon \mathcal{X} \to \mathbb{R}$ be a function. Assume that for some $\varepsilon > 0$, the function r is strictly positive in $\overline{\mathrm{B}}[p, \frac{1}{\varepsilon^2}\cdot r(p)]$ and $\varliminf_n r(x_n) > 0$ for any convergent sequence $x_n \to x \in \overline{\mathrm{B}}[p, \frac{1}{\varepsilon^2}\cdot r(p)]$.*

Then, there is a point $p^* \in \overline{\mathrm{B}}[p, \frac{1}{\varepsilon^2}\cdot r(p)]$ such that

a) $r(p^*) \leqslant r(p)$ and

b) $r(x) > (1-\varepsilon)\cdot r(p^*)$ for any $x \in \overline{\mathrm{B}}[p^*, \frac{1}{\varepsilon}\cdot r(p^*)]$.

Proof. Assume the statement is wrong. Then for any $x \in \mathrm{B}(p, \frac{1}{\varepsilon^2}\cdot r(p))$ with $r(x) \leqslant r(p)$, there is a point $x' \in \mathcal{X}$ such that

$$|x\,x'| < \tfrac{1}{\varepsilon}\cdot r(x) \quad\text{and}\quad r(x') \leqslant (1-\varepsilon)\cdot r(x).$$

Take $x_0 = p$ and consider a sequence of points (x_n) such that $x_{n+1} = x'_n$. Clearly

$$|x_{n+1}\;x_n| \leqslant \tfrac{r(p)}{\varepsilon}\cdot(1-\varepsilon)^n \quad\text{and}\quad r(x_n) \leqslant r(p)\cdot(1-\varepsilon)^n.$$

In particular, $|p\,x_n| < \frac{1}{\varepsilon^2}\cdot r(p)$. Therefore (x_n) is a Cauchy sequence, $x_n \to x \in \overline{\mathrm{B}}[p, \frac{1}{\varepsilon^2}\cdot r(p)]$ and $\lim_n r(x_n) = 0$, a contradiction. □

Proof of the globalization theorem (8.5.4). Note that since for fixed points $x, y, z \in \mathcal{L}$, the function $\kappa \mapsto \angle^{\kappa}(x\,{}^{y}_{z})$ is continuous, it is enough

to show that $\mathcal{L} \in \mathrm{CBB}[\kappa]$ for any $\kappa < \mathrm{K}$. For the rest of the proof we fix some $\kappa < \mathrm{K}$.

Exactly the same argument as in the proof of Theorem 8.2.4 shows that $\mathcal{L}$ is G-delta geodesic. By Theorem 8.5.2-6, for any hinge $[x\,{}^{p}_{y}]$ in $\mathcal{L}$ the angle $\measuredangle[x\,{}^{p}_{y}]$ is defined and moreover, if $x \in\,]yz[$ then

$$\measuredangle[x\,{}^{p}_{y}] + \measuredangle[x\,{}^{p}_{z}] \leqslant \pi.$$

Let us denote by $\mathrm{ComRad}(p, \mathcal{L})$ (which stands for *comparison radius* of $\mathcal{L}$ at p) the maximal value (possibly ∞) such that the comparison

$$\measuredangle[x\,{}^{p}_{y}] \geqslant \tilde{\measuredangle}^{\kappa}(x\,{}^{p}_{y})$$

holds for any hinge $[x\,{}^{p}_{y}]$ with $|px| + |xy| < \mathrm{ComRad}(p, \mathcal{L})$.

As follows from 8.5.2-3, $\mathrm{ComRad}(p, \mathcal{L}) > 0$ for any $p \in \mathcal{L}$ and

$$\varliminf_{n\to\infty} \mathrm{ComRad}(p_n, \mathcal{L}) > 0$$

for any converging sequence of points $p_n \to p$. That makes it possible to apply the lemma on almost minimum (8.5.8) to the function $p \mapsto$ $\mapsto \mathrm{ComRad}(p, \mathcal{L})$.

According to Short hinge lemma (8.5.6), it is enough to show that

❾ $$s_0 = \inf_{p\in\mathcal{L}} \mathrm{ComRad}(p, \mathcal{L}) \geqslant \varpi^{\kappa} \quad \text{for any} \quad p \in \mathcal{L}.$$

We argue by contradiction, assuming that ❾ does not hold.

The rest of the proof is easier for geodesic spaces and yet easier for compact spaces. Thus we give three different arguments for each of these cases.

End of the proof for a compact space. By Theorem 8.5.2-3 and compactness of $\mathcal{L}$, it's immediate that $s_0 > 0$. Take a point $p^* \in \mathcal{L}$ such that $r^* = \mathrm{ComRad}(p^*, \mathcal{L})$ is sufficiently close to s_0 (p^* such that $s_0 \leqslant r^* < \min\{\varpi^{\kappa}, \frac{3}{2}\cdot s_0\}$ will do). Then the key lemma (8.5.7) applied for p^* and ℓ slightly bigger than r^* (say, such that $r^* < \ell < \min\{\varpi^{\kappa}, \frac{3}{2}\cdot s_0\}$) implies that

$$\measuredangle[x\,{}^{p^*}_{q}] \geqslant \tilde{\measuredangle}^{\kappa}(x\,{}^{p^*}_{q})$$

for any hinge $[x\,{}^{p^*}_{q}]$ such that $|p^*x| + |xq| < \ell$. Thus $r^* \geqslant \ell$, a contradiction.

End of the proof for a geodesic space. Fix $\varepsilon = 0.0001$. Apply the lemma on almost minimum (8.5.8) to find a point $p^* \in \mathcal{L}$ such that $r^* = \mathrm{ComRad}(p^*, \mathcal{L}) < \varpi^{\kappa}$ and

❿ $$\mathrm{ComRad}(q, \mathcal{L}) > (1 - \varepsilon)\cdot r^*$$

for any $q \in \overline{\mathrm{B}}[p^*, \frac{1}{\varepsilon}\cdot r^*]$.

Applying the key lemma (8.5.7) for p^* and ℓ slightly bigger than r^* leads to a contradiction.

End of the proof, general case. Let us construct $p^* \in \mathcal{L}$ as in the previous case. Since $\mathcal{L}$ is not geodesic, we can not apply the key lemma directly. Instead, let us pass to the ultrapower $\mathcal{L}^\omega$ which is a geodesic space (see 4.3.2).

According to Theorem 8.5.2, inequality ❿ implies that Condition 8.5.2-1 holds for some fixed $R_1 = \frac{r^*}{100} > 0$ at any point $q \in \overline{\mathrm{B}}[p^*, \frac{1}{2\cdot\varepsilon}\cdot r^*] \subset \mathcal{L}$. Therefore a similar statement is true in the ultrapower $\mathcal{L}^\omega$; i.e. for any point $q_\omega \in \overline{\mathrm{B}}[p^*, \frac{1}{2\cdot\varepsilon}\cdot r^*] \subset \mathcal{L}^\omega$, Condition 8.5.2-1 holds for say $R_1 = \frac{r^*}{101}$. In particular, $\operatorname{curv}_{q_\omega}\mathcal{L}^\omega \geqslant \kappa$ for any $q_\omega \in \overline{\mathrm{B}}[p^*, \frac{1}{2\cdot\varepsilon}\cdot r^*] \subset \mathcal{L}^\omega$.

Note that $r^* \geqslant \operatorname{ComRad}(p^*, \mathcal{L}^\omega)$. Therefore we can apply the lemma on almost minimum at point p^* for space $\mathcal{L}^\omega$ with function $x \mapsto \operatorname{ComRad}(x, \mathcal{L}^\omega)$ and $\varepsilon' = \sqrt{\varepsilon} = 0.01$.

For the obtained point $p^{**} \in \mathcal{L}^\omega$, we have $r^{**} = \operatorname{ComRad}(p^{**}, \mathcal{L}) < \varpi^\kappa$ and $\operatorname{ComRad}(q_\omega, \mathcal{L}^\omega) > (1-\varepsilon')\cdot r^{**}$ for any $q_\omega \in \overline{\mathrm{B}}[p^{**}, \frac{1}{\varepsilon'}\cdot r^{**}]$. Thus applying the key lemma (8.5.7) for p^{**} and ℓ slightly bigger than r^{**} leads to a contradiction. □

8.5.9. Advanced Exercise. *Assume $\mathcal{X}$ is a non-complete geodesic space for which the (1+3)-point κ-comparison holds in a neighborhood of every point. Prove that completion of $\mathcal{X}$ is an Alexandrov's space with curvature $\geqslant \kappa$.*

8.6 Properties of geodesics and angles

Remark. All proofs in this section can be easely modified to use only the local definition of CBB-spaces without use of the globalization theorem (8.5.4).

8.6.1. Geodesics do not split. *In* CBB*-space, geodesics do not bifurcate.*

More precisely: let $\mathcal{L} \in \mathrm{CBB}$ and $[px]$, $[py]$ be two geodesics, then

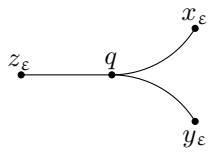
a) If there is an $\varepsilon > 0$, such that $\operatorname{geod}_{[px]}(t) = \operatorname{geod}_{[py]}(t)$ for all $t \in [0, \varepsilon)$, then $[px] \subset [py]$ or $[py] \subset [px]$.

b) If $\measuredangle[p\,{}^{x}_{y}] = 0$ then $[px] \subset [py]$ or $[py] \subset [px]$.

8.6.2. Corollary. *Let $\mathcal{L} \in \mathrm{CBB}$. Then restriction of any geodesic in $\mathcal{L}$ to a proper segment is the unique minimal geodesic joining its endpoints.*

In case $\kappa \leqslant 0$, the proof is easier, since the model triangles are always defined. To deal with $\kappa > 0$ we have to argue locally.

Proof of 8.6.1; a. Let $t_{\max}$ be the maximal value such that $\operatorname{geod}_{[px]}(t) = \operatorname{geod}_{[py]}(t)$ for all $t \in [0, t_{\max})$. Since geodesics are continuous $\operatorname{geod}_{[px]}(t_{\max}) = \operatorname{geod}_{[py]}(t_{\max})$, set



$$q = \operatorname{geod}_{[px]}(t_{\max}) = \operatorname{geod}_{[py]}(t_{\max}).$$

We have to show that $t_{\max} = \min\{|px|, |py|\}$.

If that is not true, choose sufficiently small $\varepsilon > 0$ such that points

$$x_\varepsilon = \operatorname{geod}_{[px]}(t_{\max} + \varepsilon) \quad \text{and} \quad y_\varepsilon = \operatorname{geod}_{[py]}(t_{\max} + \varepsilon)$$

are distinct. Set

$$z_\varepsilon = \operatorname{geod}_{[px]}(t_{\max} - \varepsilon) = \operatorname{geod}_{[py]}(t_{\max} - \varepsilon).$$

Clearly, $\measuredangle^\kappa(q\,{}^{z_\varepsilon}_{x_\varepsilon}) = \measuredangle^\kappa(q\,{}^{z_\varepsilon}_{y_\varepsilon}) = \pi$. Thus from (1+3)-point comparison (8.1.2), $\measuredangle^\kappa(q\,{}^{x_\varepsilon}_{y_\varepsilon}) = 0$ and thus $x_\varepsilon = y_\varepsilon$, a contradiction.

(b). From hinge comparison 8.3.1*d*

$$\measuredangle[p\,{}^{x}_{y}] = 0 \quad \Rightarrow \quad \measuredangle^\kappa\left(p\,{}^{\operatorname{geod}_{[px]}(t)}_{\operatorname{geod}_{[py]}(t)}\right) = 0$$

and thus $\operatorname{geod}_{[px]}(t) = \operatorname{geod}_{[py]}(t)$ for all small t. Therefore we can apply (a). □

8.6.3. Adjacent angle lemma. *Let $\mathcal{L} \in \mathrm{CBB}$ and $[z\,{}^{x}_{p}]$ and $[z\,{}^{y}_{p}]$ be two hinges with comon side $[zp]$ in $\mathcal{L}$. Assume that points p, x, y and z are distinct and $z \in [xy]$, then*

$$\measuredangle[z\,{}^{p}_{y}] + \measuredangle[z\,{}^{p}_{x}] = \pi.$$

Proof. From hinge comparison (8.3.1*d*) we have that both angles $\measuredangle[z\,{}^{p}_{y}]$ and $\measuredangle[z\,{}^{p}_{x}]$ are defined and

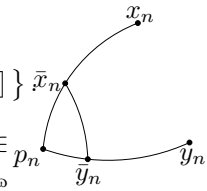
$$\measuredangle[z\,{}^{p}_{y}] + \measuredangle[z\,{}^{p}_{x}] \leqslant \pi.$$

Clearly $\measuredangle[z\,{}^{x}_{y}] = \pi$. Thus, the result follows from the triangle inequality for angles (6.3.2). □

8.6.4. Angle semicontinuity. *Let $\mathcal{L}_n \in \mathrm{CBB}[\kappa]$ and $\mathcal{L}_n \to \mathcal{L}_\omega$ as $n \to \omega$. Assume that the sequence of hinges $[p_n\,{}^{x_n}_{y_n}]$ in $\mathcal{L}_n$ converges to a hinge $[p_\omega\,{}^{x_\omega}_{y_\omega}]$ in $\mathcal{L}_\omega$. Then*

$$\measuredangle[p_\omega\,{}^{x_\omega}_{y_\omega}] \leqslant \lim_{n\to\omega} \measuredangle[p_n\,{}^{x_n}_{y_n}].$$

Proof. From 8.3.2,



$$\measuredangle[p_\omega\,{}^{x_\omega}_{y_\omega}] = \sup\left\{\, \tilde\measuredangle^\kappa(p_\omega\,{}^{\bar x_\omega}_{\bar y_\omega}) \;\middle|\; \bar x_\omega \in]p_\omega x_\omega]\,,\ \bar y_\omega \in]p_\omega x_\omega]\,\right\}.$$

For fixed $\bar x_\omega \in]p_\omega x_\omega]$ and $\bar y_\omega \in]p_\omega x_\omega]$, choose $\bar x_n \in]px_n]$ and $\bar y_n \in]py_n]$ so that $\bar x_n \to \bar x_\omega$ and $\bar y_n \to \bar y_\omega$ as $n \to \omega$. Clearly

$$\tilde\measuredangle^\kappa(p_n\,{}^{\bar x_n}_{\bar y_n}) \to \tilde\measuredangle^\kappa(p_\omega\,{}^{\bar x_\omega}_{\bar y_\omega})$$

as $n \to \omega$.

From the hinge comparison (8.3.1*d*), $\measuredangle[p_n\,{}^{x_n}_{y_n}] \geqslant \tilde\measuredangle^\kappa(p_n\,{}^{\bar x_n}_{\bar y_n})$. Hence the result. □

8.6.5. Angle continuity. *Let $\mathcal{L}_n \in \mathrm{CBB}[\kappa]$ and $\mathcal{L}_n \to \mathcal{L}_\omega$ as $n \to \omega$. Assume that sequences p_n, x_n, y_n in $\mathcal{L}_n$ converges correspondingly to the points $p_\omega, x_\omega, y_\omega$ in $\mathcal{L}_\omega$ and the following two conditions hold*

a) $p_\omega \in \mathrm{Str}(x_\omega)$,

b) $p_\omega \in \mathrm{Str}(y_\omega)$ *or* $y_\omega \in \mathrm{Str}(p_\omega)$.

Then

$$\measuredangle[p_\omega\,{}^{x_\omega}_{y_\omega}] = \lim_{n\to\omega} \measuredangle[p_n\,{}^{x_n}_{y_n}].$$

Proof. By Corollary 8.5.5, we may assume that $\kappa \leqslant 0$.

By Plaut's theorem (8.2.4), the hinge $[p_\omega\,{}^{x_\omega}_{y_\omega}]$ is uniquely defined. Therefore the hinges $[p_n\,{}^{x_n}_{y_n}]$ converge to $[p_\omega\,{}^{x_\omega}_{y_\omega}]$ as $n \to \omega$. Hence by angle semicontinuity (8.6.4), we get

$$\measuredangle[p_\omega\,{}^{x_\omega}_{y_\omega}] \leqslant \lim_{n\to\omega} \measuredangle[p_n\,{}^{x_n}_{y_n}].$$

It remains to show that

❶ $$\measuredangle[p_\omega\,{}^{x_\omega}_{y_\omega}] \geqslant \lim_{n\to\omega} \measuredangle[p_n\,{}^{x_n}_{y_n}].$$

Fix $\varepsilon > 0$. Since $p_\omega \in \mathrm{Str}(x_\omega)$, there is a point $q_\omega \in \mathcal{L}_\omega$ such that

$$\tilde\measuredangle^\kappa(p_\omega\,{}^{x_\omega}_{q_\omega}) > \pi - \varepsilon.$$

The hinge comparison (8.3.1*d*) implies that

❷ $$\measuredangle[p_\omega\,{}^{x_\omega}_{q_\omega}] > \pi - \varepsilon.$$

The triangle inequality for angles (6.3.2)

❸ $$\begin{aligned}\measuredangle[p_\omega\,{}^{x_\omega}_{y_\omega}] &\geqslant \measuredangle[p_\omega\,{}^{x_\omega}_{q_\omega}] - \measuredangle[p_\omega\,{}^{y_\omega}_{q_\omega}] > \\ &> \pi - \varepsilon - \measuredangle[p_\omega\,{}^{y_\omega}_{q_\omega}].\end{aligned}$$

Note that we can assume in addition that $q_\omega \in \mathrm{Str}(p_\omega)$. Choose $q_n \in \mathcal{L}_n$ such that $q_n \to q_\omega$ as $n \to \omega$. Note that the angle semicontinuity, again we get

❹
$$\begin{aligned}\measuredangle[p_\omega\,{}^{x_\omega}_{q_\omega}] &\leqslant \lim_{n\to\omega} \measuredangle[p_n\,{}^{x_n}_{q_n}],\\ \measuredangle[p_\omega\,{}^{y_\omega}_{q_\omega}] &\leqslant \lim_{n\to\omega} \measuredangle[p_n\,{}^{y_n}_{q_n}].\end{aligned}$$

By (1+3)-point comparison (8.1.2) and 8.3.2b, we get

$$\measuredangle[p_n\,{}^{x_n}_{y_n}] + \measuredangle[p_n\,{}^{y_n}_{q_n}] + \measuredangle[p_n\,{}^{x_n}_{q_n}] \leqslant 2\cdot\pi.$$

for all n. Together with ❹, ❷ and ❸, it implies

$$\begin{aligned}\lim_{n\to\omega} \measuredangle[p_n\,{}^{x_n}_{y_n}] &\leqslant 2\cdot\pi - \lim_{n\to\omega} \measuredangle[p_n\,{}^{x_n}_{q_n}] - \lim_{n\to\omega} \measuredangle[p_n\,{}^{y_n}_{q_n}] \leqslant\\ &\leqslant 2\cdot\pi - \measuredangle[p_\omega\,{}^{x_\omega}_{q_\omega}] - \measuredangle[p_\omega\,{}^{y_\omega}_{q_\omega}] <\\ &< \measuredangle[p_\omega\,{}^{x_\omega}_{y_\omega}] + 2\cdot\varepsilon.\end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, ❶ follows. □

8.6.6. First variation formula. *Let $\mathcal{L} \in \mathrm{CBB}$. For any point q and geodesic $[px]$ in $\mathcal{L}$, we have*

❺
$$|q\,\mathrm{geod}_{[px]}(t)| = |qp| - t\cdot\cos\varphi + o(t),$$

where φ is the infimum of angles between $[px]$ and all geodesics from p to q in the ultrapower $\mathcal{L}^\omega$.

Remark. If $\mathcal{L}$ is proper space then $\mathcal{L}^\omega = \mathcal{L}$, see Section 4.2. Therefore the infimum φ is achieved on some particular geodesic from p to q.

As a corollary we obtain the following classical result:

8.6.7. Strong angle lemma. *Let $\mathcal{L} \in \mathrm{CBB}$ and $p, q \in \mathcal{L}$ be such that there is unique geodesic from p to q in the ultrapower $\mathcal{L}^\omega$. Then for any hinge $[p\,{}^{q}_{x}]$ we have*

❻
$$\measuredangle[p\,{}^{q}_{x}] = \lim_{\substack{\bar x\to p\\ \bar x\in]px]}} \tilde\measuredangle^{\kappa}(p\,{}^{q}_{\bar x}).$$

for any $\kappa \in \mathbb{R}$ such that $|pq| < \varpi^\kappa$.

In partiqular ❻ holds if $p \in \mathrm{Str}(q)$ as well as if $q \in \mathrm{Str}(p)$

Remark.

- ⋄ The above lemma is essentially due to Alexandrov. The right hand side in ❻ is called *strong angle* of hinge $[p\,{}^{q}_{x}]$. Note that in general metric space angle and strong angle of the same hinge might differ.

- ◇ As it follows from Exercise **??**, if there is a unique geodesic $[pq]$ in the ultrapower $\mathcal{L}^\omega$ then $[pq]$ lies in $\mathcal{L}$.

Proof lemma about strong angle (8.6.7). The first statement follows directly from the first variation formula (8.6.6) and definition of model angle (see Section 6.3). The second statement follows from the Plaut's theorem (8.2.4) applyied to $\mathcal{L}^\omega$. (Note that according to Proposition 8.1.7 $\mathcal{L}^\omega \in \mathrm{CBB}$.) □

Proof of the first variation formula (8.6.6). Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$. Without loss of generality we can assume that $\kappa \leqslant 0$. The inequality

$$|q\,\mathrm{geod}_{[px]}(t)| \leqslant |qp| - t\cdot\cos\varphi + o(t)$$

is an immediate consequence of the first variation inequality (6.3.4). Thus, it is enough to show that

$$|q\,\mathrm{geod}_{[px]}(t)| \geqslant |qp| - t\cdot\cos\varphi + o(t).$$

Assume the contrary, then there is $\varepsilon > 0$, such that $\varphi + \varepsilon < \pi$ and for a sequence $t_n \to 0+$ we have

❼ $$|q\,\mathrm{geod}_{[px]}(t_n)| < |qp| - t_n\cdot\cos(\varphi - \varepsilon).$$

Set $x_n = \mathrm{geod}_{[px]}(t_n)$. Clearly

$$\measuredangle^\kappa(x_n\,{}^p_q) > \pi - \varphi + \tfrac{\varepsilon}{2}$$

for all large n.

Assume $\mathcal{L}$ is geodesic. Choose a sequence of geodesics $[x_nq]$. Let $[x_nq] \to [pq]_{\mathcal{L}^\omega}$ as $n \to \omega$ (in general $[pq]$ might lie in $\mathcal{L}^\omega$). Applying both parts of hinge comparison (8.3.1d), we get $\measuredangle[x_n\,{}^q_x] < \varphi - \frac{\varepsilon}{2}$ for all large n. According to 8.6.4, the angle between $[pq]$ and $[px]$ is at most $\varphi - \frac{\varepsilon}{2}$, a contradiction.

Finally, if $\mathcal{L}$ is not geodesic choose a sequence $q_n \in \mathrm{Str}(x_n)$, such that $q_n \to q$ and the inequality

$$\measuredangle^\kappa(x_n\,{}^p_{q_n}) > \pi - \varphi + \tfrac{\varepsilon}{2}$$

still holds. Then the same argument as above shows that $[x_nq_n]$ ω-converges to a geodesic $[pq]_{\mathcal{L}^\omega}$ from p to q having angle at most $\varphi - \frac{\varepsilon}{2}$ with $[px]$. □

8.7 On positive lower bound

In this section we consider $\mathrm{CBB}[\kappa]$ spaces for $\kappa > 0$. Appying rescaling we can assume that $\kappa = 1$.

The following theorem states that if one ignores a few exceptional spaces, then the diameter of a space with positive lower curvature bound is bounded. Note that many authors (but not us) exclude these spaces in the definition of Alexandrov space with positive lower curvature bound.

8.7.1. On diameter of space. *Let* $\mathcal{L} \in \mathrm{CBB}[1]$. *Then either*

a) $\operatorname{diam}\mathcal{L} \leqslant \pi$*;*

b) or $\mathcal{L}$ *is isometric to one of the following exceptional spaces:*

1. *real line* $\mathbb{R}$*,*
2. *a ray* $\mathbb{R}_{\geqslant 0}$*,*
3. *a closed interval* $[0,a] \in \mathbb{R}$*,* $a > \pi$*,*
4. *a circle* $\mathbb{S}^1_a$ *of length* $a > 2\cdot\pi$*.*

Proof. Assume that $\mathcal{L}$ is a geodesic space and $\operatorname{diam}\mathcal{L} > \pi$. Choose $x, y \in \mathcal{L}$ so that $|x\,y| = \pi + \varepsilon$, $0 < \varepsilon < \frac{\pi}{4}$. By moving y slightly, we can also assume that geodesic $[xy]$ is unique; to prove it, use either Plaut's theorem (8.2.4) or that geodesics do not split (8.6.1). Let z be the midpoint of a geodesic $[xy]$.

Consider the function $f = |x{*}| + |y{*}|$. As it follows from Lemma 8.7.2, f is concave in $\mathrm{B}(z, \frac{\varepsilon}{4})$. Let $p \in \mathrm{B}(z, \frac{\varepsilon}{4})$. Choose a geodesic $[zp]$. Set $h(t) = f \circ \operatorname{geod}_{[zp]}(t)$ and $\ell = |z\,p|$. Clearly h is concave. From Adjacent angle lemma (8.6.3), we have that $h^+(0) = 0$. Therefore h is nonincreasing which means that

$$|x\,p| + |y\,p| = h(\ell) \leqslant h(0) = |x\,y|.$$

Since the geodesic $[xy]$ is unique this means that $p \in [xy]$ and hence $\mathrm{B}(z, \frac{\varepsilon}{4})$ only contains points of $[xy]$. Since in CBB-spaces, geodesics do not bifurcate (8.6.1*a*), it follows that all of $\mathcal{L}$ coincides with the maximal extension of $[xy]$ as a local geodesic γ (which might be not minimizing). In other words, $\mathcal{L}$ is isometric to a 1-dimensional Riemannian manifold with possibly non empty boundary. From this, it is easy to see that $\mathcal{L}$ falls into one of the exceptional spaces described in the theorem.

Lastly, if $\mathcal{L}$ is not geodesic and $\operatorname{diam}\mathcal{L} > \pi$ then the above argument applied to $\mathcal{L}^\omega$ yields that $\mathcal{L}^\omega$ is isometric to one of the exceptional spaces. As all of those spaces are proper it means that $\mathcal{L} = \mathcal{L}^\omega$. $\square$

8.7.2. Lemma. *Let* $\mathcal{L} \in \mathrm{CBB}[1]$ *and* $p \in \mathcal{L}$. *Then* $|p{*}| \colon \mathcal{L} \to \mathbb{R}$ *is concave in* $\mathrm{B}(p,\pi)\backslash\mathrm{B}(p, \frac{\pi}{2})$.

In particular, if $\operatorname{diam}\mathcal{L} \leqslant \pi$ *then the complements* $\mathcal{L}\backslash\mathrm{B}(p,r)$ *and* $\mathcal{L}\backslash\overline{\mathrm{B}}[p,r]$ *are convex for any* $r > \frac{\pi}{2}$.

Proof. This is an immediate consequence of 8.4.1*b*. □

8.7.3. Exercise. *Let* $\mathcal{L} \in \mathrm{CBB}^m[1]$ *and* $\operatorname{diam}\mathcal{L} \leqslant \pi$. *Assume that the group* G *acts on* $\mathcal{L}$ *by isometries and has closed orbits and*

$$\operatorname{diam}(\mathcal{L}/G) > \tfrac{\pi}{2}.$$

Show that the action of G *has a fixed point in* $\mathcal{L}$.

8.7.4. On perimeter of triple. *Let* $\mathcal{L} \in \mathrm{CBB}[1]$ *and* $\operatorname{diam}\mathcal{L} \leqslant \pi$. *Then perimeter of any triple of points* $p, q, r \in \mathcal{L}$ *is at most* $2{\cdot}\pi$.

Proof. Arguing by contradiction, suppose

❶ $$|pq| + |qr| + |rp| > 2{\cdot}\pi$$

for some $p, q, r \in \mathcal{L}$. Rescaling the space slightly, we can assume that $\operatorname{diam}\mathcal{L} < \pi$, but the inequality ❶ still holds.

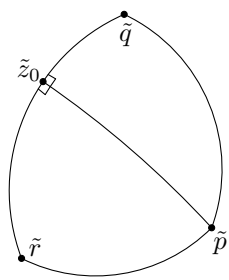
By Corollary 8.5.5, after rescaling we still have $\mathcal{L} \in \mathrm{CBB}[1]$.

Since $\mathcal{L}$ is G-delta geodesic (8.2.4), it is sufficient to consider case when there is a geodesic $[qr]$.

First note that since $\operatorname{diam}\mathcal{L} < \pi$, by 8.4.1*b* we have that the function

$$y(t) = \mathrm{md}^1|p\,\mathrm{geod}_{[qr]}(t)|$$

satisfies the differential inequality $y'' \leqslant 1 - y$.



Take $z_0 \in [qr]$ so that restriction $|p{*}||[qr]$ attains its maximum at z_0, set $t_0 = |q\,z_0|$ so $z_0 = \mathrm{geod}_{[qr]}(t_0)$. Consider the following model configuration: two geodesics $[\tilde p\tilde z_0]$, $[\tilde q\tilde r]$ in $\mathbb{S}^2$ such that

$$|\tilde p\,\tilde z_0| = |p\,z_0|, \qquad |\tilde q\,\tilde r| = |q\,r|,$$
$$|\tilde z_0\,\tilde q| = |z_0\,q|, \qquad |\tilde z_0\,\tilde r| = |z_0\,q|$$

and

$$\measuredangle[\tilde z_0\,{}^{\tilde q}_{\tilde p}] = \measuredangle[\tilde z_0\,{}^{\tilde r}_{\tilde p}] = \tfrac{\pi}{2}.$$

Clearly, $\bar y(t) = \mathrm{md}^1|\tilde p\,\mathrm{geod}_{[\tilde q\tilde r]}(t)|$ satisfies $\bar y'' = 1 - \bar y$ and $\bar y'(t_0) = 0$, $\bar y(t_0) = y(t_0)$. Since z_0 is a maximum point, $y(t) \leqslant y(t_0) + o(t - t_0)$; thus, $\bar y(t)$ is a barrier for $y(t) = \mathrm{md}^1|p\,\mathrm{geod}_{[qr]}(t)|$ at t_0 by 5.4.3*c*. From barrier inequality (5.4.3*c*), we get

$$|\tilde p\,\mathrm{geod}_{[\tilde q\tilde r]}(t)| \geqslant |p\,\mathrm{geod}_{[qr]}(t)|,$$

thus, $|\tilde{p}\tilde{q}| \geqslant |pq|$ and $|\tilde{p}\tilde{r}| \geqslant |pr|$.

Therefore, $|pq| + |qr| + |rp|$ can not exceed the perimeter of the spherical triangle $[\tilde{p}\tilde{q}\tilde{r}]$; that is,

$$|pq| + |qr| + |rp| \leqslant 2{\cdot}\pi,$$

a contradiction. □

Let $\kappa > 0$. Consider the following extension $\mathcal{Z}^{\kappa+}(*\,{}^{*}_{*})$ of the model angle function $\mathcal{Z}^{\kappa}(*\,{}^{*}_{*})$. Some authors define camparison angle to be $\mathcal{Z}^{\kappa+}(*\,{}^{*}_{*})$ but that leads to two different definition of model angles — one for CBA spaces and an other for CBB spaces.

8.7.5. Definition of extended angle. *Let p, q, r be points in a metric space such that $p \neq q$, $p \neq r$ then set*

$$\mathcal{Z}^{\kappa+}(p\,{}^{q}_{r}) = \sup\left\{\, \mathcal{Z}^{\mathrm{K}}(p\,{}^{q}_{r}) \;\middle|\; \mathrm{K} \leqslant \kappa \,\right\}.$$

The vlaue $\mathcal{Z}^{\kappa+}(p\,{}^{q}_{r})$ is called extended model angle *of triple p, q, r.*

8.7.6. Extended angle comparison. *Let $\kappa > 0$ and $\mathcal{L} \in \mathrm{CBB}[\kappa]$. Then for any hinge $[p\,{}^{q}_{r}]$ we have $\measuredangle[p\,{}^{q}_{r}] \geqslant \mathcal{Z}^{\kappa+}(p\,{}^{q}_{r})$.*

Moreover, the extended model angle can be calculated $\mathcal{Z}^{\kappa+}(p\,{}^{q}_{r})$ using the following rules:

a) $\mathcal{Z}^{\kappa+}(p\,{}^{q}_{r}) = \mathcal{Z}^{\kappa}(p\,{}^{q}_{r})$ once $\mathcal{Z}^{\kappa}(p\,{}^{q}_{r})$ is defined;
b) $\mathcal{Z}^{\kappa+}(p\,{}^{q}_{r}) = \mathcal{Z}^{\kappa+}(p\,{}^{r}_{q}) = 0$ if $|pq| + |qr| = |pr|$;
c) $\mathcal{Z}^{\kappa+}(p\,{}^{q}_{r}) = \pi$ if none of above is applicable.

Proof. From Corollary 8.5.5, $\mathrm{K} < \kappa$ implies $\mathrm{CBB}[\mathrm{K}] \supset \mathrm{CBB}[\kappa]$; thus the extended angle comparison follows from the definition.

The rules for calculating extended angle is an easy consequence of its definition. □

8.8 Remarks and open problems

The question whether the first part of 8.3.1*d* suffices to conclude that $\mathcal{L} \in \mathrm{CBB}[\kappa]$ is a long standing open problem (possibly dating back to Alexandrov), but as far as we know, it was first formally stated in print in [32] (see the footnote in subsection 4.1.5).

8.8.1. Open question. *Let $\mathcal{L}$ be a complete geodesic space (you can also assume that $\mathcal{L}$ is homeomorphic to $\mathbb{S}^2$ or $\mathbb{R}^2$) such that for any hinge $[x\,{}^{p}_{y}]$ in $\mathcal{L}$, the angle $\measuredangle[x\,{}^{p}_{y}]$ is defined and*

$$\measuredangle[x\,{}^{p}_{y}] \geqslant \mathcal{Z}^{0}(x\,{}^{p}_{y}).$$

Is it true that $\mathcal{L} \in \mathrm{CBB}[0]$?

Chapter 9

Definitions of curvature bounded above

Section 9.1 defines CBA[κ] spaces via a (2+2)-point comparison; it is an analog of CAT[κ] spaces but the definition use only distances and makes no assumptions on the presence of geodesics. Section 9.3 gives definitions in terms of angle and distance comparisons for triangles. Section 9.4 looks at thin triangles and their inheritance lemma. Section 9.5 gives a useful definitions in terms of convexity of distance functions.

Section 9.6 discusses convexity in the CBA[κ] spaces. Section 9.7 we discuss a definition via developments of the curves.

We give three types of globalization theorems: Patchwork globalization in 9.8, Lifting globalization in 9.11 and Hadamard–Cartan theorem in 9.13.

The necesury technique to prove these theorems include Reshetnyak's gluing theorem discussed in 9.9, the structure of the space of geodesics discussed in 9.10 and Reshetnyak Majorization theorem in 9.12.

Section 9.14 discusses angles and the first variation formula.

9.1 (2+2)-point comparison.

9.1.1. (2+2)-point comparison. *An quadruple of points* p^1, p^2, x^1, x^2 *in a metric space satisfies* (2+2)-point κ-comparison *if*

a) $\measuredangle^\kappa(p^1\,{}^{x^1}_{x^2}) \leqslant \measuredangle^\kappa(p^1\,{}^{p^2}_{x^1}) + \measuredangle^\kappa(p^1\,{}^{p^2}_{x^2})$, *or*

b) $\measuredangle^\kappa(p^2\,{}^{x^1}_{x^2}) \leqslant \measuredangle^\kappa(p^2\,{}^{p^1}_{x^1}) + \measuredangle^\kappa(p^2\,{}^{p^1}_{x^2})$, *or*

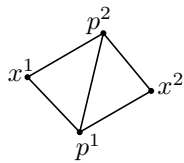
c) one of the six model angles

$$\measuredangle^{\kappa}(p^1\,{}^{x^1}_{x^2}),\ \measuredangle^{\kappa}(p^1\,{}^{p^2}_{x^1}),\ \measuredangle^{\kappa}(p^1\,{}^{p^2}_{x^2}),\ \measuredangle^{\kappa}(p^2\,{}^{x^1}_{x^2}),\ \measuredangle^{\kappa}(p^2\,{}^{p^1}_{x^1}),\ \measuredangle^{\kappa}(p^2\,{}^{p^1}_{x^2})$$

is undefined.

9.1.2. Definition. *A complete length space $\mathcal{U}$ is called a* CBA[κ] space *(also written $\mathcal{U} \in \mathrm{CBA}[\kappa]$) if every quadruple p^1, p^2, x^1, x^2 satisfies the (2+2)-point κ-comparison (9.1.1).*

We write $\mathcal{U} \in \mathrm{CBA}$ to mean: $\mathcal{U} \in \mathrm{CBA}[\kappa]$ for some κ.



The condition $\mathcal{U} \in \mathrm{CAT}[\kappa]$ should be thought of as "$\mathcal{U}$ has global curvature $\leqslant \kappa$". In Proposition 9.3.4, it will be shown that this formulation makes sense; in particular, if $\kappa \leqslant \mathrm{K}$ then

$$\mathrm{CBA}[\kappa] \subset \mathrm{CBA}[\mathrm{K}].$$

9.1.3. Exercise. *Let $\mathcal{U}$ be a complete length space. Show that $\mathcal{U} \in$ $\in \mathrm{CBA}[\kappa]$ if and only if every quadruple of points in $\mathcal{U}$ admits one labeling by (p, x^1, x^2, x^3) so that the three angles $\measuredangle^{\kappa}(p\,{}^{x^1}_{x^2})$, $\measuredangle^{\kappa}(p\,{}^{x^2}_{x^3})$ and $\measuredangle^{\kappa}(p\,{}^{x^1}_{x^3})$ satisfy all three triangle inequalities or one of these angels is undefined.*

9.1.4. Exercise. *Let $\mathcal{U}$ be a complete length space.*

Show that $\mathcal{U} \in \mathrm{CBA}[\kappa]$ if and only if for any quadruple of points p^1, p^2, x^1, x^2 in $\mathcal{U}$ such that $|p^1\,p^2|, |x^1\,x^2| \leqslant \varpi^{\kappa}$ there is a quadruple q^1, q^2, y^1, y^2 in $\mathbb{M}^m[\kappa]$ such that

$$\begin{aligned} |q^1\,q^2| &= |p^1\,p^2|, \\ |y^1\,y^2| &= |x^1\,x^2|, \\ |q^i\,y^j| &\leqslant |p^i\,x^j| \end{aligned}$$

for any i and j.

The next proposition follows directly from Definition 9.1.2 and the definitions of ultralimit and ultrapower; see Section 4.2 for the related definitions.

9.1.5. Proposition. *If $(\mathcal{U}_n, \star_n) \to (\mathcal{U}_\omega, \star_\omega)$ and $\kappa_n \to \kappa_\omega$ as $n \to \omega$, then*

$$\mathcal{U}_n \in \mathrm{CBA}[\kappa_n] \implies \mathcal{U}_\omega \in \mathrm{CBA}[\kappa_\omega].$$

Moreover, for a complete metric space $\mathcal{U}$,

$$\mathcal{U} \in \mathrm{CBA}[\kappa] \iff \mathcal{U}^\omega \in \mathrm{CBA}[\kappa].$$

9.2 Geodesics

9.2.1. Theorem on uniqueness of geodesics. *In a* CBA[κ] *space, pairs of points at distance* $< \varpi^\kappa$ *are joined by unique geodesics, and these geodesics depend continuously on their endpoint pairs.*

Proof. Let $\mathcal{U} \in \mathrm{CBA}[\kappa]$. Fix two points $p^1, p^2 \in \mathcal{U}$ such that

$$|p^1\,p^2|_{\mathcal{U}} < \varpi^\kappa.$$

Choose a sequence of approximate midpoints z_n for p^1 and p^2; i.e.

❶ $$|p^1\,z_n|, |p^2\,z_n| \to \tfrac{1}{2}\cdot|p^1\,p^2| \text{ as } n \to \infty.$$

By the law of cosines, $\measuredangle^\kappa(p^1\,{}^{z_n}_{p^2})$ and $\measuredangle^\kappa(p^2\,{}^{z_n}_{p^1})$ are arbitrarily small when n is sufficiently large.

Let us apply (2+2)-point comparison (9.1.1) to the quadruple $p^1, p^2, z_n, z_{\bar{k}}$ with large n and $\bar{k}$. We conclude that $\measuredangle^\kappa(p\,{}^{z_n}_{z_{\bar{k}}})$ is arbitrarily small when $n, \bar{k}$ are sufficiently large and p is either p^1 or p^2. By ❶ and the law of cosines, (z_n) is a Cauchy sequence.

Since $\mathcal{U}$ is complete, the sequence (z_n) converges to a midpoint of p^1 and p^2. By Lemma 2.3.6 the existence of a geodesic $[p^1p^2]$.

Now suppose $p^1_n \to p^1$, $p^2_n \to p^2$ as $n \to \infty$. Let z_n be the midpoint of a geodesic $[p^1_n p^2_n]$ and z be the midpoint of a geodesic $[p^1p^2]$.

It suffices to show that

❷ $$|z_n\,z| \to 0 \quad \text{as} \quad n \to \infty.$$

By the triangle inequality, the z_n are approximate midpoints of p^1 and p^2. Apply (2+2)-point comparison (9.1.1) to the quadruple p^1, p^2, z_n, z. For $p = p^1$ or $p = p^2$, we see that $\measuredangle^\kappa(p\,{}^{z_n}_{z})$ is arbitrarily small when n is sufficiently large. By the law of cosines, ❷ follows. □

9.2.2. Exercise. *Let* $\mathcal{U} \in \mathrm{CBA}$. *Assume* $\mathcal{U}$ *is a topological manifold. Show that any geodesic in* $\mathcal{U}$ *can be extended as a two-side infinite local geodesic.*

Moreover the same holds for any $\mathcal{U} \in \mathrm{CAT}$ *with nontrivial local homology groups at any point; in particular if* $\mathcal{U}$ *is a homological manifold.*

The following definition is analogous to the definition of CBA spaces, but more standard in the literature.

9.2.3. Definition. ϖ^κ*-geodesic lenght spaces which satisfy (2+2)-point* κ*comparison for any quadrouple of points are called* CAT[κ] spaces.

We write $\mathcal{U} \in \mathrm{CAT}$ *to mean:* $\mathcal{U} \in \mathrm{CAT}[\kappa]$ *for some* κ.

This terminology was introduced by Gromov; CAT stands for Cartan–Alexandrov–Toponogov, although Toponogov did not work with curvature bounded above. Previously, these spaces were called $\mathfrak{R}_\kappa$ *domains*, which was Alexandrov's terminology.

Unlike $\mathrm{CBA}[\kappa]$ spaces the $\mathrm{CAT}[\kappa]$ spaces maybe not complete.

From Theorem on uniqueness of geodesics 9.2.1 we get the following.

9.2.4. Corollary. *Any* $\mathrm{CBA}[\kappa]$ *space is* ϖ^κ*-geodesic. In particular*

$$\mathrm{CBA}[\kappa] \subset \mathrm{CAT}[\kappa].$$

In other words the Definition 9.2.3 is weaker than 9.1.2. On the other hand we have the following.

9.2.5. Proposition. *The completion of any* $\mathrm{CAT}[\kappa]$ *space is a* $\mathrm{CBA}[\kappa]$ *space.*

Moreover, $\mathcal{U} \in \mathrm{CAT}[\kappa]$ *if and only if there is a space* $\bar{\mathcal{U}} \in \mathrm{CBA}[\kappa]$ *which contains a* ϖ^κ*-convex dense set isometric to* $\mathcal{U}$.

Proof. By Theorem 9.2.1, in order to show that completion $\bar{\mathcal{U}}$ of any space $\mathcal{U} \in \mathrm{CAT}[\kappa]$ is a $\mathrm{CBA}[\kappa]$ space, it is sufficient to verify that the completion a length space is a length space; this is straightforward.

For the second part of proposition note that the completion $\bar{\mathcal{U}}$ contains the original space $\mathcal{U}$ as a dense ϖ^κ-convex subset and the metric on $\mathcal{U}$ coinsides with induced length metric from $\bar{\mathcal{U}}$. □

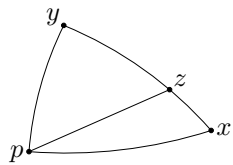
Here is a corollary from Proposition 9.2.5 and Theorem 9.2.1.

9.2.6. Corollary. *In a* $\mathrm{CAT}[\kappa]$ *space, pairs of points at distance* $< \varpi^\kappa$ *are joined by unique geodesics, and these geodesics depend continuously on their endpoint pairs.*

Proof. By Proposition 9.2.5 any $\mathrm{CAT}[\kappa]$ space is isometric to a convex dense subset of a $\mathrm{CBA}[\kappa]$ space. It remains to apply Theorem 9.2.1. □

9.3 More comparisons

Here we give a few reformulations of Definition 9.1.2.



9.3.1. Theorem. *Let $\mathcal{U}$ be a ϖ^κ-geodesic length space. Then*

a) $\mathcal{U} \in \mathrm{CAT}[\kappa]$

if and only if one of the following conditions holds for all $p, x, y \in \mathcal{U}$ of perimeter $< 2\cdot\varpi^\kappa$:

b) (adjacent-angles comparison) for any geodesic $[xy]$ and $z \in]xy[$, we have

$$\measuredangle^\kappa(z\,{}^p_x) + \measuredangle^\kappa(z\,{}^p_y) \geqslant \pi.$$

c) (point-on-side comparison) for any geodesic $[xy]$ and $z \in$ $\in]xy[$, we have

$$\measuredangle^\kappa(x\,{}^p_y) \geqslant \measuredangle^\kappa(x\,{}^p_z),$$

or equivalently,

$$|\tilde{p}\,\tilde{z}| \geqslant |p\,z|,$$

where $[\tilde{p}\tilde{x}\tilde{y}] = \tilde{\triangle}^\kappa(pxy)$, $\tilde{z} \in]\tilde{x}\tilde{y}[$, $|\tilde{x}\,\tilde{z}| = |x\,z|$.

d) (angle comparison) for any hinge $[x\,{}^p_y]$, the angle $\measuredangle[x\,{}^p_y]$ exists and

$$\measuredangle[x\,{}^p_y] \leqslant \measuredangle^\kappa(x\,{}^p_y),$$

or equivalently,

$$\tilde{\Upsilon}^\kappa[x\,{}^p_y] \leqslant |p\,y|.$$

Remark. In the following proof, the part $(a)\Rightarrow(b)$ only requires that the (2+2)-point comparison (9.1.1) hold for any quadruple, and does not require the existence of geodesics at distance $< \varpi^\kappa$. The same is true of the parts $(b)\Leftrightarrow(c)$ and $(c)\Rightarrow(d)$. Thus the conditions (b), (c) and (d) are valid for any metric space (not necessarily length space) which satisfies (2+2)-point comparison (9.1.1). The converse does not hold; for example, all these conditions are vacuously true in a totally disconnected space, while (2+2)-point comparison is not.

Proof. (a) $\Rightarrow$ (b). Since the perimeter of p, x, y is $< 2\cdot\varpi^\kappa$, so is the perimeter of any subtriple of p, z, x, y by the triangle inequality. By Alexandrov's lemma (6.2.1),

$$\measuredangle^\kappa(p\,{}^z_x) + \measuredangle^\kappa(p\,{}^z_y) < \measuredangle^\kappa(p\,{}^x_y) \text{ or } \measuredangle^\kappa(z\,{}^p_x) + \measuredangle^\kappa(z\,{}^p_y) = \pi.$$

In the former case, (2+2)-point comparison (9.1.1) applied to the quadruple p, z, x, y implies

$$\measuredangle^\kappa(z\,{}^p_x) + \measuredangle^\kappa(z\,{}^p_y) \geqslant \measuredangle^\kappa(z\,{}^x_y) = \pi.$$

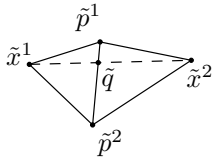
(b) $\Leftrightarrow$ (c). Follows directly from Alexandrov's lemma (6.2.1).

(c) ⇒ (d). By (c), for $\bar p \in\,]xp]$ and $\bar y \in\,]xy]$ the function $(|x\,\bar p|, |x\,\bar y|) \mapsto$ $\mapsto \measuredangle^\kappa(x\,{}^{\bar p}_{\bar y})$ is nondecreasing in each argument. In particular, $\measuredangle[x\,{}^{p}_{y}] =$ $= \inf \measuredangle^\kappa(x\,{}^{\bar p}_{\bar y})$. Thus $\measuredangle[x\,{}^{p}_{y}]$ exists and is at most $\measuredangle^\kappa(x\,{}^{p}_{y})$.

(d) ⇒ (b). By (d) and the triangle inequality for angles (6.3.2),

$$\measuredangle^\kappa(z\,{}^{p}_{x}) + \measuredangle^\kappa(z\,{}^{p}_{y}) \geqslant \measuredangle[z\,{}^{p}_{x}] + \measuredangle[z\,{}^{p}_{y}] \geqslant \pi.$$

(c) ⇒ (a). Given a quadruple p^1, p^2, x^1, x^2 whose subtriples have perimeter $< 2\cdot\varpi^\kappa$, we must verify (2+2)-point comparison (9.1.1). In $\mathbb{M}^2[\kappa]$, construct the model triangles $[\tilde p^1\tilde p^2\tilde x^1] = \tilde\triangle^\kappa(p^1p^2x^1)$ and $[\tilde p^1\tilde p^2\tilde x^2] = \tilde\triangle^\kappa(p^1p^2x^2)$, lying on either side of a common segment $[\tilde p^1\tilde p^2]$. We may suppose



$$\measuredangle^\kappa(p^1\,{}^{p^2}_{x^1}) + \measuredangle^\kappa(p^1\,{}^{p^2}_{x^2}) \leqslant \pi \quad\text{and}\quad \measuredangle^\kappa(p^2\,{}^{p^1}_{x^1}) + \measuredangle^\kappa(p^2\,{}^{p^1}_{x^2}) \leqslant \pi,$$

since otherwise (2+2)-point comparison holds trivially. Then $[\tilde p^1\tilde p^2]$ and $[\tilde x^1\tilde x^2]$ intersect, say at $\tilde q$.

By assumption, there is a geodesic $[p^1p^2]$. Choose $q \in [p^1p^2]$ corresponding to $\tilde q$; that is, $|p^1\,q| = |\tilde p^1\,\tilde q|$. Then

$$|x^1\,x^2| \leqslant |x^1\,q| + |q\,x^2| \leqslant |\tilde x^1\,\tilde q| + |\tilde q\,\tilde x^2| = |\tilde x^1\,\tilde x^2|,$$

where the second inequality follows from (c). Therefore by monotonicity of the function $a \mapsto \measuredangle^\kappa\{a; b, c\}$ (1.1.1c),

$$\measuredangle^\kappa(p^1\,{}^{x^1}_{x^2}) \leqslant \measuredangle[\tilde p^1\,{}^{\tilde x^1}_{\tilde x^2}] = \measuredangle^\kappa(p^1\,{}^{p^2}_{x^1}) + \measuredangle^\kappa(p^1\,{}^{p^2}_{x^2}).$$

□

As a corollary, we display important information from the proof of 9.3.1, namely, monotonicity of the model angle with respect to adjacent sidelengths.

9.3.2. Angle-sidelength monotonicity. *Suppose* $\mathcal{U} \in \mathrm{CAT}[\kappa]$, *and* $p, x, y \in \mathcal{U}$ *have perimeter* $< 2\cdot\varpi^\kappa$. *Then for* $\bar y \in\,]xy]$, *the function*

$$|x\,\bar y| \mapsto \measuredangle^\kappa(x\,{}^{p}_{\bar y})$$

is nondecreasing.

In particular, if $\bar p \in\,]xp]$ *then*

a) the function

$$(|x\,\bar y|, |x\,\bar p|) \mapsto \measuredangle^\kappa(x\,{}^{\bar p}_{\bar y})$$

is nondecreasing in each argument,

b) the angle $\measuredangle[x\,{}^{p}_{y}]$ *exists and*

$$\measuredangle[x\,{}^{p}_{y}] = \inf\left\{\, \measuredangle^\kappa(x\,{}^{\bar p}_{\bar y}) \;\middle|\; \bar p \in\,]xp]\,,\ \bar y \in\,]xy]\,\right\}.$$

9.3.3. Exercise. *Assume $\mathcal{U} \in \mathrm{CAT}[0]$. Show that for any two geodesic paths $\gamma, \sigma\colon [0,1] \to \mathcal{U}$ the function*

$$t \mapsto |\gamma(t)\ \sigma(t)|$$

is convex.

9.3.4. Proposition. *If $\kappa < \mathrm{K}$, then*

$$\mathrm{CBA}[\kappa] \subset \mathrm{CBA}[\mathrm{K}] \quad \textit{and} \quad \mathrm{CAT}[\kappa] \subset \mathrm{CAT}[\mathrm{K}].$$

Moreover

$$\mathrm{CBA}[\kappa] = \bigcap_{\mathrm{K}>\kappa} \mathrm{CBA}[\mathrm{K}] \quad \textit{and} \quad \mathrm{CAT}[\kappa] = \bigcap_{\mathrm{K}>\kappa} \mathrm{CAT}[\mathrm{K}].$$

Proof. The first statement follows from the adjacent-angles comparison (9.3.1*b*) and the monotonicity of the function $\kappa \mapsto \measuredangle^{\kappa}(x\,{}^{y}_{z})$ (1.1.1*d*).

The second statement follows since the function $\kappa \mapsto \measuredangle^{\kappa}(x\,{}^{y}_{z})$ is continuous. □

9.4 Thin triangles

In this section we introduce the notion of thin triangles, use it to give another definition of CAT spaces, and prove inheritance for thin triangles with respect to decomposition. The inheritance lemma will lead to two fundamental constructions: Alexandrov's patchwork globalization (9.8.2) and Reshetnyak gluing (9.9.1).

9.4.1. Definition of κ-thin triangles. *Let $[x^1x^2x^3]$ be a triangle of perimeter $< 2\cdot\varpi^{\kappa}$ in a metric space. Consider its model triangle $[\tilde{x}^1\tilde{x}^2\tilde{x}^3] = \tilde{\triangle}^{\kappa}(x^1x^2x^3)$ and the* natural map $[\tilde{x}^1\tilde{x}^2\tilde{x}^3] \to [x^1x^2x^3]$ *that sends a point $\tilde{z} \in [\tilde{x}^i\tilde{x}^j]$ to the corresponding point $z \in [x^ix^j]$ (i.e. such that $|\tilde{x}^i\,\tilde{z}| = |x^i\,z|$ and therefore $|\tilde{x}^j\,\tilde{z}| = |x^j\,z|$).*

We say the triangle $[x^1x^2x^3]$ is κ-thin *if the natural map $[\tilde{x}^1\tilde{x}^2\tilde{x}^3] \to$ $\to [x^1x^2x^3]$ is short.*

9.4.2. Exercise. *Let $\mathcal{U} \in \mathrm{CAT}[\kappa]$, $[xyz]$ is a triangle in $\mathcal{U}$ and $[\tilde{x}\tilde{y}\tilde{z}]$ is its model triangle in $\mathbb{M}^2[\kappa]$. Prove that the natural map $f\colon [\tilde{x}\tilde{y}\tilde{z}] \to$ $\to [xyz]$ is distance-preserving if and ony if one of the following conditions hold.*

a) $\measuredangle[x\,{}^{y}_{z}] = \measuredangle^{\kappa}(x\,{}^{y}_{z})$,

b) $|x\,w| = |\tilde{x}\,\tilde{w}|$ *for some* $\tilde{w} \in]\tilde{y}\tilde{z}[$ *and* $w = f(\tilde{w})$

c) $|v\,w| = |\tilde{v}\,\tilde{w}|$ *for some* $\tilde{v} \in\,]\tilde{x}\tilde{y}[$, $\tilde{w} \in\,]\tilde{x}\tilde{z}[$ *and* $v = f(\tilde{v})$, $w = f(\tilde{w})$.

9.4.3. Proposition. *Let* $\mathcal{U}$ *be a* ϖ^κ*-geodesic length space. Then* $\mathcal{U} \in \mathrm{CAT}[\kappa]$ *if and only if every triangle of perimeter* $< 2{\cdot}\varpi^\kappa$ *in* $\mathcal{U}$ *is* κ*-thin.*

Proof. "If" is immediate from point-on-side comparison (9.3.1*c*). "Only if" follows from Corollary 9.3.2*a*. □

9.4.4. Corollary. *Suppose* $\mathcal{U} \in \mathrm{CAT}[\kappa]$. *Then any local geodesic in* $\mathcal{U}$ *of length* $< \varpi^\kappa$ *is a minimizing geodesic.*

Proof. Suppose $\gamma\colon [0,\ell] \to \mathcal{U}$ is a local geodesic that is not minimizing, with $\ell < \varpi^\kappa$. Choose a to be the maximal value such that γ is minimizing on $[0,a]$. Further choose $b \in (a, \varpi^\kappa - a)$ so that γ is minimizing on $[a,b]$.

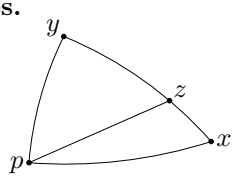
Since triangle $[\gamma(0)\gamma(a)\gamma(b)]$ is κ-thin, then

$$|\gamma(a-\varepsilon)\;\gamma(a+\varepsilon)| < 2{\cdot}\varepsilon$$

for all small $\varepsilon > 0$, a contradiction. □

Now let us formulate the main result of this section. The inheritance lemma states that in any metric space, a triangle is κ-thin if it decomposes into κ-thin triangles. In contrast, $\mathrm{CBB}[\kappa]$ comparisons are not inherited in this way.

9.4.5. Inheritance lemma for thin triangles. *In a metric space, consider a triangle* $[pxy]$ *that* decomposes *into two triangles* $[pxz]$ *and* $[pyz]$*; that is,* $[pxz]$ *and* $[pyz]$ *have common side* $[pz]$*, and the sides* $[xz]$ *and* $[zy]$ *together form the side* $[xy]$ *of* $[pxy]$.

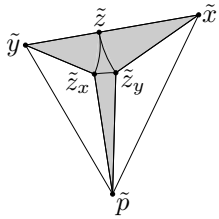


If triangle $[pxy]$ *has perimeter* $< 2{\cdot}\varpi^\kappa$ *and both triangles* $[pxz]$ *and* $[pyz]$ *are* κ*-thin, then triangle* $[pxy]$ *is* κ*-thin.*

We shall need the following model-space lemma, which is part of the proof of [145, Lemma 2].

9.4.6. Lemma. *Let* $[\tilde{p}\tilde{x}\tilde{y}]$ *be a triangle in* $\mathbb{M}^2[\kappa]$ *and* $\tilde{z} \in [\tilde{x}\tilde{y}]$. *Set* $\tilde{D} = \mathrm{Conv}[\tilde{p}\tilde{x}\tilde{y}]$. *Construct points* $\dot{p}, \dot{x}, \dot{z}, \dot{y} \in \mathbb{M}^2[\kappa]$ *such that* $|\dot{p}\,\dot{x}| = |\tilde{p}\,\tilde{x}|$, $|\dot{p}\,\dot{y}| = |\tilde{p}\,\tilde{y}|$, $|\dot{x}\,\dot{z}| = |\tilde{x}\,\tilde{z}|$, $|\dot{y}\,\dot{z}| = |\tilde{y}\,\tilde{z}|$, $|\dot{p}\,\dot{z}| \leqslant |\tilde{p}\,\tilde{z}|$ *and points* $\dot{x}$ *and* $\dot{y}$ *lie on either side of* $[\dot{p}\dot{z}]$. *Set* $\dot{D} = \mathrm{Conv}[\dot{p}\dot{x}\dot{z}] \cup \mathrm{Conv}[\dot{p}\dot{y}\dot{z}]$.

Then there is a short map $F\colon \tilde{D} \to \dot{D}$ *that maps* $\tilde{p}$, $\tilde{x}$, $\tilde{y}$ *and* $\tilde{z}$ *to* $\dot{p}$, $\dot{x}$, $\dot{y}$ *and* $\dot{z}$ *respectively.*



Proof. By Alexandrov's lemma (6.2.1), there are nonoverlapping triangles $[\tilde p\tilde x\tilde z_y] \overset{iso}{=\!=} [\dot p\dot x\dot z]$ and $[\tilde p\tilde y\tilde z_x] \overset{iso}{=\!=} [\dot p\dot y\dot z]$ inside triangle $[\tilde p\tilde x\tilde y]$.

Connect points in each pair $(\tilde z, \tilde z_x)$, $(\tilde z_x, \tilde z_y)$ and $(\tilde z_y, \tilde z)$ with arcs of circles centered at $\tilde y$, $\tilde p$, and $\tilde x$ respectively. Define F as follows.

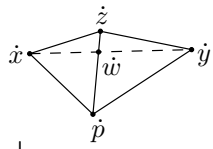
- ◇ Map $\operatorname{Conv}[\tilde p\tilde x\tilde z_y]$ isometrically onto $\operatorname{Conv}[\dot p\dot x\dot y]$; similarly map $\operatorname{Conv}[\tilde p\tilde y\tilde z_x]$ onto $\operatorname{Conv}[\dot p\dot y\dot z]$.
- ◇ If x is in one of the three circular sectors, say at distance r from center of the circle, let $F(x)$ be the point on the corresponding segment $[pz]$, $[xz]$ or $[yz]$ whose distance from the lefthand endpoint of the segment is r.
- ◇ Finally, if x lies in the remaining curvilinear triangle $\tilde z\tilde z_x\tilde z_y$, set $F(x) = z$.

By construction, F satisfies the conditions of the lemma. □

Proof of Inheritance for thin triangles (9.4.5). Construct model triangles $[\dot p\dot x\dot z] = \tilde\triangle^\kappa(pxz)$ and $[\dot p\dot y\dot z] = \tilde\triangle^\kappa(pyz)$ so that $\dot x$ and $\dot y$ lie on opposite sides of $[\dot p\dot z]$.

Suppose

$$\measuredangle^\kappa(z\,{}^p_x) + \measuredangle^\kappa(z\,{}^p_y) < \pi.$$



Then for some point $\dot w \in [\dot p\dot z]$, we have

$$|\dot x\,\dot w| + |\dot w\,\dot y| < |\dot x\,\dot z| + |\dot z\,\dot y| = |x\,y|.$$

Let $w \in [pz]$ correspond to $\dot w$; i.e. $|z\,w| = |\dot z\,\dot w|$. Since $[pxz]$ and $[pyz]$ are κ-thin, we have

$$|x\,w| + |w\,y| < |x\,y|,$$

contradicting the triangle inequality.

Thus

$$\measuredangle^\kappa(z\,{}^p_x) + \measuredangle^\kappa(z\,{}^p_y) \geqslant \pi.$$

By Alexandrov's lemma (6.2.1), this is equivalent to

❶ $$\measuredangle^\kappa(x\,{}^p_z) \leqslant \measuredangle^\kappa(x\,{}^p_y).$$

Let $[\tilde p\tilde x\tilde y] = \tilde\triangle^\kappa(pxy)$ and $\tilde z \in [\tilde x\tilde y]$ correspond to z; i.e. $|x\,z| = |\tilde x\,\tilde z|$. Inequality ❶ is equivalent to $|p\,z| \leqslant |\tilde p\,\tilde z|$. Hence Lemma 9.4.6 applies. Therefore there is a short map F that sends $[\tilde p\tilde x\tilde y]$ to $\dot D = \operatorname{Conv}[\dot p\dot x\dot z] \cup$ $\cup \operatorname{Conv}[\dot p\dot y\dot z]$ in such a way that $\tilde p \mapsto \dot p$, $\tilde x \mapsto \dot x$, $\tilde z \mapsto \dot z$ and $\tilde y \mapsto \dot y$.

By assumption, the natural maps $[\dot p\dot x\dot z] \to [pxz]$ and $[\dot p\dot y\dot z] \to [pyz]$ are short. By composition, the natural map from $[\tilde p\tilde x\tilde y]$ to $[pyz]$ is short, as claimed. □

9.5 Function comparison and development

In this section, we give analytic and geometric ways of viewing the point-on-side comparison (9.3.1c) as a convexity condition.

First we obtain a corresponding differential inequality for the distance function in $\mathcal{U}$. In particular, a geodesic space $\mathcal{U}$ is in CAT[0] if and only if for any $p \in \mathcal{U}$, the function $|p_*|^2 \colon \mathcal{U} \to \mathbb{R}$ is 2-convex; see Section 5.5 for definition.

9.5.1. Theorem. *Suppose $\mathcal{U}$ is a ϖ^κ-geodesic length space. Then the following are equivalent:*

a) $\mathcal{U} \in \mathrm{CAT}[\kappa]$,

b) for any $p \in \mathcal{U}$, the function $f = \mathrm{md}^\kappa \circ |p_*|$ satisfies

$$f'' + \kappa \cdot f \geqslant 1$$

in $\mathrm{B}(p, \varpi^\kappa)$.

Proof. For a geodesic $[xy]$ in $\mathrm{B}(p, \varpi^\kappa)$, consider the model triangle $[\tilde{p}\tilde{x}\tilde{y}] = \tilde{\triangle}^\kappa(pxy)$, and set

$$\tilde{r}(t) = |\tilde{p}\,\mathrm{geod}_{[\tilde{x}\tilde{y}]}(t)|, \qquad r(t) = |p\,\mathrm{geod}_{[xy]}(t)|.$$

Set $\tilde{f} = \mathrm{md}^\kappa \circ \tilde{r}$ and $f = \mathrm{md}^\kappa \circ r$. By Property 1.1.1$a$, we have $\tilde{f}'' = 1 - \kappa \cdot \tilde{f}$. Clearly $\tilde{f}(t)$ and $f(t)$ agree at $t = 0$ and $t = |xy|$. The point-on-side comparison (9.3.1c) is the condition $r(t) \leqslant \tilde{r}(t)$ for all $t \in [0, |xy|]$. Since md^κ is increasing on $[0, \varpi^\kappa)$, $r \leqslant \tilde{r}$ and $f \leqslant \tilde{f}$ are equivalent. Thus the claim follows by Jensen's inequality (5.4.3d). $\square$

Recall that Busemann functions defined in Proposition 6.1.1. The following exercise is analogous to Exercise 8.4.2.

9.5.2. Exercise. *Let $\mathcal{U} \in \mathrm{CBA}[\kappa]$ and $\mathrm{bus}_\gamma \colon \mathcal{U} \to \mathbb{R}$ is the Busemann function for a ray $\gamma \colon [0, \infty) \to \mathcal{L}$.*

a) If $\kappa = 0$ then the Busemann function bus_γ is convex.

b) If $\kappa = -1$ then the function

$$f = \exp \circ \mathrm{bus}_\gamma$$

satisfies

$$f'' - f \geqslant 0$$

9.6 Convex sets

The following is a corollary of Theorem 9.5.1.

9.6.1. Corollary. *Suppose $\mathcal{U} \in \mathrm{CAT}[\kappa]$. Then any ball (close or open) of radius $R < \frac{\varpi^\kappa}{2}$ in $\mathcal{U}$ is convex.*

Moreover, any open ball of radius $\frac{\varpi^\kappa}{2}$ is convex and any closed ball of radius $\frac{\varpi^\kappa}{2}$ is ϖ^κ-convex.

Proof. Suppose $p \in \mathcal{U}$, $R \leqslant \varpi^\kappa/2$ and two points x and y lie in $\overline{\mathrm{B}}[p,R]$ or $\mathrm{B}(p,R)$. By the triangle inequality, if $|xy| < \varpi^\kappa$ then any geodesic $[xy]$ lies in $\mathrm{B}(p,\varpi^\kappa)$.

By function comparison (9.5.1) the geodesic $[xy]$ lies in $\overline{\mathrm{B}}[p,R]$ and $\mathrm{B}(p,R)$ correspondingly.

We proved that any ball (close or open) of radius $R < \frac{\varpi^\kappa}{2}$ is ϖ^κ-convex. This implies convexity unless there is a pair of points in the ball on the distance at least ϖ^κ. By triangle inequality, the later is possible only for the closed ball of radius $\frac{\varpi^\kappa}{2}$. □

9.6.2. Proposition. *Any weakly ϖ^κ-convex set in a $\mathrm{CBA}[\kappa]$ space is ϖ^κ-convex.*

Proof. Follows from the uniqueness of geodesics in $\mathrm{CBA}[\kappa]$ spaces (9.2.1). □

9.6.3. Closest-point projection lemma. *Let $\mathcal{U} \in \mathrm{CBA}[\kappa]$ and $K \subset \mathcal{U}$ be a closed ϖ^κ-convex set. Assume that $|Kp| < \frac{\varpi^\kappa}{2}$ for some point $p \in \mathcal{U}$. Then there is unique point $p^* \in K$ which minimze the distance to p; that is $|p^*p| = |Kp|$.*

Proof. Fix r between $|Kp|$ and $\frac{\varpi^\kappa}{2}$. By function comparison (9.5.1) the function $f = \mathrm{md}^\kappa \circ |p_*|$ is strongly convex in $\overline{\mathrm{B}}[p,r]$.

The lemma follows from Lemma 19.1.4 applied to the subspace $K' = K \cap \overline{\mathrm{B}}[p,r]$ and the restriction $f|K'$. □

9.6.4. Exercise. *Let $\mathcal{U} \in \mathrm{CBA}[0]$ and $K \subset \mathcal{U}$ be a closed convex set. Show that the closest-point projection $\mathcal{U} \to K$ is short.*

9.6.5. Advanced exercise. *Let $\mathcal{U} \in \mathrm{CBA}[1]$ and $K \subset \mathcal{U}$ be a closed π-convex set. Assume $K \subset \overline{\mathrm{B}}(p,\frac{\pi}{2})$ for some point $p \in \mathcal{U}$. Show that there is a short retractionof $\mathcal{U}$ on K.*

9.6.6. Lemma. *Let $\mathcal{U} \in \mathrm{CBA}[\kappa]$ and $K \subset \mathcal{U}$ be a closed ϖ^κ-convex set. Set*

$$f = \mathrm{sn}^\kappa \circ |K_*|.$$

Then

$$f'' + \kappa\cdot f \geqslant 0$$

holds in $\mathrm{B}(K, \frac{\varpi^\kappa}{2})$.

Proof. It is sufficient to show that Jensen's inequality (5.4.3d) holds on a sufficiently short geodsic $[pq]$ in $\mathrm{B}(K, \frac{\varpi^\kappa}{2})$. Namely, since

$$|K\,p|, |K\,q| < \tfrac{\varpi^\kappa}{2},$$

we can assume that

❶ $$|p\,q| + |K\,p| + |K\,q| < \varpi^\kappa.$$

Namely, for each $x \in [pq]$, we need to find a value $h(x) \in \mathbb{R}$ such that for any x we have

$$h(x) \leqslant f(x), \qquad h(p) = f(p), \qquad h(q) = f(q),$$

and

❷ $$h'' + \kappa\cdot h \geqslant 0$$

along $[pq]$.

Denote by p^* and q^* the closest-point projections of p and q on K; they are provided by lemma 9.6.3. From ❶ and triangle inequality, we get

$$|p^*\,q^*| < \varpi^\kappa.$$

Since K is ϖ^κ-convex, $K \supset [pq]$; in particular

$$|K\,x| \leqslant |[p^*q^*]\;x|$$

for any $x \in \mathcal{U}$.

Consider a majorization F for quadraliteral $[pp^*q^*q]$. By Majoriztion theorem 9.12.4 and Proposition 9.12.2, the figure F is a solid convex model quadraliteral $[\tilde p\tilde p^*\tilde q^*\tilde q]$ in $\mathbb{M}^2[\kappa]$ such that

$$|\tilde p\,\tilde p^*|_{\mathbb{M}^2[\kappa]} = |p\,p^*|_{\mathcal{U}} \qquad |\tilde p\,\tilde q|_{\mathbb{M}^2[\kappa]} = |p\,q|_{\mathcal{U}}$$
$$|\tilde q\,\tilde q^*|_{\mathbb{M}^2[\kappa]} = |q\,q^*|_{\mathcal{U}} \qquad |\tilde p^*\,\tilde q^*|_{\mathbb{M}^2[\kappa]} = |p^*\,q^*|_{\mathcal{U}}$$

Given $x \in [pq]$, denote by $\tilde x$ the corresponding point on $[\tilde p\tilde q]$. According to Majorization theorem,

$$|[pq]\,x|_{\mathcal{U}} \leqslant |[\tilde p\tilde q]\,\tilde x|_{\mathbb{M}^2[\kappa]}$$

for any x. Set

$$h(x) = \mathrm{sn}^\kappa|[\tilde p\tilde q]\,\tilde x|_{\mathbb{M}^2[\kappa]}.$$

By straghtforward calculations ❷ hods and hence the staement follows. □

9.6.7. Corollary. *Let $\mathcal{U} \in \mathrm{CBA}[\kappa]$ and $K \subset \mathcal{U}$ be a closed locally convex set. Then there is an open set $\Omega \supset K$ such that the function*

$$f = \mathrm{sn}^\kappa \circ |K\,*|$$

satisfies

$$f'' + \kappa\cdot f \geqslant 0$$

in Ω.

Proof. Fix $p \in K$. By Corollary 9.6.1, all $\overline{\mathrm{B}}[p, r]$ is convex for all small $r > 0$.

Since K is locally convex there is $r_p > 0$ such that the intesection $K' = K \cap \mathrm{B}(p, r_p)$ is convex.

Note that

$$|K\,x| = |K'\,x|$$

for any $x \in \mathrm{B}(p, \frac{r_p}{2})$. Therefore the statement holds for

$$\Omega = \bigcup_{p\in K} \mathrm{B}(p, \tfrac{r_p}{2}).$$

□

9.6.8. Theorem. *Assume $\mathcal{U} \in \mathrm{CBA}[\kappa]$ and $K \subset \mathcal{U}$ is a closed connected locally convex set. Assume $|x\,y| < \varpi^\kappa$ for any $x, y \in K$. Then K is convex.*

In particular, if $\kappa \leqslant 0$ then any closed connected locally convex set in a $\mathrm{CBA}[\kappa]$ space is convex.

The following proof is inspired by the answer of Sergei Ivanov to the question of Nathan Reading, see [76].

Proof. Since K is locally convex, it is locally path connected. Since K is connected and locally path connected it is path connected.

Fix two points $x, y \in K$. Let us connect x to y by a path $\alpha\colon [0,1] \to K$. Since $|x\,\alpha(s)| < \varpi^\kappa$ for any s, Theorem 9.2.1 implies that the geodesic $[x\alpha(s)]$ uniquely defined and depends continuously on s.

Let $\Omega \supset K$ be the open set proveded by Corollary 9.6.7. If $[xy] = [x\alpha(1)]$ does not completely lie in K then there is a value $s \in [0,1]$ such that $[x\alpha(s)]$ lies in Ω, but does not completely lie in K. By Corollary 9.6.7, the finction $f = \mathrm{sn}^\kappa \circ |K\,*|_{\mathcal{U}}$ satisfies differential inequality

❸ $$f'' + \kappa\cdot f \geqslant 0$$

along $[x\alpha(s)]$.

Since

$$|x\,\alpha(s)| < \varpi^\kappa, \qquad\qquad f(x) = f(\alpha(s)) = 0,$$

the barrier inequality (5.4.3b) implies that $f(z) \leqslant 0$ for $z \in [x\alpha(s)]$; that is $[x\alpha(s)] \subset K$, a contradiction. □

9.7 Development

Geometrically, the development construction (8.4.4) translates distance comparison into a local convexity statement for subsets of $\mathbb{M}^2[\kappa]$. Recall that a curve in $\mathbb{M}^2[\kappa]$ is *(locally) concave with respect to* p if (locally) its supergraph with respect to p is a convex subset of $\mathbb{M}^2[\kappa]$ (8.4.5).

9.7.1. Development criterion. *For a ϖ^κ-geodesic length space $\mathcal{U}$, the following statements hold:*

a) For any $p \in \mathcal{U}$ and any geodesic $\gamma\colon [0,T] \to \mathrm{B}(p,\varpi^\kappa)$, suppose the κ-development $\tilde\gamma$ in $\mathbb{M}^2[\kappa]$ of γ with respect to p is locally concave. Then $\mathcal{U} \in \mathrm{CAT}[\kappa]$.

b) If $\mathcal{U} \in \mathrm{CAT}[\kappa]$, then for any $p \in \mathcal{U}$ and any geodesic $\gamma : [0,T] \to \mathcal{U}$ for which the triangle $[p\gamma(0)\gamma(T)]$ has perimeter $< 2\cdot\varpi^\kappa$, the κ-development $\tilde\gamma$ in $\mathbb{M}^2[\kappa]$ of γ with respect to p is concave.

Proof. (a). Set $\gamma = \mathrm{geod}_{[xy]}$ and $T = |x\,y|$. Let $\tilde\gamma\colon [0,T] \to \mathbb{M}^2[\kappa]$ be the concave κ-development based at $\tilde p$ of γ with respect to p. Let us show that the function

❶ $$t \mapsto \tilde\measuredangle^\kappa(x\,{}^p_{\gamma(t)})$$

is nondecreasing.

For a partition $0 = t^0 < t^1 < \ldots < t^n = T$, set

$$\tilde y^i = \tilde\gamma(t^i) \;\text{ and }\; \tau^i = |\tilde y^0\,\tilde y^1| + |\tilde y^1\,\tilde y^2| + \ldots + |\tilde y^{i-1}\,\tilde y^i|.$$

Since $\tilde\gamma$ is locally concave, for a sufficiently fine partition the broken geodesic $\tilde y^0\tilde y^1\ldots\tilde y^n$ is locally convex with respect to $\tilde p$. Alexandrov's lemma (6.2.1), applied inductively to pairs of triangles $\tilde\triangle^\kappa\{\tau^{i-1}, |p\,\tilde y^0|, |p\,\tilde y^{i-1}|\}$ and $\tilde\triangle^\kappa\{|\tilde y^{i-1}\,\tilde y^i|, |p\,\tilde y^{i-1}|, |p\,\tilde y^i|\}$, shows that the sequence $\tilde\measuredangle^\kappa\{|p\,\tilde y^i|; |p\,\tilde y^0|, \tau^i\}$ is nondecreasing.

Taking finer partitions and passing to the limit,

$$\max_i\{|\tau^i - t^i|\} \to 0,$$

we get ❶ and the point-on-side comparison (9.3.1c) follows.

(b). Consider a partition $0 = t^0 < t^1 < \ldots < t^n = T$, and set $x^i = \gamma(t^i)$. Construct a chain of model triangles $[\tilde{p}\tilde{x}^{i-1}\tilde{x}^i] = \tilde{\triangle}^\kappa(px^{i-1}x^i)$ with the direction of $[\tilde{p}\tilde{x}^i]$ turning counterclockwise as i grows. By angle comparison (9.3.1d),

❷ $$\measuredangle[\tilde{x}^i\,{}^{\tilde{x}^{i-1}}_{\tilde{p}}] + \measuredangle[\tilde{x}^i\,{}^{\tilde{x}^{i+1}}_{\tilde{p}}] \geqslant \pi.$$

Since γ is a geodesic,

❸ $$\operatorname{length}\gamma = \sum_{i=1}^{n} |x^{i-1}\,x^i| \leqslant |px^0| + |px^n|.$$

By repeated application of Alexandrov's Lemma (6.2.1), and inequality ❸,

$$\sum_{i=1}^{n} \measuredangle[\tilde{p}\,{}^{\tilde{x}^{i-1}}_{\tilde{x}^i}] \leqslant \tilde{\measuredangle}^\kappa(p\,{}^{x^0}_{x^n}) \leqslant \pi.$$

Then by ❷, the broken geodesic $\tilde{p}\tilde{x}^0\tilde{x}^1\ldots\tilde{x}^n$ is concave with respect to $\tilde{p}$.

By Lemma **??** applied to the polygons $[\tilde{x}^0\tilde{x}^1\ldots\tilde{x}^n]$, the broken geodesics $\tilde{x}^0\tilde{x}^1\ldots\tilde{x}^n$ approach the development of γ with respect to p. Hence the result. □

9.8 Patchwork globalization

The following definition gives an infinitesimal notion of upper curvature bound at a point of a space.

9.8.1. Definition. *Let $\mathcal{U}$ be a metric space and $p \in \mathcal{U}$. We say that $\mathcal{U}$* has curvature $\leqslant \kappa$ at p *(briefly,* $\operatorname{curv}_p\mathcal{U} \leqslant \kappa$*) if for any* $\mathrm{K} > \kappa$ *there is $r > 0$ such that (2+2)-point κ-comparison holds for any quadruple in* $\mathrm{B}(p,r)$.

We say that a space $\mathcal{U}$ has curvature $\leqslant \kappa$ (briefly, $\operatorname{curv}\mathcal{U} \leqslant \kappa$*) if* $\operatorname{curv}_p\mathcal{U} \leqslant \kappa$ *for any $p \in \mathcal{U}$.*

If $\mathcal{U} \in \mathrm{CBA}[\kappa]$ then $\operatorname{curv}\mathcal{U} \leqslant \kappa$; this fact follows from Proposition 9.3.4. The converse, does not hold for general complete length space.

For example, $\mathbb{S}^1$ is locally isometric to $\mathbb{R}$, and so $\operatorname{curv}\mathbb{S}^1 \leqslant 0$, but it is easy to find a quadruple of points in $\mathbb{S}^1$ which violates (2+2)-point comparison.

Nevertheless, an infinitesimal bound implies a local bound. That is, if $\mathcal{U}$ is a complete length space and $\operatorname{curv}\mathcal{U} \leqslant \kappa$, then for any p there is $r > 0$ such that $\mathrm{B}[p,r] \in \mathrm{CBA}[\kappa]$ (Corollary 9.8.3).

The following theorem, which essentially is [16, Satz 9], gives a global condition on geodesics that is necessary and sufficient for a space $\mathcal{U}$ with $\operatorname{curv}\mathcal{U} \leqslant \kappa$ to satisfy $\mathcal{U} \in \mathrm{CAT}[\kappa]$. The proof use a thin-triangle decompositions, and the inheritance lemma (9.4.5).

9.8.2. Patchwork globalization theorem. *For a length space $\mathcal{U}$, the following two statements are equivalent:*

a) $\mathcal{U} \in \mathrm{CAT}[\kappa]$.

b) $\operatorname{curv}\mathcal{U} \leqslant \kappa$*; moreover, pairs of points in $\mathcal{U}$ at distance $< \varpi^\kappa$ are joined by unique geodesics, and these geodesics depend continuously on their endpoint pairs.*

Note that the implication $(a)\Rightarrow(b)$ is immediate, being the statement of Theorem 9.2.1.

9.8.3. Corollary. *Let $\mathcal{U}$ be a complete length space and $\Omega \subset \mathcal{U}$ be an open subset. Assume $\operatorname{curv}_p\mathcal{U} \leqslant \kappa$ for any $p \in \Omega$. Then for any point $p \in \Omega$ there is $R > 0$ such that $\overline{\mathrm{B}}[p,R]$ is a convex subset of $\mathcal{U}$ and $\overline{\mathrm{B}}[p,R] \in \mathrm{CBA}[\kappa]$.*

Proof. Fix $\mathrm{K} > \kappa$. According to Definition 9.8.1, there is $R > 0$ such that (2+2)-point κ-comparison holds in $\mathrm{B}(p,R)$.

We may assume that $\mathrm{B}(p,R) \subset \Omega$ and $R < \varpi^{\mathrm{K}}$. The same argument as in the proof of theorem on uniqueness of geodesics (9.2.1) shows that any two points in $\overline{\mathrm{B}}(p,\frac{R}{2})$ can be joint by unique geodesic which depends continuously on the end points.

The same argument as in the proof of Corollary 9.6.1 shows that $\overline{\mathrm{B}}[p,\frac{R}{2}]$ is a convex set. Then$(b)\Rightarrow(a)$ of Patchwork globalization implies $\mathrm{B}(p,R) \in \mathrm{CAT}[\kappa]$. $\square$

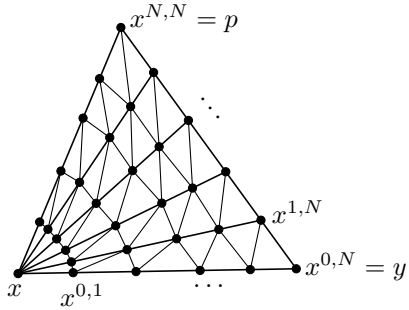
The proof of Patchwork globalization uses the following construction:

9.8.4. Definition (Line-of-sight map). *Let p be a point and α be a curve of finite length in an length space $\mathcal{X}$. Let $\bar\alpha : [0,1] \to \mathcal{U}$ be the constant-speed parameterization of α. If $\gamma_t\colon [0,1] \to \mathcal{U}$ is a geodesic from p to $\bar\alpha(t)$, we say*

$$[0,1]\times[0,1] \to \mathcal{U}\colon (t,s) \mapsto \gamma_t(s)$$

is a line-of-sight map from p to α .

Proof of Patchwork globalization theorem (9.8.2). As was already noted, it only remains to prove $(b)\Rightarrow(a)$.



Fix arbitrary $\mathrm{K} > \kappa$. Let $[pxy]$ be a triangle of perimeter $< 2{\cdot}\varpi^\kappa$ in $\mathcal{U}$. According to propositions 9.4.3 and 9.3.4, it is sufficient to show the triangle $[pxy]$ is K-thin.

Since pairs of points at distance $< \varpi^\kappa$ are joined by unique geodesics and these geodesics depend continuously on their endpoint pairs, there is a unique and continuous line-of-sight map (9.8.4) for $[xy]$ from p.

For a partition

$$0 = t^0 < t^1 < \ldots < t^N = 1,$$

set $x^{i,j} = \gamma_{t^i}(t^j)$. Since the line-of-sight map is continuous, by definition 9.8.1 we may assume each triangle $[x^{i,j}x^{i,j+1}x^{i+1,j+1}]$ and $[x^{i,j}x^{i+1,j}x^{i+1,j+1}]$ is K-thin (see Proposition 9.4.3).

Now we show that the K-thin property propagates to $[pxy]$ by repeated application of the inheritance lemma (9.4.5):

- ◇ First, for fixed i, sequentially applying the lemma shows that the triangles $[xx^{i,1}x^{i+1,2}]$, $[xx^{i,2}x^{i+1,2}]$, $[xx^{i,2}x^{i+1,3}]$, and so on are K-thin.

In particular, for each i, the long triangle $[xx^{i,N}x^{i+1,N}]$ is K-thin.

- ◇ Applying the lemma again shows that the triangles $[xx^{0,N}x^{2,N}]$, $[xx^{0,N}x^{3,N}]$, and so on are K-thin.

In particular, $[pxy] = [px^{0,N}x^{N,N}]$ is K-thin. □

9.9 Reshetnyak's gluing theorem

The following theorem was proved by Reshetnyak [144], assuming $\mathcal{U}_1$, $\mathcal{U}_2$ are proper.

9.9.1. Reshetnyak gluing. *Suppose $\mathcal{U}_1$ and $\mathcal{U}_2$ are ϖ^κ-geodesic length spaces with isometric complete ϖ^κ-convex sets $A_i \subset \mathcal{U}_i$. Let $\iota\colon A_1 \to A_2$ be an isometry. Let $\mathcal{W} = \mathcal{U}_1 \sqcup_\iota \mathcal{U}_2$; i.e., $\mathcal{W}$ is gluing of $\mathcal{U}_1$ and $\mathcal{U}_2$ along ι;see Section 2.5.*

Then

a) Both canonical mappings $\jmath_i\colon \mathcal{U}_i \to \mathcal{W}$ are isometric and the images $\jmath_i(\mathcal{U}_i)$ are ϖ^κ-convex subsets in $\mathcal{W}$.

b) If $\mathcal{U}_1, \mathcal{U}_2 \in \mathrm{CBA}[\kappa]$ then $\mathcal{W} \in \mathrm{CBA}[\kappa]$.

Proof. Part (a) follows directly from ϖ^κ-convexity of A_i.

(b). According to (a), we can identify $\mathcal{U}_i$ with its image $\jmath_i(\mathcal{U}_i)$ in $\mathcal{W}$; this way both subsets $A_i \subset \mathcal{U}_i$ will be identified and denoted further by A. Note that $A = \mathcal{U}_1 \cap \mathcal{U}_2 \subset \mathcal{W}$, therefore A is also ϖ^κ-convex set in $\mathcal{W}$.

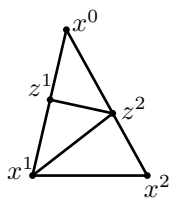
Further note that completeness of $\mathcal{U}_1$ and $\mathcal{U}_2$ implies completeness of $\mathcal{W}$. Thus part (b) can be reformulated the following way:

9.9.2. Reformulation of 9.9.1*b*. *Let $\mathcal{W}$ be a complete length space which has two closed ϖ^κ-convex sets $\mathcal{U}_1, \mathcal{U}_2 \subset \mathcal{W}$ such that $\mathcal{U}_1 \cup \mathcal{U}_1 = \mathcal{W}$ and $\mathcal{U}_1, \mathcal{U}_2 \in \mathrm{CBA}[\kappa]$. Then $\mathcal{W} \in \mathrm{CBA}[\kappa]$.*

Set $A = \mathcal{U}_1 \cap \mathcal{U}_1$. First we prove the following:

❶ *If our space $\mathcal{W}$ happened to be ϖ^κ-geodesic then $\mathcal{W} \in \mathrm{CBA}[\kappa]$*

Proof of the claim. According to 9.4.3, it is sufficient to show that any triangle $[x^0x^1x^2]$ of perimeter $< 2\cdot\varpi^\kappa$ in $\mathcal{W}$ is κ-thin. This is obviousely true if all three points x^0, x^1, x^2 lie in one of $\mathcal{U}_i$. Thus, without loss of generality, we may assume that $x^0 \in \mathcal{U}_1$ and $x^1, x^2 \in \in \mathcal{U}_2$.



Choose points $z^1, z^2 \in A$ which lie correspondingly on the sides $[x^0x^1], [x^0x^2]$. Note that all distances between any pair of points from x^0, x^1, x^2, z^1, z^2 are less than ϖ^κ. Therefore

- ⋄ triangle $[x^0z^1z^2]$ lies in $\mathcal{U}_1$,
- ⋄ both triangles $[x^1z^1z^2]$ and $[x^1z^2x^2]$ lie in $\mathcal{U}_2$.

In particular each triangle $[x^0z^1z^2]$, $[x^1z^1z^2]$ and $[x^1z^2x^2]$ is κ-thin.

Applying the inheritance lemma for thin triangles (9.4.5) twice we get that $[x^0x^1z^2]$ and consequently $[x^0x^1x^2]$ is κ-thin. △

Now we come back to the general case; that is, $\mathcal{W}$ is not necessury ϖ^κ-geodesic. According to Corollary 4.3.2, the ultrapower $\mathcal{W}^\omega$ is geodesic. It is easy to see that

- ⋄ $\mathcal{U}_i^\omega$ form weak ϖ^κ-convex sets of $\mathcal{W}^\omega$,
- ⋄ $A^\omega = \mathcal{U}_1^\omega \cap \mathcal{U}_2^\omega$ is also a weakly ϖ^κ-convex set.

From Proposition 9.1.5, $\mathcal{U}_i^\omega \in \mathrm{CBA}[\kappa]$. According to Corollary **??**, we get that A^ω is ϖ^κ-convex in both $\mathcal{U}_i$ and consequently both $\mathcal{U}_i^\omega$ are ϖ^κ-convex subsets in $\mathcal{W}^\omega$.

From Claim ❶, we get $\mathcal{W}^\omega \in \mathrm{CBA}[\kappa]$. Applying the Proposition 9.1.5 once more, we get $\mathcal{W} \in \mathrm{CBA}[\kappa]$. □

9.9.3. Exercise. *Let Q be the nonconvex subset of the plane bounded by two rays γ_1 and γ_2 with common starting point and angle α between*

them. Assume $\mathcal{U} \in \mathrm{CBA}[0]$ *and* γ_1', γ_2' *be two rays in* $\mathcal{U}$ *starting point and angle* α *between them. Show that the space glued from* Q *and* $\mathcal{U}$ *along the corresponding rays is a* $\mathrm{CBA}[0]$ *space.*

9.9.4. Exercise. *Assume* $\mathcal{U} \in \mathrm{CBA}[0]$ *and* $A \subset \mathcal{U}$ *is a closed subset. Assume that the doubling of* $\mathcal{U}$ *in* A *is* $\mathrm{CBA}[0]$*. Show that* A *is a convex set of* $\mathcal{U}$*.*

9.9.5. Exercise. *Let* $\mathcal{U} \in \mathrm{CBA}[0]$*. Assume* $\tilde{\mathcal{U}} \to \mathcal{U}$ *is a metric covering branching along a geodesic. Show that* $\tilde{\mathcal{U}} \in \mathrm{CBA}[0]$*.*

More generally, assume $A \subset \mathcal{U}$ *is a closed convex subset and* $f\colon \mathcal{X} \to \mathcal{U}\backslash A$ *is a metric cover. Denote by* $\bar{\mathcal{X}}$ *the completion of* $\mathcal{X}$ *and* $\bar{f}\colon \mathcal{X} \to \mathcal{U}$ *the continuos extension of* f*. Let* $\tilde{\mathcal{U}}$ *be the space glued from* $\bar{\mathcal{X}}$ *and* A *by identifying* x *and* $\bar{f}(x)$ *if* $\bar{f}(x) \in A$*. Show that* $\tilde{\mathcal{U}} \in \mathrm{CBA}[0]$*.*

9.10 Space of geodesics

In this section we prove a "no-conjugate-point" theorem for spaces with upper curvature bounds, and derive from it a number of statements about local geodesics. These statements will be used in the proof of the Hadamard–Cartan theorem (9.13.1) and Lifting globalization theorem (9.11.1), in much the same way as the exponential map is used in Riemannian geometry.

9.10.1. Proposition. *Let* $\mathcal{U}$ *be a metric space with* $\operatorname{curv}\mathcal{U} \leqslant \kappa$*. Let* $\gamma_n\colon [0,1] \to \mathcal{U}$ *be a sequence of local geodesic paths converging to a path* $\gamma_\infty\colon [0,1] \to \mathcal{U}$*. Then* γ_∞ *is a local geodesic path. Moreover*

$$\operatorname{length}\gamma_n \to \operatorname{length}\gamma_\infty$$

as $n \to \infty$*.*

Proof. Fix $t \in [0,1]$. By Corollary 9.8.3, we may choose R satisfying $0 < R < \varpi^{\mathrm{K}}$, and such that the ball $\mathcal{B} = \mathrm{B}(\gamma_\infty(t), R)$ is a convex subset of $\mathcal{U}$ and forms a $\mathrm{CAT}[\kappa]$ space.

A local geodesic segment with length $< R/2$, and intersecting $\mathrm{B}(\gamma_\infty(t), R/2)$, cannot leave $\mathcal{B}$ and hence is minimizing by Corollary 9.4.4. In particular, for all sufficiently large n, any subsegment of γ_n through $\gamma_n(t)$ with length $< R/2$ is a geodesic.

Since $\mathcal{B}$ is a $\mathrm{CAT}[\kappa]$ space, geodesic segments in $\mathcal{B}$ depend uniquely and continuously on their endpoint pairs by Theorem 9.2.1. Thus there is a subinterval $\mathbb{I}$ of $[0,1]$, which contains a neighborhood of t in $[0,1]$

and such that $\gamma_n|\mathbb{I}$ is minimizing for all large n. It follows that $\gamma_\infty|\mathbb{I}$ is a geodesic, and therefore γ_∞ is a local geodesic. $\square$

By analogy with Riemannian geometry, the main statement of the following theorem could be restated as: *In a space of curvature $\leqslant \kappa$, two points cannot be conjugate along a local geodesic of length $< \varpi^\kappa$.*

The following theorem was proved in [7].

9.10.2. No-conjugate-point theorem. *Suppose $\mathcal{U}$ is a locally complete metric space with $\operatorname{curv}\mathcal{U} \leqslant \kappa$. Let $\gamma\colon [0,1] \to \mathcal{U}$ be a local geodesic path with length $< \varpi^\kappa$. Then for some neighborhoods $\Omega^0 \ni \gamma(0)$ and $\Omega^1 \ni \gamma(1)$ there is a unique continuous map $\Omega^0 \times \Omega^1 \times [0,1] \to \mathcal{U}$*

$$(x,y,t) \mapsto \gamma_{xy}(t)$$

such that $\gamma_{xy}\colon [0,1] \to \mathcal{U}$ is a local geodesic path with $\gamma_{xy}(0) = x$ and $\gamma_{xy}(1) = y$ for each $(x,y) \in \Omega^0 \times \Omega^1$, and the family γ_{xy} contains γ. Moreover:

a) The map

$$(x,y,t) \mapsto \gamma_{xy}(t)\colon \Omega^0 \times \Omega^1 \times [0,1] \to \mathcal{U}$$

is $\pounds$-Lipschitz for $\pounds = \max\left\{ \frac{\operatorname{sn}^\kappa r}{\operatorname{sn}^\kappa \ell} \;\middle|\; 0 \leqslant r \leqslant \ell \right\}$.

b) There is a space $\mathcal{N} \in \mathrm{CBA}[\kappa]$, an open set $\hat\Omega \subset \mathcal{N}$, and a locally isometric immersion

$$\Phi\colon \hat\Omega \looparrowright \mathcal{U},$$

such that Ω^0 and Ω^1 are isometric images under Φ of open sets $\hat\Omega^0$ and $\hat\Omega^1$ in $\hat\Omega$, and for $\hat x \in \hat\Omega^0$, $\hat y \in \hat\Omega^1$, the geodesic $\hat\gamma_{\hat x\hat y}$ in $\mathcal{N}$ satisfies $\Phi \circ \hat\gamma_{\hat x\hat y} = \gamma_{xy}$.

The following lemma was suggested to us by Alexander Lytchak. The proof proceeds by piecing together $\mathrm{CAT}[\kappa]$ neighborhoods of points on a curve to construct a new $\mathrm{CBA}[\kappa]$ space. Exercise 9.13.8 is inspired by the original idea of the proof of No-conjugate-point theorem 9.10.2 given in [7].

9.10.3. Patchwork along a curve. *Let $\mathcal{U}$ be a locally complete metric space with $\operatorname{curv}\mathcal{U} \leqslant \kappa$, and $\alpha\colon [a,b] \to \mathcal{U}$ be a curve.*

Then there is a space $\mathcal{N} \in \mathrm{CAT}[\kappa]$ with an open set $\hat\Omega \subset \mathcal{N}$, a curve $\hat\alpha\colon [0,1] \to \hat\Omega$, and an open locally isometric immersion $\Phi\colon \hat\Omega \looparrowright \mathcal{U}$ such that $\Phi \circ \hat\alpha = \alpha$.

Moreover if α is simple, then one can assume in addition that Φ is an open embedding; thus $\hat\Omega$ is locally isometric to a neighborhood of $\Omega = \Phi(\hat\Omega)$ of α.

Proof. According to Corollary 9.8.3, for any $t \in [a,b]$ there is $r(t) > 0$ such that the closed ball $\overline{\mathrm{B}}[\alpha(t), r(t)]$ is a convex set that forms a $\mathrm{CBA}[\kappa]$ space.

Choose balls $\mathcal{B}^i = \overline{\mathrm{B}}[\alpha(t^i), r(t^i)]$ for some partition $a = t^0 < t^1 < \ldots < t^n = b$ in such a way that $\operatorname{Int}\mathcal{B}^i \supset \alpha([t^{i-1}, t^i])$ for all $i > 0$.

Consider the disjoint union $\bigsqcup_i \mathcal{B}^i = \{\, (i,x) \mid x \in \mathcal{B}^i \,\}$ with the minimal equivalence relation $\sim$ such that $(i,x) \sim (i-1,x)$ for all $i > 0$. Let $\mathcal{N}$ be the space obtained by gluing the $\mathcal{B}^i$ along $\sim$. Note that $A^i = \mathcal{B}^i \cap \mathcal{B}^{i-1}$ is convex in $\mathcal{B}^i$ and in $\mathcal{B}^{i-1}$. Applying the Reshetnyak gluing theorem (9.9.1) n times, we conclude that $\mathcal{N} \in \mathrm{CBA}[\kappa]$.

For $t \in [t^{i-1}, t^i]$, let $\hat\alpha(t)$ be the equivalence class of $(i, \alpha(t))$ in $\mathcal{N}$. Let $\hat\Omega$ be the ε-neighborhood of $\hat\alpha$ in $\mathcal{N}$, where $\varepsilon > 0$ is chosen so that $\mathrm{B}(\alpha(t), \varepsilon) \subset \mathcal{B}^i$ for all $t \in [t^{i-1}, t^i]$.

Define $\Phi\colon \hat\Omega \to \mathcal{U}$ by sending the equivalence class of (i,x) to x. It is straightforward to check that $\Phi\colon \mathcal{N} \to \mathcal{U}$, $\hat\alpha\colon [0,1] \to \mathcal{N}$ and $\hat\Omega \subset \mathcal{N}$ satisfy the conclusion of the main part of the lemma.

To prove the final statement in the lemma, we only have to choose $\varepsilon > 0$ so that in addition $|\alpha(\tau)\ \alpha(\tau')| > 2\cdot\varepsilon$ if $\tau \leqslant t^{i-1}$ and $t^i \leqslant \tau'$ for some i. □

Proof of No-conjugate-point theorem 9.10.2. Apply patchwork along γ (9.10.3). □

The No-conjugate-point theorem (9.10.2) allows us to move a local geodesic path so that its endpoints follow given trajectories. This process might terminate in two cases: when the length of the local geodesic path approaches ϖ^κ, or when the homotopy leaves a complete subset of the space. The following corollary formulates these cases more precisely.

9.10.4. Corollary. *Let $\mathcal{U}$ be a locally complete metric space with $\operatorname{curv}\mathcal{U} \leqslant \kappa$. Suppose $\gamma\colon [0,1] \to \mathcal{U}$ is a local geodesic path with length $< \varpi^\kappa$. Let $\alpha^i\colon [0,1] \to \mathcal{U}$, for $i = 0,1$, be paths starting at $\gamma(0)$ and $\gamma(1)$ respectively.*

Then there is a uniquely determined pair consisting of an interval $\mathbb{I}$ satisfying $0 \in \mathbb{I} \subset [0,1]$, and a continuous family of local geodesic paths $\gamma_t\colon [0,1] \to \mathcal{U}$ for $t \in \mathbb{I}$, such that

a) $\gamma_0 = \gamma$, $\gamma_t(0) = \alpha^0(t)$, $\gamma_t(1) = \alpha^1(t)$, and γ_t has length $< \varpi^\kappa$,

b) if $\mathbb{I} \neq [0,1]$, then $\mathbb{I} = [0,a)$, where either γ_t converges uniformly to a local geodesic γ_a of length ϖ^κ, or for some fixed $s \in [0,1]$ the curve $\gamma_t(s) : [0,a) \to \mathcal{U}$ is a Lipschitz curve with no limit as $t \to a-$.

Proof. Uniqueness follows from Theorem 9.10.2.

Let $\mathbb{I}$ be the maximal interval for which there is a family γ_t satisfying condition (a). By Theorem 9.10.2, such an interval exists and is open in $[0,1]$. Suppose $\mathbb{I} \neq [0,1]$. Then $\mathbb{I} = [0,a)$ for some $0 < a \leqslant 1$. It suffices to show that $\mathbb{I}$ satisfies condition (b).

For each fixed $s \in [0,1]$, define the curve $\alpha_s : [0,a) \to \mathcal{U}$ by $\alpha_s(t) = \gamma_t(s)$. By Theorem 9.10.2, α_s is $\pounds$-Lipschitz for some $\pounds$.

If α_s for some value of s does not converge as $t \to a-$, then condiion (b) holds. If each α_s converges as $t \to a-$, then γ_t converges as $t \to$ $\to a-$, say to γ_a. By Proposition 9.10.1, γ_a is a local geodesic path and

$$\operatorname{length}\gamma_t \to \operatorname{length}\gamma_a \leqslant \varpi^\kappa.$$

By maximality of $\mathbb{I}$, $\operatorname{length}\gamma_a = \varpi^\kappa$ and so condition (b) again holds. □

9.10.5. Corollary. *Suppose $\mathcal{U}$ is a complete length space with* $\operatorname{curv}\mathcal{U} \leqslant$ $\leqslant \kappa$ *and $\alpha\colon [0,1] \to \mathcal{U}$ is a path of length $\ell < \varpi^\kappa$. Then for all $t \in [0,1]$,*

a) there is unique homotopy of local geodesic paths $\gamma_t\colon [0,1] \to$ $\to \mathcal{U}$ such that $\gamma_0(t) = \gamma_t(0) = \alpha(0)$ and $\gamma_t(1) = \alpha(t)$,

b)
$$\operatorname{length}\gamma_t \leqslant \operatorname{length}(\alpha|[0,t]),$$
and equality holds for given t if and only if the restriction $\alpha|[0,t]$ is a reparametrization of γ_t.

Proof. Apply Corollary 9.10.4, setting $\alpha^0(t) = \alpha(0)$ for all $t \in [0,1]$, and $\alpha^1 = \alpha$. Since $\mathcal{U}$ is complete, there is an interval $\mathbb{I}$ such that statement (a) holds for all $t \in \mathbb{I}$, and either $\mathbb{I} = [0,1]$ or $\mathbb{I} = [0,a)$ where γ_t converges uniformly to a local geodesic γ_a of length ϖ^κ.

Theorem 9.10.2b implies that the values of t for which condition (b) holds form an open subset of $\mathbb{I}$ containing 0; clearly this subset is also closed in $\mathbb{I}$. Therefore (b) holds on all of $\mathbb{I}$.

Sinse $\ell < \varpi^\kappa$, $\mathbb{I} = [0,1]$ and the result follows. □

The No-conjugate-point theorem 9.10.2 immediately gives an extension of the first variation formula (**??**) from geodesics in CAT$[\kappa]$ spaces to local geodesics in spaces of curvature $\leqslant \kappa$.

9.10.6. Corollary. *Suppose $\mathcal{U}$ is a locally complete metric space with* $\operatorname{curv}\mathcal{U} \leqslant \kappa$. *For any geodesic $[px]$ in $\mathcal{U}$ and local geodesic $\gamma\colon [0,1] \to \mathcal{U}$ from p to q of length $< \varpi^\kappa$, let $\gamma_t\colon [0,1] \to \mathcal{U}$ be a continuous family of local geodesic paths with $\gamma_0 = \gamma$, $\gamma_t(0) = \operatorname{geod}_{[px]}(t)$ and $\gamma_t(1) = q$. Then*

$$\operatorname{length}(\gamma_t) = \operatorname{length}(\gamma) - t\cdot\cos\measuredangle(\gamma,[px]) + o(t).$$

9.11 Lifting globalization

The Hadamard–Cartan theorem (9.13.1) states that the universal metric cover of a complete space with curvature $\leqslant 0$ is a $\mathrm{CAT}[0]$ space. The following Lifting globalization theorem gives an appropriate generalization of the above statement to arbitrary curvature bounds.

9.11.1. Lifting globalization theorem. *Suppose $\mathcal{U}$ is a complete length space with $\operatorname{curv}\mathcal{U} \leqslant \kappa$ and $p \in \mathcal{U}$. Then there is a space $\mathcal{B} \in \mathrm{CBA}[\kappa]$, where $\mathcal{B} = \overline{\mathrm{B}}[\hat{p}, \varpi^\kappa/2]$ for some $\hat{p} \in \mathcal{B}$, and a locally isometric map $\Phi\colon \mathcal{B} \to \mathcal{U}$ with $\Phi(\hat{p}) = p$ and the following lifting property: for any path $\alpha\colon [0,1] \to \mathcal{U}$ with $\alpha(0) = p$ and $\operatorname{length}\alpha < \varpi^\kappa/2$, there is a unique path $\hat{\alpha}\colon [0,1] \to \mathcal{B}$ such that $\hat{\alpha}(0) = \hat{p}$ and $\Phi \circ \hat{\alpha} \equiv \alpha$.*

9.11.2. Corollary. *Suppose $\mathcal{U}$ is a complete length space with $\operatorname{curv}\mathcal{U} \leqslant \kappa$. Then for any $p \in \mathcal{U}$ there is $\rho_p > 0$ such that $\overline{\mathrm{B}}[p, \rho_p] \in \mathrm{CBA}[\kappa]$.*

Moreover, we can assume that $\rho_p < \frac{\varpi^\kappa}{2}$ for any p and the function $p \mapsto \rho_p$ is 1-Lipschitz.

Proof. Assume $\Phi\colon \mathcal{B} \to \mathcal{U}$ and $\hat{p} \in \mathcal{B}$ are provided by Lifting globalization theorem (9.11.1).

Since Φ is local isometry, we can choose $r > 0$ so that the restriction of Φ to $\overline{\mathrm{B}}[\hat{p}, r]$ is distance preserving. By Lifting globalization, the image $\Phi(\overline{\mathrm{B}}[\hat{p}, r])$ coincides with the ball $\overline{\mathrm{B}}[p, r]$. This proves the first part of theorem.

To prove the second part, lat us choose ρ_p to be maximal value $\leqslant \frac{\varpi^\kappa}{2}$ such that $\overline{\mathrm{B}}[p, \rho_p] \in \mathrm{CBA}[\kappa]$. By Corollary 9.6.1, the ball

$$\overline{\mathrm{B}}[q, \rho_p - |pq|]$$

is convex in $\overline{\mathrm{B}}[p, \rho_p]$. Therefore

$$\overline{\mathrm{B}}[q, \rho_p - |pq|] \in \mathrm{CBA}[\kappa]$$

for any $q \in \mathrm{B}(p, \rho_p)$. In particular $\rho_q \geqslant \rho_p - |pq|$ for any $p, q \in \mathcal{U}$. Hence the second statement follows. □

The proof of the lifting globalization theorem relies heavily on the properties of the space of local geodesic paths discussed in Section 9.10. The following lemma from [8] is a key step in the proof.

9.11.3. Radial lemma. *Let $\mathcal{U}$ be an length space with $\operatorname{curv}\mathcal{U} \leqslant \kappa$, and suppose $p \in \mathcal{U}$, $R \leqslant \varpi^\kappa/2$. Assume the ball $\overline{\mathrm{B}}[p, \bar{R}]$ is complete for all $\bar{R} < R$, and there is a unique geodesic path, $\operatorname{path}_{[px]}$, from p to any point $x \in \mathrm{B}(p, R)$, which depends continuously on x. Then $\mathrm{B}(p, R) \in \mathrm{CAT}[\kappa]$.*

Proof. Without loss of generality, we may assume $\mathcal{U} = \mathrm{B}(p, \varpi^\kappa)$.

Set $f = \mathrm{md}^\kappa \circ |p*|$. Let us show that

❶ $$f'' + \kappa\cdot f \geqslant 1.$$

Fix $z \in \mathcal{U}$, and apply Theorem 9.10.2 for the unique geodesic path γ from p to z. We use the notations Ω^i, γ_{xy}, $\mathcal{N}$, $\hat{x}$, $\hat{y}$, as in that theorem. In particular, $z \in \Omega^1$.

By assumption, $\gamma_{py} = \mathrm{path}_{[py]}$ for any $y \in \Omega^1$. Consequently, $f(y) = \mathrm{md}^\kappa|\hat{p}\,\hat{y}|_{\mathcal{N}}$. From the function comparison (9.5.1) applied in $\mathcal{N}$, we have $f'' + \kappa\cdot f \geqslant 1$ in Ω^1.

Fix $\bar{R} < R$, and consider the complete closed ball $\overline{\mathrm{B}}[p, \bar{R}] \subset \mathcal{U}$.

❷ $\overline{\mathrm{B}}[p, \bar{R}]$ *forms a convex set in* $\mathcal{U}$.

The proof of this claim takes most of the rest of proof of the theorem.

Choose arbitrary $x, z \in \overline{\mathrm{B}}[p, \bar{R}]$. First note that ❶ implies:

❸ *If* $\gamma\colon [0,1] \to \mathcal{U}$ *is a local geodesic path from* x *to* z *and* $\operatorname{length}\gamma < \varpi^\kappa$, *then* $\operatorname{length}\gamma < 2\cdot\bar{R}$ *and* γ *lies completely in* $\overline{\mathrm{B}}[p, \bar{R}]$.

Note that $|x\,z| < \varpi^\kappa$. Thus, to prove Claim ❷, it is sufficient to show that there is a geodesic path of $\mathcal{U}$ from x to z. By Corollary 9.10.5,

❹ *Given a path* $\alpha\colon [0,1] \to \mathcal{U}$ *from* x *to* z *with* $\operatorname{length}\alpha < \varpi^\kappa$, *there is a local geodesic path* γ *from* x *to* z *such that*

$$\operatorname{length}\gamma \leqslant \operatorname{length}\alpha.$$

Further, let us prove the following.

❺ *There is unique local geodesic path* γ_{xz} *in* $\overline{\mathrm{B}}[p, \bar{R}]$ *from* x *to* z.

Denote by Δ_{xz} the set of all local geodesic paths of $\overline{\mathrm{B}}[p, \bar{R}]$ from x to z. By Corollary 9.10.4, there is a bijection $\Delta_{xz} \to \Delta_{pp}$. According to ❶, Δ_{pp} contains only the constant path. Claim ❺ follows.

Note that claims ❸, ❹ and ❺ imply that γ_{xz} is minimizing; hence Claim ❷.

Further, Claim ❸ and the No-conjugate-point theorem (9.10.2) together imply that the map $(x, z) \mapsto \gamma_{xz}$ is continuous.

Therefore by the Patchwork globalization theorem (9.8.2), $\overline{\mathrm{B}}[p, \bar{R}] \in$ $\in \mathrm{CAT}[\kappa]$.

Since

$$\mathrm{B}(p, R) = \bigcup_{\bar{R}<R} \overline{\mathrm{B}}[p, \bar{R}],$$

then $\mathrm{B}(p,R)$ is convex in $\mathcal{U}$ and (2+2)-point comparison holds for any quadruple in $\mathrm{B}(p,R)$. Therefore $\mathrm{B}(p,\varpi^\kappa/2)\in\mathrm{CAT}[\kappa]$. $\square$

In the following proof, we first construct a space G_p out of local geodesic paths that start at p. The space G_p comes with a marked point $\hat p$ and the right-hand-endpoint map $\Phi\colon G_p\to\mathcal{U}$. In the Riemannian case, one can think of the map Φ as $\exp_p$, and the space G_p as the ball of radius ϖ^κ in the tangent space at p, equipped with the metric pulled back by $\exp_p$.

We are going to set $\mathcal{B}=\mathrm{B}[\hat p,\varpi^\kappa/2]\subset G_p$, and use the Radial lemma (9.11.3) to prove $\mathcal{B}\in\mathrm{CAT}[\kappa]$.

Proof of Lifting globalization theorem 9.11.1. Suppose $\hat\gamma$ is a homotopy of local geodesic paths that start at p. Thus the map

$$\hat\gamma\colon(t,\tau)\mapsto\hat\gamma_t(\tau)\colon[0,1]\times[0,1]\to\mathcal{U}$$

is continuous and for each t,

- $\hat\gamma_t(0)=p$,
- $\hat\gamma_t\colon[0,1]\to\mathcal{U}$ forms a local geodesic path in $\mathcal{U}$.

Let $\vartheta(\hat\gamma)$ denote the length of the path $t\mapsto\hat\gamma_t(1)$.

Let G_p be the set of all local geodesic paths with length $<\varpi^\kappa$ in $\mathcal{U}$ that start at p. Let $\hat p\in G_p$ be the constant path $\hat p(t)\equiv p$. Given $\alpha,\beta\in G_p$ define

$$|\alpha\,\beta|_{G_p}=\inf\{\vartheta(\hat\gamma)\},$$

with infimum taken along all homotopies $\hat\gamma\colon[0,1]\times[0,1]\to\mathcal{U}$ such that $\hat\gamma_0=\alpha$, $\hat\gamma_1=\beta$ and $\hat\gamma_t\in G_p$ for all $t\in[0,1]$.

From Theorem 9.10.2, we have $|\alpha\,\beta|_{G_p}>0$ for distinct α and β; that is,

❻ $|{*}{*}|_{G_p}$ *is a metric on* G_p.

Further, again from Theorem 9.10.2, we have

❼ *The map*

$$\Phi\colon\xi\mapsto\xi(1)\colon G_p\to\mathcal{U}$$

is a local isometry. In particular, $\operatorname{curv}G_p\leqslant\kappa$.

Let $\alpha\colon[0,1]\to\mathcal{U}$ be a path with $\operatorname{length}\alpha<\varpi^\kappa$ and $\alpha(0)=p$. The homotopy constructed in Corollary 9.10.5 can be regarded as a path in G_p, say $\hat\alpha\colon[0,1]\to G_p$, such that $\hat\alpha(0)=\hat p$ and $\Phi\circ\hat\alpha=\alpha$; in particular $\hat\alpha(1)=\alpha$. By ❼,

$$\operatorname{length}\hat\alpha=\operatorname{length}\alpha;$$

moreover, it follows that α is a local geodesic path of $\mathcal{U}$ if and only if $\hat\alpha$ is a local geodesic path of G_p.

Further, from Corollary 9.10.5, for any $\xi \in G_p$ and path $\hat\alpha\colon [0,1] \to$ $\to G_p$ from $\hat p$ to ξ, we have

$$\begin{aligned}\operatorname{length}\hat\alpha = \operatorname{length}\Phi\circ\hat\alpha \geqslant\\ \geqslant \operatorname{length}\xi = \operatorname{length}\hat\xi\end{aligned}$$

where equality holds only if $\hat\alpha$ is a reparametrization of $\hat\xi$. In particular,

❽ $$|\hat p\,\xi|_{G_p} = \operatorname{length}\xi$$

and $\hat\xi\colon [0,1] \to G_p$ is the unique geodesic path from $\hat p$ to ξ. Clearly, the map $\xi \mapsto \hat\xi$ is continuous.

From ❽ and Proposition 9.10.1, it follows

❾ *For any $\bar R < \varpi^\kappa/2$, the closed ball $\overline{\mathrm{B}}[\hat p, \bar R]$ in G_p is complete.*

Take $\mathcal{B} = \mathrm{B}(\hat p, \varpi^\kappa/2)$ and Φ constructed above. According to the Radial lemma (9.11.3), $\mathcal{B} \in \mathrm{CAT}[\kappa]$. The remaining statements are already proved. □

9.12 Reshetnyak Majorization

9.12.1. Definition. *Let $\mathcal{X}$ be a metric space, $\tilde\alpha$ be a simple closed curve of finite length in $\mathbb{M}^2[\kappa]$, and $D \subset \mathbb{M}^2[\kappa]$ be a closed region bounded by $\tilde\alpha$. A length-nonincreasing map $F\colon D \to \mathcal{X}$ is called* majorizing *if it is length-preserving on $\tilde\alpha$.*

In this case, we say that D majorizes *the curve $\alpha = F\circ\tilde\alpha$ under the map F.*

The following proposition is an immediate consequence of the definition.

9.12.2. Proposition. *Let α be a closed curve in a metric space $\mathcal{X}$. Suppose $D \subset \mathbb{M}^2[\kappa]$ majorizes α under $F\colon D \to \mathcal{X}$. Then any geodesic subarc of α is the image under F of a subarc of $\partial_{\mathbb{M}^2[\kappa]}D$ that is geodesic in the length-metric of D.*

In particular, if D is convex then the corresponding subarc is a geodesic in $\mathbb{M}^2[\kappa]$.

Proof. For a geodesic subarc $\gamma\colon [a,b] \to \mathcal{X}$ of $\alpha = F\circ\tilde\alpha$, write $\tilde\gamma = (F|\partial D)^{-1}\circ\gamma$, $s = \operatorname{length}\gamma$, $\tilde s = \operatorname{length}\tilde\gamma$, $r = |\gamma(a)\ \gamma(b)|_{\mathcal{X}}$, and $\tilde r = |\tilde\gamma(a)\ \tilde\gamma(b)|_D$. Then

$$\tilde r \geqslant r = s = \tilde s \geqslant \tilde r.$$

Therefore $\tilde{s} = \tilde{r}$. $\square$

9.12.3. Corollary. *Let $[pxy]$ be a triangle of perimeter $< 2\cdot\varpi^\kappa$ in a metric space $\mathcal{X}$. Assume a convex region $D \subset \mathbb{M}^2[\kappa]$ majorizes $[pxy]$. Then $D = \operatorname{Conv}[\tilde{p}\tilde{x}\tilde{y}]$ for a model triangle $[\tilde{p}\tilde{x}\tilde{y}] = \tilde{\triangle}^\kappa(pxy)$, and the majorizing map sends $\tilde{p}$, $\tilde{x}$ and $\tilde{y}$ respectively to p, x and y.*

Now we come to the main theorem of this section.

9.12.4. Majorization theorem. *Any closed curve α with length smaller than $2\cdot\varpi^\kappa$ in $\mathcal{U} \in \mathrm{CAT}[\kappa]$ is majorized by a convex region in $\mathbb{M}^2[\kappa]$.*

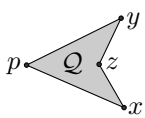
This theorem is proved by Reshetnyak in [145]; our proof also uses a trick from [24].

The theorem is not trivial even when α is a triangle, say $[pxy]$. In this case, by Proposition 9.12.3, the majorizing convex region has to be isometric to $\operatorname{Conv}[\tilde{p}\tilde{x}\tilde{y}]$, where $[\tilde{p}\tilde{x}\tilde{y}] = \tilde{\triangle}^\kappa(pxy)$.

In our proof of the theorem, we find a majorizing map for $[pxy]$ whose image W is the image of the line-of-sight map (definition 9.8.4) for $[xy]$ from p. The construction of the correct parameterization is given in lemmas 9.12.5 and 9.12.6. This map is used to construct a majorizing map for a polygon in $\mathcal{U}$, and passing to a limit we get the general case.

Finding such a parameterization $F\colon \operatorname{Conv}[\tilde{p}\tilde{x}\tilde{y}] \to W$ is not as simple as one might think. In particular, the map $F\colon \operatorname{Conv}[\tilde{p}\tilde{x}\tilde{y}] \to$ $\to W : \tilde{\gamma}_t(s) \mapsto \gamma_t(s)$ that sends the point with parameter (t, s) under the line-of-sight map for $[\tilde{x}\tilde{y}]$ from $\tilde{p}$, to the point with the same parameter under the line-of-sight map for $[xy]$ from p, is not majorizing in general. The following example illustrates this point

Example. Let $\mathcal{Q}$ be a bounded region in $\mathbb{E}^2$ formed by two congruent triangles, where the bounding quadrangle $[pxzy]$ is non-convex at z (as in the picture). Equip $\mathcal{Q}$ with the length-metric. Then $\mathcal{Q} \in \mathrm{CBA}[0]$ by Reshetnyak gluing (9.9.1). For triangle $[pxy]_{\mathcal{Q}}$ in $\mathcal{Q}$ and its model triangle $[\tilde{p}\tilde{x}\tilde{y}]$ in $\mathbb{E}^2$, we have $|\tilde{x}\tilde{y}| = |xy|_{\mathcal{Q}} = |xz| + |zy|$. Then the map F defined by matching line-of-sight parameters satisfies $F(\tilde{x}) = x$ and $|xF(\tilde{w})| > |\tilde{x}\tilde{w}|$ if $\tilde{w}$ is near the midpoint $\tilde{z}$ of $[\tilde{x}\tilde{y}]$ and lies on $[\tilde{p}\tilde{z}]$. Indeed, by the first variation formula (8.6.6), for $\varepsilon = 1 - s$ we have $|\tilde{x}\tilde{w}| = |\tilde{x}\tilde{\gamma}_{\frac{1}{2}}(s)| = |xz| + o(\varepsilon)$ and $|xF(\tilde{w})| = |x\gamma_{\frac{1}{2}}(s)| = |xz| - \varepsilon\cdot\cos\measuredangle[z\,{}^{p}_{x}] + o(\varepsilon)$. Thus F is not majorizing.



In the following proofs, $x^1 \dots x^n$ $(n \geqslant 3)$ denotes a broken geodesic with vertices $x^1, \dots, x^n$, and $[x^1 \dots x^n]$ denotes the corresponding

(closed) polygon. For a subset R of the ambient metric space, we denote by $[x^0 \dots x^n]_R$ a polygon in the length metric of R.

Our first lemma gives a model space construction based on repeated application of Lemma 9.4.6.

9.12.5. Lemma. *In $\mathbb{M}^2[\kappa]$, let β be a curve from x to y which is concave with respect to p. Let D be the subgraph of β with respect to p. Assume*

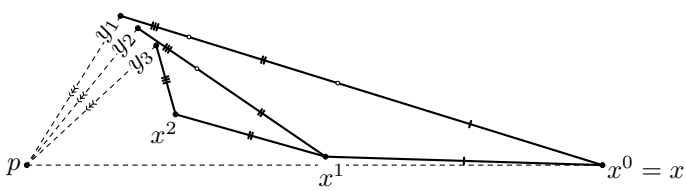
$$\operatorname{length}\beta + |px| + |py| < 2\cdot\varpi^\kappa.$$

a) Then β forms a geodesic $[xy]_D$ in D and therefore β, $[px]$ and $[py]$ form a triangle $[pxy]_D$ in the length-metric of D.

b) Let $[\tilde p\tilde x\tilde y]$ be the model triangle for triangle $[pxy]_D$. Then there is a short map $G\colon \operatorname{Conv}[\tilde p\tilde x\tilde y] \to D$ such that $\tilde p \mapsto p$, $\tilde x \mapsto x$, $\tilde y \mapsto y$, and G is length-preserving on each side of $[\tilde p\tilde x\tilde y]$. In particular, $\operatorname{Conv}[\tilde p\tilde x\tilde y]$ majorizes triangle $[pxy]_D$ in D under G.

Proof. We prove the lemma for a broken geodesic β; the general case then follows by approximation. Namely, since β is concave, it can be approximated by broken geodesics that are concave with respect to p, with their lengths converging to $\operatorname{length}\beta$. Applying Lemma **??**, we can pass to a limit to obtain the needed map G.

Suppose $\beta = x^0x^1\dots x^n$ is a broken geodesic with $x^0 = x$ and $x^n = y$. Consider a sequence of broken geodesics $\beta_i = x^0x^1\dots x^{i-1}y_i$ such that $|py_i| = |py|$ and β_i has same length as β; that is, $|x^{i-1}\,y_i| = |x^{i-1}\,x^i| + |x^i\,x^{i+1}| + \dots + |x^{n-1}\,x^n|$.



Clearly $\beta_n = \beta$. Sequentially applying Alexandrov's lemma (6.2.1) shows that each of the broken geodesics $\beta_{n-1}, \beta_{n-2}, \dots, \beta_1$ is concave with respect to p. Let D_i to be subgraph of β_i with respect to p. Applying Lemma 9.4.6 gives a short map $G_i\colon D_i \to D_{i+1}$ that maps $y_i \mapsto y_{i+1}$ and does not move p and x (in fact, G_i is the identity everywhere except on $\operatorname{Conv}[px^{i-1}y_i]$). Thus the composition $G_{n-1} \circ \circ \dots \circ G_1\colon D_1 \to D_n$ is a short map.

The result follows since $D_1 \overset{iso}{=\!=} \operatorname{Conv}[\tilde p\tilde x\tilde y]$. $\square$

9.12.6. Lemma. *In $\mathcal{U} \in \operatorname{CAT}[\kappa]$, let $[pxy]$ be a triangle in $\mathcal{U}$ of perimeter $< 2\cdot\varpi^\kappa$. In $\mathbb{M}^2[\kappa]$, let $\tilde\gamma$ be the κ-development of $[xy]$ with*

respect to p, where $\tilde\gamma$ has basepoint $\tilde p$ and subgraph D. Consider the map $H\colon D\to\mathcal{U}$ that sends the point with parameter (t,s) under the line-of-sight map for $\tilde\gamma$ from $\tilde p$, to the point with the same parameter under the line-of-sight map f for $[xy]$ from p. Then H is length-nonincreasing. In particular, D majorizes triangle $[pxy]$.

Proof. Let $\gamma=\operatorname{geod}_{[xy]}$ and $T=|xy|$. As in the proof of the development criterion (9.7.1), take a partition $0=t^0<t^1<\dots$ $\dots<t^n=T$, and set $x^i=\gamma(t^i)$. Construct a chain of model triangles $[\tilde p\tilde x^{i-1}\tilde x^i]=\tilde\triangle^\kappa(px^{i-1}x^i)$, with $\tilde x^0=\tilde x$ and the direction of $[\tilde p\tilde x^i]$ turning counterclockwise as i grows. Let D_n be the subgraph with respect to $\tilde p$ of the broken geodesic $\tilde x^0\dots\tilde x^n$.

Let δ_n be the maximum radius of a circle inscribed in any of the triangles $[\tilde p\tilde x^{i-1}\tilde x^i]$.

Now we construct a map $H_n\colon D_n\to\mathcal{U}$ that increases distances by at most $2\cdot\delta_n$.

Suppose $x\in D_n$. Then x lies on or inside some triangle $[\tilde p\tilde x^{i-1}\tilde x^i]$. Define $H_n(x)$ by first mapping x to a nearest point on $[\tilde p\tilde x^{i-1}\tilde x^i]$ (choosing one if there are several), followed by the natural map to the triangle $[px^{i-1}x^i]$.

Since triangles in $\mathcal{U}$ are κ-thin (9.4.3), the restriction of H_n to each triangle $[\tilde p\tilde x^{i-1}\tilde x^i]$ is short. Then the triangle inequality implies that the restriction of H_n to

$$U_n=\bigcup_{1\leqslant i\leqslant n}[\tilde p\tilde x^{i-1}\tilde x^i]$$

is short with respect to the length-metric on D_n. Since nearest-point projection from D_n to U_n increases the D_n-distance between two points by at most $2\cdot\delta_n$, the map H_n also increases the D_n-distance by at most $2\cdot\delta_n$.

Consider $y_n\in D_n$ with $y_n\to y\in D$ and $z_n\in D_n$ with $z_n\to$ $\to z\in D$. Since $\delta_n\to 0$ under increasingly finer partitions, and geodesics in $\mathcal{U}$ vary continuously with their endpoints (9.8.2), we have $H_n(x_n)\to H(x)$ and $H_n(y_n)\to H(y)$. Since

$$|H_n(x_n)\ H_n(y_n)|\leqslant|x_n\,y_n|_{D_n}+2\cdot\delta_n,$$

where the lefthand side converges to $|H(x)\ H(y)|$ and the righthand side converges to $|xy|_D$, it follows that H is short. □

Proof of majorization theorem (9.12.4). We begin by proving the theorem in case α is polygonal.

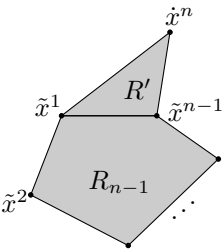
First suppose α is a triangle, say $[pxy]$. By assumption, the perimeter of $[pxy]$ is $<2\cdot\varpi^\kappa$.

Let $\tilde\gamma$ be the κ-development of $[xy]$ with respect to p, where $\tilde\gamma$ has basepoint $\tilde p$ and subgraph D. By the development criterion (9.7.1), $\tilde\gamma$ is concave. By Lemma 9.12.5, there is a short map $G\colon \operatorname{Conv}\tilde\triangle^\kappa(pxy) \to \to D$. Further, by Lemma 9.12.6, D majorizes $[pxy]$ under a majorizing map $H\colon D\to\mathcal{U}$. Clearly $H\circ G$ is a majorizing map for $[pxy]$.

Now we claim that any closed n-gon $[x^1x^2\dots x^n]$ of length $< 2\cdot\varpi^\kappa$ in a CBA$[\kappa]$ space is majorized by a convex polygonal region $R_n = \operatorname{Conv}[\tilde x^1\tilde x^2\dots\tilde x^n]$ in $\mathbb{M}^2[\kappa]$, under a map F_n such that $F_n\colon \tilde x^i\mapsto x^i$ for each i.

Assume the statement is true for $(n-1)$-gons, $n\geqslant 4$. Then $[x^1x^2\dots x^{n-1}]$ is majorized by a convex polygonal region

$$R_{n-1} = \operatorname{Conv}(\tilde x^1,\tilde x^2,\dots,\tilde x^{n-1}),$$



in $\mathbb{M}^2[\kappa]$ under a map F_{n-1} satisfying $F_{n-1}(\tilde x^i) = x^i$ for all i. Take $\dot x^n\in\mathbb{M}^2[\kappa]$ such that $[\tilde x^1\tilde x^{n-1}\dot x^n] = \tilde\triangle^\kappa(x^1x^{n-1}x^n)$ and this triangle lies on the other side of $[\tilde x^1\tilde x^{n-1}]$ from R_{n-1}. Let $\dot R = \operatorname{Conv}[\tilde x^1\tilde x^{n-1}\dot x^n]$, and $\dot F\colon\dot R\to\mathcal{U}$ be a majorizing map for $[x^1x^{n-1}x^n]$ as provided above.

Set $R = R_{n-1}\cup\dot R$, where R carries its length-metric. Since F_n and F agree on $[\tilde x^1\tilde x^{n-1}]$, we may define $F\colon R\to\mathcal{U}$ by

$$F(x) = \begin{cases} F_{n-1}(x), & x\in R_{n-1},\\ \dot F(x), & x\in\dot R.\end{cases}$$

Then F is length-nonincreasing, and is a majorizing map for $[x^1x^2\dots\dots x^n]$ (as in Definition 9.12.1).

If R is a convex subset of $\mathbb{M}^2[\kappa]$, we are done.

If R is not convex, the total internal angle of R at $\tilde x^1$ or $\tilde x^{n-1}$ or both is $>\pi$. By relabeling we may suppose the $\measuredangle\tilde x^{n-1}>\pi$. The region R is obtained by glueing R_{n-1} to $\dot R$ by $[x^1x^{n-1}]$. Thus, by Reshetnyak gluing (9.9.1), R in its length-metric is a CBA$[\kappa]$-space. Moreover $[\tilde x^{n-2}\tilde x^{n-1}]\cup[\tilde x^{n-1}\dot x^n]$ is a geodesic of R. Thus $[\tilde x^1,\tilde x^2,\dots\dots,\tilde x^{n-1}\dot x^n]_R$ is a closed $(n-1)$-gon in R, to which the induction hypothesis applies. The resulting short map from a convex region in $\mathbb{M}^2[\kappa]$ to R, followed by F, is the desired majorizing map.

Note that in fact we proved the following:

❶ *Let F_{n-1} be a majorizing map for the polygon $[x^1x^2\dots x^{n-1}]$, and $\dot F$ be a majorizing map for the triangle $[x^1x^{n-1}x^n]$. Then there is a majorizing map F_n for the polygon $[x^1x^2\dots x^n]$ such that*

$$\operatorname{Im}F_{n+1} = \operatorname{Im}F_n\cup\operatorname{Im}\dot F.$$

We now use this claim to prove the theorem for general curves.

Assume $\alpha\colon [0,\ell] \to \mathcal{U}$ is an arbitrary closed curve with natural parameter. Choose a sequence of partitions $0 = t_n^0 < t_n^2 < \ldots < t_n^n = \ell$ so that:

- ◇ The set $\{t_{n+1}^i\}_{i=0}^{n+1}$ is obtained from the set $\{t_n^i\}_{i=0}^n$ by adding one element.
- ◇ For some sequence $\varepsilon_n \to 0+$, we have $t_n^i - t_n^{i-1} < \varepsilon_n$ for all i.

Inscribe in α a sequence of polygons P_n with vertices $\alpha(t_n^i)$. Apply the claim above, to get a sequence of majorizing maps $F_n\colon R_n \to \mathcal{U}$. Note that for all $m > n$

- ◇ $\operatorname{Im} F_m$ lies in an ε_n-neighborhood of $\operatorname{Im} F_n$
- ◇ $\operatorname{Im} F_m \setminus \operatorname{Im} F_n$ lies in an ε_n-neighborhood of α.

It follows that the set

$$K = \alpha \cup \left(\bigcup_n \operatorname{Im} F_n\right)$$

is compact. Therefore the sequence (F_n) has a partial limit as $n \to \infty$; say F. Clearly F is a majorizing map for α. □

For $n = 1$, the following lemma states that in a CAT[κ] space, a sharp triangle comparison implies the presence of an isometric copy of the convex hull of the model triangle. This rigidity statement was proved in [16]. The analogue for CBB[κ]-spaces fails (see Exercise 10.7.3).

The following exercise is the rigidity case of the Majorization theorem.

9.12.7. Exercise. *Let $\mathcal{U} \in \mathrm{CAT}[\kappa]$ and $\alpha\colon [0,\ell] \to \mathcal{U}$ be a closed curve with arclength parametrization. Assume that $\ell < 2\cdot\varpi^\kappa$ and there is a closed convex curve $\tilde\alpha\colon [0,\ell] \to \mathbb{M}^2[\kappa]$ such that*

$$|\alpha(t_0)\ \alpha(t_1)|_{\mathcal{U}} = |\tilde\alpha(t_0)\ \tilde\alpha(t_1)|_{\mathbb{M}^2[\kappa]}$$

for any t_0 and t_1. Then there is a distance preserving map $F\colon \operatorname{Conv}\tilde\alpha \to \mathcal{U}$ such that $F\colon \tilde\alpha(t) \mapsto \alpha(t)$ for any t.

9.12.8. Exercise. *In $\mathcal{U} \in \mathrm{CAT}[\kappa]$, show that if a closed curve α of length $< 2\cdot\varpi^\kappa$ is not a triangle, then exactly one of these statements holds:*

a) α is majorized by non-isometric convex regions in $\mathbb{M}^2[\kappa]$;

b) the majorizing map for α is distance preserving.

9.12.9. Arm lemma. *Let $\mathcal{U} \in \mathrm{CAT}[\kappa]$, and $P = [x^0 x^1 \dots x^{n+1}]$ be a polygon of length $< 2\cdot\varpi^\kappa$ in $\mathcal{U}$. Suppose $\tilde P = [\tilde x^0 \tilde x^1 \dots \tilde x^{n+1}]$ is a convex polygon in $\mathbb{M}^2[\kappa]$ such that*

❷ $$|\tilde x^i\,\tilde x^{i\pm1}|_{\mathbb{M}^2[\kappa]} = |x^i\,x^{i\pm1}|_{\mathcal{U}} \quad \text{and} \quad \measuredangle[x^i\,{}^{x^{i-1}}_{x^{i+1}}] \geqslant \measuredangle[\tilde x^i\,{}^{\tilde x^{i-1}}_{\tilde x^{i+1}}]$$

for $1 \leqslant i \leqslant n$. Then

a)
$$|\tilde x^0\,\tilde x^{n+1}|_{\mathbb{M}^2[\kappa]} \leqslant |x^0\,x^{n+1}|_{\mathcal{U}}.$$

b) Equality holds in (a) if and only if the map $\tilde x^i \mapsto x^i$ can be extended to a distance-preserving map of $\mathrm{Conv}(\tilde x^0, \tilde x^1 \dots \dots \tilde x^{n+1})$ onto $\mathrm{Conv}(x^0, x^1 \dots x^{n+1})$.

Proof. a). By majorization (9.12.4), P is majorized by a convex region $\tilde D$ in $\mathbb{M}^2[\kappa]$. By Proposition 9.12.2 and the definition of angle, $\tilde D$ is bounded by a convex polygon $\tilde P_R = [\,y^0 \tilde y^1 \dots \tilde y^{n+1}]$ that satisfies

$$|\tilde y^i\,\tilde y^{i\pm1}|_{\mathbb{M}^2[\kappa]} = |x^i\,x^{i\pm1}|_{\mathcal{U}}, \qquad |\tilde y^0\,\tilde y^{n+1}|_{\mathbb{M}^2[\kappa]} = |x^0\,x^{n+1}|_{\mathcal{U}},$$
$$\measuredangle[\tilde y^i\,{}^{\tilde y^{i-1}}_{\tilde y^{i+1}}] \geqslant \measuredangle[x^i\,{}^{x^{i-1}}_{x^{i+1}}] \geqslant \measuredangle[\tilde x^i\,{}^{\tilde x^{i-1}}_{\tilde x^{i+1}}],$$

for $1 \leqslant i \leqslant n$, where the final inequality is by ❷.

We may increase the angle at $\tilde x_i$, $1 \leqslant i \leqslant n$, of the broken geodesic $\tilde x^0 \tilde x^1 \dots \tilde x^{n+1}$ to the angle at $\tilde y^i$ of $\tilde y^0 \tilde y^1 \dots \tilde y^{n+1}$ by moving $\tilde x^0 \tilde x^1 \dots \tilde x^i$ rigidly in $\mathbb{M}^2[\kappa]$ while fixing $\tilde x^i \tilde x^{i+1} \dots \tilde x^{n+1}$. By monotonicity (1.1.1$c$) for hinges in $\mathbb{M}^2[\kappa]$, $|\tilde x^0\,\tilde x^{n+1}|$ is no greater than the distance between the endpoints of the new broken geodesic under this move.

Applying this procedure to each vertex in turn yields a convex polygon isometric to $\tilde P_R$. It follows that $|\tilde x^0\,\tilde x^{n+1}| \leqslant |\tilde y^0\,\tilde y^{n+1}|$. Since $|\tilde y^0\,\tilde y^{n+1}| = |x^0\,x^{n+1}|$, $a)$ follows.

b). Suppose equality holds in $a)$. Then angles at the jth vertex of $\tilde P$, P and $\tilde P_R$ are equal for $1 \leqslant j \leqslant n$, and we may take $\tilde P = \tilde P_R$.

Let $F\colon \tilde D \to \mathcal{U}$ be the majorizing map for P, where $\tilde D$ is the convex region bounded by $\tilde P$, and $F|\tilde P$ is length-preserving.

❸ *Let $\tilde x, \tilde y, \tilde z$ be three vertices of $\tilde P$, and x, y, z be the corresponding vertices of P. If $|\tilde x\,\tilde y| = |x\,y|$, $|\tilde y\,\tilde z| = |x\,z|$ and $\measuredangle[\tilde y\,{}^{\tilde x}_{\tilde z}] = \measuredangle[y\,{}^{x}_{z}]$, then $F|\,\mathrm{Conv}(\tilde x, \tilde y, \tilde z)$ is distance-preserving.*

Since F is majorizing, F restricts to distance-preserving maps from $[\tilde x\tilde y]$ to $[xy]$ and $[\tilde y\tilde z]$ to $[yz]$. Suppose $\tilde p \in [\tilde x\tilde y]$ and $\tilde q \in [\tilde y\tilde z]$. Then

❹ $$|\tilde p\,\tilde q|_{\mathbb{M}^2[\kappa]} = |F(\tilde p)\;F(\tilde q)|_{\mathcal{U}}.$$

Indeed, inequality holds in one direction by majorization, and in the other direction by angle comparison (9.3.1d). By first variation (9.14.4),

it follows that each pair of corresponding angles of triangles $[\tilde{x}\tilde{y}\tilde{z}]$ and $[xyz]$ are equal. But then ❹ holds for p, q on any two sides of these triangles, so F is distance-preserving on every geodesic of $\operatorname{Conv}(\tilde{p}, \tilde{x}, \tilde{y})$. Hence the claim.

❺ *Suppose* $F|\operatorname{Conv}(\tilde{x}^{n+1}, \tilde{x}^0, \tilde{x}^1 \dots \tilde{x}^k)$ *is distance-preserving for some* k, $1 \leqslant k \leqslant n-1$. *Then* $F|\operatorname{Conv}(\tilde{x}^{n+1}, \tilde{x}^0, \tilde{x}^1 \dots \tilde{x}^{k+1})$ *is distance-preserving.*

Set $\tilde{p} = [\tilde{x}^{k-1}\tilde{x}^{k+1}] \cap [\tilde{x}^k\tilde{x}^{n+1}]$ and $p = [x^{k-1}x^{k+1}] \cap [x^k x^{n+1}]$.
The following maps are distance-preserving:
(i) $F|\operatorname{Conv}(\tilde{x}^{k-1}, \tilde{x}^k, \tilde{x}^{k+1})$,
(ii) $F|\operatorname{Conv}(\tilde{x}^{k+1}, \tilde{x}^{k-1}, \tilde{x}^{n+1})$,
(iii) $F|\operatorname{Conv}(\tilde{x}^{n+1}, \tilde{x}^k, \tilde{x}^{k+1})$.

Indeed, (i) is immediate from ❸. Therefore $|\tilde{x}^{k-1}\,\tilde{x}^{k+1}| = |x^{k-1}\,x^{k+1}|$, and so F restricts to a distance-preserving map from $[\tilde{x}^{k-1}\tilde{x}^{k+1}]$ onto $[x^{k-1}x^{k+1}]$. With the induction hypothesis, it follows that $F(\tilde{p}) = p$, hence

❻ $$\measuredangle[\tilde{x}^{k-1}\,{}^{\tilde{x}^{k+1}}_{\tilde{x}^{n+1}}] = \measuredangle[x^{n+1}\,{}^{x^{k-1}}_{x^{k+1}}].|F(\tilde{p})\;F(\tilde{q})|_{\mathcal{U}}.$$

Then (ii) follows from ❻ and ❸. Since $|\tilde{x}^k\,\tilde{x}^{n+1}| = |x^k\,x^{n+1}|$, (iii) follows from ❻ and (i).

Let $\tilde{\gamma}$ be a geodesic of $\operatorname{Conv}(\tilde{x}^{n+1}, \tilde{x}^0, \tilde{x}^1 \dots \tilde{x}^{k+1})$. Then length $\tilde{\gamma} < \varpi^\kappa$. If $\tilde{\gamma}$ does not pass through the point $[\tilde{x}^{k-1}\tilde{x}^{k+1}] \cap [\tilde{x}^k\tilde{x}^{n+1}]$, it follows from the induction hypothesis and (i) (ii), (iii) that $\gamma = F \circ \tilde{\gamma}$ is a local geodesic of length $< \varpi^\kappa$. By 9.4.4, γ is a geodesic. By continuity, $F \circ \tilde{\gamma}$ is a geodesic for all $\tilde{\gamma}$. ❺ follows and hence the lemma. □

9.12.10. Exercise. *Let* $\mathcal{U} \in \mathrm{CBA}[0]$ *and for 4 points* $x_1, x_2, x_3, x_4 \in \mathcal{U}$ *there is a convex quadrilateral* $[\tilde{x}_1\tilde{x}_2\tilde{x}_3\tilde{x}_4]$ *in* $\mathbb{E}^2$ *such that*

$$|x_i\,x_j|_{\mathcal{U}} = |\tilde{x}_i\,\tilde{x}_j|_{\mathbb{E}^2}$$

for all i *and* j. *Show that* $\mathcal{U}$ *contains an isometric copy of the* solid quadrilateral $[\tilde{x}_1\tilde{x}_2\tilde{x}_3\tilde{x}_4]$*; that is the convex hull of* $\tilde{x}_1, \tilde{x}_2, \tilde{x}_3, \tilde{x}_4$ *in* $\mathbb{E}^2$

9.13 Hadamard–Cartan theorem

The development of Alexandrov geomtry was largely influenced by Hadamard–Cartan theorem. Its original formulation states that if M is a complete Riemannian manifold with sectional curvature $\leqslant 0$ then exponential map at any point $p \in M$ is a covering; in particular it

implies that universal cover of M is diffeomorphic to the Euclidean space of the same dimension.

In this generality, theorem appeared in the lectures of Cartan, see [40]. For surfaces in the Euclidean plane, this theroem was proved by Hans von Mangoldt see [**?**] and few years later independently by Hadamard [69].

Formulations for metric spaces of different generality were proved by Busemann see [35], Rinow see [147], Gromov [63, p.119]. A detailed proof of Gromov's statement when $\mathcal{U}$ is proper was given in [22], using Birkhoff curve-shortening. The statement in the non-proper case was proved in [7]. This proof applies more generally, to *convex spaces* (see Exercise 9.13.8). It was pointed out by Kleiner (see [24, 23]) and independently by Bridson and Haefliger in [30] that the No-conjugate-point theorem (9.10.2) allows curve-shortening as in Corollary 9.10.5, and hence the Hadamard–Cartan theorem applies to length spaces as well as geodesic spaces.

The generalized Hadamard–Cartan theorem when $\mathcal{U}$ is proper was proved by Bowditch in [29].

9.13.1. Hadamard–Cartan theorem. *Let $\mathcal{U}$ be a complete, simply connected length space satisfying* $\operatorname{curv}\mathcal{U}\leqslant 0$. *Then* $\mathcal{U}\in\mathrm{CBA}[0]$.

Proof. Since $\varpi^\kappa=\infty$, the map $\Phi\colon\mathcal{B}\to\mathcal{U}$ in Theorem (9.11.1) is a metric covering. Since $\mathcal{U}$ is simply connected, $\Phi\colon\mathcal{B}\to\mathcal{U}$ is an isometry. □

To formulate Generalized Hadamard–Cartan theorem, we need the following definition.

9.13.2. Definition. *Given $\ell\in(0,\infty]$, a metric space $\mathcal{X}$ is called ℓ-simply connected if it is connected any closed curve of length $<\ell$ is null-homotopic in the class of curves of length $<\ell$ in $\mathcal{X}$.*

9.13.3. Generalized Hadamard–Cartan theorem. *A complete length space $\mathcal{U}$ is* $\mathrm{CBA}[\kappa]$ *if and only if $\mathcal{U}$ is $2\cdot\varpi^\kappa$-simply connected and* $\operatorname{curv}\mathcal{U}\leqslant\kappa$.

In the proof we will need the following lemma.

9.13.4. Lemma. *Let $\mathcal{U}$ be a complete length space,* $\kappa,\varepsilon\in\mathbb{R}$, $\varepsilon>0$, $\operatorname{curv}\mathcal{U}\leqslant\kappa$ *and* $\gamma_1,\gamma_2\colon\mathbb{S}^1\to\mathcal{U}$ *be two closed curves. Assume that*

a) $\operatorname{length}\gamma_1,\operatorname{length}\gamma_2<2\cdot\varpi^\kappa-4\cdot\varepsilon$

b) $|\gamma_1(x)\,\gamma_2(x)|<\varepsilon$ *any $x\in\mathbb{S}^1$ and the geodesic* $[\gamma_1(x)\gamma_2(x)]$ *is uniquely defined and depends continuously on x.*

c) γ_1 *is majorized by a convex region in* $\mathbb{M}^2[\kappa]$.

Then γ_2 is majorized by a convex region in $\mathbb{M}^2[\kappa]$.

Proof. Let D be a convex region in $\mathbb{M}^2[\kappa]$ which majorizes γ_1 under the map $F\colon D \to \mathcal{U}$ (see Definition 9.12.1). Denote by $\tilde\gamma_1$ the curve bounding D such that $F\circ\tilde\gamma_1 = \gamma_1$. Since

$$\begin{aligned}\operatorname{length}\tilde\gamma_1 = \operatorname{length}\gamma_1 &< \\ &< 2\cdot\varpi^\kappa - 4\cdot\varepsilon,\end{aligned}$$

there is a point $\tilde p \in D$ such that $|\tilde p\tilde\gamma(x)|_{\mathbb{M}^2[\kappa]} < \frac{\varpi^\kappa}{2} - \varepsilon$ for any $x \in$ $\in \mathbb{S}^1$. Denote by α_x the joint of the paths $F\circ\operatorname{path}_{[p\tilde\gamma_1(x)]_{\mathbb{M}^2[\kappa]}}$ and $\operatorname{path}_{[\gamma_1(x)\gamma_2(x)]}$ in $\mathcal{U}$. Note that α_x depends continuously on x and

$$\operatorname{length}\alpha_x < \tfrac{\varpi^\kappa}{2} \quad\text{and}\quad \alpha_x(1) = \gamma_2(x)$$

for any x.

Let us apply Lifting globalization theorem (9.11.1) for $p = F(\tilde p)$. We get a space $\mathcal{B} \in \mathrm{CAT}[\kappa]$ and a locally isometric map $\Phi\colon \mathcal{B} \to \mathcal{U}$ with $\Phi(\hat p) = p$ for some $\hat p \in \mathcal{B}$ and the lifting property for the curves starting at p with length $< \varpi^\kappa/2$. Applying the lifting property for α_x, we get existence of curve $\hat\gamma_2\colon \mathbb{S}^1 \to \mathcal{B}$ such that

$$\gamma_2 = \Phi\circ\hat\gamma_2.$$

Since $\mathcal{B} \in \mathrm{CAT}[\kappa]$, we can apply majorization theorem (9.12.4) for $\hat\gamma_2$. The composition of the obtained majorization with Φ is a majorization of γ_2. □

Proof. The "only if" part follows from the Reshetnyak majorization theorem (9.12.4).

Assume that γ_t, $t \in [0,1]$ be a null-homotopy of curves in $\mathcal{U}$; i.e., $\gamma_0(x) = p$ for some $p \in \mathcal{U}$ and any $x \in \mathbb{S}^1$. Assume further that $\operatorname{length}\gamma_t < 2\cdot\varpi^\kappa$ for any t. To prove "only if" part, it is sufficient to show that γ_1 is majorized by a convex region in $\mathbb{M}^2[\kappa]$.

By semicontinuity of length (5.3.1), we can choose sufficiently small $\varepsilon > 0$ such that

$$\operatorname{length}\gamma_t < 2\cdot\varpi^\kappa - 4\cdot\varepsilon$$

for all t.

By Corollary 9.11.2, we may assume in addition that

❶ $$\mathrm{B}(\gamma_t(x),\varepsilon) \in \mathrm{CAT}[\kappa]$$

for any t and x.

Choose a partition $0 = t_0 < t_1 < \ldots < t_n = 1$ so that $|\gamma_{t_i}(x)\,\gamma_{t_{i-1}}(x)| < \varepsilon$ for any i and x. According to ❶, for any i, the geodesic $[\gamma_{t_i}(x)\gamma_{t_{i-1}}(x)]$ depends continuously on x.

Note that $\gamma_0 = \gamma_{t_0}$ is majorized by a convex region in $\mathbb{M}^2[\kappa]$. Applying the lemma n times we get that so is $\gamma_1 = \gamma_{t_n}$. □

Finally note that there is a subtle difference between simply connected and ∞-simply connected spaces; the first states that any closed curve is null-homotopic while the second means that any rectifiable curve is null-homotopic in the class of rectifiable curves. The following proposition makes possible to deduce Hadamard–Cartan theorem directly from the Generalized Hadamard–Cartan theorem.

9.13.5. Proposition. *Let $\mathcal{U}$ be a complete length space and* $\operatorname{curv}\mathcal{U} \leqslant \kappa$. *Then $\mathcal{U}$ is simply connected if and only if it is ∞-simply connected.*

Proof. To prove "if" part, it is sufficient to show that any closed curve in $\mathcal{U}$ is homotopic to a closed broken geodesic.

Let γ_0 be a closed curve in $\mathcal{U}$. According to Corollary 9.11.2, there is $\varepsilon > 0$ such that

❷ $$\mathrm{B}(\gamma(x), \varepsilon) \in \mathrm{CAT}[\kappa]$$

for any x.

Choose a broken geodesic γ_1 such that $|\gamma_0\gamma_1| < \varepsilon$ for any x. By ❸, $\operatorname{path}_{[\gamma_0(x)\gamma_1(x)]}$ is uniquely defined and depends continuously on x.

Hence $\gamma_t(x) = \operatorname{path}_{[\gamma_0(x)\gamma_1(x)]}(t)$ gives a homotopy from γ_0 to γ_1.

The proof of "only if" part is similar. Assume γ_t is a homotopy between two rectifiable curves γ_0 and γ_1. Fix $\varepsilon > 0$ so that

❸ $$\mathrm{B}(\gamma_t(x), \varepsilon) \in \mathrm{CAT}[\kappa]$$

for any t and x. Choose a partition $0 = t_0 < t_1 < \ldots < t_n = 1$ so that

$$|\gamma_{t_{i-1}}(x)\ \gamma_{t_i}(x)| < \tfrac{\varepsilon}{10}$$

for any i and x. Set $\hat\gamma_{t_0} = \gamma_0$, $\hat\gamma_{t_n} = \gamma_{t_n}$. For each $0 < i < n$, approximate γ_{t_i} by a closed broken geodesic $\hat\gamma_i$.

Construct the homotopy from $\hat\gamma_{t_{i-1}}$ to $\hat\gamma_{t_i}$ setting

$$\hat\gamma_t = \operatorname{path}_{[\hat\gamma_{t_{i-1}}(x)\hat\gamma_{t_i}(x)]}(t)$$

Since ε is sufficiently small, by ??? we get

$$\operatorname{length}\hat\gamma_t < 10\cdot(\operatorname{length}\hat\gamma_{t_{i-1}} + \operatorname{length}\hat\gamma_{t_i}).$$

In particular $\hat\gamma_t$ is rectifiable for all t.

Joining the obtained homotopies for all i we get a homotoply from γ_0 to γ_1 in the class of rectifiable curves. $\square$

9.13.6. Exercise. *Let $\mathcal{X}$ be a double cover of $\mathbb{E}^3$ which is branching along two distinct lines ℓ^1 and ℓ^2. Show that $\mathcal{X} \in \mathrm{CBA}[0]$ if and only if ℓ^1 intersect ℓ^2 at right angle.*

About convex spaces. A *convex space* $\mathcal{X}$ is a geodesic space such that the function $t \mapsto |\gamma(t)\ \sigma(t)|$ is convex for any two geodesic paths $\gamma, \sigma : [0,1] \to \mathcal{X}$. A *locally convex space* is a length space in which every point has a neighborhood that is a convex space in the restricted metric.

9.13.7. Exercise. *Assume $\mathcal{X}$ is a convex space such that angle of any hinge is defined. Show that $\mathcal{X}$ is* $\mathrm{CAT}[0]$.

The following exercise gives an analog of Hadamard–Cartan theorem for locally convex spaces; see also [7].

9.13.8. Exercise. *Show that a complete, simply connected, locally convex space is a convex space.*

9.14 Angles

Recall that ω denotes a nonpriciple ultrafilter, see Section 4.2.

9.14.1. Angle semicontinuity. *Let $\mathcal{U}_n \in \mathrm{CAT}[\kappa]$ and $\mathcal{U}_n \to \mathcal{U}_\omega$ as $n \to \omega$. Assume that a sequence of hinges $[p_n\,{}^{x_n}_{y_n}]$ in $\mathcal{U}_n$ converges to a hinge $[p_\omega\,{}^{x_\omega}_{y_\omega}]$ in $\mathcal{U}_\omega$. Then*

$$\measuredangle[p_\omega\,{}^{x_\omega}_{y_\omega}] \geqslant \lim_{n\to\omega} \measuredangle[p_n\,{}^{x_n}_{y_n}].$$

Proof. By angle-sidelength monotonicity (9.3.2),

$$\measuredangle[p_\omega\,{}^{x_\omega}_{y_\omega}] = \inf\left\{\, \tilde\measuredangle^\kappa(p_\omega\,{}^{\bar x_\omega}_{\bar y_\omega}) \;\middle|\; \bar x_\omega \in\,]p_\omega x_\omega]\,,\ \bar y_\omega \in\,]p_\omega y_\omega] \,\right\}.$$

For fixed $\bar x_\omega \in\,]p_\omega x_\omega]$ and $\bar y_\omega \in\,]p_\omega x_\omega]$, choose $\bar x_n \in\,]p x_n]$ and $\bar y_n \in]p y_n]$ so that $\bar x_n \to \bar x_\omega$ and $\bar y_n \to \bar y_\omega$ as $n \to \omega$. Clearly

$$\tilde\measuredangle^\kappa(p_n\,{}^{\bar x_n}_{\bar y_n}) \to \tilde\measuredangle^\kappa(p_\omega\,{}^{\bar x_\omega}_{\bar y_\omega})$$

as $n \to \omega$.

By angle comparison (9.3.1d), $\measuredangle[p_n\,{}^{x_n}_{y_n}] \leqslant \tilde\measuredangle^\kappa(p_n\,{}^{\bar x_n}_{\bar y_n})$. Hence the result. $\square$

Now we verify that the first variation formula holds in the CAT setting. Compare first variation inequality (6.3.4) for general metric spaces and the Strong angle lemma (8.6.6) for CBB spaces.

9.14.2. Strong angle lemma. *Let $\mathcal{U} \in \mathrm{CAT}$. Then for any hinge $[p\,{}^q_x]$ in $\mathcal{U}$, we have*

❶ $$\measuredangle[p\,{}^q_x] = \lim_{\bar x\to p}\{\,\tilde\measuredangle^{\kappa}(p\,{}^q_{\bar x})\mid \bar x\in\,]px]\,\}$$

for any $\kappa\in\mathbb{R}$ such that $|pq|<\varpi^{\kappa}$.

Proof. By angle-sidelength monotonicity (9.3.2), the righthand side is defined and at least equal to the lefthand side.

By Lemma 6.3.1, we may take $\kappa=0$ in ❶. By the cosine law and the first variation inequality (6.3.4), the righthand side is at most equal to the lefthand side. □

9.14.3. First variation. *Let $\mathcal{U}\in\mathrm{CAT}[\kappa]$. For any nontrivial geodesic $[py]$ in $\mathcal{U}$ and point $q\ne p$ such that $|pq|<\varpi^{\kappa}$, we have*

$$|q\,\mathrm{geod}_{[py]}(t)| = |qp| - t\cdot\cos\measuredangle[p\,{}^q_y] + o(t).$$

Proof. The first variation equation is equivalent to the strong angle lemma (9.14.2), as follows from the Euclidean cosine law. □

9.14.4. First variation (both-endpoints version). *Let $\mathcal{U}\in\mathrm{CAT}[\kappa]$. For any nontrivial geodesics $[py]$ and $[qz]$ in $\mathcal{U}$ such that $p\ne q$ and $|pq|<\varpi^{\kappa}$, we have*

$$|\mathrm{geod}_{[py]}(t)\ \mathrm{geod}_{[qz]}(\tau)| = |qp| - t\cdot\cos\measuredangle[p\,{}^q_y] - \tau\cdot\cos\measuredangle[q\,{}^p_z] + o(t+\tau).$$

Proof. By the first variation equation (9.14.3),

$$\begin{aligned}|\mathrm{geod}_{[py]}(t)\ \mathrm{geod}_{[qz]}(\tau)| &= |q\,\mathrm{geod}_{[py]}(t)| - \tau\cdot\cos\measuredangle[q\,{}^{\mathrm{geod}_{[py]}(t)}_z] + o(\tau)\\ &= |qp| - t\cdot\cos\measuredangle[p\,{}^q_y] + o(t) - \tau\cdot\cos\measuredangle[q\,{}^{\mathrm{geod}_{[py]}(t)}_z] + o(\tau)\\ &= |qp| - t\cdot\cos\measuredangle[p\,{}^q_y] - \tau\cdot\cos\measuredangle[q\,{}^p_z] + o(t+\tau).\end{aligned}$$

Here the final equality follows from

❷ $$\lim_{t\to 0}\measuredangle[q\,{}^{\mathrm{geod}_{[py]}(t)}_z] = \measuredangle[q\,{}^p_z].$$

"$\leqslant$" in ❷ holds by angle semicontinuity (9.14.1). "$\geqslant$" holds by the triangle inequality for angles, since angle comparison (9.3.1d) gives

$$\lim_{t\to 0}\measuredangle[q\,{}^p_{\mathrm{geod}_{[py]}(t)}] = 0.$$

□

9.15 Remarks and open problems

The following question was known in folklore in the 80's, but it seems that in print it was first mentioned in [88]. We do not see any reason why it should be true, but we also cannot construct a counterexample.

9.15.1. Open question. *Let* $\mathcal{U} \in \mathrm{CBA}[0]$ *and* $K \subset \mathcal{U}$ *be a compact set. Is it true that* K *lies in a convex compact set* $\bar{K} \subset \mathcal{U}$*?*

The question can easily be reduced to the case when K is finite; so far it is not even known if any three points in a CAT[0]-space lie in a compact convex set.

Chapter 10

Kirszbraun revisited

This chapter is based on our paper [6] and earlier paper of Lang and Schroeder [94].

10.1 Short map extension definitions.

Theorems 10.1.1 and 10.1.3 give characterizations of $\mathrm{CBB}[\kappa]$ and $\mathrm{CAT}[\kappa]$.

10.1.1. Theorem. *Let $\mathcal{L}$ be a complete length space. Then $\mathcal{L} \in \mathrm{CBB}[\kappa]$ if and only if for any 3-point set V_3 and any 4-point set $V_4 \supset V_3$ in $\mathcal{L}$, any short map $f\colon V_3 \to \mathbb{M}^2[\kappa]$ can be extended to a short map $F\colon V_4 \to \mathbb{M}^2[\kappa]$ (so $f = F|V_3$).*

The proof of the "only if" part of Theorem 10.1.1 can be obtained as a corollary of Kirszbraun's theorem (10.4.1). But we present another proof, based on more elementary ideas. The "only if" part of Theorem 10.1.3 does not follow directly from Kirszbraun's theorem, since the desired extension is in $\mathcal{U}$, not just the completion of $\mathcal{U}$.

In the proof of Theorem 10.1.1, we use the following lemma in the geometry of model planes; it is useful in studying model angles determined by subtriples of a quadruple of points in a metric space. Here we say that two triangles with a common vertex *do not overlap* if their convex hulls intersect only at the common vertex.

10.1.2. Overlap lemma. *Let $[\tilde{x}^1\tilde{x}^2\tilde{x}^3]$ be a triangle in $\mathbb{M}^2[\kappa]$. Let $\tilde{p}^1, \tilde{p}^2, \tilde{p}^3$ be points such that, for any permutation $\{i, j, k\}$ of $\{1, 2, 3\}$, we have*

(i) $|\tilde{p}^i\,\tilde{x}^k| = |\tilde{p}^j\,\tilde{x}^k|$,

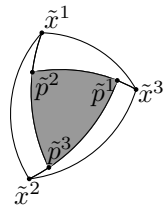
(ii) *$\tilde{p}^i$ and $\tilde{x}^i$ lie in the same closed halfspace determined by $[\tilde{x}^j\tilde{x}^k]$,*

Assume no pair of triangles $[\tilde p^i\tilde x^j\tilde x^k]$ *overlap then*

$$\measuredangle\tilde p^1+\measuredangle\tilde p^2+\measuredangle\tilde p^3>2{\cdot}\pi,$$

where $\measuredangle\tilde p^i$ *denotes* $\measuredangle[\tilde p^i\,{}^{\tilde x^k}_{\tilde x^j}]$ *for a permutation* $\{i,j,k\}$ *or* $\{1,2,3\}$.

Remarks. If $\kappa\leqslant 0$, then overlap lemma can be proved without using condition (i). This follows immediately from the formula that relates the sum of angles for the hexagon $[\tilde p^1\tilde x^2\tilde p^3\tilde x^1\tilde p^2\tilde x^3]$ and its area:



$$\measuredangle\tilde p^1-\measuredangle\tilde x^2+\measuredangle\tilde p^3-\measuredangle\tilde x^1+\measuredangle\tilde p^2-\measuredangle\tilde x^3=2{\cdot}\pi-\kappa{\cdot}\text{area}.$$

In case $\kappa>0$, condition (i) is essential. An example for $\kappa>0$ can be constructed by perturbing the degenerate spherical configuration on the picture.

Proof. Rotate the triangle $[\tilde p^3\tilde x^1\tilde x^2]$ around $\tilde x^1$ to make $[\tilde x^1\tilde p^3]$ coincide with $[\tilde x^1\tilde p^2]$. Let $\dot x^2$ denote the image of $\tilde x^2$ after rotation. Note that

$$\measuredangle[\tilde x^1\,{}^{\tilde x^3}_{\dot x^2}]=\min\{\,\measuredangle[\tilde x^1\,{}^{\tilde x^2}_{\tilde p^3}]+\measuredangle[\tilde x^1\,{}^{\tilde p^2}_{\tilde x^3}],\ 2{\cdot}\pi-(\measuredangle[\tilde x^1\,{}^{\tilde x^2}_{\tilde p^3}]+\measuredangle[\tilde x^1\,{}^{\tilde p^2}_{\tilde x^3}])\,\}.$$

By (ii), the triangles $[\tilde p^3\tilde x^1\tilde x^2]$ and $[\tilde p^2\tilde x^3\tilde x^1]$ do not overlap if and only if

❶ $$2{\cdot}\pi>\measuredangle[\tilde x^1\,{}^{\tilde x^2}_{\tilde p^3}]+\measuredangle[\tilde x^1\,{}^{\tilde p^2}_{\tilde x^3}]+\measuredangle[\tilde x^1\,{}^{\tilde x^2}_{\tilde x^3}]$$

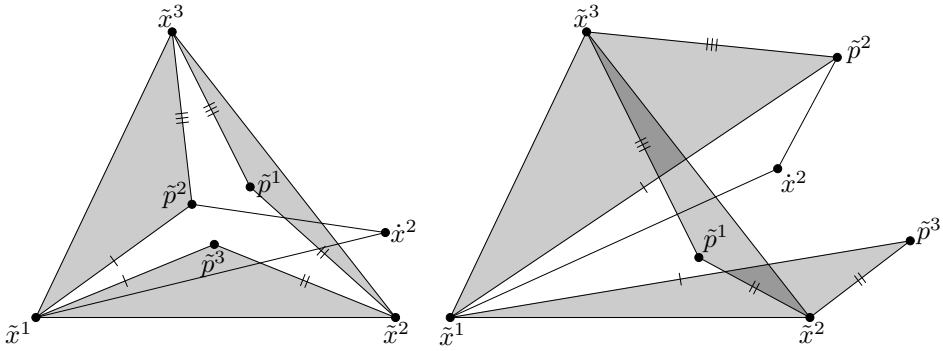
and

❷ $$\measuredangle[\tilde x^1\,{}^{\tilde x^3}_{\tilde x^2}]>\measuredangle[\tilde x^1\,{}^{\tilde x^3}_{\dot x^2}].$$

The condition ❷ holds if and only if $|\tilde x^2\,\tilde x^3|>|\dot x^2\,\tilde x^3|$, which in turn holds if and only if

❸ $$\begin{aligned}\measuredangle\tilde p^1&>\measuredangle[\tilde p^2\,{}^{\tilde x^3}_{\dot x^2}]\\&=\min\{\measuredangle\tilde p^3+\measuredangle\tilde p^2,2{\cdot}\pi-(\measuredangle\tilde p^3+\measuredangle\tilde p^2)\}.\end{aligned}$$

The inequality follows since the corresponding hinges have the same pairs of sidelengths. (The two pictures show that both possibilities for the minimum can occur.)



Now assume $\measuredangle\tilde{p}^1 + \measuredangle\tilde{p}^2 + \measuredangle\tilde{p}^3 \leqslant 2\cdot\pi$. Then ❸ implies

$$\measuredangle\tilde{p}^i > \measuredangle\tilde{p}^j + \measuredangle\tilde{p}^{\bar{k}}.$$

Since no pair of triangles overlap, the same holds for any permutation $(i, j, \bar{k})$ of $(1, 2, 3)$. Therefore

$$\measuredangle\tilde{p}^1 + \measuredangle\tilde{p}^2 + \measuredangle\tilde{p}^3 > 2\cdot(\measuredangle\tilde{p}^1 + \measuredangle\tilde{p}^2 + \measuredangle\tilde{p}^3),$$

a contradiction. □

Proof of 10.1.1; "if" part. Assume $\mathcal{L}$ is geodesic. Take $x^1, x^2, x^3 \in \mathcal{L}$ so the model triangle $[\tilde{x}^1\tilde{x}^2\tilde{x}^3] = \tilde{\triangle}^\kappa(x^1x^2x^3)$ is defined. Choose $p \in\,]x^1x^2[$; apply Kirszbrun property for $V_3 = \{x^1, x^2, x^3\}$ and $V_4 = \{x^1, x^2, x^3, p\}$ and the map $f(x^i) = \tilde{x}^i$. You obtain point-on-side comparison (8.3.1c).

In case $\mathcal{L}$ is not geodesic, pass to its ultrapower $\mathcal{L}^\omega$. Note that the short map extension property survives for $\mathcal{L}^\omega$ and recall that $\mathcal{L}^\omega$ is geodesic (see 4.3.2). Thus, from above, $\mathcal{L}^\omega \in \mathrm{CBB}[\kappa]$. By Proposition 8.1.7, we get $\mathcal{L} \in \mathrm{CBB}[\kappa]$.

"Only if" part. Assume the contrary; i.e., $x^1, x^2, x^3, p \in \mathcal{L}$, and $\tilde{x}^1, \tilde{x}^2, \tilde{x}^3 \in \mathbb{M}^2[\kappa]$ are such that $|\tilde{x}^i\,\tilde{x}^j| \leqslant |x^i\,x^j|$ for all i, j and there is no point $\tilde{p} \in \mathbb{M}^2[\kappa]$ such that $|\tilde{p}\,\tilde{x}^i| \leqslant |p\,x^i|$ for all i.

Note that in this case all comparison triangles $\tilde{\triangle}^\kappa(px^ix^j)$ are defined. That is always true if $\kappa \leqslant 0$. If $\kappa > 0$, and say $\tilde{\triangle}^\kappa(px^1x^2)$ is undefined, then

$$\begin{aligned}|px^1| + |px^2| &\geqslant 2\cdot\varpi^\kappa - |x^1\,x^2| \geqslant \\ &\geqslant 2\cdot\varpi^\kappa - |\tilde{x}^1\,\tilde{x}^2| \geqslant \\ &\geqslant |\tilde{x}^1\,\tilde{x}^3| + |\tilde{x}^2\,\tilde{x}^3|.\end{aligned}$$

It follows that we can take $\tilde p$ on $[\tilde x^1\tilde x^3]$ or $[\tilde x^2\tilde x^3]$.

For each $i\in\{1,2,3\}$, consider a point $\tilde p^i\in\mathbb{M}^2[\kappa]$ such that $|\tilde p^i\,\tilde x^i|$ is minimal among points satisfying $|\tilde p^i\,\tilde x^j|\leqslant|p\,x^j|$ for all $j\ne i$. Clearly, every $\tilde p^i$ is inside the triangle $[\tilde x^1\tilde x^2\tilde x^3]$ (i.e. in $\operatorname{Conv}(\tilde x^1,\tilde x^2,\tilde x^3)$), and $|\tilde p^i\,\tilde x^i|>|p\,x^i|$ for each i. Since the function $x\mapsto\mathcal{Z}^\kappa\{x;a,b\}$ is increasing, it follows that

(i) $|\tilde p^i\,\tilde x^j|=|p\,x^j|$ for $i\ne j$;

(ii) no pair of triangles from $[\tilde p^1\tilde x^2\tilde x^3]$, $[\tilde p^2\tilde x^3\tilde x^1]$, $[\tilde p^3\tilde x^1\tilde x^2]$ overlap in $[\tilde x^1\tilde x^2\tilde x^3]$.

As follows from Overlap lemma (10.1.2), in this case

$$\measuredangle[\tilde p^1\,{}^{\tilde x^2}_{\tilde x^3}]+\measuredangle[\tilde p^2\,{}^{\tilde x^3}_{\tilde x^1}]+\measuredangle[\tilde p^3\,{}^{\tilde x^1}_{\tilde x^2}]>2\cdot\pi.$$

Since $|\tilde x^i\,\tilde x^j|\leqslant|x^i\,x^j|$ we get

$$\measuredangle[\tilde p^k\,{}^{\tilde x^i}_{\tilde x^j}]\leqslant\mathcal{Z}^\kappa(p\,{}^{x^i}_{x^j})$$

if (i,j,k) is a permutation of $(1,2,3)$. Therefore

$$\mathcal{Z}^\kappa(p\,{}^{x^1}_{x^2})+\mathcal{Z}^\kappa(p\,{}^{x^2}_{x^3})+\mathcal{Z}^\kappa(p\,{}^{x^3}_{x^1})>2\cdot\pi;$$

the later contradicts (1+3)-comparison (8.1.2). □

10.1.3. Theorem. *Let $\mathcal{U}$ be a metric space in which any pair of points at distance $<\varpi^\kappa$ are joined by a unique geodesic. Then $\mathcal{U}\in\operatorname{CAT}[\kappa]$ if and only if for any 3-point set V_3 and 4-point set $V_4\supset V_3$ in $\mathbb{M}^2[\kappa]$, where the perimeter of V_3 is $<2\cdot\varpi^\kappa$, any short map $f\colon V_3\to\mathcal{U}$ can be extended to a short map $F\colon V_4\to\mathcal{U}$.*

In the proof of Theorem 10.1.3, we use the following lemma in the geometry of model planes:

10.1.4. Lemma. *Let $x^1,x^2,x^3,y^1,y^2,y^3\in\mathbb{M}[\kappa]$ be points such that $|x^i\,x^j|\geqslant|y^i\,y^j|$ for all i,j. Then there is a short map $\Phi\colon\mathbb{M}[\kappa]\to\mathbb{M}[\kappa]$ such that $\Phi(x^i)=y^i$ for all i; moreover, one can choose Φ so that*

$$\operatorname{Im}\Phi\subset\operatorname{Conv}(y^1,y^2,y^3).$$

We only give an idea of the proof of this lemma; alternatively, one can get the result as a corollary of Kirszbraun's theorem (10.4.1)

Idea of the proof. The map Φ can be constructed as a composition of the following folding maps: Given a halfspace H in $\mathbb{M}[\kappa]$, consider the map $\mathbb{M}[\kappa]\to H$, which is identity on H and reflects all points outside of H into H. This map is a path isometry, in particular it is short.

One can get the last part of the lemma by composing the above map with foldings along the sides of triangle $[y^1y^2y^3]$ and passing to a partial limit. $\square$

Proof of 10.1.3; "if" part. The point-on-side comparison (9.3.1c) follows by taking $V_3 = \{\tilde{x}, \tilde{y}, \tilde{p}\}$ and $V_4 = \{\tilde{x}, \tilde{y}, \tilde{p}, \tilde{z}\}$ where $z \in]xy[$. It is only necessary to observe that $F(\tilde{z}) = z$ by uniqueness of $[xy]$.

"Only if" part. Let $V_3 = \{\tilde{x}^1, \tilde{x}^2, \tilde{x}^3\}$ and $V_4 = \{\tilde{x}^1, \tilde{x}^2, \tilde{x}^3, \tilde{p}\}$.

Set $y^i = f(\tilde{x}^i)$ for all i; we need to find a point $q \in \mathcal{U}$ such that $|y^i q| \leqslant |\tilde{x}^i \tilde{p}|$ for all i.

Let D be the convex set in $\mathbb{M}^2[\kappa]$ bounded by C the model triangle $[\tilde{y}^1\tilde{y}^2\tilde{y}^3] = \tilde{\triangle}^\kappa y^1y^2y^3$; i.e., $D = \mathrm{Conv}(\tilde{y}^1, \tilde{y}^2, \tilde{y}^3)$.

Note that $|\tilde{y}^i \tilde{y}^j| = |y^i y^j| \leqslant |\tilde{x}^i \tilde{x}^j|$ for all i, j. Applying Lemma 10.1.4, we get a short map $\Phi\colon \mathbb{M}[\kappa] \to D$ such that $\Phi\colon \tilde{x}^i \mapsto \tilde{y}^i$.

Further, from majorization (9.12.4), there is a short map $F\colon D \to \mathcal{U}$ such that $\tilde{y}^i \mapsto y^i$ for all i.

Thus one can take $q = F \circ \Phi(\tilde{p})$. $\square$

10.2 (1+n)-point comparison

The following theorem gives a more sensitive analog of (1+3)-point comparison (8.1.2). In a bit more analytic form it appears in [157].

10.2.1. (1+n)-point comparison. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$. Then for any array of points $p, x^1, \dots, x^n \in \mathcal{L}$ there is a model array $\tilde{p}, \tilde{x}^1, \dots, \tilde{x}^n \in \mathbb{M}^n[\kappa]$ such that*

a) $|\tilde{p}\tilde{x}^i| = |px^i|$ for all i.

b) $|\tilde{x}^i \tilde{x}^j| \geqslant |x^i x^j|$ for all i, j.

Proof. It is enough to show that given $\varepsilon > 0$ there is a configuration $\tilde{p}, \tilde{x}^1, \dots, \tilde{x}^n \in \mathbb{M}^n[\kappa]$ such that $|\tilde{x}^i \tilde{x}^j| \geqslant |x^i x^j|$ and $\big||\tilde{p}\tilde{x}^i| - |px^i|\big| \leqslant \varepsilon$. Then one can pass to a limit configuration for $\varepsilon \to 0+$.

According to 8.2.5, the set $\mathrm{Str}(x^1, \dots, x^n)$ is dense in $\mathcal{L}$. Thus, there is a point $p' \in \mathrm{Str}(\tilde{x}^1, \dots, \tilde{x}^n)$ such that $|p'p| \leqslant \varepsilon$. According to Corollary 14.5.5, $\mathrm{T}_{p'}$ contains a subcone E isometric to a Euclidean space which contains all vectors $\log[p'x^i]$. Passing to subspace if necessary, we can assume that $\dim E \leqslant n$.

Mark a point $\tilde{p} \in \mathbb{M}^n[\kappa]$ and choose a distance preserving map $\iota\colon E \to \mathrm{T}_{\tilde{p}}\mathbb{M}^n[\kappa]$. Set

$$\tilde{x}^i = \exp_{\tilde{p}} \circ \iota(\log[p'x^i]).$$

Thus $|\tilde{p}\tilde{x}^i| = |p'x^i|$ and therefore

$$\big||\tilde{p}\tilde{x}^i| - |px^i|\big| \leqslant |pp'| \leqslant \varepsilon.$$

From the hinge comparison (8.3.1d) we have

$$\measuredangle^{\kappa}(\tilde{p}\,{}^{\tilde{x}^i}_{\tilde{x}^j}) = \measuredangle[\tilde{p}\,{}^{\tilde{x}^i}_{\tilde{x}^j}] = \measuredangle[p'\,{}^{x^i}_{x^j}] \geqslant \measuredangle^{\kappa}(p'\,{}^{x^i}_{x^j}),$$

thus

$$|\tilde{x}^i\,\tilde{x}^j| \geqslant |x^i\,x^j|.$$

□

10.2.2. Exercise. *Give an example of metric on a finite set, which satisfies the comparison inequality*

$$\measuredangle^0(p\,{}^{x_1}_{x_2}) + \measuredangle^0(p\,{}^{x_2}_{x_3}) + \measuredangle^0(p\,{}^{x_3}_{x_1}) \leqslant 2\cdot\pi$$

for any quadruple of points (p, x_1, x_2, x_3)*, but which is not isometric to a subset of Alexandrov space with curvature* $\geqslant 0$*.*

10.3 Helly's theorem

10.3.1. Helly's theorem. *Let* $\mathcal{U} \in \mathrm{CBA}[0]$ *and* $\{K_\alpha\}_{\alpha\in\mathcal{A}}$ *be an arbitrary collection of closed bounded convex subsets of* $\mathcal{U}$*.*

If

$$\bigcap_{\alpha\in\mathcal{A}} K_\alpha = \varnothing$$

then there is an index array $\alpha_1, \alpha_2, \dots, \alpha_n \in \mathcal{A}$ *such that*

$$\bigcap_{i=1}^{n} K_{\alpha_i} = \varnothing.$$

Remarks.

(i) In general, none of K_α might be compact; otherwise the statement is trivial.

(ii) If $\mathcal{U}$ is a Hilbert space (not necessarily separable), then the above result is equivalent to the statement that a convex bounded set which is closed in ordinary topology forms a compact set in the weak topology.

In fact, one can define *weak topology* on arbitrary metric space, by taking exteriors of closed ball as its prebase. Then the result above implies for $\mathcal{U} \in \mathrm{CBA}[0]$, any closed bounded convex set in $\mathcal{U}$ is compact in weak topology (compare to [115]).

We present the proof from [94].

Proof of 10.3.1. Assume the contrary. Then for any finite set $F \subset \mathcal{A}$

$$K_F \overset{def}{=\!=} \bigcap_{\alpha \in F} K_\alpha \neq \varnothing,$$

we will construct point z such that $z \in K_\alpha$ for each α. Thus we will arrive to contradiction since

$$\bigcap_{\alpha \in \mathcal{A}} K_\alpha = \varnothing.$$

Choose a point $p \in \mathcal{U}$ and set $r = \sup |K_F\, p|$ where F runs all finite subsets of $\mathcal{A}$. Set p^*_F to be the closest point on K_F from p; according to closest-point projection lemma (9.6.3), p^*_F exits and is unique.

Take a nested sequence of finite subsets $F_1 \subset F_2 \subset \ldots$ of $\mathcal{A}$, such that $|K_{F_n}\ p| \to r$.

Let us show that $p^*_{F_n}$ is a Cauchy sequence. Indeed, if not then for some fixed $\varepsilon > 0$, we can choose two subsequences (y'_n) and (y''_n) of $(p^*_{F_n})$ such that $|y'_n\, y''_n| \geqslant \varepsilon$. Set z_n to be midpoint of $[y'_n y''_n]$. From point-on-side comparison (8.3.1c), there is $\delta > 0$ such that $|p\, z_n| \leqslant \leqslant \max\{|p\, y'_n|, |p\, y''_n|\} - \delta$. Thus

$$\varlimsup_{n\to\infty} |p\, z_n| < r.$$

On the other hand, from convexity, each F_n contains all $z_{\bar{k}}$ with sufficiently large $\bar{k}$, a contradiction.

Thus, $p^*_{F_n}$ converges and we can set $z = \lim_n p^*_{F_n}$. Clearly $|p\, z| = r$.

Repeat the above arguments for the sequence $F'_n = F_n \cup \{\alpha\}$. As a result, we get another point z' such that $|p\, z| = |p\, z'| = r$ and $z, z' \in \in K_{F_n}$ for all n. Thus, if $z \neq z'$ the midpoint $\hat{z}$ of $[zz']$ would belong to all K_{F_n} and from comparison we would have $|p\, \hat{z}| < r$, a contradiction.

Thus, $z' = z$; in particular $z \in K_\alpha$ for each $\alpha \in \mathcal{A}$. □

10.4 Kirszbraun's theorem

A slightly weaker version of the following theorem appears in [94]. The Conjecture 10.6.4 (if true) gives an equivalent condition for the existence of a short extension; roughly it states that example 10.4.2 is the only obstacle.

10.4.1. Kirszbraun's theorem. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$, $\mathcal{U} \in \mathrm{CBA}[\kappa]$, $Q \subset \mathcal{L}$ be arbitrary subset and $f\colon Q \to \mathcal{U}$ be a short map. Assume that there is $z \in \mathcal{U}$ such that $f(Q) \subset \mathrm{B}[z, \frac{\varpi^\kappa}{2}]$. Then $f\colon Q \to \mathcal{U}$ can*

be extended to a short map $F\colon \mathcal{L} \to \mathcal{U}$ (that is, there is a short map $F\colon \mathcal{L} \to \mathcal{U}$ such that $F|Q = f$.)

The condition $f(Q) \subset \mathrm{B}[z, \frac{\varpi^\kappa}{2}]$ trivially holds for any $\kappa \leqslant 0$ since in this case $\varpi^\kappa = \infty$. The following example shows that this condition is needed for $\kappa > 0$.

10.4.2. Example. *Let $\mathbb{S}^m_+$ be a closed m-dimensional unit hemisphere. Denote its boundary, which is isometric to $\mathbb{S}^{m-1}$, by $\partial\mathbb{S}^m_+$. Clearly, $\mathbb{S}^m_+ \in \mathrm{CBB}[1]$ and $\partial\mathbb{S}^m_+ \in \mathrm{CBA}[1]$ but the identity map $\partial\mathbb{S}^m_+ \to \to \partial\mathbb{S}^m_+$ cannot be extended to a short map $\mathbb{S}^m_+ \to \partial\mathbb{S}^m_+$ (there is no place for the pole).*

There is also a direct generalization of this example to a hemisphere in a Hilbert space of arbitrary cardinal dimension.

First we prove this theorem in the case $\kappa \leqslant 0$ (10.4.4). In the proof of the more complicated case $\kappa > 0$, we use the case $\kappa = 0$. The following lemma is the main ingredient in the proof.

10.4.3. Finite+one lemma. *Let $\kappa \leqslant 0$, $\mathcal{L} \in \mathrm{CBB}[\kappa]$, and $\mathcal{U} \in \in \mathrm{CBA}[\kappa]$. Let $x^1, x^2, \dots, x^n \in \mathcal{L}$ and $y^1, y^2, \dots, y^n \in \mathcal{U}$ be such that $|x^i\,x^j| \geqslant |y^i\,y^j|$ for all i, j.*

Then for any $p \in \mathcal{L}$, there is $q \in \mathcal{U}$ such that $|y^i\,q| \leqslant |x^i\,p|$ for each i.

Proof. It is sufficient to prove the lemma only for $\kappa = 0$ and -1. The proofs of these two cases are identical, only the formulas differ. In the proof, we assume $\kappa = 0$ and provide the formulas for $\kappa = -1$ in the footnotes.

From $(1{+}n)$-point comparison (10.2.1), there is a model configuration $\tilde p, \tilde x^1, \tilde x^2, \dots, \tilde x^n \in \mathbb{M}^n[\kappa]$ such that $|\tilde p\,\tilde x^i| = |p\,x^i|$ and $|\tilde x^i\,\tilde x^j| \geqslant \geqslant |x^i\,x^j|$ for all i, j.

For each i, consider functions $f^i\colon \mathcal{U} \to \mathbb{R}$ and $\tilde f^i\colon \mathbb{M}^n[\kappa] \to \mathbb{R}$ defined as follows[1]

$$(A)^0 \qquad f^i = \tfrac12\cdot|y^i\,{*}|^2, \quad \tilde f^i = \tfrac12\cdot|\tilde x^i\,{*}|^2.$$

Set $\boldsymbol{f} = (f^1, f^2, \dots, f^n)\colon \mathcal{U} \to \mathbb{R}^n$ and $\tilde{\boldsymbol{f}} = (\tilde f^1, \tilde f^2, \dots, \tilde f^n)\colon \mathbb{M}^n[\kappa] \to \to \mathbb{R}^n$.

Recall that Up (superset in $\mathbb{R}^n$) is defined on page 267. Note that it is sufficient to prove that $\tilde{\boldsymbol{f}}(\tilde p) \in \mathrm{Up}\,\boldsymbol{f}(\mathcal{U})$.

[1] In case $\kappa = -1$,

$$(A)^- \qquad f^i = \cosh\circ|y^i\,{*}|, \quad \tilde f^i = \cosh\circ|\tilde x^i\,{*}|.$$

Clearly, $(f^i)'' \geqslant 1$. Thus, by the theorem on barycentric simplex (19.1.3a), the set $\operatorname{Up} \boldsymbol{f}(\mathcal{U}) \subset \mathbb{R}^n$ is convex.

Arguing by contradiction, let us assume that $\tilde{\boldsymbol{f}}(\tilde{p}) \notin \operatorname{Up} \boldsymbol{f}(\mathcal{U})$.

Then there exists a supporting hyperplane $\alpha_1 \cdot x_1 + \ldots \alpha_n \cdot x_n = c$ to $\operatorname{Up} \boldsymbol{f}(\mathcal{U})$, separating it from $\tilde{\boldsymbol{f}}(\tilde{p})$. According to Lemma 19.1.6b, $\alpha_i \geqslant 0$ for each i. So we can assume that $(\alpha_1, \alpha_2, \ldots, \alpha_n) \in \Delta^{n-1}$ and

$$\sum_i \alpha_i \cdot \tilde{f}^i(\tilde{p}) < \inf \left\{ \sum_i \alpha_i \cdot f^i(q) \;\middle|\; q \in \mathcal{U} \right\}.$$

The latter contradicts the following claim.

❶ *Given* $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \ldots, \alpha_n) \in \Delta^{n-1}$, *set*

$$h = \sum_i \alpha_i \cdot f^i \qquad h \colon \mathcal{U} \to \mathbb{R} \qquad z = \operatorname{MinPoint} h \in \mathcal{U}$$

$$\tilde{h} = \sum_i \alpha_i \cdot \tilde{f}^i \qquad \tilde{h} \colon \mathbb{M}^n[\kappa] \to \mathbb{R} \qquad \tilde{z} = \operatorname{MinPoint} \tilde{h} \in \mathbb{M}^n[\kappa]$$

Then $h(z) \leqslant \tilde{h}(\tilde{z})$.

Proof of the claim. Note that $\boldsymbol{d}_z h \geqslant 0$. Thus, for each i, we have[2]

$$(B)^0 \qquad \begin{aligned} 0 &\leqslant (\boldsymbol{d}_z h)(\operatorname{dir}[z y^i]) = \\ &= -\sum_j \alpha_j \cdot |z\,y^j| \cdot \cos \measuredangle[z\,{}^{y^i}_{y^j}] \leqslant \\ &\leqslant -\sum_j \alpha_j \cdot |z\,y^j| \cdot \cos \tilde{\measuredangle}^0(z\,{}^{y^i}_{y^j}) = \\ &= -\tfrac{1}{2 \cdot |z\,y^i|} \cdot \sum_j \alpha_j \cdot \left[|z\,y^i|^2 + |z\,y^j|^2 - |y^i\,y^j|^2\right]. \end{aligned}$$

In particular[3],

$$(C)^0 \qquad \sum_i \alpha_i \cdot \left[\sum_j \alpha_j \cdot \left[|z\,y^i|^2 + |z\,y^j|^2 - |y^i\,y^j|^2\right]\right] \leqslant 0,$$

[2]In case $\kappa = -1$, the same calculations give

$$(B)^- \qquad 0 \leqslant \ldots \leqslant -\tfrac{1}{\sinh |z\,y^i|} \cdot \sum_j \alpha_j \cdot \left[\cosh |z\,y^i| \cdot \cosh |z\,y^j| - \cosh |y^i\,y^j|\right].$$

[3]In case $\kappa = -1$, the same calculations give

$$(C)^- \qquad \sum_i \alpha_i \cdot \left[\sum_j \alpha_j \cdot \left[\cosh |z\,y^i| \cdot \cosh |z\,y^j| - \cosh |y^i\,y^j|\right]\right] \leqslant 0.$$

or[4]

$$(D)^0 \qquad 2\cdot h(z) \leqslant \sum_{i,j} \alpha_i\cdot\alpha_j\cdot|y^i\,y^j|^2.$$

Note, that if $\mathcal{U} \overset{iso}{=\!=} \mathbb{M}^n[\kappa]$, then all inequalities in (B, C, D) are sharp. Thus the same argument as above, repeated for $\tilde{x}^1, \tilde{x}^2, \dots, \tilde{x}^n \in \mathbb{M}^n[\kappa]$ gives[5]

$$(E)^0 \qquad 2\cdot\tilde{h}(\tilde{z}) = \sum_{i,j} \alpha_i\cdot\alpha_j\cdot|\tilde{x}^i\,\tilde{x}^j|^2.$$

Note that

$$|\tilde{x}^i\,\tilde{x}^j| \geqslant |x^i\,x^j| \geqslant |y^i\,y^j|$$

for all i, j. Thus, (D) and (E) imply the claim. △□

10.4.4. Kirszbraun's theorem for nonpositive bound. *Let $\kappa \leqslant 0$, $\mathcal{L} \in \mathrm{CBB}[\kappa]$, $\mathcal{U} \in \mathrm{CBA}[\kappa]$, $Q \subset \mathcal{L}$ be arbitrary subset and $f\colon Q \to \mathcal{U}$ be a short map. Then there is a short extension $F\colon \mathcal{L} \to \mathcal{U}$ of f; that is, there is a short map $F\colon \mathcal{L} \to \mathcal{U}$ such that $F|Q = f$.*

Remark. If $\mathcal{U}$ is proper, then in the following proof the Helly's theorem (10.3.1) is not needed. Everything follows directly from compactness of closed balls in $\mathcal{U}$.

Proof of 10.4.4. By Zorn's lemma, we can assume that $Q \subset \mathcal{L}$ is a maximal set; i.e. $f\colon Q \to \mathcal{U}$ does not admits a short extension to any larger set $Q' \supset Q$.

Let us argue by contradiction. Assume that $Q \neq \mathcal{L}$; choose $p \in \mathcal{L}\backslash Q$. Then

$$\bigcap_{x\in Q} \overline{\mathrm{B}}[f(x), |px|] = \varnothing.$$

Since $\kappa \leqslant 0$, the balls are convex; thus, by Helly's theorem (10.3.1), one can choose a point array $x^1, x^2, \dots, x^n \in Q$ such that

$$❷ \qquad \bigcap_{i=1}^{n} \overline{\mathrm{B}}[y^i, |x^i\,p|] = \varnothing,$$

[4]In case $\kappa = -1$,

$$(D)^- \qquad (h(z))^2 \leqslant \sum_{i,j} \alpha_i\cdot\alpha_j\cdot\cosh|y^i\,y^j|.$$

[5]In case $\kappa = -1$,

$$(E)^- \qquad (\tilde{h}(\tilde{z}))^2 = \sum_{i,j} \alpha_i\cdot\alpha_j\cdot\cosh|\tilde{x}^i\,\tilde{x}^j|.$$

where $y^i = f(x^i)$. Finally note that ❷ contradicts the Finite+one lemma (10.4.3) $\square$

Proof of Kirszbraun's theorem (10.4.1). The case $\kappa \leqslant 0$ is already proved in 10.4.4. Thus it remains to prove the theorem only in case $\kappa > 0$. After rescaling we may assume that $\kappa = 1$ and therefore $\varpi^\kappa = \pi$.

Since $\overline{\mathrm{B}}[z, \pi/2] \in \mathrm{CBA}[\kappa]$,we can assume $\mathcal{U} = \overline{\mathrm{B}}[z, \pi/2]$. In particular, any two points of $\mathcal{U}$ at distance $< \pi$ are joined by a geodesic, and $\operatorname{diam}\mathcal{U} \leqslant \pi$. If $|x\,y| = \pi$ for some $x, y \in \mathcal{U}$ then the concatenation of $[xz]$ and $[zy]$ forms a geodesic $[xy]$. Hence $\mathcal{U}$ is geodesic.

Further, we can also assume that $\operatorname{diam}\mathcal{L} \leqslant \pi$. Otherwise $\mathcal{L}$ is one-dimensional (see 8.7.1); in this case the result follows since $\mathcal{U}$ is geodesic.

Assume the theorem is false. Then there is a set $Q \subset \mathcal{L}$, a short map $f\colon Q \to \mathcal{U}$ and $p \in \mathcal{L}\backslash Q$ such that

❸ $$\bigcap_{x\in Q} \overline{\mathrm{B}}[f(x), |x\,p|] = \varnothing.$$

We will apply 10.4.4 for $\kappa = 0$ to Euclidean cones $\mathring{\mathcal{L}} = \operatorname{Cone}\mathcal{L}$ and $\mathring{\mathcal{U}} = \operatorname{Cone}\mathcal{U}$. Note that

- ⋄ $\mathring{\mathcal{U}} \in \mathrm{CBA}[0]$,
- ⋄ since $\operatorname{diam}\mathcal{L} \leqslant \pi$ we have $\mathring{\mathcal{L}} \in \mathrm{CBB}[0]$.

Further, we will view spaces $\mathcal{L}$ and $\mathcal{U}$ as unit spheres in $\mathring{\mathcal{L}}$ and $\mathring{\mathcal{U}}$ respectively. In the cones $\mathring{\mathcal{L}}$ and $\mathring{\mathcal{U}}$ we will use "$|{*}|$" for distance to the vertex, say o, "$\cdot$" for cone multiplication, "$\measuredangle(x, y)$" for $\angle[o\,{}^{x}_{y}]$ and "$\langle x, y\rangle$" for $|x|\cdot|y|\cdot\cos\angle[o\,{}^{x}_{y}]$. In particular,

- ⋄ $|x\,y|_{\mathcal{L}} = \measuredangle(x, y)$ for any $x, y \in \mathcal{L}$,
- ⋄ $|x\,y|_{\mathcal{U}} = \measuredangle(x, y)$ for any $x, y \in \mathcal{U}$,
- ⋄ for any $y \in \mathcal{U}$, we have

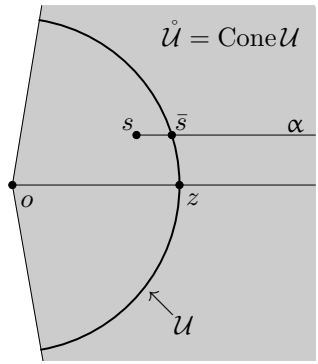
❹ $$\measuredangle(z, y) \leqslant \tfrac{\pi}{2}.$$

Set $\mathring{Q} = \operatorname{Cone} Q \subset \mathring{\mathcal{L}}$ and let $\mathring{f}\colon \mathring{Q} \to \mathring{\mathcal{U}}$ be the natural cone extension of f; i.e., $y = f(x) \Rightarrow t\cdot y = \mathring{f}(t\cdot x)$ for $t \geqslant 0$. Clearly $\mathring{f}$ is short.

Applying 10.4.4 for $\mathring{f}$, we get a short extension map $\mathring{F}\colon \mathring{\mathcal{L}} \to \mathring{\mathcal{U}}$. Set $s = \mathring{F}(p)$. Thus,

❺ $$|s\,\mathring{f}(w)| \leqslant |p\,w|$$

for any $w \in \mathring{Q}$. In particular, $|s| \leqslant 1$. Applying ❺ for $w = t\cdot x$ and $t \to \infty$ we get



❻ $\langle f(x), s\rangle \geqslant \cos\measuredangle(p,x)$

for any $x \in Q$.

According to ???, the geodesics $\mathrm{geod}_{[s\ t\cdot z]}$ converge as $t \to \infty$ to a ray, say $\alpha\colon [0,\infty) \to \mathring{\mathcal{U}}$. From ❹, we have that the function $t \mapsto \langle f(x), \alpha(t)\rangle$ is non-decreasing. From ❻, for the necessarily unique point $\bar{s}$ on the ray α such that $|\bar{s}| = 1$ we also have

$$\langle f(x), \bar{s}\rangle \geqslant \cos\measuredangle(p,x)$$

or

$$\measuredangle(\bar{s}, f(x)) \leqslant \measuredangle(p, f(x))$$

for any $x \in Q$. The latter contradicts ❸. □

10.5 (2*n*+2)-point comparison

Here we give a generalization of the (2+2)-point comparison to n points. It follows from the generalized Kirszbraun's theorem.

First let us give a reformulation of (2+2)-point comparison.

10.5.1. Reformulation of (2+2)-point comparison. *Let $\mathcal{X}$ be a metric space. A quadruple $p,q,x,y \in \mathcal{X}$ satisfies (2+2)-point comparison if one of the following holds:*

a) One of the triples (p,q,x) or (p,q,y) has perimeter $> 2\cdot\varpi^\kappa$.

b) If $[\tilde{p}\tilde{q}\tilde{x}] = \tilde{\triangle}^\kappa(pqx)$ and $[\tilde{p}\tilde{q}\tilde{y}] = \tilde{\triangle}^\kappa pqy$, then

$$|\tilde{x}\,\tilde{z}| + |\tilde{z}\,\tilde{y}| \geqslant |x\,y|,$$

for any $\tilde{z} \in [\tilde{p}\tilde{q}]$.

10.5.2. (2*n*+2)-point comparison. *Let $\mathcal{U} \in \mathrm{CBA}[\kappa]$. Consider $x,y \in \mathcal{U}$ and an array of pairs of points (p^1,q^1), $(p^2,q^2),\dots,(p^n,q^n)$ in $\mathcal{U}$, such that there is a model configuration $\tilde{x}$, $\tilde{y}$ and array of pairs $(\tilde{p}^1,\tilde{q}^1)$, $(\tilde{p}^2,\tilde{q}^2),\dots,(\tilde{p}^n,\tilde{q}^n)$ in $\mathbb{M}^3[\kappa]$ with the following properties:*

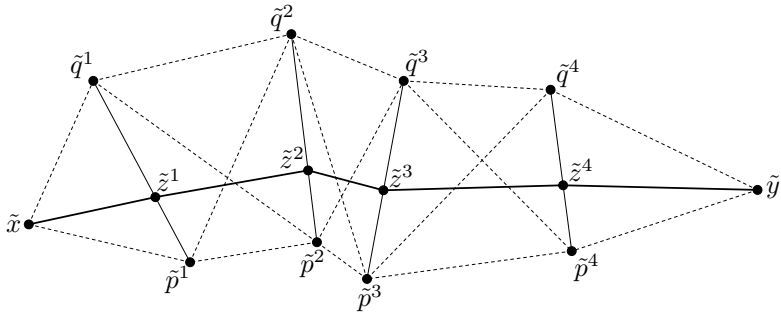
a) $[\tilde{x}\tilde{p}^1\tilde{q}^1] = \tilde{\triangle}^\kappa x p^1 q^1$ and $[\tilde{y}\tilde{p}^n\tilde{q}^n] = \tilde{\triangle}^\kappa y p^n q^n$;

b) The simplex $\tilde{p}^i\tilde{p}^{i+1}\tilde{q}^i\tilde{q}^{i+1}$ is a model simplex[6] of $p^ip^{i+1}q^iq^{i+1}$ for all i.

Then for any choice of n points $\tilde{z}^i \in [\tilde{p}^i\tilde{q}^i]$, we have

$$|\tilde{x}\,\tilde{z}^1| + |\tilde{z}^1\,\tilde{z}^2| + \ldots + |\tilde{z}^{n-1}\ \tilde{z}^n| + |\tilde{z}^n\,\tilde{y}| \geqslant |x\,y|.$$

[6] i.e., perimeter of each triple in p^i, p^{i+1}, q^i and q^{i+1} is $< 2\cdot\pi$ and $|\tilde{p}^i\,\tilde{q}^i| = |p^i\,q^i|$, $|\tilde{p}^i\,\tilde{p}^{i+1}| = |p^i\,p^{i+1}|$, $|\tilde{q}^i\,\tilde{q}^{i+1}| = |q^i\,q^{i+1}|$, $|\tilde{p}^i\,\tilde{q}^{i+1}| = |p^i\,q^{i+1}|$ and $|\tilde{p}^{i+1}\ \tilde{q}^i| = |p^{i+1}\ q^i|$.



To prove $(2n{+}2)$-point comparison, we need the following lemma, which is an easy corollary from Kirszbraun's theorem (10.4.1).

10.5.3. Lemma. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$, $\mathcal{U} \in \mathrm{CBA}[\kappa]$, and $Q \subset \mathrm{B}(p, \frac{\varpi^\kappa}{2}) \subset$ $\subset \mathcal{L}$. Then any short map $f\colon Q \to \mathcal{U}$ can be extended to a short map $F\colon \mathcal{L} \to \mathcal{U}$.*

Proof. Directly from Kirszbraun's theorem (10.4.4 or 10.4.1), we obtain the case $\kappa \leqslant 0$. Thus it remains to prove the theorem only in case $\kappa > 0$. After rescaling we may assume that $\kappa = 1$ and therefore $\varpi^\kappa = \pi$.

It is enough to prove that there is a point $z \in \mathcal{U}$ such that $|z\,f(x)| \leqslant$ $\leqslant \frac{\pi}{2}$ for all $x \in Q$; once it is proved, the statement follows from Kirszbraun's theorem (10.4.1).

Further we use the same notations as in the proof of 10.4.1.

Apply Kirszbraun's theorem (10.4.4 or 10.4.1) for $\mathring f\colon \mathring Q \to \mathring{\mathcal{U}}$ and set $q = \mathring F(p)$. Clearly,

$$\langle f(x), q\rangle \geqslant \cos\measuredangle(p,x) > 0$$

for any $x \in Q$. In particular, $|q| > 0$. Thus, for $z = \frac{1}{|q|}\cdot q \in \mathcal{U}$, we get $|z\,f(x)|_{\mathcal{U}} = \measuredangle(z, f(x)) \leqslant \frac{\pi}{2}$ for all $x \in Q$. $\square$

Proof of (2n+2)-point comparison. Direct application of 10.5.3 gives an array of short maps $f^0, f^1, \dots, f^n\colon \mathbb{M}^3[\kappa] \to \mathcal{U}$ such that

(i) $\tilde x \overset{f^0}{\longmapsto} x$, $\tilde p^1 \overset{f^0}{\longmapsto} p^1$ and $\tilde q^1 \overset{f^0}{\longmapsto} q^1$;

(ii) $\tilde p^i \overset{f^i}{\longmapsto} p^i$, $\tilde q^i \overset{f^i}{\longmapsto} q^i$ and $\tilde p^{i+1} \overset{f^i}{\longmapsto} p^{i+1}$, $\tilde q^{i+1} \overset{f^i}{\longmapsto} q^{i+1}$ for $1 \leqslant i \leqslant n-1$;

(iii) $\tilde p^n \overset{f^n}{\longmapsto} p^n$, $\tilde q^n \overset{f^n}{\longmapsto} q^n$ and $\tilde y \overset{f^n}{\longmapsto} y$.

For each $i > 0$, we have that $f^{i-1}|_{[\tilde p^i \tilde q^i]} = f^i|[\tilde p^i \tilde q^i]$ as both f^{i-1} and f^i send $[\tilde p^i \tilde q^i]$ isometrically to a geodesic $[p^i q^i]$ in $\mathcal{U}$ which is unique. Thus the curves

$$f^0([\tilde x \tilde z^1]),\ f^1([\tilde z^1 \tilde z^2]), \dots,\ f^{n-1}([\tilde z^{n-1} \tilde z^n]),\ f^n([\tilde z^n \tilde y])$$

can be joined in $\mathcal{U}$ into a curve connecting x to y with length at most

$$|\tilde{x}\,\tilde{z}^1| + |\tilde{z}^1\,\tilde{z}^2| + \ldots + |\tilde{z}^{n-1}\;\tilde{z}^n| + |\tilde{z}^n\,\tilde{y}|.$$

□

10.6 Remarks and open problems

10.6.1. Open problem. *Find a necessary and sufficient condition for a finite metric space to be isometrically embeddable into some* CBB[κ]*-space.*

A metric on a finite set $\{a^1, a^2, \ldots, a^n\}$, can be described by the matrix with components

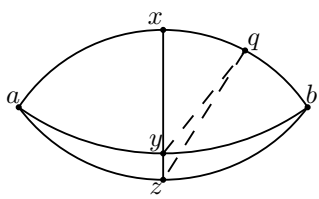
$$s^{ij} = |a^i\,a^j|^2,$$

which we will call the *decrypting matrix*. The set of decrypting matrices of all metrics that admit a distance preserving map into a CBB[0]-space form a convex cone, as follows from the fact that the product of CBB[0]-spaces is a CBB[0]-space. This convexity gives hope that the cone admits an explicit description.

The set of metrics on $\{a^1, a^2, \ldots, a^n\}$ that can be embedded into a product of spheres with different radii admits a simple description. Obviously, this gives a sufficient condition for 10.6.1. This condition is not necessary. For instance, as follows from [163, 2.2], a sufficiently dense finite subset in a generic closed positively curved manifold can not be embedded into a product of spheres.

Theorem 10.2.1 gives a necessary condition for 10.6.1, but the condition is not sufficient. One can see this in the following example constructed by Sergei Ivanov:

Example. Consider the finite set $\mathcal{F}$ formed by 6 points a, b, x, y, z and q with distances defined as follows:



1. $|a\,b| = 4$;
2. $|a\,x| = |a\,y| = |a\,z| = |b\,x| = |b\,y| = |b\,z| = 2$;
3. $|x\,y| = 2$, $|y\,z| = 1$, $|x\,z| = 3$;
4. $|x\,q| = |q\,b| = 1$ and thus $|a\,q| = 3$;
5. $\measuredangle^0(x\,{}^q_y) = \measuredangle^0(x\,{}^q_z) = \frac{\pi}{3}$; i.e., $|q\,y| = \sqrt{3}$ and $|q\,z| = \sqrt{7}$.

On the diagram the degenerate triangles are marked by solid lines. Note that if one removes from $\mathcal{F}$ the point q then the remaining part can be embedded in a sphere of intrinsic diameter 4 with poles at a and b and the points x, y, z on the equator. On the other hand, if one removes the point a from the space and changes the distance $|z\,b|$ then it can be isometrically embedded into the plane.

It is straightforward to check that this finite set satisfies the conclusion of Theorem 10.2.1 for $\kappa = 0$. However, if such a metric appeared as an inherited metric on a subset $\{a, b, x, y, z, q\} \subset \mathcal{L} \in \mathrm{CBB}[0]$ then clearly

$$\measuredangle[x\,{}^{a}_{y}] = \measuredangle[y\,{}^{a}_{z}] = \measuredangle[y\,{}^{b}_{z}] = \tfrac{\pi}{3},$$

contradicting $|b\,z| = 2$.

The following problem was mentioned in [66, 15(b)]

10.6.2. Open problem. *Describe metrics on an n-point set which are embeddable into* $\mathrm{CBA}[\kappa]$*-spaces.*

The set of metrics on $\{a^1, a^2, \dots, a^n\}$ which can be embedded into a product of trees and hyperbolic spaces admits a simple description using decrypting matrices defined above. Obviously, this gives a sufficient condition for problem 10.6.2. This condition is not necessary. The existence of a counterexample follows again from [163, 2.2]; it is sufficient to take a sufficiently dense finite subset in a ball in a generic Hadamard space.

The $(2n{+}2)$-point comparison (10.5.2) gives a necessary condition for 10.6.2 which is not sufficient. One can see this in the following example constructed by Nina Lebedava:

Consider a square $[\tilde{x}^1\tilde{y}^1\tilde{x}^2\tilde{y}^2]$ in $\mathbb{E}^3$ with yet two points in general position $\tilde{z}^1$, $\tilde{z}^2$ so that the convex hull of $\tilde{x}^1, \tilde{x}^2, \tilde{y}^1, \tilde{y}^2, \tilde{z}^1, \tilde{z}^2$ forms a nonregular octahedron with the faces formed by triangles $[\tilde{x}^i\tilde{y}^j\tilde{z}^k]$. Consider the induced metric on the 6-point set $\tilde{x}^1, \tilde{x}^2, \tilde{y}^1, \tilde{y}^2, \tilde{z}^1, \tilde{z}^2$. Note that if we increase the $|\tilde{z}^1\,\tilde{z}^2|$ slightly then in the obtained 6-point metric $\mathcal{F}_6$ space all the $(2{+}2)$ and $(4{+}2)$-point comparisons continue to hold.

Now assume we embed the points $x^1, x^2, y^1, y^2, z^1, z^2$ lie in a $\mathrm{CBA}[0]$ space $\mathcal{U}$ in such a way that all the distances except $|z^1\,z^2|$ are the same as between corresponding points in $\mathcal{F}_6$. Since $[\tilde{x}^1\tilde{y}^1\tilde{x}^2\tilde{y}^2]$ is a square, by Exercise 9.12.10, we get that $\mathcal{U}$ contains an isometric copy of a square $\mathrm{Conv}(x^1, y^1, x^2, y^2)_{\mathcal{U}} \overset{iso}{=\!=} \mathrm{Conv}(\tilde{x}^1, \tilde{y}^1, \tilde{x}^2, \tilde{y}^2)_{\mathbb{E}^3}$. Let

$$\tilde{w} \in \mathrm{Conv}(\tilde{x}^1, \tilde{y}^1, \tilde{x}^2, \tilde{y}^2)_{\mathbb{E}^3}$$

and w be corresponding point in $\mathrm{Conv}(x^1, y^1, x^2, y^2)_{\mathcal{U}}$. By point-on-side comparison (9.3.1c) we have $|z^i\,w|_{\mathcal{U}} \leqslant |\tilde{z}^i\,\tilde{w}|_{\mathbb{E}^3}$. It follows that

$$|z^1\,z^2|_{\mathcal{U}} \leqslant |\tilde{z}^1\,\tilde{z}^2|_{\mathbb{E}^3},$$

a contradiction.

10.6.3. Exercise. *Consider a metric space $\mathcal{F}_s$ on 5 point set p, q, x, y, z such that $|pq| = s$ and all the remaining distances are equal 1. Assume $\mathcal{F}_s$ admits a distance preserving map into a* CBA[0] *space. Show that $\mathcal{F}_s$ admits a distance preserving map into a* CBB[0] *space.*

The following conjecture (if true) would give the right generality for Kirszbraun's theorem (10.4.1). Roughly it states that the example 10.4.2, is the only obstacle for extending short map.

10.6.4. Conjecture. *Assume $\mathcal{L} \in \mathrm{CBB}[1]$, $\mathcal{U} \in \mathrm{CBA}[1]$, $Q \subset \mathcal{L}$ is a proper subset and $f\colon Q \to \mathcal{U}$ be a short map, which does not admit a short extension to any bigger set $Q' \supset Q$. Then*

a) Q is isometric to a sphere in a Hilbert space (of finite or cardinal dimension). Moreover, there is a point $p \in \mathcal{L}$ such that $|pq| = \frac{\pi}{2}$ for any $q \in Q$.

b) The map $f\colon Q \to \mathcal{U}$ is a distance preserving map and there is no point $p' \in \mathcal{U}$ such that $|p'q'| = \frac{\pi}{2}$ for any $q' \in f(Q)$.

10.7 Exercises

10.7.1. Exercise. *Let $\mathcal{U} \in \mathrm{CBA}[0]$ Assume there are two point arrays $x^0, x^1, \dots, x^k \in \mathcal{U}$ and $\tilde{x}^0, \tilde{x}^1, \dots, \tilde{x}^k \in \mathbb{E}^m$ such that $|x^i x^j|_{\mathcal{U}} = |\tilde{x}^i \tilde{x}^j|_{\mathbb{E}^m}$ for each i, j and for any point $z_0 \in \mathcal{U}$, we have $|z_0 x_i| \geqslant |x_0 x_i|$ for at least one $i > 0$.*

Prove that there is a subset $Q \subset \mathcal{L}$ which is isometric to a convex set in $\mathbb{E}^m$ and which contains all points x^i.

10.7.2. Exercise. *Let $\mathcal{L} \in \mathrm{CBB}[0]$, $x^0, x^1, \dots, x^k \in \mathcal{L}$ and $\tilde{x}^0, \tilde{x}^1, \dots, \tilde{x}^k \in \mathbb{E}^m$ be two point arrays such that $|x^i x^j|_{\mathcal{L}} = |\tilde{x}^i \tilde{x}^j|_{\mathbb{E}^m}$ for each i, j. Assume $\tilde{x}^0$ lies in the interior of $\mathrm{Conv}(\tilde{x}^1, \dots, \tilde{x}^k)$.*

Prove that there is a subset $Q \subset \mathcal{L}$ which is isometric to a convex set in $\mathbb{E}^m$ and which contains all points x^i.

10.7.3. Flat triangle is not flat. *Construct a space $\mathcal{L} \in \mathrm{CBB}^3[0]$ with a triangle $[xyz]$ in $\mathcal{L}$ with all angles equal to the corresponding angles of its model triangle, but which can not be filled by an isometric copy of the model triangle.*

10.7.4. Exercise. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$ and $p, a^0, a^1, \dots, a^m \in \mathcal{L}$. Assume that a point array $(a^0, a^1, \dots, a^k)$ in $\mathcal{L}$ is* κ-strutting *for a point $p \in \mathcal{L}$. Show that there are point $\tilde{p}, \tilde{a}^0, \dots, \tilde{a}^m \in \mathbb{M}^{m+1}[\kappa]$ such that*

$$|\tilde{p}\tilde{a}^i| = |pa^i| \text{ and } |\tilde{a}^i \tilde{a}^j| = |a^i a^j|$$

for all i and j.

The following statement was first observed in [77]

10.7.5. Exercise. *We say that a space $\mathcal{X}$ is a* Kirszbraun target *if for an arbitrary metric space $\mathcal{Z}$ and a subset $Q \subset \mathcal{Z}$, any short map $Q \to \mathcal{X}$ can be extended as a short map $\mathcal{Z} \to \mathcal{X}$.*

1. *Prove that any metric space $\mathcal{X}$ can be isometrically embedded into a Kirszbraun target.*
2. *Use it to construct an analog of convex hull in the category of metric spaces.*

10.7.6. Exercise. *We say that a compact space $\mathcal{X}$ is a* Kirszbrun source *if for an arbitrary complete metric space $\mathcal{Z}$ and subset $Q \subset \mathcal{X}$, any short map $Q \to \mathcal{Z}$ can be extended to a short map $\mathcal{X} \to \mathcal{Z}$. Prove that a metric space $\mathcal{X}$ is a Kirszbrun source if and only if it satisfies ultratriangle inequality for all triples of points; i.e.*

$$|x\,z| \leqslant \max\{|x\,y|, |y\,z|\}$$

for any $x, y, z \in \mathcal{X}$.

10.7.7. Exercise. *Assume that a Riemannian manifold $\mathcal{R}$ satisfies the following condition: for an arbitrary subset $Q \subset \mathcal{R}$, any short map $Q \to \mathcal{R}$ can be extended to a short map $\mathcal{R} \to \mathcal{R}$. Show that $\mathcal{R}$ has constant sectional curvature.*

10.7.8. Exercise. *Assume $\mathcal{X}$ is a complete length space which satisfy the following condition: any 4-point subset can be isometrically embedded in Euclidean 3-space.*

Prove that $\mathcal{X}$ is isometric to a closed convex substet of a Hilbert space.

Part III

Constructions

Chapter 11

Warped products

Warped products is a construction which produce a new metric space denoted as $\mathcal{B} \times_f \mathcal{F}$ from two metric spaces provide $\mathcal{B}$, $\mathcal{F}$ which are called base and fiber of the warped product and a function $f\colon \mathcal{B} \to \to \mathbb{R}_{\geqslant 0}$.

Many important construction, as direct product, cone, spherical suspension and joints will be defined using warped products.

11.1 Definitions

Consider length spaces $\mathcal{B}$ and $\mathcal{F}$, and a continuous function $f : \mathcal{B} \to \to \mathbb{R}_{\geqslant 0}$.

Given a path γ in $\mathcal{B} \times \mathcal{F}$, denote by $\gamma_{\mathcal{B}}$ and $\gamma_{\mathcal{F}}$ its projections to $\mathcal{B}$ and $\mathcal{F}$; so $\gamma(t) = (\gamma_{\mathcal{B}}(t), \gamma_{\mathcal{F}}(t)) \in \mathcal{B} \times \mathcal{F}$ for any $t \in [0, 1]$.

According to Rademacher's theorem **??**, speed of any Lipschitz curve is defined almost everywhere.

For a Lipschitz path γ, let us define its f-speed as

$$\operatorname{speed}_f \gamma(t) \overset{def}{=\!=} \sqrt{[\operatorname{speed} \gamma_{\mathcal{B}}(t)]^2 + [f(\gamma_{\mathcal{B}}(t)) \cdot \operatorname{speed} \gamma_{\mathcal{F}}(t)]^2}.$$

and set

❶ $$\operatorname{length}_f \gamma \overset{def}{=\!=} \int\limits_0^1 \operatorname{speed}_f \gamma(t) \cdot \boldsymbol{d}t$$

Consider the f-pseudometric on $\mathcal{B} \times \mathcal{F}$ defined as the infimum of f-lengths of Lipschitz curves connecting the points. That is

$$|x\,y|_f \overset{def}{=\!=} \inf\{\, \operatorname{length} \gamma \mid \gamma(0) = x, \gamma(1) = y \,\}$$

Note that $|xy|_f = 0$ if and only if $x = y$ or both $x, y \in b \times \times \mathcal{F}$ for some $b \in \mathcal{B}$ such that $f(b) = 0$. Set $x \sim y$ iff $|xy|_f = 0$. The pseudometric $|{*}{*}|_f$ becomes a metric on the equivalence classes of "$\sim$". The obtained metric space will be called warped product of $\mathcal{B}$ and $\mathcal{F}$ with the warping function f; it will be denoted by $\mathcal{B} \times {}_f\mathcal{F}$.

From Fiber-independence theorem (11.2.3) follows that if $\iota\colon A \to \check{A}$ is an isometry between two subsets $A \subset \mathcal{F}$ and $\check{A} \subset \check{\mathcal{F}}$ in the length spaces than for any warping function $f\colon \mathcal{B} \to \mathbb{R}_{\geqslant 0}$, the map ι induces an isometry between sets $A \times {}_f\mathcal{B} \subset \mathcal{F} \times {}_f\mathcal{B}$ and $\check{A} \times {}_f\mathcal{B} \subset \check{\mathcal{F}} \times {}_f\mathcal{B}$.

The later observation makes possible to define warped product $\mathcal{F} \times \times {}_f\mathcal{B}$ where the fiber $\mathcal{F}$ does not have length metric. Ideed according to ???, any metric space is a subspace $\mathcal{F}$ of a length space say $\mathcal{F}'$. Therefore we can take the warped product $\mathcal{F}' \times {}_f\mathcal{B}$ and identify $\mathcal{F} \times {}_f\mathcal{B}$ with its subspace formed all pairs (φ, b) such that $\varphi \in \mathcal{F}$. According to Fiber-independence theorem (11.2.3) the resultig space does not depend on the chaice of $\mathcal{F}'$.

11.2 Basic propetries

We will refer to $\mathcal{B}$ and $\mathcal{F}$ as *base* and *fiber* respectively; and to $\mathcal{B} \times \times \{\varphi_0\}$ for $\varphi_0 \in \mathcal{F}$ and $\{p_0\} \times \mathcal{F}$ for $p_0 \in \mathcal{B} - f^{-1}(0)$ as *horizontal* and *vertical leaves.*

11.2.1. Proposition. *The warped product $\mathcal{B} \times {}_f\mathcal{F}$ satisfies:*

a) The projection $(p, \varphi_0) \mapsto p$ of any horizontal leaf $\mathcal{B} \times \{\varphi_0\}$, with its length-metric, is an isometry onto $\mathcal{B}$.

b) The projection $(p_0, \varphi) \mapsto \varphi$ of any vertical leaf $\{p_0\} \times \times \mathcal{F}$, with its length-metric, is a homothety onto $\mathcal{F}$ with multiplier $1/f(p_0)$.

c) Each horizontal leaf $\mathcal{B} \times \{\varphi_0\}$ is isometrically embedded in $\mathcal{B} \times {}_f\mathcal{F}$.

d) If f achieves its minimum at p_0, then the vertical leaf $\{p_0\} \times \mathcal{F}$ is isometrically embedded in $\mathcal{B} \times {}_f\mathcal{F}$.

Proof. Claims (a), (b) and (d) are immediate from the length formula ❶.

Also by (❶), the projection of $\mathcal{B} \times {}_f\mathcal{F}$ onto $\mathcal{B} \times \{\varphi_0\}$ given by $(p, \varphi) \mapsto (p, \varphi_0)$ is length-nonincreasing, as is the projection onto $\{p_0\} \times \mathcal{F}$ given by $(p, \varphi) \mapsto (p_o, \varphi)$ if p_0 is a local minimum point of f. Hence (c) and (d). □

A horizontal leaf need not be convex even if $\mathcal{B} \times {}_f\mathcal{F}$ is a geodesic space, since vanishing of the warping function f allows geodesics to bifurcate into distinct horizontal leaves (see Proposition **??** (**??**)). For

instance, suppose $\alpha : [0,1] \to \mathcal{B}$ is a geodesic of $\mathcal{B}$ satisfying $\{0,1\} \subset \subset \alpha^{-1}(Z) \neq [0,1]$, where Z is the zero set of f. Then for any distinct $\varphi_1, \varphi_2 \in \mathcal{F}$, the geodesic (α, φ_2) of $\mathcal{B} \times_f \mathcal{F}$ has its endpoints in $\mathcal{B} \times \times \{\varphi_1\}$ but does not lie in $\mathcal{B} \times \{\varphi_1\}$.

11.2.2. Exersice. *Show that orbits of isometric cohomogenity 1 group action on Riemannian manifold form a vertical fiber of a warped product.*

Distance in a warped product is fiber-independent, in the sense that distances may be calculated by substituting for $\mathcal{F}$ a different length space:

11.2.3. Fiber-independence theorem. *Consider length spaces $\mathcal{B}$, $\mathcal{F}$ and $\check{\mathcal{F}}$, and a continuous function $f : \mathcal{B} \to \mathbb{R}_{\geqslant 0}$. Assume $p, q \in \mathcal{B}$, $\varphi, \psi \in \mathcal{F}$ and $\check{\varphi}, \check{\psi} \in \check{\mathcal{F}}$: Then*

$$|\varphi\,\psi|_{\mathcal{F}} \geqslant |\check{\varphi}\,\check{\psi}|_{\check{\mathcal{F}}}$$

❶ $$\Downarrow$$

$$|(p,\varphi)\ (q,\psi)|_{\mathcal{B}\times_f\mathcal{F}} \geqslant |(p,\check{\varphi})\ (q,\check{\psi})|_{\mathcal{B}\times_f\check{\mathcal{F}}}$$

Proof. Fix a path $\gamma = (\gamma_{\mathcal{B}}, \gamma_{\mathcal{F}})$ from (p,φ) to (q,ψ).

Since $|\varphi\,\psi|_{\mathcal{F}} \geqslant |\check{\varphi}\,\check{\psi}|_{\check{\mathcal{F}}}$, there is a Lipschitz path $\gamma_{\check{\mathcal{F}}}$ from $\check{\varphi}$ to $\check{\psi}$ in $\check{\mathcal{F}}$ such that

$$(\operatorname{speed}\gamma_{\mathcal{F}})(t) \geqslant (\operatorname{speed}\gamma_{\check{\mathcal{F}}})(t)$$

for almost all $t \in [0,1]$.

Consider the path $\check{\gamma} = (\gamma_{\mathcal{B}}, \gamma_{\check{\mathcal{F}}})$ from $(p,\check{\varphi})$ to $(q,\check{\psi})$ in $\mathcal{B} \times_f \check{\mathcal{F}}$. Clearly

$$\operatorname{length}_f \gamma \geqslant \operatorname{length}_f \check{\gamma}.$$

Hence the statement follows. □

11.2.4. Exercise. *Let $\mathcal{B}$ and $\mathcal{F}$ be two length spaces and $f, g\ \mathcal{B} \to \to \mathbb{R}_{\geqslant}$ be two continuous nonnegative function. Assume $f(b) \leqslant g(b)$ for any $b \in \mathcal{B}$. Show that $\mathcal{B} \times_f \mathcal{F} \leqslant \mathcal{B} \times_g \mathcal{F}$; that is there is a distance noncontructing map $\mathcal{B} \times_f \mathcal{F} \to \mathcal{B} \times_g \mathcal{F}$.*

11.3 Examples

Direct product. The simplest example is the *direct product* $\mathcal{B} \times \mathcal{F}$ which could be also written as a warped product $\mathcal{B} \times_1 \mathcal{F}$. For $p, q \in \mathcal{B}$ and $x, y \in \mathcal{F}$, the direct product metric simplifies to

$$|(p,x)\ (q,y)| = \sqrt{|p\,q|^2 + |x\,y|^2}.$$

This is taken as the defining formula for the direct product of two arbitrary metric spaces $\mathcal{B}$ and $\mathcal{F}$.

Cones. The *Euclidean cone* $\operatorname{Cone}\mathcal{F}$ over space $\mathcal{F}$ can be defined as a warped product $[0,\infty)\times{}_{\mathrm{id}}\mathcal{F}$. For $s,t\in[0,\infty)$ and $x,y\in\mathcal{F}$, the cone metric is given by cosine rule

$$|(p,x)\ (q,y)| = \sqrt{s^2+t^2-2\cdot s\cdot t\cdot\cos\alpha},$$

where $\alpha=\max\{\pi,|x\,y|\}$.

Instead Euclidean cosine rule, one use the cosine rule in $\mathbb{M}^2[\kappa]$

$$|(p,x)\ (q,y)| = \tilde{\Upsilon}^{\kappa}\{\alpha;s,t\}.$$

This way we get so called κ-cones over $\mathcal{F}$ denoted by $\operatorname{Cone}^{\kappa}\mathcal{F}=[0,\infty)\times{}_{\operatorname{sn}^{\kappa}}\mathcal{F}$ for $\kappa\leqslant 0$ and $\operatorname{Cone}^{\kappa}\mathcal{F}=[0,\varpi^{\kappa}]\times{}_{\operatorname{sn}^{\kappa}}\mathcal{F}$ for $\kappa>0$.

For 1-cone $\operatorname{Cone}^1\mathcal{F}$ is also called *spherical suspension* over $\mathcal{F}$ and it is also denoted $\operatorname{Susp}\mathcal{F}$. That is

$$\operatorname{Susp}\mathcal{F}=[0,\pi]\times{}_{\sin}\mathcal{F}.$$

11.3.1. Exercise. *Let $\mathcal{F}$ be a length space and $A\subset\mathcal{F}$. Show that* $\operatorname{Cone}^{\kappa}A$ *is convex in* $\operatorname{Cone}^{\kappa}\mathcal{F}$ *if and only if A is π convex in $\mathcal{F}$.*

The elements of the Euclidean cone $\mathcal{K}=\operatorname{Cone}\mathcal{F}$ will be ofter refered as *vector*. The tip of the cone $\mathcal{K}$ usually denoted as o or $o_{\mathcal{K}}$ The *absolute value* $|w|$ of the vector w is defiened as $|o\,w|_{\mathcal{K}}$; that is the distance from w to the origin. The *scalar product* $\langle v,w\rangle$ of two vectors $v,w\in\mathcal{K}$ are defined as

$$\langle v,w\rangle \overset{def}{=\!=} |v\,w|^2_{\mathcal{K}}-|v|^2-|w|^2.$$

Doubling. The Doubling space $\mathcal{W}$ of space $\mathcal{V}$ is a closed subset $A\subset\subset\mathcal{V}$ can be also defined as a special type of warped product. Consider 2-point space $\mathcal{F}$ with distance π from each other. Then $\mathcal{W}$ is isometric to the warped product with base $\mathcal{V}$, fiber $\mathcal{F}$ and warping function $|A_*|$; that is

$$\mathcal{W}\overset{iso}{=\!=}\mathcal{F}\times{}_{|A_*|}\mathcal{V}.$$

11.4 1-dimensional base

The following theorem is a baby case of Theorem ???; it provides conditions for the spaces and functions in the warped product with 1-dimensional base.

11.4.1. Theorem.

a) If $\mathcal{L} \in \mathrm{CBB}[1]$ *and* $\operatorname{diam}\mathcal{L} \leqslant \pi$ *then*

$$\begin{aligned}\operatorname{Susp}\mathcal{L} &= \mathcal{L} \times {}_{\sin}[0,\pi] \in \mathrm{CBB}[1],\\ \operatorname{Cone}\mathcal{L} &= \mathcal{L} \times {}_{\mathrm{id}}[0,\infty) \in \mathrm{CBB}[0],\\ \operatorname{Cone}^{-1}\mathcal{L} &= \mathcal{L} \times {}_{\sinh}[0,\infty) \in \mathrm{CBB}[-1].\end{aligned}$$

Moreover the converse also holds in each of three cases.

b) If $\mathcal{L} \in \mathrm{CBB}[0]$ *then*

$$\begin{aligned}\mathcal{L} \times \mathbb{R} &\in \mathrm{CBB}[0],\\ \mathcal{L} \times {}_{\exp}\mathbb{R} &\in \mathrm{CBB}[-1].\end{aligned}$$

Moreover the converse also holds in each of two cases.

c) If $\mathcal{L} \in \mathrm{CBB}[-1]$ *then*

$$\mathcal{L} \times {}_{\cosh}\mathbb{R} \in \mathrm{CBB}[-1].$$

Moreover the converse also holds.

11.4.2. Theorem. *Let* $\mathcal{L}$ *be a length space.*

a) If $\mathcal{L} \in \mathrm{CBA}[1]$ *then*

$$\begin{aligned}\operatorname{Susp}\mathcal{L} &= \mathcal{L} \times {}_{\sin}[0,\pi] \in \mathrm{CBA}[1],\\ \operatorname{Cone}\mathcal{L} &= \mathcal{L} \times {}_{\mathrm{id}}[0,\infty) \in \mathrm{CBA}[0],\\ \operatorname{Cone}^{-1}\mathcal{L} &= \mathcal{L} \times {}_{\sinh}[0,\infty) \in \mathrm{CBA}[-1].\end{aligned}$$

Moreover, the converse also holds in each of three cases.

b) If $\mathcal{L} \in \mathrm{CBA}[0]$ *then*

$$\begin{aligned}\mathcal{L} \times \mathbb{R} &\in \mathrm{CBA}[0],\\ \mathcal{L} \times {}_{\exp}\mathbb{R} &\in \mathrm{CBA}[-1].\end{aligned}$$

Moreover the converse also holds in each of two cases.

c) If $\mathcal{L} \in \mathrm{CBA}[-1]$ *then*

$$\mathcal{L} \times {}_{\cosh}\mathbb{R} \in \mathrm{CBA}[-1].$$

Moreover the converse also holds.

In the proof of the above two theorems we will use the following proposition.

11.4.3. Proposition.

a)

$$\begin{aligned}\operatorname{Susp}\mathbb{S}^{m-1} &= \mathbb{S}^{m-1}\times_{\sin}[0,\pi] \overset{iso}{=\!=} \mathbb{S}^m,\\ \operatorname{Cone}\mathbb{S}^{m-1} &= \mathbb{S}^{m-1}\times_{\mathrm{id}}[0,\infty) \overset{iso}{=\!=} \mathbb{E}^m,\\ \operatorname{Cone}^{-1}\mathbb{S}^{m-1} &= \mathbb{S}^{m-1}\times_{\sinh}[0,\infty) \overset{iso}{=\!=} \mathbb{M}^m[-1].\end{aligned}$$

b)

$$\begin{aligned}\mathbb{E}^{m-1}\times\mathbb{R} &\overset{iso}{=\!=} \mathbb{E}^m,\\ \mathbb{E}^{m-1}\times_{\exp}\mathbb{R} &\overset{iso}{=\!=} \mathbb{M}^m[-1].\end{aligned}$$

c)

$$\mathbb{M}^{m-1}[-1]\times_{\cosh}\mathbb{R} \overset{iso}{=\!=} \mathbb{M}^m[-1].$$

To prove is left to the reader.

Proof of 11.4.2. Let us prove the last statement in (a); the rest of the statements are similar; each proof is based on Fiber-independence theorem 11.2.3 and the correspondg statement in the Proposition 11.4.3.

Choose arbitrary quadruple of points

$$(p,s),(x^1,t^1),(x^2,t^2),(x^3,t^3)\in\mathcal{L}\times_{\sinh}[0,\infty).$$

Since $\operatorname{diam}\mathcal{L}\leqslant\pi$, $(1{+}n)$-point comparison (10.2.1) provides a quadruple of points $q,y^1,y^2,y^3\in\mathbb{S}^3$ such that

$$|q\,y^i|_{\mathbb{S}^3}=|p\,x^i|_{\mathcal{L}}$$

and

$$|y^i\,y^j|_{\mathbb{S}^3}\geqslant|x^i\,x^j|_{\mathcal{L}}$$

for all i and j.

According to Proposition 11.4.3a,

$$\operatorname{Cone}^{-1}\mathbb{S}^2=\mathbb{S}^2\times_{\sinh}[0,\infty)\overset{iso}{=\!=}\mathbb{M}^3[-1].$$

Consider the quadruple points

$$(q,s),(y^1,t^1),(y^2,t^2),(y^3,t^3)\in\mathbb{S}^2\times_{\sinh}[0,\infty)\overset{iso}{=\!=}\mathbb{M}^3[-1].$$

By Fiber-independence theorem 11.2.3

$$|(q,s)\ (y^i,t^i)|_{\mathbb{S}^2}=|(p,s)\ (x^i,t^i)|_{\mathcal{L}}$$

and

$$|(y^i,t^i)\ (y^j,t^j)|_{\mathbb{S}^2}\geqslant|(x^i,t^i)\ (x^j,t^j)|_{\mathcal{L}}$$

for all i and j. It remains to apply Exercise 8.1.3. $\square$

Chapter 12

Subsets

12.1 Motivating examples

Consider three subgraphs of different paraboloids

$$\begin{aligned} A &= \left\{ (x,y,z) \in \mathbb{E}^3 \;\middle|\; z \leqslant x^2 + y^2 \right\}, \\ B &= \left\{ (x,y,z) \in \mathbb{E}^3 \;\middle|\; z \leqslant -x^2 - y^2 \right\}, \\ C &= \left\{ (x,y,z) \in \mathbb{E}^3 \;\middle|\; z \leqslant x^2 - y^2 \right\}. \end{aligned}$$

Here is a question which motivates the rest of the chapter.

12.1.1. Question. *Which of the sets A, B and C, if equipped with the induced length metric, form a* CBA[0] *space and why?*

The answers is given below, but it is instructive to think about this question before reading further.

A. No, $A \notin \mathrm{CBA}[0]$.

The boundary ∂A is the parabolid described by $z = x^2 + y^2$ in particular it bounds an open convex set in $\mathbb{E}^3$ which complement is A. The closest point projection of $A \to \partial A$ is short (see 9.6.3). It follows that ∂A is a convex set in A equipped with the induced length metric.

Therefore if A forms a CBA[0] then so is ∂A. The latter is not true, ∂A is a smooth convex surface; by Gauss formula it has strictly positive curvature.

B. Yes, $B \in \mathrm{CBA}[0]$.

Evidently B is a convex closed set in $\mathbb{E}^3$. Therefore length metric on B coinsides with Euclidean metric and (2+2)-comparison holds.

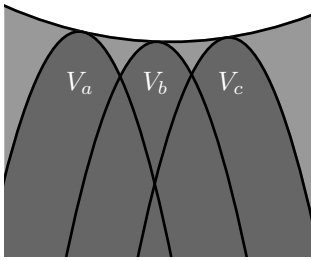
C. Yes, $C \in \mathrm{CBA}[0]$, but the proof is not as easy as before. We give a sketch here; a complete proof of more general statement is given in Section 12.3.

Consider the one-parameter family of sets

$$V_t = \left\{ (x,y,z) \in \mathbb{E}^3 \;\middle|\; z \leqslant x^2 - y^2 - 2\cdot(x-t)^2 \right\}.$$

Each set V_t is closed, convex and

$$C = \bigcup_t V_t.$$



Further note that

❶ $$V_b \supset V_a \cap V_c$$

for any choice real numbers $a < b < c$.

The inclusion ❶ makes possible to apply recursively Reshetnyak gluing theorem 9.9.1 and show that any finite union of V_t forms a CBA[0] space. By approximation, the (2+2)-point comparison holds for any 4 points in C; that is, $C \in \mathrm{CBA}[0]$.

Remark. The set C is not convex, but it is two-convex as defined in the next section. As you will see two-convexity of subsets is closely related to the inheritance of upper curvature bound by a subset.

12.2 Two-convexity

The following definition is closely related to the one given by Gromov in [65, §$\frac{1}{2}$], see also [121].

12.2.1. Definition. *We say that a subset $K \subset \mathbb{E}^m$ is* two-convex *if the following condition holds for any plane $W \subset \mathbb{E}^m$. If γ is a closed curve in $W \cap K$ which is null-homotopic in K then it is null-homotopic in $W \cap K$.*

The following proposition follows immediately from the definition.

12.2.2. Proposition. *Any subset in $\mathbb{E}^2$ is 2-convex.*

The following proposition describes closed two-convex sets with smooth boundary.

12.2.3. Proposition. *Let $K \subset \mathbb{E}^m$ be a closed subset.*

Assume that the boundary of K forms a smooth hypersurface S. Consider the orthonormal vector field ν on S which points outside of K. Denote by $k_1 \leqslant \ldots \leqslant k_{m-1}$ the scalar fields of principal curvatures on S with respect to ν.

Then K is two-convex if and only if $k_2(p) \geqslant 0$ at any point $p \in S$. Moreover, if $k_2(p) < 0$ at some point p then K the Definition 12.2.1 fails for a curve γ forming a triangle.

Proof; "if" part. If $k_2(p) < 0$ for some $p \in S$, consider the plane W through p which spanned by the first two principle directions. Choose a small triangle in W which surrounds p and move it slightly in the direction of $\nu(p)$. We get a triangle $[xyz]$ which is null-homotopic in K, but the solid triangle $\Delta = \operatorname{Conv}\{x, y, x\}$ bounded by $[xyz]$ does not lie in K completely. Therefore K is not two-convex.

"Only-if" part. Recall that a smooth function $f\colon \mathbb{E}^m \to \mathbb{R}$ is called *strongly convex* if its Hessinan is positive definte at each point.

Fix a smooth strongly convex function $f\colon \mathbb{E}^m \to \mathbb{R}$, such that the restriction $f|_S$ is a Morse function. Note that generic smooth strongly convex function $f\colon \mathbb{E}^m \to \mathbb{R}$ has this property.

For a critical point p of $f|_S$ the outer normal vector $\nu(p)$ is parallel to the gradient $\nabla_p f$; we say that p is *positive critical point* if $\nu(p)$ and $\nabla_p f$ point in the same direction and *negative* otherwise. (If $\nabla_p f = 0$, that is p is the minimum point of f, then the sign is undefined.)

Since S is two-convex and f is convex, the negative critical points of $f|_S$ have index at most 1.

Given a real value s, set

$$K_s = \{\, x \in K \mid f(x) < s \,\}.$$

Assume $\varphi_0\colon \mathbb{D} \to K$ is a continuous map of the disc $\mathbb{D}$ such that $\varphi(\partial\mathbb{D}) \subset K_s$. By Morse Lemma, there is a homotopy $\varphi_t\colon \mathbb{D} \to K$ such that $\varphi_1(\mathbb{D}) \subset K_s$.

Consider a closed curve $\gamma\colon \mathbb{S}^1 \to K$ which is null-homotopic in K. Note that the distance function $f_0(x) = |\operatorname{Conv}\gamma\ x|_{\mathbb{E}^m}$ is convex. Therefore f_0 can be approximated by smooth strongly convex function f such that the restriction $f|_S$ is a Morse function. From above the there is a disc in K with boundary γ which lies arbitrary close to $\operatorname{Conv}\gamma$. Since K is closed, the statement follows. □

12.2.4. Proposition. *Intersection of arbitrary collection of two-convex sets in $\mathbb{E}^m$ is two-convex.*

12.2.5. Proposition. *Interior of any two-convex set in $\mathbb{E}^m$ is a two-convex set.*

Proofs. The Proposition 12.2.4 is evident.

To prove Proposition 12.2.5, apply the Definition 12.2.1 for parallel translations of the curve γ by small vectors. □

12.2.6. Definition. *Given a subset $K \subset \mathbb{E}^m$ define its two-convex hull (briefly $\operatorname{Conv}_2 K$) as the intersection of all two-convex subsets containing K.*

Note that by Proposition 12.2.4, two-convex hull of any set is two-convex. Further, by Proposition 12.2.5, two-convex hull of open set is open.

12.3 Sets with smooth boundary

In this section we characterize the subsets in $\mathbb{E}^m$ with smooth boundary which form CBA[0] spaces.

12.3.1. Theorem. *Let K be a closed connected subset in $\mathbb{E}^m$ equipped with the induced length metric. Assume K is bounded by a smooth hypersurface. Then $K \in \mathrm{CBA}[0]$ if and only if K is two-convex.*

This theorem is a baby case of the main result in [3], it is discussed in the end of chapter.

Proof. Denote by S and by Ω the boundary and the interior of K. Since K is connected and S is smooth, Ω is also connected.

Denote by $k_1(p) \leqslant \ldots \leqslant k_{m-1}(p)$ the principle curvatures of S at $p \in S$ with respect to the normal vector pointing out of K. By Proposition 12.2.3, K is two-convex if and only if $k_2(p) \geqslant 0$ at any $p \in S$.

"Only-if" part. Assume K is not two-convex. Then by Proposition 12.2.3, there is a triangle $[xyz]$ in K which is null-homotopic in K, but the solid triangle $\Delta = \mathrm{Conv}\{x, y, z\}$ does not lie in K completely.

If $K \in \mathrm{CBA}[0]$, then by Reshetnyak majorization theorem, there is a short map $f\colon \Delta \to K$ which preserves x, y and z.

Note that the map f has to be identity. In particular $\Delta \subset K$, a contradiction.

"If" part. Since K is simply connected, by Cartan–Hadamard theorem 9.13.1, it is sufficient to show that $\mathrm{curv}_p K \leqslant 0$ at any point $p \in K$.

If $p \in \mathrm{Int}\, K$ then it admits a neighborhood isometric to a subset of $\mathbb{E}^m$; therefore $\mathrm{curv}_p K \leqslant 0$.

Fix $p \in S$. Assume that $k_2(p) > 0$. Fix sufficiently small $\varepsilon > 0$ and set $K' = K \cap \overline{\mathrm{B}}[p, \varepsilon]$. Let us show that

❶ $$K' \in \mathrm{CBA}[0].$$

Consider the coordinate system with the origin at p and the principle directions and $\nu(p)$ as the coordinate directions. For small $\varepsilon > 0$, the set K' can be described as a subgraph

$$K' = \left\{ (x_1, \ldots, x_m) \in \overline{\mathrm{B}}[p, \varepsilon] \;\middle|\; x_m \leqslant f(x_1, \ldots, x_{m-1} \right\}.$$

Moreover, since ε is small and $k_2(p) > 0$, for any fixed $s \in [-\varepsilon, \varepsilon]$ the function

$$(x_2, \dots, x_{m-1}) \mapsto f(s, x_2, \dots, x_{m-1})$$

is concave for $|x_i| < 2\cdot\varepsilon$ for each i

Fix a negative real value $\lambda < k_1(p)$. Given $s \in (-\varepsilon, \varepsilon)$, consider the set

$$V_s = \left\{ (x_1, \dots, x_m) \in K' \;\middle|\; x_m \leqslant f(x_1, \dots, x_{m-1}) + \lambda\cdot(x_1 - s)^2 \right\}.$$

Note that the function

$$(x_1, \dots, x_{m-1}) \mapsto f(x_1, \dots, x_{m-1}) + \lambda\cdot(x_1 - s)^2$$

is concave near the origin. Since ε is small, we can assume that V_s forms a convex set in $\mathbb{E}^m$ for any s.

Further note that

$$K' = \bigcup_{s\in[-\varepsilon,\varepsilon]} V_s$$

and

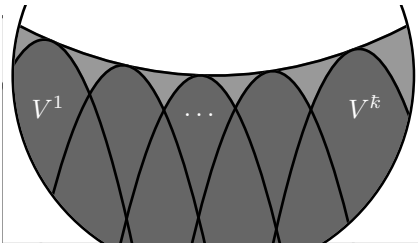
❷ $$V_b \supset V_a \cap V_c$$

for any choice real numbers $a < b < c$.

Given an array of values $s^1 < \dots < s^k$ in $[-\varepsilon, \varepsilon]$, set $V^i = V_{s^i}$ and consider the unions

$$W^i = V^1 \cup \dots \cup V^i$$

equipped with the induced intrinsic metric.



Note that the array (s^n) can be chosen in such a way that W^k is arbitrary close to K'.

Therefore, in order to prove ❶, it is sufficient to show the following.

❸ *All W^i form* CBA[0] *spaces.*

This claim is proved by induction. Base: $W^1 = V^1$ is a CBA[0] space as a convex subset in $\mathbb{E}^m$.

Step: assume that $W^i \in \mathrm{CBA}[0]$. According to ❷

$$V^{i+1} \cap W^i = V^{i+1} \cap V^i.$$

Moreover, this set forms a convex set in $\mathbb{E}^m$ and therefore it is a convex set in W^i and in V^{i+1}. By Reshetnyak's gluing theorem, $W^{i+1} \in$ $\in \mathrm{CBA}[0]$. Hence the claim follows. △

Note that we proved the following

❹ $K \in \mathrm{CBA}[0]$ *if* K *is* strongly two-convex*; that is* $k_2(p) > 0$ *at any point* $p \in S$.

It remains to show that $\operatorname{curv}_p K \leqslant 0$ in the case $k_2(p) = 0$.

Choose a coordinate system $(x_1, \dots, x_n)$ as above, so $(x_1, \dots, x_{m-1})$-coordinate hyperplane forms the tangent subspace to S at p.

Fix $\varepsilon > 0$ so that a neighborhood of p in S forms a graph

$$x_m = f(x_1, \dots, x_{m-1})$$

for a function f defined at the open ball B of radius ε centered at the origin in the $(x_1, \dots, x_{m-1})$-hyperplane. Fix a smooth positive strongly convex function $\varphi\colon B \to \mathbb{R}_+$ such that $\varphi(x) \to \infty$ as x approaches the boundary B. Note that for $\delta > 0$, the subgraph K_δ defined by the inequality

$$x_m \leqslant f(x_1, \dots, x_{m-1}) - \delta \cdot \varphi(x_1, \dots, x_{m-1})$$

is strongly two-convex. By ❹, $K_\delta \in \mathrm{CBA}[0]$.

Finally as $\delta \to 0$, the closed ε-neighborhoods of p in K_δ converges to the closed ε-neighborhood of p in K. Therefore $\operatorname{curv}_p K \leqslant 0$. □

The following CBB-analog of the above theorem; its proof is simpler.

12.3.2. Exercise. *Let K be a closed connected subset in $\mathbb{E}^m$ equipped with the induced length metric. Assume that the* $\operatorname{Int} K$ *is dense in K. Show that $K \in \mathrm{CBB}[0]$ if and only if K is convex.*

12.4 Open sets of Euclidean plane

In this section we will consider inheritance of upper curvature bound by subsets of Euclidean plane.

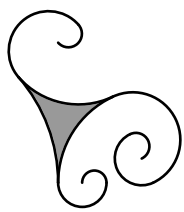
12.4.1. Theorem. *Let Ω be an open simply connected subset of $\mathbb{E}^2$. Equip Ω with induced length metric and denote by by K its completion. Show that $K \in \mathrm{CBA}[0]$.*

The assumption that the set Ω is not critical, instead one ca assume that the induced length metric takes finite values at all points of Ω. We sketch the proof from Bishop's note [28] and suggest to finish the details as an exercise.

Sketch of proof. It is sufficient to show that any triangle in K is thin, as defined in 9.4.1.

Note that K admits a length-preserving map to $\mathbb{E}^2$ which extends the embedding $\Omega \hookrightarrow \mathbb{E}^2$. Therefore the triangle $[xyz]$ can be mapped to the plane in a length-preserving way. Since Ω is simply connected, any open region, say Δ, surrounded by the image of $[xyz]$ completely lies in Ω.

Note that in each triangle $[xyz]$ in K the sides $[xy]$, $[yz]$ and $[zx]$ intersect each other along a geodesic starting at common vertex; possibly a one-point geodesic. In other words, every triangle in K look like one on the diagram.



Assuming contrary, there will be a diangle in K; that is two minimizing geodesics with common ends, but no other common points. The image of this diangle in the plane have to bound an open region of Ω and both sides of the region have to be concave, otherwise one could shorten the sides by pushing them into Ω. Evidently, there is no plane diangle with concave sides, a contradiction.

Note that it is enough to consider only injective triangle $[xyz]$; that is the sides $[xy]$, $[yz]$ and $[zx]$ intersect each other only at the common vertexes. If this is not the case, chopping the overlapping part of sides reduces to the injective case.

Again, the open region, say Δ, bounded by the image of $[xyz]$ has concave sides, otherwise one could shorten the sides by pushing them into Ω. It remains to show that any triangle with concave sides is thin. The latter is left to the reader. $\square$

Here is a spherical analog of above lemma; which can be proved along the same lines. It will be used in the next section.

12.4.2. Exercise. *Let Θ be an open connected subset of the unit sphere $\mathbb{S}^2$ which does not contain a closed hemisphere. Equip Θ with induced length metric. Let $\tilde{\Theta}$ be a metric covering of Θ such that any closed curve in $\tilde{\Theta}$ of length smaller than $2{\cdot}\pi$ is contactable.*

Show that the completion of $\tilde{\Theta}$ is a CBA[1] *space.*

12.5 Open sets of Euclidean space

The material below is inspired by work of Samuel Shefel who seems to be very intrigued by survival of metric properties under affine transformation. To describe an instance of such phenomena, note that two-convexity survives under affine transformations of Euclidean space. Therefore, as a consequence of Theorem 12.3.1, the following holds.

12.5.1. Corollary. *Let K be closed connected subset of Euclidean space equipped with the induced length metric. Assume K is bounded by a smooth hypersurface and it forms* CBA[0] *space. Then any affine transformation of K also forms* CBA[0] *space.*

By Corollary 12.5.4, an analogous statement holds for sets bounded by Lipschitz surfaces in three-dimensional Euclidean space. In higher dimensions this is not longer true.

Here is the main theorem of the section.

12.5.2. Shefel's Theorem. *Let Ω be an open set $\mathbb{E}^3$. Equipp Ω with the induced length metric and denote by K the completion of universal metric cover of Ω. Then K is a* CBA[0] *space if and only if Ω is two-convex.*

The following exercise shows that analogous statement does not hold in higher dimensions.

12.5.3. Exercise. *Let Π_1, Π_2 be two planes in $\mathbb{E}^4$ intersecting at single point. Consider the complement $\Omega = \mathbb{E}^4 \backslash (\Pi_1 \cup \Pi_2)$ equipped with induced length metric denote by K the completion of universal metric cover of Ω.*

Show that

a) Ω is two-convex.

b) $K \in$ CBA[0] *if and only if $\Pi_1 \perp \Pi_2$.*

Before coming to the proof of Shefel's theorem, let us formulate few corollaries. The following corollary look more like a generalization of Theorem 12.3.1 for three-dimensional Euclidean space.

12.5.4. Corollary. *Let K be a closed subset in $\mathbb{E}^3$ bounded by a Lipschitz hypersurface. Then K with the induced length metric is a* CBA[0] *space if and only if the interior of K is two-convex and simply connected.*

Proof. Set $\Omega = \operatorname{Int} K$. Since K is simply connected and bounded by a surface, Ω is also simply connected.

Apply Shefel's Theorem to Ω. Note that the completion of Ω equipped with the induced length metric is isometric to K with induced intrinsic metric. Hence the result follows. $\square$

The following corollary is the main statement in Shefel's original paper [153].

Let U be an open set in $\mathbb{R}^2$. Recall, that a continuous function $f\colon U \to \mathbb{R}$ is called *saddle* if for any linear function $\ell\colon \mathbb{R}^2 \to \mathbb{R}$ the difference $f - \ell$ does not have local maxima and minima in U. Equivalently, open subgraph and epigraph of f

$$\left\{\, (x,y,z) \in \mathbb{E}^3 \;\middle|\; z < f(x,y),\ (x,y) \in U \,\right\},$$
$$\left\{\, (x,y,z) \in \mathbb{E}^3 \;\middle|\; z > f(x,y),\ (x,y) \in U \,\right\}$$

form a two-convex subsets in $\mathbb{E}^3$

12.5.5. Corollary. *Let $f\colon \mathbb{D} \to \mathbb{R}$ be a Lipschitz function which is saddle in the interior of the closed unit disc $\mathbb{D}$. Then the graph $z = f(x,y)$ in $\mathbb{E}^3$ equipped with induced length metric is a* CBA[0] *space.*

Proof. Since the function f is Lipschitz, its graph Γ with the induced length metric is bi-Lipschitz to the Euclidean metric on $\mathbb{D}$.

Consider the sequence of sets

$$K_n = \left\{\, (x,y,z) \in \mathbb{E}^3 \;\middle|\; z \lesseqgtr f(x,y) \pm \tfrac{1}{n},\ (x,y) \in \mathbb{D} \,\right\}.$$

Note that each K_n is closed, simply connected and two-convex. Moreover the boundary of K_n forms a Lipschitz surface.

Equip K_n with induced length metric. By Corollary 12.5.4, $K_n \in$ $\in$ CBA[0]. It remains to note that as $n \to \infty$ the sequence of space K_n converges to Γ in the sence of Gromov–Hausdorff. $\square$

Now we are back to the proof of Shefel's theorem 12.5.2.

Recall, that a subset P of $\mathbb{E}^m$ is called *polytope* if it can be presented as a union of finite number of simplices. Similarly, a *spherical polytope* is a union of finite number of simplices in $\mathbb{S}^m$.

Note any polytope admits a finite triangulation. Therefore any polytope equipped with induced intrinsic metric forms a Euclidean polyhedral space as defined in 13.1.1.

Let P be a polytope and Ω its interior, both considered with induced intrinsic metric. Typically the completion K of Ω is isometric to P. However in general we only have locally distance preserving map $K \to P$; it does not have to be onto and it may be not injective. An example can be guessed from the picture.

It is easy to see that K is a polyhedral spaces.

12.5.6. Lemma. *Shefel's Theorem 12.5.2 holds if the set Ω is formed by interior of a polytope.*

The statement might look obvious, but there is an underwater stone in the proof.

Proof. Denote by $\tilde\Omega$ the universal metric cover of Ω. Let $\tilde K$ and K be the completion of $\tilde\Omega$ and Ω correspondingly.

Note that K is a polyhedral space and the covering $\tilde\Omega\to\Omega$ extends to a covering map $\tilde K\to K$ which might be branching at some vertexes.

Fix a point $\tilde p\in\tilde K\backslash\tilde\Omega$; denote by p the image of $\tilde p$ in K. By Hadamard–Cartan theorem (9.13.1) it is sufficient to show that

❶ $$\operatorname{curv}_{\tilde p}\tilde K\leqslant 0.$$

Recall that $\Sigma_{\tilde p}\tilde K$ denotes the space of directions at $\tilde p$. Note that a small neighborhood of $\tilde p$ in $\tilde K$ isometric to an open set in the cone over $\Sigma_{\tilde p}\tilde K$. By Theorem 11.4.2*a*, ❶ follows if

❷ $$\Sigma_{\tilde p}\tilde K\in\mathrm{CBA}[1].$$

By rescaling, we can assume every face of K which does not conatin p lies on the distance at least 2 from p. Denote by $\mathbb{S}^2$ the unit sphere centered at p, set $\Theta=\mathbb{S}^2\cap\Omega$. Note that Σ_pK is isomeric to the completion of Θ and $\Sigma_{\tilde p}\tilde K$ is a completion of a regular metric covering $\tilde\Theta$ of Θ which is induced by universal metric cover $\tilde\Omega\to\Omega$.

By Exercise 12.4.2, it remains to show that the following.

❸ *Any closed curve in $\tilde\Theta$ of length less than $2\cdot\pi$ is contactable.*

Consider a colsed curve $\tilde\gamma$ of length $<2\cdot\pi$ in $\tilde\Theta$. Its projection γ in $\Theta\subset\mathbb{S}^2$ has the same length. Therefore by Hemisphere lemma 1.2.1, γ lies in an open hemisphere. Then for a plane Π close to p, the central projection γ' of γ to Π is defined and lies in Ω. From two-convexity of Ω, the curve γ' is contractible in $\Pi\cap\Omega$. It follows that γ is contractible in Θ and therefore $\tilde\gamma$ is contractible in $\tilde\Theta$. □

The following proposition describes a construction which was essentially given by Shefel in [153]. It produce $\operatorname{Conv}_2\Omega$ for an open set $\Omega\subset\mathbb{E}^3$.

12.5.7. Proposition. *Let $\Pi_1,\Pi_2\dots$ be an everywhere dense sequence of planes in $\mathbb{E}^3$. Given an open set Ω consider the recucevly defined sequence of open sets $\Omega=\Omega_0\subset\Omega_1\subset\dots$ such that Ω_n is the union of Ω_{n-1} and all the bounded compontnets of $\mathbb{E}^3\backslash(\Pi_n\cup\Omega_{n-1})$. Then*

$$\operatorname{Conv}_2\Omega=\bigcup_n\Omega_n.$$

Proof. Set

$$\Omega' = \bigcup_n \Omega_n.$$

Note that Ω' is a union of open set, in particular it is open.

The inclusion $\operatorname{Conv}_2 \Omega \supset \Omega'$ is evident.

It remains to show that Ω' is two-convex. Assume contrary; that is, there is a plane Π and a closed curve $\gamma\colon \mathbb{S}^1 \to \Pi \cap \Omega'$ which is null-homotopic in Ω', but not null-homotopic in $\Pi \cap \Omega'$.

By approximation we can assume that $\Pi = \Pi_n$ for a large enough n and that γ lies in Ω_{n-1}. The latter contradicts the n-th step in the construction. □

12.5.8. Key lemma. *The two-convex hull of the interior of polytope in $\mathbb{E}^3$ is an interior of a polytope.*

Proof. Fix a polytope P in $\mathbb{E}^3$. Set $\Omega = \operatorname{Int} P$, we can assume that Ω is dense in P Denote by $F_1, \dots, F_m$ the facets of P.

Set $\Omega' = \operatorname{Conv}_2 \Omega$ and let P' be the closure of Ω'. Further, for each i, set $F_i' = F_i \backslash \Omega'$. In other words, F_i' is the subset of facet F_i which remains on the boundary of K'.

From the construction of two-convex hull (12.5.7)

❹ *F_i' is convex subset of F_i.*

Further, since Ω' is two-convex, we get the following.

❺ *Each connected component of the complement $F_i \backslash F_i'$ is convex.*

Indeed, assume a connected component A of $F_i \backslash F_i'$ fails to be convex. Then there is a suppoiting line ℓ to A touching A at a single point in the interior of F_i. Then one could rotate the plane of F_i slightly arounf ℓ and move it parallel to cut a hat from the complement of Ω. The latter means that Ω is not two-convex, a contradiction. △

From ❹ and ❺, we get that

❻ *F_i' is a convex polygon for each i.*

Consider the complement $\mathbb{E}^3 \backslash \Omega$ equipped with the length metric. By construction of two-convex hull (12.5.7), the complement $L = \mathbb{E}^3 \backslash (\Omega' \cup K)$ is locally convex; that is, any points of L admits a convex neighborhood.

Summarizing (1) Ω' is a 2-convex open set, (2) the boundary $\partial\Omega'$ contains a finite number of polygons F_i' and the remaing part is locally concave. It remains to show that (1) and (2) imply that $\partial\Omega'$ is piecewise linear.

The proof of the last statement is left to the reader. □

Proof of 12.5.2. Note that it is sufficient to show that (2+2)-comparison holds for any 4 points $x^1, x^2, x^3, x^4 \in \Omega$.

Fix $\varepsilon > 0$. Choose six broken lines connecting all the pairs of points x^1, x^2, x^3, x^4 such that length of each at most ε bigger than the distance between its ends in the length metric on Ω. Choose a polytope P in Ω such that the interior $\operatorname{Int} P$ is simply connected and it contains all these six broken lines.

Denote by Ω' the two-convex hull of the interior of P. According to Key Lemma (12.5.8) Ω' is an interior of a polytope.

Equip Ω' with the induced length metric. Consider the universal metric cover $\tilde{\Omega}'$ of Ω'. (The covering $\tilde{\Omega}' \to \Omega'$ might be nontrivial; despite that $\operatorname{Int} P$ is simply connected, its two-convex hull Ω' might be not simply connected.) Denote by $\tilde{K}'$ the completion of $\tilde{\Omega}'$.

By Lemma 12.5.6, $\tilde{K}' \in \mathrm{CBA}[0]$.

Since $\operatorname{Int} P$ is simply connected, the embedding $\operatorname{Int} P \hookrightarrow \Omega'$ admits a lifting $\iota\colon \operatorname{Int} P \hookrightarrow \tilde{K}'$. By construction, ι almost preserves the distances between the points x^1, x^2, x^3, x^4; namely

$$|\iota(x^i)\ \iota(x^j)|_L \gtrless |x^i x^j|_{\operatorname{Int} P} \pm \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary and (2+2)-comparison holds in $\tilde{K}'$, we get that (2+2)-comparison holds in Ω for x^1, x^2, x^3, x^4.

The statement follows since the quadruple $x^1, x^2, x^3, x^4 \in \Omega$ is arbitrary. □

12.5.9. Exercise. *Assume $K \subset \mathbb{E}^m$ is a closed set bounded by a Lipschitz hypersurface. Equip K with the induced length metric. Show that if $K \in \mathrm{CBA}[0]$ then K is two-convex.*

The following exercise is analogous to Exercise 12.5.3. It provides a counterexample to the analog of Corollary 12.5.4 in higher dimensions.

12.5.10. Exercise. *Let $K = W \cap W'$, where*

$$W = \left\{\, (x, y, z, t) \in \mathbb{E}^4 \;\middle|\; z \geqslant -x^2 - y^2 \,\right\}$$

and $W' = \iota(W)$ for some motion $\iota\colon \mathbb{E}^4 \to \mathbb{E}^4$.

Show that K is always two-convex and one can choose ι so that K with the induced length metric is not $\mathrm{CBA}[0]$.

12.6 Convex hypersurfaces in Riemannian manifolds

12.6.1. Theorem. *Let M be an m-dimensional Riemannian manifold with sectional curvature $\geqslant \kappa$ and $F \subset M$ be a convex hypersurface*

equipped with the length-metric; that is, F bounds a convex set. Then $F \in \mathrm{CBB}^{m-1}[\kappa]$.

This theorem, in a slightly weaker form was proved in by Buyalo in [38], and then its statement was made more exact in [5].

12.6.2. Lemma. *Let M be a Riemannian manifold and $f\colon M \looparrowright \mathbb{R}$ be a locally Lipshits λ-concave subfunction. Then there is a sequence of smooth subfunctions $f_n\colon M \looparrowright \mathbb{R}$ and $\lambda_n \in \mathbb{R}$ such that $f_n \to f$ and $\lambda_n \to \lambda$ as $n \to \infty$ and each f_n is λ_n-concave.*

This lemma is a slight generalization of [57, Theorem 2] by Greene and Wu; it can be proves the same way.

Proof of theorem 12.6.1. Without loss of generality one can assume that

1. $\kappa \geqslant -1$,
2. F bounds a compact convex set C in M,
3. there is a function h defined in a neighborhood of C such that $h'' \leqslant -2$ and $|h(x)| < 1/10$ for any $x \in C$,
4. there is unique geodesic between any two points in C.

(If not, rescale and pass to the boundary of the convex piece cut by F from a small convex ball centered at $x \in F$, taking $h = -10\cdot|x_*|$.)

Consider the function $f = |F_*|$. By Rauch comparison (as it is stated in Petersen's book [132, 11.4.8]), for any unit-speed geodesic γ in the interior of C, $(f\circ\gamma)''$ is bounded in the barrier sense by the corresponding value in the model case — when $M \overset{iso}{=\!=} \mathbb{M}^2[-1]$ and F is a geodesic. In particular,

$$(f\circ\gamma)'' \leqslant f\circ\gamma.$$

Therefore $f + \varepsilon\cdot h$ is $(-\varepsilon)$-concave in $\Omega_\varepsilon = f^{-1}((0,\varepsilon)) \cap C$. Take

$$K_\varepsilon = f^{-1}([\tfrac{1}{3}\cdot\varepsilon, \tfrac{2}{3}\cdot\varepsilon]) \cap C.$$

Applying Lemma 12.6.2, we can find a smooth $(-\frac{\varepsilon}{2})$-concave function f_ε which is arbitrarily close to $f+\varepsilon\cdot h$ on K_ε and which is defined on a neighborhood of K_ε. Take a regular value $\vartheta_\varepsilon \approx \frac{1}{2}\cdot\varepsilon$ of f_ε. (In fact one can take $\vartheta_\varepsilon = \frac{1}{2}\cdot\varepsilon$, but it requires a little work.) Since $|(h|C)| < 1/10$, the level set $F_\varepsilon = f_\varepsilon^{-1}(\vartheta_\varepsilon)$ will lie entirely in K_ε. Therefore F_ε forms a smooth closed convex hypersurface.

Let us denote by ρ and ρ_ε the length-metrics on correspondingly F and F_ε. By the Gauss formula, $(F_\varepsilon, \rho_\varepsilon)$ has curvature $\geqslant \kappa$. Further, F_ε bounds a compact convex set C_ε and $F_\varepsilon \to F$, $C_\varepsilon \to C$ in Hausdorff sense as $\varepsilon \to 0$. By property (4), the restricted metrics from M to C and to C_ε are length-metrics. Thus, C_ε is an Alexandrov space

with F_ε as boundary, that converges in Gromov–Hausdorff sense to C. It follows from ??? that $(F_\varepsilon, \rho_\varepsilon)$ converges in Gromov–Hausdorff sense to (F, ρ). Therefore (F, ρ) is an Alexandrov space with curvature $\geqslant \kappa$. □

12.7 Comments and open problems

Sets with smooth boundary. In [3] Alexander, Bishop and Berg gave the exact upper bound on Alexandrov's curvature for the Riemannian manifolds with boundary. Namely they show the following.

12.7.1. Theorem. *Let M be a Riemannian manifold with boundary ∂M. A direction tangent to the boundary will be called concave if there is a short geodesic in this direction which leaves the boundary and goes into interior of M. A sectional direction will be called concave if all the directions in it are concave.*

Then $\operatorname{curv} M \leqslant \kappa$, *where κ is the infimum of sectional curvatures in M and the curvatures of ∂M in the concave directions.*

12.7.2. Corollary. *Let M be a Riemannian manifold with boundary ∂M. Assume all the sectional curvatures of M and ∂M are bounded from above by κ. Then* $\operatorname{curv} M \leqslant \kappa$.

A surface S in $\mathbb{E}^3$ is called saddle if for any linear function $f\colon \mathcal{X} \to \mathbb{R}$, the restriction $f|S$ has no local maxima or maxima in the interior of S.

Nonsmooth sets.

12.7.3. Shefel's conjecture. *Any saddle hypersurface in $\mathbb{R}^3$ equipped with the length-metric has curvature $\leqslant 0$ at any point.*

From Corollary 12.5.4, it follows that if there is a counterexample then arbitrary neighborhood of some point of the surface can not be presented as graph in any coordinate system.

Further more, if a counterexample exists, then it forms a piece of boundary of a two-convex set Ω with the following property. There is a point $p \in \partial\Omega$ and $\varepsilon > 0$ such that for any $\delta > 0$ we have

$$\mathrm{B}(p, \varepsilon) \subset \operatorname{Conv}_2[\Omega \cup \mathrm{B}(p, \delta)].$$

None of such examples of Ω are known so far.

A surface S in a metric space. We say that S is saddle if for any convex function $f\colon \mathcal{X} \to \mathbb{R}$, the restriction $f|S$ has no local maxima in the interior of S.

12.7.4. Generalized Shefel's conjecture. *Any 2-dimensional saddle surface in a* $\mathrm{CAT}[\kappa]$ *equipped with the length-metric has curvature* $\leqslant \kappa$ *at any point.*

A subset K is called strongly two-convex if any null-homotopic circle $\gamma\colon \mathbb{S}^1 \to K$ is also null-homotopic in $K \cap \mathrm{Conv}[\gamma(\mathbb{S}^1)]$.

12.7.5. Question. *Is it true that any closed strongly two-convex set with dense interior bounded by a Lipschitz hypersurface in* $\mathbb{E}^m$ *forms a* $\mathrm{CBA}[0]$ *space?*

Metric minimizing surfaces. In [**?**] was shown that discs in $\mathrm{CBA}[0]$ spaces which do not admit a length-decreasing deformations with fixed boundary form $\mathrm{CBA}[0]$ spaces.

Any metric minimizing disc forms a saddle surface as defined above. There are saddle surfaces which are not metric minimizing globally and it is expected that there saddle surfaces which are not metric minimizing in arbitrary neighborhood of a fixed point. Constructing such examples could shed light on the Shefel's question.

On convex hypersurfaces in Alexandrov space. We are not aware of any proof of theorem 12.6.1 which is not based on the Gauss formula.

Even if the ambient space is Euclidean, we do not know any proof which avoids approximation by simpler spaces; the synthetic argument of Milka in [113] is very intersting, but still it requires approximation.

Finding such a proof would be interesting on its own, and also could lead to the generalization of theorem 12.6.1 to the case when M is an Alexandrov space. The latter problem is one of the first questions in CBB spaces which is still open.

12.7.6. Question. *Let* $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$ *and its boundary* $\partial\mathcal{L}$ *is not empty. Is it true that* $\partial\mathcal{L}$ *equipped with induced length metric forms a* $\mathrm{CBB}^{m-1}[\kappa]$ *space?*

Equivalently one can ask, is it true that convex hypersurface in a $\mathrm{CBB}^m[\kappa]$ space without boundary forms a $\mathrm{CBB}^{m-1}[\kappa]$ space?

Chapter 13

Polyhedral spaces

13.1 Definitions

13.1.1. Definition. *A length space $\mathcal{P}$ is called* $\mathbb{M}[\kappa]$-polyhedral space *(briefly $\mathcal{P} \in \mathbf{P}\mathbb{M}[\kappa]$, where* $\mathbf{P}$ *stands for* piecewise*) if it admits a finite triangulation τ such that an arbitrary simplex σ in τ is isometric to a simplex in the model space $\mathbb{M}^m[\kappa]$, where m is the dimension of σ.*

If we do not wish to specify κ, we will say that $\mathcal{P}$ is a polyhedral space*.*

By a triangulation of a polyhedral space *we will understand the triangulation as in the definition.*

a) The $\mathbb{M}[1]$-polyhedral spaces will also be called spherical polyhedral spaces*;*

b) The $\mathbb{M}[0]$-polyhedral spaces will also be called Euclidean polyhedral spaces*;*

c) The $\mathbb{M}[-1]$-polyhedral spaces will also be called hyperbolic polyhedral space*.*

Note that according to the above definition, all polyhedral spaces are compact. However, most of the statements below admit straightforward generalizations to *locally polyhedral spaces*; that is, complete length spaces, any point of which admits a closed neighborhood isometric to a polyhedral space. The latter class of spaces includes in particular the infinite covers of polyhedral spaces.

The dimension of a polyhedral space $\mathcal{P}$ is defined as the maximal dimension of the simplex in one (and therefore any) triangulation of $\mathcal{P}$. The class of $\mathbf{P}\mathbb{M}[\kappa]$ spaces of dimension m will be denoted as $\mathbf{P}\mathbb{M}^m[\kappa]$.

Links. Let $\mathcal{P}$ be a polyhedral space and σ be a simplex in a triangulation τ of $\mathcal{P}$.

The simplices which contain σ form an abstract simplicial complex called the *link* of σ, denoted by Link_σ. If m is the dimension of σ then the set of vertices of Link_σ is formed by the $(m+1)$-simplices which contain σ; the set of its edges are formed by the $(m+2)$-simplices which contain σ, and so on.

The link Link_σ can be identified with the subcomplex of τ formed by all the simplices σ' such that $\sigma \cap \sigma' = \varnothing$ but both σ and σ' are faces of the same simplex.

The points in Link_σ can be identified with the normal directions to σ at a point in σ. The angle metric between directions makes Link_σ into a spherical polyhedral space. We will always consider the link with this metric.

Tangent space and space of directions. Let $\mathcal{P}$ be a polyhedral space and let τ be a triangulation of $\mathcal{P}$. If a point $p \in \mathcal{P}$ lies in the interior of a $\bar{k}$-simplex σ of τ then the tangent space $\mathrm{T}_p\mathcal{P}$ is naturally isometric to

$$\mathbb{E}^{\bar{k}} \times (\mathrm{Cone}\,\mathrm{Link}_\sigma).$$

Equivalently, the space of directions Σ_p can be isometrically identified with the $\bar{k}$-th spherical suspension over Link_σ; that is,

$$\Sigma_p \overset{iso}{=\!=} \mathrm{Susp}^{\bar{k}}(\mathrm{Link}_\sigma).$$

If $\mathcal{P}$ is an m-dimensional polyhedral space, then for any $p \in \mathcal{P}$ the space of directions Σ_p is a spherical polyhedral space of dimension at most $m-1$.

In particular, for any point p in σ, the isometry class of Link_σ together with $\bar{k} = \dim \sigma$ determines the isometry class of Σ_p and the other way around.

A small neighborhood of p is isometric to a neighborhood of the tip of the κ-cone over Σ_p. In fact, if this propery holds at any point of a compact length space $\mathcal{P}$ then $\mathcal{P} \in \mathbf{P}\mathbb{M}[\kappa]$; see [98].

Curvature bounded below. The following theorem provides a combinatorial description of polyhedral spaces with curvature bounded below.

13.1.2. Theorem. *Let $\mathcal{P} \in \mathbf{P}\mathbb{M}^m[\kappa]$ and τ be a triangulation of $\mathcal{P}$. Then $\mathcal{P} \in \mathrm{CBB}[\kappa]$ if and only if the following conditions hold.*

a) τ is pure*; that is, any simplex in τ is a face of some simplex of dimension exactly m.*

b) The link of any simplex of dimension $m-1$ is formed by single point or two points.

c) The link of any simplex of dimension $\leqslant m-2$ is connected.

d) Any link of any simplex of dimension $m-2$ has diameter at most π.

The condition (c) can be reformulated in the following way:

c′) Any path $\gamma\colon [0,1] \to \mathcal{P}$ can be approximated by paths $\gamma_n\colon [0,1] \to \mathcal{P}$ which cross only simplexes of dimension m and $m-1$.

Further, modulo the other conditions, the condition (d) is equivalent to the following:

d′) The link of any simplex of dimension $m-2$ is isometric to a circle of length $\leqslant 2\cdot\pi$ or a closed real interval of length $\leqslant \pi$.

Proof. We apply induction on m. The base case $m=1$ follows from the assumption (b).

Induction Step. Assume that the theorem is proved for polyhedral spaces of $\dim < m$. Suppose $\dim \mathcal{P} = m$.

We will only give a proof of the "if" direction. The "only if" direction is similar and is left to the reader.

According to the Globalization theorem (8.5.4), it is sufficient to show that

$$\operatorname{curv}_p \mathcal{P} \geqslant \kappa$$

for any point $p \in \mathcal{P}$.

Fix $p \in \mathcal{P}$. Note that a spherical neighborhood of p is isometric to a spherical neighborhood of the tip of the tangent κ-cone

$$\mathrm{T}_p^\kappa = \operatorname{Cone}^\kappa(\Sigma_p).$$

Hence it is sufficient to show that

❶ $$\mathrm{T}_p^\kappa \in \mathrm{CBB}[\kappa]$$

for any $p \in \mathcal{P}$.

By Theorem 11.4.1a, the latter is equivalent to

❷ $\operatorname{diam} \Sigma_p \leqslant \pi$ *and* $\Sigma_p \in \mathrm{CBB}[1]$.

If $m=2$ then ❷ follows from (b).

To prove the case $m \geqslant 3$, note that Σ_p is an $(m-1)$-dimensional spherical polyhedral space and all the conditions of the theorem hold for Σ_p. It remains to apply the induction hypothesis. □

13.1.3. Exercise. *Assume* $\mathcal{P} \in \mathbf{P}\mathbb{M}[\kappa]$ *and* $\dim \mathcal{P} \geqslant 2$. *Show that if* $\mathcal{P} \in \mathrm{CBB}[\kappa']$ *then* $\kappa' \leqslant \kappa$ *and* $\mathcal{P} \in \mathrm{CBB}[\kappa]$.

Curvature bounded above. Recall that the definition of ℓ-simply connected space is given in 9.13.2.

The following theorem provides a combinatorial description of polyhedral spaces with curvature bounded above.

13.1.4. Theorem. *Let $\mathcal{P} \in \mathbf{P}\mathbb{M}[\kappa]$ and τ be a triangulation of $\mathcal{P}$. Then*

a) $\operatorname{curv}\mathcal{P} \leqslant \kappa$ *if and only if any connected component of the link of any simplex σ in τ is $(2{\cdot}\pi)$-simply connected. Equivalently, if and only if any closed local geodesic in $\operatorname{Link}_\sigma$ has length at least $2{\cdot}\pi$.*

b) $\mathcal{P} \in \operatorname{CBA}[\kappa]$ *if and only if $\mathcal{P}$ is $(2{\cdot}\varpi^\kappa)$-simply connected and any connected component of the link of any simplex σ in τ is $(2{\cdot}\pi)$-simply connected.*

Proof. Let us apply induction on $\dim\mathcal{P}$. The base case $\dim\mathcal{P} = 0$ is evident.

Step. Assume that the theorem is proved in the case $\dim\mathcal{P} < m$. Suppose $\dim\mathcal{P} = m$.

We will only give a proof of the "if" direction.

Fix a point $p \in \mathcal{P}$. A neighborhood of p is isometric to the κ-cone over the Σ_p. By Theorem 11.4.1a, it is sufficient to show that

❸ $$\Sigma_p \in \operatorname{CAT}[1].$$

Note that Σ_p is a spherical polyhedral space and its links are isometric to links of $\mathcal{P}$. By the induction hypothesis, $\operatorname{curv}\Sigma_p \leqslant 1$. Applying Generalized Hadamard–Cartan theorem (9.13.3), we get ❸.

To prove (b) apply Generalized Hadamard–Cartan theorem (9.13.3) to $\mathcal{P}$. □

13.1.5. Exercise. *Show that if in a Euclidean polyhedral space $\mathcal{P}$ any two points can be connected by a unique geodesic then $\mathcal{P} \in \operatorname{CBA}[0]$.*

13.1.6. Exercise. *Assume $\mathcal{P} \in \mathbf{P}\mathbb{M}[\kappa]$ and $\dim\mathcal{P} \geqslant 2$. Show that if $\mathcal{P} \in \operatorname{CBA}[\kappa']$ then $\kappa' \geqslant \kappa$ and $\mathcal{P} \in \operatorname{CBA}[\kappa]$.*

13.2 Flag complexes

13.2.1. Definition. *A simplicial complex $\mathcal{S}$ is called* flag *if whenever $\{v_0, \dots, v_{\bar k}\}$ is a set of distinct vertices of $\mathcal{S}$ which are pairwise joined by edges, then the vertexes $v_0, \dots, v_{\bar k}$ span a $\bar k$-simplex in $\mathcal{S}$.*

If the above condition is satisfied only for $\bar k = 2$, then we say $\mathcal{S}$ satisfies the no-triangle condition.

Note that every flag complex is determined by its 1-skeleton.

13.2.2. Exercise. *Show that the barycentric subdivision of any simplicial complex is a flag complex. Conclude that any simplicial complex is homeomorphic to a* CBA[1] *space.*

13.2.3. Proposition. *A simplicial complex $\mathcal{S}$ is flag if and only if $\mathcal{S}$ as well as all the links of all its simplices satisfies the no-triangle condition.*

From the definition of flag complex we get the following.

13.2.4. Lemma. *Any link of a flag complex is flag.*

Proof of Proposition 13.2.3. By Lemma 13.2.4, the no-triangle condition holds for any flag complex and all its links.

Now assume a complex $\mathcal{S}$ and all its links satisfy the no-triangle condition. It follows that $\mathcal{S}$ includes a 2-simplex for each triangle. Applying the same observation for each edge we get that $\mathcal{S}$ includes a 3-simplex for any complete graph with 4 vertices. Repeating this observation for triangles, 4-simplexes, 5-simplices and so on we get that $\mathcal{S}$ is flag. □

All-right triangulation. A triangulation of a spherical polyhedral space is called an *all-right triangulation* if each simplex of the triangulation is isometric to a spherical simplex all of whose angles are right. Similarly, we say that a simplicial complex is equipped with an *all-right spherical metric* if it is a length metric and each simplex is isometric to a spherical simplex all of whose angles are right.

Spherical polyhedral CBA[1] spaces glued from of right-angled simplices admit the following characterization discovered by Gromov [63, p. 122].

13.2.5. Flag condition. *Assume that a spherical polyhedral space $\mathcal{P}$ admits an all-right triangulation τ. Then $\mathcal{P} \in$ CBA[1] if and only if τ is flag.*

Proof; "only-if" part. Assume there are three vertices v_1, v_2 and v_3 of τ that are pairwise joined by edges but do not span a simplex. Note that in this case

$$\measuredangle[v_1\,{}^{v_2}_{v_3}] = \measuredangle[v_2\,{}^{v_3}_{v_1}] = \measuredangle[v_3\,{}^{v_1}_{v_1}] = \pi.$$

Equivalently,

❶ *The join of the geodesics $[v_1v_2]$, $[v_2v_3]$ and $[v_3v_1]$ forms a locally geodesic loop in $\mathcal{P}$.*

Now assume that $\mathcal{P} \in$ CBA[1]. Then by 11.4.1*a*, $\operatorname{Link}_\sigma \mathcal{P} \in\in$ $\in$ CBA[1] for every simplex σ in τ.

Each of these links is an all-right spherical complex and by Theorem 13.1.4, none of these links can contain a geodesic circle of length less than $2\cdot\pi$.

Therefore Proposition 13.2.3 and ❶ imply the "only-if" part.

"If" part. By Lemma 13.2.4 and Theorem 13.1.4, it is sufficient to show that any closed local geodesic γ in a flag complex $\mathcal{S}$ with all-right metric has length at least $2\cdot\pi$.

Fix a flag complex $\mathcal{S}$. Recall that the *star* of a vertex v (briefly $\overline{\mathrm{Star}}_v$) is formed by all the simplices containing v. Similarly, Star_v, the open star of v, is the union of all simplices containing v with faces opposite v removed.

Choose a simplex σ which contains a point of γ. Let v be a vertex of σ. Set $f(t) = \cos|v\gamma(t)|$. Note that

$$f''(t) + f(t) = 0$$

if $f(t) > 0$. Since the zeroes of f are π apart, γ spends time π on every visit to Star_v.

After leaving Star_v, the local geodesic γ has to enter another simplex, say σ', which has a vertex v' not joined to v by an edge.

Since τ is flag, we have that the stars Star_v and $\mathrm{Star}_{v'}$ do not overlap. The same argument as above shows that γ spends time π on every visit to $\mathrm{Star}_{v'}$. Therefore the total length of γ is at least $2\cdot\pi$. $\square$

13.2.6. Exercise. *Assume that a spherical polyhedral space $\mathcal{P}$ admits a triangulation τ such that all dihedral angles of all simplexes are at least $\frac{\pi}{2}$. Show that $\mathcal{P} \in \mathrm{CBA}[1]$ if τ is flag.*

13.2.7. Exercise. *Let $\mathcal{U} \in \mathrm{CBA}[0]$ and $\varphi_1, \varphi_2, \dots, \varphi_k \colon \mathcal{U} \to \mathcal{U}$ be commuting short retractions; that is*

- $\varphi_i \circ \varphi_i = \varphi_i$ *for each* i;
- $\varphi_i \circ \varphi_j = \varphi_j \circ \varphi_i$ *for any* i *and* j;
- $|\varphi_i(x)\ \varphi_i(y)|_{\mathcal{U}} \leqslant |x\,y|_{\mathcal{U}}$ *for each* i *and any* $x, y \in \mathcal{U}$.

Set $A_i = \operatorname{Im}\varphi_i$ for all i; note that each A_i is a weakly convex set.

Assume Γ is a finite graph (without loops and multiple edges) with edges labeled by $1, 2, \dots, n$. Denote by $\mathcal{U}^\Gamma$ the space obtained by taking a copy of $\mathcal{U}$ for each vertex of Γ and gluing two such copies along A_i if the corresponding vertices are joint by an edge labeled by i.

Show that $\mathcal{U}^\Gamma \in \mathrm{CBA}[0]$

The space of trees. The following construction is given by Billera, Holmes and Vogtmann in [27].

Let $\mathcal{T}_n$ be the set of all metric trees with n end-vertices labeled by $a_1, \dots, a_n$. To describe one tree in $\mathcal{T}_n$ we may fix a topological tree τ

with end vertices $a_1, \dots, a_n$ and all the other verices of degree 3 and prescribe the lengths of $2{\cdot}n-3$ edges. If the lengh of an edge is 0, we assume that edge degenerates; such a tree can be also decribed using a different topological tree τ'. The subset of $\mathcal{T}_n$ corresponding to the given topological tree τ can be identified with a convex closed cone in $\mathbb{R}^{2\cdot n-3}$. Equip each such subset with the metric induced from $\mathbb{R}^{2\cdot n-3}$ and consider the legth metric on $\mathcal{T}_n$ induced by these metrics.

13.2.8. Exercise. *Show that $\mathcal{T}_n$ with the described metric is a* CBA[0] *space.*

13.3 Cubical complexes

The definition of a cubical complex mostly repeats the definition of a simplicial complex, with simplices replaced by cubes.

Formally, a *cubical complex* is defined as a subcomplex of the unit cube in the Euclidean space of large dimension; that is, a collection of faces of cube such that together with each face it contains all its sub-faces. Each cube face in this collection will be called a *cube* of the cubical complex.

Note that according to this definition, any cubical complex is finite, that is, contains a finite number of cubes.

The union of all the cubes in a cubical complex $\mathcal{Q}$ will be called its *underlying space*; it will be denoted by $\mathcal{Q}$ or by $\underline{\mathcal{Q}}$ if we need to emphasize that we are talking about a set, not a complex. A homeomorphism from $\underline{\mathcal{Q}}$ to a topological space $\mathcal{X}$ is called a *cubulation of* $\mathcal{X}$.

The underlying space of a cubical complex $\mathcal{Q}$ will be always considered with the length metric induced from $\mathbb{R}^N$. In particular, with this metric, each cube of $\mathcal{Q}$ is isometric to the unit cube of the same dimension.

It is straightforward to construct a triangulation of $\underline{\mathcal{Q}}$ such that each simplex is isometric to a Euclidean simplex. In particular $\underline{\mathcal{Q}}$ is a Euclidean polyhedral space.

The link of each cube in a cubical complex admits a natural all-right triangulation; each simplex corresponds to an adjusted cube.

Cubical analog of a simplicial complex. Let $\mathcal{S}$ be a simplicial complex and $\{v_1, \dots, v_N\}$ be the set of its vertexes.

Consider $\mathbb{R}^N$ with the standard basis $\{e_1, \dots, e_N\}$. Denote by $\square^N$ the standard unit cube in $\mathbb{R}^N$; that is

$$\square^N = \left\{ (x_1, \dots, x_n) \in \mathbb{R}^N \;\middle|\; 0 \leqslant x_i \leqslant 1 \text{ for each } i \right\}.$$

Given a $\bar{k}$-dimensional simplex $\langle v_{i_0}, \dots, v_{i_{\bar{k}}} \rangle$ in $\mathcal{S}$, mark the $(\bar{k}+1)$-dimensional faces in $\square^N$ (there are $2^{N-\bar{k}}$ of them) which are parallel to the coordinate $(k+1)$-plane spanned by $e_{i_0}, \dots, e_{i_{\bar{k}}}$.

Note that the set of all marked faces of $\square^N$ forms a cubical complex; it will be called the *cubical analog* of $\mathcal{S}$ and will be denoted as $\square_{\mathcal{S}}$.

Note that if a simplicial complex is connected then so is its cubical analog.

13.3.1. Proposition. *Let $\mathcal{S}$ be a connected simplicial complex and $\mathcal{Q} = \square_{\mathcal{S}}$ be its cubical analog. Then $\mathcal{Q}$ is connected and the link of any vertex of $\mathcal{Q}$ is isometric to $\mathcal{S}$ equipped with the spherical right-angled metric.*

In particular, if $\mathcal{S}$ is a flag complex then $\mathcal{Q}$ is a locally CBA[0] *space and therefore its universal cover $\tilde{\mathcal{Q}}$ is a* CBA[0] *space.*

Proof. The first part of the proposition follows from the construction above.

If $\mathcal{S}$ is flag, then by Flag condition (13.2.5) the link of any cube in $\mathcal{Q}$ is a CBA[1] space. Therefore, by cone construction (11.4.1*a*) $\mathcal{Q}$ is locally CBA[0] space. It remains to apply Hadamard–Cartan theorem 9.13.1. $\square$

From Proposition 13.3.1, it follows that the cubical analog of any flag complex is aspherical. The following exercise states that the converse also holds, see [46, 5.4].

13.3.2. Exercise. *A simplicial complex is flag if and only if its cubical analog is aspherical.*

13.4 Exotic aspherical manifolds

By Hadamard–Cartan theorem (9.13.1) any CBA[0] space is contractible. Therefore any complete locally CBA[0] space is *aspherical*; that is, they have contractible universal covers. This observation can be used to construct examples of aspherical spaces.

Let $\mathcal{X}$ be a proper topological space. Recall that $\mathcal{X}$ is called *simply connected at infinity* if for any compact set $K \subset \mathcal{X}$ there is a bigger compact set $K' \supset K$ such that $\mathcal{X} \backslash K'$ is path connected and any loop which lies in $\mathcal{X} \backslash K'$ is null-homotopic in $\mathcal{X} \backslash K$.

Recall that path connected spaces are not empty by definition. Therefore compact spaces are not simply connected at infinity.

The following statement was proved by Michael Davis in [45].

13.4.1. Proposition. *For any $m \geqslant 4$ there is a closed aspherical m-dimensional piecewise linear manifold whose universal cover is not simply connected at infinity.*

In particular, the universal cover of this manifold is not homeomorphic to the m-dimensional Euclidean space.

The proof requires the following lemma.

13.4.2. Lemma. *Let $\mathcal{S}$ be a flag complex, $\mathcal{Q} = \square_{\mathcal{S}}$ be its cubical analog and $\tilde{\mathcal{Q}}$ be the universal cover of $\mathcal{Q}$.*

Assume $\tilde{\mathcal{Q}}$ is simply connected at infinity. Then $\mathcal{S}$ is simply connected.

The proof below can be modified to prove the following extension.

13.4.3. Exercise. *Under the assumptions of the Lemma 13.4.2, for any vertex v in $\mathcal{S}$ the complement $\mathcal{S}\backslash\{v\}$ is simply connected.*

Proof of Lemma 13.4.2. Assume $\mathcal{S}$ is not simply connected. Choose a shortest noncontractible circle $\gamma\colon \mathbb{S}^1 \to \mathcal{S}$ formed by the edges of $\mathcal{S}$.

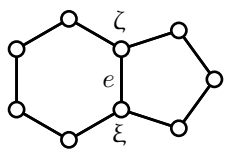
Note that γ forms a 1-dimensional subcompelex of $\mathcal{S}$ which is a closed local geodesic. Denote by G the subcomplex of $\mathcal{Q}$ which corresponds to γ.

Fix a vertex $v \in G$; let G_v be the connected component of G containing v. Let $\tilde{G}$ be the inverse image of G_v in $\tilde{\mathcal{Q}}$ for the universal cover $\tilde{\mathcal{Q}} \to \mathcal{Q}$. Fix a point $\tilde{v} \in \tilde{G}$ in the inverse image of v.

Note that

❶ *$\tilde{G}$ forms a convex set in $\tilde{\mathcal{Q}}$.*

Indeed, according to Proposition 13.3.1, $\tilde{\mathcal{Q}}$ is CBA[0]. By ???, it is sufficient to show that $\tilde{G}$ is locally convex in $\tilde{\mathcal{Q}}$, or equivalently G is locally convex in $\mathcal{Q}$.



The latter can only fail if γ passes through two vertices, say ξ to ζ in $\mathcal{S}$, which are joined by an edge not in γ; denote this edge by e.

Each edge of $\mathcal{S}$ has length $\frac{\pi}{2}$. Therefore each of two circles formed by e and an arc of γ from ξ to ζ is shorter that γ. Moreover, at least one of them is noncontractable since γ is not. That is, γ is not a shortest noncontractible circle — a contradiction. △

Further, note that $\tilde{G}$ is homeomorphic to the plane, since $\tilde{G}$ is a 2-dimensional manifold without boundary which by the above is CBA[0] and hence is contractible.

Denote by C_R the circle of radius R in $\tilde{G}$ centered at $\tilde{v}$. All C_R are homotopic to each other in $\tilde{G}\backslash\{\tilde{v}\}$ and therefore in $\tilde{\mathcal{Q}}\backslash\{\tilde{v}\}$.

Note that the map $\tilde{\mathcal{Q}}\backslash\{\tilde{v}\} \to \mathcal{S}$ defined by $x \mapsto \operatorname{dir}[\tilde{v}x]$ maps C_R to a circle homotopic to γ. Therefore C_R is not contractible in $\tilde{\mathcal{Q}}\backslash\{\tilde{v}\}$.

In particular, C_R is not contactable in $\tilde{\mathcal{Q}}\backslash K$ if $K \supset \tilde{v}$. If R is large, the circle C_R lies outside of any compact set K' in $\tilde{\mathcal{Q}}$. It follows that $\tilde{\mathcal{Q}}$ is not simply connected at infinity, a contradiction. □

Proof of Proposition 13.4.1. Let Σ^{m-1} be an $(m-1)$-dmensional smooth homology sphere which is not simply connected and bounds a contractible smooth compact m-dimensional manifold $\mathcal{W}$.

For $m \geqslant 5$ the existence of such $(\mathcal{W}, \Sigma)$ follows from [83]. For $m = 4$ it follows from the construction in [106].

Pick any smooth triangulation τ of W and let $\mathcal{S}$ be the resulting subcomplex which triangulates Σ.

We can assume that $\mathcal{S}$ is flag; otherwise pass to the barycentric subdivision of τ and apply Exercise 13.2.2.

Let $\mathcal{Q} = \square_{\mathcal{S}}$ be the cubical analog of $\mathcal{S}$.

By Proposition 13.3.1, $\mathcal{Q}$ is a homology manifold. It follows that $\mathcal{Q}$ is a piecewise linear manifold with a finite number of singularities at its vertices.

Removing a small neighborhood of each vertex in $\mathcal{Q}$, we can obtain a piecewise linear manifold which boundary is formed by several copies of Σ. To each copy of Σ, glue a copy of $\mathcal{W}$ along its boundary. This results in a closed piecewise linear manifold $\mathcal{M}$ which is homotopically equivalent to $\mathcal{Q}$.

Finally, by the Lemma 13.4.2, the universal cover $\tilde{\mathcal{Q}}$ of $\mathcal{Q}$ is not simply connected at infinity.

The same holds for the universal cover $\tilde{\mathcal{M}}$ of $\mathcal{M}$. The later follows since homotopy equivalences $f\colon \mathcal{Q} \to \mathcal{M}$ and $g\colon \mathcal{M} \to \mathcal{Q}$ lift to proper maps $\tilde{f}\colon \tilde{\mathcal{Q}} \to \tilde{\mathcal{M}}$ and $\tilde{g}\colon \tilde{\mathcal{M}} \to \tilde{\mathcal{Q}}$. □

13.5 Comments

Generalized flag condition. It would be interesting to get an analog of the flag condition for other simplices which (not necessary right-angled). Such a condition for some spherical Coxeter simplex could resolve the following problem; see [**?**] for more details.

13.5.1. Braid space problem. *Consider $\mathbb{C}^n$ with coordinates $z_1, z_2, \dots$ $\dots, z_n$. Let us remove from $\mathbb{C}^n$ the complex hyperplanes $z_i = z_j$, for all $i \neq j$, pass to the universal caver and consider the completion $\mathcal{B}_n$ of the obtained space.*

Is it true that $\mathcal{B}_n \in \mathrm{CAT}[0]$ for any n.

The above question has an affirmative answer for $n \leqslant 3$ and open for all $n \geqslant 4$. This and number of related questions are discussed in [122]

More exotic constructions. The construction used in the proof of Proposition 13.4.1 does many other interesting thins; some of them discussed in the survey [46] by Michael Davis.

The following Proposition was proved by Fredric Ancel, Michael Davis and Craig Guilbault in [21].

13.5.2. Proposition. *Given $m \geqslant 5$ there is a Euclidean polyhedral space $\mathcal{P}$ such that*

a) *$\mathcal{P}$ is homeomorphic to a closed m-dimensional manifold.*

b) $\operatorname{curv}\mathcal{P} \leqslant 0$

c) *The universal cover of $\mathcal{P}$ is not simply connected at infinity.*

Proof. Apply Exercise 13.4.3 to the barycentric subdivision of the simplicial complex $\mathcal{S}$ provided by Exercise 13.5.3. □

13.5.3. Exercise. *Given a positive integer $m \geqslant 5$ construct an $(m-1)$-dimensional simplicial complex $\mathcal{S}$ such that* $\operatorname{Cone}\mathcal{S}$ *is homeomorphic to $\mathbb{E}^m$ and $\pi_1(\mathcal{S}\backslash\{v\}) \neq 0$ for some vertex v in $\mathcal{S}$.*

It worth to note that any complete simply connected Riemannian manifold with nonpositive curvature is homeomorphic to the Euclidean space of the same dimension. In fact by Hadamard–Cartan theorem (9.13.1), the exponential map at one point of such manifold is a homeomorphism. In particular there is no Riemannian analog of the proposition. Moreover, according to a theorem of Stone, see [158, 47], there is no piecewise linear analog of the proposition; that is the homeomorphism to a manifold in Proposition 13.5.2 can not be made piecewise linear.

A similar construction was used by Michael Davis, Tadeusz Januszkiewicz and Jean-François Lafont [48] to construct a closed smooth 4-dimensional manifold M with universal cover $\tilde{M}$ diffeomorphic to $\mathbb{R}^4$ such that (1) M admits a polyhedral metric which is locally CBA[0] but it does not admit a Riemannian metric with nonpositive sectional curvature.

Part IV

Structure and tools

Chapter 14

First order differentiation

14.1 Space of directions and tangent space

Recall that according to the corollaries 8.3.2*b* and 9.3.2*b*, the CBB and CBA spaces have *defined angles*; that is the angle $\measuredangle[p\,{}^{x}_{y}]$ is defined for any hinge $[p\,{}^{x}_{y}]$ in the space.

Let $\mathcal{X}$ be a metric space with defined angles. Fix a point $p \in \mathcal{X}$.

Consider the set $\mathfrak{S}_p$ of all nontrivial unit-speed geodesics which start at p. By 6.3.2 the triangle inequality holds for $\measuredangle$ on $\mathfrak{S}_p$; that is, $(\mathfrak{S}_p, \measuredangle)$ forms a pseudometric space.

The metric space corresponding to $(\mathfrak{S}_p, \measuredangle)$ is called *space of geodesic directions* at p and denoted as Σ'_p or $\Sigma'_p\mathcal{X}$. The elements of Σ'_p are called *geodesic directions* at p. Each geodesic direction is formed by an equivalence class of geodesics starting from p for the equivalence relation

$$[px] \sim [py] \quad \Longleftrightarrow \quad \measuredangle[p\,{}^{x}_{y}] = 0.$$

14.1.1. Exercise. *Assume $\mathcal{L} \in$ CBB, and $[px]$, $[py]$ be two geodesics which correspond to the same geodesic direction $\xi \in \Sigma'_p$. Show that*

$$[px] \subset [py] \quad \text{or} \quad [px] \supset [py].$$

14.1.2. Exercise. *Assume $\mathcal{U} \in$ CBA be a space with extandable geodesics; that is for any geodesic in $\mathcal{U}$ is an arc in a both-side infinite local geodesic.*

Show that Σ'_p is complete for any $p \in \mathcal{U}$.

The completion of Σ'_p is called *space of directions* at p and is denoted as Σ_p or $\Sigma_p\mathcal{X}$. The elements of Σ_p are called *directions* at p.

14.1.3. Exercise. *Let $\mathcal{L} \in \mathrm{CBB}[1]$. Show that $\mathcal{L}$ contains at most 3 points with space of directions $\leqslant \frac{1}{2}\cdot\mathbb{S}^n$.*

The Euclidean cone $\operatorname{Cone}\Sigma_p$ over the space of directions Σ_p is called tangent space at p and denoted as T_p or $\mathrm{T}_p\mathcal{X}$.

The tangent space T_p could be also defined directly, without introducing the space of direction. To do so consider the set $\mathfrak{T}_p$ of all geodesics starting at p, with arbitrary speed. Given $\alpha, \beta \in \mathfrak{T}_p$, set

❶ $$|\alpha\,\beta|_{\mathfrak{T}_p} = \lim_{\varepsilon\to 0} \frac{|\alpha(\varepsilon)\ \beta(\varepsilon)|_{\mathcal{X}}}{\varepsilon}$$

Since the angles in $\mathcal{X}$ are defined, ❶ defines a pseudometric on $\mathfrak{T}_p$.

The corresponding metric space admits a natuaral isometric identification with the cone $\mathrm{T}'_p = \operatorname{Cone}\Sigma'_p$. The elements of T'_p are formed by the equivalence classes for the realtion

$$\alpha \sim \beta \quad\Longleftrightarrow\quad |\alpha(t)\ \beta(t)|_{\mathcal{X}} = o(t).$$

The completion of T'_p is therefore natuaral isometric to T_p.

The elements of T_p will be called tangent vector at p, despite that T_p is only cone — not a vector space. The elements of T'_p will be called geodesic tangent vector at p.

14.2 Velocity of curves

14.2.1. Definition. *Assume $\alpha\colon [0,a) \to \mathcal{X}$ for some $a > 0$ is a function, not necessary continuous, such that $\alpha(0) = p$. We say that $v \in \mathrm{T}_p$ is the right derivative of α at 0, breifly $\alpha^+(0) = v$ if there is a squence of vectors $v_n \in \mathrm{T}'_p$ with corresponding geodesics γ_n, such that $v_n \to v$ as $n \to \infty$ and*

$$\varlimsup_{\varepsilon\to 0+} \frac{|\alpha(\varepsilon)\ \gamma_n(\varepsilon)|_{\mathcal{X}}}{\varepsilon} \to 0 \quad \text{as} \quad n \to \infty.$$

Note that if the condition in the Definition 14.2.1 holds for one choice of the sequences (v_n) and (γ_n) then it holds for any other choice of that type.

Note that if γ is a geodesic starting at p and the tangent vector $v \in \mathrm{T}'_p$ corresponds to γ then $\gamma^+(0) = v$.

14.2.2. Exercise. *Assume $\mathcal{X}$ is a metric space with defined angles and $\alpha, \beta\colon [0,a) \to \mathcal{X}$ be two maps such that the right derivatives $\alpha^+(0)$, $\beta^+(0)$ are defined and $\alpha^+(0) = \beta^+(0)$. Show that*

$$|\alpha(t)\ \beta(t)|_{\mathcal{X}} = o(t).$$

14.2.3. Proposition. *Let $\mathcal{X}$ be a metric space with defined angles and $p \in \mathcal{X}$. Then for any tangent vector $v \in \mathrm{T}_p\mathcal{X}$ there is a map $\alpha\colon [0,\varepsilon) \to \mathcal{X}$ such that $\alpha^+(0) = v$.*

Proof. If $v \in \mathrm{T}'_p$ then for the corresponding geodesic α we have $\alpha^+(0) = v$.

Given $v \in \mathrm{T}_p$ construct a sequence $v_n \in \mathrm{T}'_p$ such that $v_n \to v$ and let γ_n be the sequence of corresponding geodesic.

The needed map α can be found among the maps such that $\alpha(0) = p$ and

$$\alpha(t) = \gamma_n(t) \quad \text{if} \quad \varepsilon_{n+1} \leqslant t < \varepsilon_n,$$

where (ε_n) is a decreasing sequence converging to 0 as $n \to \infty$. In order to satify the condition one has to choose the sequence ε_n converging to 0 very fast. □

We define right and left derivatives $\alpha^+(t_0)$ and $\alpha^-(t_0)$ of α at $t_0 \in \mathbb{I}$ as

$$\alpha^\pm(t_0) = \check{\alpha}^+(0),$$

where $\check{\alpha}(t) = \alpha(t_0 \pm t)$. The sign conversion is not quite standard, say if α is a smooth curve in a Riemannian manifold then we have

$$\alpha^+(t) = -\alpha^-(t).$$

14.2.4. Definition. *Let $\mathcal{X}$ be a metric space and $\alpha\colon \mathbb{I} \to \mathcal{X}$ be a curve.*

For $t_0 \in \mathbb{I}$, if $\alpha^+(t_0)$ or $\alpha^-(t_0)$ or both are defined, we say correspondingly that α is right *or* left *or* both-sided differentiable *at t_0. In the exceptional cases where t_0 is the left (respectively right) end of $\mathbb{I}$, α is by definition left (respectively right) differentiable at t_0.*

14.2.5. Exercise. *Assume $\mathcal{X}$ is a metric space with defined angles Show that any geodesic $\gamma\colon \mathbb{I} \to \mathcal{X}$ is both-sided differentiable everywhere.*

14.3 Differential

14.3.1. Definition. *Let $\mathcal{X}$ be a metric space with defined angles and $f\colon \mathcal{X} \to \mathbb{R}$ be a subfunction, $p \in \operatorname{Dom} f$ and $\mathbb{I}$ be a real interval. A function $\varphi\colon \mathrm{T}_p \to \mathbb{R}$ is called differential of f at p (briefly $\varphi = \boldsymbol{d}_p f$) if for any map $\alpha\colon \mathbb{I} \to \mathcal{X}$ such that $\alpha(0) = p$ and $\alpha^+(0)$ is defined, we have*

$$(f \circ \alpha)^+(0) = \varphi(\alpha^+(0)).$$

14.3.2. Proposition. *Let $f\colon \mathcal{X} \to \mathbb{R}$ be a locally Lipschitz semiconcave subfunction. Then differential $\boldsymbol{d}_p f$ is uniquely defined for any $p \in \operatorname{Dom} f$. Moreover,*

a) The differential $\boldsymbol{d}_p f\colon \mathrm{T}_p \to \mathbb{R}$ is Lipschitz and the Lipschitz constant of $\boldsymbol{d}_p f\colon \mathrm{T}_p \to \mathbb{R}$ does not exceed the Lipschitz constant of f in a neighborhood of p.

b) $\boldsymbol{d}_p f\colon \mathrm{T}_p \to \mathbb{R}$ is a positive homogenius function; that is, for any $\mathfrak{z} \geqslant 0$ and $v \in \mathrm{T}_p$ we have

$$\mathfrak{z}\cdot \boldsymbol{d}_p f(v) = \boldsymbol{d}_p f(\mathfrak{z}\cdot v).$$

Proof. Passing to a subdomain of f if nesessary, we can assume that f is $\pounds$-Lipschitz and λ-concave for some $\pounds, \lambda \in \mathbb{R}$.

Take a geodessic γ starting at p which lies in $\operatorname{Dom} f$. Since $f \circ \gamma$ is semiconcave, the rigth derivative $(f \circ \gamma)^+(0)$ is defined. Since f is $\pounds$-Lipschitz, we have

$$❶ \qquad |(f \circ \gamma)^+(0) - (f \circ \gamma_1)^+(0)| \leqslant \pounds\cdot|\gamma^+(0)\ \gamma_1^+(0)|$$

for any other geodesic γ_1 starting at p.

Define $\varphi\colon \mathrm{T}'_p \to \mathbb{R}\colon \gamma^+(0) \mapsto (f \circ \gamma)^+(0)$. From ❶, φ is a $\pounds$-Lipschtz function defined on T'_p. Thus, we can extend φ to a whole T_p as a $\pounds$-Lipschitz function.

It remains to show that φ is differential of f at p. Assume $\alpha\colon [0, a) \to \mathcal{X}$ is a map such that $\alpha(0) = p$ and $\alpha^+(0) = v \in \mathrm{T}_p$. Let $\gamma_n \in \Gamma_p$ be a sequence of geodesics as in the definition 14.2.1; that is, if

$$v_n = \gamma_n^+(0) \quad \text{and} \quad a_n = \varlimsup_{t \to 0+} |\alpha(t)\ \gamma_n(t)|/t$$

then $a_n \to 0$ and $v_n \to v$ as $n \to \infty$. Then

$$\varphi(v) = \lim_{n \to \infty} \varphi(v_n),$$

$$f \circ \gamma_n(t) = f(p) + \varphi(v_n)\cdot t + o(t),$$

$$|f \circ \alpha(t) - f \circ \gamma_n(t)| \leqslant \pounds\cdot|\alpha(t)\ \gamma_n(t)|.$$

Hence

$$f \circ \alpha(t) = f(p) + \varphi(v)\cdot t + o(t)$$

□

14.4 Ultratangent space

Recall that we assume that ω is a once for all fixed choice of a selective ultrafilter.

For a metric space $\mathcal{X}$ and a positive real number $\mathfrak{z}$, we will denote by $\mathfrak{z}\cdot\mathcal{X}$ its $\mathfrak{z}$-*blowup*, which is a metric space with the same underlying set as $\mathcal{X}$ and the metric multiplied by $\mathfrak{z}$. The tautological bijection $\mathcal{X} \to \mathfrak{z}\cdot\mathcal{X}$ will be denoted as $x \mapsto x^{\mathfrak{z}}$, so

$$|x^{\mathfrak{z}}\,y^{\mathfrak{z}}| = \mathfrak{z}\cdot|x\,y|$$

for any $x, y \in \mathcal{X}$.

The ω-blowup $\omega\cdot\mathcal{X}$ of $\mathcal{X}$ is defined as the ω-limit of $n\cdot\mathcal{X}$; that is,

$$\omega\cdot\mathcal{X} \overset{def}{=\!=} \lim_{n\to\omega} n\cdot\mathcal{X}.$$

Given a point $x \in \mathcal{X}$ we can consider the sequence $x^n \in n\cdot\mathcal{X}$; it corresponds to a point $x^\omega \in \omega\cdot\mathcal{X}$. Note that if $x \neq y$ then

$$|x^\omega\,y^\omega|_{\omega\cdot\mathcal{X}} = \infty;$$

that is, x^ω and y^ω belong to different metric components of $\omega\cdot\mathcal{X}$.

The metric component of x^ω in $\omega\cdot\mathcal{X}$ is called ultratangent space of $\mathcal{X}$ at x and it is denoted as $\mathrm{T}^\omega_x\mathcal{X}$.

Equivalently, ultratangent space $\mathrm{T}^\omega_x\mathcal{X}$ can be defined the following way. Consider all the sequences of points $x_n \in \mathcal{X}$ such that the sequence $\ell_n = n\cdot|x\,x_n|_{\mathcal{X}}$ is bounded. Define the pseudistance between two such sequences as

$$|(x_n)\ (y_n)| = \lim_{n\to\omega} n\cdot|x_n\,y_n|_{\mathcal{X}}.$$

Then $\mathrm{T}^\omega_x\mathcal{X}$ is the corresponding metric space.

Both, tangent space and ultratangent space, generalize the notion of tangent space of Riemannian manifold. In the simplest cases these two notions define the same space. In general, they are different and both useful — often lack of a property in one is compensated by the other.

It is clear from the definition that tangent space has cone structure. On the other hand, in general, ultratangent space does not have a cone structure; a Hilbert's cube $\prod_{n=1}^{\infty}[0, 2^{-n}]$ which is CBB[0] as well as CBA[0] gives an example.

14.4.1. Theorem.

a) If $\mathcal{L} \in \mathrm{CBB}[\kappa]$ and $p \in \mathcal{L}$ then $\mathrm{T}^\omega_p \in \mathrm{CBB}[0]$.

b) If $\mathcal{U} \in \mathrm{CBA}[\kappa]$ and $p \in \mathcal{U}$ then $\mathrm{T}^\omega_p \in \mathrm{CBA}[0]$.

The proofs of both parts are nearly identical.

Proof; (a). Since $\mathcal{L} \in \mathrm{CBB}[\kappa]$, then for its blowup $n{\cdot}\mathcal{L}$, we have $n{\cdot}\mathcal{L} \in$ $\in \mathrm{CBB}[\kappa/n^2]$. Since $n{\cdot}(\mathcal{L}, p) \to (\mathrm{T}_p^\omega, o)$ as $n \to \omega$, Proposition 8.1.7 implies $\mathrm{T}_p^\omega \in \mathrm{CBB}[0]$.

(b). Since $\mathcal{U} \in \mathrm{CBA}[\kappa]$, then for its blowup $n{\cdot}\mathcal{U}$, we have $n{\cdot}\mathcal{L} \in$ $\in \mathrm{CBA}[\kappa/n^2]$. Since $n{\cdot}(\mathcal{U}, p) \to (\mathrm{T}_p^\omega, o)$ as $n \to \omega$, Proposition 9.1.5 implies $\mathrm{T}_p^\omega \in \mathrm{CBA}[0]$. □

The next theorem shows that T_p can be (and often will be) considered as a subset of T_p^ω. That also implies that (1+3)-point comparison (8.1.2) as well as n-point comparison (10.2.1) holds in T_p.

14.4.2. Theorem. *Let $\mathcal{X}$ be a metric space with defined angles. Then for any $p \in \mathcal{L}$, there is an distance preserving map*

$$\iota : \mathrm{T}_p \hookrightarrow \mathrm{T}_p^\omega$$

such that for any geodesic γ starting at p we have $\gamma^+(0) \mapsto \lim_{n\to\omega}[\gamma(\frac{1}{n})]^n$.

Proof. Given $v \in \mathrm{T}'_p$ choose a geodesic γ which starts at p and $\gamma^+(0) = v$. Set

$$v^\omega = \lim_{n\to\omega}[\gamma(\tfrac{1}{n})]^n.$$

Note that the value $v^\omega \in \mathrm{T}_p^\omega$ does not depend on choice of γ. Indeed, if γ_1 is an other geodesic starting at p such that $\gamma_1^+(0) = v$ then

$$|\gamma(t)\ \gamma_1(t)|_{\mathcal{X}} = o(t).$$

Therefore

$$\lim_{n\to\omega}[\gamma(\tfrac{1}{n})]^n = \lim_{n\to\omega}[\gamma_1(\tfrac{1}{n})]^n.$$

Set $\iota(v) = v^\omega$. Since angles between geodesics in $\mathcal{X}$ are defined, for any $v, w \in \mathrm{T}'_p$ we have $n{\cdot}|v_n\, w_n| \to |v\, w|$. Thus $|v_\omega\, w_\omega| = |v\, w|$; that is, ι is a global isometry of T'_p.

Since T'_p is dense in T_p, we can extend ι to a global isometry $\mathrm{T}_p \to \mathrm{T}_p^\omega$. □

14.5 Linear subspace of tangent space

14.5.1. Definition. *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$, $p \in \mathcal{L}$ and $u, v \in \mathrm{T}_p$. We say that vectors u and v are* opposite *to each other, (briefly, $u + v = 0$) if $|u| = |v| = 0$ or $\measuredangle(u, v) = \pi$ and $|u| = |v|$.*

The subcone

$$\mathrm{Lin}_p = \{v \in \mathrm{T}_p \mid \exists\, w \in \mathrm{T}_p \ \ \textit{such that} \ \ w + v = 0\}$$

will be called linear subcone *of* T_p.

The reason for choosing the name "linear subcone" will be evident after proving Theorem 14.5.4.

Let $u \in \mathrm{Lin}_p$; that is $u + v = 0$ for some $v \in \mathrm{T}_p$. Given $s < 0$, set $s{\cdot}u = (-s){\cdot}v$. This way we define multiplication of any vector in Lin_p by any real number (positive and negative). Proposition 14.5.3 implies that such multiplication is uniquely defined.

14.5.2. Proposition. *Let* $\mathcal{L} \in \mathrm{CBB}$ *and* $p \in \mathcal{L}$. *Given two vectors* $u, v \in \mathrm{T}_p$, *the following statements are equivalent:*

a) $u + v = 0$;

b) $\langle u, x\rangle + \langle v, x\rangle = 0$ *for any* $x \in \mathrm{T}_p$;

c) $\langle u, \xi\rangle + \langle v, \xi\rangle = 0$ *for any* $\xi \in \Sigma_p$.

Proof. The condition $u + v = 0$ is equivalent to $\langle u, u\rangle = -\langle u, v\rangle = \langle v, v\rangle$; thus $(b)\Rightarrow(a)$. Since T_p is isometric to a subset of T_p^ω, the splitting theorem (16.9.1) applied for T_p^ω gives $(a)\Rightarrow(b)$. The equivalence $(b)\Leftrightarrow(c)$ is trivial. □

14.5.3. Proposition. *Let* $\mathcal{L} \in \mathrm{CBB}$ *and* $p \in \mathcal{L}$. *Then for any three vectors* $u, v, w \in \mathrm{T}_p$, $u + v = 0$ *and* $u + w = 0$ *implies* $v = w$.

Proof. From Proposition 14.5.2 both v and w satisfy the condition in corollary 16.3.2. Hence the result. □

14.5.4. Theorem. *Let* $\mathcal{L} \in \mathrm{CBB}[\kappa]$ *and* $p \in \mathcal{L}$. *Then* Lin_p *forms a subcone of* T_p *isometric to a Hilbert space.*

Before comming to the proof of the theorem, let us give its corollary.

14.5.5. Corollary. *Let* $\mathcal{L} \in \mathrm{CBB}[\kappa]$ *and* $p \in \mathrm{Str}(x_1, x_2, \dots, x_n)$. *Then there is a subcone* $E \subset \mathrm{T}_p$ *which isometric to a Euclidean space such that* $\log[px_i] \in E$ *for every* i.

Proof. By the definition of Str (8.2.3), $\log[px_i] \in \mathrm{Lin}_p$ for each i. It remains to apply Theorem 14.5.4. □

The main difficulty in the proof of Theorem 14.5.4 comes from the fact that in general $\mathrm{T}_p \notin \mathrm{CBB}[0]$; see Habeisen's example (Section 14.7). Otherwise the statement would follow directly from the Splitting theorem (16.9.1). In fact the proof of this theorem is a far

walk around, in the proof we use construction of gradient, as well as Splitting theorem, proof of which use gradient flow. Thus in order to understand our proof completely one needs to read most of Chapter 16.

First we give a construction of a tangent vector w for given two vectors u, v. If the tangent space is Euclidean, then $w = -u - v$.

Proof of 14.5.4. First we show that $\mathrm{Lin}_p \in \mathrm{CBB}[0]$.

Note that $\mathrm{T}_p^\omega \in \mathrm{CBB}[0]$ (see 14.4.1a) and Lin_p is a closed subset of T_p^ω. Thus, it is sufficient to show that the metric on Lin_p inherited from T_p^ω is a length-metric.

Fix two vectors $x, y \in \mathrm{Lin}_p$. Let u and v be such that $u + \frac{1}{2}\cdot x = 0$ and $v + \frac{1}{2}\cdot y = 0$. Let us apply Lemma 16.3.1 for vectors u and v to obtain vector $w \in \mathrm{T}_p$.

❶ *w is a midpoint of $[xy]$.*

Indeed, according to Lemma 16.3.1,

$$\begin{aligned}|w|^2 &= -\langle w, u\rangle - \langle w, v\rangle = \\ &= \tfrac{1}{2}\cdot\langle w, x\rangle + \tfrac{1}{2}\cdot\langle w, y\rangle.\end{aligned}$$

Therefore,

$$\begin{aligned}|xw|^2 + |wy|^2 &= 2\cdot|w|^2 + |x|^2 + |y|^2 - 2\cdot\langle w, x\rangle - 2\cdot\langle w, y\rangle = \\ &= |x|^2 + |y|^2 - \langle w, x\rangle - \langle w, y\rangle \leqslant \\ &\leqslant |x|^2 + |y|^2 + \langle u, x\rangle + \langle v, x\rangle + \langle u, y\rangle + \langle v, y\rangle = \\ &= \tfrac{1}{2}\cdot|x|^2 + \tfrac{1}{2}\cdot|y|^2 - \langle x, y\rangle = \\ &= \tfrac{1}{2}\cdot|xy|^2.\end{aligned}$$

Thus $|xw| = |wy| = \frac{1}{2}\cdot|xy|$ and ❶ follows. △

Note that for any $v \in \mathrm{Lin}_p$ there is a line ℓ passing trough v and o, thus applying 16.9.2, we get that Lin_p is isometric to a Hilbert space. □

14.6 Differential and ultradifferential

Note that $|p_*|\colon \mathcal{L} \to \mathbb{R}$ is semiconcave in $\mathcal{L}\backslash\{p\}$. Thus for any $q \neq p$ one can define differential. The following claim follows directly from first variation formula (8.6.6).

14.6.1. Theorem???. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$ and $p, q \in \mathcal{L}$ be distinct points. Let $\mathrm{Dir}_p q$ denotes the set of all directions of geodesics from p to q in $\mathcal{L}^\omega$. Then*

$$(\boldsymbol{d}_q|p_*|)(v) = -\sup\{\,\langle\xi, v\rangle \mid \xi \in ???\mathrm{Dir}_q p\,\}.$$

In particular, for any geodesic $[qx]$ *we have*

$$(\boldsymbol{d}_q|p_*|)(\mathrm{dir}[qx]) \geqslant -\cos \measuredangle^{\kappa}(q\,{}^{x}_{p})$$

if the right hand side is defined.

14.6.2. Theorem. *Let* $\mathcal{L} \in \mathrm{CBB}[\kappa]$ *and* $p, q \in \mathcal{L}$ *be to distinct points such that there is unique geodesic* $[pq]$ *in* $\mathcal{L}^{\omega}$. *Then for any* $v \in \mathrm{T}_q$, *we have*

$$(\boldsymbol{d}_q|p_*|)(v) = -\langle \mathrm{dir}[qp], v\rangle.$$

In particular, the above identity holds if $q \in \mathrm{Str}(p)$ *and if* $p \in \mathrm{Str}(q)$.

In general, as it shown in Halbeisen's example (see section 14.7), T_p might be not a length space; thus concavity of $\boldsymbol{d}_p f$ is meaningless. Nevertheless, as the following lemma shows, differential $\boldsymbol{d}_p f$ of a semiconcave function allways satisfy the following weaker property similar to concavity (compare [140, 136], [118, 4.2]). Note that in finite dimensional case $\boldsymbol{d}_p f$ is concave (see ???), but again, we use gradient to prove some basic properties of dimension.

14.6.3. Lemma. *Let* $\mathcal{L} \in \mathrm{CBB}[\kappa]$, $f\colon \mathcal{L} \looparrowright \mathbb{R}$ *be a locally Lipschitz semiconcave subfunction and* $p \in \mathrm{Dom}\, f$. *Then*

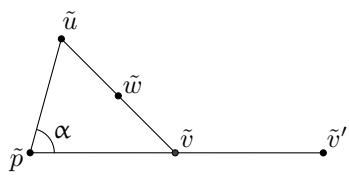
$$\sup\{\, (\boldsymbol{d}_p f)(\xi) \mid \xi \in \Sigma_p \,\} \geqslant \frac{(\boldsymbol{d}_p f)(u) + (\boldsymbol{d}_p f)(v)}{\sqrt{|u|^2 + 2\cdot\langle u, v\rangle + |v|^2}}$$

for any $u, v \in \mathrm{T}_p$.

Proof of 14.6.3. We can assume that $\alpha = \measuredangle(u, v) > 0$, otherwise the statement is trivial. Moreover, since $\mathrm{T}'_p = \mathrm{Cone}(\Sigma'_p)$ is dense in T_p and $\boldsymbol{d}_p f\colon \mathrm{T}_p \to \mathbb{R}$ is Lipschitz, we can assume that $u, v \in \mathrm{T}'_p$; that is, $\exp_p(t\cdot u)$ and $\exp_p(t\cdot v)$ are defined for all small $t > 0$.

Prepare a model configuration of five points: $\tilde p, \tilde u, \tilde v, \tilde v', \tilde w \in \mathbb{E}^2$ such that

- $\measuredangle[\tilde p\,{}^{\tilde u}_{\tilde v}] = \alpha$,
- $|\tilde p\,\tilde u| = |u|$,
- $|\tilde p\,\tilde v| = |v|$,
- $\tilde v'$ lies on the extension of $[\tilde p\tilde v]$ so that $\tilde v$ is midpoint of $[\tilde p\tilde v']$,
- $\tilde w$ is the midpoint for $\tilde u$ and $\tilde v$.



Note that

$$|\tilde{p}\,\tilde{w}| = \tfrac{1}{2}\cdot\sqrt{|u|^2 + 2\cdot\langle u, v\rangle + |v|^2}.$$

Assume that $\mathcal{L}$ is geodesic.

For all small $t > 0$, construct points $u_t, v_t, v'_t, w_t \in \mathcal{L}$ the following way:

(a) $v_t = \exp_p(t\cdot v)$, $v'_t = \exp_p(t\cdot v')$

(b) $u_t = \exp_p(t\cdot u)$.

(c) w_t is the midpoint of $[u_t v_t]$.

Clearly $|p\,u_t| = t|u|$, $|p\,v_t| = t|v|$, $|p\,v'_t| = t|v'|$. Since $\measuredangle(u, v)$ is defined, we have $|u_t\,v_t| = t|\tilde{u}\,\tilde{v}| + o(t)$ and $|u_t\,v'_t| = t|\tilde{u}\,\tilde{v}'| + o(t)$ (see Theorem 8.3.1d and Section 8.1).

From point-on-side and hinge comparison (8.3.1c+d), we get that

$$\measuredangle^{\kappa}(v_t\,{}^{p}_{w_t}) \geqslant \measuredangle^{\kappa}(v_t\,{}^{p}_{u_t}) \geqslant \measuredangle[\tilde{v}\,{}^{\tilde{p}}_{\tilde{u}}] + \tfrac{o(t)}{t}$$

and

$$\measuredangle^{\kappa}(v_t\,{}^{v'_t}_{w_t}) \geqslant \measuredangle^{\kappa}(v_t\,{}^{v'_t}_{u_t}) \geqslant \measuredangle[\tilde{v}\,{}^{\tilde{v}'}_{\tilde{u}}] + \tfrac{o(t)}{t}.$$

Clearly, $\measuredangle[\tilde{v}\,{}^{\tilde{p}}_{\tilde{u}}] + \measuredangle[\tilde{v}\,{}^{\tilde{x}}_{\tilde{u}'}] = \pi$. From adjacent angle comparison (8.3.1b), $\measuredangle^{\kappa}(v_t\,{}^{p}_{v_t}) + \measuredangle^{\kappa}(v_t\,{}^{u_t}_{v'_t}) \leqslant \pi$. Hence $\measuredangle^{\kappa}(v_t\,{}^{p}_{w_t}) \to \measuredangle[\tilde{v}\,{}^{\tilde{p}}_{\tilde{v}}]$ as $t \to 0+$ and thus

$$|p\,w_t| = t|\tilde{p}\,\tilde{w}| + o(t).$$

Since f is λ-concave we have

$$\begin{aligned} 2\cdot f(w_t) &\geqslant f(u_t) + f(v_t) + \tfrac{\lambda}{4}\cdot|u_t\,v_t|^2 = \\ &= 2\cdot f(p) + t\cdot[(\boldsymbol{d}_p f)(u) + (\boldsymbol{d}_p f)(v)] + o(t). \end{aligned}$$

Applying λ-concavity of f, we get

❶ $$(\boldsymbol{d}_p f)(\mathrm{dir}[p w_t]) \geqslant \frac{t\cdot[(\boldsymbol{d}_p f)(u) + (\boldsymbol{d}_p f)(v)] + o(t)}{2\cdot t\cdot|\tilde{p}\,\tilde{w}| + o(t)}$$

and the lemma follows.

Finally, if $\mathcal{L}$ is not geodesic one needs to make two adjustments in the above constructions. Namely:

(i) For geodesic $[u_t v_t]$ to be defined, one has to take in (b) $u_t \in \mathrm{Str}(v_t)$, $u_t \approx \exp_p(t\cdot u)$. More precicely,

$$|u_t \exp_p(t\cdot u)| = o(t).$$

Thus instead of $|p\,u_t| = t\cdot|u|$ we have

$$|p\,u_t| = t|u| + o(t),$$

but it is enough for the rest of proof.

(ii) The direction $\operatorname{dir}[pw_t]$ might be undefined. Thus, in the estimate ❶, instead of $\operatorname{dir}[pw_t]$, one should take $\operatorname{dir}[pw'_t]$ for some point $w'_t \in \operatorname{Str}(p)$ near w_t (that is, $|w_t\,w'_t| = o(t)$) □

Ultradifferential. Given a function $f\colon \mathcal{L} \to \mathbb{R}$, consider sequence of functions $f_n\colon n{\cdot}\mathcal{L} \to \mathbb{R}$, defined by

$$f_n(x^n) = n{\cdot}(f(x) - f(p)),$$

here $x^n \in n{\cdot}\mathcal{L}$ is the point corresopnding to $x \in \mathcal{L}$. While $n{\cdot}(\mathcal{L}, p) \to \to (\mathrm{T}^\omega, o)$ as $n \to \omega$, functions f_n converge to ω-differential of f at p. It will be denoted by $\boldsymbol{d}^\omega_p f$;

$$\boldsymbol{d}^\omega_p f\colon \mathrm{T}^\omega_p \to \mathbb{R}, \quad \boldsymbol{d}^\omega_p f = \lim_{n\to\omega} f_n.$$

Clearly, the ω-differential of a locally Lipschitz subfunction is defined at each point of definition.

14.6.4. Proposition. *Assume $\mathcal{X}$ is a* CBB *or* CBA *space and $f\colon \mathcal{X} \looparrowright \mathbb{R}$ be a semiconcave locally Lipscitz subfunction. Then for any $p \in \operatorname{Dom} f$, the ultradifferential $\boldsymbol{d}^\omega_p\colon \mathrm{T}^\omega_p \to \mathbb{R}$ is a concave function.*

Proof. Fix a geodesic $[x^\omega y^\omega]$ in T^ω_p. It is sufficient to show that for any subarc $[\bar{x}^\omega \bar{y}^\omega]$ of $[x^\omega y^\omega]$ which does not contains the ends there is a sequence of geodesics $[\bar{x}^n \bar{y}^n]$ in $n{\cdot}\mathcal{X}$ which converges to $[\bar{x}^\omega \bar{y}^\omega]$.

Choose any sequences $\bar{x}^n, \bar{y}^n \in n{\cdot}\mathcal{X}$ such that $\bar{x}^n \to \bar{x}^\omega$ and $\bar{y}^n \to \to \bar{y}^\omega$ as $n \to \omega$. Note taht $[\bar{x}^n \bar{y}^n]$ converges to $[\bar{x}^\omega \bar{y}^\omega]$ as $n \to \omega$. The latter holds tivially in case CBA and in CBB case follows from ???. □

14.7 Halbeisen's example

Here we give a construction of the following

14.7.1. Halbeisen's example. *There is a space $\check{\mathcal{L}} \in \mathrm{CBB}^\infty$ with a point $p \in \check{\mathcal{L}}$ such that the space of directions $\Sigma_p\check{\mathcal{L}}$ and therefore the tangent space $\mathrm{T}_p\check{\mathcal{L}}$ is not a length space.*

Our construction is just a minor variation of one in [70]. If the dimension is finite, such examples do not exist, see **??**; for proper spaces the question is open, see **??**.

Construction. Let $\mathbb{H}$ be a Hilbert space formed by infinite sequences of real numbers $\boldsymbol{x} = (x_0, x_1, \dots)$ with norm $|\boldsymbol{x}|^2 = \sum_i (x_i)^2$. Fix $\varepsilon = 0.001$ and consider two functions $f, \check{f} : \mathbb{H} \to \mathbb{R}$

$$f(\boldsymbol{x}) = |\boldsymbol{x}|,$$

$$\check{f}(\boldsymbol{x}) = \max\left\{|\boldsymbol{x}|, \max_{n\in\mathbb{N}}\{(1+\varepsilon)\cdot x_n - \tfrac{1}{n}\}\right\}.$$

Both of these functions are convex and Lipschitz, therefore their graphs in $\mathbb{H}\times\mathbb{R}$ equipped with its length-metric form infinite dimensional Alexandrov spaces, say $\mathcal{L}$ and $\check{\mathcal{L}}$ (it is proved formally in 14.7.2). Set p to be the origin of $\mathbb{H}\times\mathbb{R}$.

Note that $\check{\mathcal{L}}\cap\mathcal{L}$ is starshaped in $\mathbb{H}$ with center at p. Further $\check{\mathcal{L}}\backslash\mathcal{L}$ consist of countable number of disjointed sets

$$\Omega_n = \left\{\, (\boldsymbol{x}, \check{f}(\boldsymbol{x})) \in \check{\mathcal{L}} \;\middle|\; (1+\varepsilon)\cdot x_n - \tfrac{1}{n} > |\boldsymbol{x}| \,\right\}$$

and $|\Omega_n p| > \frac{1}{n}$. It follows that for any geodesic $[pq]$ in $\check{\mathcal{L}}$, a small subinterval $[p\bar{q}] \subset [pq]$ coinsides with a straight segment in $\mathbb{H}\times\mathbb{R}$, which is also a geodesic in $\mathcal{L}$. Thus we can treat $\Sigma_p\mathcal{L}$ and $\Sigma_p\check{\mathcal{L}}$ as one set, with two angle metrics $\measuredangle$ and $\check{\measuredangle}$; let us denote by $\measuredangle_{\mathbb{H}\times\mathbb{R}}$ the angle in $\mathbb{H}\times\mathbb{R}$.

The space $\mathcal{L}$ is isometric to the Euclidean cone over $\Sigma_p\mathcal{L}$ with vertex at p; $\Sigma_p\mathcal{L}$ is isometric to a sphere in Hilbert space with radius $\frac{1}{\sqrt{2}}$. In particular, $\measuredangle$ is length-metric of $\Sigma_p\mathcal{L}$ in $\measuredangle_{\mathbb{H}\times\mathbb{R}}$.

Thus, to show that $\check{\measuredangle}$ does not define a length-metric on $\Sigma_p\mathcal{L}$ it is sufficient to construct a pair of directions (ξ_+, ξ_-) such that

$$\check{\measuredangle}(\xi_+, \xi_-) < \measuredangle(\xi_+, \xi_-).$$

Set $\boldsymbol{e}_0 = (1,0,0,\dots)$, $\boldsymbol{e}_1 = (0,1,0,\dots), \dots \in \mathbb{H}$. Consider the following two rays in $\mathbb{H}\times\mathbb{R}$

$$\gamma_+(t) = \tfrac{t}{\sqrt{2}}\cdot(\boldsymbol{e}_0, 1) \;\text{ and }\; \gamma_-(t) = \tfrac{t}{\sqrt{2}}\cdot(-\boldsymbol{e}_0, 1), \quad t\in[0,+\infty).$$

They form unit-speed geodesics in both, $\mathcal{L}$ and $\check{\mathcal{L}}$. Let $\xi_\pm$ be the directions of $\gamma_\pm$ at p. Denote by σ_n the half-planes in $\mathbb{H}$ spanned by $\boldsymbol{e}_0$ and $\boldsymbol{e}_n$; that is, $\sigma_n = \{\, x\cdot\boldsymbol{e}_0 + y\cdot\boldsymbol{e}_n \mid y \geqslant 0 \,\}$. Consider sequence of 2-dimensional sectors $Q_n = \check{\mathcal{L}}\cap(\sigma_n\times\mathbb{R})$. For each n, the sector Q_n passes through Ω_n and it is bounded by two geodesic rays $\gamma_\pm$. Note that $Q_n \xrightarrow[\text{GH}]{} Q$, where Q is a solid Euclidean angle in $\mathbb{E}^2$ with angle measure $\beta < \measuredangle(\xi_+, \xi_-) = \frac{\pi}{\sqrt{2}}$. Indeed, Q_n is isometric to the subset of $\mathbb{E}^3$ described by

$$y \geqslant 0 \;\text{ and }\; z = \max\left\{\sqrt{x^2+y^2}, (1+\varepsilon)\cdot y - \tfrac{1}{n}\right\}$$

with length-metric. Thus, its limit Q is isometric to the subset of $\mathbb{E}^3$ described by

$$y \geqslant 0 \;\text{ and }\; z = \max\left\{\sqrt{x^2+y^2}, (1+\varepsilon)\cdot y\right\}$$

with length-metric. In particular, for any $t, \tau \geqslant 0$,

$$|\gamma_+(t)\ \gamma_-(\tau)|_{\tilde{\mathcal{L}}} \leqslant \lim_{n\to\infty} |\gamma_+(t)\ \gamma_-(\tau)|_{Q_n} = \\ = \tilde{\Upsilon}^0\{\beta; t, \tau\}.$$

I.e., $\check{\measuredangle}(\xi_+, \xi_-) \leqslant \beta < \measuredangle(\xi_+, \xi_-)$. □

14.7.2. Lemma. *Let $\mathbb{H}$ be a Hilbert space, $f\colon \mathbb{H} \to \mathbb{R}$ be a convex Lipschitz function and $S \subset \mathbb{H} \times \mathbb{R}$ be the graph of f equipped with length-metric. Then $S \subset \mathrm{CBB}[0]$.*

Proof. Recall that a subset $X \subset \mathbb{H} \times \mathbb{R}$, we will denote by $|{*}{*}|_X$ the length-metric on X.

Note that according to 12.6.1 any convex hypersurface in a Euclidean space equipped with length-metric is non-negatively curved. Thus it is enough to show that for any 4-point set $\{x_0, x_1, x_2, x_3\} \subset \subset S$, there is a finite-dimensional subspace $E \subset \mathbb{H} \times \mathbb{R}$, such that $\{x_i\} \in E$ and $|x_i x_j|_{S\cap E}$ is arbitrary close to $|x_i x_j|_S$.

Clearly $|x_i x_j|_{S\cap E} \geqslant |x_i x_j|_S$; thus it is enough to show that for given $\varepsilon > 0$ one can choose E, so that

❶ $$|x_i x_j|_{S\cap E} < |x_i x_j|_S + \varepsilon$$

For each pair (x_i, x_j) choose a broken line β_{ij} connecting x_i, x_j, which lies under S (that is, outside of $\operatorname{Conv} S$) in $\mathbb{H}\times\mathbb{R}$ and has length at most $|x_i x_j|_S + \varepsilon$. Take E to be the affine hull of all the vertexes in all β_{ij}. Thus,

$$|x_i x_j|_{S\cap E} \leqslant \operatorname{length} \beta_{ij}$$

and ❶ follows. □

14.7.3. Exercise. *Construct a non-compact space $\mathcal{L} \in \mathrm{CBB}[0]$ which contains no rays.*

Chapter 15

Dimension

In Section 15.1, we give definitions of different type of dimension-like invariants which have sense for metric spaces and state general relations between them. In sections 15.2 and 15.3, we prove equivalence of all reasonable dimension-like invariants for CBB spaces. In Section 15.5 we discuss the dimension of CBA spaces.

15.1 Preliminaries.

The proof of most of the statements in this section can be found in [74], the rest follows directly from the definitions.

15.1.1. Hausdorff dimension. *Let $\mathcal{X}$ be a metric space. Its Hausdorff dimension is defined as*

$$\operatorname{HausDim}\mathcal{X} = \sup\bigl\{\alpha \in \mathbb{R} \ \mid \operatorname{HausMes}_\alpha(\mathcal{X}) > 0\bigr\},$$

where $\operatorname{HausMes}_\alpha$ *denotes* α*-dimensional Hausdorff measure; it is defined in Section 7.1.*

Let $\mathcal{X}$ be a metric space and $\{V_\beta\}_{\beta\in\mathcal{B}}$ be its open cover. Let us recall two notions in general topology:

- ◇ The *order* of $\{V_\beta\}$ is supremum of all integers n such that there is a collection of $n+1$ elements of $\{V_\beta\}$ which has nonempy intersection.
- ◇ An open cover $\{W_\alpha\}_{\alpha\in\mathcal{A}}$ of $\mathcal{X}$ is called *refinement* of $\{V_\beta\}_{\beta\in\mathcal{B}}$ if for any $\alpha\in\mathcal{A}$ there is $\beta\in\mathcal{B}$ such that $W_\alpha\subset V_\beta$.

15.1.2. Topological dimension. *Let $\mathcal{X}$ be a metric space. the topological dimension of $\mathcal{X}$ is infimum of numbers $\mathbb{Z}_{\geqslant}$, such that for*

any finite open cover of $\mathcal{X}$ there is a finite open refinement with order n.

If no such n exists, the topological dimension of $\mathcal{X}$ is infinite.

The topological dimension of $\mathcal{X}$ will be denoted by $\operatorname{TopDim}\mathcal{X}$.

The following statements (15.1.3, 15.1.4, 15.1.5) we call "axioms" since an invariant which satisfies these conditions is commonly called "dimension".

15.1.3. Normalization axiom. *For any $m \in \mathbb{Z}_{\geqslant 0}$,*

$$\operatorname{TopDim}\mathbb{E}^m = \operatorname{HausDim}\mathbb{E}^m = m.$$

15.1.4. Cover axiom. *If $\{A_n\}_{n=1}^{\infty}$ is a countable closed cover of $\mathcal{X}$ then*

$$\operatorname{TopDim}\mathcal{X} = \sup_n\{\operatorname{TopDim}A_n\} \quad \textit{and} \quad \operatorname{HausDim}\mathcal{X} = \sup_n\{\operatorname{HausDim}A_n\}.$$

15.1.5. Product axiom. *For any metric spaces $\mathcal{X}$ and $\mathcal{Y}$,*

$$\operatorname{TopDim}(\mathcal{X}\times\mathcal{Y}) \leqslant \operatorname{TopDim}\mathcal{X} + \operatorname{TopDim}\mathcal{Y},$$

$$\operatorname{HausDim}(\mathcal{X}\times\mathcal{Y}) \leqslant \operatorname{HausDim}\mathcal{X} + \operatorname{HausDim}\mathcal{Y}.$$

Note that both inequalities in 15.1.5 might be strict, such examples were constructed in [143] and [104]. The following theorem follows from [74, theorems V 8 and VII 2].

15.1.6. Szpilrajn's theorem. *Let $\mathcal{X}$ be a separable metric space. Assume* $\operatorname{TopDim}\mathcal{X} \geqslant m$. *Then* $\operatorname{HausMes}_m\mathcal{X} > 0$.

In particular, $\operatorname{TopDim}\mathcal{X} \leqslant \operatorname{HausDim}\mathcal{X}$.

Remark. In fact it is true that for any separable metric space $\mathcal{X}$ we have

$$\operatorname{TopDim}\mathcal{X} = \inf\{\operatorname{HausDim}\mathcal{Y}\},$$

where infimum is taken over all metric spaces $\mathcal{Y}$ which are homeomorphic to $\mathcal{X}$.

15.1.7. Definition. *Let $\mathcal{X}$ be a metric space and $F\colon \mathcal{X}\to\mathbb{R}^m$ be a continuous map. A point $\boldsymbol{z}\in\operatorname{Im}F$ is called a* stable value *of F if there is $\varepsilon>0$ such that $\boldsymbol{z}\in\operatorname{Im}F'$ for any continuous map $F'\colon\mathcal{X}\to\mathbb{R}^m$ which is ε-close to F; i.e., $|F'(x)-F(x)|<\varepsilon$ for all $x\in\mathcal{X}$.*

The following theorem follows from [74, theorems VI 1&2]. (This theorem also holds for non-separable metric spaces, see [116, ???] or [52, 3.2.10]).

15.1.8. Stable value theorem. *Let $\mathcal{X}$ be a separable metric space. Then* $\operatorname{TopDim}\mathcal{X} \geqslant m$ *if and only if there is a map $F\colon \mathcal{X} \to \mathbb{R}^m$ with a stable value.*

15.1.9. Proposition. *Suppose $\mathcal{X}$ and $\mathcal{Y}$ are metric spaces and $\Phi\colon \mathcal{X} \to \to \mathcal{Y}$ satisfies*

$$|\Phi(x)\ \Phi(x')| \geqslant \varepsilon\cdot|x\,x'|$$

for some fixed $\varepsilon > 0$ and any pair $x, x' \in \mathcal{X}$. Then

$$\operatorname{HausDim}\mathcal{X} \leqslant \operatorname{HausDim}\mathcal{Y}.$$

In particular, if there is a Lipschitz onto map $\mathcal{Y} \to \mathcal{X}$ then

$$\operatorname{HausDim}\mathcal{X} \leqslant \operatorname{HausDim}\mathcal{Y}.$$

In addition to HausDim and TopDim, we will use yet another one, which we call *linear dimension.*

Recal that cone map is defined in **??**.

15.1.10. Definition of linear dimension. *Let $\mathcal{X}$ be a metric space. The* linear dimension *of $\mathcal{X}$ (denoted by* $\operatorname{LinDim}\mathcal{X}$*) is defined by*

$$\operatorname{LinDim}\mathcal{X} = \sup\{m \in \mathbb{Z}_{\geqslant 0} \mid \exists\, p \in \mathcal{X} \ \text{ which admits an isometric cone embedding } \ \mathbb{E}^m \hookrightarrow \mathrm{T}_p\mathcal{X}\},$$

where $\mathbb{E}^m$ denotes m-dimensional Euclidean space and $\mathrm{T}_p\mathcal{X}$ denotes tangent space of $\mathcal{X}$ at p (defined in **??***).*

Note that LinDim takes values in $\mathbb{Z}_{\geqslant 0} \cup \{\infty\}$.

For general metric spaces, LinDim has almost no relations to HausDim and TopDim. Also, LinDim does not satisfy the cover axiom (15.1.4). For LinDim, an inverse of the product axiom (15.1.5) holds; i.e.,

$$\operatorname{LinDim}(\mathcal{X} \times \mathcal{Y}) \geqslant \operatorname{LinDim}\mathcal{X} + \operatorname{LinDim}\mathcal{Y} \qquad ❶$$

for any two metric spaces $\mathcal{X}$ and $\mathcal{Y}$; that is easy to check. According to the following exercise, he inequality ❶ might be strict for some spaces, see also [54].

15.1.11. Exercise. *Construct two norms on $\mathbb{R}^n$ such that non of the corresponding metric spaces $\mathcal{X}$ and $\mathcal{Y}$ have a isometric copy of $\mathbb{E}^2$ but $\mathcal{X} \times \mathcal{Y}$ has isometric copy of $\mathbb{E}^{10}$.*

The linear dimension will be applied only to Alexandrov spaces and to their open subsets (in both cases of curvature bounded below

and above). As we shall see, in all these cases, LinDim behaves nicely and is easy to work with.

Remarks. Linear dimension was first introduced in [140] under the name *local dimension*. *Geometric dimension* introduced in [87] is closely related; it coincides with the linear dimension for CBB and CBA spaces.

One could modify the definition of linear dimension by taking an arbitrary n-dimensional Banach space instead of the Euclidean n-space. Such a definition makes more sense for general metric spaces. This dimension is additive with respect to the direct product (i.e. we always have equality in ❶).

For Alexandrov spaces (either CBB or CBA) this modification is equivalent to the definition we use in this book (compare to Exercise **??**).

15.2 Dimension theorem for CBB spaces

As the main dimension-like invariant, we will use the linear dimension LinDim; see Definition 15.1.10. We write

$$\begin{aligned} \mathcal{L} \in \mathrm{CBB}^m \qquad & \text{if } \mathcal{L} \in \mathrm{CBB} \text{ and } \operatorname{LinDim} \mathcal{L} = m \in \mathbb{Z}_{\geqslant 0}, \\ \mathcal{L} \in \mathrm{CBB}^\infty \qquad & \text{if } \mathcal{L} \in \mathrm{CBB} \text{ and } \operatorname{LinDim} \mathcal{L} = \infty. \end{aligned}$$

Thus, once we write $\mathcal{L} \in \mathrm{CBB}^m$ we automatically assume $m < \infty$.

The following theorem is the main result of this section.

15.2.1. Theorem. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$, $q \in \mathcal{L}$, $R > 0$ and $m \in \mathbb{Z}_{\geqslant 0}$. Then the following statements are equivalent:*

A) $\operatorname{LinDim} \mathcal{L} \geqslant m$.

B) There is a point $p \in \mathcal{L}$ which admits a κ-strutting array $(b, a^1, \dots, a^m) \in \mathcal{L}^{m+1}$.

C) The set

$$\mathrm{Euk}^m = \{p \in \mathcal{L} \mid \textit{there is an isometric cone embedding } \mathbb{E}^m \hookrightarrow \mathrm{T}_p\}$$

contains a dense G-delta set in $\mathcal{L}$.

D) There is a $C^{\frac{1}{2}}$-embedding, i.e. bi-Hölder with exponent $\frac{1}{2}$,

$$\overline{\mathrm{B}}[1]_{\mathbb{E}^m} \hookrightarrow \mathrm{B}(q, R).$$

.

E)

$$\operatorname{pack}_\varepsilon \mathrm{B}(q, R) > \frac{c}{\varepsilon^m}$$

for some fixed $c > 0$ and any $\varepsilon > 0$.

In particular:

(i) If $\operatorname{LinDim}\mathcal{L}=\infty$*, then all the statements (C), (D) and (E) are satisfied for all* $m\in\mathbb{Z}_{\geqslant 0}$*.*

(ii) If the statement (D) or (E) is satisfied for some choice of $q\in\mathcal{L}$ *and* $R>0$*, then it also is satisfied for any other choice of* q *and* R*.*

For finite-dimensional spaces, Theorem 15.3.1 gives a stronger version of the theorem above.

The proof of the above theorem with exception of statement (D) was given in [139]. At that time, it was not known whether for any $\mathcal{L}\in\mathrm{CBB}[\kappa]$,

$$\operatorname{LinDim}\mathcal{L}=\infty\quad\Rightarrow\quad\operatorname{TopDim}\mathcal{L}=\infty.$$

Perelman proved this implication, by combining an idea of Plaut with the technique of gradient flow (see [130]). The statement 15.2.1D which we prove is somewhat stronger.

To prove Theorem 15.2.1 we will need the following three propositions.

15.2.2. Proposition. *Let* $\mathcal{L}\in\mathrm{CBB}[\kappa]$ *and* $p\in\mathcal{L}$*. Assume there is an isometric cone embedding* $\iota\colon\mathbb{E}^m\hookrightarrow\mathrm{T}_p\mathcal{L}$*. Then either*

a) $\operatorname{Im}\iota=\mathrm{T}_p\mathcal{L}$*, or*

b) there is a point p' *arbitrarily close to* p *such that there is an isometric cone embedding* $\iota'\colon\mathbb{E}^{m+1}\hookrightarrow\mathrm{T}_{p'}\mathcal{L}$*.*

Proof. Assume $\iota(\mathbb{E}^m)$ is a proper subset of $\mathrm{T}_p\mathcal{L}$. Equivalently, there is a direction $\xi\in\Sigma_p\backslash\iota(\mathbb{S}^{m-1})$, where $\mathbb{S}^{m-1}$ is the unit sphere in $\mathbb{E}^m$.

Fix $\varepsilon>0$ so that $\measuredangle(\xi,\sigma)>\varepsilon$ for any $\sigma\in\iota(\mathbb{S}^{m-1})$. Choose a *maximal* ε*-packing* in $\iota(\mathbb{S}^{m-1})$; i.e. an array of directions $\zeta^1,\zeta^2,\dots$ $\dots,\zeta^n\in\iota(\mathbb{S}^{m-1})$ so that $n=\operatorname{pack}_\varepsilon\mathbb{S}^{m-1}$ and $\measuredangle(\zeta^i,\zeta^j)>\varepsilon$ for any $i\neq j$.

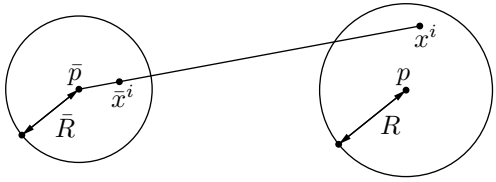
Choose an array of points $x,z^1,z^2,\dots,z^n\in\mathcal{L}$ so that $\operatorname{dir}[px]\approx\xi$, $\operatorname{dir}[pz^i]\approx\zeta^i$; here write "$\approx$" for "sufficiently close". We can choose this array on such a way that $\tilde{\measuredangle}^\kappa(p\,{}^{x}_{z^i})>\varepsilon$ for all i and $\tilde{\measuredangle}^\kappa(p\,{}^{z^i}_{z^j})>\varepsilon$ for all $i\neq j$. Applying Corollary 14.5.5, we can find a point p' arbitrary close to p so that all directions $\operatorname{dir}[p'x]$, $\operatorname{dir}[p'z^1]$, $\operatorname{dir}[p'z^2],\dots,\operatorname{dir}[p'z^n]$ belong to an isometric copy of $\mathbb{S}^{\bar{k}-1}$ in $\Sigma_{p'}$. In addition, we can assume that $\tilde{\measuredangle}^\kappa(p'\,{}^{x}_{z^i})>\varepsilon$ and $\tilde{\measuredangle}^\kappa(p'\,{}^{z^i}_{z^j})>\varepsilon$. From hinge comparison (8.3.1d), $\measuredangle(\operatorname{dir}[p'x],\operatorname{dir}[p'z^i])>\varepsilon$ and $\measuredangle(\operatorname{dir}[p'z^i],\operatorname{dir}[p'z^j])>\varepsilon$; i.e.

$$\operatorname{pack}_\varepsilon\mathbb{S}^{\bar{k}-1}\geqslant n+1>\operatorname{pack}_\varepsilon\mathbb{S}^{m-1}.$$

Hence $\bar{k}>m$. $\square$

15.2.3. Proposition. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$ then for any two poins $p, \bar{p} \in \mathcal{L}$ and any $R, \bar{R} > 0$ there is a contsant $\delta = \delta(\kappa, R, \bar{R}, |p\bar{p}|) > 0$ such that*

$$\operatorname{pack}_{\delta\cdot\varepsilon} \mathrm{B}(\bar{p}, \bar{R}) \geqslant \operatorname{pack}_{\varepsilon} \mathrm{B}(p, R).$$



Proof. According to 8.5.5, we can assume that $\kappa \leqslant 0$.

Let $n = \operatorname{pack}_{\varepsilon} \mathrm{B}(p, R)$ and $x^1, x^2, \dots, x^n \in \mathrm{B}(p, R)$ be a maximal ε-packing; i.e., $|x^i x^j| > \varepsilon$ for all $i \neq j$. Without loss of generality we can assume that $x^1, x^2, \dots, x^n \in \mathrm{Str}(\bar{p})$. Thus, for each i there is a unique geodesic $[\bar{p}x^i]$ (see 8.2.4). Choose factor $s > 0$, so that $\bar{r} > s\cdot(|p\bar{p}| + R)$. For each i, take $\bar{x}^i \in [\bar{p}x^i]$ so that $|\bar{p}\bar{x}^i| = s\cdot(|px^i|)$. From 8.3.2*a*,

$$\measuredangle^{\kappa}\left(\bar{p}\,{}^{\bar{x}^i}_{\bar{x}^j}\right) \geqslant \measuredangle^{\kappa}(\bar{p}\,{}^{x^i}_{x^j}).$$

Applying cosine rule, we get a constant $\delta = \delta(\kappa, R, \bar{R}, |p\bar{p}|) > 0$ such that

$$|\bar{x}^i \bar{x}^j| > \delta\cdot(|x^i x^j|) > \delta\cdot\varepsilon$$

for all $i \neq j$. Hence the statement follows. □

15.2.4. Proposition. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$, $r < \varpi^{\kappa}$ and $p \in \mathcal{L}$. Assume that*

❶ $$\operatorname{pack}_{\varepsilon} \mathrm{B}(p, r) > \operatorname{pack}_{\varepsilon} \overline{\mathrm{B}}[r]_{\mathbb{M}^{\bar{k}}[\kappa]}$$

for some $\varepsilon > 0$. Then there is a G-delta set $A \subset \mathcal{L}$, which is dense in a neighborhood of p, such that $\dim \mathrm{Lin}_q > m$ for any $q \in A$.

Proof. Choose a maximal ε-packing in $\mathrm{B}(p, r)$; i.e. an array of points $x^1, x^2, \dots, x^n \in \mathrm{B}(p, r)$ so that $n = \operatorname{pack}_{\varepsilon} \mathrm{B}(p, r)$ and $|x^i x^j| > \varepsilon$ for any $i \neq j$. Choose a neighborhood $\Omega \ni p$, such that $|qx^i| < r$ for any $q \in \Omega$ and all i. Set

$$A = \Omega \cap \mathrm{Str}(x^1, x^2, \dots, x^n).$$

According to Theorem 8.2.4, A is a G-delta set which is dense in Ω.

Assume $\bar{k} = \dim \mathrm{Lin}_q \leqslant m$ for some $q \in A$. Consider an array of vectors $v^1, v^2, \dots, v^n \in \mathrm{Lin}_q$, $v^i = \log[qx^i]$. Clearly $|v^i| = |qx^i| < r$

and from hinge comparison (8.3.1d) we have $\tilde{\Upsilon}^{\kappa}[o\,{}^{v^i}_{v^j}] \geqslant |x^i\,x^j| > \varepsilon$. Note that $\mathrm{B}(o,r) \subset \mathrm{Lin}_q$ equipped with the metric $\rho(v,w) = \tilde{\Upsilon}^{\kappa}[o\,{}^{v}_{w}]$ is isometric to $\overline{\mathrm{B}}[r]_{\mathbb{M}^{\bar{k}}[\kappa]}$. Thus,

$$\operatorname{pack}_\varepsilon \overline{\mathrm{B}}[r]_{\mathbb{M}^{\bar{k}}[\kappa]} \geqslant \operatorname{pack}_\varepsilon \mathrm{B}(p,r),$$

which contradicts $\bar{k} \leqslant m$ and ❶. □

Proof of 15.2.1. The proof is essentially done in 15.2.2, 15.2.3, 15.2.4, **??**, **??**; here we only assemble the proof from these parts.

We prove implications

$$(C) \Rightarrow (A) \Rightarrow (B) \Rightarrow (E) \Rightarrow (C) \Rightarrow (D) \Rightarrow (E).$$

The implication $(C)\Rightarrow(A)$ is trivial. The proof of $(D)\Rightarrow(E)$ is valid for general metric spaces; it is based on general relations between topological dimension, Hausdorff measure and $\operatorname{pack}_\varepsilon$.

(A)⇒(B). Choose a point $p \in \mathcal{L}$ such that $\dim \mathrm{Lin}_p \geqslant m$. Clearly one can choose an array of directions $\xi^0, \xi^1, \dots, \xi^m \in \mathrm{Lin}_p$ so that $\measuredangle(\xi^i, \xi^j) > \frac{\pi}{2}$ for all $i \neq j$. Choose an array of points $x^0, x^1, \dots, x^m \in \mathcal{L}$ so that each $\operatorname{dir}[px^i]$ is sufficiently close to ξ^i; in particular, we have $\measuredangle[p\,{}^{x^i}_{x^j}] > \frac{\pi}{2}$. Choose points $a^i \in \left]px^i\right]$ sufficiently close to p. One can do it so that each $\measuredangle^{\kappa}(p\,{}^{a^i}_{a^j})$ is arbitrary close to $\measuredangle[p\,{}^{a^i}_{a^j}]$, in particular, $\measuredangle^{\kappa}(p\,{}^{a^i}_{a^j}) > \frac{\pi}{2}$. Finally, set $b = a^0$.

(B)⇒(E). Let $p \in \mathcal{L}$ be a point which admits a κ-strutting array $b, a^1, \dots, a^m \in \mathcal{L}$. The right-inverse mapping map theorem (**??**) implies that the distance map $|\boldsymbol{a}_*| \colon \mathcal{L} \to \mathbb{R}^m$,

$$|\boldsymbol{a}_*| \colon x \mapsto (|a^1\,x|, |a^2\,x|, \dots, |a^n\,x|)$$

is open in a neighborhood of p. Since the distance map $|\boldsymbol{a}_*|$ is Lipschitz, for any $r > 0$, there is $c > 0$ such that

$$\operatorname{pack}_\varepsilon \mathrm{B}(p,r) > \frac{c}{\varepsilon^m}.$$

Applying 15.2.3, we get similar inequality for any other ball in $\mathcal{L}$; i.e., for any point $q \in \mathcal{L}$, $R > 0$ there is $c' > 0$ such that

$$\operatorname{pack}_\varepsilon \mathrm{B}(q,R) > \frac{c'}{\varepsilon^m}.$$

(E)⇒(C). Note that for any $q' \in \mathcal{L}$ and $R' > |q\,q'| + R$ we have

$$\begin{aligned}\operatorname{pack}_\varepsilon \mathrm{B}(q',R') &\geqslant \operatorname{pack}_\varepsilon \mathrm{B}(q,R) \geqslant \\ &\geqslant \frac{c}{\varepsilon^m} > \\ &> \operatorname{pack}_\varepsilon \overline{\mathrm{B}}[R']_{\mathbb{M}^{m-1}[\kappa]}.\end{aligned}$$

for all small enough $\varepsilon > 0$. Applying 15.2.4, we get that

$$\mathrm{Euk}^m = \{p \in \mathcal{L} \mid \text{there is an isometric cone embedding } \iota : \mathbb{E}^m \hookrightarrow \mathrm{T}_p\}$$

contains a G-delta set which is dense in a neighborhood of any point $q' \in \mathcal{L}$.

(C)⇒(D). Since Euk^m contains a dense G-delta set in $\mathcal{L}$, we can choose $p \in \mathrm{B}(q,R)$ with an isometric cone embedding $\iota\colon \mathbb{E}^m \hookrightarrow \mathrm{T}_p$. Thus, the same way as in (A)⇒(B), we can construct a κ-strutting array $b, a^1, \dots, a^m \in \mathcal{L}$ for p. Applying the right-inverse theorem (**??**), we obtain a $C^{\frac{1}{2}}$-submap

$$\Phi\colon \mathbb{R}^m \multimap \mathrm{B}(q,R)$$

which is a right inverse for $|\boldsymbol{a}*|\colon \mathcal{L} \to \mathbb{R}^m$ and $\Phi(|\boldsymbol{a}p|) = p$. In particular, Φ is a $C^{\frac{1}{2}}$-embedding of $\operatorname{Dom}\Phi$.

(D)⇒(E). Let $W \subset \mathrm{B}(q,R)$ be the image of the embedding. Since $\operatorname{TopDim} W = m$, Szpilrajn's theorem (15.1.6) implies that

$$\operatorname{HausMes}_m W > 0.$$

Given $\varepsilon > 0$, consider a maximal ε-packing of W; i.e. an array of points $(x^1, x^2, \dots, x^n)$ in W such that $n = \operatorname{pack}_\varepsilon W$ and $|x^i x^j| > \varepsilon$ for all $i \neq j$. Note that W is covered by balls $\mathrm{B}(x^i, 2\cdot\varepsilon)$. Thus, for all small $\varepsilon > 0$,

$$\operatorname{pack}_\varepsilon W \geqslant \frac{c}{\varepsilon^m} \cdot \operatorname{HausMes}_m W.$$

Thus, (E) follows. □

15.3 Equivalence of dimensions for CBB spaces

In this section we will show that all reasonable notions of dimension coinside on the class of Alexandrov spaces with curvature bounded below.

First we will prove a stronger version of Theorem 15.2.1 for the finite-dimensional case.

15.3.1. Theorem. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$, $0 < R \leqslant \varpi^\kappa$, $q \in \mathcal{L}$ and $m \in \mathbb{Z}_{\geqslant 0}$. Then the following statements are equivalent:*

a) $\operatorname{LinDim}\mathcal{L} = m < \infty$*;*

b) m is the maximal integer, such that there is a point $p \in \mathcal{L}$ which admits a κ-strutting array $b, a^1, \dots, a^m$.

c) $\mathrm{T}_p \overset{iso}{=\!=} \mathbb{E}^m$ *for any point p in a dense G-delta set of $\mathcal{L}$.*

d) There is an open bi-Lipschitz embedding

$$\overline{\mathrm{B}}[1]_{\mathbb{E}^m} \hookrightarrow \mathrm{B}(q,R) \subset \mathcal{L};$$

e) For any $\varepsilon > 0$,

$$\operatorname{pack}_\varepsilon \overline{\mathrm{B}}[R]_{\mathbb{M}^m[\kappa]} \geqslant \operatorname{pack}_\varepsilon \mathrm{B}(q,R),$$

moreover, there is $c = c(q,R) > 0$ *such that*

$$\operatorname{pack}_\varepsilon \mathrm{B}(q,R) > \frac{c}{\varepsilon^m}.$$

The above theorem was essentially proved in [34]. The next two corollaries follow directly from 15.3.1*e*.

15.3.2. Corollary. *Any space in* $\mathrm{CBB}^m[\kappa]$ *is proper and geodesic.*

15.3.3. Corollary. *Let* $\mathcal{L}_n \in \mathrm{CBB}^m[\kappa]$ *and* $\mathcal{L}_n \to \mathcal{L}$ *as* $n \to \omega$. *Then* $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$.

15.3.4. Corollary. *Let* $\mathcal{L} \in \mathrm{CBB}[\kappa]$. *Then for any open* $\Omega \subset \mathcal{L}$, *we have*

$$\operatorname{LinDim}\mathcal{L} = \operatorname{LinDim}\Omega = \operatorname{TopDim}\Omega = \operatorname{HausDim}\Omega,$$

where TopDim *and* HausDim *denotes topological dimension (15.1.2) and Hausdorff dimension (15.1.1) correspondingly.*

In particular, $\mathcal{L}$ *is dimension homogeneous; i.e. any open set has the same linear dimension.*

Remark. Using theorems 15.2.1 and 15.3.1, one can show that linear dimension is equal to many different types of dimensions, such as *small* and *big inductive dimension* and *upper* and *lower Minkowski dimension* (which is also called *rough dimension* and *box counting dimension*), *homological dimension* and so on.

Proof of 15.3.4. The equality

$$\operatorname{LinDim}\mathcal{L} = \operatorname{LinDim}\Omega$$

follows from 15.2.1*A*&*C*.

If $\operatorname{LinDim}\mathcal{L} = \infty$ then applying 15.2.1*D*, for $\mathrm{B}(q,R) \subset \Omega$, we get that there is a compact subset $K \subset \Omega$ with arbitrary large $\operatorname{TopDim} K$ therefore

$$\operatorname{TopDim}\Omega = \infty.$$

From Szpilrajn's theorem (15.1.6), $\operatorname{HausDim} K \geqslant \operatorname{TopDim} K$. Thus, we also have

$$\operatorname{HausDim}\Omega = \infty.$$

If $\operatorname{LinDim}\mathcal{L} = m < \infty$ then first inequality in 15.3.1*e* implies that

$$\operatorname{HausDim}\mathrm{B}(q,R) \leqslant m.$$

According to Corollary 15.3.2, $\mathcal{L}$ is proper and in particular it has countable base. Thus applying Szpilrajn theorem again, we get

$$\operatorname{TopDim}\Omega \leqslant \operatorname{HausDim}\Omega \leqslant m.$$

Finally, 15.3.1d implies that $m \leqslant \operatorname{TopDim}\Omega$. □

Proof of Theorem 15.3.1. Note that equivalence $(a)\Leftrightarrow(b)$ follows from 15.2.1.

(a)⇒(c). If $\operatorname{LinDim}\mathcal{L} = m$, then from Theorem 15.2.1, we get that Euk^m contains a dense G-delta set in $\mathcal{L}$. From 15.2.2, it follows that T_p is isometric to $\mathbb{E}^m$ for any $p \in \operatorname{Euk}^m$.

(c)⇒(d). It is proved exactly the same way as implication *(C)⇒(D)* of theorem 15.2.1, but we have to apply existence of distance chart (**??**) instead of the right-inverse theorem.

(d)⇒(e). From (d), it follows that, there is a point $p \in \mathrm{B}(q,R)$ and $r > 0$ such that $\mathrm{B}(p,r) \subset \mathcal{L}$ is bi-Lipschit homeomorphic to a bounded open set of $\mathbb{E}^m$. Thus, there is $c > 0$ such that

❶ $$\operatorname{pack}_\varepsilon \mathrm{B}(p,r) > \frac{c}{\varepsilon^m}.$$

Applying 15.2.3, we get that inequality ❶, with different constants, holds for any other ball, in particular for $\mathrm{B}(q,R)$.

Applying 15.2.4, we get the first inequality.

(e)⇒(a). From theorem 15.2.1, we have $\operatorname{LinDim}\mathcal{L} \geqslant m$. Applying theorem 15.2.1 again, we get that if $\operatorname{LinDim}\mathcal{L} \geqslant m+1$ then for some $c > 0$ and any $\varepsilon > 0$

$$\operatorname{pack}_\varepsilon \mathrm{B}(q,R) \geqslant \frac{c}{\varepsilon^{m+1}},$$

but

$$\frac{c'}{\varepsilon^m} \geqslant \operatorname{pack}_\varepsilon \mathrm{B}(q,R),$$

for any $\varepsilon > 0$, a contradiction. □

The following exercise was suggested by Alexander Lytchak.

15.3.5. Exercise. *Let $\mathcal{L} \in \mathrm{CBB}$ and $\Sigma_p\mathcal{L}$ is compact for any $p \in \mathcal{L}$. Prove that $\mathcal{L}$ is finite dimensional.*

15.4 One-dimensional CBB spaces

15.4.1. Theorem. *Let $\mathcal{L} \in \mathrm{CBB}^1[\kappa]$. Then $\mathcal{L}$ is isometric to a connected complete Riemannian 1-dimensional manifold with possibly non-empty boundary.*

Proof. Clearly $\mathcal{L}$ is connected. It remains to show the following.

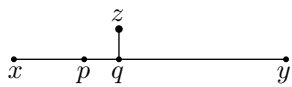
❶ *For any point $p \in \mathcal{L}$ there is $\varepsilon > 0$ such that $\mathrm{B}(p,\varepsilon)$ is isometric to either $[0,\varepsilon)$ or $(-\varepsilon,\varepsilon)$.*

First let us show that

❷ *If $p \in\,]xy[$ for some x, $y \in \mathcal{L}$ and $\varepsilon < \min\{|px|,|py|\}$ then $\mathrm{B}(p,\varepsilon) \subset\,]xy[$. In particular $\mathrm{B}(p,\varepsilon) \overset{iso}{=\!=} (-\varepsilon,\varepsilon)$.*

Assume the contrary; i.e., there is

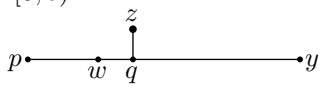
$$z \in \mathrm{B}(p,\varepsilon)\backslash\,]xy[\,.$$



Consider a geodesic $[pz]$, let $q \in [pz] \cap [xy]$ be the point which maximizes the distance $|pq|$. At q, we have three distinct directions: to x, to y and to z. Moreover, $\measuredangle[q\,{}^{x}_{y}] = \pi$. Thus, according to Proposition 15.2.2, $\operatorname{LinDim}\mathcal{L} > 1$, a contradiction.

Now assume there is no geodesic passing though p. Since $\operatorname{LinDim}\mathcal{L} = 1$ there is a point $x \neq p$. Take a positive $\varepsilon < |px|$. Let us show that

❸ $\mathrm{B}(p,\varepsilon) \subset [px]$*; in particular* $\mathrm{B}(p,\varepsilon) \overset{iso}{=\!=} [0,\varepsilon)$.



Assume the contrary; let $z \in \mathrm{B}(p,\varepsilon)\backslash[py]$. Choose a point $w \in\,]py[$ such that $|pw| + |pz| < \varepsilon$. Consider geodesic $[wz]$, let $q \in [py]\cap[wz]$ be the point which maximizes the distance $|wq|$. Since no geodesics pass through p, we have $p \neq q$. As above, $\measuredangle[q\,{}^{p}_{y}] = \pi$ and $\operatorname{dir}[qz]$ is distinct from $\operatorname{dir}[qp]$ and $\operatorname{dir}[qp]$. Thus, according to Proposition 15.2.2, $\operatorname{LinDim}\mathcal{L} > 1$, a contradiction.

Clearly ❷ + ❸ $\Rightarrow$ ❶; hence the result. □

15.5 Dimension of CBA spaces

We start with two examples. The first example shows that the dimension of a CBA space is not local; i.e. a CBA space might have open sets with different linear dimension. Such an example can be constructed by gluing at one point two Euclidean spaces of different dimensions. (According to Reshetnyak's gluing theorem (9.9.1), this construction gives a CBA[0] spaces.)

The second example provides a CBA space with topological dimension one and arbitrary large Hausdorff dimension. Thus, for CBA spaces, one should not expect any relations between topological and Hausdorff dimensions except for the one provided by Szpilrajn's theorem (15.1.6).

To construct the second type of examples, note that the completion of any metric tree has topological dimension one and is $\mathrm{CBA}[\kappa]$ for any κ. Start with a binary tree Γ, and a sequence $\varepsilon_n > 0$ such that $\sum_n \varepsilon_n < \infty$. Define the metric on Γ by prescribing length of edge from level n to the level $n+1$ to be ε_n. For an appropriately chosen sequence ε_n, the completion of Γ will contain a Cantor set of arbitrary large dimension.

The following is a version of the theorem proved by Kleiner in [87], which is improved using the generalized Rademacher's theorem (**??**) proved by Lytchak in [101].

15.5.1. Theorem. *For any $\mathcal{U} \in \mathrm{CBA}[\kappa]$, the following statements are equivalent:*

a) $\mathrm{LinDim}\,\mathcal{U} \geqslant m$*;*

b) For some $z \in \mathcal{U}$ there is an array of $m+1$ balls $B^i = \mathrm{B}(a^i, r^i)$ with $a^0, a^1, \dots, a^m \in \mathrm{B}(z, \frac{\varpi^\kappa}{2})$ such that $\bigcap_i B^i = \varnothing$, $\bigcap_{i\neq j} B^i \neq \varnothing$ for each j

c) There is a $C^{\frac{1}{2}}$-embedding $\Phi\colon \overline{\mathrm{B}}[1]_{\mathbb{E}^m} \hookrightarrow \mathcal{U}$; i.e. Φ is bi-Hölder with exponent $\frac{1}{2}$.

d) There is a closed separable set $K \subset \mathcal{U}$ such that $\mathrm{TopDim}\,K \geqslant m$.

Remarks. Theorem 19.3.1 gives a stronger version of part (c) in finite dimensional case. Namely, a CBA space with linear dimension m admits a bi-Lipschitz embedding Φ of an open set of $\mathbb{R}^m$. Moreover, according to ??? the Lipschitz constants of Φ can be made arbitrary close to 1.

15.5.2. Corollary. *For any separable $\mathcal{U} \in \mathrm{CBA}$, we have*

$$\mathrm{TopDim}\,\mathcal{U} = \mathrm{LinDim}\,\mathcal{U}.$$

To prove Theorem 15.5.1 we will need the followng lemma. Any simplicial complex can be equiped with an length-metric such that each simplex Δ^k is isometric to the standard k-simplex with the metric induced by the ℓ^1-norm on $\mathbb{R}^{k+1}$. This metric will be called the ℓ^1-*metric* on the simplicial complex.

15.5.3. Lemma. *Let $\mathcal{U} \in \mathrm{CBA}[\kappa]$, ρ be a continuous positive function on $\mathcal{U}$. Then there is a simplicial complex $\mathcal{N}$ equiped with ℓ^1-metric, a locally Lipschitz map $\Phi\colon \mathcal{U} \to \mathcal{N}$ and a Lipschitz map $\Psi\colon \mathcal{N} \to \mathcal{U}$ such that*

a) *The* displacement *of composition* $\Psi\circ\Phi\colon\mathcal{U}\to\mathcal{U}$ *is bounded by* ρ*; i.e.,*

$$|x\,\Psi\circ\Phi(x)|<\rho(x)$$

for any $x\in\mathcal{U}$

b) *If* $\operatorname{LinDim}\mathcal{U}\leqslant m$ *then the* Ψ*-image of any closed simplex* Δ *in* $\mathcal{N}$ *coinsides with the image of its* m*-skeliton.*

Proof. Without loss of generality, we can assume that for any x we have $\rho(x)<\rho_0$ for some fixed $\rho_0<\frac{\varpi^\kappa}{2}$.

By Stone's theorem, any metric space is paracompact. Thus, we can choose a locally finite covering $\{\,\Omega_\alpha\mid\alpha\in\mathcal{A}\,\}$ of $\mathcal{U}$ such that $\Omega_\alpha\subset\subset\mathrm{B}(x,\frac{1}{3}\cdot\rho(x))$ for any $x\in\Omega_\alpha$.

Denote by $\mathcal{N}$ the nerve of the covering $\{\Omega_\alpha\}$; i.e., $\mathcal{N}$ is an abstract simplicial complex with set of vertexes formed by $\mathcal{A}$ such that $\{\alpha^0,\alpha^1,\dots,\alpha^n\}\subset\mathcal{A}$ are vertexes of a simplex if and only if $\Omega_{\alpha^0}\cap\cap\,\Omega_{\alpha^1}\cap\ldots\cap\Omega_{\alpha^n}\neq\varnothing$.

Further, choose a Lipschitz partition of unity $\varphi_\alpha\colon\mathcal{U}\to[0,1]$, subordinate to $\{\Omega_\alpha\}$. One may think of partition $\{\varphi_\alpha\}$ as of a locally Lipschitz map $\Phi\colon\mathcal{U}\to\mathcal{N}$; the α-s barycentric coordinate of $\Phi(p)$ is $\varphi_\alpha(p)$. Clearly Φ-preimage of any open simplex in $\mathcal{N}$ lies in Ω_α for some $\alpha\in\mathcal{A}$.

For each $\alpha\in\mathcal{A}$, choose $x_\alpha\in\Omega_\alpha$. Let us extend the map $\alpha\mapsto\mapsto x_\alpha$ to a map $\Psi\colon\mathcal{N}\to\mathcal{U}$ which is κ-barycentric on each simplex; according to Theorem 19.2.1*a*, this extension exists, Ψ is Lipschtz and its Lipschitz constant depends only on ρ_0 and κ.

(a) Fix $x\in\mathcal{U}$. Denote by Δ the minimal simplex which contains $\Phi(x)$; and let $(\alpha^0,\alpha^1,\dots,\alpha^n)$ be the vertexes of Δ. Note that α is a vertex of Δ if and only if $\varphi_\alpha(x)>0$. Thus,

$$|x\,x_{\alpha^i}|<\tfrac{1}{3}\cdot\rho(x)$$

for any i. Therefore

$$\operatorname{diam}\Psi(\Delta)\leqslant\max_{i,j}\{|x_{\alpha^i}\,x_{\alpha^j}|\}<\tfrac{2}{3}\cdot\rho(x).$$

In particular

$$|x\,\Psi\circ\Phi(x)|\leqslant|x\,x_{\alpha^0}|+\operatorname{diam}\Psi(\Delta)<\rho(x).$$

(b) Assume the contrary; i.e., $\Psi(\mathcal{N})$ is not inculded in the Ψ-image of m-skeleton of $\mathcal{N}$. Then for some $\bar{k}>m$, there is a $\bar{k}$-simplex $\Delta^{\bar{k}}$ in $\mathcal{N}$, such that the baricentric simplex $\sigma=\Psi|\Delta^{\bar{k}}$ is nondegenrate; i.e.,

$$W=\Psi(\Delta^{\bar{k}})\backslash\Psi(\partial\Delta^{\bar{k}})\neq\varnothing.$$

Applying Corollary 19.2.4, we get $\operatorname{LinDim}\mathcal{U} \geqslant \mathfrak{k}$, a contradiction. $\square$

Proof of Theorem 15.5.1. Note that

- ⋄ The implication $(b)\Rightarrow(c)$ follows directly from Lemma 19.2.5 and Theorem 19.2.1c.
- ⋄ The implication $(c)\Rightarrow(d)$ is trivial.

(d)⇒(a). According to Theorem 15.1.8, there is a continuous map $f\colon K \to \mathbb{R}^m$ with a stable value. By Tietze extension theorem, it is possible to extend f to a continuous map $F\colon \mathcal{U} \to \mathbb{R}^m$.

Fix $\varepsilon > 0$. Since F is continuous, there is a continuous positive function ρ defined on $\mathcal{U}$ such that

$$|xy| < \rho(x) \quad\Rightarrow\quad |F(x) - F(y)| < \tfrac{1}{3}\cdot\varepsilon.$$

Apply Lemma 15.5.3 for the function ρ. For the obtained simplicial complex $\mathcal{N}$, and maps $\Phi\colon \mathcal{U} \to \mathcal{N}$, $\Psi\colon \mathcal{N} \to \mathcal{U}$, we have

$$|F \circ \Psi \circ \Phi(x) - F(x)| < \tfrac{1}{3}\cdot\varepsilon$$

for any $x \in \mathcal{U}$.

According to Lemma 5.6.2, there is a locally Lipschitz map $F_\varepsilon : \mathcal{U} \to \to \mathbb{R}^{m+1}$ such that $|F_\varepsilon(x) - F(x)| < \frac{1}{3}\cdot\varepsilon$ for any $x \in \mathcal{U}$.

Note that $\Phi(K)$ is contained in a countable subcomplex of $\mathcal{N}$, say $\mathcal{N}'$. Indeed, since K is separable, there is a countable dense collection of points $\{x_n\}$ in K. Denote by Δ_n the minimal simplex of $\mathcal{N}$ which contains $\Phi(x_n)$. Then $\Phi(K) \subset \bigcup_i \Delta_n$.

Arguing by contradiction, assume $\operatorname{LinDim}\mathcal{U} < m$. By Lemma 15.5.3$b$, the image $F_\varepsilon \circ \Psi \circ \Phi(K)$ lies in the F_ε-image of $(m-1)$-skeliton of $\mathcal{N}'$; in particular it can be covered by countable collection of Lipschitz images of $(m-1)$-simplexes. Hence $\mathbf{0} \in \mathbb{R}^m$ is not a stable value of the restriction $F_\varepsilon \circ \Psi \circ \Phi|K$. Since $\varepsilon > 0$ is arbitrary, we get that $\mathbf{0} \in \mathbb{R}^m$ is not a stable value of $f = F|K$, a contradiction.

(a)⇒(b). The following claim is almost trivial.

❶ *Let $q \in \mathcal{U}$ and $\dot x^1, \dot x^2, \dots, \dot x^n \in \mathrm{T}_q$. Then given $\delta > 0$, there is a point array $x^1, x^2, \dots, x^n$ in $\mathcal{U}$ such that*

$$\measuredangle(\dot x^i, \log[q x^i]) < \delta$$

and for some fixed $\mathfrak{z} > 0$ we have

$$\tfrac{1}{\mathfrak{z}}\cdot|q\,x^i| = |\dot x^i| \quad \text{and} \quad |\tfrac{1}{\mathfrak{z}}\cdot|x^i\,x^j| - |\dot x^i\,\dot x^j|| < \delta$$

for all i and j.

Moreover the value $\mathfrak{z}$ can be taken arbitrary small.

Proof of the Claim. For each i choose a geodesic γ^i from q which goes almost in the directions of $\dot{x}^i$. Then take the point x^i on γ^i on the distance $\mathfrak{z}\cdot|\dot{x}^i|$ from q. $\triangle$

Choose $q \in \mathcal{U}$ such that T_q contains a subcone E isometric to m-dimensional Euclidean space. Note that one can choose $\varepsilon > 0$ and a point arrray $(\dot{a}^0, \dot{a}^1, \dots, \dot{a}^m)$ in $E \subset \mathrm{T}_q$ such that $\bigcap_i \overline{\mathrm{B}}[\dot{a}^i, 1+\varepsilon] = \varnothing$ and $\bigcap_{i\ne j} \overline{\mathrm{B}}[\dot{a}^i, 1-\varepsilon] \ne \varnothing$ for each j.

Applying Claim ❶, we get a point array $(a^0, a^1, \dots, a^m)$ in $\mathcal{U}$ such that $\bigcap_i \overline{\mathrm{B}}[a^i, \mathfrak{z}] = \varnothing$ and $\bigcap_{i\ne j} \overline{\mathrm{B}}[a^i, \mathfrak{z}] \ne \varnothing$ for each j. Since $\mathfrak{z} > 0$ can be chousen arbitrary small, it implies (b). $\square$

15.6 Remarks and open problems

The following conjecture (in an equivalent form) appers in [87] (see also [64, p. 133]) remains open for nonseparable CBA spaces; for separable spaces it is true by Corollary 15.5.2.

15.6.1. Conjecture. *If* $\mathcal{U} \in \mathrm{CBA}$*, then*

$$\mathrm{TopDim}\,\mathcal{U} = \mathrm{LinDim}\,\mathcal{U}.$$

Chapter 16

Gradient flow

The technique of gradient flow takes it roots in Sharafutdinov's retraction, which was introduced in [156] and used widely in comparison geometry since then. In CBB-spaces, it was first used by in [130]. Bit later, independently Jost and Mayer in [79] and [105] used the gradient flow in CBA-spaces; this approach has been refined in [4]. Later, Lytchak in [102], unified and generalized these two approaches to a wide class of metric spaces and then it was yet developed further by Ohta in [118] and by Sevaré in [149].

Gradient flow provides a useful tool in Alexandrov's geometry, which we will use everywhere in the book.

This chapter can be divided in few parts.

- ⋄ Sections 16.1–16.4 describes "gradient vector field" and its properties.
- ⋄ Sections 16.5–16.12 describe gradient curves and gradient flow; gradient-like curves are needed to introduce gradient curves.
- ⋄ Sections 17.1–17.3 describe radial curves and gradient exponent. It is analog of exponential map which is defined for CBB-spaces; it has many important properties of the ordinary exponential map and in certain respects it is "better", even in Riemannian case.

In sections 16.11– we describe how to modify the proofs so it will work for general semiconcave functions, not necessary locally Lipschitz.

One of the technical difficulties in Alexandrov's geometry comes from nonextendability of geodesics. In particular, for $\mathcal{L} \in \mathrm{CBB}$ and a point $p \in \mathcal{L}$, the exponential map, $\exp_p \colon \mathrm{T}_p \to \mathcal{L}$, if defined the usual way, can be undefined in an arbitrary small neighborhood of the origin

in T_p. Here we construct its analog — the *gradient exponential map*, $\mathrm{gexp}_p\colon \mathrm{T}_p \to \mathcal{L}$, which practically solves this problem. It has many important properties of the ordinary exponential map and in certain respects it is "better", even in Riemannian case.

16.1 Gradient

16.1.1. Definition of gradient. *Let $\mathcal{L} \in \mathrm{CBB}$, $f\colon \mathcal{L} \looparrowright \mathbb{R}$ be a subfunction and for a point $p \in \mathrm{Dom}\, f$ the differential $\boldsymbol{d}_p f\colon \mathrm{T}_p \to \mathbb{R}$ is defined.*

A tangent vector $g \in \mathrm{T}_p$ is called a gradient of f at p *(briefly, $g = \nabla_p f$) if*

a) $(\boldsymbol{d}_p f)(w) \leqslant \langle g, w\rangle$ for any $w \in \mathrm{T}_p$, and

b) $(\boldsymbol{d}_p f)(g) = \langle g, g\rangle$.

16.1.2. Existence and uniqueness of the gradient. *Assume $\mathcal{L} \in \mathrm{CBB}[\kappa]$ and $f\colon \mathcal{L} \looparrowright \mathbb{R}$ be locally Lipschitz and semiconcave subfunction. Then for any point $p \in \mathrm{Dom}\, f$, there is unique gradient $\nabla_p f \in \mathrm{T}_p$.*

Proof; (uniqueness). If $g, g' \in \mathrm{T}_p$ are two gradients of f then

$$\langle g, g\rangle = (\boldsymbol{d}_p f)(g) \leqslant \langle g, g'\rangle, \qquad \langle g', g'\rangle = (\boldsymbol{d}_p f)(g') \leqslant \langle g, g'\rangle.$$

Therefore,

$$|g\,g'|^2 = \langle g, g\rangle - 2\cdot\langle g, g'\rangle + \langle g', g'\rangle = 0;$$

that is, $g = g'$.

(Existence). Note first that if $\boldsymbol{d}_p f \leqslant 0$ then one can take $\nabla_p f = o$.

Otherwise, if $s = \sup\{\, (\boldsymbol{d}_p f)(\xi) \mid \xi \in \Sigma_p \,\} > 0$, it is sufficient to show that there is $\overline{\xi} \in \Sigma_p$ such that

❶ $$(\boldsymbol{d}_p f)\big(\overline{\xi}\big) = s.$$

Indeed, if $\overline{\xi}$ exists, then applying Lemma 14.6.3 for $u = \overline{\xi}$, $v = \varepsilon\cdot w$ with $\varepsilon \to 0+$, we get

$$(\boldsymbol{d}_p f)(w) \leqslant \langle w, s\cdot\overline{\xi}\rangle$$

for any $w \in \mathrm{T}_p$; that is, $s\cdot\overline{\xi}$ is the gradient at p.

Take a sequence of directions $\xi_n \in \Sigma_p$, such that $(\boldsymbol{d}_p f)(\xi_n) \to s$. Yet once applying Lemma 14.6.3 for $u = \xi_n$, $v = \xi_m$, we get

$$s \geqslant \frac{(\boldsymbol{d}_p f)(\xi_n) + (\boldsymbol{d}_p f)(\xi_m)}{\sqrt{2 + 2\cdot\cos\measuredangle(\xi_n, \xi_m)}}.$$

Therefore $\measuredangle(\xi_n, \xi_m) \to 0$ as $n, m \to \infty$; that is, (ξ_n) is a Cauchy sequence. Clearly $\overline{\xi} = \lim_n \xi_n$ is satisfies ❶. □

16.2 Calculus of gradient

The next lemma roughly states that the gradient points in the direction of maximal slope; moreover if the slope in the given direction is almost maximal then it is almost direction of the gradient.

16.2.1. Lemma. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$, $f\colon \mathcal{L} \looparrowright \mathbb{R}$ be locally Lipschitz and semiconcave and $p \in \operatorname{Dom} f$.*

Assume $|\nabla_p f| > 0$, set $\overline{\xi} = \frac{1}{|\nabla_p f|}\cdot\nabla_p f$ then

a) If for $v \in \mathrm{T}_p$, we have $|v| \leqslant 1+\varepsilon$ and $(\boldsymbol{d}_p f)(v) > |\nabla_p f|\cdot(1-\varepsilon)$, then

$$|\overline{\xi}v| < 100\cdot\sqrt{\varepsilon}.$$

b) If $v_n \in \mathrm{T}_p$ be a sequence of vectors such that

$$\varlimsup_{n\to\infty} |v_n| \leqslant 1 \quad \text{and} \quad \varliminf_{n\to\infty} (\boldsymbol{d}_p f)(v_n) \geqslant |\nabla_p f|$$

then

$$\lim_{n\to\infty} v_n = \overline{\xi}.$$

c) $\overline{\xi}$ is the unique maximum direction for the restriction $\boldsymbol{d}_p f|_{\Sigma_p}$. In particular,

$$|\nabla_p f| = \sup\{\, \boldsymbol{d}_p f \mid \xi \in \Sigma_p f \,\}.$$

Proof. According to definition of gradient,

$$\begin{aligned}|\nabla_p f|\cdot(1-\varepsilon) < (\boldsymbol{d}_p f)(v) &\leqslant \\ &\leqslant \langle v, \nabla_p f\rangle = \\ &= |v|\cdot|\nabla_p f|\cdot\cos\measuredangle(\nabla_p f, v).\end{aligned}$$

Thus $|v| > 1-\varepsilon$ and $\cos\measuredangle(\nabla_p f, v) > \frac{1-\varepsilon}{1+\varepsilon}$. Hence (a).

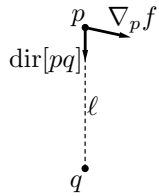
Statements (b) and (c) follow directly from (a). □

As a corollary of the above lemma and Proposition 5.5.3 we get the following:

16.2.2. Chain rule. *Let $\mathcal{L} \in \mathrm{CBB}$, $f\colon \mathcal{L} \looparrowright \mathbb{R}$ be a semiconcave function and $\varphi\colon \mathbb{R} \to \mathbb{R}$ be a non-decreasing semiconcave function. Then $\varphi\circ f$ is semiconcave and $\nabla_x(\varphi\circ f) = \varphi^+(f(x))\cdot\nabla_x f$ for any $x \in \operatorname{Dom} f$.*

The following inequalities describe an important property of the "gradient vector field" which will be used throughout this paper.

16.2.3. Lemma. *Let $\mathcal{L} \in \mathrm{CBB}$, $f\colon \mathcal{L} \looparrowright \mathbb{R}$ satisfies $f'' + \kappa\cdot f \leqslant \lambda$ for some $\kappa, \lambda \in \mathbb{R}$, $[pq] \subset \operatorname{Dom} f$ and $\ell = |pq|$. Then*



$$\langle \mathrm{dir}[pq], \nabla_p f\rangle \geqslant \frac{f(q) - f(p)\cdot \mathrm{cs}^\kappa \ell - \lambda\cdot \mathrm{md}^\kappa \ell}{\mathrm{sn}^\kappa \ell}.$$

In particular,

a) if $\kappa = 0$,

$$\langle \mathrm{dir}[pq], \nabla_p f\rangle \geqslant \big(f(q) - f(p) - \tfrac{\lambda}{2}\cdot \ell^2\big)/\ell;$$

b) if $\kappa = 1$, $\lambda = 0$ *we have*

$$\langle \mathrm{dir}[pq], \nabla_p f\rangle \geqslant (f(q) - f(p)\cdot \cos \ell)\,/\,\sin \ell;$$

c) if $\kappa = -1$, $\lambda = 0$ *we have*

$$\langle \mathrm{dir}[pq], \nabla_p f\rangle \geqslant (f(q) - f(p)\cdot \cosh \ell)\,/\,\sinh \ell;$$

Proof of 16.2.3. Note that $\mathrm{geod}_{[pq]}(0) = p$, $\mathrm{geod}_{[pq]}(\ell) = q$, $(\mathrm{geod}_{[pq]})^+(0) = \mathrm{dir}[pq]$. Thus,

$$\begin{aligned}\langle \mathrm{dir}[pq], \nabla_p f\rangle &\geqslant d_p f(\mathrm{dir}[pq]) = \\ &= (f\circ \mathrm{geod}_{[pq]})^+(0) \geqslant \\ &\geqslant \frac{f(q) - f(p)\cdot \mathrm{cs}^\kappa \ell - \lambda\cdot \mathrm{md}^\kappa \ell}{\mathrm{sn}^\kappa \ell}.\end{aligned}$$

□

The following corollary states that gradient vector field is monotonic in the sense similar to definition of *monotone operators*; see for example [138].

16.2.4. Monotonicity of gradient. *Let* $\mathcal{L} \in \mathrm{CBB}[\kappa]$, $f\colon \mathcal{L} \looparrowright \mathbb{R}$ *be locally Lipschitz and* λ*-concave and* $[pq] \subset \mathrm{Dom}\, f$. *Then*

$$\langle \mathrm{dir}[pq], \nabla_p f\rangle + \langle \mathrm{dir}[qp], \nabla_q f\rangle \geqslant -\lambda\cdot |pq|.$$

Proof. Add two inequalities from 16.2.3*a*. □

16.2.5. Lemma. *Let* $\mathcal{L} \in \mathrm{CBB}[\kappa]$, $f, g\colon \mathcal{L} \looparrowright \mathbb{R}$ *and* $p \in \mathrm{Dom}\, f \cap \cap \mathrm{Dom}\, g$.

Then

$$|\nabla_p f\ \nabla_p g|^2_{\mathrm{T}_p} \leqslant (|\nabla_p f| + |\nabla_p g|)\cdot \sup\{\, |(\boldsymbol{d}_p f)(\xi) - (\boldsymbol{d}_p f)(\xi)| \mid \xi \in \Sigma_p \,\}.$$

In particular, if $f_n\colon \mathcal{L} \looparrowright \mathbb{R}$ *is a sequence of locally Lipschitz and semiconcave subfunctions,* $p \in \mathrm{Dom}\, f_n$ *for each* n *and* $\boldsymbol{d}_p f_n$ *converges uniformly on* Σ_p *then sequence* $\nabla_p f_n \in \mathrm{T}_p$ *converges.*

Proof. Set

$$s = \sup\{\ |(\boldsymbol{d}_p f)(\xi) - (\boldsymbol{d}_p g)(\xi)|\ |\ \xi \in \Sigma_p\ \}.$$

Clearly for any $v \in \mathrm{T}_p$, we have

$$|(\boldsymbol{d}_p f)(v) - (\boldsymbol{d}_p g)(v)| \leqslant s\cdot|v|.$$

From the definition of gradient (16.1.1) we have:

$$\begin{aligned}(\boldsymbol{d}_p f)(\nabla_p g) &\leqslant \langle \nabla_p f, \nabla_p g\rangle, & (\boldsymbol{d}_p g)(\nabla_p f) &\leqslant \langle \nabla_p f, \nabla_p g\rangle,\\ (\boldsymbol{d}_p f)(\nabla_p f) &= \langle \nabla_p f, \nabla_p f\rangle, & (\boldsymbol{d}_p g)(\nabla_p g) &= \langle \nabla_p g, \nabla_p g\rangle.\end{aligned}$$

Therefore,

$$\begin{aligned}|\nabla_p f\ \nabla_p g| &= \langle \nabla_p f, \nabla_p f\rangle + \langle \nabla_p g, \nabla_p g\rangle - 2\cdot\langle \nabla_p f, \nabla_p g\rangle \leqslant\\ &\leqslant (\boldsymbol{d}_p f)(\nabla_p f) + (\boldsymbol{d}_p g)(\nabla_p g) - (\boldsymbol{d}_p f)(\nabla_p g) - (\boldsymbol{d}_p g)(\nabla_p f) \leqslant\\ &\leqslant s\cdot(|\nabla_p f| + |\nabla_p g|).\end{aligned}$$

□

16.3 Polar vectors

The next is a corollary of the above lemma. It will be used to prove basic properties of tangent space. Note that the notion of gradient does not appear in the formulation.

16.3.1. Anti-sum lemma. *Let $\mathcal{L} \in$ CBB and $p \in \mathcal{L}$.*

Given two vectors $u, v \in \mathrm{T}_p$ there is unique vector $w \in \mathrm{T}_p$ such that

$$\langle u, x\rangle + \langle v, x\rangle + \langle w, x\rangle \geqslant 0$$

for any $x \in \mathrm{T}_p$ and

$$\langle u, w\rangle + \langle v, w\rangle + \langle w, w\rangle = 0.$$

To obtain the following corollary, applying the above lemma for $u = v$. In finite-dimensional case, a stronger statement holds, see Milka's lemma (20.5.2).

16.3.2. Existence of polar vector. *Let $\mathcal{L} \in$ CBB and $p \in \mathcal{L}$. Given a vector $u \in \mathrm{T}_p$ there is unique vector $u^* \in \mathrm{T}_p$ such that $\langle u^*, u^*\rangle + \langle u, u^*\rangle = 0$ and u^* is* polar *to u; that is, $\langle u^*, x\rangle + \langle u, x\rangle \geqslant 0$ for any $x \in \mathrm{T}_p$.*

In particular, for any vector $u \in \mathrm{T}_p$ there is a polar vector $u^* \in \mathrm{T}_p$ such that $|u^*| \leqslant |u|$.

Proof of 16.3.1. Choose two sequences of points $a_n, b_n \in \mathrm{Str}(p)$ such that $\mathrm{dir}[pa_n] \to u/|u|$ and $\mathrm{dir}[pb_n] \to v/|v|$. Consider sequence of functions

$$f_n = |u|\cdot|a_n *| + |v|\cdot|b_n *|.$$

According to Theorem 14.6.2,

$$(\boldsymbol{d}_p f_n)(x) = -|u|\cdot\langle \mathrm{dir}[pa_n], x\rangle - |v|\cdot\langle \mathrm{dir}[pb_n], x\rangle.$$

Thus, we have the following uniform convergence for all $x \in \Sigma_p$

$$(\boldsymbol{d}_p f_n)(x) \xrightarrow[n\to\infty]{} -\langle u, x\rangle - \langle v, x\rangle.$$

According to Lemma 16.2.5, sequence $\nabla_p f_n$ converges. Set $w = \lim_n \nabla_p f_n$.

From definition of gradient

$$\begin{aligned}\langle w, w\rangle &= \lim_{n\to\infty}\langle \nabla_p f_n, \nabla_p f_n\rangle = \\ &= \lim_{n\to\infty}(\boldsymbol{d}_p f_n)(\nabla_p f_n) = \\ &= -\langle u, w\rangle - \langle v, w\rangle,\end{aligned}$$

$$\begin{aligned}\langle w, x\rangle &= \lim_{n\to\infty}\langle \nabla_p f_n, x\rangle \geqslant \\ &\geqslant \lim_{n\to\infty}(\boldsymbol{d}_p f_n)(x) = \\ &= -\langle u, x\rangle - \langle v, x\rangle.\end{aligned}$$

□

16.4 Semicontinuity of |gradient|

In this section we collect number of corollaries of the following lemma.

16.4.1. Ultra limit of |gradient|. *Assume*

- ◇ $\mathcal{L}_n \in \mathrm{CBB}[\kappa]$ *and* $(\mathcal{L}_n, p_n) \to (\mathcal{L}_\omega, p_\omega)$ *as* $n \to \omega$*;*
- ◇ $f_n\colon \mathcal{L}_n \looparrowright \mathbb{R}$ *and* $f_\omega\colon \mathcal{L}_\omega \looparrowright \mathbb{R}$ *are locally Lipschitz and* λ*-concave and* $f_n \to f_\omega$ *as* $n \to \omega$*;*
- ◇ $x_n \in \mathrm{Dom}\, f_n$ *and* $x_n \to x_\omega \in \mathrm{Dom}\, f_\omega$ *as* $n \to \omega$*.*

Then

$$|\nabla_{x_\omega} f_\omega| \leqslant \lim_{n\to\omega}|\nabla_{x_n} f_n|.$$

Remarks. To construct an example such that $|\nabla_{x_\omega} f_\omega| < \lim_{n\to\omega}|\nabla_{x_n} f_n|$; one can take $\mathcal{L}_n = \mathcal{L}_\omega = \mathbb{R}$, $f_n(x) = f_\omega(x) = -|x|$ and $x_n \to 0+$.

From the convergence of gradient curves (proved later in 16.8.3), one can deduce the following slightly stronger statement.

16.4.2. Proposition. *Assume*

- ◇ $\mathcal{L}_n \in \mathrm{CBB}[\kappa]$ *and* $(\mathcal{L}_n, p_n) \to (\mathcal{L}_\omega, p_\omega)$ *as* $n \to \omega$*;*
- ◇ $f_n\colon \mathcal{L}_n \looparrowright \mathbb{R}$ *and* $f_\omega\colon \mathcal{L}_\omega \looparrowright \mathbb{R}$ *are locally Lipschitz and* λ*-concave and* $f_n \to f_\omega$ *as* $n \to \omega$*.*

Then

$$|\nabla_{x_\omega} f_\omega| = \inf\{\lim_{n\to\omega} |\nabla_{x_n} f_n|\},$$

where infimum is taken for all sequences $x_n \in \operatorname{Dom} f_n$ *such that* $x_n \to$ $\to x_\omega \in \operatorname{Dom} f_\omega$ *as* $n \to \omega$.

Proof of 16.4.1. Fix an $\varepsilon > 0$ and choose $y_\omega \in \operatorname{Dom} f_\omega$ sufficiently close to x_ω such that

$$|\nabla_{x_\omega} f_\omega| - \varepsilon < \frac{f_\omega(y_\omega) - f_\omega(x_\omega)}{|x_\omega y_\omega|}.$$

Choose $y_n \in \mathcal{L}_n$ such that $y_n \to y_\omega$ as $n \to \omega$. Sinse $|x_\omega y_\omega|$ is sufficiently small, the λ-concavity of f_n implies that

$$|\nabla_{x_\omega} f_\omega| - 2\cdot\varepsilon < (\boldsymbol{d}_{x_n} f_n)(\operatorname{dir}[x_n y_n]),$$

for ω-almost all n. Hence,

$$|\nabla_{x_\omega} f_\omega| - 2\cdot\varepsilon \leqslant \lim_{n\to\omega} |\nabla_{x_n} f_n|.$$

Sinse $\varepsilon > 0$ is arbitrary, the proposition follows. □

Note that the distance preserving map $\iota\colon \mathcal{L} \hookrightarrow \mathcal{L}^\omega$ induces an embedding

$$\boldsymbol{d}_p\iota\colon \mathrm{T}_p\mathcal{L} \hookrightarrow \mathrm{T}_p\mathcal{L}^\omega.$$

Thus, we can (and will) consider $\mathrm{T}_p\mathcal{L}$ as a subcone of $\mathrm{T}_p\mathcal{L}^\omega$.

16.4.3. Corollary. *Let* $\mathcal{L} \in \mathrm{CBB}$ *and* $f\colon \mathcal{L} \looparrowright \mathbb{R}$ *be locally Lipschitz semiconcave subfunction. Then for any point* $p \in \operatorname{Dom} f$ *we have*

$$\nabla_p f = \nabla_p f^\omega.$$

Proof. Note that $\mathcal{L} \supset \operatorname{Dom} f \subset \operatorname{Dom} f^\omega \subset \mathcal{L}^\omega$. Applying 16.4.1 for $\mathcal{L}_n = \mathcal{L}$ and $x_n = x$, we get $|\nabla_x f| \geqslant |\nabla_x f^\omega|$.

On the other hand, $f = f^\omega|\mathcal{L}$, hence $\boldsymbol{d}_p f = \boldsymbol{d}_p f^\omega|\mathrm{T}_p\mathcal{L}$. Thus, from 16.2.1*c*, $|\nabla_x f| \leqslant |\nabla_x f^\omega|$. Therefore

❶ $$|\nabla_x f| = |\nabla_x f^\omega|.$$

Further,

$$\begin{aligned}|\nabla_x f|^2 &= (\boldsymbol{d}_p f)(\nabla_x f)\\ &= \boldsymbol{d}_p f^\omega(\nabla_x f) \leqslant\\ &\leqslant \langle \nabla_x f^\omega, \nabla_x f\rangle =\\ &= |\nabla_x f^\omega|\cdot|\nabla_x f|\cdot \cos\measuredangle(\nabla_x f^\omega, \nabla_x f).\end{aligned}$$

Together with ❶, this implies $\measuredangle(\nabla_x f^\omega, \nabla_x f) = 0$ and the statement follows. □

In particular, we have lower-semicontinuity of the function $x \mapsto$ $\mapsto |\nabla_x f|$:

16.4.4. Semicontinuity of |gradient|. *Let $\mathcal{L} \in \mathrm{CBB}$ and $f\colon \mathcal{L} \looparrowright$ $\looparrowright \mathbb{R}$ be locally Lipschitz semiconcave subfunction. Then the function $x \mapsto |\nabla_x f|$ is lower-continuous; that is for any sequence $x_n \to x \in$ $\in \operatorname{Dom} f$, we have*

$$|\nabla_x f| \leqslant \varliminf_{n\to\infty} |\nabla_{x_n} f|.$$

Proof. According to 16.4.3, $|\nabla_x f| = |\nabla_x f^\omega|$. Applying 16.4.1 for $x_n \to x$, we get that

$$\lim_{n\to\omega} |\nabla_{x_n} f| \geqslant |\nabla_x f^\omega| = |\nabla_x f|.$$

Passing to arbitrary subsequence of (x_n) we obtain the result. □

16.5 Gradient-like curves

Gradient-like curves provide a technical tool which will be used later in the construction of gradient curves. The later appear to be a special reparametrization of gradient-like curves.

16.5.1. Definition. *Let $\mathcal{L} \in \mathrm{CBB}$ and $f\colon \mathcal{L} \looparrowright \mathbb{R}$ be locally Lipschitz semiconcave subfunction.*

A Lipschitz curve $\hat\alpha\colon [s_{\min}, s_{\max}) \to \operatorname{Dom} f$ will be called f-gradient-like curve if

$$\hat\alpha^+ = \tfrac{1}{|\nabla_{\hat\alpha} f|}\cdot\nabla_{\hat\alpha} f;$$

that is, for any $s \in [s_{\min}, s_{\max})$, $\hat\alpha^+(s)$ is defined and $\hat\alpha^+(s) = \frac{1}{|\nabla_{\hat\alpha(s)} f|}\cdot\nabla_{\hat\alpha(s)} f$.

Note that in particular this definition implies that $|\nabla_p f| > 0$ for any point p on $\hat\alpha$. The next theorem shows that a weaker condition gives an equivalent definition.

16.5.2. Theorem. *Let $\mathcal{L} \in \mathrm{CBB}$, $f\colon \mathcal{L} \looparrowright \mathbb{R}$ be locally Lipschitz semiconcave subfunction and $|\nabla_p f| > 0$ for any $p \in \operatorname{Dom} f$.*

A curve $\hat\alpha\colon [s_{\min}, s_{\max}) \to \operatorname{Dom} f$ is an f-gradient-like curve if and only if it is 1-Lipschitz and

❶ $$\varliminf_{s\to s_0+} \frac{f\circ\hat\alpha(s) - f\circ\hat\alpha(s_0)}{s - s_0} \geqslant |\nabla_{\hat\alpha(s_0)} f|$$

for almost all $s_0 \in [s_{\min}, s_{\max})$.

Proof. The "only if" part follows directly from definition. To prove the "if" part, note that for any $s_0 \in [s_{\min}, s_{\max})$ we have

$$\varliminf_{s\to s_0+} \frac{f\circ\hat\alpha(s) - f\circ\hat\alpha(s_0)}{s-s_0} \geqslant \varliminf_{s\to s_0+} \oint_{s_0}^{s} |\nabla_{\hat\alpha(\mathfrak{s})} f|\cdot \boldsymbol{d}\mathfrak{s} \geqslant$$
$$\geqslant |\nabla_{\hat\alpha(s_0)} f|;$$

the first inequality follows from ❶ and the second from lower-semicontinuity of function $x \mapsto |\nabla_x f|$, see 16.4.4. Form 16.2.1, we have

$$\hat\alpha^+(s_0) = \tfrac{1}{|\nabla_{\hat\alpha(s_0)} f|}\cdot \nabla_{\hat\alpha(s_0)} f;$$

hence the result. □

The following theorem is similar to [105, 2.36] and [118, 5.7].

16.5.3. Theorem. *Let* $\mathcal{L} \in \mathrm{CBB}$ *and* $f\colon \mathcal{L} \looparrowright \mathbb{R}$ *be locally Lipschitz and* λ*-concave. Assume* $\hat\alpha\colon [0, s_{\max}) \to \operatorname{Dom} f$ *is an* f*-gradient-like curve then*

$$(f\circ\hat\alpha)'' \leqslant \lambda$$

everywhere on $[0, s_{\max})$.

16.5.4. Corollary. *Let* $\mathcal{L} \in \mathrm{CBB}$, $f\colon \mathcal{L} \looparrowright \mathbb{R}$ *be a locally Lipschitz and semiconcave function and* $\hat\alpha\colon [0, s_{\max}) \to \operatorname{Dom} f$ *be an* f*-gradient-like curve. Then function* $s \mapsto |\nabla_{\hat\alpha(s)} f|$ *is right-continuous; that is, for any* $s_0 \in [0, s_{\max})$ *we have*

$$|\nabla_{\hat\alpha(s_0)} f| = \lim_{s\to s_0+} |\nabla_{\hat\alpha(s)} f|.$$

Proof . Applying 16.5.3 locally, we have that $f\circ\hat\alpha(s)$ is semiconcave. The statement follows since

$$(f\circ\hat\alpha)^+(s) = (\boldsymbol{d}_p f)\left(\tfrac{1}{|\nabla_{\hat\alpha(s)} f|}\cdot \nabla_{\hat\alpha(s)} f\right) = |\nabla_{\hat\alpha(s)} f|.$$

□

Proof of 16.5.3. For any $s > s_0$,

$$\begin{aligned}(f\circ\hat\alpha)^+(s_0) &= |\nabla_{\hat\alpha(s_0)} f| \geqslant \\ &\geqslant (d_{\hat\alpha(s_0)} f)(\operatorname{dir}[\hat\alpha(s_0)\hat\alpha(s)]) \geqslant \\ &\geqslant \frac{f\circ\hat\alpha(s) - f\circ\hat\alpha(s_0)}{|\hat\alpha(s)\ \hat\alpha(s_0)|} - \tfrac{\lambda}{2}\cdot|\hat\alpha(s)\ \hat\alpha(s_0)|.\end{aligned}$$

Set $\lambda_+ = \max\{0, \lambda\}$. Since $s - s_0 \geqslant |\hat\alpha(s)\ \hat\alpha(s_0)|$, for any $s > s_0$ we have

❷ $$(f\circ\hat\alpha)^+(s_0) \geqslant \frac{f\circ\hat\alpha(s) - f\circ\hat\alpha(s_0)}{s - s_0} - \tfrac{\lambda_+}{2}\cdot(s - s_0).$$

Thus $f\circ\hat\alpha$ is λ_+-concave. That finishes the proof for $\lambda \geqslant 0$; in case $\lambda < 0$ we get only that $f\circ\hat\alpha$ is 0-concave.

Note that $|\hat\alpha(s)\ \hat\alpha(s_0)| = s - s_0 - o(s - s_0)$, thus

❸ $$(f\circ\hat\alpha)^+(s_0) \geqslant \frac{f\circ\hat\alpha(s) - f\circ\hat\alpha(s_0)}{s - s_0} - \tfrac{\lambda}{2}\cdot(s - s_0) + o(s - s_0).$$

Together, ❷ and ❸ imply that $f\circ\hat\alpha$ is λ-concave. □

16.5.5. Proposition. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$, $p, q \in \mathcal{L}$. Assume $\hat\alpha\colon [s_{\min}, s_{\max}) \to \mathcal{L}$ be a $|p_*|$-gradient-like curve such that $\hat\alpha(s) \to z \in\,]pq[$ as $s \to s_{\max}+$ then α is a unit-speed geodesic which lies in $[pq]$.*

Proof. Clearly,

❹ $$\tfrac{d^+}{dt}|q\,\hat\alpha(t)| \geqslant -1$$

On the other hand,

❺ $$\begin{aligned}\tfrac{d^+}{dt}|p\,\hat\alpha(t)| &\geqslant (\boldsymbol{d}_{\hat\alpha(t)}|p_*|)(\mathrm{dir}[\hat\alpha(t)q]) \geqslant \\ &\geqslant -\cos\measuredangle^\kappa(\hat\alpha(t)\,{}^p_q).\end{aligned}$$

Inequalities ❹ and ❺ imply that the function $t \mapsto \measuredangle^\kappa(q\,{}^{\hat\alpha(t)}_p)$ is nondecreasing. Hence the result. □

16.6 Gradient curves: Definition

In this section we define gradient curves and tie them tightly to gradient-like curves which were introduced in Section 16.5.

16.6.1. Definition. *Let $\mathcal{L} \in \mathrm{CBB}$ and $f\colon \mathcal{L} \looparrowright \mathbb{R}$ be a locally Lipschitz and semiconcave function.*

A locally Lipschitz curve $\alpha\colon [t_{\min}, t_{\max}) \to \mathrm{Dom}\, f$ will be called f-gradient curve *if*

$$\alpha^+ = \nabla_\alpha f;$$

that is, for any $t \in [t_{\min}, t_{\max})$, $\alpha^+(t)$ is defined and $\alpha^+(t) = \nabla_{\alpha(t)} f$.

The next lemma states that gradient and gradient-like curves are special reparametrizations of each-other.

16.6.2. Lemma. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$ and $f\colon \mathcal{L} \looparrowright \mathbb{R}$ be a locally Lipschitz semiconcave subfunction such that $|\nabla_p f| > 0$ for any $p \in \operatorname{Dom} f$.*

Assume $\alpha\colon [0, t_{\max}) \to \operatorname{Dom} f$ be a locally Lipschitz curve and $\hat\alpha\colon [0, s_{\max}) \to \operatorname{Dom} f$ be its reparametrization by arc-length, so $\alpha = \hat\alpha \circ \sigma$ for some homeomorphism $\sigma\colon [0, t_{\max}) \to [0, s_{\max})$, then

$$\alpha^+ = \nabla_\alpha f \qquad \Leftrightarrow \qquad \hat\alpha^+ = \frac{1}{|\nabla_{\hat\alpha} f|}\cdot\nabla_{\hat\alpha} f \quad \textit{and} \quad \sigma^{-1}(s) = \int\limits_0^s \frac{\boldsymbol{d}\mathfrak{s}}{(f\circ\hat\alpha)'(\mathfrak{s})}.$$

Proof; ($\Rightarrow$). According to 5.3.2,

❶ $$\sigma'(t) \overset{a.e.}{=\!=} |\alpha^+(t)| = |\nabla_{\alpha(t)} f|.$$

Note that

$$(f\circ\alpha)'(t) \overset{a.e.}{=\!=} (f\circ\alpha)^+(t) = |\nabla_{\alpha(t)} f|^2.$$

Setting $s = \sigma(t)$, we have

$$(f\circ\hat\alpha)'(s) \overset{a.e.}{=\!=} \frac{(f\circ\alpha)'(t)}{\sigma'(t)} \overset{a.e.}{=\!=} |\nabla_{\alpha(t)} f| = |\nabla_{\hat\alpha(s)} f|.$$

From 16.5.2, it follows that $\hat\alpha(t)$ is an f-gradient-like curve; that is,

$$\hat\alpha^+ = \frac{1}{|\nabla_{\hat\alpha} f|}\cdot\nabla_{\hat\alpha} f.$$

In particular, $(f\circ\hat\alpha)^+(s) = |\nabla_{\hat\alpha^+(s)} f|$ and from ❶,

$$\sigma^{-1}(s) = \int\limits_0^s \frac{1}{|\nabla_{\hat\alpha(\mathfrak{s})} f|}\cdot\boldsymbol{d}\mathfrak{s} = \int\limits_0^s \frac{1}{(f\circ\hat\alpha)'(\mathfrak{s})}\cdot\boldsymbol{d}\mathfrak{s}.$$

($\Leftarrow$). Clearly,

$$\sigma(t) = \int\limits_0^t (f\circ\hat\alpha)^+(\sigma(\mathfrak{t}))\cdot\boldsymbol{d}\mathfrak{t} = \int\limits_0^t |\nabla_{\alpha(\mathfrak{t})} f|\cdot\boldsymbol{d}\mathfrak{t}.$$

According to 16.5.4, the function $s \mapsto |\nabla_{\hat\alpha(s)} f|$ is right-continuous. Therefore the same is true for function $t \mapsto |\nabla_{\hat\alpha\circ\sigma(t)} f| = |\nabla_{\alpha(t)} f|$. Hence, for any $t_0 \in [0, t_{\max})$ we have

$$\sigma^+(t_0) = \lim_{t\to t_0+} \oint\limits_{t_0}^t |\nabla_{\alpha(\mathfrak{t})} f|\cdot\boldsymbol{d}\mathfrak{t} = |\nabla_{\alpha(t_0)} f|.$$

Thus, we have

$$\alpha^+(t_0) = \sigma^+(t_0)\cdot\hat\alpha^+(\sigma(t_0)) = \nabla_{\alpha(t_0)}f.$$

□

16.6.3. Exercise. *Let $\mathcal{L} \in \mathrm{CBB}$, $f\colon \mathcal{L} \to \mathbb{R}$ be a locally Lipschitz semiconcave function. Assume $\alpha\colon \mathbb{I} \to \mathcal{L}$ is a Lipschitz curve such that*

$$\begin{aligned}\alpha^+(t) &\leqslant |\nabla_{\alpha(t)}f|,\\ (f\circ\alpha)^+(t) &\geqslant |\nabla_{\alpha(t)}f|\end{aligned}$$

for almost all t. Show that α is an f-gradient curve.

16.7 Gradient curves: distance estimates

16.7.1. First distance estimate. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$, $f\colon \mathcal{L} \to \mathbb{R}$ be a locally Lipschitz and λ-concave function. Assume $\alpha,\beta\colon [0,t_{\max}) \to \mathcal{L}$ be two f-gradient then for any $t \in [0,t_{\max})$*

$$|\alpha(t)\ \beta(t)| \leqslant e^{\lambda\cdot t}\cdot|\alpha(0)\ \beta(0)|.$$

Moreover, the same conclusion holds for a for locally Lipschitz and λ-concave subfunction $f\colon \mathcal{L} \circ\!\!\to \mathbb{R}$ if for any $t \in [0,t_{\max})$ there is a geodesic $[\alpha(t)\,\beta(t)]$ in $\operatorname{Dom} f$.

Proof. If $\mathcal{L}$ is not geodesic space, let us pass to its ultrapower $\mathcal{L}^\omega$.

Fix a choice of geodesic $[\alpha(t)\,\beta(t)]$ for each t.

Set $\ell(t) = |\alpha(t)\ \beta(t)|$, from the first variation inequality (6.3.4) and the estimate in 16.2.4 we get

$$\ell^+(t) \leqslant -\langle \mathrm{dir}[\alpha(t)\beta(t)], \nabla_{\alpha(t)}f\rangle - \langle \mathrm{dir}[\beta(t)\alpha(t)], \nabla_{\beta(t)}f\rangle \leqslant \lambda\cdot\ell(t).$$

Hence the result. □

The following result will not be needed in this chapter, it just goes naturally with the previous one.

16.7.2. Second distance estimate. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$, $\varepsilon > 0$ and $f,g\colon \mathcal{L} \to \mathbb{R}$ be two λ-concave locally Lipschitz function such that $|f - g| < \varepsilon$. Assume $\alpha,\beta\colon [0,t_{\max}) \to \mathcal{L}$ are correspondingly f- and g-gradient curves such that $\alpha(0) = \beta(0)$. Then*

$$|\alpha(t)\ \beta(t)| \leqslant \sqrt{\tfrac{1}{2\cdot\varepsilon\cdot\lambda}\cdot\left(e^{\frac{t\cdot\lambda}{\varepsilon}} - 1\right)}$$

for any $t \in [0, t_{\max})$. *In particular, if* $t_{\max} < \infty$, *then*

$$|\alpha(t)\ \beta(t)| \leqslant c\cdot\sqrt{\varepsilon\cdot t}$$

for some constant $c = c(t_{\max}, \lambda)$.

Moreover, the same conclusion holds for locally Lipschitz and λ-*concave subfunctions* $f, g\colon \mathcal{L} \hookrightarrow \mathbb{R}$ *if for any* $t \in [0, t_{\max})$ *there is a geodesic* $[\alpha(t)\,\beta(t)]$ *in* $\operatorname{Dom} f \cap \operatorname{Dom} g$.

Proof. Set $\ell = \ell(t) = |\alpha(t)\ \beta(t)|$. Fix arbitrary fixed t, set $p = \alpha(t)$ and $q = \beta(t)$. From the first variation formula and 16.2.3,

$$\begin{aligned}\ell^+ &\leqslant -\langle \operatorname{dir}[pq], \nabla_p f\rangle - \langle \operatorname{dir}[qp], \nabla_q g\rangle \leqslant \\ &\leqslant -\Big(f(q) - f(p) - \lambda\cdot\tfrac{\ell^2}{2}\Big)/\ell - \Big(g(p) - g(q) - \lambda\cdot\tfrac{\ell^2}{2}\Big)/\ell \leqslant \\ &\leqslant \lambda\cdot\ell + \tfrac{2\cdot\varepsilon}{\ell}.\end{aligned}$$

Integrating the above estimate, we get

$$\ell(t) \leqslant \sqrt{\tfrac{1}{2\cdot\varepsilon\cdot\lambda}\cdot\Big(e^{\frac{t\cdot\lambda}{\varepsilon}} - 1\Big)}.$$

□

16.8 Gradient curves: existence, uniqueness and convergence

In general, "past" of gradient curves can not be determined by present. For example, consider concave function $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = -|x|$; two curves $\alpha(t) = \min\{0, t\}$ and $\beta(t) = 0$ are f-gradient and $\alpha(t) = \beta(t) = 0$ for all $t \geqslant 0$, however $\alpha(t) \neq \beta(t)$ for all $t < 0$.

The next theorem shows that "future" gradient curve is unique.

16.8.1. Picard's theorem. *Let* $\mathcal{L} \in \mathrm{CBB}$ *and* $f\colon \mathcal{L} \hookrightarrow \mathbb{R}$ *be semi-concave subfunction. Assume* $\alpha, \beta\colon [0, t_{\max}) \to \operatorname{Dom} f$ *be two* f*-gradient curves such that* $\alpha(0) = \beta(0)$ *then* $\alpha(t) = \beta(t)$ *for any* $t \in [0, t_{\max})$.

Proof. Follows directly from the first distance estimate (16.7.1). □

16.8.2. Local existence. *Let* $\mathcal{L} \in \mathrm{CBB}[\kappa]$ *and* $f\colon \mathcal{L} \hookrightarrow \mathbb{R}$ *be locally Lipschitz* λ*-concave subfunction. Then for any* $p \in \operatorname{Dom} f$

a) if $|\nabla_p f| > 0$ *then for some* $\varepsilon > 0$, *there is an* f*-gradient-like curve* $\hat\alpha\colon [0, \varepsilon) \to \mathcal{L}$ *which starts at* p *(that is* $\hat\alpha(0) = p$*);*

b) for some $\delta > 0$, *there is an* f*-gradient curve* $\alpha\colon [0, \delta) \to \mathcal{L}$ *which starts at* p *(that is* $\alpha(0) = p$*).*

This theorem was proved in [130]; here we present a simplified proof from [102].

Proof. Note that if $|\nabla_p f| = 0$ then one can take constant curve $\alpha(t) = p$. Otherwise, take $\varepsilon > 0$, such that $\mathrm{B}(p,\varepsilon) \subset \operatorname{Dom} f$, the restriction $f|_{\mathrm{B}(p,\varepsilon)}$ is Lipschitz and $|\nabla_x f| > \varepsilon$ for all $x \in \mathrm{B}(p,\varepsilon)$; that is possible due to semicontinuity of |gradient| (16.4.4).

The curves $\hat\alpha$ and α will be constructed in the following three steps. First we construct an f^ω-gradient-like curve $\hat\alpha_\omega\colon [0,\varepsilon) \to \mathcal{L}^\omega$ as an ω-limit of certain sequence of broken geodesics in $\mathcal{L}$. Second, we parametrize $\hat\alpha_\omega$ as in 16.6.2, to obtain an f^ω-gradient curve α_ω in $\mathcal{L}^\omega$. Third, applying Picard's theorem (16.8.1) together with Lemma 4.2.3, we obtain that α_ω lies in $\mathcal{L} \subset \mathcal{L}^\omega$ and therefore one can take $\alpha = \alpha_\omega$ and $\hat\alpha = \hat\alpha_\omega$.

Note that if $\mathcal{L}$ is proper then $\mathcal{L} = \mathcal{L}^\omega$ and $f^\omega = f$. Thus, in this case, the third step is not necessary.

Step 1. Given $n \in \mathbb{N}$, by open-close argument, we can construct a unit-speed curve $\hat\alpha_n\colon [0,\varepsilon] \to \mathcal{L}$ starting at p, with a partition of $[0,\varepsilon)$ into countable number of half-open intervals $[\sigma_i, \bar\sigma_i)$ so that for each i we have

(i) $\hat\alpha_n([\sigma_i,\bar\sigma_i])$ is a geodesic and $\bar\sigma_i - \sigma_i < \frac{1}{n}$.

(ii) $f\circ\hat\alpha_n(\bar\sigma_i) - f\circ\hat\alpha_n(\sigma_i) > (\bar\sigma_i - \sigma_i)\cdot(|\nabla_{\hat\alpha_n(\sigma_i)} f| - \frac{1}{n})$.

Pass to a subsequence of $(\hat\alpha_n)$ such that $f\circ\hat\alpha_n$ uniformly converges; set

$$h(s) = \lim_{n\to\infty} f\circ\hat\alpha_n(s).$$

Set $\hat\alpha_\omega = \lim_{n\to\omega}\hat\alpha_n$, it is a curve in $\mathcal{L}^\omega$ which starts at $p \in \mathcal{L} \subset \mathcal{L}^\omega$. Clearly $\hat\alpha_\omega$ is 1-Lipschitz. From (ii) and 16.4.1, we get

$$(f^\omega\circ\hat\alpha_\omega)^+(\sigma) \geqslant |\nabla_{\hat\alpha_\omega(\sigma)} f^\omega|.$$

According to 16.5.2, $\hat\alpha_\omega\colon [0,\varepsilon) \to \mathcal{L}^\omega$ is an f^ω-gradient-like curve.

Step 2. Clearly $h(s) = f^\omega\circ\alpha_\omega$. Therefore, according to 16.5.3, h is λ-concave. Thus, we can define a homeomorphism $\sigma\colon [0,\delta] \to [0,\varepsilon]$ by

❶ $$\sigma^{-1}(s) = \int_0^s \frac{1}{h'(\mathfrak{s})}\cdot \boldsymbol{d}\mathfrak{s},$$

According to 16.6.2, $\alpha(t) = \hat\alpha\circ\sigma(t)$ is an f^ω-gradient curve in $\mathcal{L}^\omega$.

Step 3. Clearly, $\nabla_p f = \nabla_p f^\omega$ for any $p \in \mathcal{L} \subset \mathcal{L}^\omega$; more formally if $\iota\colon \mathcal{L} \hookrightarrow \mathcal{L}^\omega$ is the natural embedding then $(\boldsymbol{d}_p\iota)(\nabla_p f) = \nabla_p f^\omega$. Thus, it is sufficient to show that $\alpha_\omega \subset \mathcal{L}$. Assume the contrary, then according to 4.2.3 there is a subsequence $\hat\alpha_{n_k}$ such that

$$\hat\alpha_\omega \neq \hat\alpha'_\omega \overset{def}{=\!=} \lim_{k\to\omega}\hat\alpha_{n_k}.$$

Clearly $h(s) = f^\omega \circ \hat\alpha_\omega = f^\omega \circ \hat\alpha'_\omega$. Thus, for $\sigma\colon [0,\delta] \to [0,\varepsilon]$ defined by ❶, we have that both curves $\hat\alpha_\omega \circ \sigma$ and $\hat\alpha'_\omega \circ \sigma$ are f^ω-gradient. Thus, from Picard's theorem (16.8.1), we get $\hat\alpha_\omega \circ \sigma = \hat\alpha'_\omega \circ \sigma$ and therefore $\hat\alpha_\omega = \hat\alpha'_\omega$, a contradiction □

16.8.3. Ultra limit of gradient curves. *Assume*

- ◇ $\mathcal{L}_n \in \mathrm{CBB}[\kappa]$ *and* $(\mathcal{L}_n, p_n) \to (\mathcal{L}_\omega, p_\omega)$ *as* $n \to \omega$*;*
- ◇ $f_n\colon \mathcal{L}_n \looparrowright \mathbb{R}$ *are* $\pounds$*-Lipschitz and* λ*-concave* $f_n \to f_\omega$ *as* $n \to \omega$ *and* $p_\omega \in \operatorname{Dom} f_\omega$*.*

Then

a) If $|\nabla_{p_\omega} f_\omega| > 0$ *then there is* $\varepsilon > 0$ *such that, the* f_n*-gradient-like curves* $\hat\alpha_n\colon [0,\varepsilon) \to \mathcal{L}_n$ *are defined for* ω*-almost all* n*. And moreover, a curve* $\hat\alpha_\omega\colon [0,\varepsilon) \to \mathcal{L}_\omega$ *is a gradient-like curve which starts at* p_ω *if and only if* $\hat\alpha_n(s) \to \hat\alpha_\omega(s)$ *as* $n \to \omega$ *for all* $s \in [0,\varepsilon)$*.*

b) For some $\delta > 0$*, the* f_n*-gradient curves* $\alpha_n\colon [0,\delta) \to \mathcal{L}_n$ *are defined for* ω*-almost all* n*. And moreover, a curve* $\alpha_\omega\colon [0,\delta) \to \mathcal{L}_\omega$ *is a gradient curve which starts at* p_ω *if and only if* $\alpha_n(t) \to \alpha_\omega(t)$ *as* $n \to \omega$ *for all* $t \in [0,\delta)$*.*

The idea in the proof is the same as in the proof of local existence (16.8.2).

Proof. According to **??**, f_ω is λ-concave.

"if"-part of (a). Take $\varepsilon > 0$ so small that $\mathrm{B}(p_\omega,\varepsilon) \subset \operatorname{Dom} f_\omega$ and $|\nabla_{x_\omega} f_\omega| > 0$ for any $x_\omega \in \mathrm{B}(p_\omega,\varepsilon)$ (that is possible due to 16.4.4).

Clearly $\hat\alpha_\omega$ is 1-Lipschitz. From 16.4.1, we get $(f_\omega \circ \hat\alpha_\omega)^+(s) \geqslant |\nabla_{\hat\alpha_\omega(s)} f^\omega|$. According to 16.5.2, $\hat\alpha_\omega\colon [0,\varepsilon) \to \mathcal{L}^\omega$ is an f_ω-gradient-like curve.

"if"-part of (b). Assume first $|\nabla_{p_\omega} f_\omega| > 0$ — so we can apply "if"-part of (a). Let $h_n = f_n \circ \hat\alpha_n\colon [0,\varepsilon) \to \mathbb{R}$ and $h_\omega = f_\omega \circ \hat\alpha_\omega$. From 16.5.3, h_n are λ-concave and clearly $h_n \to h_\omega$ as $n \to \omega$. Let us define reparametrizations

$$\sigma_n^{-1}(s) = \int_0^s \frac{1}{h'_n(\mathfrak{s})}\cdot \boldsymbol{d}\mathfrak{s}, \qquad \sigma_\omega^{-1}(s) = \int_0^s \frac{1}{h'_\omega(\mathfrak{s})}\cdot \boldsymbol{d}\mathfrak{s}.$$

The λ-convexity of h_n implies that $\sigma_n \to \sigma_\omega$ as $n \to \omega$. By Lemma 16.6.2, $\alpha_n = \hat\alpha_n \circ \sigma_n$. Applying "if"-part of (a) together with Lemma 16.6.2, we get that $\alpha_\omega = \hat\alpha_\omega \circ \sigma_\omega$ is gradient curve.

The remaining case $|\nabla_{p_\omega} f_\omega| = 0$, can be reduced to the one above using the following trick. Consider sequence of spaces $\mathcal{L}_n^\times = \mathcal{L}_n \times \mathbb{R}$, with the sequence of subfunction $f_n^\times\colon \mathcal{L}_n^\times \to \mathbb{R}$, defined by $f_n^\times(p,t) = f_n(p) + t$. According to Theorem **??**, $\mathcal{L}_n^\times \in \mathrm{CBB}[\kappa_-]$

for $\kappa_- = \min\{\kappa, 0\}$ and clearly, and $f_n^\times$ are λ_+-concave, for $\lambda_+ = \max\{\lambda, 0\}$. Analogously define $\mathcal{L}_\omega^\times = \mathcal{L}_\omega \times \mathbb{R}$, and $f_\omega^\times(p,t) = f_\omega(p) + t$.

Clearly $\mathcal{L}_n^\times \to \mathcal{L}_\omega^\times$, $f_n^\times \to f_\omega^\times$ as $n \to \omega$ and $|\nabla_x f_\omega^\times| > 0$ for any $x \in \operatorname{Dom} f_\omega^\times$. Thus, for the sequence $f_n^\times \colon \mathcal{L}_n^\times \looparrowright \mathbb{R}$, we can apply "if"-part of (a). It is easy to see that $\alpha_\omega^\times(t) = (\alpha_\omega(t), t)$ is a $f_\omega^\times$-gradient curve in $\mathcal{L}_\omega^\times$ if and only if $\alpha_\omega(t)$ is a f_ω-gradient curve.

"only if"-part of (b) and (a). The "only if"-part of (b) follows from the "if"-part of (b) and Picard's theorem (16.8.1). Applying Lemma 16.6.2, we get "only if"-part of (a). □

Directly form local existence (16.8.2) and distance estimates (16.7.1), we get the following:

16.8.4. Global existence. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$ and $f \colon \mathcal{L} \looparrowright \mathbb{R}$ be locally Lipschitz and λ-concave subfunction. Then for any $p \in \operatorname{Dom} f$, there is $t_{\max} \in (0, \infty]$ such that there is an f-gradient curve $\alpha \colon [0, t_{\max}) \to \mathcal{L}$ with $\alpha(0) = p$. Moreover, for any sequence $t_n \to t_{\max}-$, the sequence $\alpha(t_n)$ does not have a limit point in* $\operatorname{Dom} f$.

The following theorem gurantees the existance of gradient curves for all times for the special type of semiconcave functions which play important role in the theory. It follows from 16.8.4, 16.5.3 and 16.6.2.

16.8.5. Theorem. *Let $\mathcal{L} \in \mathrm{CBB}$ and $f \colon \mathcal{L} \to \mathbb{R}$ satisfies*

$$f'' + \kappa \cdot f \leqslant \lambda$$

for some real values κ and λ. Then f has complete gradient*; that is for any $x \in \mathcal{L}$ there is a f-gradient curve $\alpha \colon [0, \infty) \to \mathcal{L}$ which starts at x.*

16.9 Line splitting theorem

Let $\mathcal{X}$ be a metric space and $A, B \subset \mathcal{X}$. We will write

$$\mathcal{X} = A \oplus B$$

If there projections $\operatorname{proj}_A \colon \mathcal{X} \to A$ and $\operatorname{proj}_B \colon \mathcal{X} \to B$ such that

$$|x\,y|^2 = |\operatorname{proj}_A(x)\ \operatorname{proj}_A(y)|^2 + |\operatorname{proj}_B(x)\ \operatorname{proj}_B(y)|^2$$

for any two points $x, y \in \mathcal{X}$.

Note that if

$$\mathcal{X} = A \oplus B$$

then

- ⋄ A intersects B at single point,
- ⋄ both sets A and B are convex sets in $\mathcal{X}$.

Recall that a line in a metric space is a geodesic which is minimizing on each segment.

16.9.1. Line splitting theorem. *Let $\mathcal{L} \in \mathrm{CBB}[0]$ and γ be a line in $\mathcal{L}$. Then*

$$\mathcal{L} = X \oplus \gamma(\mathbb{R})$$

for some subset $X \subset \mathcal{L}$.

For smooth 2-dimensional surfaces, this theorem was proved by Cohn-Vossen in [43]. For the Riemannian manifolds of higher dimensions it was proved by Toponogov in [159]. Then it was generalized by Milka in [109] to Alexandrov's spaces, almost the same prove is given in [32, 1.5].

Further generalizations of Splitting theorem for Riemannian manifolds with nonnegative Ricci curvature were obtained by Cheeger and Gromoll in [41]. Eshenburg in [51] obtained an analogous result for the Lorentzian manifolds, that is, pseudo-Riemannian manifolds of signature $(1, n)$.

We present a proof which use gradient flow for Busemann's functions, it is close in the spirit to the proof in [41].

16.9.2. Corollary. *Let $\mathcal{L} \in \mathrm{CBB}[0]$. Then there is an isometric splitting*

$$\mathcal{L} = \mathcal{L}' \oplus H$$

where $H \subset \mathcal{L}$ is a subset isometric to a Hilbert space and $\mathcal{L}' \subset \mathcal{L}$ be a convex subset which contains no lines.

16.9.3. Corollary. *Let $\mathcal{K} \in \mathrm{CBB}[0]$ be a cone and $v_+, v_- \in \mathcal{K}$ be a pair of opposite vectors (i.e., $v_+ + v_- = 0$). Then there is an isometry $\iota\colon K \to K' \times \mathbb{R}$, where $K' \in \mathrm{CBB}[0]$ is a cone with vertex o' such that $\iota : v_\pm \mapsto (o', \pm|v_\pm|)$.*

16.9.4. Corollary. *Assume $\mathcal{L} \in \mathrm{CBB}^m[1]$, $m \geqslant 2$ and $\operatorname{rad}\mathcal{L} = \pi$ then*

$$\mathcal{L} \overset{iso}{=\!=} \mathbb{S}^m.$$

Proof. Consider two Busemann's functions bus_+ and bus_- asociated with rays $\gamma : [0, \infty) \to \mathcal{L}$ and $\gamma : (-\infty, 0] \to \mathcal{L}$, i.e.

$$\mathrm{bus}_\pm(x) = \lim_{t\to\infty} |\gamma(\pm t)\ x| - t.$$

According to Exercise 8.4.2, both functions $\mathrm{bus}_\pm$ are concave.

Fix $x \in \mathcal{L}$. Note that since γ is a line, we have

$$\mathrm{bus}_+(x) + \mathrm{bus}_-(x) \geqslant 0.$$

On the other hand, by 8.4.1*b* $f(t) = |x*|^2 \circ \gamma(t)$ is 2-concave. In particular, $f(t) \leqslant t^2 + at + b$ for some constants $a, b \in \mathbb{R}$. Passing to the limit as $t \to \pm\infty$, we get

$$\mathrm{bus}_+(x) + \mathrm{bus}_-(x) \leqslant 0.$$

Hence

$$\mathrm{bus}_+(x) + \mathrm{bus}_-(x) = 0$$

for any $x \in \mathcal{L}$. In particular the functions $\mathrm{bus}_\pm$ are affine; that is they are convex and concave at the same time.

It follows that for any x,

$$\begin{aligned} |\nabla_x \mathrm{bus}_\pm| &= \sup\{\, \boldsymbol{d}_x \mathrm{bus}_\pm(\xi) \mid \xi \in \Sigma_x \,\} = \\ &= \sup\{\, -\boldsymbol{d}_x \mathrm{bus}_\mp(\xi) \mid \xi \in \Sigma_x \,\} \equiv \\ &\equiv 1. \end{aligned}$$

By Exercise 16.6.3, 1-Lipschitz curve α, such that $\mathrm{bus}_\pm(\alpha(t)) = t + c$ is a $\mathrm{bus}_\pm$-gradient curve. In particular, $\alpha(t)$ is a bus_+-gradient curve if and only if $\alpha(-t)$ is a bus_--gradient curve. It follows that for any $t > 0$, the $\mathrm{bus}_\pm$-gradient flows commute; that is

$$\mathfrak{d}^t_{\mathrm{bus}_+} \circ \mathfrak{d}^t_{\mathrm{bus}_-} = \mathrm{id}_{\mathcal{L}}\,.$$

Set

$$\mathfrak{d}^t = \begin{cases} \mathfrak{d}^t_{\mathrm{bus}_+} & \text{if } t \geqslant 0 \\ \mathfrak{d}^t_{\mathrm{bus}_-} & \text{if } t < 0 \end{cases}$$

it defines an $\mathbb{R}$-action on $\mathcal{L}$.

Set $\mathcal{L}' = \mathrm{bus}_+^{-1}(0) = \mathrm{bus}_-^{-1}(0) \subset \mathcal{L}$, it is a closed convex set, therefore it forms an Alexandrov's space. Consider map $h\colon \mathcal{L}' \times \mathbb{R} \to \mathcal{L}$ defined by $h\colon (x,t) \mapsto \mathfrak{d}^t(x)$. It is easy to see that h is onto. Applying distance estimate 16.7.1 for $\mathfrak{d}^t_{\mathrm{bus}_+}$ and $\mathfrak{d}^t_{\mathrm{bus}_-}$, we get that h is a short and non-contracting at the same time; i.e. h is an isometry. $\square$

16.10 Gradient curves of a family of functions

Here we extend notion of gradient curves to a family of functions.

Let $\mathcal{L} \in \mathrm{CBB}$. A subfunction $f\colon \mathbb{R} \times \mathcal{L} \looparrowright \mathbb{R}$, will be called *family of subfunctions*; in this case we prefer the notation $f\colon (t,p) \mapsto f_t(p)$. A

family of subfuncrions f_t is called *locally Lipschitz* if the corresponding subfunction $(t,p)\mapsto f_t(p)$ is locally Lipschitz.

Let $\lambda\colon\mathbb{R}\multimap\mathbb{R}\colon t\mapsto\lambda_t$ be a continuous subfunction. A family of subfunctions $f_t\colon\mathcal{L}\to\mathbb{R}$, will be called λ_t-*concave* and for each fixed t, the subfunction $f_t\colon\mathcal{L}\multimap\mathbb{R}$ is λ_t-concave.

Let $\mathbb{I}$ be a real inetrval, and $\alpha\colon\mathbb{I}\to\mathcal{L}$ be locally Lipschitz??? curve. We will write $\alpha^+(t)=\nabla f_t$ if for any $t\in\mathbb{I}$, the right tangent vector $\alpha^-(t)$ is defined and $\alpha^+(t)=\nabla_{\alpha(t)}f_t$. The same way we can consider equation $\alpha^-(t)=\nabla_{\alpha(t)}f_t$ for the left tangent vector. The solutions of $\alpha^+(t)=\nabla f_t$ will be also called f_t-gradient curves.

The following proposition extends the existence 16.8.4, the distance estimates 16.7.1, 16.7.2 and passage to the limit 16.8.3 to the gradient curves for a family of semiconcave functions. The proofs are straightforward, and left to the reader.

16.10.1. Proposition-definition. *Let $\mathcal{L}\in\mathrm{CBB}^m[\kappa]$, $\mathbb{I}$ be an open real interval, $\lambda\colon t\mapsto\lambda_t\colon\mathbb{I}\to\mathbb{R}$ be a continuous function and $f_t\colon\mathcal{L}\to\to\mathbb{R}$, $t\in\mathbb{I}$ be locally Lipschitz λ_t-concave family of subfunctions.*

Then for any $t^0\in\mathbb{I}$ and any $x^0\in\operatorname{Dom}f_t$ there is an f_t-gradient curve α which is defined in a neighborhood of t^0 and such that $\alpha(t^0)=x^0$. Moreover

a) If $\alpha,\beta\colon\mathbb{I}\to\mathcal{L}$ are f_t-gradient then for any $t^0,t^1\in\mathbb{I}$, $t^0\leqslant t^1$,

$$|\alpha(t^1)\ \beta(t^1)|\leqslant\pounds\cdot|\alpha(t^0)\ \beta(t^0)|,$$

where

$$\pounds=\exp\left(\int\limits_{t^0}^{t^1}\lambda(t)\cdot\boldsymbol{d}t\right).$$

b) Let $\varepsilon>0$ and $f_t,g_t\colon\mathcal{L}\to\mathbb{R}$ be two λ_t-concave locally Lipschitz family of function such that $|f-g|<\varepsilon$. Assume $\alpha,\beta\colon[0,t_{\max})\to\mathcal{L}$ are correspondingly f_t- and g_t-gradient curves such that $\alpha(0)=\beta(0)$. Then

$$|\alpha(t)\ \beta(t)|\leqslant???\sqrt{\tfrac{1}{2\cdot\varepsilon\cdot\lambda}\cdot\left(e^{\frac{t\cdot\lambda}{\varepsilon}}-1\right)}$$

for any $t\in[0,t_{\max})$. In particular, if $t_{\max}<\infty$, then

$$|\alpha(t)\ \beta(t)|\leqslant???c\cdot\sqrt{\varepsilon\cdot t}$$

for some constant $c=c(t_{\max},\lambda)$.

Moreover, the same conclusion holds for locally Lipschitz and λ_t-concave subfunctions $f_t,g_t\colon\mathcal{L}\multimap\mathbb{R}$ if for any $t\in\in[0,t_{\max})$ there is a geodesic $[\alpha(t)\,\beta(t)]$ in $\operatorname{Dom}f\cap\operatorname{Dom}g$.

c) ???convergence

16.11 Gradient curves for non-Lipschitz functions

In this book, we only consider gradient curves for locally Lipschitz semiconcave subfunctions; it turns out to be sufficient for all our needs. However, instead of Lipschitz semi-concave subfunctions, it is more natural to consider upper semi-continuous semi-concave functions with target in $[-\infty,\infty)$ and to assume in addition that the function take finite values at a dense set in the domain of definition. The set of such subfunctions on a CBB space $\mathcal{L}$ will be denoted as $\mathrm{LCC}(\mathcal{L})$ for **l**ower semi-**c**ontinous and semi-**c**oncave.

In this section we describe the adjustments needed to construct gradient curves in for the subfunctions in $\mathrm{LCC}(\mathcal{L})$.

There is one place where this type of functions appears; it is the entropy and some other closely related functionals on the Wasserstein space over a CBB[0] space. The gradient flow for these function play an important role in the theory of optimal transport, see [162] and references there in.

Differential. First we need to extend the definition of differential (**??**) to LCC subfunctions.

Let $\mathcal{L}\in\mathrm{CBB}$ and $f\in\mathrm{LCC}(\mathcal{L})$. Given a point $p\in\operatorname{Dom}f$, and a geodesic direction $\xi=\operatorname{dir}[pq]$, set $\hat{\boldsymbol{d}}_pf(\xi)=(f\circ\operatorname{geod}_{[pq]})^+(0)$. Since f is semiconcave the value $\hat{\boldsymbol{d}}_pf(\xi)$ is defined if $f\circ\operatorname{geod}_{[pq]}(t)$ is finite at all sufficiently small values $t>0$, but $\hat{\boldsymbol{d}}_pf(\xi)$ may take value ∞. Note that $\hat{\boldsymbol{d}}_pf$ is defined on a dense subset of Σ_p.

Set

$$\boldsymbol{d}_pf(\zeta)=\varlimsup_{\xi\to\zeta}\hat{\boldsymbol{d}}_pf(\xi)$$

and then set $\boldsymbol{d}_pf(v)=|v|\cdot\boldsymbol{d}_pf(\xi)$ if $v=|v|\cdot\xi$ for some $\xi\in\Sigma_p$.

In other words, we define differential as the smallest upper semi-continuous positive-homogeneous function $\boldsymbol{d}_pf\colon\mathrm{T}_p\to\mathbb{R}$ such that $\boldsymbol{d}_pf(\xi)\geqslant\hat{\boldsymbol{d}}_pf(\xi)$ if $\hat{\boldsymbol{d}}_pf(\xi)$ is defined.

Existence and uniqueness of the gradient. Note that in the proof of 16.1.2, we used Lipschitz condition just once, to show that

$$\begin{aligned}s&=\sup\{\,(\boldsymbol{d}_pf)(\xi)\mid\xi\in\Sigma_p\,\}=\\&=\varlimsup_{x\to p}\frac{f(x)-f(p)}{|xp|}<\\&<\infty.\end{aligned}$$

The value s above will be denoted as $|\nabla|_pf$. Note that if the gradient ∇_pf is defined then $|\nabla|_pf=|\nabla_pf|$ and otherwise $|\nabla|_pf=\infty$.

Summarizing the discussion above, we get the following.

16.1.2′ Existence and uniqueness of the gradient. *Assume* $\mathcal{L} \in \in \mathrm{CBB}[\kappa]$ *and* $f \in \mathrm{LCC}(\mathcal{L})$. *Then for any point* $p \in \operatorname{Dom} f$ *either there is unique gradient* $\nabla_p f \in \mathrm{T}_p$ *or* $|\nabla|_p f = \infty$

Further, in all the results of Section 16.2, we can only assume that $f \in \mathrm{LCC}(\mathcal{L})$ and its gradient is defined at the points in the consideration; the proofs are the same.

In sections 16.4–16.8 require almost no changes; mainly, where it is appropriate, one needs to exchange $|\nabla_p f|$ to $|\nabla|_p f$ and/or assume that the gradient is defined at the points of interest. Also one has to take ❶ in Theorem 16.5.2 as the definition of gradient-like curve. Then the theorem states that any gradient-like curve $\alpha\colon \mathbb{I} \to \mathcal{L}$ satisfies the definition 16.5.1 at $t \in \mathbb{I}$ if $\nabla_{\hat\alpha(s)} f$ is defined. Further in the Definition 16.6.1, should be changed to the following

16.6.1′. Definition. *Let* $\mathcal{L} \in \mathrm{CBB}$ *and* $f \in \mathrm{LCC}(\mathcal{L})$.

A curve $\alpha\colon [t_{\min}, t_{\max}) \to \operatorname{Dom} f$ *will be called* f-gradient curve *if*

$$\alpha^+(t) = \nabla_{\alpha(t)} f$$

if $\nabla_{\alpha(t)} f$ *is defined and*

$$(f \circ \alpha)^+(t) = \infty$$

otherwise.

In the proof of Local existence (16.8.2), the condition (ii) has to be exchanged to the following condition

(ii)′ $f \circ \hat\alpha_n(\bar\sigma_i) - f \circ \hat\alpha_n(\sigma_i) > (\bar\sigma_i - \sigma_i) \cdot \max\{n, |\nabla|_{\hat\alpha_n(\sigma_i)} f - \frac{1}{n})\}$.

Any gradient curve $\alpha[0, \ell) \to \mathcal{L}$ for a subfunction $f \in \mathrm{LCC}(\mathcal{L})$ satisfy the equation

$$\alpha^+(t) = \nabla_{\alpha(t)} f$$

at all values t with possible exception $t = 0$. In particular, the gradient of f is defined at all points of any f-gradient curve with the exception for the initial point.

16.12 Gradient flow

In this section we define gradient flow for semiconcave subfunctions and reformulate theorems obtained earlier in this chapter using this new terminology.

Let $\mathcal{L} \in \mathrm{CBB}$ and $f\colon \mathcal{L} \looparrowright \mathbb{R}$ be a locally Lipschitz semiconcave subfunctions. For any $t \geqslant 0$ write $\mathfrak{d}_f^t(x) = y$ if there is a f-gradient

curve α such that $\alpha(0) = x$ and $\alpha(t) = y$. The defined map $\mathfrak{d}_f^t\colon \mathcal{L} \to \mathcal{L}$ is called *f-gradient flow* for time t.

From 16.7.2, it follows that for any $t \geqslant 0$, the domain of definition of $\mathfrak{d}_f^t$ is open subset of $\mathcal{L}$; that is, $\mathfrak{d}_f^t$ is a submap. Moreover, if f is defined on whole $\mathcal{L}$ and $f'' + \mathrm{K}\cdot f \leqslant \lambda$ for some $\mathrm{K}, \lambda \in \mathbb{R}$, then according to 16.8.5, $\mathfrak{d}_f^t(x)$ is defined for all pairs $(x,t) \in \mathcal{L} \times \mathbb{R}_{\geqslant 0}$.

Clearly $\mathfrak{d}_f^{t_1+t_2} = \mathfrak{d}_f^{t_1} \circ \mathfrak{d}_f^{t_2}$; in other words gradient flow is given by an action of semigroup $(\mathbb{R}_{\geqslant 0}, +)$.

From the first distance estimate 16.7.1, we get the following.

16.12.1. Proposition. *Let $\mathcal{L} \in \mathrm{CBB}$ and $f\colon \mathcal{L} \to \mathbb{R}$ be a semiconcave functions. Then the map $x \mapsto \mathfrak{d}_f^t(x)$ is locally Lipschitz.*

Moreover, if f is λ-concave then $\mathfrak{d}_f^t$ is $e^{\lambda\cdot t}$-Lipschitz.

Next proposition states that gradient flow is stable under Gromov–Hausdorff convergence. It follows directly from the proposition on ultra limit of gradient curves 16.8.3.

16.12.2. Proposition. *If $\mathcal{L}_n \in \mathrm{CBB}^m[\kappa]$, $\mathcal{L}_n \xrightarrow{\tau} \mathcal{L}$, $f_n\colon \mathcal{L}_n \to \mathbb{R}$ is a sequence of λ-concave functions which converges to $f\colon \mathcal{L} \to \mathbb{R}$ then $\mathfrak{d}_{f_n}^t\colon \mathcal{L}_n \to \mathcal{L}_n$ converges to $\mathfrak{d}_f^t\colon \mathcal{L} \to \mathcal{L}$.*

On gradient flow for semiconcave functions on Riemannian manifolds. The gradient flow for general semiconcave functions on smooth Riemannian manifold can be introduced much cheaper. To do this note that the distance estimates proved in the Section 16.7 can be proved the same way for gradient curves of smooth semiconcave subfunctions. By Greene–Wu Lemma (see ???), given a λ-concave function f, a compact set $K \subset \operatorname{Dom} f$ and $\varepsilon > 0$ there is a smooth $(\lambda - \varepsilon)$-concave function which is ε-close to f in the C^0-topology on K. Hence one can apply the smoothing and pass to the limit as $\varepsilon \to 0$. Note that by the second distance estimate the obtained limit curve does not depend on the smoothing.

Chapter 17

Gradient exponent

17.1 Radial curves: definition

The radial curves are specially reparametrized gradient curves for distance functions. This parametrization makes them to behave as unit-speed geodesics in a natural comparison sense.

17.1.1. Definition. *Assume* $\mathcal{L} \in \mathrm{CBB}$, $\kappa \in \mathbb{R}$, *and* $p \in \mathcal{L}$. *A curve*

$$\sigma\colon [s_{\min}, s_{\max}) \to \mathcal{L}$$

is called (p,κ)-radial curve *if*

$$s_{\min} = |p\,\sigma(s_{\min})| \in (0, \tfrac{\varpi^\kappa}{2}),$$

and it satisfies the following differential equation

❶ $$\sigma^+(s) = \frac{\mathrm{tg}^\kappa |p\,\sigma(s)|}{\mathrm{tg}^\kappa s} \cdot \nabla_{\sigma(s)} |p\,*|.$$

for any $s \in [s_{\min}, s_{\max})$, *here* $\mathrm{tg}^\kappa x = \frac{\mathrm{sn}^\kappa x}{\mathrm{cs}^\kappa x}$.

If $x = \sigma(s_{\min})$, *we say that* σ starts in x.

Note that according to the definition $s_{\max} \leqslant \frac{\varpi^\kappa}{2}$.

Further you will see that the (p,κ)-radial curves work best for $\mathrm{CBB}[\kappa]$-spaces.

17.1.2. Definition. *Let* $\mathcal{L} \in \mathrm{CBB}$ *and* $p \in \mathcal{L}$. *A unit-speed geodesic* $\gamma\colon \mathbb{I} \to \mathcal{L}$ *is called* p-radial geodesic *if* $|p\,\gamma(s)| \equiv s$.

The proof of the following two propositions follows directly from the definitions:

17.1.3. Proposition. *Let $\mathcal{L} \in \mathrm{CBB}$ and $p \in \mathcal{L}$. Assume $\frac{\varpi^\kappa}{2} \geqslant s_{\max}$. Then any p-radial geodesic $\gamma\colon [s_{\min}, s_{\max}) \to \mathcal{L}$ is a (p,κ)-radial curve.*

17.1.4. Proposition. *Let $\mathcal{L} \in \mathrm{CBB}$, $p \in \mathcal{L}$ and $\sigma\colon [s_{\min}, s_{\max}) \to \mathcal{L}$ be a (p,κ)-radial curve. Then for any $s \in [s_{\min}, s_{\max})$, we have $|p\,\sigma(s)| \leqslant s$.*

Moreover, if for some s_0 we have $|p\,\sigma(s_0)| = s_0$ then the restriction $\sigma|[s_{\min}, s_0]$ is a p-radial geodesic.

17.1.5. Existence and uniqueness. *Let $\mathcal{L} \in \mathrm{CBB}$, $\kappa \in \mathbb{R}$, $p \in \mathcal{L}$, and $x \in \mathcal{L}$. Assume $0 < |px| < \frac{\varpi^\kappa}{2}$. Then there is unique (p,κ)-radial curve $\sigma\colon [|px|, \frac{\varpi^\kappa}{2}) \to \mathcal{L}$ which starts in x.*

Proof; existence. Set

$$\mathrm{itg}^\kappa\colon [0, \tfrac{\varpi^\kappa}{2}) \to \mathbb{R}, \quad \mathrm{itg}^\kappa(t) = \int\limits_0^t \mathrm{tg}^\kappa \mathfrak{t}\cdot \boldsymbol{d}\mathfrak{t}.$$

Clearly itg^κ is smooth and increasing. From 5.5.3 it follows that the composition

$$f = \mathrm{itg}^\kappa \circ |p_*|$$

is semiconcave in $\mathrm{B}(p, \frac{\varpi^\kappa}{2})$.

According to 16.8.2, there is an f-gradient curve $\alpha\colon [0, t_{\max}) \to \mathcal{L}$ defined on the maximal interval such that $\alpha(0) = x$.

Now consider solution of differential equation $\tau(t)$, $\tau' = (\mathrm{tg}^\kappa \tau)^2$ and $\tau(0) = r$. Note that $\tau(t)$ is also a gradient curve for function itg^κ defined on $[0, \frac{\varpi^\kappa}{2})$. Direct calculations show that composition $\alpha \circ \tau^{-1}$ forms an (p,κ)-radial curve.

Uniqueness. Assume σ^1, σ^2 be two (p,κ)-radial curves which starts in x. Then compositions $\sigma^i \circ \tau$ both give f-gradient curves. By Picard's theorem (16.8.1), we have $\sigma^1 \circ \tau \equiv \sigma^2 \circ \tau$. Therefore $\sigma^1(s) = \sigma^2(s)$ for any $s \geqslant r$ such that both sides are defined. $\square$

17.2 Radial comparisons

In this section we show that radial curves in some comparison sense behave as unit-speed geodesics.

17.2.1. Radial monotonicity. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$ and p, q be distinct points in $\mathcal{L}$. Assume $\sigma\colon [s_{\min}, \frac{\varpi^\kappa}{2}) \to \mathcal{L}$ is a (p,κ)-radial curve. Then the function*

$$s \mapsto \mathcal{Z}^\kappa\{|q\,\sigma(s)|; |pq|, s\}$$

is nonincreasing in all the domain of definition.

From Radial monotonicity, by straightforward calculations one gets the following.

17.2.2. Corollary. *Let $\kappa \leqslant 0$, $\mathcal{L} \in \mathrm{CBB}[\kappa]$ and $p, q \in \mathcal{L}$. Assume $\sigma\colon [s_{\min}, \frac{\varpi^\kappa}{2}) \to \mathcal{L}$ is a (p,κ)-radial curve. Then for any $w \geqslant 1$, the function*

$$s \mapsto \mathcal{Z}^\kappa\{|q\,\sigma(s)|; |pq|, w\cdot s\}$$

is non-increasing in whole domain of definition.

17.2.3. Radial comparison. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$ and $p \in \mathcal{L}$. Assume $\rho\colon [r_{\min}, \frac{\varpi^\kappa}{2}) \to \mathcal{L}$ and $\sigma\colon [s_{\min}, \frac{\varpi^\kappa}{2}) \to \mathcal{L}$ are two (p,κ)-radial curves. Set*

$$\varphi_{\min} = \mathcal{Z}^\kappa\left(p\,{}^{\rho(r_{\min})}_{\sigma(s_{\min})}\right).$$

Then for any $r \in [r_{\min}, \frac{\varpi^\kappa}{2})$ and $s \in [s_{\min}, \frac{\varpi^\kappa}{2})$, we have

$$\mathcal{Z}^\kappa\{|\rho(r)\ \sigma(s)|; r, s\} \leqslant \varphi_{\min},$$

or equivalently,

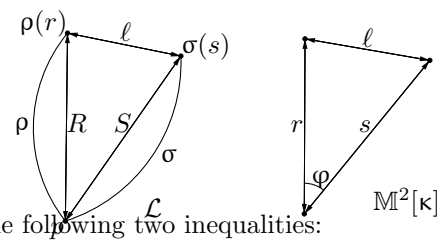
$$|\rho(r)\ \sigma(s)| \leqslant \tilde{\Upsilon}^\kappa\{\varphi_{\min}; r, s\}.$$

We prove both of the theorems simultaneously. The proof is an application of 16.2.3 plus trigonometric manipulations. We give a prove first in the simplest case $\kappa = 0$ and then the harder case $\kappa \neq 0$.

We proof case $\kappa = 0$ separately since it is easier to follow. The arguments for both cases are nearly the same, but the case $\kappa \neq 0$ require an extra idea. In fact once the case $\kappa \neq 0$ is proved, the case $\kappa = 0$ can be obtained by a limit procedure.

Proof of 17.2.1 and 17.2.3 in case $\kappa = 0$. Set

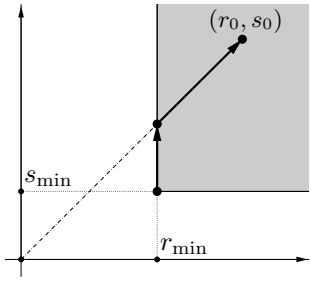
$$\begin{aligned} R &= R(r) = |p\,\rho(r)|,\\ S &= S(s) = |p\,\sigma(s)|,\\ \ell &= \ell(r,s) = |\rho(r)\ \sigma(s)|,\\ \varphi &= \varphi(r,s) = \mathcal{Z}^0\{\ell(r,s); r, s\}. \end{aligned}$$



It will be sufficient to prove the following two inequalities:

$$(*)^0_\varphi \qquad \frac{\partial^+\varphi}{\partial r}(s_{\min}, r) \leqslant 0, \qquad \frac{\partial^+\varphi}{\partial s}(s, r_{\min}) \leqslant 0$$

$$(**)^0_\varphi \qquad s\cdot\frac{\partial^+\varphi}{\partial s} + r\cdot\frac{\partial^+\varphi}{\partial r} \leqslant 0.$$



Once they are proved, the radial monotonicity follows from $(*)^0_\varphi$. The radial comparison follows from both $(*)^0_\varphi$ and $(**)^0_\varphi$. Indeed, one can connect $(s_{\min}, r_{\min})$ and (s_0, r_0) in $[s_{\min}, \infty) \times [r_{\min}, \infty)$ by a join of coordinate line and a segment defined by $r/s = r_0/s_0$. According to $(*)^0_\varphi$ and $(**)^0_\varphi$, the value of φ does not increase while pair (r, s) moving along this join. Thus $\varphi(r_0, s_0) \leqslant$ $\leqslant \varphi(r_{\min}, s_{\min}) = \varphi_{\min}$.

It remains to show $(*)^0_\varphi$ and $(**)^0_\varphi$. First let us rewrite the inequalities $(*)^0_\varphi$ and $(**)^0_\varphi$ in an equivalent form:

$$(*)^0_\ell \qquad \begin{aligned} &\frac{\partial^+}{\partial s}\ell(s, r_{\min}) \leqslant \cos \measuredangle^0\{r_{\min}; s, \ell\},\\ &\frac{\partial^+}{\partial r}\ell(s_{\min}, r) \leqslant \cos \measuredangle^0\{s_{\min}; r, \ell\},\end{aligned}$$

$$(**)^0_\ell \qquad s\cdot\frac{\partial^+\ell}{\partial s} + r\cdot\frac{\partial^+\ell}{\partial r} \leqslant s\cdot \cos \measuredangle^0\{r; s, \ell\} + r\cdot \cos \measuredangle^0\{s; r, \ell\} = \ell.$$

Set

$$(A)^0 \qquad f = \tfrac{1}{2}\cdot|p_*|^2.$$

Clearly f is 1-concave and

$$(B)^0 \qquad \rho^+(r) = \tfrac{1}{r}\cdot\nabla_{\rho(r)}f \quad\text{and}\quad \sigma^+(s) = \tfrac{1}{s}\cdot\nabla_{\sigma(s)}f.$$

Thus from 16.2.3, we have

$$(C)^0 \qquad \frac{\partial^+\ell}{\partial r} = -\tfrac{1}{r}\cdot\langle\nabla_{\rho(r)}f, \operatorname{dir}[\rho(r)\sigma(s)]\rangle \leqslant \frac{\ell^2 + R^2 - S^2}{2\cdot\ell\cdot r}.$$

Since $R(r) \leqslant r$ and $S(s_{\min}) = s_{\min}$, we get

$$(D)^0 \qquad \begin{aligned}\frac{\partial^+\ell}{\partial r}(r, s_{\min}) &\leqslant \frac{\ell^2 + r^2 - s_{\min}^2}{2\cdot\ell\cdot r} =\\ &= \cos \measuredangle^0\{s_{\min}; r, \ell\},\end{aligned}$$

which is the first inequality in $(*)^0_\ell$. By switching places of ρ and σ we obtain the second inequality in $(*)^0_\ell$. Further, summing together $(C)^0$ with its mirror-inequality for $\frac{\partial^+\ell}{\partial s}$, we get

$$(E)^0 \qquad r\cdot\frac{\partial^+\ell}{\partial r} + s\cdot\frac{\partial^+\ell}{\partial s} \leqslant \frac{\ell^2 + R^2 - S^2}{2\cdot\ell} + \frac{\ell^2 + S^2 - R^2}{2\cdot\ell} = \ell$$

which is $(**)^0_\ell$. $\square$

Proof of 17.2.1 and 17.2.3 in case $\kappa \neq 0$. Set as before

$$\begin{aligned} R &= R(r) = |p\,\rho(r)|, & \ell &= \ell(r,s) = |\rho(r)\ \sigma(s)| \\ S &= S(s) = |p\,\sigma(s)|, & \varphi &= \varphi(r,s) = \measuredangle^\kappa\{\ell(r,s); r, s\}. \end{aligned}$$

The statement follows from the following three inequalities:

$$(*)^\pm_\varphi \qquad \frac{\partial^+\varphi}{\partial r}(s_{\min}, r) \leqslant 0 \quad \frac{\partial^+\varphi}{\partial s}(s, r_{\min}) \leqslant 0$$

$$(**)^\pm_\varphi \qquad \operatorname{sn}^\kappa s\cdot\operatorname{cs}^\kappa S\cdot\frac{\partial^+\varphi}{\partial s} + \operatorname{sn}^\kappa r\cdot\operatorname{cs}^\kappa R\cdot\frac{\partial^+\varphi}{\partial r} \leqslant 0$$

Once they are proved, the radial monotonicity follows from $(*)^\pm_\varphi$. The radial comparison follows from both $(*)^0_\varphi$ and $(**)^\pm_\varphi$. Indeed, functions $s \mapsto \operatorname{sn}^\kappa s\cdot\operatorname{cs}^\kappa S$ and $r \mapsto \operatorname{sn}^\kappa r\cdot\operatorname{cs}^\kappa R$ are Lipschitz. Thus there is a solution for differential equation

$$(r', s') = (\operatorname{sn}^\kappa s\cdot\operatorname{cs}^\kappa S, \operatorname{sn}^\kappa r\cdot\operatorname{cs}^\kappa R)$$

with any initial data. $(r_0, s_0) \in [r_{\min}, \frac{\varpi^\kappa}{2}) \times [s_{\min}, \frac{\varpi^\kappa}{2})$. (Unlike case $\kappa = 0$ the solution can not be written explicitly.) Since $\operatorname{sn}^\kappa s\cdot\operatorname{cs}^\kappa S$, $\operatorname{sn}^\kappa r\cdot\operatorname{cs}^\kappa R > 0$, this solution $(r(t), s(t))$ must meet one of coordinate rays $\{r_{\min}\} \times [s_{\min}, \frac{\varpi^\kappa}{2})$ or $[r_{\min}, \frac{\varpi^\kappa}{2}) \times \{s_{\min}\}$. I.e., one can connect pair $(s_{\min}, r_{\min})$ to (s_0, r_0) by a join of coordinate line and the solution $(r(t), s(t))$. According to $(*)^\pm_\varphi$ and $(**)^\pm_\varphi$, the value of φ does not increase while pair (r, s) moving along this join. Thus $\varphi(r_0, s_0) \leqslant$ $\leqslant \varphi(r_{\min}, s_{\min})$.

As before we rewrite the inequalities $(*)^\pm_\varphi$ and $(**)^\pm_\varphi$ in terms of ℓ:

$$(*)^\pm_\ell \qquad \begin{aligned} &\frac{\partial^+\ell}{\partial s}(s, r_{\min}) \leqslant \cos\measuredangle^\kappa\{r_{\min}; s, \ell\}, \\ &\frac{\partial^+\ell}{\partial r}(s_{\min}, r) \leqslant \cos\measuredangle^\kappa\{s_{\min}; r, \ell\}, \end{aligned}$$

$$(**)^\pm_\ell \qquad \begin{aligned} &\operatorname{sn}^\kappa s\cdot\operatorname{cs}^\kappa S\cdot\frac{\partial^+\ell}{\partial s} + \operatorname{sn}^\kappa r\cdot\operatorname{cs}^\kappa R\cdot\frac{\partial^+\ell}{\partial r} \leqslant \\ &\qquad \leqslant \operatorname{sn}^\kappa s\cdot\operatorname{cs}^\kappa S\cdot\cos\measuredangle^\kappa\{r; s, \ell\} + \operatorname{sn}^\kappa r\cdot\operatorname{cs}^\kappa R\cdot\cos\measuredangle^\kappa\{s; r, \ell\} \end{aligned}$$

Further, set

$$(A)^\pm \qquad f = -\tfrac{1}{\kappa}\cdot\operatorname{cs}^\kappa \circ |p_*| = \operatorname{md}^\kappa \circ |p_*| - \tfrac{1}{\kappa}.$$

Clearly $f'' + \kappa\cdot f \leqslant 0$ and

$$(B)^{\pm} \qquad \begin{aligned} \rho^+(r) &= \frac{1}{\operatorname{tg}^\kappa r\cdot \operatorname{cs}^\kappa R}\cdot\nabla_{\rho(r)} f \\ \sigma^+(s) &= \frac{1}{\operatorname{tg}^\kappa s\cdot \operatorname{cs}^\kappa S}\cdot\nabla_{\sigma(s)} f. \end{aligned}$$

Thus from 16.2.3, we have

$$(C)^{\pm} \qquad \begin{aligned} \frac{\partial^+\ell}{\partial r} &= -\frac{1}{\operatorname{tg}^\kappa r\cdot \operatorname{cs}^\kappa R}\cdot\langle\nabla_{\rho(r)} f, \operatorname{dir}[\rho(r)\sigma(s)]\rangle \leqslant \\ &\leqslant \frac{1}{\operatorname{tg}^\kappa r\cdot \operatorname{cs}^\kappa R}\cdot\frac{\operatorname{cs}^\kappa S - \operatorname{cs}^\kappa R\cdot\operatorname{cs}^\kappa \ell}{\kappa\cdot\operatorname{sn}^\kappa \ell} = \\ &= \frac{\frac{\operatorname{cs}^\kappa S}{\operatorname{cs}^\kappa R} - \operatorname{cs}^\kappa \ell}{\kappa\cdot\operatorname{tg}^\kappa r\cdot\operatorname{sn}^\kappa \ell}. \end{aligned}$$

Note that for all $\kappa \neq 0$, the function $x \mapsto \frac{1}{\kappa\cdot\operatorname{cs}^\kappa x}$ is increasing. Thus, since $R(r) \leqslant r$ and $S(s_{\min}) = s_{\min}$, we get

$$(D)^{\pm} \qquad \begin{aligned} \frac{\partial^+\ell}{\partial r}(r, s_{\min}) &\leqslant \frac{\frac{\operatorname{cs}^\kappa s_{\min}}{\operatorname{cs}^\kappa r} - \operatorname{cs}^\kappa \ell}{\kappa\cdot\operatorname{tg}^\kappa r\cdot\operatorname{sn}^\kappa \ell} = \\ &= \frac{\operatorname{cs}^\kappa s_{\min} - \operatorname{cs}^\kappa \ell\cdot\operatorname{cs}^\kappa r}{\kappa\cdot\operatorname{sn}^\kappa r\cdot\operatorname{sn}^\kappa \ell} = \\ &= \cos \measuredangle^\kappa\{s_{\min}; r, \ell\}, \end{aligned}$$

which is the first inequality in $(*)^{\pm}_{\ell}$ for $\kappa \neq 0$. By switching places of ρ and σ we obtain the second inequality in $(*)^{\pm}_{\ell}$. Further, summing together $(C)^{\pm}$ with its mirror-inequality for $\frac{\partial^+\ell}{\partial s}$, we get

$$(E)^{\pm} \qquad \begin{aligned} \operatorname{sn}^\kappa r\cdot\operatorname{cs}^\kappa R\cdot\frac{\partial^+\ell}{\partial r} &+ \operatorname{sn}^\kappa s\cdot\operatorname{cs}^\kappa S\cdot\frac{\partial^+\ell}{\partial s} \leqslant \\ &\leqslant \frac{\operatorname{cs}^\kappa S\cdot\operatorname{cs}^\kappa r - \operatorname{cs}^\kappa \ell\cdot\operatorname{cs}^\kappa R\cdot\operatorname{cs}^\kappa r}{\kappa\cdot\operatorname{sn}^\kappa \ell} + \\ &\quad + \frac{\operatorname{cs}^\kappa R\cdot\operatorname{cs}^\kappa s - \operatorname{cs}^\kappa \ell\cdot\operatorname{cs}^\kappa S\cdot\operatorname{cs}^\kappa s}{\kappa\cdot\operatorname{sn}^\kappa \ell} = \\ &= \operatorname{sn}^\kappa r\cdot\operatorname{cs}^\kappa R\cdot\frac{\operatorname{cs}^\kappa s - \operatorname{cs}^\kappa \ell\cdot\operatorname{cs}^\kappa r}{\kappa\cdot\operatorname{sn}^\kappa r\cdot\operatorname{sn}^\kappa \ell} + \\ &\quad + \operatorname{sn}^\kappa s\cdot\operatorname{cs}^\kappa S\cdot\frac{\operatorname{cs}^\kappa r - \operatorname{cs}^\kappa \ell\cdot\operatorname{cs}^\kappa s}{\kappa\cdot\operatorname{sn}^\kappa s\cdot\operatorname{sn}^\kappa \ell} = \\ &= \operatorname{sn}^\kappa r\cdot\operatorname{cs}^\kappa R\cdot\cos \measuredangle^\kappa\{r; s, \ell\} + \\ &\quad + \operatorname{sn}^\kappa s\cdot\operatorname{cs}^\kappa S\cdot\cos \measuredangle^\kappa\{s; r, \ell\} \end{aligned}$$

which is $(**)^{\pm}_{\ell}$. $\square$

Analogs or radial curves for negative curvature bound. Let $\kappa \geqslant 0$. Assume that for some function ψ, the curves defined by equation

$$\sigma^+(s) = \psi(s, |p\,\sigma(s)|)\cdot\nabla_{\sigma(s)}|p_*|$$

satisfy radial comparison 17.2.3. Then in fact $\sigma(s)$ are radial curves; that is,

$$\psi(s, |p\,\sigma(s)|) = \frac{\mathrm{tg}^{\mathrm{K}}|p\,\sigma(s)|)}{\mathrm{tg}^{\mathrm{K}}s},$$

see exercise 17.5.6.

In case $\kappa < 0$, such ψ is not unique. In particular one can take curves defined by simpler equation

$$\sigma^+(s) = \frac{\mathrm{sn}^\kappa|p\,\sigma(s)|}{\mathrm{sn}^\kappa s}\cdot\nabla_{\sigma(s)}|p_*| = \frac{1}{\mathrm{sn}^\kappa s}\cdot\nabla_{\sigma(s)}(\mathrm{md}^\kappa\circ|p_*|).$$

Among all curves of that type, the radial curves for curvature κ as defined in 17.1.1 maximize the growth of $|p\,\sigma(s)|$.

17.3 Gradient exponent

Let $\mathcal{L}\in\mathrm{CBB}[\kappa]$, $p\in\mathcal{L}$ and $\xi\in\Sigma_p$. Consider a sequence of points $x_n\in$ $\in\mathcal{L}$ so that $\mathrm{dir}[px_n]\to\xi$. Set $r_n=|px_n|$; denote by $\sigma_n\colon[r_n,\frac{\varpi^\kappa}{2})\to\mathcal{L}$ be the (p,κ)-radial curve which starts in x_n.

Directly from radial comparison (17.2.3), we have that $\sigma_n\colon[r_n,\frac{\varpi^\kappa}{2})\to$ $\to\mathcal{L}$ converge to a curve $\sigma_\xi\colon(0,\frac{\varpi^\kappa}{2})\to\mathcal{L}$ and this limit is independent from the choice of the sequence x_n. Set $\sigma_\xi(0)=p$ and if $\kappa>0$ define

$$\sigma_\xi(\tfrac{\varpi^\kappa}{2}) = \lim_{t\to\frac{\varpi^\kappa}{2}}\sigma_\xi(t).$$

The obtained curve σ_ξ will be called (p,κ)-*radial curve in direction* ξ.

Let us define *gradient exponent* as

$$\mathrm{gexp}_p^\kappa\colon r\cdot\xi\mapsto\sigma_\xi(r)\colon\overline{\mathrm{B}}[o,\tfrac{\varpi^\kappa}{2}]\subset\mathrm{T}_p\to\mathcal{L}.$$

Here are properties of radial curves reformulated in terms of gradient exponent:

17.3.1. Theorem. *Let $\mathcal{L}\in\mathrm{CBB}[\kappa]$. Then*

a) If $p,q\in\mathcal{L}$ be points such that $|pq|\leqslant\frac{\varpi^\kappa}{2}$ then for any geodesic $[pq]$ in $\mathcal{L}$, we have

$$\mathrm{gexp}_p^\kappa(\log[pq]) = q$$

b) For any $v, w \in \overline{\mathrm{B}}[o, \frac{\varpi^\kappa}{2}] \subset \mathrm{T}_p$,

$$|\mathrm{gexp}_p^\kappa v \ \mathrm{gexp}_p^\kappa w| \leqslant \tilde{\Upsilon}^\kappa[o\,{}^v_w].$$

In other words, if we denote by $\mathcal{T}_p^\kappa$ *the set* $\overline{\mathrm{B}}[o, \frac{\varpi^\kappa}{2}] \subset \mathrm{T}_p$ *equipped with metric* $|v\,w|_{\mathcal{T}_p^\kappa} = \tilde{\Upsilon}^\kappa[o\,{}^v_w]$, *then*

$$\mathrm{gexp}_p^\kappa : \mathcal{T}_p^\kappa \to \mathcal{L}$$

is a short map.

c) Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$, $p, q \in \mathcal{L}$ *and* $|pq| \leqslant \frac{\varpi^\kappa}{2}$. *If* $v \in \mathrm{T}_p$, $|v| \leqslant 1$ *and*

$$\sigma(t) = \mathrm{gexp}_p^\kappa(t{\cdot}v)$$

then the function

$$s \mapsto \mathcal{Z}^\kappa(\sigma|_0^s, q) \overset{def}{=\!=} \mathcal{Z}^\kappa\{|q\,\sigma(s)|; |q\,\sigma(0)|, s\}$$

is non-increasing in whole domain of definition.

Proof. Follows directly from construction of gexp_p^κ and the radial comparison (17.2.3). □

Applying the theorem above together with 15.3.1*c*, we obtain the following.

17.3.2. Corollary. *Let* $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$, $p \in \mathcal{L}$ *and* $0 < R \leqslant \frac{\varpi^\kappa}{2}$. *Then there is a short map* $f\colon \overline{\mathrm{B}}[R]_{\mathbb{M}^m[\kappa]} \to \mathcal{L}$ *such that* $\operatorname{Im} f = \overline{\mathrm{B}}[p, R] \subset \mathcal{L}$.

17.4 Radial curves for sets

Here we generalize the constructions of radial curves and gradient exponent. Roughly, we show that one can use a distance function $|A_*|$ to any closed set A instead of the distant function to one point. We only give the corresponding definitions and state the results, the proofs are straightforward generalization of corresponding one-point-set version.

First we give a more general form of the definition of radial curves (17.1.1) and the definition of radial geodesic (17.1.2):

17.4.1. Definition. *Assume* $\mathcal{L} \in \mathrm{CBB}$, $\kappa \in \mathbb{R}$, *and* $A \subset \mathcal{L}$ *be a closed subset. A curve* $\sigma\colon [s_{\min}, s_{\max}) \to \mathcal{L}$ *is called* (A, κ)-radial curve *if* $s_{\min} = |A\,\sigma(s_{\min})| \in (0, \frac{\varpi^\kappa}{2})$, *and it satisfies the following differential equation*

$$\sigma^+(s) = \frac{\mathrm{tg}^\kappa|p\,\sigma(s)|}{\mathrm{tg}^\kappa s}{\cdot}\nabla_{\sigma(s)}|A_*|.$$

for any $s \in [s_{\min}, s_{\max})$, *here* $\operatorname{tg}^\kappa x = \frac{\operatorname{sn}^\kappa x}{\operatorname{cs}^\kappa x}$.

If $x = \sigma(s_{\min})$, *we say that* σ starts in x.

17.4.2. Definition. *Let* $\mathcal{L} \in \mathrm{CBB}$ *and* $A \subset \mathcal{L}$ *be a closed subset. A unit-speed geodesic* $\gamma\colon \mathbb{I} \to \mathcal{L}$ *is called* A-radial geodesic *if* $|A\gamma(s)| \equiv s$.

The following propositions are analogous to the propositions 17.1.3 and 17.1.4. Their proofs follow directly from the definitions:

17.4.3. Proposition. *Let* $\mathcal{L} \in \mathrm{CBB}$, $A \subset \mathcal{L}$ *be a closed subset. Assume that* $\frac{\varpi^\kappa}{2} \geqslant s_{\max}$. *Then any* $|A*|$*-radial geodesic* $\gamma\colon [s_{\min}, s_{\max}) \to \mathcal{L}$ *is an* (A,κ)*-radial curve.*

17.4.4. Proposition. *Let* $\mathcal{L} \in \mathrm{CBB}$, $A \subset \mathcal{L}$ *be a closed subset and* $\sigma\colon [s_{\min}, s_{\max}) \to \mathcal{L}$ *be a* (A,κ)*-radial curve. Then for any* $s \in [s_{\min}, s_{\max})$, *we have* $|A\,\sigma(s)| \leqslant s$.

Moreover, if for some s_0 *we have* $|A\,\sigma(s_0)| = s_0$ *then the restriction* $\sigma|_{[s_{\min}, s_0]}$ *is a* A*-radial geodesic.*

Here is the corresponding generalization of existence and uniqueness for (A,κ)-radial curves; it can be proved the same way as 17.1.5

17.4.5. Existence and uniqueness. *Let* $\mathcal{L} \in \mathrm{CBB}$, $\kappa \in \mathbb{R}$, $A \subset \mathcal{L}$ *be a closed subset, and* $x \in \mathcal{L}$. *Assume* $0 < |Ax| < \frac{\varpi^\kappa}{2}$. *Then there is unique* (A,κ)*-radial curve* $\sigma\colon [|Ax|, \frac{\varpi^\kappa}{2}) \to \mathcal{L}$ *which starts in* x.

Next we formulate radial monotonicity and radial comparison for (A,κ)-radial curves. The proof of these two statements are almost exactly the same as the proof of 17.2.1 and 17.2.3.

17.4.6. Radial monotonicity. *Let* $\mathcal{L} \in \mathrm{CBB}[\kappa]$, $A \subset \mathcal{L}$ *be a closed subset and* $q \in \mathcal{L}\backslash A$. *Assume* $\sigma\colon [s_{\min}, \frac{\varpi^\kappa}{2}) \to \mathcal{L}$ *is an* (A,κ)*-radial curve. Then the function*

$$s \mapsto \measuredangle^\kappa\{|q\,\sigma(s)|; |Aq|, s\}$$

is nonincreasing in all the domain of definition.

To formulate generalized radial comparison, we need to introduce a short cut notation. Given a set A and two points x and y in a metric space define

$$\measuredangle^\kappa\left(A\,{}^{x}_{y}\right) \overset{def}{=\!=} \measuredangle^\kappa\{|xy|; |Ax|, |Ay|\}$$

Note that distances $|xy|$, $|Ax|$ and $|Ay|$ might not satisfy the triangle inequality. Therefore the model angle $\measuredangle^\kappa\left(A\,{}^{x}_{y}\right)$ might be undefined even for $\kappa \leqslant 0$.

17.4.7. Radial comparison. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$ and $A \subset \mathcal{L}$ be a closed set. Assume $\rho\colon [r_{\min}, \frac{\varpi^\kappa}{2}) \to \mathcal{L}$ and $\sigma\colon [s_{\min}, \frac{\varpi^\kappa}{2}) \to \mathcal{L}$ are two $|A_*|$-radial curves for curvature κ. Assume further that*

$$\varphi_{\min} = \measuredangle^\kappa\left(A\,{}^{\rho(r_{\min})}_{\sigma(s_{\min})}\right)$$

is defined. Then for any $r \in [r_{\min}, \frac{\varpi^\kappa}{2})$ and $s \in [s_{\min}, \frac{\varpi^\kappa}{2})$, we have

$$|\rho(r)\ \sigma(s)| \leqslant \tilde{\Upsilon}^\kappa\{\varphi_{\min}; r, s\}.$$

Finally, assume p be an isolated point of a closed set A in $\mathcal{L} \in$ $\in \mathrm{CBB}$. Applying the same limiting procedure as in Section 17.3, for any $\xi \in \Sigma_p$ one can construct an (A, κ)-radial curve σ_ξ such that $\sigma_\xi(0) = p$ and $\sigma^+(0) = \xi$. This way we obtain a map $\mathrm{gexp}^\kappa_A\colon \mathrm{T}_p \looparrowright \mathcal{L}$, $r\cdot\xi \mapsto \sigma_\xi(r)$. For the constructed map, the following analog of 17.3.1 holds; the proof is straightforward.

17.4.8. Theorem. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$ and $A \subset \mathcal{L}$ is a closed subset with an isolated point $p \in A$. Then*

a) Assume $|Aq| = |pq| \leqslant \frac{\varpi^\kappa}{2}$ be an A-radial geodesic then

$$\mathrm{gexp}^\kappa_A(\log[pq]) = q$$

b) For any $v, w \in \overline{\mathrm{B}}[o, \frac{\varpi^\kappa}{2}] \subset \mathrm{T}_p$,

$$|\mathrm{gexp}^\kappa_p v\ \mathrm{gexp}^\kappa_p w| \leqslant \tilde{\Upsilon}^\kappa[o\,{}^v_w].$$

In other words, if we denote by $\mathcal{T}^\kappa_p$ the set $\overline{\mathrm{B}}[o, \frac{\varpi^\kappa}{2}] \subset \mathrm{T}_p$ equipped with metric $|vw|_{\mathcal{T}^\kappa_p} = \tilde{\Upsilon}^\kappa[o\,{}^v_w]$, then

$$\mathrm{gexp}^\kappa_p : \mathcal{T}^\kappa_p \to \mathcal{L}$$

is a short map.

c) Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$, $p, q \in \mathcal{L}$ and $|pq| \leqslant \frac{\varpi^\kappa}{2}$. If $v \in \mathrm{T}_p$, $|v| \leqslant 1$ and

$$\sigma(t) = \mathrm{gexp}^\kappa_p(t\cdot v)$$

then the function $s \mapsto \measuredangle^\kappa(\sigma|^s_0, q)$ is non-increasing in whole domain of definition.

17.5 Exercises

17.5.1. Exercise. *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$ and a, b, p are mutually distinct points in $\mathcal{L}$. Prove that*

$$(d_p|a_*|)(\nabla_p|b_*|) + \cos\measuredangle^\kappa(p\,{}^a_b) \leqslant 0.$$

17.5.2. Exercise. *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$, the function $f\colon \mathcal{L} \to \mathbb{R}$ be semiconcave and locally Lipschitz and $\alpha\colon \mathbb{I} \to \mathcal{L}$ be a Lipschitz curve. Show that*

$$\langle \nabla_{\alpha(t)} f, \alpha^+(t)\rangle = (\boldsymbol{d}_{\alpha(t)} f)(\alpha^+(t))$$

for almost all $t \in \mathbb{I}$.

17.5.3. Exercise. *Let $\mathcal{L} \in \mathrm{CBB}$ and $f\colon \mathcal{L} \to \mathbb{R}$ be a concave locally Lipschitz function. Show that $\alpha\colon \mathbb{R} \to \mathcal{L}$ is an f-gradient curve if and only if*

$$|x\,\alpha(t_1)|^2_{\mathcal{L}} - |x\,\alpha(t_0)|^2_{\mathcal{L}} \leqslant 2\cdot(t_1 - t_0)\cdot(f \circ \alpha(t_1) - f(x))$$

for any $t_1 > t_0$ and $x \in \mathcal{L}$.

17.5.4. Sharfutdinov's retruction. *Let K_t be a one parameter famity of convex sets in $\mathbb{E}^m$ which is nested; that is, $K_{t_1} \supset K_{t_2}$ if $t_1 \leqslant t_2$. Show that there is a family of short maps $\varphi_t\colon \mathbb{E}^m \to K_t$ such that $\varphi_t|K_t = \mathrm{id}$ and $\varphi_{t_2} \circ \varphi_{t_1} = \varphi_{t_2}$ for all $t_1 \leqslant t_2$.*

17.5.5. Exercise. *Let $\mathcal{L} \in \mathrm{CBB}^m$, $\partial\mathcal{L} = \varnothing$, $K \subset \mathcal{L}$ be a compact subset and $f\colon \mathcal{L} \to \mathbb{R}$ be semiconcave function. Assume that for some $t > 0$ the gradient flow $\mathfrak{d}^t_f$ is defined everywhere in K, prove that*

$$\partial_{\mathcal{L}} \mathfrak{d}^t_f K \subset \mathfrak{d}^t_f \partial K.$$

17.5.6. Exercise. *Let $\mathcal{L} \subset \mathbb{E}^2$ be Euclidean halfplane. Clearly $\mathcal{L} \in$ $\in \mathrm{CBB}^2[0]$. Given a point $x \in \mathbb{E}^2$ let us denote by $\mathrm{proj}(x)$ the closest point on $\mathcal{L}$.*

Apply the Radial comparison (17.2.3) to show that for any interior point $p \in \mathcal{L}$ we have

$$\mathrm{gexp}_p\, v = \mathrm{proj}(p + v).$$

17.5.7. Exercise. *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$ and $\mathrm{rad}\,\mathcal{L} = R$. Prove that there is a $(\mathrm{sn}^\kappa R)$-Lipschitz map $\Phi\colon \mathbb{S}^{m-1} \to \mathcal{L}$ such that $\mathrm{Im}\,\Phi \supset \partial\mathcal{L}$.*

17.5.8. Exercise. *Let $\mathcal{L} \in \mathrm{CBB}[\kappa]$ and $x, y, z \in \mathcal{L}$. Assume*

$$\measuredangle^\kappa(z\,{}^x_y) = \pi.$$

Show that there is a geodesic $[xy]$ which pass through z.

Compare to Exercise 10.7.2

Chapter 18

Extremal sets

Imagine that you want to move a heavy box inside an empty room by pushing it around. If the box is located in the middle of the room, you can push it in any direction. But once it is pushed against a wall you can not push it back to the center; and once it is pushed into a corner you cannot push it anywhere anymore. The same happens in Alexandrov spaces when one tries to move a point by pushing it along a gradient flow, but the role of walls and corners is played by extremal subsets.

Extremal subsets first appeared in the study of their special case — the boundary of an Alexandrov's space. They were introduced in [129], and were studied further in [134], [127].

The presence of extremal subsets in a CBB space makes it behave as something new, intersting and different from Riemannian manifolds. It gives an additional combinatoric structure which reflects geometry and topology of the space itself, as well as of nearby spaces.

18.1 New Def

Let $\mathcal{L} \in \mathrm{CBB}^m$ and $E \subset \mathcal{L}$ be a closed subset. Recall that tangent cone T_pE is defined in ???.

18.1.1. Definition. *Let $\mathcal{L} \in \mathrm{CBB}^m$. A closed subset $E \subset \mathcal{L}$ is called* extremal *if for any $p \in E$ and any $v \in \mathrm{T}_p$ there is $w \in \mathrm{T}_pE$ such that*

$$\langle v, x\rangle \geqslant \langle w, x\rangle$$

for any $x \in \mathrm{T}_p$.

18.1.2. Exercise. *Let $\mathcal{L} \in \mathrm{CBB}^m$ and $p \in \mathcal{L}$. Show that one-point set $\{p\}$ forms an extremal set if and only if* $\operatorname{diam}\Sigma_p \leqslant \frac{\pi}{2}$.

18.1.3. Lemma. *Let*

18.2 The definition

18.2.1. Definition. *Let $\mathcal{L} \in \mathrm{CBB}^m$. A closed subset $E \subset \mathcal{L}$ is called* extremal *if for any point $p \notin E$ the following condition fulfilled. If $q \in E$ is a point which minimize the distance to p then $\measuredangle[q\,{}^{p}_{x}] \leqslant \frac{\pi}{2}$ for any $x \in \mathcal{L}$. Equivalently, q is a critical point of the distance function $|p*|$, see Definition* **??**.

18.2.2. Proposition. *Let $\mathcal{L} \in \mathrm{CBB}^m$. Then given a point $p \in \mathcal{L}$ there is $\varepsilon > 0$ such that if $|Ep| < \varepsilon$ then $E \ni p$ for any extremal set $E \subset \mathcal{L}$.*

Proof. According to ???, there is $\varepsilon > 0$ such that all the values in $(0, \varepsilon)$ are regular for the distance function $|p*|$. It remains to apply the definition of extremal set. □

18.2.3. Proposition. *The union of arbitrary collection of extremal sets is extremal.*

Proof. Let $\{E_\alpha\}_{\alpha\in\mathcal{A}}$ be a collection of extremal sets and

$$E = \bigcup_\alpha E_\alpha.$$

By Proposition 18.2.2, E is closed.

Now assume $p \notin E$; and let $q \in E$ is a point which minimize the distance to p. Choose α so that $q \in E_\alpha$. Since E_α is extremal, the point q is critical for the distance function $|p*|$. Hence the result follows. □

18.2.4. Definition. *An extremal set E will be called* primitive *if it can not be presented as a union of extremal sets which are proper subsets of E.*

Examples. Note that according to the definition whole space and empty set are extremal sets.

Let $\mathcal{Q}$ be unit square in $\mathbb{E}^2$. Note that $\mathcal{Q} \in \mathrm{CBB}^2[0]$.

The empty set, whole $\mathcal{Q}$ as well as each vertex and each side of $\mathcal{Q}$ forms a primitive extremal subset. In addition the unions of these primitive extremal sets are also extremal. (The total number of extremal sets of square should be 47 and 10 of them are primitive.)

On the other hand, the regular pentagon $\mathcal{P}$ in $\mathbb{E}^2$ has only 3 extremal sets: $\varnothing$, $\partial\mathcal{P}$ and $\mathcal{P}$ and all these sets are primitive.

18.3 Extremal sets of the quotient spaces

Recall that quotient spaces were considered in Section 5.2. The following two theorems provide an important sourse of extremal subsets.

18.3.1. Theorem. *Let $\mathcal{L} \in \mathrm{CBB}^m$ and $G \curvearrowright \mathcal{L}$ is an isometric group action. Assume A is a connected component of the fixed point set of the action. Then* $\mathrm{proj}(E)$ *forms an extremal subset in $\mathcal{L}/G$.*

18.3.2. Theorem. *Let $\mathcal{L} \in \mathrm{CBB}^m$. Assume $G \curvearrowright \mathcal{L}$ is an isometric group action; and* $\mathrm{proj}\colon \mathcal{L} \to \mathcal{L}/G$ *is the projection to the orbit space.*

Then for any extremal set $E \subset \mathcal{L}$, its projection $\mathrm{proj}(E) \subset \mathcal{L}/G$ *forms an extremal set.*

18.4 Topology of Extremal sets

The following proposition gives yet big number of examples.

18.4.1. Proposition. *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$. Given a cone K consider set $S_K \subset \mathcal{L}$ of all points $p \in \mathcal{L}$ such that there is a homeomorphism $\mathrm{T}_p \to K$ which sends tip of T_p to the tip of K. Let Z be a path-connected components of S_K. Then the closure of Z forms an extremal subset.*

Note that according to ???, the set $S_K \subset \mathcal{L}$ can be also defined as set of all points such that its spherical neigborhood is homeomorphic to K.

Proof. Fix a path-connected component Z of S_K and let E be the closure of Z. Choose arbitrary point $p \notin E$ and let $q \in E$ be a point which minimize the distance $|pq|$.

Assume $\measuredangle[q\,{}^p_x] > \frac{\pi}{2}$; in other words the q is a regular point for the function $|p{*}|$. Then by Morse lemma (**??**), there is a neighborhood $\Omega \ni q$ which splits along $|p{*}|$; i.e., there is a homeomorphism $h\colon \mathbb{I} \times \times \Lambda \to \Omega$, where $\mathbb{I}$ is an open real inteval containing $|pq|$ such that $|ph(t,x)| = t$ for any $(t,x) \in \mathbb{I} \times \Lambda$.

Choose a point $q' = h(t_0, x_0) \in \Omega \cap Z$. Clearly the curve $t \mapsto h(t,x)$ belongs to Z. Choose $t_1 \in \mathbb{I}$ such that $t_1 < |pq|$ then $q'' = h(t_1, x_0) \in \in Z$ and $|pq''| < |pq|$, a contradiction. $\square$

18.5 Invariance with respect to gradient flow.

Extremal subsets can be defined as "ideals" of the gradient flow. Recall that $\mathfrak{d}_f^t$ denotes the f-gradient flow for time t, see Chapter **??**.

18.5.1. Definition. *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$. A subset $E \subseteq \mathcal{L}$ is an* extremal subset*, if for any native semiconcave function f on $\mathcal{L}$, $t \geqslant 0$ and $x \in E$, we have $\mathfrak{d}_f^t(x) \in E$.*

Note that according to the definition whole space and empty set are extremal subsets. Any other extremal subset of the space will be called *proper*.

18.5.2. Proposition. *Extremal subsets are closed.*

Moreover, if $\mathcal{L} \in \mathrm{CBB}^m$ then for any $p \in \mathcal{L}$ there is $\varepsilon > 0$ such that if an extremal set $E \subset \mathcal{L}$ intersects ε-neighborhood of p then $p \in E$.

18.5.3. Proposition. *Let $\mathcal{L} \in \mathrm{CBB}^m$ and $E \subset \mathcal{L}$ be an extremal subset. Then the length-metric of E is locally finite. More over, for any point $p \in E$ and $\varepsilon > 0$ there is $\delta > 0$ such that any point $q \in E \cap \cap \mathrm{B}(p, \delta)$ can be joint to p by a curve which lies in E and has length $< \varepsilon$.*

Both propositions above follow from the following lemma.

18.5.4. Lemma. *Let $\mathcal{L} \in \mathrm{CBB}^m$ and $p \in \mathcal{L}$. Then there is a native convex subfunction $f\colon \mathcal{L} \looparrowright \mathbb{R}$ such that $d_p f(\xi) < -1$ for any $\xi \in \Sigma_p$.*

Proof. Fix small $\varepsilon > 0$ (any $\varepsilon < \frac{\pi}{2}$ will do).

Choose a finite set of geodesic directions $\mathrm{dir}[pa_1], \mathrm{dir}[pa_2], \dots, \mathrm{dir}[pa_n] \in \in \Sigma_p$ such that for any direction $\xi \in \Sigma_p$ there is a direction ξ_i such that $\measuredangle(\xi, \mathrm{dir}[pa_i]) < \varepsilon$. Without loss of generality, we can assume a_i lie on the same distance from p, say $|pa_i| = r$ for all i and fixed $r > 0$.

Consider the function

$$f(x) = \min_i \{|a_i x|\}.$$

Choose $\varepsilon > 0$ so that the restriction $f|\mathrm{B}(p, \varepsilon)$ is λ-concave and ...

Let $x(t) = \mathfrak{d}_f^t x$. If $x(t) \neq p$ for some t then

$$f(p) - f(x(t)) \geqslant -\boldsymbol{d}_p f(\mathrm{dir}[px]) \cdot |px| - \lambda \cdot |px|^2 \geqslant ???$$

Therefore $(\boldsymbol{d}_x |p*|)(\nabla_x f) \leqslant ???$ if $0 < |px| < ???$.

If follows that if $x(t) = \mathfrak{d}_f^t$ then $\frac{\boldsymbol{d}}{\boldsymbol{d}t}|px(t)| \leqslant ???$. I.e., $\mathfrak{d}_f^t$ shrinks ε-neighborhood to p in time ???.

Note that f is a native semiconcave in a neighborhood of p. Moreover for some $d_p f(\xi) \leqslant -\delta$ for some fixed $\delta > 0$ and any $\xi \in \Sigma_p$.

Let us pass to a spherical neighborhood □

(i) An Alexandrov's space itself, as well as the empty set, forms an extremal subsets.

(ii) A point $p \in \mathcal{L}$ forms a one-point extremal subset if its space of directions Σ_p has a diameter $\leqslant \frac{\pi}{2}$

(iii) Let $\mathcal{L}/G$ be a factor of an Alexandrov's space by an isometry group and $S_H \subset \mathcal{L}$ be the set of points with stabilizer conjugate to a subgroup $H \subset G$ (or its connected component). Then the closure of the projection of S_H in $\mathcal{L}/G$ forms an extremal subset.
WHy???
For example: A cube can be presented as a quotient of a flat torus by a discrete isometry group, and each face of the cube forms an extremal subset.

The following theorem gives an equivalence of our definition of extremal subset and the definition given in [129]:

18.5.5. Theorem. *A closed subset E in an Alexandrov's space $\mathcal{L}$ is extremal if and only if for any $q \in \mathcal{L}\backslash E$, the following condition is fulfilled:*

If $|q_*|$ has a local minimum on E at a point p, then p is a critical point of $|q_*|$ on $\mathcal{L}$; i.e., $\nabla_p|q_*| = o$.

Proof. For the "only if" part, note that if $p \in E$ is not a critical point of $|q_*|$, then one can find a point $x \in \mathrm{Str}(p)$ (thus $\mathrm{dir}[px]$ is uniquely defined) such that

$$(\boldsymbol{d}_p|q_*|)(\mathrm{dir}[px]) > 0.$$

Since $\nabla_p|x_*|$ is polar to $\mathrm{dir}[px]$ (see ???) we get

$$(\boldsymbol{d}_p|q_*|)(\nabla_p|x_*|) < 0,$$

see ???. Hence, the gradient flow $\mathfrak{d}^t_{|x*|}$ pushes the point p closer to q, which contradicts the fact that p is a minimum point $|q_*|$ on E.

To prove the "if" part, it is enough to show that if $F \subset \mathcal{L}$ satisfies the condition of the theorem, then for any $p \in F$, and any semiconcave function f, either $\nabla_p f = o$ or $\frac{\nabla_p f}{|\nabla_p f|} \in \Sigma_p F$. If so, an f-gradient curve can be obtained as a limit of broken lines with vertexes on F, and from uniqueness, any gradient curve which starts at F lives in F.

Let us use induction on $\dim \mathcal{L}$. Note that if $F \subset \mathcal{L}$ satisfies the condition, then the same is true for $\Sigma_p F \subset \Sigma_p$, for any $p \in F$. Then using the inductive hypothesis we get that $\Sigma_p F \subset \Sigma_p$ is an extremal subset.

If p is isolated, then clearly $\operatorname{diam} \Sigma_p \leqslant \frac{\pi}{2}$ and therefore $\nabla_p f = o$, so we can assume $\Sigma_p F \neq \varnothing$.

Note that $\varphi = \boldsymbol{d}_p f|\Sigma_p$ satisfies $\varphi'' + \varphi \leqslant 0$. Take $\xi = \frac{\nabla_p f}{|\nabla_p f|}$, so $\xi \in \Sigma_p$ is the maximal point of φ. Let $\eta \in \Sigma_p F$ be a direction closest to ξ. Then $\measuredangle(\xi, \eta) \leqslant \frac{\pi}{2}$; otherwise F would not satisfy the condition

in the theorem for a point q with $\operatorname{dir}[pq] \approx \xi$. Hence, since $\Sigma_p F \subset \Sigma_p$ is an extremal subset, $\nabla_\eta \varphi \in \Sigma_\eta \Sigma_p F$ and therefore

$$\boldsymbol{d}_\eta \varphi(\operatorname{dir}[\eta\xi]) \leqslant \langle \nabla_\eta \varphi, \operatorname{dir}[\eta\xi]\rangle \leqslant 0.$$

Hence, $\varphi(\eta) \geqslant \varphi(\xi)$, and therefore $\xi = \eta$; i.e. $\frac{\nabla_p f}{|\nabla_p f|} \in \Sigma_p F$. □

From this theorem it follows that in the definition of extremal subset (18.5.1), one has to check only squares of distance functions. Namely:

18.5.6. Corollary. *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$. Then $E \subset \mathcal{L}$ is an extremal subset, if for any point $p \in \mathcal{L}$, and any $x \in E$, we have $\mathfrak{d}^t_{|p*|^2}(x) \in E$ for any $t \geqslant 0$.*

In particular, applying Lemma **??** we get

18.5.7. Lemma. *The limit of extremal subsets is an extremal subset.*

Namely, if $\mathcal{L}_n \in \mathrm{CBB}^m[\kappa]$, $\mathcal{L}_n \xrightarrow[\mathrm{GH}]{} \mathcal{L}$ and $E_n \subset \mathcal{L}_n$ is a sequence of extremal subsets such that $E_n \to E \subset \mathcal{L}$ then E is an extremal subset of $\mathcal{L}$.

The following is yet another important technical lemma:

18.5.8. Lemma. [129, 3.1(2)] *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$ be compact. Then there is $\varepsilon > 0$ such that $|E*|$ has no critical values in $(0, \varepsilon)$. Moreover,*

$$|\nabla_x |E*|| > \varepsilon \quad \text{if} \quad 0 < |Ex| < \varepsilon.$$

For a non-compact $\mathcal{L}$, the same holds for the restriction $|E*||\Omega$ to any bounded open $\Omega \subset \mathcal{L}$.

Proof. Follows from Lemma **??** and Theorem 18.5.5. □

18.6 Tangent cone

Note that the tangent cone $\mathrm{T}_p E$ of an extremal subset $E \subset \mathcal{L}$ is defined; i.e. for any $p \in E$, subsets $\mathfrak{z}E$ in $(\mathfrak{z}\mathcal{L}, p)$ converge to a subcone of $\mathrm{T}_p E \subset \mathrm{T}_p \mathcal{L}$ as $\mathfrak{z} \to \infty$. Indeed, assume $E \subset \mathcal{L}$ is an extremal subset and $p \in E$. For any $\xi \in \Sigma_p E$[1], the radial curve $\operatorname{gexp}(t\cdot\xi)$ lies in E.[2] In particular, there is a curve which goes in any tangent direction of E. Therefore, as $\mathfrak{z} \to \infty$, $(\mathfrak{z}E \subset \mathfrak{z}\mathcal{L}, p)$ converges to a subcone $\mathrm{T}_p E \subset \mathrm{T}_p \mathcal{L}$, which is simply cone over $\Sigma_p E$ (see also [129, 3.3])

[1]For a closed subset $X \subset \mathcal{L}$, and $p \in X$, $\Sigma_p X \subset \Sigma_p$ denotes the set of tangent directions to X at p, i.e. the set of limits of $\operatorname{dir}[pq_n]$ for $q_n \to p$, $q_n \in X$.

[2]that follows from the fact that the curves $t \mapsto \operatorname{gexp}(t\cdot\operatorname{dir}[pq_n])$ starting with q_n belong to E and their converge to $\operatorname{gexp}(t\cdot\xi)$

???The same argument shows that for any $p \in \mathcal{L}\backslash E$ the fuction $\inf_{\xi\in\Sigma_x E}\{\measuredangle(\mathrm{dir}[xp],\xi)\}$ is locally bounded away from zero.

Next we list some properties of tangent cones of extremal subsets, compare [129, 1.4]

18.6.1. Theorem. *A closed subset $E \subset \mathcal{L}$ is extremal if and only if at any point $p \in E$, its tangent cone $\mathrm{T}_pE \subset \mathrm{T}_p\mathcal{L}$ is defined, and it is an extremal subset of the tangent cone $\mathrm{T}_p\mathcal{L}$.*

Here is an equivalent formulation in terms of the space of directions:

18.6.2. Theorem. *For any $p \in E$, either (a) $\Sigma_pE = \varnothing$ and $\operatorname{diam}\Sigma_p \leqslant \frac{\pi}{2}$ or (b) $\Sigma_pE = \{\xi\}$ is one point extremal subset and $\overline{\mathrm{B}}[\xi,\frac{\pi}{2}] = \Sigma_p$ or (c) Σ_pE is extremal subset of Σ_p with at least two points.)*

Proof.??? T_pE is extremal as a limit of extremal subsets, see Lemma 18.5.7. On the other hand for any semiconcave function f and $p \in E$, the differential $\boldsymbol{d}_pf : \mathrm{T}_p \to \mathbb{R}$ is concave and since $\mathrm{T}_pE \subset \mathrm{T}_p$ is extremal we have $\nabla_pf \in \mathrm{T}_pE$. I.e. gradient curves can be approximated by broken geodesics with vertices on E, see page **??**. □

The following theorem was prove in [129, 3.4–5].

18.6.3. Theorem. *If E and F are extremal sets*

a) The set $E \cap F$ is extremal and

$$\mathrm{T}_p(E \cup F) = \mathrm{T}_pE \cup \mathrm{T}_pF$$

for any $p \in E \cap F$.

b) The set $E \cup F$ is extremal and

$$\mathrm{T}_p(E \cap F) = \mathrm{T}_pE \cap \mathrm{T}_pF$$

for any $p \in E \cup F$

c) The set $\overline{E\backslash F}$ is extremal and

$$\mathrm{T}_p(\overline{E\backslash F}) = \overline{\mathrm{T}_pE\backslash \mathrm{T}_pF}$$

for any $p \in \overline{E\backslash F}$.

In particular, if $\mathrm{T}_pE = \mathrm{T}_pF$ then E and F coincide in a neighborhood of p.

Proof.??? The properties (a) and (b) are obvious. The property (c) follows from Property 18.6.1 and Lemma 18.5.8. □

18.7 Branching coverings

Here is a partial converse to Theorem ???

18.7.1. Theorem. *Let $\mathcal{X}$ be a legth space with isometric action $\mathbb{Z}_2 \curvearrowright \mathcal{X}$. Assume that the quotient space $\mathcal{L} = \mathcal{X}/\mathbb{Z}_2$ is $\mathrm{CBB}^m[\kappa]$; denote by $E \subset \mathcal{L}$ the projection of fixed point set of $\mathbb{Z}_2 \curvearrowright \mathcal{X}$.*

Then $\mathrm{spc}X \in \mathrm{CBB}^m[\kappa]$ if and only if E is an extremal set in $\mathcal{L}$.

18.8 Intrisic geometry of extremal sets

1. Let $\mathcal{L}_n \in \mathrm{CBB}^m[\kappa]$, $\mathcal{L}_n \xrightarrow[\mathrm{GH}]{} \mathcal{L}$ without collapse (i.e. $\dim \mathcal{L} = m$) and $E_n \subset \mathcal{L}_n$ be extremal subsets. Assume $E_n \to E \subset \subset \mathcal{L}$ as subsets. Then

   (i) [82, 9.1] For all large n, there is a homeomorphism of pairs $(\mathcal{L}_n, E_n) \to (\mathcal{L}, E)$. In particular, for all large n, E_n is homeomorphic to E,

   (ii) [134, 1.2] $E_n \xrightarrow[\mathrm{GH}]{} E$ as metric spaces with the length-metrics from $\mathcal{L}_n$ and $\mathcal{L}$.

   The first property is a coproduct of the proof of Perelman's stability theorem. The proof of the second is an application of quasigeodesics.

Let us denote by $\mathrm{Ext}(x)$ the minimal extremal subset which contains a point $x \in \mathcal{L}$. Extremal subsets which can be obtained this way will be called *primitive*. Set

$$\mathrm{Ext}^\circ(x) = \{y \in \mathrm{Ext}(x) \mid \mathrm{Ext}(y) = \mathrm{Ext}(x)\};$$

let us call $\mathrm{Ext}^\circ(x)$ the *main part* of $\mathrm{Ext}(x)$. $\mathrm{Ext}^\circ(x)$ is the same as $\mathrm{Ext}(x)$ with its proper extremal subsets removed. From Property 18.6.3c on page 263, $\mathrm{Ext}^\circ(x)$ is open and everywhere dense in $\mathrm{Ext}(x)$. Clearly the main parts of primitive extremal subsets form a disjoint covering of M.

1. [129, 3.8] *Stratification.* The main part of a primitive extremal subset is a topological manifold. In particular, the main parts of primitive extremal subsets stratify Alexandrov's space into topological manifolds.
   This follows from Theorem 18.5.5 and the Morse lemma (property **??** page **??**); see also example **??**, page **??**.

18.8.1. Lieberman's Lemma. *Let $\mathcal{L} \in \mathrm{CBB}^m$ and $E \subset \mathcal{L}$ be extremal set. Then any unit-speed geodesic for the length-metric on E is a quasigeodesic in $\mathcal{L}$.*

???ADJUST THE PROOF TO THE NEW DEF OF QG???

Proof. Let $\gamma : \mathbb{I} \to E$ be unit-speed geodesic in an extremal subset $E \subset \mathcal{L}$ and f be a native λ-concave subfunction such that $\gamma \subset \operatorname{Dom} f$. Assume $f \circ \gamma$ is not λ-concave, then there is a non-negative Lipschitz function τ with support in (a, b) such that

$$\int_a^b [(f \circ \gamma)' \cdot \tau' + \lambda \cdot \tau] \cdot \boldsymbol{d}s < 0$$

Then as follows from Lemma **??**, for small $t \geqslant 0$

$$\gamma_t(s) = \mathfrak{d}_f^{t \cdot \tau(s)} \circ \gamma_0(s)$$

gives a length-contracting homotopy of curves relative to ends and according to Definition 18.5.1, it stays in E, a contradiction. □

The next theorem implies that extrinsic and length-metrics on extrimal subset are bi-Lipschitz equivalent.

18.8.2. Theorem [129, 3.2(3)]. *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$, $E \subset \mathcal{L}$ be an extremal set and $\Omega \subset \mathcal{L}$ be a bounded open subset.*

Then there is $\varepsilon > 0$ such that for any $x, y \in E \cap \Omega$, we have

$$|x\,y|_E \leqslant \tfrac{1}{\varepsilon} \cdot |x\,y|.$$

Moreover $\varepsilon = \varepsilon(m, \kappa, \operatorname{diam} \Omega, \operatorname{vol} \Omega)$.

Proof. ???From Lemma **??**, it follows that for two sufficiently close points $x, y \in E$ near p there is a point z so that $\langle \nabla_x |z\,*|, \operatorname{dir}[xy] \rangle > \varepsilon$ or $\langle \nabla_y |z\,*|, \operatorname{dir}[yx] \rangle > \varepsilon$. Then, for the corresponding point, say x, the gradient curve $t \to \mathfrak{d}^t_{|z\,*|}(x)$ lies in E, it is 1-Lipschitz and the distance $|\mathfrak{d}^t_{|z\,*|}(x)\ y|$ is decreasing with the speed of at least ε. Hence the result. □

18.8.3. Theorem. *Let $\mathcal{L}_n \in \mathrm{CBB}^m[\kappa]$ be a conerging non-collapsing sequence, $\mathcal{L}_n \xrightarrow[\mathrm{GH}]{} \mathcal{L}$ and $E_n \subset \mathcal{L}_n$ be a sequence extremal subsets which converges $E_n \to E$ as subsets.*

Equip subsets $E_n \subset \mathcal{L}_n$ and $E \subset \mathcal{L}$ with length-metrics, then $E_n \xrightarrow[\mathrm{GH}]{} E$.

We first prove that inner and external on extremal subsets are locally bi-Lipschtz:

Proof. Assume the contrary, then, after passing to a subsequence, one can choose two sequences p_n, $q_n \in E_n$, $p_n \to p \in E$ and $q_n \to q \in E$ such that

$$|p\,q|_F < \lim_{n \to \infty} |p_n\,q_n|_{F_n}.$$

Therefore we need to show only that $|pq|_F \geqslant \overline{\lim}_{n\to\infty} |p_n q_n|_{F_n}$. Set $\|pq\|_F = \overline{\lim}_{n\to\infty} |p_n q_n|_{F_n}$; this is easily seen to be a metric. From the previous paragraph, $\|pq\|$ does not depend on the choice of sequences (p_n) and (q_n), and we have $\|pq\| < \frac{1}{\varepsilon}\cdot|pq|$, because from above ε can be found uniformly for all $\mathcal{L}_n$ in the absence of collapse.

Let $\gamma\colon [a,b] \to E$ be a shortest path in E between p and q which is parametrized by arc-length. Assume $|pq|_F < \|pq\|_F$. Then, from [36, 5.14], for some $t_0 \in [a,b]$ and $\varepsilon > 0$ there is a sequence $t_i \to t_0\pm$ such that

$$\|\gamma(t_0)\ \gamma(t_i)\| \geqslant (1+\varepsilon)\cdot|t_i - t_0|.$$

Setting $r = \gamma(t_0)$ and $s = \gamma(t_i)$, take sequences r_n, $s_n \in E_n$ such that $r_n \to r$ and $s_n \to s$. Let γ_i in E be the limit curve to the shortest paths between r_n and s_n in E_n. By [130, 2.3(3)]??? and the generalized Lieberman lemma, γ_i is a quasigeodesic between $\gamma(t_0)$ and $\gamma(t_i)$. From above, $\operatorname{length}\gamma_i \geqslant (1+\varepsilon)\cdot|t_i - t_0|$. Now consider the limit $(\frac{1}{|t_0-t_i|}*\mathcal{L}, r) \to (\mathrm{T}_r, o)$. Consider the curve in T_r given by $\gamma^*(t) = \lim_{i\to\infty}\gamma_i(\frac{t-t_0}{t_1-t0})$, where ??? denotes the image of γ_i in $\frac{1}{|t_0-t_i|}\mathcal{L}$. Then γ^* is a quasigeodesic between o and the tangent vector $\gamma^+(t_0)$ which has length not less then $1+\varepsilon$. This is a contradiction since $|\gamma^\pm(t_0)| = 1$ by [Perelman and Petrunin 1994, 2.3(2)].

18.8.4. The first variation formula[134, 1.4]. *Assume $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$ and $E \subset \mathcal{L}$ is an extremal subset. Recall that $|{*}{*}|_E$ denotes its length-metric. Let $p, q \in E$ and $\alpha(t)$ be a curve in E starting from p in direction $\alpha^+(0) \in \Sigma_p E$. Then*

$$|\alpha(t)\ q|_E = |pq|_E - \cos\varphi\cdot t + o(t).$$

where φ is the minimal length-distance in $\Sigma_p E$ between $\alpha^+(0)$ and a direction of a shortest path in E from p to q (if $\varphi > \pi$, we assume $\cos\varphi = -1$).

Proof.??? □

18.9 Exercises

A CBB space is called *orientable* if its main part is an orientable manifold.

18.9.1. Exercise. *Show that for any $\mathcal{L} \in \mathrm{CBB}$ there is an orientable space $\bar{\mathcal{L}} \in \mathrm{CBB}$ without boundary and an involution $\iota\colon \bar{\mathcal{L}} \to \bar{\mathcal{L}}$ such that $\mathcal{L}$ is isometric to the quotient space $\bar{\mathcal{L}}/\iota$.*

Chapter 19

Webs and barycenters

The material of this chapter is used mostly for CBA spaces, although the results in section 19.1 find some applications for finite dimensional CBB spaces as well. The analogous material for CBB spaces is described in Chapter **??**.

19.1 The case of complete geodesic spaces

The following construction gives a $\bar{k}$-dimensional submanifold for a given "nondegenerate" array of $\bar{k}+1$ strongly convex functions. This construction was introduced by Kleiner in [87].

Webs. Let us introduce a partial order $\succcurlyeq$ on $\mathbb{R}^{\bar{k}+1}$.

19.1.1. Definition. *For two real atrrays $\boldsymbol{v}$, $\boldsymbol{w} \in \mathbb{R}^{\bar{k}+1}$, $\boldsymbol{v} = (v^0, v^1, \dots$ $\dots, v^{\bar{k}})$ and $\boldsymbol{w} = (w^0, w^1, \dots, w^{\bar{k}})$, we will write $\boldsymbol{v} \succcurlyeq \boldsymbol{w}$ if $v^i \geqslant w^i$ for each i.*

Given a subset $Q \subset \mathbb{R}^{\bar{k}+1}$, denote by $\operatorname{Up} Q$, the smallest upper set containing Q and $\operatorname{Min} Q$, the set of minimal elements of Q with respect to $\succcurlyeq$; i.e.,

$$\begin{aligned}\operatorname{Up} Q &= \left\{\, \boldsymbol{v} \in \mathbb{R}^{\bar{k}+1} \;\middle|\; \exists\, \boldsymbol{w} \in Q \text{ such that } \boldsymbol{v} \succcurlyeq \boldsymbol{w} \,\right\},\\ \operatorname{Min} Q &= \left\{\, \boldsymbol{v} \in Q \,\middle|\, \text{if } \boldsymbol{v} \succcurlyeq \boldsymbol{w} \in Q \text{ then } \boldsymbol{w} = \boldsymbol{v} \,\right\}.\end{aligned}$$

19.1.2. Definition. *Let $\mathcal{X}$ be a metric space and $\boldsymbol{f} = (f^0, f^1, \dots$ $\dots, f^{\bar{k}})\colon \mathcal{X} \to \mathbb{R}^{\bar{k}+1}$ be a function array. The set*

$$\operatorname{Web} \boldsymbol{f} \overset{def}{=\!=} \boldsymbol{f}^{-1}[\operatorname{Min} \boldsymbol{f}(\mathcal{X})] \subset \mathcal{X}$$

will be called the web *on $\boldsymbol{f}$.*

Given an array $\boldsymbol{f} = (f^0, f^1, \dots, f^k)$, we denote by $\boldsymbol{f}^{\ne i}$ the subarray of $\boldsymbol{f}$ with f^i removed; i.e.,

$$\boldsymbol{f}^{\ne i} \overset{def}{=\!=} (f^0, \dots, f^{i-1}, f^{i+1}, \dots, f^k).$$

Clearly $\operatorname{Web} \boldsymbol{f}^{\ne i} \subset \operatorname{Web} \boldsymbol{f}$. Define the *inner web* of $\boldsymbol{f}$ as

$$\operatorname{InWeb} \boldsymbol{f} = \operatorname{Web} \boldsymbol{f} \backslash \left(\bigcup_i \operatorname{Web} \boldsymbol{f}^{\ne i} \right).$$

We say that a function array is *nondegenerate* if $\operatorname{InWeb} \boldsymbol{f} \ne \varnothing$.

Example. If $\mathcal{X}$ is a geodesic space, then $\operatorname{Web}(|x*|, |y*|)$ is the union of all geodesics from x to y and $\operatorname{InWeb}(|x*|, |y*|) = \operatorname{Web}(|x*|, |y*|) \backslash \{x, y\}$.

Barycenters. Let us denote by $\Delta^k \subset \mathbb{R}^{k+1}$ the *standard k-simplex*; i.e. $\boldsymbol{x} = (x^0, x^1, \dots, x^k) \in \Delta^k$ if $\sum_{i=0}^k x^i = 1$ and $x^i \geqslant 0$ for all i.

Let $\mathcal{X}$ be a metric space and $\boldsymbol{f} = (f^0, f^1, \dots, f^k) \colon \mathcal{X} \to \mathbb{R}^{k+1}$ be a function array. Consider the map $\mathfrak{S}_{\boldsymbol{f}} \colon \Delta^k \to \mathcal{X}$, defined by

$$\mathfrak{S}_{\boldsymbol{f}}(\boldsymbol{x}) = \operatorname{MinPoint} \sum_{i=0}^k x^i \cdot f^i,$$

where $\operatorname{MinPoint} f$ denotes a point of minimum of f. The map $\mathfrak{S}_{\boldsymbol{f}}$ will be called a *barycentric simplex* of $\boldsymbol{f}$. Note that for general function array $\boldsymbol{f}$, the value $\mathfrak{S}_{\boldsymbol{f}}(\boldsymbol{x})$ might be undefined or nonuniquely defined.

It is clear from the definition, that $\mathfrak{S}_{\boldsymbol{f}^{\ne i}}$ coincides with the restriction of $\mathfrak{S}_{\boldsymbol{f}}$ to the corresponding face of Δ^k.

19.1.3. Theorem. *Let $\mathcal{X}$ be a complete geodesic space and $\boldsymbol{f} = (f^0, f^1, \dots, f^k) \colon \mathcal{X} \to \mathbb{R}^{k+1}$ be an array of strongly convex and locally Lipschitz functions. Then $\boldsymbol{f}$ defines a $C^{\frac{1}{2}}$-embedding $\operatorname{Web} \boldsymbol{f} \hookrightarrow \mathbb{R}^{k+1}$.*

Moreover,

a) $W = \operatorname{Up}[\boldsymbol{f}(\mathcal{X})]$ is a convex closed subset of $\mathbb{R}^{k+1}$ and $S = \partial_{\mathbb{R}^{k+1}} W$ is a convex hypersurface in $\mathbb{R}^{k+1}$.

b) $\boldsymbol{f}(\operatorname{Web} \boldsymbol{f}) = \operatorname{Min} W \subset S$ and $\boldsymbol{f}(\operatorname{InWeb} \boldsymbol{f}) = \operatorname{Int}_S(\operatorname{Min} W)$.

c) The barycentric simplex $\mathfrak{S}_{\boldsymbol{f}} \colon \Delta^k \to \mathcal{X}$ is a uniquely defined Lipshitz map and $\operatorname{Im} \mathfrak{S}_{\boldsymbol{f}} = \operatorname{Web} \boldsymbol{f}$.
In particular $\operatorname{Web} \boldsymbol{f}$ is compact.

d) Let us equip Δ^k with metric induced by ℓ^1-norm on $\mathbb{R}^{k+1}$. Then the Lipschitz constant of $\mathfrak{S}_{\boldsymbol{f}} \colon \Delta^k \to \mathcal{U}$ can be estimated in terms of positive lower bounds on $(f^i)''$ and Lipshitz constants of f^i in a neigborhood of $\operatorname{Web} \boldsymbol{f}$ for all i.

In particular, from (a) and (b), it follows that $\operatorname{InWeb}\boldsymbol{f}$ *is* $C^{\frac{1}{2}}$*-homeomorphic to an open set of* $\mathbb{R}^k$.

The proof is preceded by few preliminary statements.

19.1.4. Lemma. *Assume* $\mathcal{X}$ *is a complete geodesic space and let* $f\colon \mathcal{X} \to \mathbb{R}$ *be a locally Lipschitz, strongly convex function. Then the minimum point of* f *is uniquely defined.*

Proof. Without loss of generality, we can assume that f is 1-convex; in particular the following clim holds.

❶ *if* z *is a midpoint of* $[xy]$ *then*

$$s \leqslant f(z) \leqslant \tfrac{1}{2}\cdot f(x) + \tfrac{1}{2}\cdot f(y) - \tfrac{1}{8}\cdot |x\,y|^2.$$

Uniqueness. Assume that x and y are distinct minimum points of f. From ❶, we get

$$f(z) < f(x) = f(y),$$

a contradiction.

Existence. Fix a point $p \in \mathcal{X}$; let $\pounds \in \mathbb{R}$ be a Lipschitz constant of f in a neighborhood of p.

Consider function $\varphi(t) = f \circ \operatorname{geod}_{[px]}(t)$. Clearly φ is 1-convex and $\varphi^+(0) \geqslant -\pounds$. Setting $\ell = |px|$, we get

$$\begin{aligned} f(x) = \varphi(\ell) &\geqslant \\ &\geqslant f(p) - \pounds\cdot\ell + \tfrac{1}{2}\cdot\ell^2 \geqslant \\ &\geqslant f(p) - \tfrac{1}{2}\cdot\pounds^2. \end{aligned}$$

In particular,

$$s \overset{def}{=\!=} \inf\{\, f(x) \mid x \in \mathcal{X} \,\} \geqslant f(p) - \tfrac{1}{2}\cdot\pounds^2.$$

Choose a sequence of points $p_n \in \mathcal{X}$ such that $f(p_n) \to s$. Applying ❶, for $x = p_n$, $y = p_m$, we get that (p_n) is a Cauchy sequence. Hence, p_n converges to a minimum point of f. □

19.1.5. Definition. *Let* Q *be a closed subset of* $\mathbb{R}^{k+1}$. *A vector* $\boldsymbol{x} = (x^0, x^1, \dots, x^k) \in \mathbb{R}^{k+1}$ *is* subnormal *to* Q *at point* $\boldsymbol{v} \in \partial_{\mathbb{R}^{k+1}} Q$ *if*

$$\langle \boldsymbol{x}, \boldsymbol{w} - \boldsymbol{v} \rangle \overset{def}{=\!=} \sum_i x^i\cdot(w^i - v^i) \geqslant 0$$

for any $\boldsymbol{w} \in Q$.

19.1.6. Lemma. *Let* $\mathcal{X}$ *be a complete geodesic space and* $\boldsymbol{f} = (f^0, f^1, \dots$ $\dots, f^k)\colon \mathcal{X} \to \mathbb{R}^{k+1}$ *be an array of strongly convex and locally Lipschitz functions. Set* $W = \operatorname{Up}\boldsymbol{f}(\mathcal{X})$. *Then:*

a) *W is a closed convex set which is bounded below with respect to $\succcurlyeq$.*
b) *If $\boldsymbol{x}$ is a subnormal vector to W, then $\boldsymbol{x} \succcurlyeq \mathbf{0}$.*
c) *$S = \partial_{\mathbb{R}^{k+1}} W$ is a complete convex hypersurface in $\mathbb{R}^{k+1}$.*

Proof. Denote by $\bar{W}$ the closure of W.

Convexity of f^i implies that for any two points $p, q \in \mathcal{X}$ and $t \in$ $\in [0,1]$ we have

❷ $$(1-t)\cdot \boldsymbol{f}(p) + t\cdot \boldsymbol{f}(q) \succcurlyeq \boldsymbol{f} \circ \operatorname{path}_{[pq]}(t).$$

where $\operatorname{path}_{[pq]}$ denotes a geodesic path from p to q. Therefore W as well as $\bar{W}$ are convex sets in $\mathbb{R}^{k+1}$.

Set
$$w^i = \min\left\{\, f^i(x) \;\middle|\; x \in \mathcal{X} \,\right\}.$$
By Lemma 19.1.4, w^i is finite for each i. Clearly $\boldsymbol{w} = (w^0, w^1, \dots, w^k)$ is a lower bound of $\bar{W}$ with respect to $\succcurlyeq$.

It is clear that W has nonempty interior and $W \neq \mathbb{R}^{k+1}$ since it is bounded below. Therefore $S = \partial_{\mathbb{R}^{k+1}} W = \partial_{\mathbb{R}^{k+1}} \bar{W}$ is a complete convex hypersurface in $\mathbb{R}^{k+1}$.

Since $\bar{W}$ is closed and bounded below, we also have

❸ $$\bar{W} = \operatorname{Up}[\operatorname{Min} \bar{W}].$$

Choose arbitrary $\boldsymbol{v} \in S$. Let $\boldsymbol{x} \in \mathbb{R}^{k+1}$ be a subnormal vector to $\bar{W}$ at $\boldsymbol{v}$. In particular, $\langle \boldsymbol{x}, \boldsymbol{y} \rangle \geqslant 0$ for any $\boldsymbol{y} \succcurlyeq \mathbf{0}$; i.e. $\boldsymbol{x} \succcurlyeq \mathbf{0}$.

Further, according to Lemma 19.1.4, the function $\sum_i x^i \cdot f^i$ has uniquely defined minimum point, say p. Clearly

❹ $$\boldsymbol{v} \succcurlyeq \boldsymbol{f}(p) \quad \text{and} \quad \boldsymbol{f}(p) \in \operatorname{Min} W.$$

Note that for any $\boldsymbol{u} \in \bar{W}$ there is $\boldsymbol{v} \in S$ such that $\boldsymbol{u} \succcurlyeq \boldsymbol{v}$. Therefore ❹ implies that
$$\bar{W} \subset \operatorname{Up}[\operatorname{Min} W] \subset W.$$
Hence $\bar{W} = W$; i.e. W is closed. □

Proof of Theorem 19.1.3. Without loss of generality we may assume that all f^i are 1-convex.

Given $\boldsymbol{v} = (v^0, v^1, \dots, v^k) \in \mathbb{R}^{k+1}$, consider the function $h_{\boldsymbol{v}} \colon \mathcal{X} \to$ $\to \mathbb{R}$ defined as
$$h_{\boldsymbol{v}}(p) = \max_i \{ f^i(p) - v^i \}.$$
Note that $h_{\boldsymbol{v}}$ is 1-convex. Set
$$\Phi(\boldsymbol{v}) \overset{def}{=\!=} \operatorname{MinPoint} h_{\boldsymbol{v}}.$$

According to Lemma 19.1.4, $\Phi(\boldsymbol{v})$ is uniquely defined.

From the definition of web (19.1.2) we have $\Phi \circ \boldsymbol{f}(p) = p$ for any $p \in \operatorname{Web} \boldsymbol{f}$; i.e., Φ is a left inverse to the restriction $\boldsymbol{f}|\operatorname{Web}\boldsymbol{f}$. In particular,

❺ $$\operatorname{Web}\boldsymbol{f} = \operatorname{Im}\Phi.$$

Given $\boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^{k+1}$, set $p = \Phi(\boldsymbol{v})$ and $q = \Phi(\boldsymbol{w})$. Since $h_{\boldsymbol{v}}$ and $h_{\boldsymbol{w}}$ are 1-convex, we have

$$h_{\boldsymbol{v}}(q) \geqslant h_{\boldsymbol{v}}(p) + \tfrac{1}{2}\cdot|pq|^2, \qquad h_{\boldsymbol{w}}(p) \geqslant h_{\boldsymbol{w}}(q) + \tfrac{1}{2}\cdot|pq|^2.$$

Therefore,

$$\begin{aligned}|pq|^2 &\leqslant 2\cdot \sup_{x\in\mathcal{X}}\{|h_{\boldsymbol{v}}(x) - h_{\boldsymbol{w}}(x)|\} \leqslant \\ &\leqslant 2\cdot \max_i\{|v^i - w^i|\}.\end{aligned}$$

In particular, Φ is $C^{\frac{1}{2}}$-continuous, or $\boldsymbol{f}|\operatorname{Web}\boldsymbol{f}$ is a $C^{\frac{1}{2}}$-embedding.

As in Lemma 19.1.6, set $W = \operatorname{Up}\boldsymbol{f}(\mathcal{X})$ and $S = \partial_{\mathbb{R}^{k+1}} W$. Recall that S is a convex hypersurface in $\mathbb{R}^{k+1}$. Clearly $\boldsymbol{f}(\operatorname{Web}\boldsymbol{f}) = \operatorname{Min} W \subset S$. From the definition of inner web we have $\boldsymbol{v} \in \boldsymbol{f}(\operatorname{InWeb}\boldsymbol{f})$ if and only if $\boldsymbol{v} \in S$ and for any i there is $\boldsymbol{w} = (w^0, w^1, \dots, w^k) \in W$ such that $w^j < v^j$ for all $j \neq i$. Thus, $\boldsymbol{f}(\operatorname{InWeb}\boldsymbol{f})$ is open in S. I.e., $\operatorname{InWeb}\boldsymbol{f}$ is $C^{\frac{1}{2}}$-homeomorphic to an open set in a convex hypersurface $S \subset \mathbb{R}^{k+1}$.

(c)+(d). Since f^i is 1-convex, for any $\boldsymbol{x} = (x^0, x^1, \dots, x^k) \in \Delta^k$, the convex combination

$$\left(\sum_i x^i\cdot f^i\right) : \mathcal{X} \to \mathbb{R}$$

is also 1-convex. Therefore, according to Lemma 19.1.4, $\mathfrak{S}_{\boldsymbol{f}}(\boldsymbol{x})$ is defined for any $\boldsymbol{x} \in \Delta^k$.

For $\boldsymbol{x}, \boldsymbol{y} \in \Delta^k$, set

$$\begin{aligned} f_{\boldsymbol{x}} &= \sum_i x^i\cdot f^i, & f_{\boldsymbol{y}} &= \sum_i y^i\cdot f^i,\\ p &= \mathfrak{S}_{\boldsymbol{f}}(\boldsymbol{x}), & q &= \mathfrak{S}_{\boldsymbol{f}}(\boldsymbol{y}),\\ \ell &= |pq| \end{aligned}$$

Note the following:

- ⋄ The function $\varphi(t) = f_{\boldsymbol{x}} \circ \operatorname{geod}_{[pq]}(t)$ has minimum at 0. Therefore $\varphi^+(0) \geqslant 0$

◇ The function $\psi(t) = f_{\boldsymbol{y}} \circ \operatorname{geod}_{[pq]}(t)$ has minimum at ℓ. Therefore $\psi^-(\ell) \geqslant 0$.

From 1-convexity of $f_{\boldsymbol{y}}$, we have $\psi^+(0) + \psi^-(\ell) + \ell \leqslant 0$.

Let $\pounds$ be a Lipschitz constant for all f^i in a neighborhood $\Omega \ni p$. Then

$$\psi^+(0) \leqslant \varphi^+(0) + \pounds \cdot \|\boldsymbol{x} - \boldsymbol{y}\|_1,$$

where $\|\boldsymbol{x}-\boldsymbol{y}\|_1 = \sum_{i=0}^{k} |x^i - y^i|$. I.e., given $\boldsymbol{x} \in \Delta^k$, there is a constant $\pounds$ such that

$$\begin{aligned} |\mathfrak{S}_{\boldsymbol{f}}(\boldsymbol{x})\ \mathfrak{S}_{\boldsymbol{f}}(\boldsymbol{y})| &= \ell \leqslant \\ &\leqslant \pounds \cdot \|\boldsymbol{x} - \boldsymbol{y}\|_1 \end{aligned}$$

for any $\boldsymbol{y} \in \Delta^k$. In particular, there is $\varepsilon > 0$ such that if $\|\boldsymbol{x} - \boldsymbol{y}\|_1$, $\|\boldsymbol{x} - \boldsymbol{z}\|_1 < \varepsilon$, then $\mathfrak{S}_{\boldsymbol{f}}(\boldsymbol{y})$, $\mathfrak{S}_{\boldsymbol{f}}(\boldsymbol{z}) \in \Omega$. Thus, the same argument as above implies

$$|\mathfrak{S}_{\boldsymbol{f}}(\boldsymbol{y})\ \mathfrak{S}_{\boldsymbol{f}}(\boldsymbol{z})| = \ell \leqslant \pounds \cdot \|\boldsymbol{y} - \boldsymbol{z}\|_1$$

for any $\boldsymbol{y}$ and $\boldsymbol{z}$ sufficiently close to $\boldsymbol{x}$, i.e., $\mathfrak{S}_{\boldsymbol{f}}$ is locally Lipschitz. Since Δ^k is compact, $\mathfrak{S}_{\boldsymbol{f}}$ is Lipschitz.

Clearly $\mathfrak{S}_{\boldsymbol{f}}(\Delta^k) \subset \operatorname{Web} \boldsymbol{f}$; it remains to show that $\mathfrak{S}_{\boldsymbol{f}}(\Delta^k) \supset \supset \operatorname{Web} \boldsymbol{f}$. According to Lemma 19.1.6, $W = \operatorname{Up} \boldsymbol{f}(\mathcal{X})$ is a closed convex set in $\mathbb{R}^{k+1}$. Let $p \in \operatorname{Web} \boldsymbol{f}$; clearly $\boldsymbol{f}(p) \in \operatorname{Min} W \subset S = \partial_{\mathbb{R}^{k+1}} W$. Let $\boldsymbol{x}$ be a subnormal vector to W at $\boldsymbol{f}(p)$. According to Lemma 19.1.6, $\boldsymbol{x} \succcurlyeq \boldsymbol{0}$. Without loss of generality we may assume that $\sum_i x^i = 1$; i.e. $\boldsymbol{x} \in \Delta^k$. By Lemma 19.1.4, p is the unique minimum point of $\sum_i x^i \cdot f^i$; i.e., $p = \mathfrak{S}_{\boldsymbol{f}}(\boldsymbol{x})$. □

19.2 The case of CAT spaces

Let $\boldsymbol{a} = (a^0, a^1, \dots, a^k)$ be a point array in a metric space $\mathcal{U}$. Recall that $|\boldsymbol{a}_*|$ denotes the distance map $(|a^0{}_*|, |a^1{}_*|, \dots, |a^k{}_*|) \colon \mathcal{U} \to \to \mathbb{R}^{k+1}$ which can be also regarded as a function array. The *radius of the point array* $\boldsymbol{a}$ is defined to be the radius of the set $\{a^0, a^1, \dots, a^k\}$; i.e.,

$$\operatorname{rad} \boldsymbol{a} = \inf \left\{ r > 0 \ \middle|\ a^i \in \mathrm{B}(z, r) \text{ for all } i \text{ and some } z \in \mathcal{U} \right\}.$$

Fix $\kappa \in \mathbb{R}$. Let $\boldsymbol{a} = (a^0, a^1, \dots, a^k)$ be a point array of radius $< \frac{\varpi^\kappa}{2}$ in a metric space $\mathcal{U}$. Consider function array $\boldsymbol{f} = (f^0, f^1, \dots \dots, f^k)$ where

$$f^i(x) = \operatorname{md}^\kappa |a^i x|.$$

Assuming the barycentric simplex $\mathfrak{S}_{\boldsymbol{f}}$ is defined, then $\mathfrak{S}_{\boldsymbol{f}}$ is called the κ-*barycentric simplex* for the point array $\boldsymbol{a}$; it will be denoted by $\mathfrak{S}^{\kappa}_{\boldsymbol{a}}$. The points $a^0, a^1, \dots, a^{\bar{k}}$ are called *vertexes of* κ-*barycentric simplex*. Note that once we say the κ-barycentric simplex is defined, we automatically assume that $\operatorname{rad} \boldsymbol{a} < \frac{\varpi^{\kappa}}{2}$.

19.2.1. Theorem. *Let $\mathcal{U} \in \mathrm{CBA}[\kappa]$ and $\boldsymbol{a} = (a^0, a^1, \dots a^{\bar{k}})$ be a point array with radius $< \frac{\varpi^{\kappa}}{2}$. Then:*

a) The κ-barycentric simplex $\mathfrak{S}^{\kappa}_{\boldsymbol{a}} \colon \Delta^{\bar{k}} \to \mathcal{U}$ is defined. Moreover, $\mathfrak{S}^{\kappa}_{\boldsymbol{a}}$ is a Lipschitz map, and if $\Delta^{\bar{k}}$ is equipped with the ℓ^1-metric then its Lipschitz contant can be estimated in terms of κ and the radius of $\boldsymbol{a}$ (in particular it does not depend on $\bar{k}$).

b) $\operatorname{Web}|\boldsymbol{a}_*| = \operatorname{Im} \mathfrak{S}^{\kappa}_{\boldsymbol{a}}$. Moreover, if a closed convex set K contains all a^i then $\operatorname{Web}|\boldsymbol{a}_*| \subset K$.

c) The restriction[1] $|\boldsymbol{a}^{\neq 0}{}_*|\,|\operatorname{InWeb}|\boldsymbol{a}_*|$ is an open $C^{\frac{1}{2}}$-embedding in $\mathbb{R}^{\bar{k}}$. Thus there is an inverse of $|\boldsymbol{a}^{\neq 0}{}_*|\,|\operatorname{InWeb}|\boldsymbol{a}_*|$, say $\Phi \colon \mathbb{R}^{\bar{k}} \looparrowright \mathcal{U}$.

The subfunction $f = |a^0{}_*| \circ \Phi$ is semiconvex and locally Lipschitz. Moreover, if $\kappa \leqslant 0$, then f is convex.

In particular, $\operatorname{Web}|\boldsymbol{a}_*|$ is compact and $\operatorname{InWeb}|\boldsymbol{a}_*|$ is $C^{\frac{1}{2}}$-homeomorphic to an open subset of $\mathbb{R}^{\bar{k}}$.

19.2.2. Definition. *The submap $\Phi \colon \mathbb{R}^{\bar{k}} \looparrowright \mathcal{X}$ as in Theorem 19.2.1c will be called the* $|\boldsymbol{a}_*|$-web embedding *with* brace $|a^0{}_*|$.

19.2.3. Definition. *Let $\mathcal{U} \in \mathrm{CBA}[\kappa]$ and $\boldsymbol{a} = (a^0, a^1, \dots a^{\bar{k}})$ be a point array with radius $< \frac{\varpi^{\kappa}}{2}$. If $\operatorname{InWeb}|\boldsymbol{a}_*| \neq \varnothing$, then the point array $\boldsymbol{a}$ is called* nondegenerate.

Lemma 19.2.5 will provide examples of nondegenerate point arrays, which can be used in Theorem 19.2.1*c*.

19.2.4. Corollary. *Let $\mathcal{U} \in \mathrm{CBA}[\kappa]$, $\boldsymbol{a} = (a^0, a^1, \dots a^m)$ be a nondegenerate point array of radius $< \frac{\varpi^{\kappa}}{2}$ in $\mathcal{U}$ and $\sigma = \mathfrak{S}^{\kappa}_{\boldsymbol{a}}$ be corresponding κ-baricentric simplex. Then for some $\boldsymbol{x} \in \Delta^m$ the differential $\boldsymbol{d}_{\boldsymbol{x}}\sigma$ is linear and the image $\operatorname{Im} \boldsymbol{d}_{\boldsymbol{x}}\sigma$ forms a subcone of the tangent cone $\mathrm{T}_{\sigma(\boldsymbol{x})}$ which is isometric to an m-dimensional Euclidean space.*

Proof. For shortness, denote by $\tau \colon \mathcal{U} \to \mathbb{R}^m$ the distance map $|\boldsymbol{a}^{\neq 0}{}_*|$.

According to Theorem 19.2.1, σ is Lipschitz and the distance map τ, gives an open embedding of $\operatorname{InWeb}|\boldsymbol{a}_*| = \sigma(\Delta^m) \backslash \sigma(\partial \Delta^m)$. Clearly τ its Lipschitz. According to Rademacher's theorem (**??**) the differential $\boldsymbol{d}_{\boldsymbol{x}}(\tau \circ \sigma)$ is linear for almost all $\boldsymbol{x} \in \Delta^m$. Further since

[1] Recall that $|\boldsymbol{a}^{\neq 0}{}_*|$ denotes array $(|a^1{}_*|, \dots, |a^{\bar{k}}{}_*|)$.

$\operatorname{InWeb}|\boldsymbol{a}_*| \neq \varnothing$, the area formula ??? implies that $\boldsymbol{d}_{\boldsymbol{x}}(\tau\circ\sigma)$ is surjective on a set of positive masure of points $\boldsymbol{x}\in\Delta^m$.

Note that $\boldsymbol{d}_{\boldsymbol{x}}(\tau\circ\sigma) = (\boldsymbol{d}_{\sigma(\boldsymbol{x})}\tau)\circ(\boldsymbol{d}_{\boldsymbol{x}}\sigma)$. Applying Rademacher's theorem again, we get that $\boldsymbol{d}_{\boldsymbol{x}}\sigma$ is linear for almost all $\boldsymbol{x}\in\Delta^m$, at these points $\operatorname{Im}\boldsymbol{d}_{\boldsymbol{x}}\sigma$ forms a subcone of $\mathrm{T}_{\sigma(\boldsymbol{x})}$ which is isometric to a Euclidean space. Clearly the dimension of $\operatorname{Im}\boldsymbol{d}_{\boldsymbol{x}}(\tau\circ\sigma)$ is at least as big as dimension of $\operatorname{Im}\boldsymbol{d}_{\boldsymbol{x}}\sigma$. Hence the result. □

Proof of Theorem 19.2.1. Fix $z\in\mathcal{U}$ and $r<\frac{\varpi^\kappa}{2}$ such that $|z\,a^i|<r$ for all i. Note that the set $K\cap\overline{\mathrm{B}}[z,r]$ is convex, closed, and contains all a^i. Applying the theorem on short retract (**??**), we get the second part of (b).

The rest of the statements are proved first in case $\kappa\leqslant 0$ and then the remaining case $\kappa>0$ is reduced to the case $\kappa=0$.

Case $\kappa\leqslant 0$. Consider function array $f^i=\mathrm{md}^\kappa\circ|a^i{}_*|$. From the definition of web (19.1.2), it is clear that $\operatorname{Web}|\boldsymbol{a}_*| = \operatorname{Web}\boldsymbol{f}$. Further, from the definition of κ-barycentric simplex, $\mathfrak{S}^\kappa_{\boldsymbol{a}}=\mathfrak{S}_{\boldsymbol{f}}$.

All the functions f^i are strongly convex (see 9.5.1b). Thus, (19.2.1a), (19.2.1b) and the first statements in (19.2.1b) follow from Theorem 19.1.3.

Case $\kappa>0$. Applying rescaling, we can assume $\kappa=1$; so $\varpi^\kappa=\varpi^1=\pi$.

Set $\mathring{\mathcal{U}}=\operatorname{Cone}\mathcal{U}$; according to ???, $\mathring{\mathcal{U}}\in\mathrm{CBA}[0]$. Let us denote by ι the natural embedding of $\mathcal{U}$ as the unit sphere in $\mathring{\mathcal{U}}$ and $\operatorname{proj}\colon\mathring{\mathcal{U}}\circ\!\!\to\mathcal{U}$ the submap, defined as $\operatorname{proj}(v)=\iota^{-1}(v/|v|)$ for all $v\neq o$ in $\mathring{\mathcal{U}}$. Note that there is $z\in\mathcal{U}$ and $\varepsilon>0$ such that the set

$$K_\varepsilon=\left\{\,v\in\mathring{\mathcal{U}}\;\middle|\;\langle\iota(z),v\rangle\geqslant\varepsilon\,\right\}$$

contains all $\iota(a^i)$. Note that $o\notin K_\varepsilon$ and the set K_ε is closed and convex. The later follows from Theorem **??**, since $v\mapsto-\langle\iota(z),v\rangle$ is a Busemann function.

Denote by $\iota(\boldsymbol{a})$ the point array $(\iota(a^0),\iota(a^1),\dots,\iota(a^k))$ in $\mathring{\mathcal{U}}$. From the case $\kappa=0$, we get $\operatorname{Im}\mathfrak{S}^0_{\iota(\boldsymbol{a})}\subset K_\varepsilon$. In particular $\operatorname{Im}\mathfrak{S}^0_{\iota(\boldsymbol{a})}\not\ni o$ and thus $\operatorname{proj}\circ\mathfrak{S}^0_{\iota(\boldsymbol{a})}$ is defined. Direct calculations show

$$\mathfrak{S}^1_{\boldsymbol{a}}=\operatorname{proj}\circ\mathfrak{S}^0_{\iota(\boldsymbol{a})}\quad\text{and}\quad\operatorname{Web}|\boldsymbol{a}_*|=\operatorname{proj}[\operatorname{Web}|\iota(\boldsymbol{a})_{*|}].$$

This reduces the case $\kappa=1$ of the theorem to the case $\kappa=0$, which is proved already. □

19.2.5. Lemma. *Let $\mathcal{U}\in\mathrm{CBA}[\kappa]$, $\boldsymbol{a}=(a^0,a^1,\dots a^k)$ be an array of radius $<\frac{\varpi^\kappa}{2}$ and $B^i=\overline{\mathrm{B}}[a^i,r^i]$ for some array of positive reals*

$(r^0, r^1, \dots, r^{\bar{k}})$. *Assume that* $\bigcap_i B^i = \varnothing$, *but* $\bigcap_{i\neq j} B^i \neq \varnothing$ *for any* j. *Then* $\boldsymbol{a}$ *is nondegenerate.*

Proof. Without loss of generality we can assume that $\mathcal{U}$ is geodesic and $\operatorname{diam}\mathcal{U} < \varpi^\kappa$. If not, choose a point $z \in \mathcal{U}$ and $r < \frac{\varpi^\kappa}{2}$ such that $|z\,a^i| \leqslant r$ for each i and consider $\overline{\mathrm{B}}[z,r]$ instead of $\mathcal{U}$. The later can be done since $\overline{\mathrm{B}}[z,r]$ is convex and closed, thus $\overline{\mathrm{B}}[z,r] \in \mathrm{CBA}[\kappa]$ and $\operatorname{Web}|\boldsymbol{a}_*| \subset \overline{\mathrm{B}}[z,r]$; see 9.6.1 and 19.2.1*b*.

By Theorem 19.2.1, $\operatorname{Web}|\boldsymbol{a}_*|$ is a compact set, therefore there is a point $p \in \operatorname{Web}|\boldsymbol{a}_*|$ minimizing the function

$$f(x) = \max_i\{|B^i\,x|\} = \max\{0, |a^0\,x| - r^0, \dots, |a^{\bar{k}}\,x| - r^{\bar{k}}\}.$$

By the definition of web (19.1.2), p is also the minimum point of f on $\mathcal{U}$. Now let us show the following claim.

❶ $p \notin B^j$ *for any* j.

Indeed, assume the contrary, i.e.,

❷ $$p \in B^j$$

for some j. Then p is a point of local minimum for the function

$$h^j(x) = \max_{i\neq j}\{|B^i\,x|\}.$$

Hence

$$\max_{i\neq j}\{\measuredangle[p\,{}^{x}_{a^i}]\} \geqslant \frac{\pi}{2}$$

for any $x \in \mathcal{U}$. From angle comparison (9.3.1*d*), it follows that p is a global minimum of h^j and hence

$$p \in \bigcap_{i\neq j} B^i.$$

The latter and ❷ contradict $\bigcap_i B^i = \varnothing$. △

From the definition of web, it also follows that

$$\operatorname{Web}|\boldsymbol{a}^{\neq j}{}_*| \subset \bigcup_{i\neq j} B^i.$$

Therefore, the claim implies that $p \notin \operatorname{Web}|\boldsymbol{a}^{\neq j}{}_*|$ for each j; i.e., $p \in$ $\in \operatorname{InWeb}|\boldsymbol{a}_*|$. □

19.3 The case of finite-dimensional CAT spaces

Recall that web embedding and its brace is defined in 19.2.2.

19.3.1. Theorem. *Let $\mathcal{U} \in \mathrm{CBA}[\kappa]$, $\mathrm{LinDim}\,\mathcal{U} = m$, and $\boldsymbol{a} = (a^0, a^1, \dots a^m)$ be a point array in $\mathcal{U}$ with radius $< \frac{\varpi^\kappa}{2}$. Then the $|\boldsymbol{a}_*|$-web embedding $\Phi\colon \mathbb{R}^m \hookrightarrow \mathcal{U}$ with brace $|a^0{}_*|$ is locally Lipschitz.*

Note that if $\boldsymbol{a}$ is degenerate, i.e., if $\mathrm{InWeb}\,|\boldsymbol{a}_*| = \varnothing$, then the domain of web embedding Φ above is empty and hence the conclusion of the theorem trivially holds.

As will be seen in Section **??**, the web embedding as in the above theorem is much better than bi-Lipschitz. In particular, it can be used to construct bi-Lipschitz embeddings with constants arbitrarily close to 1.

19.3.2. Lemma. *Let $\mathcal{U} \in \mathrm{CBA}[\kappa]$, and $\boldsymbol{a} = (a^0, a^1, \dots a^k)$ be a point array with radius $< \frac{\varpi^\kappa}{2}$. Then for any $p \in \mathrm{InWeb}\,|\boldsymbol{a}_*|$, there is $\varepsilon > 0$ such that if for some $q \in \mathrm{Web}\,|\boldsymbol{a}_*|$ and $b \in \mathcal{U}$ we have $|pq| < \varepsilon$, $|pb| < \varepsilon$ and $\measuredangle[q\,{}^{b}_{a^i}] < \frac{\pi}{2} + \varepsilon$, then the array $b, a^0, a^1, \dots, a^m$ is nondegenerate.*

Proof. Without loss of generality, we may assume that $\mathcal{U}$ is geodesic and $\mathrm{diam}\,\mathcal{U} < \varpi^\kappa$. If not, consider instead of $\mathcal{U}$, a ball $\overline{\mathrm{B}}[z, r] \subset \mathcal{U}$ for some $z \in \mathcal{U}$ and $r < \frac{\varpi^\kappa}{2}$ such that $|z a^i| \leqslant r$ for each i.

From angle comparison (9.3.1d), it follows that $p \in \mathrm{InWeb}\,\boldsymbol{a}$ if and only if both of the following conditions hold:

1. $\max_i\{\measuredangle[p\,{}^{a^i}_{u}]\} \geqslant \frac{\pi}{2}$ for any $u \in \mathcal{U}$,
2. for each i there is $u^i \in \mathcal{U}$ such that $\measuredangle[p\,{}^{a^j}_{u^i}] < \frac{\pi}{2}$ for all $j \neq i$.

Due to the semicontinuity of angles (9.14.1), there is $\varepsilon > 0$ such that for any $x \in \mathrm{B}(p, 10\cdot\varepsilon)$ we have

❶ $$\measuredangle[x\,{}^{a^j}_{u^i}] < \tfrac{\pi}{2} - 10\cdot\varepsilon \quad \text{for all} \quad j \neq i.$$

Now assume for sufficiently small $\varepsilon > 0$ there are points $b \in \mathcal{U}$ and $q \in \mathrm{Web}\,|\boldsymbol{a}_*|$ such that

❷ $$|pq| < \varepsilon, \quad |pb| < \varepsilon, \quad \measuredangle[q\,{}^{b}_{a^i}] < \tfrac{\pi}{2} + \varepsilon \quad \text{for all} \quad i.$$

Acording to Theorem 19.2.1b, for all small $\varepsilon > 0$, we have

$$\mathrm{rad}\{b, a^0, a^1, \dots, a^k\} < \tfrac{\varpi^\kappa}{2}.$$

Fix sufficiently small $\delta > 0$ and set

$$v^i = \operatorname{geod}_{[qu^i]}(\tfrac{1}{3}\cdot\delta) \quad \text{and} \quad w^i = \operatorname{geod}_{[v^ib]}(\tfrac{2}{3}\cdot\delta).$$

Clearly,

$$\begin{aligned}|bw^i| &= |bv^i| - \tfrac{2}{3}\cdot\delta \leqslant \\ &\leqslant |bq| - \tfrac{1}{3}\cdot\delta.\end{aligned}$$

Further, the inequalities ❶ and ❷ imply

$$\begin{aligned}|a^j w^i| &< |a^j v^i| + \tfrac{2}{3}\cdot\varepsilon\cdot\delta < \\ &< |a^i q| - \varepsilon\cdot\delta < \\ &< |a^i q|\end{aligned}$$

for all $i \neq j$.

Set $B^i = \overline{\mathrm{B}}[a^i, |a^i q|]$ and $B^{m+1} = \overline{\mathrm{B}}[b, |a^i q| - \frac{1}{3}\cdot\delta]$. Clearly

$$\bigcap_{i\neq m+1} B^i = \{q\},$$

$$\bigcap_{i\neq j} B^i \ni w^j \quad \text{for} \quad j \neq m+1.$$

$$\bigcap_{i} B^i = \{q\} \cap B^{m+1} = \varnothing.$$

Lemma 19.2.5 finishes the proof. □

Proof of 19.3.1. From 19.2.2, it is sufficient to show that the Φ is Lipshitz. Assume the contrary; i.e., there are sequences $\boldsymbol{y}_n, \boldsymbol{z}_n \to \boldsymbol{x} \in$ $\in \operatorname{Dom}\Phi$ such that

❸ $$\frac{|\Phi(\boldsymbol{y}_n)\ \Phi(\boldsymbol{z}_n|)}{|\boldsymbol{y}_n - \boldsymbol{z}_n|} \to \infty \quad \text{as} \quad n \to \infty.$$

Set $p = \Phi(\boldsymbol{x})$, $q_n = \Phi(\boldsymbol{y}_n)$, and $b_n = \Phi(\boldsymbol{z}_n)$. From 19.2.2, p, q_n, $b_n \in$ $\in \operatorname{InWeb}|\boldsymbol{a}_*|$ and $q_n, b_n \to p$ as $n \to \infty$. Fix arbitrary $\varepsilon > 0$. Note that ❸ implies

$$\measuredangle[q_n\, {}^{a^i}_{b_n}] < \tfrac{\pi}{2} + \varepsilon$$

for all $i > 0$ and all large n. Further, according to 19.2.2, the subfunction $(|\boldsymbol{a}^0{}_*|) \circ \Phi$ is locally Lipschitz. Therefore we also have

$$\measuredangle[q_n\, {}^{a^0}_{b_n}] < \tfrac{\pi}{2} + \varepsilon$$

for all large n. According to Lemma 19.3.2, the point array $b_n, a^0, \ldots$ $\ldots, a^k$ for large n is nondegenerate.

Applying Corollary 19.2.4, we get a contradiction □

Chapter 20

Quasigeodesics

The class of quasigeodesics[1] generalizes the class of geodesics to non-smooth metric spaces. This class of curves was first considered for 2-dimensional convex hypersurfaces in the Euclidean space by Alexandrov in [13]. Alexandrov defined quasigeodesics as curves which "turn" right and left simultaneously. This type of curves was studied further in [141], [20], [112]. Later, it was generalized to surfaces with bounded integral curvature [14] and to multidimensional polyhedral spaces [110], [111]. A. D. Milka gave a characterization of quasigeodesics using notion of development (8.4.4). This characterization made it possible to define quasigeodesics multi-dimensional Alexandrov's spaces; this was first done in Petrunin's master thesis and developed further in [130].

In Alexandrov's spaces, quasigeodesics behave more naturally than geodesics, mainly:

- ⋄ There is a quasigeodesic starting in any direction from any point (20.5.1).
- ⋄ The limit of quasigeodesics is a quasigeodesic (20.2.3c).

These two property make quasigeodesics to be a useful technical tool. Also due to the generalized Lieberman lemma (18.8.1), they are useful in the study of length-metric of extremal subsets, in particular the boundary of Alexandrov's space. There is an overlap in the applications of quasigeodesics and gradient exponent, defined in Section 17.3. A good example is the proof of Theorem **??**, see footnote **??**, page **??**. In this type of argument, radial curves give a simpler tool, which is also superior since it works in infinitely dimensional Alexandrov's spaces.

[1] It should be noted that the class of quasigeodesics described here has nothing to do with the Gromov's quasigeodesics in δ-hyperbolic spaces.

20.1 Convex curves and quasigeodesics

In this secion we define κ-*convex curves* and κ-*quasigeodesics*. Roughly, a curve is called κ-convex if it satisfy the same comparison as geodesic in $\mathrm{CBB}[\kappa]$-space. If in addition it is a unit-speed curve, then it is called κ-quasigeodesic.

20.1.1. Definition. *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$ and $\beta\colon \mathbb{I} \to \mathcal{L}$ be a curve. We say that β is* κ-convex *if for any $p \in \mathcal{L}$ the function*

$$y_p(t) = \mathrm{md}^\kappa|p\,\beta(t)|$$

satisfies differential inequality $y_p'' + \kappa\cdot y_p \leqslant 1$ for all $t \in \mathbb{I}$ such that $|p\,\beta(t)| < \varpi^\kappa$.

If in addition β is a unit-speed curve then it is called κ-quasigeodesic.

In Section 20.3, we give an invariant definition of quasigeodesics; once equivalence of these definitions is proved, the parameter κ in the definition of κ-quasigeodesic can be omited. In Theorem 20.1.4, we will give a number of reformulations of the above definition in terms of angle comparison and development.

Let us list some examples of κ-convex curves and κ-quasigeodesics:

- ◇ From function comparison 8.4.1, any local geodesic in $\mathrm{CBB}^m[\kappa]$-space is a κ-quasigeodesic as well as κ-convex curve.
- ◇ If $\operatorname{diam}\mathcal{L} \leqslant \frac{\varpi^\kappa}{2}$ then any constant curve in $\mathcal{L}$ is κ-convex. In particular, if $\kappa \leqslant 0$ then any constant curve is κ-convex.
- ◇ Let $\mathcal{L}$ be isometric to a convex figure in a Euclidean plane. Then any billiard trajectory in $\mathcal{L}$ as well as arc of its boundary equipped with a unit-speed parametrization forms a quasigeodesic in $\mathcal{L}$.
- ◇ A joint of two rays in a 2-dimensional cone is a quasigeodesic if they cut form the cone two angles with angle measure $\leqslant \pi$.

Now let us turn to differentiability of κ-convex curves, see Definition 14.2.4. Let us remind that two vectors v and w in the tangent space T_p are called *opposite* (briefly $v + w = 0$) if $|v| = |w|$ and $\measuredangle(v, w) = \pi$, see Definition 14.5.1. To formulate the next theorem, we need to define the following weak analog of opposite vectors.

20.1.2. Definition. *Let $\mathcal{L} \in \mathrm{CBB}$, $p \in \mathcal{L}$ and $v, w \in \mathrm{T}_p$. We say that v is* polar *to w if*

$$\langle x, v\rangle + \langle x, w\rangle \geqslant 0$$

for any vector $x \in \mathrm{T}_p$.

20.1.3. Theorem. *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$ and $\beta\colon \mathbb{I} \to \mathcal{L}$ be a κ-convex curve, then*

a) β *is* both-side differentiable*; in particular,* $\beta^+(t_0)$ *and* $\beta^-(t_0)$ *are defined for any interior value* t_0 *of* $\mathbb{I}$*,*

b) β *is differentiable at all but countable subset of values in* $\mathbb{I}$*; i.e., for some countable set* $A \subset \mathbb{I}$*, we have* $\beta^+(t_0) + \beta^-(t_0) = 0$ *for any* $t \in \mathbb{I}\backslash A$*.*

c) For each interior value t_0 *of* $\mathbb{I}$*, vector* $\beta^+(t_0)$ *is polar to* $\beta^-(t_0)$*.*

d) β *is a* 1*-Lipschitz curve.*

In particular, $|\beta^\pm(t_0)| \leqslant 1$*.*

Proof. Set $p = \beta(t_0)$ and $y_q(t) = \mathrm{md}^\kappa(|q\,\beta(t)|)$ for $q \in \mathcal{L}$.

(d). If t_0 is an interior value of $\mathbb{I}$, from the barrier inequality (5.4.3*b*), for any $|t - t_0| < \varpi^\kappa$ we have $y_p(t) \leqslant \mathrm{md}^\kappa(t - t_0)$ or, equivalently $|\beta(t_0)\ \beta(t)| \leqslant |t - t_0|$.

(a)+(b). Since the function y_q is semiconcave, it is both-side differentiable at any t such that $0 < |q\,\beta(t)| < \varpi^\kappa$. Moreover, it is differentable at all but countable subset of such values t. Clearly, the same remains true for the function $|q\,*| \circ \beta$.

It remains to apply **??** for a dense countable subset Q of $\mathrm{B}(p, \varpi^\kappa)$; the existance of such Q follows from 15.3.2.

(c). It is sufficient to prove that for any geodesic direction ξ we have

$$\langle \xi, \beta^+\rangle + \langle \xi, \beta^+\rangle \geqslant 0. \qquad ❶$$

Choose $q \in \mathrm{Str}(p)$ so that $\xi = \mathrm{dir}[pq]$ is uniquely defined. From 14.6.2, we have $(\boldsymbol{d}_p|q\,*|)(x) = -\langle \xi, x\rangle$. Since y_q is semiconcave, $y_q^+(t_0) + y_q^-(t_0) \leqslant 0$. Further,

$$y_q^\pm(t_0) = -\mathrm{sn}^\kappa|pq|\cdot\langle \xi, \beta^\pm(t_0)\rangle$$

and ❶ follows. □

Arc-point model angle. Let us introduce a new shortcut notation for model angle. Assume $\alpha\colon \mathbb{I} \to \mathcal{X}$ is a 1-Lipschitz curve $t_0, t \in \mathbb{I}$ and p is a point in a metric space $\mathcal{X}$. Consider a triangle in $\mathbb{M}^2[\kappa]$ with sides $|t - t_0|$, $|q\,\alpha(t_0)|$ and $|q\,\alpha(t)|$ and denote by $\measuredangle^\kappa(\alpha|_{t_0}^t, p)$ the angle at the vertex corresponding to $\alpha(t_0)$. It will be called *arc-point model angle* for arc $\alpha|_{t_0}^t$ and point p.

In other words,

$$\measuredangle^\kappa(\alpha|_{t_0}^t, p) \overset{def}{=\!=} \measuredangle^\kappa\{|p\,\alpha(t)|; |p\,\alpha(t_0)|, |t - t_0|\},$$

where $\measuredangle^\kappa\{a; b, c\}$ is defined in Section 1.1. Note that

- ◇ If the angle $\measuredangle^\kappa(\alpha|_{t_0}^{t_1}, p)$ is defined then $\measuredangle^\kappa(\alpha|_{t_0}^{t}, p)$ is defined for any t between t_0 and t_1.
- ◇ If one fixes p and t_0 such that $|p\gamma(t_0)| < \varpi^\kappa$, then the angle $\measuredangle^\kappa(\alpha|_{t_0}^{t}, p)$ is defined once $|t - t_0| > 0$ is sufficiently small.
- ◇ If $\varphi = \lim_{t\to t_0+} \measuredangle^\kappa(\alpha|_{t_0}^{t}, p)$ is defined then

$$\cos\varphi = -(|p_*| \circ \alpha)^+(t_0).$$

20.1.4. Theorem. *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$ and $\beta\colon \mathbb{I} \to \mathcal{L}$ be a 1-Lipschitz curve. Then the following conditions are equivalent:*

a) *β is κ-convex.*

b) *For any $t_0 \in \mathbb{I}$ and $p \neq \beta(t_0)$, the subfunction*

$$\tau \mapsto \measuredangle^\kappa(\beta|_{t_0}^{t_0+\tau}, p)$$

is non-increasing for $\tau > 0$.

c) *β is both-side differentiable, and vector $\beta^+(t_0)$ is polar to $\beta^-(t_0)$ at each interior value $t_0 \in \mathbb{I}$. Moreover, for any $t_0 \in \mathbb{I}$ and $p \neq \beta(t_0)$, inequalities*

$$\langle \mathrm{dir}[\beta(t_0)p], \beta^\pm(t_0)\rangle \leqslant \cos\measuredangle^\kappa(\beta|_{t_0}^{t_0\pm\tau}, p)$$

hold for $\tau > 0$; once the right hand side is defined.

d) *For any subinterval $[a, b] \subset \mathbb{I}$ and point $p \in \mathcal{L}$, if the κ-developing of the restriction $\beta|[a, b]$ with respect to p is defined then it is locally convex.*

Moreover, if in addition β is a unit-speed curve then each of the conditions (a)–(d) imply that β is a κ-quasigeodesic. In this case the inequality in (c) can be also rewritten the following way:

c′) $\measuredangle(\mathrm{dir}[\beta(t_0)p], \beta^\pm(t_0)) \geqslant \measuredangle^\kappa(\beta|_{t_0}^{t_0\pm\tau}, p)$.

Remark. The inequality (c') states that in a "triangle", one side of which is a quasigeodesic and two other sides are geodesics, one has usual comparison for the angle adjacent to the quasigeodesic. Note however, that it gives no controle on the angle between geodesic sides in this triangle.

Proof. Equivalence *(a)⇔(b)* follows from Alexandrov's lemma (6.2.1) and description of solution $y'' + \kappa\cdot y \leqslant 1$ through Jensen inequality (5.4.3*d*).

The equivalence *(a)⇔(c)* is essentially description of solution $y'' + \kappa\cdot y \leqslant 1$ trough barrier inequality (5.4.3*b*).

The implementations *(c)⇒(d)⇒(b)* can be proved the same way as Development comparison (8.4.6).

The last statement will follow from 20.2.4. □

20.2 Passage to the limit

The following proposition follows directly from the definition of κ-convex curve (20.1.1) and Theorem 20.1.3*d*.

20.2.1. Proposition. *Let $\mathcal{L}_n \in \mathrm{CBB}^m[\kappa_n]$, $\mathcal{L}_n \xrightarrow[\mathrm{GH}]{} \mathcal{L}_\infty$ and $\kappa_n \to \kappa_\infty \in \mathbb{R}$.*

Assume $\beta_n\colon \mathbb{I} \to \mathcal{L}_n$ be a sequence of κ_n-convex curves and $\beta_n(t) \to \beta_\infty(t)$ for all $t \in \mathbb{I}$. Then β_∞ is a κ_∞-convex curve in $\mathcal{L}_\infty$.

Here is the main theorem in this section.

20.2.2. Theorem. *Let $\mathcal{L}_n \in \mathrm{CBB}^m[\kappa_n]$, $\mathcal{L}_n \xrightarrow[\mathrm{GH}]{} \mathcal{L}_\infty$ be noncollapsing (i.e., $\dim \mathcal{L}_\infty = m$) and $\kappa_n \to \kappa_\infty \in \mathbb{R}$.*

Assume $\beta_n\colon \mathbb{I} \to \mathcal{L}_n$ be a sequence of κ_n-convex curves and $\beta_n(t) \to \beta_\infty(t)$ for all $t \in \mathbb{I}$. Then if $\beta_\infty\colon \mathbb{I} \to \mathcal{L}_\infty$ is differentiable at an interior value $t_0 \in \mathbb{I}$ then

$$|\beta^+(t_0)| = |\beta^-(t_0)| = \lim_{n\to\infty} |\beta_n^\pm(t_0)|.$$

Before giving a proof, let us give couple of corollaries.

20.2.3. Corollaries. *Let $\mathcal{L}_n \in \mathrm{CBB}^m[\kappa_n]$, $\mathcal{L}_n \xrightarrow[\mathrm{GH}]{} \mathcal{L}_\infty$ be noncollapsing (i.e., $\dim \mathcal{L}_\infty = m$) and $\kappa_n \to \kappa_\infty \in \mathbb{R}$.*

Assume $\beta_n\colon \mathbb{I} \to \mathcal{L}_n$ be a sequence of κ_n-convex curves and $\beta_n(t) \to \beta_\infty(t)$ for all $t \in \mathbb{I}$. Then

a)
$$|\beta_\infty^\pm(t)| = \lim_{n\to\infty} |\beta_n^\pm(t)|$$
for all but countable set of values $t \in \mathbb{I}$.

b) $\operatorname{length} \beta_n \to \operatorname{length} \beta_\infty$ *as $n \to \infty$.*

c) If in addition $\beta_n\colon \mathbb{I} \to \mathcal{L}_n$ be a sequence of κ_n-quasigeodesics then $\beta_\infty\colon \mathbb{I} \to \mathcal{L}$ is a κ_∞-quasigeodesic.

Proof. Follows from Theorem 20.2.2, Proposition 20.2.1 and Theorem 20.1.3*b*. □

20.2.4. Corollary. *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$ and $\beta\colon \mathbb{I} \to \mathcal{L}$ be a κ-convex curve. Then for any $t_0 \in \mathbb{I}$, we have*

$$|\beta^+(t_0)| = \lim_{t\to t_0+} |\beta^\pm(t)|, \qquad |\beta^-(t_0)| = \lim_{t\to t_0-} |\beta^\pm(t)|.$$

In particular, if $\beta\colon \mathbb{I} \to \mathcal{L}$ is a κ-quasigeodesic then $|\beta^\pm(t)| \equiv 1$ for all $t \in \mathbb{I}$ with the exception for the ends of $\mathbb{I}$, where one of derivatives is not defined.

Proof. We will prove the first identity. Without loss of generality we can assume $t_0 = 0$ and $\mathbb{I} = [0,1]$. Set $p = \beta(0)$.

From 20.1.3*d*, $|\beta^\pm(t)| \leqslant 1$ for any t. Thus, it is sufficient to show that for any sequence $t_n \to 0+$ such that $|\beta^\pm(t_n)|$ converges we have

$$|\beta^+(0)| = \lim_{n\to\infty} |\beta^\pm(t_n)|.$$

Consider sequence of blowups $\mathcal{L}_n = \frac{1}{t_n}*\mathcal{L}$ with sequence of curves

$$\beta_n\colon [0, \tfrac{1}{t_n}] \to \mathcal{L}_n\colon s \mapsto \tfrac{1}{t_n}*\beta(t_n\cdot s).$$

Note that $\mathcal{L}_n \in \mathrm{CBB}^m[(t_n)^2\cdot\kappa]$ and $\beta_n(s)$ is $(t_n)^2\cdot\kappa$-convex curve.

For the canonical convergence $\mathcal{L}_n \xrightarrow[\mathrm{GH}]{} \mathrm{T}_p$ defined in **??**, the sequence of curves $\beta_n\colon [0, \frac{1}{t_n}] \to \mathcal{L}_n$ converges to

$$\beta_\infty\colon [0,\infty) \to \mathrm{T}_p\colon s \mapsto s\cdot\beta^+(0).$$

Note that $|\beta^\pm_\infty(1)| = |\beta^+(0)|$ and β_∞ is differentiable at 1. Thus applying Theorem 20.2.2, we get

$$|\beta^\pm(t_n)| = |\beta^\pm_n(1)| \xrightarrow[n\to\infty]{} |\beta^\pm_\infty(1)| = |\beta^+(0)|.$$

Now if β is a κ-quasigeodesic then it is a unit-speed curve. From **??**, $|\beta^\pm(t)| \overset{a.e.}{=\!=} 1$. Applying the proved identity, we get that the identity $|\beta^\pm(t)| \equiv 1$ holds everywhere. □

In the proof of the theorem we will use the followng lemma.

20.2.5. Lemma. *Let $K = \operatorname{Cone}\Sigma \in \mathrm{CBB}^m[0]$ and $v, w \in K$.*

Assume that equality

$$\langle v, x\rangle = \langle w, x\rangle$$

holds for x in a subset $E \subset K$ of positive volume then $v = w$.

Proof. To be written. □

Proof Theorem 20.2.2. It is sufficient to show that $|\beta^\pm_n(t_0)| \to |\beta^\pm_\infty(t_0)|$ as $n \to \omega$ for some selective utrafilter ω.

Set

(a) $p_\infty = \beta_\infty(t_0)$, $p_n = \beta_n(t_0)$;
(b) $(K_\omega, 0) = \lim_{n\to\omega}(\mathrm{T}_{p_n}, 0)$;
(c) $\beta^\pm_n(t_0) \to w_\pm \in K_\omega$ as $n \to \omega$.

We will construct a map $\Phi\colon \mathrm{T}_{p_\infty} \to K_\omega$ which satisfies the following properties:

(i) It is a *conic map*; i.e., $\Phi(r\cdot x) = r\cdot\Phi(x)$ for all $r \geqslant 0$ and $x \in \mathrm{T}_{p_\infty}$.
(ii) It is norm-preserving; i.e., $|\Phi(x)| = |x|$ for any $x \in \mathrm{T}_{p_\infty}$.
(iii) It is non-contracting; i.e. $|\Phi(x)\,\Phi(y)| \geqslant |xy|$ for any $x, y \in$ $\in \mathrm{T}_{p_\infty}$.
(iv) for any $z \in \Phi(\mathrm{T}_{p_\infty})$, we have

$$\langle z, w_\pm\rangle = \langle z, \Phi(\beta^\pm_\infty(t_0))\rangle.$$

Once Φ is constructed, Lemma 20.2.5 and properties (iii) and (iv) imply

$$\Phi(\beta^\pm_\infty(t_0)) = w_\pm.$$

By property (ii),

$$|w_\pm| = |\beta^\pm_\infty(t_0)|.$$

Further, by (c),

$$|\beta^\pm_n(t_0)| \xrightarrow[n\to\omega]{} |\beta^\pm_\infty(t_0)|;$$

i.e., once the map $\Phi\colon \mathrm{T}_{p_\infty} \to K_\omega$ is constructed, the theorem follows.

Construction of Φ. Set

$$\Sigma_\omega = \lim_{n\to\omega} \Sigma_{p_n};$$

clearly $K_\omega = \operatorname{Cone}\Sigma_\omega$. According to Corollary **??**, $\Sigma_{p_n} \leqslant \mathbb{S}^{m-1}$ for each n; thereofre $\Sigma_\omega \leqslant \mathbb{S}^{m-1}$. In particular Σ_ω is compact.

Choose $q_\infty \in \operatorname{Str}(p_\infty)$, so the geodesic $[p_\infty q_\infty]$ is uniquely defined. Set $\xi_\infty = \operatorname{dir}[p_\infty q_\infty]$. Further, choose a sequence of points $q_n \in \operatorname{Str}(p_n)$ so that $q_n \xrightarrow[\mathrm{GH}]{} q_\infty$ and set $\xi_n = \operatorname{dir}[p_n q_n]$. Finally set

$$\Phi(\xi_\infty) = \lim_{n\to\omega} \xi_n.$$

The value $\Phi(\xi_\infty) \in \Sigma_\omega$ depends on choice of point q_∞ and the choice of sequence (q_n). But we can fix such choices for each direction $\xi_\infty \in \Sigma'_p$; recal that Σ'_p denotes the set of geodesic directions in Σ_p.

Since $[p_\infty q_\infty]$ is unique, we have $[p_n q_n] \to [p_\infty q_\infty]$. From angle semicontinuity (8.6.4), the map $\Phi\colon \Sigma'_{p_\infty} \to \Sigma_\omega$ is non-contracting.

For a non-geodesic direction $\zeta \in \Sigma_{p_\infty}$, fix a sequence of geodesic directions $\zeta_n \to \zeta$ and set $\Phi(\zeta)$ to be a partial limit of $\Phi(\zeta_n)$ as $n \to$ $\to \infty$. The pratial limit exists since Σ_ω is compact. This way we extended the map Φ to Σ_{p_∞} keeping it noncontracting.

Finally, let us extend Φ to a conic map $\Phi\colon \mathrm{T}_{p_\infty} \to K_\omega$ by setting

$$\Phi(r\cdot\zeta) = r\cdot\Phi(\zeta)$$

for all $r \geqslant 0$ and $\zeta \in \Sigma_{p_\infty}$.

Now let us show that the constructed map Φ satisfies the conditions. The conditions (i)–(iii) follow directly from the construction. To prove (iv), it is sufficient to show that equality

$$\langle \Phi(\xi_\infty), w_\pm\rangle = \langle \Phi(\xi_\infty), \Phi(\beta^\pm_\infty(t_0))\rangle$$

holds for any geodesic directions ξ_∞.

Let q_∞, (q_n) and (ξ_n) be as in the construction of Φ above. Set

$$f_\infty(t) = |q_\infty\,\beta_\infty(t)|, \qquad\qquad f_n(t) = |q_n\,\beta_n(t)|.$$

Clearly, $f_n \to f_\infty$. Since $q_\infty \in \mathrm{Str}(p_\infty)$ and $q_n \in \mathrm{Str}_n(p_n)$, 14.6.2 implies

$$(\boldsymbol{d}_{p_\infty}|q_\infty\,*|)(x) \equiv -\langle \xi_\infty, x\rangle, \qquad\qquad (\boldsymbol{d}_{p_n}|q_n\,*|)(x) \equiv -\langle \xi_n, x\rangle.$$

Thus, the functions f_∞ is differentiable at t_0 and

$$f'_\infty(t_0) = \mp\langle \xi_\infty, \beta^\pm_\infty(t_0)\rangle, \qquad\qquad f^\pm_n(t_0) = -\langle \xi_n, \beta^\pm_n(t_0)\rangle.$$

Set $y_\infty = \mathrm{md}^\kappa \circ f_\infty$ and $y_n = \mathrm{md}^\kappa \circ f_n$. Since β_∞ is κ_∞-convex and β_n are κ_n-convex, we have

$$y''_\infty + \kappa_\infty\cdot y_\infty \leqslant 1 \qquad\qquad y''_n + \kappa_n\cdot y_n \leqslant 1$$

Therefore, according to the lemma on equilibrium (5.4.4), $y^\pm_n(t_0) \to \to \pm y'(t_0)$ and hence $f^\pm_n(t_0) \to \pm f'_\infty(t_0)$ as $n \to \omega$. Summarizing, all above we get

$$\begin{aligned}\langle \Phi(\xi_\infty), w_\pm\rangle &= \lim_{n\to\omega}\langle \xi_n, \beta^\pm_n(t_0)\rangle = \\ &= -\lim_{n\to\omega} f^\pm_n(t_0) = \\ &= \mp f'_\infty(t_0) = \\ &= \langle \xi_\infty, \beta^\pm_\infty(t_0)\rangle.\end{aligned}$$

Therefore it remains to show that

$$\langle \Phi(\xi_\infty), \Phi(\beta^\pm_\infty(t_0))\rangle = \langle \xi_\infty, \beta^\pm_\infty(t_0)\rangle.$$

Note that since Φ is non-contracting and norm-preserving, we have

❶
$$\begin{aligned}\measuredangle\big(\Phi(\xi_\infty), \Phi(\beta^\pm_\infty(t_0))\big) &\geqslant \measuredangle\big(\xi_\infty, \beta^\pm_\infty(t_0)\big)\,,\\ \measuredangle\big(\Phi(\beta^+_\infty(t_0)), \Phi(\beta^-_\infty(t_0))\big) &\geqslant \measuredangle\big(\beta^+_\infty(t_0), \beta^-_\infty(t_0)\big)\,.\end{aligned}$$

Since $\beta^+_\infty(t_0) + \beta^-_\infty(t_0) = 0$,

$$\measuredangle(\xi_\infty, \beta^+_\infty(t_0)) + \measuredangle(\xi_\infty, \beta^-_\infty(t_0)) = \pi, \qquad \measuredangle(\beta^+_\infty(t_0), \beta^-_\infty(t_0)) = \pi.$$

Therefore we have equalities in both inequalites of ❶. Hence

$$\langle \Phi(\xi_\infty), \Phi(\beta^\pm_\infty(t_0))\rangle = |\beta^\pm_\infty(t_0)|\cdot\cos\measuredangle(\xi_\infty, \beta^\pm_\infty(t_0)) = \langle \xi_\infty, \beta^\pm_\infty(t_0)\rangle. \qquad \square$$

20.3 Invariant description of quasigeodesics

Recall that a semiconcave subfunction $f\colon \mathcal{L} \to \mathbb{R}$ on a CBB space $\mathcal{L}$ is called *native* if at each point $p \in \operatorname{Dom} f$ there is a vector $w \in \mathrm{T}_p$ such that

$$d_p f(x) + \langle w, x\rangle \leqslant 0$$

for any $x \in \mathrm{T}_p$; see Definition **??**. Note, that according to **??**, if $\partial\mathcal{L} = \varnothing$ then any semiconcave function is native.

20.3.1. Theorem. *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$ and $\gamma\colon \mathbb{I} \to \mathcal{L}$ be a curve. Then the following conditions are equivalent:*

a) γ is a κ-quasigeodesic;

b) for any $\lambda \in \mathbb{R}$ and any native λ-concave locally Lipschitz subfunction $f\colon \mathcal{L} \looparrowright \mathbb{R}$ the composition $f \circ \gamma$ is λ-concave.

Before going ito proof, let us show that the word *native* is necessury in the formulation of this theorem.

Example. The unit disc $\mathbb{D}$ in $\mathbb{E}^2$ forms a $\mathrm{CBB}^2[0]$ space. Let γ be an arc of boundary of $\mathbb{D}$ equipped with unit-speed parametrization. Note that γ is a quasigeodesic in $\mathbb{D}$.

On the other hand any linear function $f\colon \mathbb{E}^2 \to \mathbb{R}$ is also a concave function on $\mathbb{D}$ and it is easy to choose f so that the composition $f \circ \gamma$ is not concave.

The theorem implies that we can talk about quasigeodesics in CBB-spaces without ambiguity; more presicely:

20.3.2. Corollary. *Let $\mathrm{K} > \kappa$ and $\mathcal{L} \in \mathrm{CBB}^m[\mathrm{K}]$. Then the classes of K-quasigeodesics and κ-quasigeodesics in $\mathcal{L}$ coincide.*

Proof of 20.3.1, (a) $\Leftarrow$ (b). First, let us define class of *probation curves* which includes all curves satisfying condition (b) for the functions with controlled concavity. Thus, it will be sufficient to show that any probation curve γ is a κ-quasigeodesic.

20.3.3. Definition. *A curve γ in $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$ is called* probation curve *if for any $\lambda \in \mathbb{R}$ and any subfunction $f\colon \mathcal{L} \looparrowright \mathbb{R}$ of controlled concavity type (λ, κ), we have that $f \circ \gamma$ is λ-concave.*

First, let us briefly discuss properties of probation curves. Note that for any $p, q \in \mathcal{L}$, function $f_p = \mathrm{md}^\kappa \circ |p_*|$ has controlled concavity type $(1 - \kappa\cdot f_p(q), \kappa)$ at q. Thus, any probation curve is κ-convex.

Since lifting of functions with given controlled concavity type has the same controlled concavity type (see **??**), we have that limit of probation curves is a probation curve. More precisely:

❶ *Let* $\kappa_n \to \kappa$, $\mathcal{L}_n \in \mathrm{CBB}^m[\kappa_n]$, $\mathcal{L}_n \xrightarrow[\mathrm{GH}]{\tau} \mathcal{L}$ *and* $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$ *(i.e.* $\mathcal{L}_n$ *is not a collapsing).*

Assume $\gamma_n\colon \mathbb{I} \to \mathcal{L}$ *is a sequence of probation curves and* $\gamma_n(t) \xrightarrow{\tau} \gamma(t)$ *for any* $t \in \mathbb{I}$. *Then* γ *is a probation curve.*

Now, let us continue the proof; assume $\gamma\colon \mathbb{I} \to \mathcal{L}$ is a probation curve. Since γ is κ-convex, it is 1-Lipschitz (see 20.1.3). Therfore, according to **??**, it is enough to show that

❷ $$\gamma^+(t_0) + \gamma^-(t_0) = 0 \quad \Rightarrow \quad |\gamma^\pm(t_0)| \geqslant 1.$$

for any $t_0 \in \mathbb{I}$.

Without loss of generality, we can assume that $t_0 = 0$. Set $p = \gamma(0)$. Consider curves $\gamma_n(t) \overset{def}{=\!=} n * \gamma(t/n)$ in the blowup $n * \mathcal{L}$. For the canonical convergence $n * \mathcal{L} \xrightarrow[\mathrm{GH}]{\tau_p} \mathrm{T}_p$ defined in **??**, we have

$$\gamma_n(t) \xrightarrow{\tau_p} \gamma_\infty(t) = \pm t{\cdot}\gamma^\pm(0).$$

Since $\gamma^+(0) + \gamma^-(0) = 0$ we can multiply $\gamma^\pm(0)$ by all real numbers — thus the last expression has sence. From Claim ❶, we get that γ_∞ is a probation curve in T_p.

Take $f\colon \mathrm{T}_p \to \mathbb{R}$ as in Lemma **??**. Then $f \circ \gamma_\infty(s)$ is (-1)-concave. On the other hand $\gamma^\pm(0) \in \mathrm{Lin}_p$, therefore

$$\begin{aligned} f \circ \gamma_\infty(t) &= f\big(t{\cdot}\gamma^+(0)\big) = \\ &= -\tfrac{1}{2}|\gamma^\pm(0)|^2{\cdot}t^2 + o(t^2). \end{aligned}$$

Thus, ❷ follows.

(a) ⇒ (b). We rewrite the proof given in [130, 6.1], it is based on the same idea as **??** and **??**.

In the proof we will use so called *Hamilton–Jacobi shifts* of f. It is a function analog of equidistant for hypersurface. It is named this way since the family of functions f_s can be also regarded as a generalized solution for the Hamilton–Jacobi equations $\frac{\partial f_s}{\partial s} + \frac{1}{2}{\cdot}|\nabla f_s|^2 = 0$.

❸ Hamilton–Jacobi shifts. *Let* $\mathcal{L}$ *be a metric space and* $f\colon \mathcal{L} \looparrowright \mathbb{R}$ *be a subfunction. A Hamilton–Jacobi shift of* f *for time* $s > 0$ *is a subfunction* $f_s\colon \mathcal{L} \looparrowright \mathbb{R}$ *defined as*

$$f_s(y) \overset{def}{=\!=} \min_{x \in \mathrm{Dom}\, f} f(x) + \tfrac{1}{2{\cdot}s}{\cdot}|x\,y|^2;$$

with $\mathrm{Dom}\, f_s$ *to be the maximal open set such that the minimum is defined.*

Let us list its basic properties. Note first that $f_{s_1+s_2} = (f_{s_1})_{s_2}$; i.e., using a more fancy language, $f \mapsto f_s$ is an *action of semigroup* $(\mathbb{R}_{\geqslant 0}, +)$.

Assume that f is $\pounds$-Lipschitz. Then for any $x \in \operatorname{Dom} f$, we have $f_s(x)$ is defined for all sufficiently small $s > 0$ and moreover,

$$0 \leqslant f(x) - f_s(x) \leqslant \tfrac{1}{2}\cdot \pounds^2 \cdot s.$$

In particular, $f_s(x) \to f(x)$ as $s \to 0+$.

By the definition, for any $y \in \operatorname{Dom} f_s$ there is *foot point* $x \in \operatorname{Dom} f$ such that

$$f_s(y) = f(x) + \tfrac{1}{2\cdot s}\cdot |xy|^2;$$

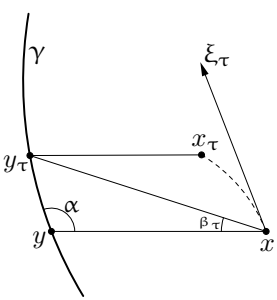
i.e., the function $f + \frac{1}{2\cdot s}\cdot |y*|^2$ has minimum at x. Clearly

❹ $$|xy| \leqslant \pounds \cdot s.$$

If we assume in addition that $\mathcal{L} \in \mathrm{CBB}^m$, then f_s is defined as a minimum of semiconcave functions. Thus, f_s is semiconcave for all $s > 0$. If in addition if f is native, then subfunction $f + \frac{1}{2\cdot s}\cdot |x*|^2$ is native, see **??**. Thus, if x is a foot point of $y \in \operatorname{Dom} f_s$ then, from the existence of supporting vector, we have $\boldsymbol{d}_y f + \frac{1}{2\cdot s}\cdot \boldsymbol{d}_y |x*|^2 = 0$. It follows that the geodesic $[xy]$ is uniquely defined, $\log[xy] \in \operatorname{Lin}_y$ and $(\boldsymbol{d}_x f)(v) = \frac{1}{s}\cdot \langle \log[xy], v\rangle$ for any $v \in \mathrm{T}_x$.

Now, we ready for the proof. First note that without loss of generality, we can assume that f is $\pounds$-Lipschitz for some $\pounds \in \mathbb{R}$; otherwise restrict f to a smaller domain. It is sufficient to show that for any interior value $t_0 \in \mathbb{I}$, $s > 0$ and all sufficiently small τ, we have

❺ $$f_s \circ \gamma(t_0 + \tau) \leqslant f_s \circ \gamma(t_0) + c\cdot\tau + \tfrac{1}{2}\cdot[\lambda - O(s)]\cdot\tau^2.$$



Set $y = \gamma(t_0)$ and $y_\tau = \gamma(t_0 + \tau)$. Let x be a foot point of y; i.e. the function $f + \frac{1}{2\cdot s}\cdot |y*|^2$ has a minimum at x (we can assume $x \neq y$, otherwise ❺ follows since $f_s \leqslant f$, $\boldsymbol{d}_x f \equiv 0$ and $|y y_\tau|^2 \leqslant \tau$).

Set $\alpha = \measuredangle(\log[yx], \gamma^+(0))$ and $\beta_\tau = \measuredangle[x\,{}^{y}_{y_\tau}]$. Since T_p splits in the direction of $\log[xy]$, there is direction $\xi_\tau \in \Sigma_x$ such that

$$\measuredangle(\xi_\tau, \operatorname{dir}[xy]) = \measuredangle(\xi_\tau, \operatorname{dir}[xy_\tau]) + \beta_\tau = \pi - \alpha.$$

Define $x_\tau = \operatorname{gexp}^0_x(\tau\cdot\xi_\tau)$.

First note that

❻ $$|x_\tau y_\tau|^2 \leqslant |xy|^2 + [1 + O(s^2)]\cdot\tau^2.$$

Indeed, for any $p \in \mathcal{L}$, the function $\frac{1}{2}\cdot|p_*|^2$ is $\big[1+O(r^2)\big]$-concave in $\mathrm{B}(p,r)$. Thus, from ❹ and **??**, for all small τ, we have

❼
$$\begin{aligned}|x_\tau y_\tau|^2 &\leqslant |x y_\tau|^2 + 2\cdot\tau\cdot|x y_\tau|\cdot\cos(\alpha+\beta_\tau) + \big[1+O(s^2)\big]\cdot\tau^2,\\ |x y_\tau|^2 &\leqslant |x y|^2 - 2\cdot\tau\cdot|x y|\cdot\cos\alpha + \big[1+O(s^2)\big]\cdot\tau^2\end{aligned}$$

Note also that $|x y_\tau| = |x y| - \tau\cdot\cos\alpha + o(\tau)$ and $|y y_\tau| = \tau + o(\tau)$; therefore

❽
$$\beta_\tau > \frac{\tau\cdot\sin\alpha}{\mathrm{sn}^\kappa|x y|} + o(t).$$

From ❼ and ❽, we get ❻.

Further, from **??**,

❾
$$\begin{aligned}f(x_\tau) &\leqslant f(x) + (\boldsymbol{d}_x f)(\xi_\tau)\cdot\tau + \tfrac{\lambda}{2}\cdot\tau^2 =\\ &= f(x) - \tfrac{|x y|}{s}\cdot(\cos\alpha)\cdot\tau + \tfrac{\lambda}{2}\cdot\tau^2.\end{aligned}$$

Since,

$$f_s(y_\tau) \leqslant f(x_\tau) + \tfrac{1}{2\cdot s}\cdot|x_\tau y_\tau|^2,$$

❹, ❻ and ❾ imply ❺. □

20.4 Pre-quasigeodesics

Pre-quasigeodesic is an intermediate class of curves — it only used to construct quasigeodesics. It was introduced by Perelman, he first proved existence of pre-quasigeodesics and only few year later existence of quasigeodesics.

The main result of this section is existence of pre-quasigeodesics — theorem 20.4.2. We also prove a number of properties of pre-quasigeodesics which will be needed the construction of quasigeodesics in the next section.

20.4.1. Definition. *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$. A κ-convex curve $\gamma\colon \mathbb{I} \to \mathcal{L}$ is called a* pre-quasigeodesic *if for any $a \in \mathbb{I}$ such that $|\gamma^+(a)| > 0$, the curve γ^a defined by*

$$\gamma^a(t) \overset{def}{=\!=} \gamma\left(a + \tfrac{t}{|\gamma^+(a)|}\right)$$

is κ-convex for $t \geqslant 0$, and if $|\gamma^+(a)| = 0$ then $\gamma(t) = \gamma(a)$ for all $t \geqslant a$.

Formally speaking, this definition of depends on κ, but, in this and the next section we fix some $\kappa \leqslant 0$ such that $\mathcal{L} \in \mathrm{CBB}[\kappa]$, thus it will be no ambiguity.

20.4.2. Existence of pre-quasigeodesics. *Let* $\kappa \leqslant 0$, $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$, $x \in \mathcal{L}$, $v \in \mathrm{T}_x$ *and* $|v| \leqslant 1$. *Then there is a pre-quasigeodesic* $\gamma\colon [0,\infty) \to \mathcal{L}$ *such that* $\gamma^+(0) = v$.

Pre-quasigeodesics will be constructed in two steps. First we construct κ-convex curves out of radial curves, which are κ-*monotonic* in the sense of the following definition. Second we construct pre-quasigeodesics out of κ-convex.

20.4.3. Definition. *Let* $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$. *A* 1-*Lipschitz curve* $\alpha\colon \mathbb{I} \to \mathcal{L}$ *is called* κ-monotonic *with respect to* $t_0 \in \mathbb{I}$ *if for any* $p \in \mathcal{L}$ *the function* $t \mapsto \tilde{\angle}^\kappa(\alpha|_{t_0}^t, p)$ *is non-increasing for* $t > t_0$ *in the domain of definition.*

Here is a construction which makes possible to join κ-monotonic curves as well as κ-convex curves.

20.4.4. Extention procedure I. *Let* $\kappa \leqslant 0$, $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$ *and* $\alpha_1\colon [a,\infty) \to \mathcal{L}$, $\alpha_2\colon [b,\infty) \to \mathcal{L}$ *be two* κ-*monotonic curves with respect to* a *and* b *respectively. Assume*

$$a \leqslant b, \quad \alpha_1(b) = \alpha_2(b) \quad \text{and} \quad \alpha_1^+(b) = \alpha_2^+(b).$$

Then its joint

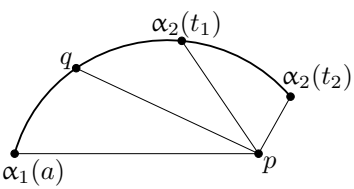
$$\beta\colon [a,\infty) \to \mathcal{L}, \quad \beta(t) = \begin{bmatrix} \alpha_1(t) & \text{if} & t < b \\ \alpha_2(t) & \text{if} & t \geqslant b \end{bmatrix}$$

is κ-*monotonic with respect to* a *and* b.

Moreover, if in addition, α_1 *and* α_2 *are* κ-*convex curves then so is* β.

Proof. It is enough to show that for any $p \in \mathcal{L}$,

❶ the function $t \mapsto \tilde{\angle}^\kappa(\beta|_a^t, p)$ is non-increasing for $t \geqslant b$.



Set $q = \beta(b) = \alpha_1(b) = \alpha_2(b)$. Applying Alexandrov's lemma twice, the statement ❶ can be reduced to the following two inequalities for $b < t_1 < t_2$:

❷ $\tilde{\angle}^\kappa(\alpha_2|_{t_1}^b, p) + \tilde{\angle}^\kappa(\alpha_2|_{t_1}^{t_2}, p) \leqslant \pi$,

❸ $$\tilde{\angle}^\kappa(\alpha_1|_b^a, p) + \tilde{\angle}^\kappa(\alpha_2|_b^{t_1}, p) \leqslant \pi.$$

Inequality ❷ follows directly from κ-monotonicity of α_2 — again Alexandrov's lemma.

To prove ❸, set $\varphi = \sup\{\, \angle^{\kappa}(\alpha_2|_b^t, p) \mid t > b \,\}$. Since α_2 is κ-monotonic w.r.t. b, we have $\cos\varphi = -(\boldsymbol{d}_q|p_*|)(v)$, where $v = \alpha_1^+(b) = \alpha_2^+(b)$. In particular,

$$\varphi = \lim_{t\to b+} \angle^{\kappa}(\alpha_1|_b^t, p).$$

Yet applying Alexandrov's lemma, from κ-monotonicity of α_1 w.r.t. a, we get

$$\angle^{\kappa}(\alpha_1|_b^a, p) + \varphi \leqslant \pi$$

and ❸ follows.

The last part follows from 20.1.4*b*. □

20.4.5. Lemma. *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$. Assume $\alpha_n\colon [0,a] \to \mathcal{L}$ be a sequence of κ-monotonic curves w.r.t. 0 and $\alpha_n(t) \to \alpha(t)$ for each $t \in [0,a]$. Then α is κ-monotonic.*

Moreover, if for some $x \in \mathcal{L}$ we have $\alpha_n(0) = x$ and $|\alpha^+(0)| = 1$ then $\alpha_n^+(0) \to \alpha^+(0)$.

Proof of 20.4.5. The first part follows directly from Definition 20.4.3. Note that for any p,

$$(|p_*| \circ \alpha)^+(0) = -\cos\left(\lim_{t\to 0+} \angle^{\kappa}(\alpha|_0^t, p)\right).$$

From monotonicity, the limit is defined. Hence $(|p_*| \circ \alpha)^+(0)$ and, from **??**, also $\alpha^+(0)$ are defined.

Set $\xi_n = \alpha^+(0)$. Passing to a subsequence if nesesurily, we can assume that $\xi_n \to \xi \in \Sigma_x$. For any geodesic $[xp]$, we have

$$\varlimsup_{t\to t_0+} \angle^{\kappa}(\alpha_n|_0^t, p) \leqslant \measuredangle(\xi_n, \mathrm{dir}[xp]).$$

Since, α_n is κ-monotonic, we have that $\angle^{\kappa}(\alpha_n|_0^t, p) \leqslant \measuredangle(\xi_n, \mathrm{dir}[xp])$ and hence $\angle^{\kappa}(\alpha|_0^t, p) \leqslant \measuredangle(\xi, \mathrm{dir}[xp])$, for any $t > 0$. In other words,

❹ $$\langle \alpha^+(0), \mathrm{dir}[xp]\rangle \geqslant \cos\measuredangle(\xi, \mathrm{dir}[xp]).$$

Note that $|\alpha^+(0)| \leqslant 1$ — monotonic curves are by definition 1-Lipschitz. Thus, applying ❹ for $\mathrm{dir}[xp] \approx \xi$, we get $\alpha^+(0) = \xi$. □

20.4.6. Existence of convex curve. *Let $\kappa \leqslant 0$, $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$, $x \in \mathcal{L}$ and $v \in \mathrm{T}_x$ be such that $|v| \leqslant 1$. Then there is a κ-convex curve $\beta_v\colon [0,\infty) \to \mathcal{L}$ with $\beta_v^+(0) = v$.*

Proof. For $z \in \mathcal{L}$ and $w \in \mathrm{T}_z$ such that $|w| \leqslant 1$, consider the curve $\alpha_w\colon [0,\infty) \to \mathcal{L}$ defined as

$$\alpha_w(t) = \mathrm{gexp}_z^{\kappa}\, t{\cdot}w.$$

Directly from the construction of gradient exponential map (section 17.3), we have $\alpha_w^+(t)$ is defined for all $t \geqslant 0$ and α_w is 1-Lipschitz. According to Theorem 17.3.1*c*, α_w is κ-monotonic.

Fix $\varepsilon > 0$. Given $\xi \in \Sigma_x$, consider the following recursively defined sequence of curves $\alpha_{v_n}(t)$ such that $v_0 = \xi$ and $v_n = \alpha_{v_{n-1}}^+(\varepsilon)$. Then joint all $\alpha_{v_n}|[0,\varepsilon]$ into a new curve

$$\beta_{\xi,\varepsilon}(t) = \alpha_{v_n}(t - n\cdot\varepsilon), \quad \text{where} \quad n = \lfloor t/\varepsilon \rfloor.$$

Applying extension procedure I (20.4.4), we get that $\beta_{\xi,\varepsilon}\colon [0,\infty) \to \mathcal{L}$ is κ-monotonic with respect to every parameter value of the form $n\varepsilon$, $n \in \mathbb{Z}_{\geqslant 0}$. Passing to a partial limit, $\beta_{\xi,\varepsilon} \to \beta_\xi$ as $\varepsilon \to 0$ we get a curve β_ξ which is κ-monotonic w.r.t. any value of parameter. Thus, according to 20.1.4*b*, β_ξ is κ-convex and from 20.4.5, $\beta_\xi^+(0) = \xi$.

Finally, for $v = r\cdot\xi$, set $\beta_v(t) = \beta_\xi(r\cdot t)$. If $|v| \leqslant 1$, the curve β_v is κ-convex; that follows directly from definition 20.1.1 since $\kappa \leqslant 0$. □

Proof of 20.4.2. For each point $z \in \mathcal{L}$ and $w \in \mathrm{T}_z$ such that $|w| \leqslant 1$, choose a κ-convex curve $\beta_w\colon [0,\infty) \to \mathcal{L}$ with $\beta_w^+(0) = w$; it exists due to 20.4.6.

Let us construct a κ-convex curve $\gamma_\varepsilon\colon [0,\infty) \to \mathcal{L}$ with $\gamma_\varepsilon^+(0) = v$, such that there is a representation of $[0,\infty)$ as a countable union of disjoint half-open intervals $[a_i, b_i)$, such that $|a_i - b_i| \leqslant \varepsilon$ and for any $t \in [a_i, b_i)$ we have

$$|\gamma_\varepsilon^+(a_i)| \geqslant |\gamma_\varepsilon^+(t)| \geqslant (1-\varepsilon)|\gamma_\varepsilon^+(a_i)|; \tag{❺}$$

moreover, for each i, the curve

$$\gamma_\varepsilon^{a_i}(t) = \gamma_\varepsilon\left(a_i + \tfrac{t}{|\gamma_\varepsilon^+(a_i)|}\right)$$

is also κ-convex for $t \geqslant 0$.

Construction of γ_ε. Assume we already can construct γ_ε in the interval $[0,a)$, and cannot extend it any further. Since γ_ε is 1-Lipschitz, we can extend it continuously to $[0,a]$. Use 16.3.2 to construct a vector w polar to $\gamma_\varepsilon^-(a)$ with $|w| \leqslant |\gamma_\varepsilon^-(a)|$ (in case $a = 0$ take $\gamma(0) = x$ and $w = v$). Joint γ_ε with a short half-open segment of β_w. According to the extension procedure I (20.4.4) the new curve is convex. From 20.2.4 we get that the added segment satisfies ❺, a contradiction.

Choose a sequence $\varepsilon_n \to 0+$ such that γ_{ε_n} converges to a curve γ. From Corollary 20.2.3???, $|\gamma^+(t)| \overset{a.e.}{=\!=} \lim |\gamma_{\varepsilon_n}^+(t)|$. Combining this with inequality ❺ and 20.2.4, we get that γ^a is κ-convex for any $a \geqslant 0$. Applying 20.4.5 for $\gamma_n^0 \to \gamma^0$, we get $\gamma^+(0) = v$. □

Entropy of pre-quasigeodesics. Let us define *entropy* of pre-quasigeodesic; it measures "how far" a pre-quasigeodesic is from being a quasigeodesic.

20.4.7. Definition. *Let γ be a pre-quasigeodesic defined on real interval $\mathbb{I}$. The* entropy *of γ, is a measure μ_γ defined on the interior of $\mathbb{I}$ such that*

$$\mu_\gamma(a,b) = \ln|\gamma^+(a)| - \ln|\gamma^-(b)|.$$

for any open subinterval $(a,b) \subset \mathbb{I}$.

Note that since γ^a is κ-convex, we have that for any $b > a$, $|\gamma^+(a)| \geqslant |\gamma^\pm(b)|$. In particular entropy is non-negative (i.e. it is indeed a measure). From 20.2.4, it follows that for any interior value $a \in \mathbb{I}$, we have

$$\mu_\gamma\{a\} = \ln|\gamma^-(a)| - \ln|\gamma^+(a)|.$$

Directly from the definition we have that if $|\gamma^+(a)| = 1$ and $\mu_\gamma(a,b) = 0$ then $\gamma|[a,b]$ is a quasigeodesic.

The following proposition follows directly from Corollary 20.2.2???, Lemma 20.4.5 and above discussion.

20.4.8. Proposition. *Let $\kappa \leqslant 0$, $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$ and $\gamma_n\colon \mathbb{I} \to \mathcal{L}$ is a sequence of pre-quasigeodesics. Assume $\gamma_n(t) \to \gamma(t)$ for each $t \in \mathbb{I}$, then γ is a pre-quasigeodesic. Moreover, $\mu_{\gamma_n} \rightharpoonup \mu_\gamma$; i.e., the entropies of γ_n weakly converge to the entropy of γ.*

In particular, if $\mu_{\gamma_n}(a,b) \to 0$ and $|\gamma_n^+(a)| = 1$ then the restriction $\gamma|[a,b]$ is a quasigeodesic.

The next construction is similar to the extension procedure I (20.4.4) and it can be proved the same way.

20.4.9. Extention procedure II. *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$, $\gamma_1\colon [a,\infty) \to \mathcal{L}$ and $\gamma_2\colon [b,\infty) \to \mathcal{L}$ be two pre-quasigeodesics. Assume*

$$a \leqslant b,\ \ \gamma_1(b) = \gamma_2(b),\ \ \gamma_1^-(b) \text{ is polar to } \gamma_2^+(b) \text{ and } |\gamma_1^-(b)| \geqslant |\gamma_2^+(b)|$$

then its joint

$$\gamma\colon [a,\infty) \to \mathcal{L},\quad \gamma(t) = \begin{cases} \gamma_1(t) & \text{if } t \leqslant b \\ \gamma_2(t) & \text{if } t \geqslant b \end{cases}$$

is a pre-quasigeodesic. Moreover, its entropy is defined by

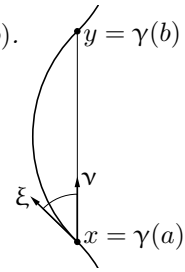
$$\mu_\gamma|(a,b) = \mu_{\gamma_1},\ \ \mu_\gamma|(b,\infty) = \mu_{\gamma_2} \text{ and } \mu_\gamma(\{b\}) = \ln|\gamma^-(b)| - \ln|\gamma^+(b)|.$$

20.4.10. Chopping procedure. *Let $\kappa \leqslant 0$, $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$, $\gamma\colon [0,\infty) \to \mathcal{L}$ be a pre-quasigeodesic. Assume for $a \geqslant 0$ we have $|\gamma^+(a)| > 0$,*

then for any $\varepsilon > 0$ *there is* $b > a$ *such that*

$$\mu_\gamma(a,b) < \varepsilon[\measuredangle(\xi,\nu) + b - a], \quad b - a < \varepsilon, \quad \text{and} \quad \measuredangle(\xi,\nu) < \varepsilon,$$

where $\xi = \frac{\gamma^+(a)}{|\gamma^+(a)|}$, $\nu = \operatorname{dir}[xy]$, $x = \gamma(a)$ *and* $y = \gamma(b)$.



Proof. Set $s = b - a$. Clearly, if $s > 0$ is sufficiently small, we have $\measuredangle(\xi,\nu) < \varepsilon$. Further, from κ-convexity of γ^a it follows that

$$\mu_\gamma(a, a + \tfrac{s}{3}) < c\cdot[\measuredangle(\xi,\nu)]^2\,.$$

The following exercise completes the proof. □

20.4.11. Exercise. *Let* $\varphi,\psi\colon \mathbb{R}_> \to \mathbb{R}_>$ *be two functions such that*

a) $\varphi(s) \to 0$ *as* $s \to 0+$*;*

b) for any sufficiently small $s > 0$ *we have*

$$\psi(\tfrac{s}{3}) \leqslant (\varphi(s))^2.$$

Show that for any $\varepsilon > 0$ *there is* $s > 0$ *such that*

$$\psi(s) < 10(\varphi(s) + s)^2 \quad \text{and} \quad \varphi(s) + s \leqslant \varepsilon.$$

20.5 Existence of quasigeodesics

Here is the main result of this section. It was proved by Perelman around 1992.

20.5.1. Existence of quasigeodesics. *Let* $\mathcal{L} \in \mathrm{CBB}^m$. *Then for any point* $x \in \mathcal{L}$, *and any direction* $\xi \in \Sigma_x$ *there is a quasigeodesic* $\gamma\colon \mathbb{R} \to \mathcal{L}$ *such that* $\gamma^+(0) = \xi$.

The prove relies heavily on the results in the previous section.

Proof. Fix $\kappa \leqslant 0$ such that $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$.

Note that base case $m = 1$ is trivial. Thus, making an induction argument, we can assume that 20.5.1 holds in all dimensions smaller than m. This assumption is used just once, to prove the following lemma; the proof is taken from [110].

20.5.2. Milka's lemma. *Let* $m \geqslant 2$, $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$, $x \in \mathcal{L}$ *and* $\xi \in \Sigma_x$. *Then there is a direction* $\xi^* \in \Sigma_x$ *polar to* ξ*; i.e. such that*

$$\langle \xi, v\rangle + \langle \xi^*, v\rangle \geqslant 0$$

for any $v \in \mathrm{T}_x$.

Proof. Note that $\Sigma_x \in \mathrm{CBB}^{m-1}[1]$ (see **??**). Applying existence of quasigeodesics (20.5.1) in dimension $m-1$, for any $\xi \in \Sigma_x$ there is a quasigeodesic[2] γ in Σ_x with $\gamma(0) = \xi$. Set $\xi^* = \gamma(\pi)$; the quasigeodesic comparison (20.1.4c') implies that

$$|\xi\eta| + |\eta\,\xi^*| = \measuredangle(\xi,\eta) + \measuredangle(\eta,\xi^*) \le \pi$$

for any $\eta \in \Sigma_x$. Thus, ξ is polar to ξ^*. □

Let $\Theta \subset \mathcal{L}$ be a maximal open set which satisfies the following property: *for any* $x \in \Theta$ *and any* $\xi \in \Sigma_x$ *there is a quasigeodesic* $\gamma\colon [0,r] \to \mathcal{L}$ *with* $\gamma^+(0) = \xi$ *and* $r > 0$.

❶ Definition. *Let* $\Omega \subset \mathcal{L}$ *be an open subset. A pre-quasigeodesic* γ *in* $\mathcal{L}$ *is called* Ω-pre-quasigeodesic *if its entropy vanishes on* Ω*; i.e.* $\mu_\gamma(\gamma^{-1}(\Omega)) = 0$.

❷ *For any* $x \in \mathcal{L}$ *and any* $v \in \mathrm{T}_x$ *such that* $|v| \leqslant 1$ *there is an* Θ*-pre-quasigeodesic* $\gamma\colon [0,\infty) \to \mathcal{L}$ *with* $\gamma^+(0) = v$.

❸ *In order to prove 20.5.1, it is sufficient to show that* $\mathcal{L} = \Theta$

Proof of claims ❷ *and* ❸. Fix $\varepsilon > 0$ and set $\Theta_\varepsilon = \{\, x \in \Theta \mid \overline{\mathrm{B}}[x,\varepsilon] \subset \Theta \,\}$. Clearly, Θ_ε is an open set.

Let us construct a Θ_ε-pre-quasigeodesic $\gamma_\varepsilon\colon [0,\infty) \to \mathcal{L}$ with $\gamma_\varepsilon^+(0) = v$. Assume we have a Θ_ε-pre-quasigeodesic $\gamma_\varepsilon\colon [0,a) \to \mathcal{L}$ with $\gamma_\varepsilon^+(0) = \xi$. We can extend γ_ε to $[0,a]$ by setting $\gamma_\varepsilon(a) = \lim_{t\to a-} \gamma_\varepsilon(t)$; this limit exists since κ-convex curves are 1-Lipschitz, see 20.1.3d (if $a = 0$ set $\gamma_\varepsilon(0) = x$).

For $\gamma_\varepsilon\colon [0,a] \to \mathcal{L}$, applying Milka's lemma, we can find a vector w which is polar to $\gamma_\varepsilon^-(a)$ and such that $|w| = |\gamma_\varepsilon^-(a)|$ (if $a = 0$, set $w = v$). Further:

- ◇ If $|w| = 0$, set $\gamma_\varepsilon(t) = \gamma_\varepsilon(a)$ for all $t \geqslant a$.

Otherwise:

- ◇ If $\gamma_\varepsilon(a) \in \Theta$ then, from the assumption of the claim, there is a quasigeodesic $\bar\gamma\colon [0,r] \to \mathcal{L}$ with $\bar\gamma^+(0) = w/|w|$. Let us extend γ_ε behind a as $\gamma_\varepsilon(a+t) = \bar\gamma(|w|\cdot t)$.
- ◇ If $\gamma_\varepsilon(a) \notin \Theta$, according to 20.4.2, we can find a pre-quasigeodesic $\bar\gamma\colon [0,\varepsilon] \to \mathcal{L}$ with $\bar\gamma^+(0) = w$. Then extend γ_ε behind a as $\gamma_\varepsilon(a+t) = \bar\gamma(t)$, clearly $\gamma_\varepsilon([a,a+\varepsilon]) \subset \mathcal{L}\backslash\Theta_\varepsilon$.

[2]Note that according to 20.3.1, it is also 1-quasigeodesic.

According to extension procedure II (20.4.9), in all above cases, the extended γ_ε is still Θ_ε-pre-quasigeodesic. Thus, the maximal interval of definition of γ_ε, must be open and closed in $[0,\infty)$; hence the desired γ_ε exists.

Now take a converging sequence $\gamma_{\varepsilon_n}\to\gamma$ as $\varepsilon_n\to 0+$. Appling Lemma 20.4.5 for $\gamma_n^0\to\gamma^0$, we get $\gamma^+(0)=v$. From 20.4.8, γ is Θ-pre-quasigeodesic.

In case $\Theta=\mathcal{L}$ and $|v|=1$, the constructed pre-quasigeodesic $\gamma\colon[0,\infty)\to\mathcal{L}$ is also a quasigeodesic. It only remains to extend γ to negative part of real line. To do this we can apply open-close argument as above in a backward direction. $\triangle$

According to claim ❸, the following claim implies the theorem.

❹ Key claim. *If $\Theta\ne\mathcal{L}$ then there is $p\in\mathcal{L}\backslash\Theta$ and $r>0$ such that for any $x\in\mathrm{B}(p,\frac{r}{6})\backslash\Theta$ and $\xi\in\Sigma_x$ there is a quasigeodesic $\gamma\colon[0,\frac{r}{6}]\to\mathcal{L}$ with $\gamma^+(0)=\xi$.*

First let us list the properties of $p\in\mathcal{L}$ and $r>0$ which are sufficient for the key claim to hold.

Set $\mathfrak{C}=\mathcal{L}\backslash\Theta$. The following claim is an extract from Lemma 20.5.3.

❺ *Given a nonempty closed subset $\mathfrak{C}\subset\mathcal{L}$, there is a point $p\in\mathfrak{C}$, a point array $q_1,q_2,\dots,q_{\bar k}$ in $\mathcal{L}$ and sufficiently small $r>0$ (say $r<\frac{1}{100\cdot(1+|\kappa|)}$ and $r<\frac{1}{100}\cdot|pq_i|$ for each i) such that*

a) For each $x\in\mathfrak{C}\cap\mathrm{B}(p,r)$, the set $\{\mathrm{dir}[xq_i]\}$ forms an $\frac{\pi}{10}$-net of Σ_x.

b) For each $x\in\mathfrak{C}\cap\mathrm{B}(p,r)$, if $\xi\in\Sigma_x\mathfrak{C}$, $\nu\in\Sigma_x$ and $\measuredangle(\xi,\nu)<\frac{\pi}{10}$, then for some i there is a choice of geodesic $[xq_i]$ such that

$$\measuredangle(\xi,\nu)\leqslant 2[\measuredangle(\xi,\mathrm{dir}[xq_i])-\measuredangle(\nu,\mathrm{dir}[xq_i])],\quad \tfrac{3}{5}\cdot\pi<\measuredangle(\xi,\mathrm{dir}[xq_i])<\tfrac{4}{5}\cdot\pi.$$

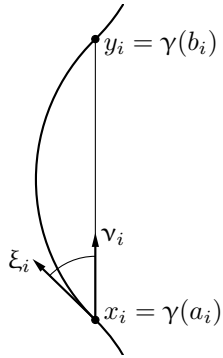
Proof of claim ❺. Fix $\varepsilon>0$ and choose $p\in\mathfrak{C}$ and $r>0$ as in Lemma 20.5.3.

Choose a point array $q_1,q_2,\dots,q_{\bar k}$ such that directions $\mathrm{dir}[pq_i]$ form a ε-net in Σ_p. We can assume that for any $x\in\mathrm{B}(p,r)$, the values $\tilde\measuredangle^\kappa(x\,{}^{q_i}_{q_j})$, $\tilde\measuredangle^\kappa(p\,{}^{q_i}_{q_j})$ and $\measuredangle[p\,{}^{q_i}_{q_j}]$ are ε-close to each-other (otherwise move q_i to p along geodesics and pass to smaller $r>0$). Since $\measuredangle[x\,{}^{q_i}_{q_j}]\geqslant\tilde\measuredangle^\kappa(x\,{}^{q_i}_{q_j})$ and $\mathrm{D}_{\mathsf{GH}}(\Sigma_x,\Sigma_p)<\varepsilon$, the set of directions $\mathrm{dir}[xq_i]$ forms a $10\cdot\varepsilon$-net in Σ_x; hence (**??**).

For any $x\in\mathrm{B}(p,r)\cap\mathfrak{C}$ and $\xi\in\Sigma_x\mathfrak{C}$ we have $\mathrm{D}_{\mathsf{GH}}(\Sigma_\xi\mathrm{T}_x,\Sigma_p)<\varepsilon$. Thus, if $\varepsilon>0$ is small enough, one can choose a point q_i such that

$$\tfrac{3}{5}\cdot\pi<\measuredangle(\xi,\mathrm{dir}[xq_i])<\tfrac{4}{5}\cdot\pi\quad\text{and}\quad\measuredangle[\xi\,{}^{\mathrm{dir}[xq_i]}_{\nu}]<\tfrac{\pi}{10}.$$

Hence (**??**). $\triangle$



Proof of key claim ❹. Given $\varepsilon > 0$, we first construct a Θ-pre-quasigeodesic $\gamma_\varepsilon\colon [0, \frac{r}{6}] \to \mathcal{L}$ with $\gamma_\varepsilon^+(0) = \xi$ and an open set $\Upsilon_\varepsilon \subset (0, \frac{r}{6})$ formed by countable disjoined union of intervals (a_i, b_i) such that the enthropy μ_{γ_ε} vanish on $(0, \frac{r}{6})\backslash\Upsilon_\varepsilon$ and for each i we have

(i) $\xi_i = \frac{\gamma^+(a_i)}{|\gamma^+(a_i)|} \in \Sigma_{\gamma(a_i)}\mathfrak{C}$;

(ii) If $x_i = \gamma(a_i)$, $y_i = \gamma(b_i)$ and $\nu_i = \operatorname{dir}[x_i y_i]$ then $\measuredangle(\xi_i, \nu_i) < \varepsilon$ and

$$\mu_\gamma(a_i, b_i) \leqslant \varepsilon\cdot[\measuredangle(\xi_i, \nu_i) + b_i - a_i]\,.$$

After that we will show that $\mu_{\gamma_\varepsilon}(0, \frac{r}{6}) \to 0$ as $\varepsilon \to 0$.

Finally, we consider γ, a partial limit of γ_ε as $\varepsilon \to 0$. From 20.4.8, γ is a quasigeodesic and according to 20.4.5, $\gamma^+(0) = \xi$, which proves the claim.

The construction of γ_ε. It almost repeats the construction in the proof of Claim ❷. Assume that we have constructed γ_ε on an interval $[0, a]$, $a < \frac{r}{6}$ and cannot extend it any further. Applying Milka's lemma (20.5.2), we can find a vector $w \in \mathrm{T}_p$ which is polar to $\gamma_\varepsilon^-(a)$ and such that $|w| = |\gamma_\varepsilon^-(a)|$. Then extend γ by a Θ-pre-quasigeodesic with $\gamma^+(a) = w$; it exists according to claim ❷. Clearly,

$$\mu_{\gamma_\varepsilon}\{a\} = \ln|\gamma^-(a)| - \ln|\gamma^+(a)| = 0.$$

Further:

- ◇ If $\frac{\gamma^+(a)}{|\gamma^+(a)|} \in \Sigma_{\gamma(a_i)}\mathfrak{C}$, apply chopping procedure (20.4.10) starting from a. In this case include the chopped interval (a, b) in Υ_ε.
- ◇ If $\frac{\gamma^+(a)}{|\gamma^+(a)|} \notin \Sigma_{\gamma(a_i)}\mathfrak{C}$, then for on some $b > a$ the curve interval $\gamma|(a, b)$ lies in Θ, therefore its entropy vanish.

In both cases, we get a longer curve $\gamma\colon [0, b] \to \mathcal{L}$ with the desired property, a contradiction

Vanishing of entropy: From the condition (ii) above, we have that

$$\mu_{\gamma_\varepsilon}(0, \tfrac{r}{6}) < \varepsilon\cdot\left[\tfrac{r}{6} + \sum_i \measuredangle(\xi_i, \nu_i)\right].$$

Therefore, to show that $\mu_{\gamma_\varepsilon}(0, \frac{r}{6}) \to 0$, it only remains to show that $\sum_i \measuredangle(\xi_i, \nu_i)$ is bounded above by a constant independent of ε.

Since $\xi_i \in \Sigma_{x_i}\mathfrak{C}$, according to claim ❺, for each i there is q_j such that

❻ $$\measuredangle(\xi_i, \nu_i) < 2\cdot[\measuredangle(\operatorname{dir}[x_i q_j], \xi_i) - \measuredangle(\operatorname{dir}[x_i q_j], \nu_i)]$$

and moreover,

❼ $$\tfrac{3}{5}\cdot\pi < \measuredangle(\operatorname{dir}[x_iq_j],\xi_i) < \tfrac{4}{5}\cdot\pi \quad\text{and}\quad \measuredangle(\xi_i,\nu_i) < \varepsilon < \tfrac{\pi}{10}.$$

In particular, $\measuredangle(\operatorname{dir}[x_iq_j],\nu_i) > \frac{\pi}{2}$. Note that, $\gamma \subset \mathrm{B}(p,\frac{r}{3})$ and thus, any geodesic $[x_iy_i]$ lies in $\mathrm{B}(p,r)$. Set $\lambda = 1/r$, since r is small, we have

$$\begin{aligned}|q_j\,y_i| - |q_j\,x_i| &\leqslant -|x_i\,y_i|\cdot\cos\measuredangle(\operatorname{dir}[x_iq_j],\nu_i) + \tfrac{\lambda}{2}\cdot|x_i\,y_i|^2 \leqslant\\ &\leqslant -(b_i-a_i)\cdot\cos\measuredangle(\operatorname{dir}[x_iq_j],\nu_i) + \tfrac{\lambda}{2}\cdot(b_i-a_i)^2.\end{aligned}$$

Since γ is κ-convex,

$$|q_j\,x_i| - |q_j\,y_i| \leqslant -(b_i-a_i)\cdot\langle\operatorname{dir}[y_iq_j],\gamma^-(b_i)\rangle + \tfrac{\lambda}{2}\cdot(b_i-a_i)^2.$$

Therefore,

❽ $$\cos\measuredangle(\operatorname{dir}[x_iq_j],\nu_i) \leqslant -\langle\operatorname{dir}[y_iq_j],\gamma^-(b_i)\rangle + \lambda\cdot(b_i-a_i).$$

Since $\gamma^+(a_i)$ is polar to $\gamma^-(a_i)$, we have

❾ $$\begin{aligned}\cos\measuredangle(\operatorname{dir}[x_iq_j],\xi_i) &= \tfrac{1}{|\gamma^+(a_i)|}\cdot\langle\operatorname{dir}[x_iq_j],\gamma^+(a_i)\rangle \geqslant\\ &\geqslant -\tfrac{1}{|\gamma^+(a_i)|}\cdot\langle\operatorname{dir}[x_iq_j],\gamma^-(a_i)\rangle\end{aligned}$$

Note that $\gamma^+(a_i) \geqslant \frac{1}{2}$; it follows from κ-convexity of γ since there is j such that $\measuredangle(\xi,\operatorname{dir}[xq_j]) < \frac{\pi}{10}$, see property a. Set

$$\vartheta_{q_j}[0,b) = 1 - \langle\operatorname{dir}[y_iq_j],\gamma^-(b)\rangle + \lambda\cdot b.$$

Clearly, $b \mapsto -\langle\operatorname{dir}[y_iq_j],\gamma^-(b)\rangle + \lambda\cdot b$ is non-decreasing; thus ϑ_{q_j} defines a positive measure on $[0,\frac{r}{6})$. Using ❼, ❽ and ❾, we can continue ❻:

$$\begin{aligned}\measuredangle(\xi_i,\nu_i) &< 2\cdot[\measuredangle(\operatorname{dir}[x_iq_j],\xi_i) - \measuredangle(\operatorname{dir}[x_iq_j],\nu_i)] \leqslant\\ &\leqslant 10\cdot[\cos\measuredangle(\operatorname{dir}[x_iq_j],\nu_i) - \cos\measuredangle(\operatorname{dir}[x_iq_j],\xi_i)] \leqslant\\ &\leqslant 20\cdot\big[\langle\operatorname{dir}[x_iq_j],\gamma^-(a_i)\rangle - \langle\operatorname{dir}[y_iq_j],\gamma^-(b_i)\rangle + \lambda(b_i-a_i)\big] =\\ &= 20\cdot\vartheta_{q_j}[a_i,b_i)\end{aligned}$$

In the exceptional case $a_i = 0$, although $\gamma^-(0)$ might be undefined, the last inequality holds. Since the sistance function $|q_j\,*|$ is 1-Lipschitz, we get $\vartheta_{q_j}[0,\frac{r}{6}) \leqslant 2 + \lambda r$. Therefore,

$$\sum_i \measuredangle(\xi_i,\nu_i) \leqslant 20\cdot\sum_j \vartheta_{q_j}[0,\tfrac{r}{6}) \leqslant 20\cdot\bar{k}\cdot(2+\lambda\cdot r).$$

$\triangle\square$

Informally, the following lemma says that if in any closed set $\mathfrak{C}$ of CBB-space, one can find a special point p such that any near by point $x \in \mathfrak{C}$ has *almost the same* tangent space, and moreover, T_x *almost splits* in direction of any $\xi \in \Sigma_x\mathfrak{C}$.

20.5.3. Lemma. *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$ and $\mathfrak{C} \subset \mathcal{L}$ be a closed subset. Then, given $\varepsilon > 0$, there is a point $p \in \mathfrak{C}$, and $r > 0$ such that*

a) for any $x \in \mathfrak{C} \cap \mathrm{B}(p,r)$ we have $\mathrm{D}_{\mathsf{GH}}(\Sigma_x, \Sigma_p) < \varepsilon$ and moreover,

b) for any $\xi \in \Sigma_x\mathfrak{C}$, $\mathrm{D}_{\mathsf{GH}}(\Sigma_\xi \mathrm{T}_x, \Sigma_p) < \varepsilon$.

Proof. Note that as follows from volume continuity (**??**), for any $\varepsilon > 0$ there is $\delta = \delta(m, \varepsilon) > 0$ such that if $\Sigma_x + \delta \geqslant \Sigma_p$ and $\operatorname{vol}\Sigma_x - \delta \leqslant \leqslant \operatorname{vol}\Sigma_p$ then $\mathrm{D}_{\mathsf{GH}}(\Sigma_p, \Sigma_x) < \varepsilon$.

Let us choose p with δ-maximal volume; i.e. such that $\operatorname{vol}\Sigma_x - \delta < \operatorname{vol}\Sigma_p$ for any $x \in \mathfrak{C}$.

Now, let us apply semicontinuity of space of directions (**??**) several times:

- First, there is $r > 0$ such that $\Sigma_x + \delta \geqslant \Sigma_p$ for any $x \in \in \mathrm{B}(p,r)$. Hence $\mathrm{D}_{\mathsf{GH}}(\Sigma_x, \Sigma_p) < \varepsilon$.
- Second, $\Sigma_\xi \mathrm{T}_x \geqslant \Sigma_x$ (here we apply **??** for T_x).
- Third, if Σ is a partial limit of Σ_{x_i} for $x_i \in \mathfrak{C} \cap \mathrm{B}(p,r)$, then $\Sigma_\xi \mathrm{T}_p \leqslant \Sigma$. Thus, $\mathrm{D}_{\mathsf{GH}}(\Sigma_\xi \mathrm{T}_x, \Sigma_p) < \varepsilon$. □

20.6 Quasigeodesics in extremal sets.

Here we give an extended version of Theorem 20.5.1. The proof repeats the proof of existence of quasigeodesics (20.6.1); it only require a stronger analog of Milka's lemma (20.5.2).

20.6.1. Quasigeodesic in extremal set. *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$, $E \subset \subset \mathcal{L}$ be an extremal set. Then for any point $x \in E$, and any direction $\xi \in \Sigma_x E$ there is a quasigeodesic $\gamma\colon \mathbb{R} \to E$ such that $\gamma(0) = x$ and $\gamma^+(0) = \xi$.*

Sketch of proof. Note that according to ??? as a result of construction in 20.4.6 we get a κ-convex curve in E. Further, construction in 20.4.2 gives a pre-quasigeodesic in E. According to ???, for any $p \in E$, the tangent space $\mathrm{T}_p E$ is defined and it is an extremal set of T_p. Thus if one use extended Milka's lemma 20.6.2 instead of 20.5.2, one can keep in E all the constructions in the proof of 20.5.1. □

20.6.2. Extremal Milka's lemma. *Let $\mathcal{L} \in \mathrm{CBB}^m[\kappa]$, $p \in \mathcal{L}$ and $E \subset \mathrm{T}_p$ be an extremal set of a tangent cone. Then for any vector $v \in E$ there is a polar vector $v^* \in E$ such that $|v| = |v^*|$.*

Proof. Set $X = E \cap \Sigma_p$. If $\Sigma_\xi X \neq \varnothing$ then the proof is the same as for the standard Milka's lemma; it is enough to choose a direction in $\Sigma_\xi X$ and shoot a quasigeodesic γ in X of length π in this direction such that $\gamma \subset X$; such γ exists from the induction hypothesis.

If $X = \{\xi\}$ then from ???, we have $\overline{\mathrm{B}}[\xi, \frac{\pi}{2}] = \Sigma_p$. Thus, ξ is polar to itself.

Otherwise, if $\Sigma_\xi X = \varnothing$ and X contains at least two points, choose ξ^* to be closest point in $X \backslash \xi$ from ξ. Since $X \subset \Sigma_p$ is extremal (???), for any $\eta \in \Sigma_p$ we have $\measuredangle[\xi^* {}^{\eta}_{\xi}] \leqslant \frac{\pi}{2}$ and since $\Sigma_\xi X = \varnothing$ we have $\measuredangle[\xi {}^{\eta}_{\xi^*}] \leqslant \frac{\pi}{2}$. Therefore, in the model spherical triangle $[\tilde\xi\tilde\eta\tilde\xi^*] = \tilde\triangle^1(\xi\eta\xi^*)$, we have $\measuredangle\tilde\xi \leqslant \frac{\pi}{2}$ and $\measuredangle\tilde\xi^* \leqslant \frac{\pi}{2}$. Hence

$$\measuredangle(\xi, \eta) + \measuredangle(\eta, \xi^*) = |\tilde\xi\tilde\eta| + |\tilde\eta\,\tilde\xi^*| \leqslant \pi.$$

□

20.7 Remarks and open problems

Is there an analog of the Liouville's theorem for "quasigeodesic flow"? Namely:

20.7.1. Question. *Let $\mathcal{L} \in \mathrm{CBB}^m$ and Γ be the space of all quasigeodesics $\gamma\colon \mathbb{R} \to \mathcal{L}$. Is it possible to define measure μ on Γ such that*

a) μ is invariant with respect to natural group action on Γ generated by $\gamma \mapsto \gamma \circ \sigma$, where $\sigma(t) = t + a$ or $\sigma(t) = -t$

b) if e_t denotes evaluation map $e_t\colon \gamma \mapsto \gamma(t)$ then

$$e_t\#\mu = \mathrm{vol}_m$$

where # stays for push-forward.

If the answer to the following question is positive, it could give an alternative proof of existence of quasigeodesics (20.5.1).

20.7.2. Question. *Is it true that in a* CBB*-space without boundary there is an infinitely long geodesic?*

As it was noted by A. Lytchak, the these two questions can be reduced to the following: Assume $\mathcal{L} \in \mathrm{CBB}^m$ and $\partial\mathcal{L} = \varnothing$. Set

$$V(r) = \int\limits_{\mathcal{L}} \mathrm{vol}_m \mathrm{B}(x, r) \cdot \boldsymbol{d}_x \,\mathrm{vol}_m,$$

then

$$V(r) = \mathrm{vol}_m \mathcal{L} \cdot c_m \cdot r^m + o(r^{m+1}).$$

The technique of *tight maps* makes it possible to prove only that

$$V(r) = \operatorname{vol}_m \mathcal{L}\cdot c_m\cdot r^m + O(r^{m+1}).$$

Note that if $\mathcal{L}$ is a Riemannian manifold with boundary then

$$V(r) = r^m\cdot c_m\cdot \operatorname{vol}_m \mathcal{L} + r^{m+1}\cdot c'_m\cdot \operatorname{vol}_{m-1} \partial\mathcal{L} + o(r^{m+1}).$$

Variation of turn of a broken geodesic $a_0a_2\dots a_n$ can be defined as sum of angle excesses

$$\sum_{i=1}^{n-1}\left(\pi - \measuredangle[a_i\,{}^{a_{i+1}}_{a_{i-1}}]\right).$$

Further, variation of turn for a curve can be defined as lower limit of turns of broken geodesics which converge to a given curve (you can choose in which sense).

20.7.3. Question. *Let $\mathcal{L}\in \mathrm{CBB}^m$. Is it true that any quasigeodesic in $\mathcal{L}$ defined on a bounded interval has bounded variation of turn.*

If the all tangent spaces of $\mathcal{L}$ are sufficiently close to $\mathbb{E}^m$ then the answer is "yes". In this case variation of turn can be estimated though *total turn to a point*, which defined as follows:

$$\sum_{i=1}^{n-1}\max\left\{0, \pi - \measuredangle^{\kappa}(a_i\,{}^{a_{i-1}}_{p}) - \measuredangle^{\kappa}(a_i\,{}^{a_{i+1}}_{p})\right\}.$$

One could look at the following class of curves in general metric space. This class of curves should be interesting in the metric spaces, which has "a lot" of semiconcave functions.

20.7.4. Definition. *Let $\mathcal{X}$ be a metric space. A curve $\gamma\colon \mathbb{I}\to\mathcal{X}$ is called* funny quasigeodesic *if for any $\lambda\in\mathbb{R}$ and any λ-concave subfunction $f\colon \mathcal{X}\to\mathbb{R}$ the composition $f\circ\gamma$ is λ-concave.*

An analog of existence theorem (???) does not hold for this type of curves, once funny quasigeodesic comes to the boundary it terminates. On the other hand it has a nice property: if $\mathcal{X}$ is a convex closed subset of $\mathcal{L}\in\mathrm{CBB}^m$ then any funny quasigeodesic in $\mathcal{X}$ is a funny quasigeodesic in $\mathcal{L}$. Also, it is easy to see that in CBA-spaces, the class of funny quasigeodesics coinsides with class of geodesics.

20.8 Exercises

20.8.1. Exercise. *Show that if* $\mathcal{L} \in \mathrm{CBB}^m[1]$ *and* $\partial\mathcal{L} \neq \varnothing$ *then* $\operatorname{rad}\mathcal{L} \leqslant \frac{\pi}{2}$.

20.8.2. Exercise. *Let* $\mathcal{L} \in \mathrm{CBB}^m[1]$, *and* $p_1, p_2, \cdots, p_m \in \mathcal{L}$. *Show that there is* $x \in \mathcal{L}$ *such that* $|p_n\,x| \leqslant \frac{\pi}{2}$ *for each* n.

20.8.3. Exercise. *Let* $\mathcal{L} \in \mathrm{CBB}^m[1]$. *Assume* $\operatorname{rad}\mathcal{L} > \frac{\pi}{2}$. *Show that*

$$\operatorname{rad}\Sigma_p > \tfrac{\pi}{2}$$

for any $p \in \mathcal{L}$.

20.8.4. Exercise. *Construct* $\mathcal{L} \in \mathrm{CBB}^2[0]$ *with* $p \in \mathcal{L}$ *and* $\xi \in \Sigma_p$ *such that there are two quasigeodesics* $\gamma_1, \gamma_2\colon [0,\infty) \to \mathcal{L}$ *with initial data* $\gamma_1^+(0) = \gamma_2^+(0) = \xi$, *such that* $\gamma_1(t) \neq \gamma_2(t)$ *for any* $t > 0$.

20.8.5. Exercise. *Construct* $\mathcal{L} \in \mathrm{CBB}^3[0]$, $\partial\mathcal{L} \neq \varnothing$ *and a quasigeodesic* γ *in* $\mathcal{L}$ *which is not a projection of any quasigeodesic in doubling of* $\mathcal{L}$.

Show that there is no such example in $\mathrm{CBB}^2[0]$.

Part V

Results

Chapter 21

Nonnegatively curved spaces

21.1 Soul theorem

In Riemannian case, the following theorem was proved in [42]. The last statement was proved in [156].

21.1.1. Soul theorem. *Let $\mathcal{L} \in \mathrm{CBB}^m[0]$. Then there is a convex compact subset $S \subset \mathcal{L}$ (which is called* soul *of $\mathcal{L}$), such that $S \in$ $\in \mathrm{CBB}^{\bar{k}}[0]$, $\bar{k} \leqslant m$, S has no boundary, S is a strict short deformational retuct of $\mathcal{L}$, i.e. there one parameter of short maps $h_t\colon \mathcal{L} \to \mathcal{L}$ such that $h_0 = \mathrm{id}$, $h_1(\mathcal{L}) = S$, $h_t(s) = s$ for any $s \in S$ and $t \in [0,1]$.*

More over, S is uniquely defined up to isometry.

If $\mathcal{L}$ is a Riemannian manifold then in fact $\mathcal{L}$ is diffeomorphic to the total space of the normal bundle of $\iota(S)$ in $\mathcal{L}$. For Alexandrov spaces a similar strengthening is false as the following example shows:

Example Consider product space $\mathbb{S}^2 \times \mathbb{E}^2$ equiped with isometric $\mathbb{Z}_2$-action, which acts by cental simmetry on each $\mathbb{S}^2$ and $\mathbb{E}^2$; thus it has exactly two fixed points v and w on $\mathbb{S}^2 \times 0$.

According to ???, the factor-space $\mathcal{L} = \mathbb{S}^2 \times \mathbb{E}^2/\mathbb{Z}_2$ is a 4-dimensional Alexandrov space with curvature $\geqslant 0$. It has exactly two singular points, the orbits of v and w, a neigborhood at each is homeomorphic to $\operatorname{Cone}\mathbb{RP}^3$. The image of $\mathbb{S}^2 \times 0$ in $\mathcal{L}$ is homeomorphic to $\mathbb{S}^2$ and it forms an Alexandrov space without boundary.

Thus, the soul of $\mathcal{L}$ is homeomorphic to $\mathbb{S}^2$. Since $\mathcal{L}$ has exactly two topological singularities it can not be presented as a fiber bunndle over $\mathbb{S}^2$.

The proof of is based on ???, the rest of the proof repeats the proof of classical soul theorem [42] and [156].

Proof.??? □

21.2 Shioya's splitting

For Riemannian case, corollary 21.2.2 was proved in [107], it was conjectured in [155] where a weaker statement had been proved. The followng more general theorem was obtained in [127] independently.

Given $\mathcal{L} \in \mathrm{CBB}^m[0]$ and $p \in \mathcal{L}$, set $(\mathrm{T}_\infty, 0) = \mathsf{GH}\text{-}\lim_{\varepsilon\to 0} \varepsilon*(\mathcal{L}, p)$. Clearly T_∞ does not depends on the choice of p. From ??? it is clear that $\mathrm{T}_\infty\mathcal{L} \in \mathrm{CBB}^{\bar{k}}[0]$ for $\bar{k} \leqslant m$. The space $\mathrm{T}_\infty\mathcal{L}$ is called *cone at infinity*, it has natural cone structure WHy???. Thus, there is space $\Sigma_\infty\mathcal{L} \in \mathrm{CBB}^{\bar{k}-1}[1]$ such that $\mathrm{T}_\infty\mathcal{L} \overset{iso}{=\!=} \mathrm{Cone}\,\Sigma_\infty\mathcal{L}$. The space $\Sigma_\infty\mathcal{L}$ is called *ideal boundary* of $\mathcal{L}$.

21.2.1. Theorem. *Let $\mathcal{L} \in \mathrm{CBB}^m[0]$ and $\dim \mathrm{T}_\infty = \bar{k}$. Assume T_∞ has no extremal subsets then $\mathcal{L}$ can be isometrically splitted into product*

$$\mathcal{L} = K \oplus \mathcal{L}'$$

where $K, \mathcal{L}' \subset \mathcal{L}$ are closed convex subsets, $\dim \mathcal{L}' = \bar{k}$ and K is copmact.

21.2.2. Corollary. *Let $\mathcal{L} \in \mathrm{CBB}^m[0]$ and $\dim \Sigma_\infty\mathcal{L} = \bar{k}-1$. Assume $\mathrm{rad}\,\Sigma_\infty\mathcal{L} > \frac{\pi}{2}$, then $\mathcal{L}$ can be isometrically splitted into product*

$$\mathcal{L} = K \oplus \mathcal{L}'$$

where $K, \mathcal{L}' \subset \mathcal{L}$ are closed convex subsets, $\dim \mathcal{L}' = \bar{k}$ and K is copmact.

Note that if $\mathrm{rad}\,\Sigma_\infty\mathcal{L} = \frac{\pi}{2}$, the conclusion of the corollary does not hold even in Riemannian category. In fact by taking a product of flat open Möbius band with $\mathbb{R}$ we have a counterexample.

21.3 Erdős' problem rediscovered

The following theorem was proved in ???.

21.3.1. Theorem. *Let $\mathcal{L} \in \mathrm{CBB}^m[0]$. Then it has at most 2^m one-point extremal sets.*

The proof is a translation of proof of classical problem in combinatoric geometry to Alexandrov's language.

21.3.2. Erdős' problem. *Let F be a set of points in $\mathbb{E}^m$ such that any triangle formed by three distinct points in F has no obtuse angles. Then number of elements in F can not exeed 2^m.*

Moreover, if $|F| = 2^m$ then F consists of vertexes of right paralelepiped.

This problem was posted in [50] and solved in [44]. The question of classifing all $\mathcal{L}$ with the maximal number one-point extremal sets turns out to be more delicate see [95]. Let us describe a class of examples. Let $\mathbb{T}^m$ be standart torus and Γ be a subgroup generated by all coordinte reflections. Clearly, Γ is isomorphic to $\mathbb{Z}_2^m$ and it has exactly 2^m fixed points in $\mathbb{T}^m$. Let $\Gamma' \subset \Gamma$ be a subgroup which has the same fixed point set as Γ and g be a flat Γ'-ivariant metric on $\mathbb{T}^m$. Then, $(\mathbb{T}^m, g)/\Gamma' \in \mathrm{CBB}^m[0]$ and it has exactly 2^m one-point extremal subsets; it follows from ??? and ???. The above construction does not describes all $\mathrm{CBB}^m[0]$-spaces with 2^m one-point extremal subsets, but it is close to the correct answer.

The finding resonable estimates for maximal number of extraml one-point subsets in a $\mathrm{CBB}^m[1]$-space is completely open; it analogous to the follolwing problem: finding maximal number $n(m)$ such that there are points $p_1, p_2, \dots p_n$ in $\mathbb{E}^m$ such that any triangle $[p_i p_j p_k]$ is acute. It is expected $n(m) \ll 2^m$, but so far it is only known that

1. $n(m) \geqslant 2m$ — the configuration s slight perturbation of vertexes of m-octahedra.
2. $n(m) \leqslant$???

Proof of 21.3.1. Let $\{p_i\}$, $i \in \{1, 2, \dots, N\}$ be the one-point extreaml sets. For each p_i consider its open Voronoi domain V_i; i.e.

$$V_i = \{\, x \in \mathcal{L} \mid |p_i\,x| < |p_j\,x| \text{ for any } j \neq i \,\}.$$

Clearly $V_i \cap V_j = \varnothing$ if $i \neq j$. Note that $\mathrm{vol}_m V_i > \frac{1}{2^m} \mathrm{vol}_m \mathcal{L}$.

Indeed, fix i and for given $\alpha \in (0,1)$, consider α-homothety $\Phi_\alpha \colon \mathcal{L} \to \to \mathcal{L}$ with center at p_i; i.e. for each point $x \in \mathcal{L}$ choose a geodesic $[p_i x]$ and set $\Phi_\alpha x = \mathrm{geod}_{[p_i x]}(\alpha |p_i\,x|)$. From comparison we have that $\mathrm{vol}(\Phi_\alpha \mathcal{L}) \geqslant \alpha^m \mathrm{vol}\,\mathcal{L}$. For any $x \in \mathcal{L}$ and all $\alpha < \frac{1}{2}$ we have $\Phi_\alpha x \in V_i$. Assume $x' = \Phi_\alpha x \notin V_i$, then threre is p_j such that $|p_i\,x'| \geqslant |p_j\,x'|$. Then from comparison, we have $\measuredangle^0(p_j\,{}^{p_i}_{x}) > \frac{\pi}{2}$; i.e. p_j does not form a one-point extremal set.??? $\square$

21.4 Circle actions on 4-dimensional manifolds

The following result was obtained by Hsiang and Kleiner in [75]. The original proof used Alexandrov geometry implicitly; in fact at the mo-

ment Alexandrov spaces with lower curvature bound were defined only in dimension 2.

21.4.1. Theorem. *Let M be a connected complete Riemannian manifold and $\mathbb{S}^1 \curvearrowright M$ be isometric effective circle action. Then*

a) If M has nonnegative sectional curvature then the action has at most 4 isolated fixed points.

b) If M has positive sectional curvature then the action has at most 3 isolated fixed points.

Proof. Consider the quotient space $\mathcal{L} = M/\mathbb{S}^1$.

Assume x is a fixed point of $\mathbb{S}^1$-action; denote by y the projection of x in $\mathcal{L}$. Note that

❶ $\Sigma_y \leqslant \frac{1}{2}\cdot\mathbb{S}^2$*; that is there is a noncontracting map* $\Sigma_y \to \frac{1}{2}\cdot\mathbb{S}^2$*. In particular,*

$$\measuredangle[y\,{}^{z^1}_{z^2}] + \measuredangle[y\,{}^{z^2}_{z^3}] + \measuredangle[y\,{}^{z^3}_{z^1}] \leqslant \pi$$

for any triple of points $z^1, z^2, z^3 \in \mathcal{L}$.

Indeed, $\Sigma_x \overset{iso}{=\!=} \mathbb{S}^3$. Therefore $\Sigma_y \overset{iso}{=\!=} \mathbb{S}^3/\mathbb{S}^1$, for an isometric $\mathbb{S}^1$-action. Note that we can identify $\mathbb{S}^3$ as the unit sphere in $\mathbb{C}^2$ in such a way that the $\mathbb{S}^1$-action is given by diagonal matrices $\left(\begin{smallmatrix} z^p & 0 \\ 0 & z^q \end{smallmatrix}\right)$ for some realatively prime integers p, q and $z \in \mathbb{S}^1 \subset \mathbb{C}$.

Consider the torus action $\mathbb{T}^2 \curvearrowright \mathbb{S}^3$ by diagonal matrices $\left(\begin{smallmatrix} v & 0 \\ 0 & w \end{smallmatrix}\right)$. Note that $[0, \frac{\pi}{2}] \overset{iso}{=\!=} \mathbb{S}^3/\mathbb{T}^2$ and Σ_w is isometric to the warped product $\mathbb{S}^1 \times_{f_{p,q}} [0, \frac{\pi}{2}]$ (compare with Exercise 11.2.2). Let us compute the warping function $f_{p,q}$. If $t \in [0, \frac{\pi}{2}]$ is the projection of $x \in \mathbb{S}^3$ then

$$\begin{aligned} f_{p,q}(t) &= \frac{\operatorname{area}[\mathbb{T}^2\cdot x]}{\operatorname{length}[\mathbb{S}^1\cdot x]} \\ &= \frac{\sin t\cdot\cos t}{\sqrt{(p\cdot\sin t)^2 + (q\cdot\cos t)^2}} \end{aligned}$$

Note that $f_{1,1}(t) \geqslant f_{p,q}(t)$ for any pair (p, q) of relatively prime inegers and any $t \in [0, \frac{\pi}{2}]$ and

$$\tfrac{1}{2}\cdot\mathbb{S}^2 \overset{iso}{=\!=} \mathbb{S}^1 \times_{f_{1,1}} [0, \tfrac{\pi}{2}],$$

(In other words $\frac{1}{2}\cdot\mathbb{S}^2$ is isometric to the orbit space of Hopf fibration on $\mathbb{S}^3$.) Applying Exercise 11.2.4, we get ❶. △

(b). Assume the circle action has 4 isolated points. Denote by y_1, y_2, y_3, y_4 their projections in $\mathcal{L}$.

According to ???, $\mathcal{L} \in \mathrm{CBB}[\kappa]$ for some $\kappa > 0$. According to Claim clm:Sigma=¡sphere/2,

$$\measuredangle[y_i\,{}^{y_j}_{y_k}] \leqslant \tfrac{\pi}{2}$$

for any distinct i, j and k. In particular non of 4 triangles $[y^1y^2y^3]$, $[y^1y^2y^4]$, $[y^1y^3y^4]$, $[y^2y^3y^4]$ is degenerate. Therefore The sum of the angles in each triuangle is strictly larger than π. Denote by ω the sum of all 12 angles of these 4 triangles. We get

❷ $$\omega > 4{\cdot}\pi.$$

From Claim ❶, the sum of 3 angles among these 12 which share one vertex is at most π. Therefore

$$\omega \leqslant 4{\cdot}\pi.$$

The later contradicts ❷.

(a). Assume there are 5 fixed points of the circle action. Denote by y_1, y_2, y_3, y_4, y^5 their projections in $\mathcal{L}$.

□

The result above was used to classify positively and nonnegatively curved mainifolds which admit isometric circle action. This classification was improved later by Grove and Wilking in [68]. Namely they show the following.

21.4.2. Theorem. *A closed nonnegatively curved complete simply connected 4-dimensional manifold M with an isometric circle action is diffeomorphic to $\mathbb{S}^4$, $\mathbb{CP}^2$, $\mathbb{S}^2 \times \mathbb{S}^2$ or one of $\mathbb{CP}^2\#(\pm\mathbb{CP}^2)$ and the action extends to a smooth torus action.*

In particular, if M is positively curved then the circle action is equivariantly diffeomorphic to a linear action on $\mathbb{S}^4$, $\mathbb{RP}^2$ or $\mathbb{CP}^2$.

21.5 Remarks and open questions

In [126], it was proved that if $\mathcal{L}$ is non-negatively curved Riemannian manifold and $S \subset \mathcal{L}$ is its soul then Sharafundinov's returction $\mathrm{Shar}\colon \mathcal{L} \to S$ is a submetry; i.e. for any $x \in \mathcal{L}$ and $y \in S$ there is $y' \in \mathrm{Shar}^{-1}(y)$ such that $|\mathrm{Shar}(x)\ y| = |x\,y'|$.

This statement is likely remains true for non-negatively curved Alexandrov space, but the proof can not be generalized directly.

21.6 Exercises

21.6.1. Exercise. *Let* $\mathcal{L} \in \mathrm{CBB}[0]$, $b_1, \dots, b_{\bar{k}}$ *be an array of Busemann's functions and* $\lambda_1, \dots, \lambda_{\bar{k}}$ *be an array of positive real numbers. Assume*

$$\lambda_1 \cdot b_1 + \ldots + \lambda_{\bar{k}} \cdot b_k$$

admits minimum in $\mathcal{L}$. *Show that* $\mathcal{L}$ *admits splitting as* $\mathcal{L}' \oplus H$, *where* H *is isometric to a Hilbert space and for*

Chapter 22

Positively curved spaces

22.1 Sphere theorems

22.1.1. Diameter-suspension theorem. *Let* $\mathcal{L} \in \mathrm{CBB}^m[1]$. *Assume* $\operatorname{diam}\mathcal{L} > \frac{\pi}{2}$ *and* $p, q \in \mathcal{L}$ *be two points such that* $|pq| = \operatorname{diam}\mathcal{L}$. *then* $\mathcal{L}$ *is homeomorphic to* $\operatorname{Susp}\Sigma_p$.

Moreover Σ_p *is homeomorphic to* Σ_q.

Proof. Clearly for any $x \in \mathcal{L}\backslash\{p, q\}$ we have $\measuredangle[x\,{}^p_q] \geqslant \tilde{\measuredangle}^1(x\,{}^p_q) > \frac{\pi}{2}$. Thus $|p_*|$ has only two critical points p and q. Thus according to Morse lemma, $\mathcal{L}$ is homeomorphic to $\operatorname{Susp} Z_t$, where $Z_t = |p_*|^{-1}(t)$, $0 < t < |pq|$. Finally, according to ???, for all sufficiently small t we have Σ_p is homeomorphic to Z_t.

To prove the second statement ???. □

22.1.2. Sphere theorem. *Assume* $\mathcal{L} \in \mathrm{CBB}^m[1]$, $\mathcal{L}$ *does not have proper extremal subsets and* $\operatorname{diam}\mathcal{L} > \frac{\pi}{2}$ *then* $\mathcal{L}$ *is homeomorphic to* $\mathbb{S}^m$.

Proof. Assume $p, q \in \mathcal{L}$ realize the diameter of $\mathcal{L}$. Since $\mathcal{L}$ has no extremal subsets, from Example **??**, it follows that a small spherical neighborhood of $p \in \mathcal{L}$ is homeomorphic to $\mathbb{R}^m$. From angle comparison, $|p_*|$ has only two critical points p and q. Therefore, this theorem follows from the Morse lemma (**??**) applied to $|p_*|$. □

22.1.3. Radius-sphere theorem. *Assume* $\mathcal{L} \in \mathrm{CBB}^m[1]$ *and* $\operatorname{rad}\mathcal{L} > \frac{\pi}{2}$ *then* $\mathcal{L}$ *is homeomorphic to a sphere.*

We give two proofs, both are build on the following lemma. The first a direct proof; the second is a reduction to theorem 22.1.2.

22.1.4. Lemma. *Let* $\mathcal{L} \in \mathrm{CBB}^m[1]$ *and* $\operatorname{rad}\mathcal{L} > \frac{\pi}{2}$. *Then*

$$\operatorname{rad}\Sigma_p > \tfrac{\pi}{2}$$

for any $p \in \mathcal{L}$.

Proof. Assume $\operatorname{rad}\Sigma_p \leqslant \frac{\pi}{2}$ for some $p \in \mathcal{L}$. Let $\xi \in \Sigma_p$ be a direction, such that $\overline{\mathrm{B}}[\xi, \frac{\pi}{2}] = \Sigma_p$. Draw a quasigeodesic γ from p in direction ξ.

Set $q = \gamma(\frac{\pi}{2})$. For any point $x \in \mathcal{L}$ we have $\measuredangle(\xi, \operatorname{dir}[px]) \leqslant \frac{\pi}{2}$. Therefore, by the comparison for γ (20.1.4c'), $|x\,q| \leqslant \frac{\pi}{2}$. I.e. $\overline{\mathrm{B}}[q, \frac{\pi}{2}] = \mathcal{L}$, a contradiction. □

Proof №1. From Lemma 22.1.4, $\operatorname{rad}\Sigma_p > \frac{\pi}{2}$. Since $\dim\Sigma_p < m$, by the induction hypothesis we have $\Sigma_p \simeq \mathbb{S}^{m-1}$. Now the Morse lemma (**??**) for $|p{*}|\colon \mathcal{L} \to \mathbb{R}$ gives that $\mathcal{L} \simeq \operatorname{Susp}\Sigma_p \simeq \mathbb{S}^m$, here $\operatorname{Susp}\Sigma_p$ denotes a spherical suspension over Σ_p. □

Proof №2. Assume the contrary, then according to Sphere theorem (22.1.2), $\mathcal{L}$ contains a proper extremal subset $E \subset \mathcal{L}$. Thus according to 18.5.5, there is a point[1] $q \in E$ such that $\operatorname{rad}\Sigma_q \leqslant \pi$, which contradicts Lemma 22.1.4. □

22.2 Collapsing with positive curvature bound

The following theorem was proved in [127]. The corollary was obtained earlier: it follows from more delicate result 22.2.4, proved in [58]; also it was proved directly in [133].

22.2.1. Limit of collapse. *Assume* $\mathcal{L}_n \in \mathrm{CBB}^m[1]$ *is a collapsing sequence and* $\mathcal{L}_n \to \mathcal{L}$ *then either* $\mathcal{L}$ *has an extremal subset or* $\operatorname{diam}\mathcal{L} \leqslant \frac{\pi}{2}$.

Proof. The conclusion of the theorem is equivalent to the fact that $\operatorname{Cone}\mathcal{L}$ has an extremal set. ??? □

22.2.2. Corollary. *Let* $\mathcal{L}_n \in \mathrm{CBB}^m[1]$ *be a collapsing sequence and* $\mathcal{L}_n \xrightarrow[\mathrm{GH}]{} \mathcal{L}$. *Then* $\operatorname{rad}\mathcal{L} \leqslant \frac{\pi}{2}$.

The direct proof below will use the following

22.2.3. Exercise. *Let* $\mathcal{L} \in \mathrm{CBB}^m[1]$ *and* $\operatorname{rad}\mathcal{L} > \frac{\pi}{2}$. *Show that there is an array of* $m+2$ *points* $p, a_0, a_1, \dots, a_m \in \mathcal{L}$ *such that for any* $\xi \in \Sigma_p$ *there is* i *such that*

$$\measuredangle(\xi, \operatorname{dir}[pa_i]) > \tfrac{\pi}{2}.$$

[1] In fact, according to ???, it is true for any $q \in E$.

Hint: For smooth nondecreasing function $\varphi:\mathbb{R}\to\mathbb{R}$ such that $\varphi(x)=0$ for all $x\leqslant\frac{\pi}{2}$ and $\varphi(x)>0$ for all $x>\frac{\pi}{2}$, take p to be the minimum point of

$$f(y)=\int\limits_{\mathcal{L}}\varphi|x\,y|\cdot\boldsymbol{d}x.$$

Direct proof. From ???, we have $\mathcal{L}\in\mathrm{CBB}^{\bar{k}}[1]$ and since $\mathcal{L}_n$ is collapsing, $\bar{k}<m$.

Choose $p,\{a_i\}_{i=0}^{\bar{k}}\in\mathcal{L}$ as in the exercise. Let $p_n,\{a_{i,n}\}_{i=0}^{\bar{k}}$ in $\mathcal{L}_n$ be such that $p_n\to p$ and $a_{i,n}\to a_i$ as $\mathcal{L}_n\xrightarrow[\mathrm{GH}]{}\mathcal{L}$. Since $m>\bar{k}$ there is a direction $\xi_n\in\Sigma_{p_n}$ such that $\measuredangle(\xi_n,\mathrm{dir}[p_n a_{i,n}])\leqslant\frac{\pi}{2}$. Let us shoot quasigeodesics $\gamma_n\colon[0,\frac{\pi}{2}]\to\mathcal{L}_n$ with the initial data $\gamma^+(0)=\xi_n$ and pass to its partial limit $\gamma\colon[0,\frac{\pi}{2}]\to\mathcal{L}$. According to ??? γ is a quasigeodesic, thus $\gamma^+(0)$ is defined and $|\gamma^+(0)|=1$. According to ???, $|a_{i,n}\gamma_n(t)|\leqslant|a_{i,n}p_n|$ for all t; thus we have $|a_i\gamma(t)|\leqslant|a_i p|$ for all t. Therefore $\measuredangle(\gamma^+(0),\mathrm{dir}[pa_i])\leqslant\frac{\pi}{2}$ for each i, which contradicts with the exercise. □

22.2.4. Volume-radius gap. *$\mathcal{L}\in\mathrm{CBB}^m[1]$ and $\operatorname{rad}\mathcal{L}_n\leqslant\frac{\pi}{2}$ then $\operatorname{vol}\mathcal{L}>\varepsilon_m$, where $\varepsilon_m=$???.*

This fact much stronger than ??? and its proof use only Morse lemma from all Alexandrov geometry. The hart of it lie in Gromov's Filling radius estimate???

Proof. For any $p\in\mathcal{L}$, we can find $q\in\mathcal{L}$ such that $|pq|>\frac{\pi}{2}$. Hence, if $|x\,p|<\frac{\pi}{2}$ has no have critical values in $(0,\frac{\pi}{2})$; by Morse lemma (???), all balls $\mathrm{B}(p,\rho)$, where $\rho<\frac{\pi}{2}$ are contractible.

In terminology of [62], that means that $\operatorname{ConRad}\mathcal{L}\geqslant\frac{\pi}{2}$, where $\operatorname{ConRad}\mathcal{L}$ is contractibility radius of $\mathcal{L}$. According to **??** $[\mathcal{L}_n]\neq 0\in$ $\in H_m(\mathcal{L}_n)$. Thus, applying [62, ???], the filling radius can be estimated from contractibility, namely

$$\operatorname{FillRad}\mathcal{L}>\frac{1}{2\cdot(m+2)}\cdot\operatorname{ConRad}\mathcal{L}\geqslant\frac{\pi}{4\cdot(m+2)}.$$

Finally applying celebrated Gromov's result [62, 1.2.A], we have

$$(\operatorname{vol}_m\mathcal{L})^{\frac{1}{m}}>c_m\operatorname{FillRad}\mathcal{L},$$

for some $c_m\gg\frac{1}{m^{3m}}$; and the result follows. □

22.2.5. Open question. *Assume M_n be a sequence of Riemannian manifolds with curvature $\geqslant 1$ which is collapsing to Riemannian manifold M of positive dimension. Is it true that*

$$\dim M_n=\dim M+1$$

for all large n?

22.3 On area of boundary

The following probelem was suggested by A. Lytchak. It looks simple, but we could not make a proof without use of gradient exponent.

This problem would have followed from Conjecture **??** (that boundary of an Alexandrov's space is an Alexandrov's space), but before this conjecture has been proved, any partial result is of some interest, see also remark on page 317.

22.3.1. Claim. *Let* $\mathcal{L} \in \mathrm{CBB}^m[1]$. *Then*

$$\mathrm{vol}_{m-1}\,\partial\mathcal{L} \leqslant \mathrm{vol}_{m-1}\,\mathbb{S}^{m-1}.$$

Let us first prepare a proposition:

22.3.2. Proposition. *Let* $\mathcal{L} \in \mathrm{CBB}^m$, $p, q \in \mathcal{L}$. *Then for any point* $x \in\,]pq[$ *the inverse of the gradient exponential map* $\mathrm{gexp}^{-1}_{p;\kappa}\,x$ *is uniquely defined.*

Proof. Let $\gamma\colon [0,t_0] \to \mathcal{L}$ be a unit-speed geodesic, $\gamma(0) = p$, $\gamma(t_0) = q$. From the angle comparison we get that $|\nabla_x|p_*|| \geqslant -\cos \measuredangle^\kappa(x\,{}^p_q)$. Therefore, for any ζ we have

$$\tfrac{d^+}{dt}|p\,\alpha_\zeta(t)| \geqslant -|\alpha^+_\zeta(t)|\cdot\cos\measuredangle^\kappa(\alpha_\zeta(t)\,{}^p_q) \quad\text{and}\quad \tfrac{d^+}{dt}|\alpha_\zeta(t)\;q| \geqslant -|\alpha^+_\zeta(t)|.$$

It follows that $\measuredangle^\kappa(q\,{}^{\alpha_\zeta(t)}_p)$ is nondecreasing in t, hence the result. $\square$

Proof of 22.3.1. Let $z \in \mathcal{L}$ be the point at maximal distance from $\partial\mathcal{L}$; in particular z realizes maximum of $f = \sin\circ|\partial\mathcal{L}_*|$. From Theorem **??**, $f(z) \leqslant 1$ and $f'' + f \leqslant 0$.

Note that $\mathcal{L} \subset \overline{\mathrm{B}}[z, \frac{\pi}{2}]$, otherwise if $y \in \mathcal{L}$ with $|y\,z| > \frac{\pi}{2}$, then appling inequality $f''+f \leqslant 0$ to the geodesic $[zy]$, we get $(\boldsymbol{d}_z f)(\mathrm{dir}[zy]) > 0$, i.e. z is not a maximum of f.

From this it follows that gradient exponent

$$\mathrm{gexp}^1_z : (\overline{\mathrm{B}}[o, \tfrac{\pi}{2}], \mathfrak{s}) \to \mathcal{L}$$

is a short onto map.

Moreover,

$$\partial\mathcal{L} \subset \mathrm{gexp}^1_z(\partial\overline{\mathrm{B}}[o, \tfrac{\pi}{2}]).$$

Indeed, gexp gives a homotopy equivalence $\partial\overline{\mathrm{B}}[o, \frac{\pi}{2}]) \to \mathcal{L}\backslash\{z\}$. Clearly, $\Sigma_z = \partial(\overline{\mathrm{B}}[o, \frac{\pi}{2}], \mathfrak{s})$ has no boundary, therefore $H_{m-1}(\partial\mathcal{L}, \mathbb{Z}_2) \neq 0$, see [58, lemma 1]. Hence for any point $x \in \partial\mathcal{L}$, any geodesic $[zx]$ must have a point of the image $\mathrm{gexp}^1_z(\partial\overline{\mathrm{B}}[o, \frac{\pi}{2}])$ but, as it is shown in Proposition 22.3.2, it can only be its end x.

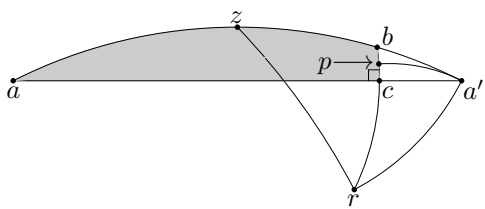
Now since

$$\operatorname{gexp}^1_z : (\overline{\mathrm{B}}[o, \tfrac{\pi}{2}], \mathfrak{s}) \to \mathcal{L}$$

is short and $(\partial\overline{\mathrm{B}}[o, \frac{\pi}{2}], \mathfrak{s})$ is isometric to $\Sigma_z\mathcal{L}$ we get $\operatorname{vol}\partial\mathcal{L} \leqslant \operatorname{vol}\Sigma_z\mathcal{L}$ and clearly, $\operatorname{vol}\Sigma_z\mathcal{L} \leqslant \operatorname{vol}\mathbb{S}^{m-1}$. □

Remark. Among other corollaries of Conjecture **??**, it is expected that if $\mathcal{L} \in \mathrm{CBB}^m[1]$ then $\partial\mathcal{L}$, equipped with length-metric, admits a noncontracting map to $\mathbb{S}^{m-1}$. In particular, its intrinsic diameter is at most π and intrinsic perimeter of any triangle in $\partial\mathcal{L}$ is at most $2{\cdot}\pi$. This does not follow from the proof above, since in general $\operatorname{gexp}^1_z \partial\overline{\mathrm{B}}[o, \frac{\pi}{2}] \not\subset \partial\mathcal{L}$, i.e. $\operatorname{gexp}_{z;1} \partial\overline{\mathrm{B}}[o, \frac{\pi}{2}]$ might have some creases left inside of $\mathcal{L}$, which might be used as a shortcut for curves with ends in $\partial\mathcal{L}$.

The existence of such creases one can see already in dimension 2: Cut a spherical triangle $[abc]$], with right angle at c and $|a\,b| > \frac{\pi}{2}$. Let us glue space $\mathcal{L}$ from two copies of $[abc]$ along sides $[ab]$ and $[bc]$.



According to doubling theorem ???, $\mathcal{L} \in \mathrm{CBB}^2[1]$ and its boundary consists of copies of side $[ac]$. The point z, the farest point from the boundary, lies on side $[ab]$ and $|a\,z| = \frac{\pi}{2}$. Let us show that

$$p = \operatorname{gexp}^1_z(\tfrac{\pi}{2}{\cdot}\operatorname{dir}[zb]) \notin \partial\mathcal{L}.$$

Clearly $p \in [bc]$, thus we need to show only that $p \neq c$. One can check it by calculations, but also one can see it using the following additional construction in $\mathbb{S}^2$: Let a' be the opposite point to a in $\mathbb{S}^2$ and r be a point on the extension of $[bc]$ behind c such that $|z\,r| = \frac{\pi}{2}$. Then $p \in [bc]$ and from the fact that gexp^1_z is short, one can see that $|p\,r| = |a'\,r|$. Clearly $|a'\,r| > |c\,r|$; therefore $p \neq c$; i.e. $p \notin \partial\mathcal{L}$.

22.4 Petersen's problem

22.4.1. Petersen's problem. *Let $\mathcal{L}$ be a smoothable Alexandrov's m-space; i.e. there is a sequence of Riemannian m-manifolds M_n with curvature $\geqslant \kappa$ such that $M_n \xrightarrow[\mathrm{GH}]{} \mathcal{L}$.*

Prove that the space of directions $\Sigma_x\mathcal{L}$ for any point $x \in \mathcal{L}$ is homeomorphic to the standard sphere.

Note that Perelman's stability theorem (see [124], [82]) only gives that $\Sigma_x\mathcal{L}$ has to be homotopically equivalent to the standard sphere.

Sketch of the proof: Fix a big negative λ and construct a function $f\colon \mathcal{L}\to\mathbb{R}$ with $\boldsymbol{d}_p f(x)\approx -|x|$ and controlled concavity of type (λ,κ). From **??**, the liftings $f_n\colon M_n\to\mathbb{R}$ of f (see **??**) are strictly concave for large n. Let us slightly smooth the functions f_n keeping them strictly concave. Then the level sets $f_n^{-1}(a)$, for values of a, which are little below the maximum of f_n, have strictly positive curvature and are diffeomorphic to the standard sphere[2].

Let us denote by $p_n\in M_n$ a maximum point of f_n. Then it is not hard to choose a sequence $\{a_n\}$ and a sequence of rescalings $\{\mathfrak{z}_n\}$ so that $(\mathfrak{z}_n M_n, p_n)\xrightarrow[\text{GH}]{}(\mathrm{T}_p,0)$ and $\mathfrak{z}_n f_n^{-1}(a_n)\subset\mathfrak{z}_n M_n$ converge to a convex hypersurface S close to $\Sigma_p\subset\mathrm{T}_p$. Then, from Perelman's stability theorem, it follows that S and therefore Σ_p is homeomorphic to the standard sphere. $\square$

Remark. From this proof it follows that Σ_p is itself smoothable. Moreover, there is a non-collapsing sequence of Riemannian metrics g_n on $\mathbb{S}^{m-1}$ such that $(\mathbb{S}^{m-1},g_n)\xrightarrow[\text{GH}]{}\Sigma_p$. This observation makes possible to proof a similar statement for iterated spaces of directions of smoothable Alexandrov space.

22.5 Codimension of collaps

In the case of collapsing, the liftings f_n of a function f with controlled concavity type do not have the same controlled concavity type.

Nevertheless, the liftings are semiconcave and moreover, as was noted in [81], if M_n is a sequence of $m+k$-dimensional Riemannian manifolds with curvature $\geqslant\kappa$, $M_n\xrightarrow[\text{GH}]{}\mathcal{L}$, $\dim\mathcal{L}=m$, then one has a good control over the sum of $k+1$ maximal eigenvalues of their Hessians. In particular, a construction as in the proof of Theorem **??** gives a strictly concave function on $\mathcal{L}$ for which the liftings f_n on $\mathcal{L}_n$ have Morse index $\leqslant k$. It follows that one can retract an ε-neighborhood of p_n to a k-dimensional CW-complex[3], where $p_n\in\mathcal{L}_n$ is a maximum point of f_n and ε does not depend on n. This observation gives a lower bound for the *codimension of a collapse*[4] to particular spaces. For example, for any lower curvature bound κ, the codimension of a

[2]Since f has only one critical value above a and it is a local maximum.

[3]it is unknown whether it could be retracted to an k-submanifold. If true, it would give some interesting applications

[4]in our case, it is k; the difference between the dimension of spaces from the collapsing sequence and the dimension of the limit space

collapse to $\operatorname{Susp}\mathbb{HP}^m$[5] is at least 3, and for $\operatorname{Susp} Ca\mathrm{P}^2$ is at least 8 (it is expected to be ∞). In addition, it yields the following theorem, which seems to be the only sphere theorem which does not assume positiveness of curvature.

22.5.1. Funny sphere theorem. *If a $4\cdot(m+1)$ Riemannian manifold M with sectional curvature $\geqslant \kappa$ is sufficiently close*[6] *to* $\operatorname{Susp}\mathbb{HP}^m$*, then it is homeomorphic to a sphere.*

22.6 Convrgence

22.6.1. ???Theorem. *Let $\mathcal{L}_n \in \mathrm{CBB}^m[1]$ and $\mathcal{L}_n \xrightarrow[\mathrm{GH}]{} \mathcal{L}$. Assume that there is an array of points $\tilde{\boldsymbol{p}} = (\tilde{p}_0, \tilde{p}_1, \dots, \tilde{p}_k)$ in $\mathbb{S}^k$ and a sequence of arrays $\boldsymbol{p}_n = (p_{n;0}, p_{n;1}, \dots, p_{n;k})$ in $\mathcal{L}_n$ such that for all i, j we have $|p_{n;i}\,p_{n;j}| \to |p_i\,p_j|$ as $n \to \infty$ and for any n and $i \neq j$, we have $|p_{n;i}\,p_{n;j}| > \frac{\pi}{2}$. Assume $\boldsymbol{p}$ be the limit array in $\mathcal{L}$ then there is an isometri embedding $\mathbb{S}^k \hookrightarrow \mathcal{L}$ which contains all points of array $\boldsymbol{p}$ in the image.*

[5] i.e. a spherical suspension over $\mathbb{HP}^m$

[6] i.e. ε-close for some $\varepsilon = \varepsilon(\kappa, m)$

Part VI

Remarks

Chapter 23

Solutions

Exercise 9.9.3. Consider the angle A in the plane of measure $\pi - \alpha$. Note that A is CAT[0]. Therefore by Reshetnyak gluing theorem 9.9.1, by gluing a side of A to γ_1 in $\mathcal{U}$ we obtain a CAT[0] space, say $\mathcal{U}'$.

Note that γ_2 together with the other side of A forms a both sides infinite geodesic, say γ in $\mathcal{U}'$. In particular, γ is a convex set isometric to $\mathbb{R}$.

Glue a half-plane along its boundary to γ. By Reshetnyak gluing theorem 9.9.1 the obtained space is CAT[0].

It remains to note that this space can be obtained directly by gluing $\mathcal{U}$ to with Q along γ_1 and γ_2.

Exercise 1.2.2. Let α be a closed curve in $\mathbb{S}^2$ of length $\leqslant 2\pi$. We wish to prove that it's contained in a hemisphere in $\mathbb{S}^2$. By approximation it's clearly enough to prove this for smooth curves of length $< 2\pi$ with transverse self-intersections. Furthermore, by changing such a curve near its self-intersection points it can be approximated with respect to the Hausdorff distance by simple closed curves. Thus, without loss of generality we can assume that α is a simple closed curve of length $< 2\pi$. By Crofton's formula we have that

$$L(\alpha) = \frac{1}{4}\int_{\mathbb{S}^2} \#(\alpha \cap v^\perp)\boldsymbol{d}_v \operatorname{vol}_2$$

Obviously, if $\#(\alpha \cap v^\perp) = 0$ then α is contained in one of the hemispheres determined by $v^\perp$. By the intermediate value theorem the same holds true if $\#(\alpha \cap v^\perp) = 1$. Suppose $\#(\alpha \cap v^\perp) \geqslant 2$ for all $v \in \mathbb{S}^2$. Then Crofton's formula implies that $L(\alpha) \geqslant \frac{1}{4}\int_{\mathbb{S}^2} 2 = 2\pi$. $\square$

Exercise 2.3.4.

Given a metric graph Γ define $P_k(\Gamma)$ as follows. Let Γ^b be the barycentric subdivision of Γ with the natural metric. For any two

adjacent vertices $p, q \in \Gamma^b$ substitute the edge $[pq]$ by a countable collection of intervals $\{I_i\}_{i\geqslant 1}$ of length $|pq| + \frac{|pq|}{2^k i}$ where one end of each I_i is glued to p and the other to q. Note that the resulting space $P_k(\Gamma)$ is again a metric graph with an inner metric.

Let $\mathcal{X}_0 = [0,1]$ and define $\mathcal{X}_k$ for $k \geqslant 1$ inductively as $\mathcal{X}_k = P_k(\mathcal{X}_{k-1})$.

Let $\mathcal{Y}_k$ be the set of vertices of $\mathcal{X}_k$ with the induced metric. By construction the inclusion $\mathcal{Y}_k \subset \mathcal{Y}_{k+1}$ is distance preserving.

Let $\mathcal{Y}_\infty = \cup_{k\geqslant 1}\mathcal{Y}_k$ with the obvious metric and let $\mathcal{Y} = \bar{\mathcal{Y}}_\infty$ be its metric completion. Then $\mathcal{Y}$ is a length space since it satisfies the almost midpoint property. But it is not hard to see that no two distinct points in $\mathcal{Y}$ can be connected by a shortest geodesic. $\square$

Exercise 2.3.9. The following example is from [30].

Consider the following subset of $\mathbb{R}^2$:

$$\mathcal{X} = (0,1] \times \{0\} \cup (0,1] \times \{1\} \cup {}_{n\geqslant 1}\{1/n\} \times [0,1]$$

Consider the induced inner metric on $\mathcal{X}$. It's obviously locally compact and geodesic. However, it's immediate to check that its metric completion $\bar{\mathcal{X}} = [0,1] \times \{0\} \cup [0,1] \times \{1\} \cup {}_{n\geqslant 1}\{1/n\} \times [0,1]$ is neither. $\square$

Besicovitch inequality 7.8.2. Fix $\varepsilon > 0$ and let $f_1, f_2 \ldots : \mathbb{E}^m \looparrowright \mathcal{X}$ be the short submaps such that

$$\operatorname{Im}\Phi \subset \bigcup_n \operatorname{Im} f_n$$

and

$$\sum_n \operatorname{vol}_m(\operatorname{Dom} f_n) < \operatorname{LongMes}_m(\operatorname{Im}\Phi) + \varepsilon.$$

Consider the functions $\psi^i = |A^i{}_*|$ and the map $\boldsymbol{\psi} = (\psi^1, \ldots \ldots, \psi^n)\colon \mathcal{X} \to \mathbb{R}^n$. Note that for each i and n, the composition $\psi^i \circ f_n$ is 1-Lipschitz. It follows that

$$\begin{aligned}|[\boldsymbol{d}_x(\boldsymbol{\psi} \circ f_n)]^{\wedge n}| &\leqslant \operatorname{lip}(\psi^1 \circ f_n) \cdots \operatorname{lip}(\psi^m \circ f_n) \leqslant \\ &\leqslant 1.\end{aligned}$$

for almost all $x \in \operatorname{Dom} f_n$.

Note that

$$\operatorname{Im}\boldsymbol{\psi} \circ \Phi \supset [0, a^1] \times \ldots \times [0, a^n].$$

Applying Federer's area formula, for $\boldsymbol{\psi}\circ f_n\colon \mathbb{R}^m \looparrowright \mathbb{R}^m$, we get

$$\begin{aligned}\operatorname{vol}_n(\operatorname{Dom} f_n) &= \int\limits_{\operatorname{Im}\boldsymbol{\psi}\circ f_n} |[\boldsymbol{d}_x(\boldsymbol{\psi}\circ f_n)]^{\wedge n}|\cdot \boldsymbol{d}_x \operatorname{vol}_m \leqslant \\ &\leqslant \operatorname{vol}_n(\operatorname{Im}\boldsymbol{\psi}\circ f_n)\end{aligned}$$

$$\begin{aligned}\operatorname{LongMes}_m \mathcal{X} + \varepsilon &\geqslant \sum_n \operatorname{vol}_n(\operatorname{Dom} f_n) \geqslant \\ &\geqslant \sum_n \int\limits_{\operatorname{Dom} f_n} \big|[\boldsymbol{d}_x(\boldsymbol{\psi}\circ f_n)]^{\wedge n}\big|\cdot \boldsymbol{d}_x \operatorname{vol}_n = \\ &= \sum_n \operatorname{vol}_n(\operatorname{Im}\boldsymbol{\psi}\circ f_n) \geqslant \\ &\geqslant \operatorname{vol}_n[\boldsymbol{\psi}(\mathcal{X})].\end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, the result follows. □

Exercise ??. It is sufficient to show that midpoint z of $[pq]$ lies in $\mathcal{X}$. Take a sequnce of points $z_n \in \mathcal{X}$ such that $z_n \to z$ as $n \to \omega$.

Arguing by contradiction, assume that $z_\omega \notin \mathcal{X}$. Then, according to Lemma 4.2.3, there is a subsequence (z'_n) of (z_n) such that $z'_n \to z' \neq z$ as $n \to \omega$. Clearly z' is a midpoint for p and q.

According to 4.3.2 $\mathcal{X}^\omega$ is geodesic. Choose two geodesics $[pz']$ and $[z'q]$; together they form a geodesic from p to q in $\mathcal{X}^\omega$ which is distinct from $[pq]$, a contradiction. □

Exercise ??. Applying partiiton of unity, we may assume that support of f lies in the domain Ω which admits a bi-Lipschitz distance embedding $\boldsymbol{a}\colon \mathcal{L} \to \mathbb{R}^k$. Choose sufficiently big constant; $c \geqslant$??? will do. Define

$$\varphi(\boldsymbol{x}) = \min\{\, c\cdot||\boldsymbol{a}p| - \boldsymbol{x}| + f(p) \mid p \in \Omega \,\}.$$

Exercise ??. Let $\varphi \overset{can}{=\!=} F\circ|\boldsymbol{a}_*|$ and $\psi \overset{can}{=\!=} G\circ|\boldsymbol{b}_*|$; clearly,

$$\begin{aligned}\boldsymbol{d}_p\varphi(v) &= \sum_i \partial_i F\cdot(\boldsymbol{d}_p|a^i{}_*|)(v),\\ \boldsymbol{d}_p\psi(v) &= \sum_i \partial_j G\cdot(\boldsymbol{d}_p|b^j{}_*|)(v).\end{aligned}$$

Applying the definition of gradient (16.1.1), Theorem 14.6.1 and the identities above, we get that for any choice of geodesics $[pa^i]$ the fol-

lowing holds

$$\begin{aligned}
\boldsymbol{d}_p\varphi(\nabla_p\psi) &= \sum_i \partial_i F\cdot(\boldsymbol{d}_p|a^i{}_*|)(\nabla_p\psi) \leqslant \\
&\leqslant -\sum_i \partial_i F\cdot\langle \mathrm{dir}[pa^i], \nabla_p\psi\rangle \leqslant \\
&\leqslant -\sum_i \partial_i F\cdot\boldsymbol{d}_p\psi(\mathrm{dir}[pa^i]) = \\
&= -\sum_{i,j} \partial_i F\cdot\partial_j G\cdot(\boldsymbol{d}_p|b^j{}_*|)(\mathrm{dir}[pa^i]) \leqslant \\
&\leqslant \sum_{i,j} \partial_i F\cdot\partial_j G\cdot \cos\measuredangle^{\kappa}(p\,{}^{a^i}_{b^j}) = \\
&= \tilde{\boxdot}^{\kappa}_p(\varphi,\psi)
\end{aligned}$$

□

Exercise 17.5.2. Recall that given two vectors $v, w \in \mathrm{T}_p$ we write $v + w = 0$ if $|v| = |w|$ and $\measuredangle(v, w) = \pi$.

According to ???, for almost all $t \in \mathbb{I}$, the right and left derivatives $\alpha^+(t), \alpha^-(t) \in \mathrm{T}_{\alpha(t)}$ are defined and $\alpha^+(t) + \alpha^-(t) = 0$. In particular,

$$\langle \nabla_p f, \alpha^+(t)\rangle + \langle \nabla_p f, \alpha^-(t)\rangle \overset{a.e.}{=\!=} 0$$

By ???, $f \circ \alpha$ is differentiable for almost all $t \in \mathbb{I}$. Therefore

$$\boldsymbol{d}_p f(\alpha^+(t)) + \boldsymbol{d}_p f(\alpha^-(t)) \overset{a.e.}{=\!=} 0.$$

Since

$$\langle \nabla_p f, \alpha^\pm(t)\rangle \geqslant \boldsymbol{d}_p f(\alpha^\pm(t)),$$

we get the result.

Exercise ??. Consider space $\hat{\mathcal{L}} = \mathcal{L} \times \{-1, +1\}$ with involution $\psi\colon (x, s) \mapsto (x, -s)$. Let $\sim$ be the minimal equivalence relation on $\hat{\mathcal{L}}$ such that $(x, s) \approx (\iota(x), -s)$ for any $x \in \partial\mathcal{L}$.

According to the Gluing theorem (**??**), $\hat{\mathcal{L}}/\sim \in \mathrm{CBB}^m[\kappa]$. Note that ψ induce an isometry on $\hat{\mathcal{L}}/\sim$.

Finally notice that $\mathcal{L}/\iota = (\hat{\mathcal{L}}/\sim)/\psi$ and apply Theorem 8.1.9.

Exercises 8.4.2 and 9.5.2 By the definition of Busemann function,

$$\begin{aligned}
\exp(\sqrt{-\kappa}\cdot \mathrm{bus}_\gamma) &= \exp \lim_{t\to\infty} \sqrt{-\kappa}\cdot(|\gamma(t)_{*|-t}) \\
&= \lim_{t\to\infty} \left(\exp\sqrt{-\kappa}(d_{\gamma(t)} - t) + \exp\sqrt{-\kappa}\cdot(-d_{\gamma(t)} - t)\right) \\
&= \lim_{t\to\infty} 2\cosh\sqrt{-\kappa}d_{\gamma(t)} \exp\sqrt{-\kappa}\cdot(-t).
\end{aligned}$$

By the function comparison definitions of CAT[κ] (9.5.1b) or CBB[κ] (8.4.1b), for any $p \in \mathcal{U}$ the function $f = \cosh\sqrt{-\kappa} \circ |p{*}|$ satisfies $f'' + \kappa{\cdot}f \geqslant 0$ (respectively $f'' + \kappa{\cdot}f \leqslant 0$). The result follows.

Exercise 9.6.5. Without loss of generality, we may assume that $p \in$ $\in K$.

If $|K\,x| \geqslant \pi$ then set $\Psi(x) = p$.

Otherwise, if $|K\,x| < \pi$, by Closest-point projection lemma 9.6.3, there is unique point $x^* \in K$ that minimizes distance to x; that is, $|x^*\,x| = |K\,x|$. Let us define ℓ_x, φ_x and ψ_x using the floowing identities:

$$\begin{aligned} \ell_x &= |p\,x^*|, \\ \varphi_x &= \tfrac{\pi}{2} - |x^*\,x|, \\ \sin\psi_x &= \sin\varphi_x{\cdot}\sin\ell_x, \quad 0 \leqslant \psi_x \leqslant \tfrac{\pi}{2}. \end{aligned}$$

Set

$$\Psi(x) = \operatorname{geod}_{[p\,x^*]}(\psi_x).$$

Note that Ψ is a retraction to K; that is, $\Psi(x) \in K$ for any $x \in \mathcal{U}$ and $\Psi(a) = a$ for any $a \in K$.

Let us show that Ψ is short. Given $x, y \in \mathrm{B}(K, \frac{\pi}{2})$, set

$$\begin{aligned} x' &= \Psi(x) & y' &= \Psi(y) \\ r &= |x\,y| & r' &= |x'\,y'| \\ d &= |x^*\,y^*| & \alpha &= \measuredangle^1(p\,{}^{x^*}_{y^*}) \end{aligned}$$

Note that

❶ $$\cos r \leqslant \cos\varphi_x{\cdot}\cos\varphi_y - \cos d{\cdot}\sin\varphi_x{\cdot}\sin\varphi_y.$$

Indeed, if $x, y \notin K$, then $\measuredangle[x^*\,{}^{x}_{y*}], \measuredangle[y^*\,{}^{y}_{x*}] \geqslant \frac{\pi}{2}$ and the inequality ❶ follows from the Arm lemma (9.12.9). If $x \in K$ and $y \notin K$, we get ❶, by angle comparison (9.3.1d) since $\measuredangle[y^*\,{}^{y}_{x*}] \geqslant \frac{\pi}{2}$. The same way ❶ is proved in case $x \notin K$ and $y \in K$. Finally, if $x, y \in K$, $\varphi_x = \varphi_y = \frac{\pi}{2}$ and $r = d$; i.e., the inequality trivially holds.

Further note that

$$\cos\alpha = \frac{\cos d - \cos\ell_x{\cdot}\cos\ell_y}{\sin\ell_x{\cdot}\sin\ell_y}.$$

Applying angle-sidelength monotonicity (9.3.2) we get

$$\begin{aligned} \cos r' &\geqslant \cos\psi_x{\cdot}\cos\psi_y - \cos\alpha{\cdot}\sin\psi_x{\cdot}\sin\psi_y = \\ &= \cos\psi_x{\cdot}\cos\psi_y - (\cos d - \cos\ell_x{\cdot}\cos\ell_y){\cdot}\sin\varphi_x{\cdot}\sin\varphi_y \geqslant \\ &\geqslant \cos\psi_x{\cdot}\cos\psi_y - \cos d{\cdot}\sin\varphi_x{\cdot}\sin\varphi_y \end{aligned}$$

Note that $\psi_x \leqslant \varphi_x$ and $\psi_y \leqslant \varphi_y$; in particular,

$$\cos\varphi_x \cdot \cos\varphi_y \leqslant \cos\psi_x \cdot \cos\psi_y.$$

Hence

$$\cos r' \geqslant \cos r;$$

i.e., the restriction $\Psi|\mathrm{B}(K, \frac{\pi}{2})$ is short. Clearly Ψ is continuous, since the complement of $\mathrm{B}(K, \frac{\pi}{2})$ is mapped to p, we get that Ψ is short; i.e.,

❷ $$r' \leqslant r$$

for any $x, y \in \mathcal{U}$.

If we have equality in ❷ then

$$\cos\ell_x \cdot \cos\ell_y \cdot \sin\varphi_x \cdot \sin\varphi_y = 0.$$

If $K \subset \mathrm{B}(p, \frac{\pi}{2})$ then $\ell_x, \ell_y < \frac{\pi}{2}$; which implies that $x \in K$ or $y \in K$. Without loss of generality we may assume that $x \in K$.

It remains to show that if $y \notin K$ then the inequality ❷ is strict. If $|K\,y| \geqslant \frac{\pi}{2}$ then ❷ holds since the left hand side is $< \frac{\pi}{2}$, while right hand side is $\geqslant \frac{\pi}{2}$. If $|K\,y| < \frac{\pi}{2}$ then $\varphi_y > 0$ and clearly $\psi_y < \varphi_y$, hence the inequality ❷ is strict. □

We fail to find a transparent geometric proof of the statement above. Below you will find a geometric way to think about the construction; compare to the construction in the proof of Kirszbraun's theorem (10.4.1).

Geometric interpretation of the map Ψ. Set $\mathring{\mathcal{U}} = \mathrm{Cone}\,\mathcal{U}$; denote by $\mathring{K}$ the subcone of $\mathring{\mathcal{U}}$ spanned by K. The space $\mathcal{U}$ can be naturally identified with the unit sphere in $\mathring{\mathcal{U}}$; i.e., the set

$$\left\{ z \in \mathring{\mathcal{U}} \;\middle|\; |z| = 1 \right\}.$$

According to ??? $\mathring{\mathcal{U}} \in \mathrm{CBA}[0]$. Note that $\mathring{K}$ forms a convex closed subset of $\mathring{\mathcal{U}}$. According to ???, for any point x there is unique point $\hat{x} \in \mathring{K}$ which minimize the distance to x; i.e., $|\hat{x}\,x| = |K\,x|$. (If $|\hat{x}| \neq 0$ then in the notations above we have $x^* = \frac{1}{|\hat{x}|} \cdot \hat{x}$.)

Consider the ray $t \mapsto t \cdot p$ in $\mathring{\mathcal{U}}$. According to ???, for given $s \in \mathring{\mathcal{U}}$ the geodesics $\mathrm{geod}_{[s\ t\cdot p]}$ converge as $t \to \infty$ to a ray, say $\alpha_s \colon [0, \infty) \to$ $\to \mathring{\mathcal{U}}$.

Note that if $|x| = 1$ then $|\hat{x}| \leqslant 1$. By assumption for any $a \in K$ the function $t \mapsto |\alpha_a(t)|$ is monotonicity increasing. Therefore there

is unique value $t_x \geqslant 0$ such that $|\alpha_{\hat{x}}(t_x)| = 1$. Consider the map $\Psi\colon \mathcal{U} \to K$ defined as

$$\Psi(x) = \alpha_{\hat{x}}(t_x).$$

Exercise 9.12.7. *(Easier way.)* Let $(t,s) \mapsto \gamma_t(s)$ be the line-of-sight map for α from $\alpha(0)$, and $(t,s) \mapsto \tilde{\gamma}_t(s)$ be the line-of-sight map for $\tilde{\alpha}$ from $\tilde{\alpha}(0)$. Consider the map $F\colon \operatorname{Conv}\tilde{\alpha} \to \mathcal{U}$ such that $F\colon \tilde{\gamma}_t(s) \mapsto$ $\mapsto \gamma_t(s)$.

Show that F majorizes α and conclude that F is distance-preserving.

(Harder way.) Prove and apply the following lemma together with the Majorization theorem.

23.0.1. Lemma. *Let α and β be two convex curves in $\mathbb{M}^2[\kappa]$. Assume*

$$\operatorname{length}\alpha = \operatorname{length}\beta < 2\cdot\varpi^\kappa$$

and there is a short bijecction $f\colon \alpha \to \beta$. Then f is an isometry.

Exercise 8.3.5. Read [97].

Exercise 13.5.3. In the proof we apply the following lemma from [49]; it follows from the disjoint discs property.

23.0.2. Lemma. *Let $\mathcal{S}$ be a simplicial complex which is an m-dimensional homology manifold for some $m \geqslant 5$. Assume all the vertices of $\mathcal{S}$ have simply connected links. Then $\mathcal{S}$ is a topological manifold.*

It is sufficient to construct a simplicial complex $\mathcal{S}$ such that

- ⋄ $\mathcal{S}$ is a closed $(m-1)$-dimensional homology manifold;
- ⋄ $\pi_1(\mathcal{S}\backslash\{v\}) \neq 0$ for some vertex v in $\mathcal{S}$;
- ⋄ $\mathcal{S} \sim \mathbb{S}^{m-1}$; that is, $\mathcal{S}$ is homotopy equivalent to $\mathbb{S}^{m-1}$.

Indeed, assume such $\mathcal{S}$ is constructed. Then the suspension $\mathcal{R} =$ $= \operatorname{Susp}\mathcal{S}$ is an m-dimensional homology manifold with a natural triangulation coming from $\mathcal{S}$. By Lemma 23.0.2, $\mathcal{R}$ is a topological manifold. According to generalized Poincaré conjecture, $\mathcal{R} \simeq \mathbb{S}^m$; that is $\mathcal{R}$ is homeomorphic to $\mathbb{S}^m$. Since $\operatorname{Cone}\mathcal{S} \simeq \mathcal{R}\backslash\{s\}$ where s denotes a south pole of the suspension and $\mathbb{E}^m \simeq \mathbb{S}^m\backslash\{p\}$ for any point $p \in \mathbb{S}^m$ we get

$$\operatorname{Cone}\mathcal{S} \simeq \mathbb{E}^m.$$

Let us construct $\mathcal{S}$. Fix an $(m-2)$-dimensional homology sphere Σ with a triangulation such that $\pi_1\Sigma \neq 0$. According to [83] an example of that type exists for any $m \geqslant 5$.

Remove from Σ one $(m-2)$-simplex. Denote the obtained complex by Σ'. Since $m \geqslant 5$, we have $\pi_1\Sigma = \pi_1\Sigma'$.

Consider the product $\Sigma' \times [0,1]$. Attach to it the cone over its boundary $\partial(\Sigma' \times [0,1])$. Denote by $\mathcal{S}$ the obtained simplicial complex and by v the tip of the attached cone.

Note that $\mathcal{S}$ is homotopy equivalent to the spherical suspension over Σ which is a simply connected homology sphere and hence is homotopy equivalent to $\mathbb{S}^{m-1}$. Hence $\mathcal{S} \sim \mathbb{S}^{m-1}$.

The complement $\mathcal{S}\backslash\{v\}$ is homotopy equivalent to Σ'. Therefore

$$\pi_1(\mathcal{S}\backslash\{v\}) = \pi_1\Sigma' = \pi_1\Sigma \neq 0.$$

That is, $\mathcal{S}$ satisfies the conditions above.

Exercise 8.3.4. See the construction of Urysohn's space [61, $3.11\frac{3}{2}_+$].

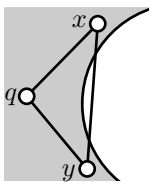
Exercise 12.3.2. Denote by Ω the interior of K; that is $\Omega = K\backslash S$. Since K is connected and its boundary is smooth then so is Ω.

Recall that $k_1(p), \dots, k_{m-1}(p)$ denote the principle curvatures taken in the nondecreasing order of the surface S at the point p.

"if"-part. Note that if S is convex then K is locally convex; that is any point $p \in K$ admits a neighborhood $U \ni p$ such that $U \cap K$ is convex.

Since K is connected, by Theorem 9.6.8, K is convex. It follows that induced length metric on K coincides with the Euclidean metric. In particular (3+1)-point comparison holds for any quadruple of points in K.

"Only-if"-part. If S is not convex then there is a triangle $[qxy]$ in $\mathbb{E}^m$ such that two sides $[qx]$ and $[qy]$ lie in Ω, but the side $[xy]$ does not completely lie in K. Such a triangle can be found in the plane spanned by the normal vector $\nu(p)$ and the first principle direction at a point $p \in S$ where $k_1(p) < 0$.



Clearly

❸ $$|xy|_K > |xy|_{\mathbb{E}^m}.$$

On the other hand $[qx]$ and $[qy]$ form geodesics in K and $\mathbb{E}^m$. Since q lies in the interior of K,

$$\measuredangle[q\,{}^{x}_{y}]_K = \measuredangle[q\,{}^{x}_{y}]_{\mathbb{E}^m}.$$

If K forms a CBB[0] space then by hinge comparison (8.3.1d) we have

$$|xy|_K \leqslant |xy|_{\mathbb{E}^m}.$$

The latter contradicts ❸. □

Exercise ??. If Ω is not two-convex then there is a plane Π in $\mathbb{E}^3$ which pass through a vertex v of K such that punctured neightborhood of v in Π lies in Ω. Choose a plane Π' parallel and very close to Π

which cuts from the complement of Ω a little piramid S with vertex v. Consider a small triangle $\triangle$ in Π' wich surrounds the base of S. Note that $\triangle$ is a geodesic triangle in Ω^* for which the point-on-side comparison 9.3.1c fails. That is, $\operatorname{curv}_v \Omega^* \not\leq 0$.

"if" part. Since Ω is two-convex, by Proposition **??**, any point v on the boundary of K admits a conic neighborhood U in K such that the intersection $U \cap \Omega$ is formed by a finite collection of simply connected components.

It follows that any point $\Omega^* \backslash \Omega$ is locally isometric to a cone over spherical polygons. Moreover since Ω is two-convex, each polygon does not contain a closed hemisphere in its interior. By Lemma ??? each of these spherical polygon is CBA[1]. Therefore, by cone construction (11.4.1a) we get that Ω^* is locally CBA[0]. $\square$

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Chapter 24

Remarks for AKP

1. Eventially, everywhere lower index in the arrays of points (and other things) will be changed to upper index. The reasons (1) coordinates in Riemannian geometry usually come with upper idexes and it is more natural to write $x^i = |a^i{}_*|$ (2) some times we need sequences of arrays, then it is better to write a^i_n than $a_{i,n}$. There is a little problem once you want to use prime — I would suggest to use $\acute{a}$ instead of a' in such cases.
2. I plan to change the notation of GH-convergence from $X_n \xrightarrow[\text{GH}]{} X$ to $X_n \xrightarrow{\tau} X$; here τ stays for the topology on $\bigsqcup X_i$ which defines Gromov–Hausdorff convergence. The main reason is that for induced convergences we use $\to$, or $\rightrightarrows$ $\rightharpoonup$ and it looks very unnatural.
3. I removed \olim and \oto everywhere. Instead we write $\lim_{n\to\omega}$ or $x_n \to x_\omega$ as $n \to \omega$.
4. DONE: \underline $\to$ \ushort
5. DONE: change "induced" to "inherited/intrinsic"
6. T to ℓ.
7. At the moment proj (\proj) and ϖ (\varpi) are nearly identical...
8. $\sup_{x\in X}\{f(x)\}$ to $\sup\{\, f(x) \mid x \in X \,\}$; the same for inf, max, min and so on.
9. extremal subset $\to$ extremal set
10. remove "native"
11. mid-points OR midpoints?
    S: midpoints.
12. We seem to agree to use terms "geodesic" mean minimizing geodesic; "local geodesic" means geodesic; "unit-speed

geodesic" and the term geodesic can be also used for the image of minimizing geodesic the class of all constant speed reparametrizations of a geodesic

13. I think it is better to change everywhere $\mathbb{Z}_{\geqslant 0}$ to $\mathbb{Z}_+$ and $\mathbb{R}_{\geqslant 0}$ to $\mathbb{R}_+$ it suits better with $\mathbb{S}_+$ (and $\mathbb{S}_{\geqslant 0}$ does not have sense)
14. Should we change to Bourbaki-notation of real intervals? As $]a, b[$ instead of (a, b)?
15. We use ∂ for the set of supporting vectors, for the boundary, and S. wants to use it for frontier(=rel. boundary). We also use Int for interior — it seems more natural to use some letters for frontier...
16. Stephanie suggests to change ∂ to ∂; in positve curvature might be a bit of confusion between rel. boundary and boundary. We might use $\partial_{\mathcal{X}} A$ for rel boundary of $A \subset \mathcal{X}$ and $\partial\mathcal{X}$ for boundary. I also use $\partial_p f$ for subgradient...
17. in the end we need to change $\left]x^1 x^2\right[$ to something closer to $]x^1 x^2[$.
18. DONE: $G_\delta \to$ G-delta
19. DONE: Some people write $\mathrm{B}(p, R)$ and $\overline{\mathrm{B}}[p, R]$. Maybe it is better this way — ")" reminds open interval and "]" closed...
20. in the end check usage of CBA and CBB...
21. me might write $(1{+}3)^\kappa$-point comparison or point-on-side$^\kappa$ comparison???
    S: I think we need "κ-comparison".
22. (2+2)-comparison $\to$ (2+2)-point comparison.
23. We might want to rename ultratangent cone to ultrablowup
24. I want to rename the general construction in baricentric simplexex to something else and use the name "κ-barycentric simplex" for only for arays modified distance functions $(\mathrm{md}^\kappa \circ |x^i{}_*|)$.
    I think of a good name for general construction, but can not come up with anything better than "argmin map".
    Any suggestions?
    S: "minpoint map".
25. $\notin$ and others split wrong at the end of the line...
26. I think to change expressions like $|x - y| < \varepsilon$ to $x \approx y \pm \varepsilon$. (I noticed that the notation $x = y \pm \varepsilon$ is used for $x = y$ with absolute error ε, but we use "$\pm$" for "$+$" ar "$-$", so "$\approx$" instead of "$=$" will remind that meaning is different).
    I would also use $\approx$ with little o notation.

A. hates $o(x)$ notation. We need a notation for $o(x)/x$. One candidate is $\dot{o}_x$, so for $o(x\cdot \ln x)$ we could write $\dot{o}_x\cdot x\cdot \ln x$ Perelman use $\varkappa(x)$. Can you suggest anything better?

A possible solution is to use $\lesseqgtr$ or $\fallingdotseq$for example

$$x \lesseqgtr y \pm \varepsilon(t)$$

$$x \fallingdotseq y \pm \varepsilon(t)$$

27. use term "linearly differentiable" if differential is defined and linear.
28. vol^m or vol_m?
29. $\imath \to \iota$?
30. change supporting vector to maybe subnormal?
31. subergraph $\to$ epigraph; subgraph $\to$ hypograph?
32. in line use either $\frac{\varpi^\kappa}{2} \to \varpi^\kappa/2$
33. what is better?

$$\measuredangle[p\,{}^q_x] = \lim_{\substack{\bar{x}\to p\\ \bar{x}\in\,]px]}} \tilde{\measuredangle}^\kappa(p\,{}^q_{\bar{x}})$$

or

$$\measuredangle[p\,{}^q_x] = \lim_{\bar{x}\to p}\{\,\tilde{\measuredangle}^\kappa(p\,{}^q_{\bar{x}})\mid \bar{x}\in\,]px]\,\}$$

34. polygon line OR broken geodesic
35. ε-Hausdorff approximation OR ε-isometry — BBI use ε-isometry.
36. metion authors around cites
37. unitspeed $\to$ unit-speed and constantspeed $\to$ constant-speed?
38. do we need pregeodesic?
39. better name for Str?
40. Fix pagebrake before \qedsf and \claimqedsf
41. Define distance maps and their liftings
42. What if we change Gromov–Hausdorff to Hausdorff? By the way, Hausdorff defined so called closed and open convergences. They are better than Hausdorff convergence, but the later became mor popular by stupid reason. It also made Gromov define convergence wrong way.
43. vertexes $\to$ verteces
44. subnormal $\to$ normal to supporting hyperplane
45. sequence is a "lifting" or "liftings"?
46. CHECK Lipschitz/locally Lipschitz and co-Lipschitz/ locally co-Lipschitz.

47. Find a better way to say "noncollapsing", $\mathcal{L}_n \xrightarrow[\text{GH}]{} \mathcal{L}$ such that $\mathcal{L}_n, \mathcal{L} \in \mathrm{CBB}^m[\kappa]$ does not emphasize sufficiently that $\dim \mathcal{L}_n = \dim \mathcal{L}$.
48. unify "if" and "only if" parts.
49. good name for radial curves — base point, curvature, and starting point???
50. Remove new par after claim
51. $A \times {}_f B$ to $A \times_f B$.
52. Tangent space to tangent cone — we use tangent κ-cone sometimes.
53. semi-continuous semi-concave OR semicontinuous semiconcave.
54. N.B. or Pay attention?
55. i.e. or that is it?
56. Reshetnyak's puth pastry $\to$ Reshetnyak's building
57. $[xy] \to [x, y]$.
58. Chauchy sequence $\to$ sequence conveges in itself
59. What about $\mathrm{CBA}[-\infty]$ spaces?
60. Do we need R-length spaces?
61. Eventually we will need to change notation for the subfunctions. They are essentially continous partially defined functions. I would say "function" for "partially defined functions" and "total function" for "function".
    I propose the following system of notations:
    $f\colon X \to Y$ for partially defined function
    $f\colon X \mapsto Y$ for function
    $f\colon X \mapsto\!\!\!\to Y$ for onto functon
    $f\colon X \leftrightarrow Y$ for injective partially defined function
    $f\colon X \mapsto\!\!\!\leftrightarrow Y$ for bijection
    and so on.

Notations:

1. p' usually denotes $p' \approx p$; and $\acute{p}^i \approx p^i$
2. $\bar{x}$ denotes a point on $]px]$ (for an evident choice of p)
3. The origin of tangent cone T_p is denoted by o or o_p. BUT it would be more intuiteve to denote it by something like $\vec{p}$ or $\dot{p}$ — you may also write $\log[pp]$, but that is really stupid. It would be also good to use the same modifier for shortcut like $\dot{q} = \lg pq$
4. \map and \map[1] gives Φ, \map[2] gives Ψ, \map[3] gives Θ.
5. $\mathrm{B}(S, R)$ to denote an R-neighborhood of a set S.